Mass Transfer-II

K. A. Gavhane

NIRALI PRAKASHAN

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MASS TRANSFER – II

DEGREE COURSE IN CHEMICAL ENGINEERING / PETROCHEMICAL ENGINEERING OF ALL UNIVERSITIES IN INDIA

REVISED EDITION

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PREFACE TO THE NINTH EDITION

I am very happy to present the ninth edition of the book **MASS TRANSFER-II** to the students of degree course in Chemical Engineering. The matter is presented in a simple and lucid language to understand the concepts underlying each topic clearly. A fairly large number of illustrative examples with graphical solutions (wherever necessary) are given on each topic.

The book includes 117 illustrative examples which will positively help the students to bridge the gap between theory and applications.

I am thankful to my colleagues, friends and students for appreciating and liking my books.

I am grateful to the publisher Shri Dineshbhai Furia and Shri Jigneshbhai Furia, and entire staff of Nirali Prakashan for their efforts to bring out this book in such a neat and presentable form.

All constructive suggestions for further improvement of this book will be gratefully appreciated.

I hope that the book would fill a long felt need for a comprehensive book on Mass Transfer - II.

I am sure that students as well as staff members will find this book very easy and useful.

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Chapter ... 1

DISTILLATION

Distillation is *a unit operation in which the constituents of a liquid mixture (solution) are separated using thermal energy.* Basically, the difference in **vapour pressures** of different constituents at the same temperature is responsible for such a separation. This unit operation is also termed as **fractional distillation** or **fractionation**. With this technique it is possible to separate a liquid mixture into its components in almost pure form and this fact has made distillation perhaps the most important of all mass transfer operations.

In distillation, the phases involved are : liquid and vapour or gas (the vapour phase is created by supplying heat to the liquid) and mass is transferred from both the phases to one another, by vaporisation from the liquid phase and by condensation from the vapour phase. The net effect is an increase in composition of the more volatile component in the vapour and of the less volatile component in the liquid. The basic requirement for a separation of components by distillation is that the composition of the vapour be different from the composition of the liquid with which it is in equilibrium - the vapour is always richer in the more volatile component than the liquid from which it is formed. If the vapour composition is the same as the liquid composition, distillation technique will not effect a separation.

Distillation is commonly encountered in chemical and petroleum industries as a means of separating the liquid mixture into its component parts. The separation of a mixture of ethanol and water, production of absolute alcohol from 95% ethyl alcohol using benzene, the separation of petroleum crude into gasoline, kerosene, fuel oils, etc. are **typical examples of distillation.**

Evaporation is concerned with the separation of a solution containing a non-volatile solute and a volatile solvent whereas distillation is concerned with the separation of solutions where all the components are appreciably volatile. Thus, the separation of a brine into salt and water is an evaporation whereas the separation of a mixture of alcohol and water into its components is a distillation.

Boiling Point :

For any given pressure, a pure liquid when heated will boil or vaporise at a certain single temperature known as the boiling point of the liquid. The boiling point of a liquid increases with increase in pressure and vice versa. Boiling points at a given pressure vary greatly for different liquids. For example, the boiling point of water is 373 K (100° C), that of toluene is 383.6 K (110.6° C) and that of methanol is 337.7 K (64.7° C) at one atmosphere (101.325 kPa). [The normal boiling point is the temperature at which boiling takes place under a pressure of 101.325 kPa (1 atm)].

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In a binary mixture (a two-component system), the component with the lower boiling point or with the higher vapour pressure at a given temperature is termed as the **more volatile or lighter** and the component with the higher boiling point or with the lower vapour pressure at a given temperature is termed as the **less volatile or heavier**. In a binary system containing components A and B when we say A to be more volatile it means that the vapour pressure of pure A is higher than that of pure B at any given temperature. The composition of a binary mixture, by convention, will be expressed in terms of the concentration of the more volatile component.

Thus, in the case of a binary mixture of methanol and water, methanol is a more volatile component and water is a less volatile component. The whole mixture of methanol and water will boil somewhere between 337.7 K (64.7° C) and 373 K (100° C) at the atmospheric pressure (101.325 kPa).

Equilibrium :

- Equilibrium refers to the absence of any tendency for a change to take place/occur.
- It is a static condition in which the net transfer of material between the phases ceases for a given set of operating conditions.

Driving force :

When two phases (not at equilibrium) are brought into intimate contact, the phases will tend to approach equilibrium due to a tendency for a change to take place. The difference between the existing condition and the equilibrium condition is the driving force which causes a change. A concentration difference is the driving force for mass transfer analogous to a temperature difference for heat transfer.

Equilibrium stage :

One class of mass transfer devices consists of a series of stages through which phases flow, from one to the other, in a countercurrent fashion.

A stage consists of two steps :

(i) **Intimate contacting of two insoluble phases :** Whereby mass transfer occurs between the phases tending to bring them to equilibrium (various diffusing substances distribute themselves between the phases).

(ii) **Mechanical separation of the phases :** So any device or combination of devices wherein two insoluble phases are first brought into intimate contact whereby mass transfer occurs between the phases and the phases are then mechanically separated is referred to as a stage. A process carried out in this manner is a single-stage process. For separations requiring greater changes in concentration, the phases flow countercurrent to each through a series of stages (in each stage the phases are brought into contact, mixed, and then separated) and the process carried out in this manner is called a multistage process. Such a multistage arrangement is called a cascade. In such a case for establishing a standard to measure the performance, the equilibrium, ideal or theoretical stage concept is introduced. The ideal or theoretical stage is the one from which the phases leaving (the effluent phases)

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are in equilibrium with each other, so that no additional change in the composition results by providing a further time of contact between the phases.

Equilibrium stage is defined as the one in which two insoluble phases not at equilibrium are brought into contact, sufficient time is provided to attain equilibrium, the phases are separated and streams leave the stage in equilibrium. For a given set of operating conditions, an equilibrium stage gives maximum possible composition change, so it is also known as the ideal or theoretical stage. In actual practice, equilibrium is not achieved and hence stage efficiencies are always less than 100 per cent.

Vapour - Liquid Equilibrium :

The basic data for distillation problems are the equilibria existing between the vapour and liquid phases of the system under consideration. The equilibrium in vapour-liquid systems is governed by phase rule. In case of a binary mixture subjected to distillation, the components (C) involved are two, the phases (P) involved are two (vapour and liquid), so according to the phase rule [F = C – P + 2], the degree of freedom (F) or the number of intensive variables that can be varied independently are two. In distillation, there are four variables : temperature, pressure, vapour phase composition and liquid phase composition. Thus, if the pressure is fixed then only one variable say, for example, the liquid phase composition can be changed independently, and the temperature and vapour phase composition then follow. (Thus, the system is completely defined.)

Constant-Pressure Vapour-Liquid Equilibria

Vapour-liquid equilibrium refers to systems where in a single liquid phase is in equilibrium with its vapour. We restrict our discussion to binary systems (systems containing two chemical species).

Equilibrium in vapour-liquid systems is usually expressed in terms of the mole fraction of the more volatile component in the vapour phase (y) and the mole fraction of the same component in the liquid phase (x).

Vapour-liquid equilibrium data for a binary system at constant total pressure are represented by means by

(i) the temperature-composition diagram (the T-x-y diagram or boiling point diagram).

(ii) the equilibrium curve (x-y plot/diagram or distribution diagram).

Boiling Point Diagram :

The boiling point diagram is a plot of temperature as ordinate against compositions of liquid and vapour phases as abscissa. So, this diagram is used to show how the equilibrium vapour and liquid phase compositions vary/change with temperature. The composition of the liquid phase is expressed in terms of the mole fraction of the more volatile component in the liquid. The composition of the vapour phase is expressed in terms of the mole fraction of the more volatile component in the vapour phase.

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With the help of an equilibrium still one can determine experimentally the composition of the vapour in equilibrium with the liquid of a given composition and the results thus obtained can be shown on the temperature - composition diagram. Let us follow the whole process of boiling a binary mixture in an equilibrium still. For the situation under consideration refer to Fig. 1.1. Suppose that the liquid mixture handled consists of benzene (the more volatile component) and toluene (the less volatile/heavier component). The composition of the mixture is plotted as abscissa in terms of the mole fraction of the more volatile component and temperature of the mixture is plotted as ordinate. The mixture presented by point A is at a temperature of T_1°C and contains 50% of benzene. When we apply heating to the mixture it will start boiling at temperature T_2°C. The vapour given off by the mixture on boiling will contain more of the more volatile component so the first quantity of the vapour issuing from the binary mixture B will have the composition indicated by point C, i.e., this vapour will contain 91% benzene at temperature T_2 °C. The vapour at the point C is in equilibrium with the liquid at the point B and thus a horizontal line BC is known as a tie line. Hence, a considerable enrichment in benzene has taken place from 50% to 91%. If we remove this portion of vapours and condense it completely in a condenser then we would obtain the liquid represented by point D still containing 91% benzene. If we reheat this portion of the liquid, it will boil at T_3 °C at point E ($T_3 < T_2$). The first portion of the vapours issuing from boiling liquid at point E will contain more of the more volatile component, composition of it is represented by point F. If we condense this part of the vapour completely, we get the liquid containing almost 100% of benzene indicating further enrichment in the vapour.

If we further heat the liquid at the point B the boiling point of liquid rises to T' because of the removal of a majority of the more volatile component by vaporisation. At this temperature, the liquid phase composition is represented by point L and the vapour phase composition by point N and further heating to T'' all the liquid get vaporised to give the vapour represented by point H.

Alternatively, consider a mixture of vapours represented by point Q, when we cool it down to temperature T'', condensation will start at the point H. The first drop of the liquid formed will have the composition represented by point K (i.e., the less volatile component will tend to condense first). If we further cool the vapours to T' we will get the liquid represented by the point L and the vapour represented by the point N. This means that the vapour is richer in the more volatile component than the liquid. Hence we can say that partial vaporisation as well as partial condensation will both tend to enrich the vapour in the more volatile component.

In the process of boiling, the mixture boils over a temperature range from B to K (from the bubble point to the dew point), so the term boiling point has no meaning for the mixture and the correct term is the bubble point. **Bubble point** is the temperature at which a liquid mixture of a given composition starts to vaporise as the temperature is increased. The liquid represented by any point on the lower curve is at its bubble point and hence the lower curve is called as the saturated liquid curve or the **bubble-point (temperature) curve** (T v/s x). The temperature at which a vapour mixture on cooling first begins to condense is called as the dew point. Condensation starts at any point on the upper curve and hence it is termed as the saturated vapour curve or the **dew-point (temperature) curve** (T v/s y). The mixture between the bubble-point and dew-point curves cannot exist as a single phase, but it spontaneously separates into saturated liquid and vapour phases that are in equilibrium.

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For any pure liquid, bubble point and dew point are identical and equal to the boiling point of the pure liquid. Whereas the dew point and bubble point for a binary system are functions of its composition. The dew point and bubble point are identical at x, y = 0 and x, y = 1 as these compositions indicate pure toluene and pure benzene.

For systems which follow Raoult's law, the boiling point diagram can be constructed from the pure components vapour pressure data.

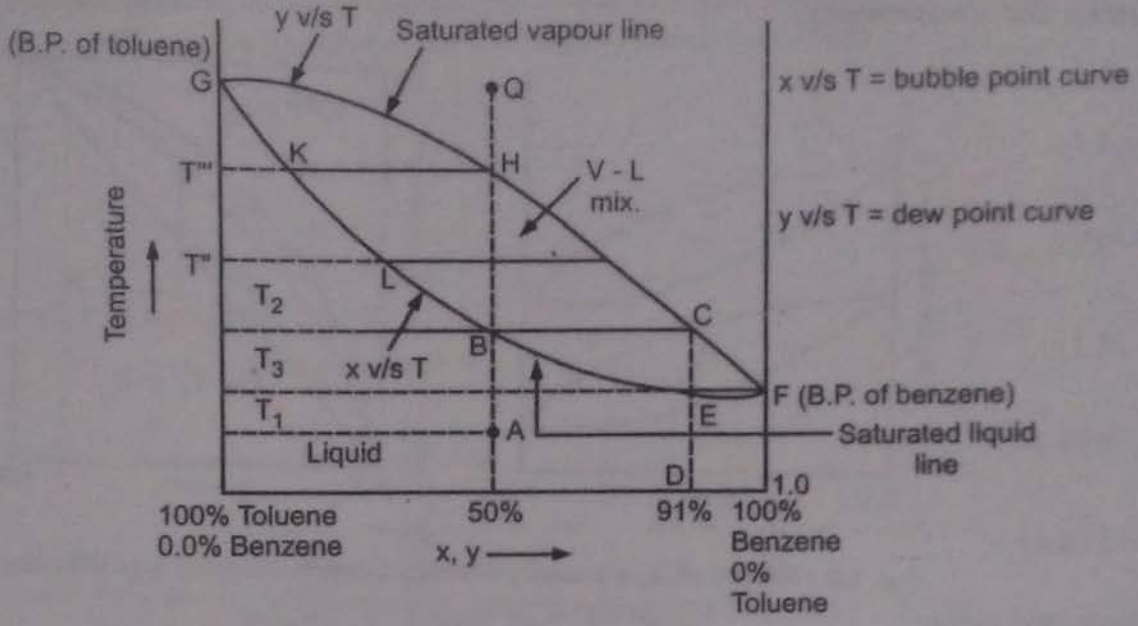

Fig. 1.1 : Boiling point diagram
(Constant pressure vapour-liquid equilibrium)

As the total pressure of the system is increased say, for example, from P_{t1} to P_{t2}, the spread between the bubble point curve and the dew point curve decreases, and the separability by distillation becomes less at high pressure, owing to decrease in the value of the relative volatility at higher pressure. The effect of increase in pressure is shown in Fig. 1.3.

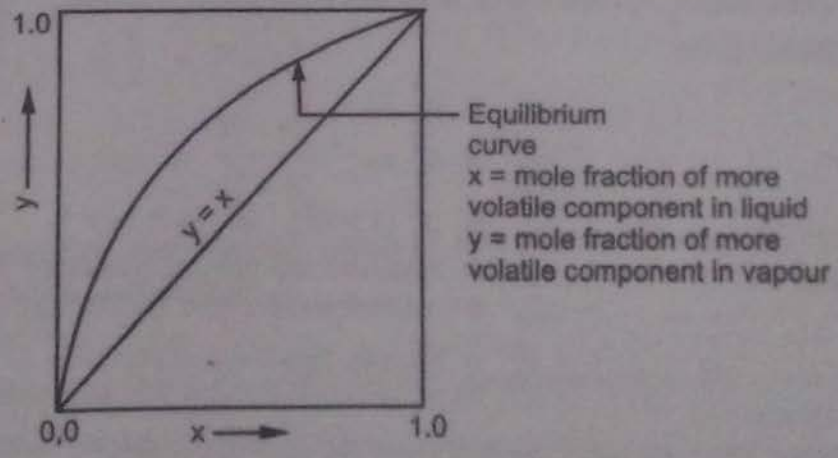

Fig. 1.2 : Equilibrium diagram (Normal)
(Constant pressure vapour-liquid equilibrium)

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A large majority of distillation operations are carried out at a constant total pressure. For distillation calculations, the equilibrium vapour-liquid composition can also be plotted as shown in Fig. 1.2 wherein the vapour phase composition (y) is plotted as an ordinate and the liquid phase composition (x) is plotted as an abscissa. Such a diagram is called as an equilibrium diagram or distribution diagram. As the vapour is richer in the more volatile component than the liquid, the equilibrium curve lies above the 45° diagonal line which is drawn for comparison.

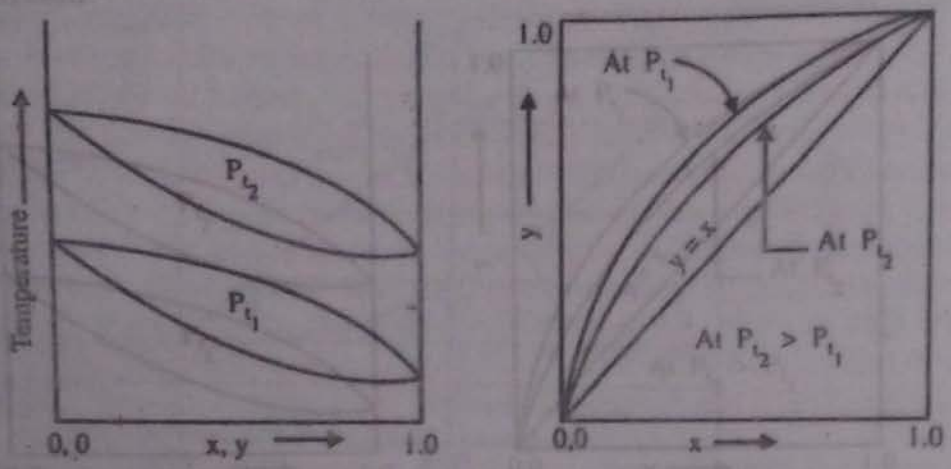

Fig. 1.3 : Effect of increased pressure : vapour-liquid equilibrium

Raoult's law :

It is commonly used for predicting the vapour-liquid equilibrium for an ideal solution in equilibrium with an ideal gas mixture from the pure component vapour-pressure data. It states that the *equilibrium partial pressure of a constituent/component in a solution at a given temperature is equal to the product of its vapour pressure in the pure state and its mole fraction in the liquid phase.*

Thus, for a binary (two component) system, if p_A is the equilibrium partial pressure of A, p_A° is the vapour pressure of A in the pure state and x_A is the mole fraction of A in the liquid phase, then

$$p_A = p_A^\circ x_A \qquad \ldots (1.1)$$

and

$$p_B = p_B^\circ x_B \qquad \ldots (1.2)$$

$$= p_B^\circ (1 - x_A), \text{ as } x_A + x_B = 1$$

where p_B = equilibrium partial pressure of B
x_B = mole fraction of B in liquid phase
p_B° = vapour pressure of pure B.

Dalton's law :

It states that *the total pressure exerted by a gas/vapour mixture is equal to the sum of the partial pressures of the components present in the gas/vapour mixture.* Thus, it expresses the additive nature of the partial pressures.

Mathematically, for a binary system :

$$P = p_A + p_B \qquad \ldots (1.3)$$

where P is the total pressure.

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For an ideal gas or vapour, the partial pressure is related to the mole fraction of the component in the gas or vapour phase by the relation :

Partial pressure = Mole fraction × Total pressure

Thus, for component A,

$$p_A = y_A \cdot P \qquad \ldots (1.4)$$

where y_A is the mole fraction of A in the vapour phase.

Knowing the vapour pressures of components A (more volatile) and B at various values of temperatures, x - y data can be generated for an ideal solution as follows :

$$p_A = p_A^o \cdot x_A \qquad \ldots (1.5)$$

$$p_B = p_B^o \cdot (1 - x_A) \qquad \ldots (1.6)$$

$$p_A = y_A \cdot P \qquad \ldots (1.7)$$

$$p_B = y_B \cdot P \qquad \ldots (1.8)$$

From Equations (1.5) and (1.7), we get

$$p_A^o \cdot x_A = y_A \cdot P \qquad \ldots (1.9)$$

$$y_A = \frac{p_A^o}{P} \cdot x_A \qquad \ldots (1.10\ a)$$

Similarly,

$$y_B = \frac{p_B^o}{P} \cdot x_B \qquad \ldots (1.10\ b)$$

But $y_B + y_B = 1$ and $x_B = 1 - x_A$

$$y_A + y_B = 1 \qquad \ldots (1.11)$$

Putting values of y_A and y_B from Equations (1.10) and (1.11), and of x_B in terms of x_A, in Equation (1.11), we get

$$\frac{p_A^o}{P} \cdot x_A + \frac{p_B^o}{P}(1 - x_A) = 1 \qquad \ldots (1.12)$$

$$p_A^o \cdot x_A + p_B^o - p_B^o \cdot x_A = P \qquad \ldots (1.13)$$

$$\therefore \quad x_A\left(p_A^o - p_B^o\right) = P - p_B^o \qquad \ldots (1.14)$$

$$\therefore \quad x_A = \frac{P - p_B^o}{p_A^o - p_B^o} \qquad \ldots (1.15)$$

Knowing x_A, the corresponding equilibrium value of the vapour phase concentration (y_A) is obtained with the help of Equation (1.10 a).

Relative volatility :

Volatility of A : It is defined as the *ratio of the partial pressure of A to the mole fraction of A in the liquid phase.*

$$\text{Volatility of A} = \frac{p_A}{x_A} \qquad \ldots (1.16)$$

Similarly,

$$\text{Volatility of B} = \frac{p_B}{x_B} \qquad \ldots (1.17)$$

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Relative volatility of component A with respect to B is *the ratio of the volatility of A (the more volatile component) to that of B*. It is also known as the volatility of A with respect to B and is denoted by the symbol α_{AB}.

$$\therefore \quad \alpha_{AB} = \frac{p_A x_B}{x_A p_B} \qquad \ldots (1.18)$$

but $P y_A = p_A$ and $P y_B = p_B$

$$\therefore \quad \alpha_{AB} = \frac{y_A x_B}{y_B x_A} \qquad \ldots (1.19)$$

$$\alpha_{AB} = \frac{y_A / y_B}{x_A / x_B} \qquad \ldots (1.20)$$

Thus, the relative volatility is the *ratio of the concentration ratio of A to B in the vapour phase to that in the liquid phase.*

$$\alpha_{AB} = \frac{y_A x_B}{y_B x_A}$$

We have, $y_B = 1 - y_A \qquad \ldots (1.21)$

$$x_B = 1 - x_A \qquad \ldots (1.22)$$

$$\therefore \quad \alpha_{AB} = \frac{y_A (1 - x_A)}{(1 - y_A) x_A} \qquad \ldots (1.23)$$

$$\alpha_{AB} (1 - y_A) x_A = y_A (1 - x_A)$$

$$\alpha_{AB} x_A - \alpha_{AB} y_A x_A = y_A - y_A x_A$$

$$\alpha_{AB} x_A = y_A + y_A x_A (\alpha_{AB} - 1)$$

$$\alpha_{AB} x_A = y_A [1 + (\alpha_{AB} - 1) x_A]$$

$$\therefore \quad y_A = \frac{\alpha_{AB} x_A}{1 + (\alpha_{AB} - 1) x_A} \qquad \ldots (1.24)$$

Dropping subscripts, we get

$$y = \frac{\alpha x}{1 + (\alpha - 1) x} \qquad \ldots (1.25)$$

$$\alpha = \frac{y/(1 - y)}{x/(1 - x)} \qquad \ldots (1.26)$$

From Equation (1.25), knowing α for a given binary system, x – y data (equilibrium data) can be generated by taking x = 0, 0.1 ..... to 1 and evaluating the corresponding values of y (equilibrium vapour phase composition).

For an ideal system, volatility is equal to the vapour pressure of the pure component.

Therefore, the relative volatility of two substances forming an ideal solution is the ratio of their vapour pressures.

$$\text{Relative volatility} = \alpha_{AB} = \alpha = \frac{p_A^0}{p_B^0} \qquad \ldots (1.27)$$

The relative volatility of component A with respect to B is the ratio of the pure component vapour pressure of A to that of B at the same temperature.

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Relative volatility is a measure of the separability by distillation. When $\alpha = 1$, a separation by distillation is not possible. The separation by distillation is possible for relative volatility values greater than one. Larger the value of the relative volatility, the easier is the separation by distillation.

THE METHODS OF DISTILLATION - BINARY SYSTEMS :

Basically, distillation is carried out in two ways :

1. The liquid mixture to be separated is heated to create a vapour. The vapours formed are condensed in a condenser and withdrawn as product. As there is no reflux, products of relatively low purities are obtained.

2. The liquid mixture to be separated is heated to create a vapour, the vapours formed are condensed in a condenser. A part of the condensed liquid is returned to the distillation still (as reflux) and the remaining part is withdrawn as product. In this method, the liquid and vapour are brought into intimate contact for a number of times and almost pure product can be achieved. The part of the vapour returned as liquid to the distillation unit/still is called as reflux and the operation is called rectification or fractionation. The term **rectification** originated in the alcohol industry whereas the term **fractionation** is popular in the petroleum industry.

Common methods used in distillation practice are :

1. Differential or simple distillation,
2. Flash or equilibrium distillation,
3. Rectification or fractionation.

Out of these three methods, distillation with rectification or simply called rectification is the most important. The first two methods are carried out without reflux and the third one is carried out with reflux (– which is nothing but returning a part of the condensed liquid back to the distillation system).

Differential or Simple Distillation :

In this distillation technique, a known quantity of a liquid mixture is charged into a jacketed kettle or still. The jacket is provided for heating the mass in the still with the help of a heating media such as steam. The charge is boiled slowly, vapours formed are withdrawn and fed to a condenser where they are liquified and collected in a receiver as a distillate. In the early stage of distillation, vapours leaving the still are richest in the more volatile component and as the distillation proceeds the liquid in the still becomes lean with respect to the more volatile component. The composition of the less volatile component thereby increases and hence the boiling point increases. The product (distillate) from such units can be collected in several receivers, called cuts, to give products of various purities over the length of distillation period. The distillation is continued till the boiling point of the liquid reaches a predetermined value and the contents of the still are finally removed as residual liquid containing majority of the less volatile component.

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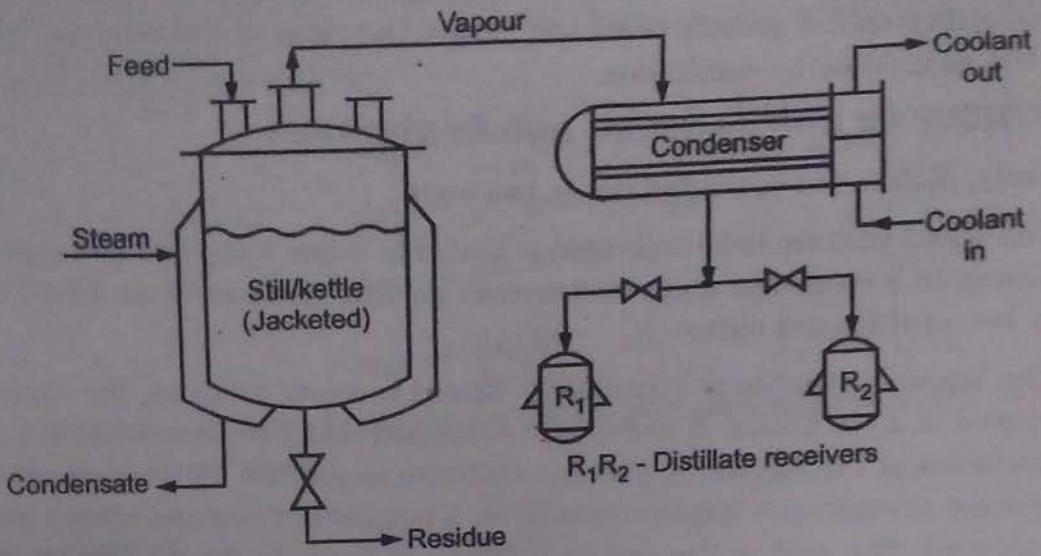

Fig. 1.4 : Simple distillation unit

Material balance - binary mixtures :

As the composition of the vapour issuing from the distillation still and that of the liquid remaining in it changes during the course of operation, the mathematical approach should be differential.

Let F be the kmol of a liquid mixture (A + B) containing x_F mole fraction of A which is charged to a distillation still. Let D be the kmol of distillate and W be the kmol of residual liquid in the still which are obtained at the end of operation. Let $y_{D\,avg}$ and x_W be the mole fractions of A in the distillate and the bottom residual liquid.

Let L be the kmol of liquid in the still at any time during the course of distillation and let x be the mole fraction of A in the liquid. Let a very small amount dD kmol of the distillate of the composition y in equilibrium with the liquid is vaporised. Then, the composition and the quantity of liquid decreases from x to x – dx and L to L–dL respectively.

Then,

The overall material balance at any time is :

$$L = L - dL + dD \quad \ldots (1.28)$$

$$\therefore \quad dL = dD \quad \ldots (1.29)$$

The material balance of the component A yields

$$Lx = (L - dL)(x - dx) + y\,dD \quad \ldots (1.30)$$

$$Lx = Lx - Ldx - xdL + dLdx + ydD$$

dLdx being very small can be neglected.

$$0 = -Ldx - xdL + ydD \quad \ldots (1.31)$$

Put the value of dD as dL from Equation (1.29)

$$0 = -Ldx - xdL + ydL \quad \ldots (1.32)$$

$$Ldx = (y - x)\,dL \quad \ldots (1.33)$$

$$\therefore \quad \frac{dL}{L} = \frac{dx}{y - x} \quad \ldots (1.34)$$

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Integrating the above equation between two extreme conditions :

$$L = F, \quad x = x_F$$

and

$$L = W, \quad x = x_W$$

$$\int_W^F \frac{dL}{L} = \int_{x_W}^{x_F} \frac{dx}{y - x} \qquad \ldots (1.35)$$

$$\ln \frac{F}{W} = \int_{x_W}^{x_F} \frac{dx}{y - x} \qquad \ldots (1.36)$$

(y is in equilibrium with x i.e. y is the vapour phase composition in equilibrium with the liquid phase composition (x)).

Equation (1.36) is known as the Rayleigh equation. It is used to determine F, W, x_F or x_W when three of these are known. The RHS of Equation (1.36), i.e., the integral on the RHS side is evaluated graphically by plotting $1/(y - x)$ as an ordinate and x as an abscissa, i.e., plotting $1/(y - x)$ v/s x and determining the area under the curve between $x = x_F$ and $x = x_W$. The required data for the above procedure are taken from the vapour-liquid equilibrium relationship.

If $y_{D,\,avg.}$ is the composition of the composited distillate, then,

The material balance of the component A is :

$$x_F F = y_{D,\,avg.} D + x_W W \qquad \ldots (1.37)$$

Though the simple or differential distillation as a method of separation is not effective, many such units are used, especially where (i) the components to be separated have widely different boiling points and (ii) methods giving sharp separations are not necessary.

Constant Relative Volatility :

During the course of distillation, although the temperature in the distillation still/kettle increases, the relative volatility does not change much and so some average volatility α can be used, over the concentration range involved, which makes the direct integration of RHS of Equation (1.36) possible and graphical integration can thus be avoided.

The relationship between the composition of vapour (y) and that of liquid (x), in equilibrium, in terms of relative volatility is

$$y = \alpha x / 1 + (\alpha - 1)\, x$$

So y from the above equation can be substituted in Equation (1.36).

$$\ln \frac{F}{W} = \int_{x_W}^{x_F} \frac{dx}{y - x}$$

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$$\ln \frac{F}{W} = \int_{x_W}^{x_F} \frac{dx}{\frac{\alpha x}{1+(\alpha-1)x} - x}$$

$$\ln \frac{F}{W} = \int_{x_W}^{x_F} \frac{[1+(\alpha-1)x]\,dx}{\alpha x - x[1+(\alpha-1)x]} \quad \ldots (1.38)$$

Consider the term $\frac{1+(\alpha-1)x}{\alpha x - x[1+(\alpha-1)x]}$ of the RHS of Equation (1.38).

$$\frac{1+(\alpha-1)x}{\alpha x - x[1+(\alpha-1)x]} = \frac{1+\alpha x - x}{\alpha x - x - \alpha x^2 + x^2}$$

$$= \frac{(1-x)+\alpha x}{x(\alpha-1) - x^2(\alpha-1)}$$

$$= \frac{(1-x)+\alpha x}{(\alpha-1)(x-x^2)}$$

$$= \frac{(1-x)+\alpha x}{(\alpha-1)x(1-x)}$$

$$= \frac{1}{\alpha-1}\left[\frac{(1-x)}{x(1-x)} + \frac{\alpha x}{x(1-x)}\right]$$

$$= \frac{1}{\alpha-1}\left[\frac{1}{x} + \frac{\alpha}{1-x}\right]$$

Putting the above value of RHS back into Equation (1.38), we get

$$\ln \frac{F}{W} = \frac{1}{\alpha-1}\left[\int_{x_W}^{x_F} \frac{1}{x}\,dx + \alpha \int_{x_W}^{x_F} \frac{dx}{1-x}\right]$$

Integrating, we get

$$\boxed{\ln \frac{F}{W} = \frac{1}{\alpha-1}\left[\ln \frac{x_F}{x_W} - \alpha \ln \frac{1-x_F}{1-x_W}\right]} \quad \ldots (1.39)$$

Equation (1.39) relates F, W, x_F and x_W through α.

Equation (1.39) can be arranged in another useful form as :

Add and subtract $\ln [(1-x_W)/(1-x_F)]$ in bracket on RHS of Equation (1.39).

$$\ln \frac{F}{W} = \frac{1}{\alpha-1}\left[\ln \frac{x_F}{x_W} + \ln \frac{1-x_W}{1-x_F} - \ln \frac{1-x_W}{1-x_F} - \alpha \ln \frac{1-x_F}{1-x_W}\right]$$

Combining and rearranging the terms in bracket,

$$\ln \frac{F}{W} = \frac{1}{\alpha-1}\left[\ln \frac{x_F(1-x_W)}{x_W(1-x_F)} - \ln \frac{1-x_W}{1-x_F} + \alpha \ln \frac{1-x_W}{1-x_F}\right]$$

$$\ln \frac{F}{W} = \frac{1}{\alpha-1}\left[\ln \frac{x_F(1-x_W)}{x_W(1-x_F)} + (\alpha-1)\ln \frac{1-x_W}{1-x_F}\right]$$

$$\boxed{\ln \frac{F}{W} = \frac{1}{\alpha-1}\ln \frac{x_F(1-x_W)}{x_W(1-x_F)} + \ln \frac{1-x_W}{1-x_F}} \quad \ldots (1.40)$$

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This can be rearranged to another form as :

$$\ln \frac{F}{W} - \ln \frac{1 - x_W}{1 - x_F} = \frac{1}{\alpha - 1} \ln \frac{x_F (1 - x_W)}{x_W (1 - x_F)}$$

$$\ln \frac{F (1 - x_F)}{W (1 - x_W)} = \frac{1}{\alpha - 1} \ln \frac{x_F (1 - x_W)}{x_W (1 - x_F)}$$

$$(\alpha - 1) \ln \frac{F (1 - x_F)}{W (1 - x_W)} = \ln \frac{x_F (1 - x_W)}{x_W (1 - x_F)}$$

$$\alpha \ln \frac{F (1 - x_F)}{W (1 - x_W)} - \ln \frac{F (1 - x_F)}{W (1 - x_W)} = \ln \frac{x_F (1 - x_W)}{x_W (1 - x_F)}$$

$$\alpha \ln \frac{F (1 - x_F)}{W (1 - x_W)} = \ln \frac{x_F (1 - x_W)}{x_W (1 - x_F)} + \ln \frac{F(1 - x_F)}{W (1 - x_W)}$$

$$\alpha \ln \frac{F (1 - x_F)}{W (1 - x_W)} = \ln \frac{x_F (1 - x_W)}{x_W (1 - x_F)} \cdot \frac{F(1 - x_F)}{W (1 - x_W)}$$

$$\alpha \ln \frac{F (1 - x_F)}{W (1 - x_W)} = \ln \frac{F x_F}{W x_W}$$

$$\therefore \quad \boxed{\ln \frac{F x_F}{W x_W} = \alpha \ln \frac{F (1 - x_F)}{W (1 - x_W)}} \qquad \ldots (1.41)$$

Equation (1.41) relates the number of moles of A remaining in the residue, W x_W to that of B remaining, W $(1 - x_W)$.

$F x_F$ = number of moles of A initially, let it be N_{OA}.

$F (1 - x_F)$ = number of moles of B initially, let it be N_{OB}.

$W x_W$ = number of moles of A remaining, let it be N_A (after distillation is over)

$W (1 - x_W)$ = number of moles of B remaining, let it be N_B.

$$\therefore \quad \ln \frac{N_{OA}}{N_A} = \alpha \ln \frac{N_{OB}}{N_B}$$

$$\frac{N_{OA}}{N_A} = \left(\frac{N_{OB}}{N_B}\right)^{\alpha}$$

$$\left(\frac{N_B}{N_{OB}}\right)^{\alpha} = \frac{N_A}{N_{OA}}$$

$$\frac{N_B}{N_{OB}} = \left(\frac{N_A}{N_{OA}}\right)^{1/\alpha} \qquad \ldots (1.42)$$

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Differential Condensation :

In this case, a vapour feed is slowly condensed and the condensate is drawn in equilibrium with the vapour residue. Let F be the moles of vapour of composition y_F and D be the moles of vaporous residue of composition y_D then the treatment similar to that described above yields

$$\ln \frac{F}{W} = \int_{y_F}^{y_D} \frac{dy}{y - x^*} \quad \ldots (1.43)$$

Flash or Equilibrium distillation :

Flash distillation is normally carried out in a continuous manner. In this method, a liquid mixture is partially vaporised, the vapour and liquid are allowed to attain equilibrium and finally withdrawn separately.

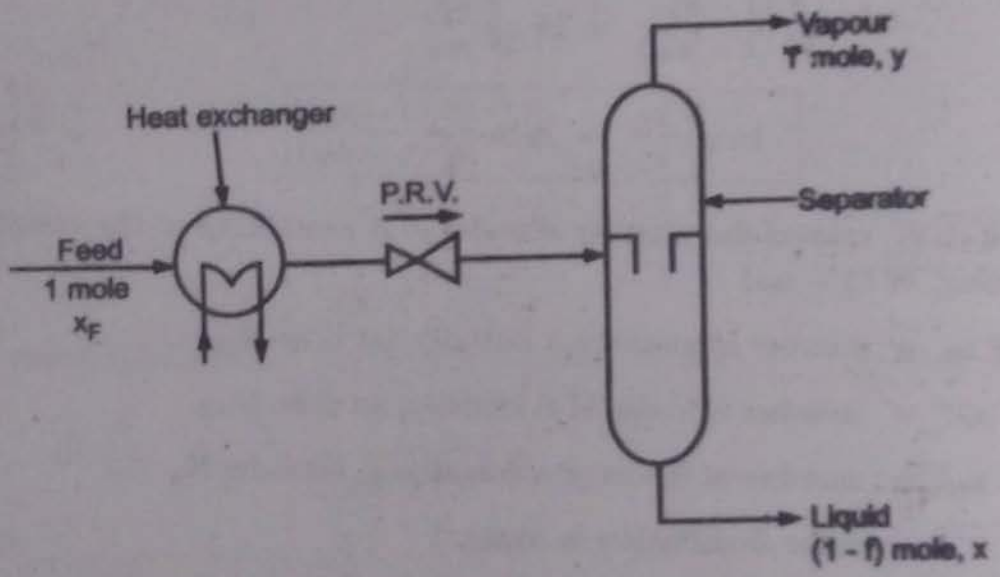

Fig. 1.5 : Flash/Equilibrium distillation

Feed is heated in a tubular heat exchanger. The hot liquid mixture is then fed to a separator via a pressure reducing valve whereby the pressure is reduced and a vapour is formed at the expense of liquid adiabatically. The liquid is withdrawn from the bottom of the separator and the equilibrium vapour leaves the separator from the top which is then liquified in a condenser. Flash distillation is commonly used in petroleum industry, handling multicomponent system in the pipe stills.

Consider one mole of a liquid mixture, having x_F mole fraction of the more volatile component, is fed to a flash distillation unit. Let f be the fraction of the feed that is vaporised and is of composition y. Then (1 – f) will be the moles of the residual liquid obtained. Let x be the mole fraction of the more volatile component in the liquid.

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Then, a material balance of the more volatile component gives

$$x_F = f \cdot y + (1 - f)\, x \quad \ldots (1.44)$$

$$f y = -(1 - f)\, x + x_F \quad \ldots (1.45)$$

$$y = -\frac{(1 - f)}{f}\, x + \frac{x_F}{f} \quad \ldots (1.46)$$

Equation (1.46) is the material balance/operating line for flash distillation with a slope equal to $-(1 - f)/f$ and an intercept equal to x_F/f.

The point of intersection of the operating line and the diagonal ($x = y$) is,

$$y = -\frac{(1 - f)\, x}{f} + \frac{x_F}{f}$$

$$x = y$$

$$\therefore \quad x = -\frac{(1 - f)\, x}{f} + \frac{x_F}{f}$$

$$xf = -x + xf + x_F$$

$$\therefore \quad x = x_F$$

and $y = x_F$

For $f = 1$, feed totally vaporised (feed 100 mole per cent vaporised)

$$\text{Slope} = -\frac{(1 - f)}{f} = 0$$

and hence the operating line is parallel to x-axis through the point (x_F, x_F) on the diagonal.

For $f = 0$, – no feed is vaporised.

Slope $= \infty$ and hence the operating line will be parallel to y-axis through the point (x_F, x_F) on the diagonal.

The operating line for various values of f on the equilibrium diagram is as shown in Fig. 1.6.

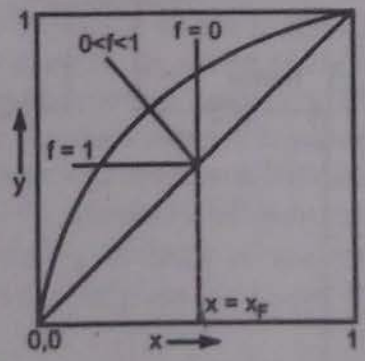

Fig. 1.6 : Plotting the material balance line (operating line) for flash distillation on equilibrium diagram

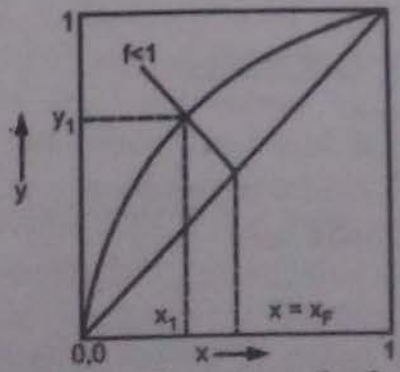

Fig. 1.7 : Determination of x and y for a given f

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Fig. 1.7 shows a method of obtaining the equilibrium compositions of vapour (y_1) and liquid (x_1) for a given f.

Continuous Rectification – Binary Systems :

Rectification is commonly encountered in industrial practice as it is possible to get almost pure product by this method.

- The enrichment of the vapour stream as it passes through the column in contact with reflux is termed as rectification.
- Rectification is the separation of the components of a liquid mixture by successive distillations - successive vaporisations and condensations.
- Continuous rectification or fractionation is a multistage countercurrent distillation.
- With this separation technique, separations into almost pure components are possible.

In this separation method, a part of the condensed liquid is returned back as reflux to the still and a maximum enrichment of the more volatile component in the vapour is obtained by successive partial vaporisation and condensation by a multistage contact of the vapour and the liquid. This is achieved in a single unit called as a fractionating column.

Fractionating Column :

A fractionating column or fractionator consists of (i) a cylindrical shell divided into sections by a series of perforated trays, (ii) a reboiler and (iii) a condenser. The liquid mixture to be separated is introduced in the cylindrical column more or less centrally. The column itself is divided into two sections - rectifying and stripping section. The section above the feed plate or tray is called the **rectifying section**, wherein the vapour is washed to remove the less volatile component with the liquid returned to the column from the top

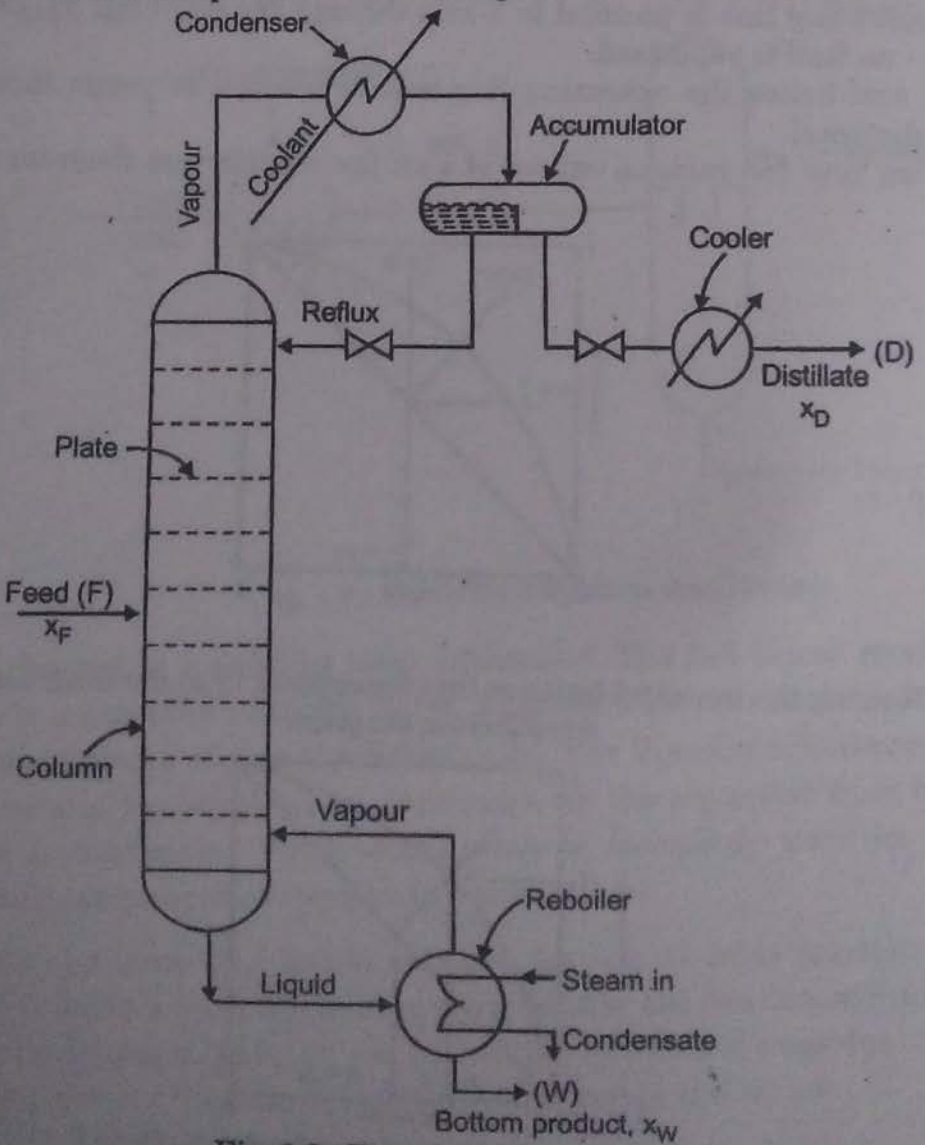

Fig. 1.8 : Fractionating column

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(known as reflux). The rectifying section is also called the **enriching section** as the feed is enriched in this section, owing to which the top product/distillate is richer in the more volatile component than the feed. The portion below the feed plate including feed plate is called the **stripping section** wherein the liquid is stripped off more volatile component by the rising vapour. Perforated trays are nothing but gas-liquid contacting devices on which gas/vapour and liquid streams are brought into intimate contact for mass transfer to occur.

Vapours are generated in a reboiler (generally steam heated) and are fed to the bottom of the column. The liquid removed from the fractionator rich in the less volatile component is called the bottoms or bottom product. The vapour issuing from the top of the column is fed to a condenser where the latent heat is removed with the help of a coolant circulated through the condenser. A part of the condensed liquid is returned to the column (reflux) and the remaining part is withdrawn as the top product or distillate which is rich in the more volatile component. In the column, the liquid and vapour streams are brought into intimate contact for a number of times, so that the vapour stream tends to vaporise the low-boiling constituent/the more volatile component from the descending liquid, and the liquid stream tends to condense the high-boiling component/the less volatile component from the vapour. As we move up the column, the vapour becomes richer and richer in the more volatile component and as we move down the column, the liquid becomes richer and richer in the less volatile component. As the liquid is at its bubble point and the vapour is at its dew point, temperature is maximum at the bottom and minimum at the top.

Rectification on ideal plate :

Refer to Fig. 1.9 wherein the plates are numbered serially from the top down. Consider the n^{th} plate from the top in the cascade. The plate n–1 is immediately above the plate n and the plate n + 1 is immediately below the plate n. On every plate, two different fluid streams not at equilibrium are brought into intimate contact, mass transfer takes place, the phases are separated, and finally the two fluid streams leave the plate in equilibrium with each other. Thus, the plate under consideration will receive a liquid stream of L_{n-1} moles/h of composition x_{n-1} and a vapour stream of V_{n+1} moles/h of composition y_{n+1} from plates n–1 and n+1 respectively. The liquid stream L_n moles/h of composition x_n and the vapour stream V_n moles/h of composition y_n leave this plate for plates n+1 and n–1 respectively in equilibrium with each other.

When the vapour from plate n+1 is brought into intimate contact on the plate under consideration with the liquid from plate n–1 and (the diffusing substances distribute themselves between the phases) their compositions tend to change to attain equilibrium values. During the interchange process, some of the more volatile component is vaporised from the liquid L_{n+1}, decreasing the liquid concentration from x_{n-1} to x_n; and some of the less volatile component is condensed from the vapour V_{n+1}, increasing the vapour concentration from y_{n+1} to y_n. In the column, as the liquid is at its bubble point and the vapour is at its dew point, the heat to vaporise the more volatile component from the liquid is supplied by the heat released in the condensation of the less volatile component from the vapour. Thus the net effect is that the more volatile component is transferred to the vapour rising up from the liquid running down the column, while the less volatile component is transferred to the liquid running down from the vapour rising up the column. Liquid and vapour phase compositions increase along the column height with respect to the more

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volatile component and therefore temperature decreases. The temperature of plate n is less than that of plate n + 1 and greater than that of plate n – 1.

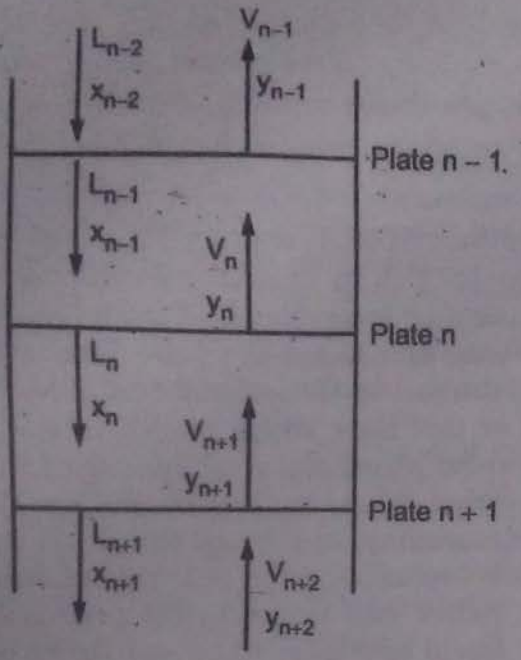

Fig. 1.9 : Material balance over ideal plate

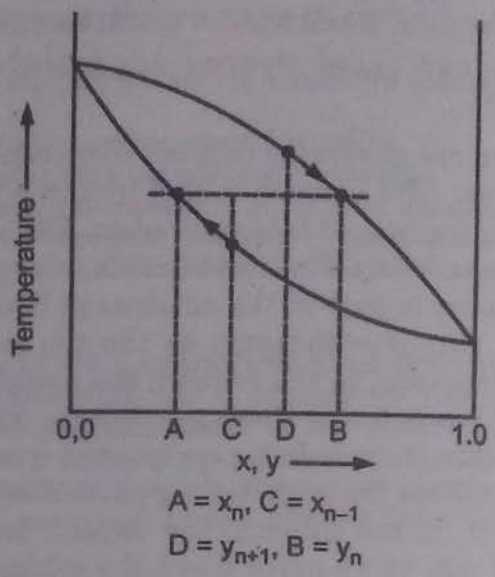

Fig. 1.10 : Boiling point diagram showing rectification on an ideal plate

Material balance :

Consider a column fed with F moles/time of a liquid mixture (feed) with x_F mole fraction of the more volatile component. Let D moles/time of composition x_D are withdrawn as a distillate and W moles/time of composition x_W are withdrawn as a bottom product from the column.

Overall material balance :

$$\text{Feed} = \text{distillate} + \text{bottom product}$$

$$F = D + W \qquad \ldots (1.47)$$

Material balance of the more volatile component (A) over the column/fractionator :

$$\text{A in feed} = \text{A in distillate} + \text{A in bottom product}$$

$$x_F\ F = x_D\ D + x_W\ W \qquad \ldots (1.48)$$

With the help of Equations (1.47), and (1.48), it is possible to calculate D and W knowing F, x_F, x_D and x_W.

Eliminating W from Equations (1.47) and (1.48), we get

$$D/F = (x_F - x_W)/(x_D - x_W) \qquad \ldots (1.49\ a)$$

Eliminating D gives

$$W/F = (x_D - x_F)/(x_D - x_W) \qquad \ldots (1.49\ b)$$

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Let L be the moles/time of liquid returned to the top of column then,

$$\text{Reflux ratio} = R = \frac{L}{D}$$

This reflux ratio is also known as the external reflux ratio.

If V is the (moles/time) molal flow rate of vapour leaving the top of the column then, the ratio L/V is known as the internal reflux ratio.

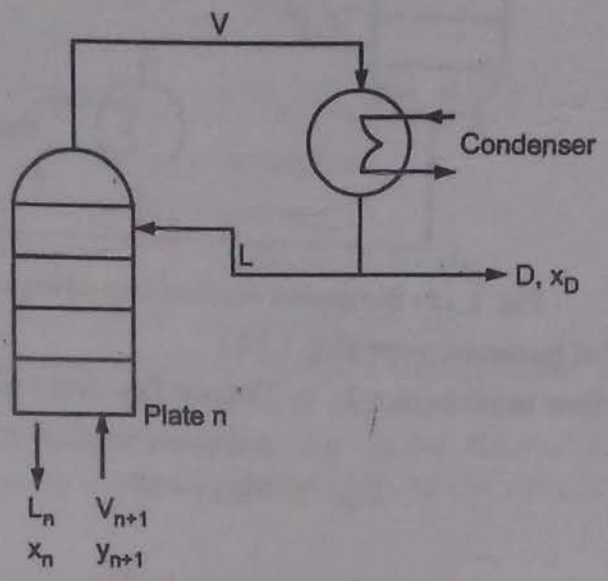

Fig. 1.11 : Rectifying section including condenser

Material balance around the condenser :

$$V = L + D$$

$$\therefore \quad L = V - D \qquad \ldots (1.50)$$

and $$D = V - L$$

Overall material balance over the Fig. 1.11 :

Vapour flow to plate n = liquid flow from plate n + distillate withdrawn

$$V_{n+1} = L_n + D \qquad \ldots (1.51)$$

Similarly, **material balance of the more volatile component (A) over the Fig. 1.11 :**

$$V_{n+1} y_{n+1} = L_n x_n + D x_D \qquad \ldots (1.52)$$

$$\therefore \quad y_{n+1} = \frac{L_n}{V_{n+1}} x_n + \frac{D x_D}{V_{n+1}}$$

$$\therefore \quad y_{n+1} = \frac{L_n}{V_{n+1}} x_n + \frac{D x_D}{V_{n+1}} \qquad \ldots (1.53)$$

From Equation (1.51),

$$V_{n+1} = L_n + D$$

Therefore Equation (1.53) becomes :

$$y_{n+1} = \frac{L_n}{L_n + D} x_n + \frac{D x_D}{L_n + D} \qquad \ldots (1.54)$$

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Equation (1.54) is known as the material balance line or **operating line of the rectifying section.**

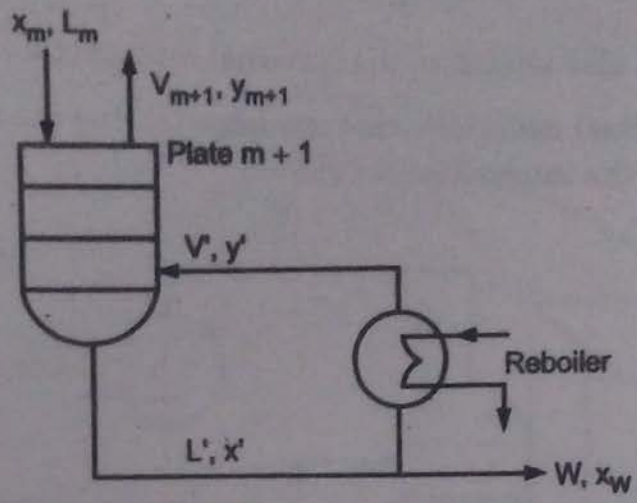

Fig. 1.12 : Stripping section including reboiler

Overall material balance over Fig. 1.12 :

Liquid flow to plate m + 1 = Vapour flow from plate m + 1 + Bottom product withdrawn

$$L_m = V_{m+1} + W \quad \ldots (1.55)$$

Similarly,

Material balance of more volatile component (A) :

$$L_m x_m = V_{m+1} y_{m+1} + W x_W \quad \ldots (1.56)$$

Rearranging the Equation (1.56),

$$y_{m+1} = \frac{L_m}{V_{m+1}} x_m - \frac{W x_W}{V_{m+1}} \quad \ldots (1.57)$$

We know from the Equation (1.55), that $V_{m+1} = L_m - W$

$$\therefore \quad y_{m+1} = \frac{L_m}{L_m - W} x_m - \frac{W x_W}{L_m - W} \quad \ldots (1.58)$$

The Equation (1.58) is known as the material balance line or **operating line of the stripping section.**

Analysis of fractionating columns :

The theoretical/ideal stages/plates required for a given degree of separation can be obtained by :

1. McCabe – Thiele method.
2. Lewis – Sorel method.
3. Ponchon – Savarit method.

1. McCabe – Thiele Method :

It is a graphical procedure of obtaining theoretical plates.

Assumptions underlying this method are :

1. Constant molal heat of vaporisation.
2. No heat losses.
3. No heat of mixing.

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Consider any plate n in the rectifying section.

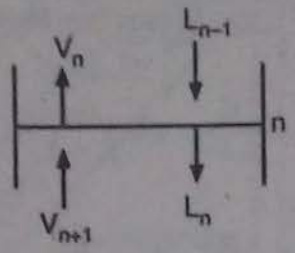

Fig. 1.13

A heat balance across plate n is

$$L_{n-1}\ H_{L_{n-1}} + V_{n+1}\ H_{V_{n+1}} = L_n\ H_{L_n} + V_n\ H_{V_n} + \text{heat losses} + \text{heat of mixing}$$

Heat losses are negligible for good insulation.

Heat of mixing is zero for an ideal system.

$$\therefore \quad L_{n-1}\ H_{L_{n-1}} + V_{n+1}\ H_{V_{n+1}} = L_n\ H_{L_n} + V_n\ H_{V_n}$$

H_L and H_V – enthalpies of liquid and vapour per mol.

If the molar heat of vaporisation is constant (independent of the composition) then one mol of vapour V_{n+1} on condensing releases sufficient heat to liberate one mol of vapour V_n. It follows that $V_n = V_{n+1}$. Therefore, the molar flow of vapour up the column is constant. As the temperature change from one plate to the next is small, H_{L_n} can be taken equal to $H_{L_{n-1}}$.

$$\therefore \quad L_{n-1}\ H_{L_{n-1}} + V_{n+1}\ H_{V_{n+1}} = V_{n+1} H_{V_{n+1}} + L_n\ H_{L_{n-1}}$$

$$\therefore \quad L_{n-1} = L_n$$

So the moles of liquid reflux in this section are also constant.

Therefore, L_n and V_{n+1} are constant in the rectifying section and V_{m+1} and L_m are constant in the stripping section.

i.e. molar flow rates of vapour and liquid are constant in each section of the column.

$$L_n = L_{n-1} = L, \quad L_m = L_{m+1} = L'$$

$$V_n = V_{n+1} = V, \quad V_{m+1} = V_m = V'$$

[The above cited assumptions lead to a concept of constant molal vapour flow and constant molal liquid flow in any section of the column. In other words, for one mole of vapour condensed, one mole of liquid is vaporised. Thus $V_n = V_{n+1}$, $L_n = L_{n+1}$ ... etc.]

The subscripts n, n + 1, m, m + 1 ... etc. may be dropped and the operating lines then plotted on the equilibrium diagram are straight.

The operating line of the rectifying section becomes :

$$y = \frac{L}{L+D}\,x + \frac{D\,x_D}{L+D} \qquad \ldots (1.59)$$

As $R = \frac{L}{D}$, the above equation becomes :

$$y = \frac{R}{R+1}\,x + \frac{x_D}{R+1} \qquad \ldots (1.60)$$

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The point of intersection of the operating line for the rectifying section and the 45° diagonal (x = y) obtained as,

$$y = \frac{R}{R+1}x + \frac{x_D}{R+1}$$

Putting $y = x$

$$\therefore \quad x = \frac{R}{R+1}x + \frac{x_D}{R+1}$$

$$(R+1)x = Rx + x_D$$

$$x = x_D$$

and $y = x_D$

Thus, the operating line of the rectifying section is to be drawn on the equilibrium diagram through the point (x_D, x_D) on the 45° diagonal and with a slope equal to $R/(R+1)$ or an intercept on y-axis equal to $x_D/(R+1)$.

The equation of the operating line of the stripping section becomes :

$$y = \frac{L'}{L'-W}x - \frac{W\,x_W}{L'-W} \qquad \ldots (1.61)$$

The point of intersection of the operating line of the stripping section and the 45° diagonal is obtained as follows :

$$y = \frac{L'}{L'-W}x - \frac{W\,x_W}{L'-W}$$

Diagonal : $y = x$

$$\therefore \quad x = \frac{L'}{L'-W}x - \frac{W\,x_W}{L'-W}$$

$$\therefore \quad L'\cdot x - W\cdot x = L'\cdot x - W\,x_W$$

$$\therefore \quad x = x_W$$

and $y = x_W$

Thus, the operating line of the stripping section is to be drawn through the point (x_W, x_W) on the diagonal and with a slope equal to $\frac{L'}{L'-W}$.

Step-wise procedure for obtaining theoretical plates :

1. By material balances, evaluate the terms D, W, L, etc.
2. Draw the equilibrium curve and the diagonal with the help of x – y data given. If the relative volatility is given, generate the x – y data.
3. Draw the operating line of the rectifying section through the point P (x_D, x_D) on the diagonal and with an intercept equal to $x_D/(R+1)$ or $Dx_D/(L+D)$ or a slope equal to $R/(R+1)$.
4. Draw the operating line of the stripping section through the point M (x_W, x_W) on the diagonal and with a slope equal to $\frac{L'}{L'-W}$.
5. Starting from the point (x_D, x_D) on the diagonal, draw a horizontal line to meet the equilibrium curve at a point 'a'. Drop a vertical from the point 'a' to meet the operating line at point 1.

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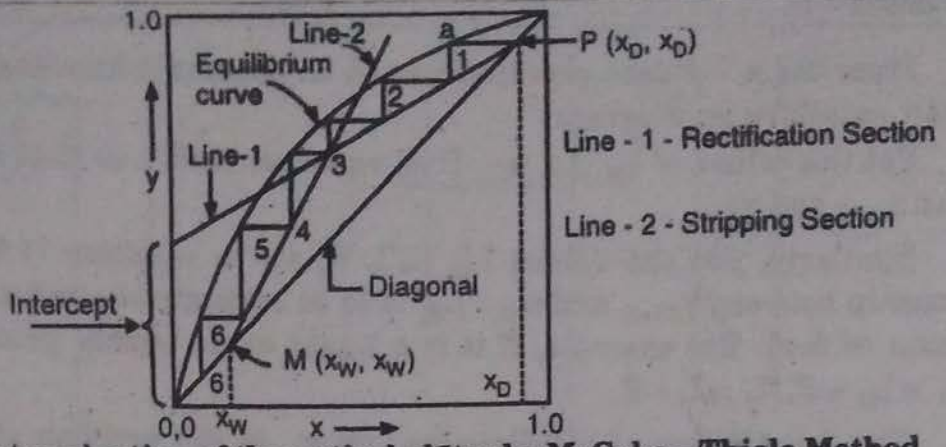

Fig. 1.14 : Determination of theoretical plates by McCabe – Thiele Method

6. Proceed in this way, that is constructing the triangles (representing the stages) between the equilibrium curve and the operating line of rectifying section, till we are above the point of intersection of two operating lines. Once we cross this point of intersection, construct the triangles between the equilibrium curve and the operating line of the stripping section, i.e., dropping verticals on the operating line of the stripping section.
7. Proceed in the same manner till we reach exactly or cover the point M (x_W, x_W).
8. Count the triangles constructed between x_D and x_W. Each triangle on the x – y diagram represents a theoretical plate.
9. If the number of triangles constructed are 'n', then 'n' represents the theoretical number of plates including reboiler (since the vapour and liquid leaving the reboiler are in equilibrium, the reboiler is equivalent to one theoretical plate) and n – 1 represents the number of theoretical plates in the column to achieve a desired degree of separation.

The graphical step-by-step construction utilising alternately the operating lines and the equilibrium curve to find the number of ideal plates/stages for a given separation is known as the **McCabe-Thiele method**.

Limitations of McCabe – Thiele Method :

1. It is not used when the relative volatility is less than 1.3 or greater than 5.
2. It is not used when more than 25 theoretical stages/plates are required, and
3. It is not used when the operating reflux ratio is less than 1.1 times the minimum reflux ratio.

2. Lewis – Sorel Method of Determination of Number of Theoretical Stages/Plates :

The equation of the operating line of the rectifying section is :

$$y_{n+1} = \frac{L_n}{L_n + D} x_n + \frac{D\, x_D}{L_n + D} \qquad \ldots (1.62)$$

The equation of the operating line of the stripping section is :

$$y_{m+1} = \frac{L_m}{L_m - W} x_m - \frac{W\, x_W}{L_m - W} \qquad \ldots (1.63)$$

1. From the data cited in a given problem, evaluate the terms - D, W, L etc.
2. From the x – y data provided (or can be generated knowing the relative volatility), draw an equilibrium diagram.

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2. From the x – y data provided (or can be generated knowing the relative volatility), draw an equilibrium diagram.

3. Put the values of L_n (L), x_D, D in equation (1.62), so that it will be a relationship between y_{n+1} and x_n.

4. Similarly, put the values L_m (L'), W, x_W in equation (1.63), so that it will be a relationship between y_{m+1} and x_m. L_m is to be evaluated by taking into consideration the condition of feed. For example, if it is a liquid at its bubble point, $L_m = L_n + qF$, $q = 1$
$\therefore L_m = L_n + F$, $L' = L + F$.

5. The composition of distillate (x_D) given in the problem statement represents the composition of vapour (y_n) as it is obtained by condensing the vapour leaving the top of column.

6. From the vapour phase composition – $y_n = x_D$, find the liquid phase composition x_{n+1} from the equilibrium diagram and put this value of x_{n+1} in the equation of the operating line of the rectifying section to get y_{n+2}.

7. Find x_{n+2} value from the equilibrium diagram corresponding to y_{n+2}.

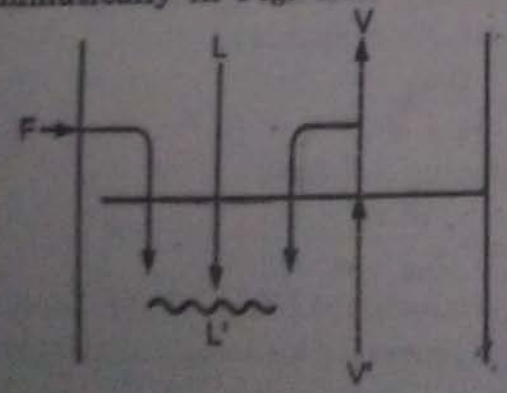
8. Calculate y_{n+3} and proceed with the same equation till we reach the point when the liquid phase composition equal to or less than x_F.

9. Then make use of the material balance equation or operating line of the stripping section and proceed in the same way as described above till we get the liquid phase composition equal to or below x_W. Suppose we end with y_{n+10}, [m = n, for convenience] it means that there will be 10 plates in the column.

Feed Plate and Feed Line :

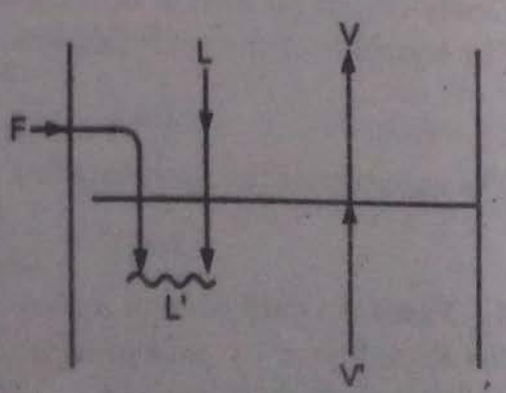
The plate on which feed is introduced is called the **feed plate**. The feed to the column may be introduced as : (a) cold liquid, (b) liquid at its bubble point (saturated-liquid), (c) partially vaporised (partly vapour and partly liquid), (d) saturated vapour at its dew point and (e) superheated vapour. The condition of feed introduced on the feed plate alters the phase flow rates. Introduction of the feed as a cold liquid and liquid at its bubble point increases the liquid flow rate in the stripping section. Feed - partially vaporised increases the vapour flow rate in the rectifying section and also the liquid flow rate in the stripping section. The feed as a saturated vapour and a superheated vapour increases vapour flow rate in the rectifying section.

Phase flow rates into and out of the feed plate for various feed conditions are shown diagrammatically in Fig. 1.15.



L' > L and V < V'

(a) Feed - cold liquid



L' > L and V' = V

(b) Feed – saturated liquid

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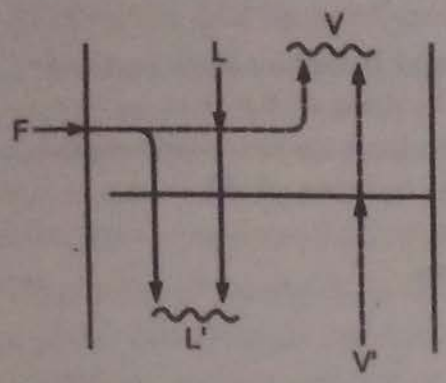
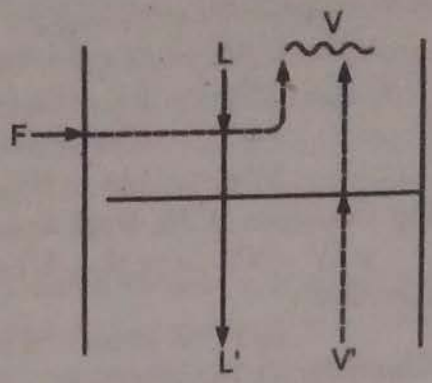


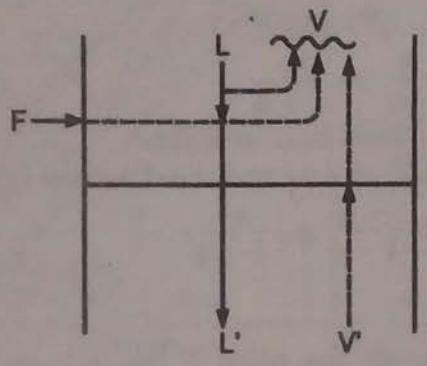
L' > L and V > V'

(c) Feed-partially vaporised



L' = L and V > V'

(d) Feed-saturated vapour



L' < L and V > V'

(e) Feed superheated vapour

Fig. 1.15 : Flow through feed plate for various feed conditions

The phase flow rates are shown based on the fact that the column liquid is always at its bubble point and the vapour is always at its dew point. To calculate the change in phase flow rates by the introduction of feed, a factor 'q' is introduced. The 'q' is a measure of the thermal condition of the feed and is defined as the *number of moles of saturated liquid resulting in the stripping section for each mole of feed introduced.* Thus for a feed F, we get :

$$L' = L + qF \quad \ldots (1.64)$$

and

$$V = V' + (1 - q)F \quad \ldots (1.65)$$

q = fraction of feed that is liquid

The values of 'q' for the various thermal conditions of feed are :

(a) Cold liquid : $q > 1$

(b) Saturated liquid : $q = 1$

(c) Feed partially vapour (partially flashed, i.e., vapour-liquid mixture) : $0 < q < 1$

(d) Saturated vapour feed : $q = 0$, and

(e) Superheated vapour feed : $q < 0$.

The value of q for any particular feed condition can be calculated from :

$$q = \frac{\text{Energy to convert 1 mole of feed to saturated vapour}}{\text{Molal latent heat of vaporisation}}$$

The liquid flow in the stripping section is :

$$L' = L + qF \quad \ldots (1.66)$$

$$\therefore \quad L' - L = qF \quad \ldots (1.67)$$

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Similarly, the vapour flow in the rectifying section is :

$$V = V' + (1 - q) F \quad \ldots (1.68)$$

$$\therefore \quad V - V' = (1 - q) F \quad \ldots (1.69)$$

$$V = L + D - \text{(overall material balance upper section)} \quad \ldots (1.70)$$

$$\therefore \quad V_y = Lx + Dx_D - \text{(material balance of A)} \quad \ldots (1.71)$$

and $$V' = L' - W - \text{(overall material balance lower section)} \quad \ldots (1.72)$$

$$V'y = L'x - Wx_W - \text{(material balance of A)} \quad \ldots (1.73)$$

Subtracting Equation (1.73) from Equation (1.71),

$$\therefore \quad y (V - V') = x (L - L') + D x_D + W x_W \quad \ldots (1.74)$$

But we know that :

$$x_F F = D x_D + W x_W \quad \ldots (1.75)$$

$\therefore$ The Equation (1.74) becomes :

$$y (V - V') = x (L - L') + x_F F \quad \ldots (1.76)$$

Putting the values of $V - V'$ and $L - L'$ from Equations (1.69) and (1.67) into Equation (1.76),

$$y (1 - q) F = x (- qF) + x_F F \quad \ldots (1.77)$$

$$\therefore \quad y = \frac{- q}{1 - q} x + \frac{x_F}{1 - q} \quad \ldots (1.78)$$

Equation (1.78) is known as the **feed line or q-line.**

The point of intersection of the feed line and the diagonal ($y = x$) is :

$$y = x = \frac{- q}{1 - q} x + \frac{x_F}{1 - q}$$

$$\therefore \quad x = x_F$$

and $$y = x_F$$

The feed line thus passes through the point N (x_F, x_F) on the 45° diagonal and has a slope equal to $\frac{-q}{1-q}$.

For $q = 1$, slope $= \infty$ and hence the feed line is parallel to y-axis through the point N (x_F, x_F) on the diagonal on the equilibrium diagram.

For $q = 0$, slope $= 0$ and hence the feed line is parallel to the x-axis through the point N (x_F, x_F) on the diagonal.

Typical feed line or q-line constructions for the various values of 'q' are shown in Fig. 1.16.

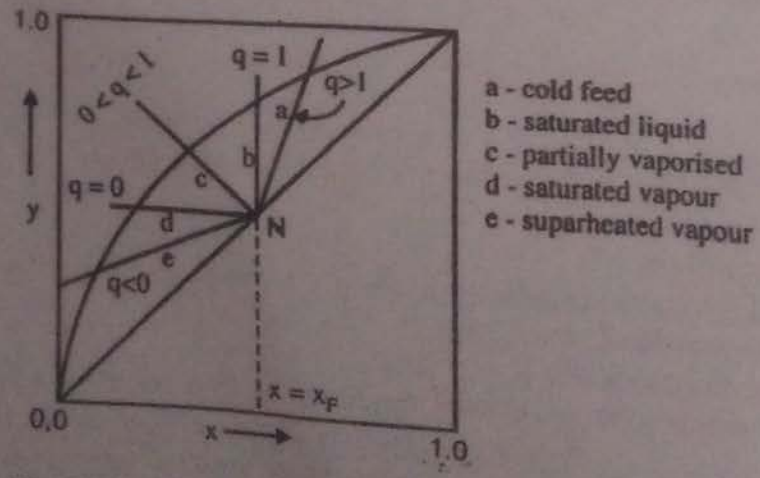

Fig. 1.16 : Effect of feed condition on feed line

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Plotting the feed line and operating lines to determine the number of stages :

Draw (1) the feed line on the equilibrium diagram, and draw (2) the operating line of the rectifying section as usual, it will cut the feed line at certain point then (3) the operating line of the stripping section can be easily plotted. This line is to be plotted through the point M (x_W, x_W) on the diagonal and it will pass through the point of intersection of the feed line and the operating line of the rectifying section.

The point of intersection of lines – 1, – 2 and 3 falls in a triangle which represents a feed plate. The triangle representing the feed plate has one corner on the operating line of the rectifying section and one on the operating line of the stripping section. Feed should be

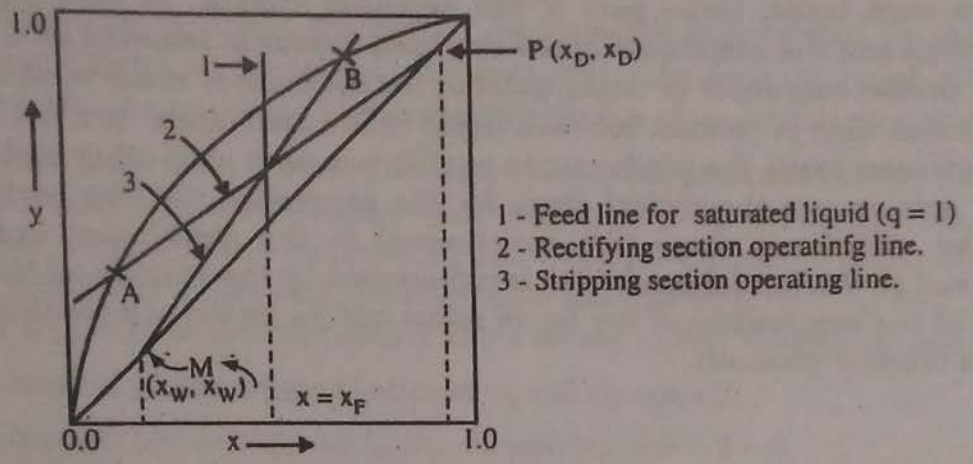

Fig. 1.17 : Plotting feed line and operating lines on x–y diagram for determining number of theoretical plates

introduced at any point between A and B but at the same time we have to keep in mind the maximum enrichment and minimum number of plates. Introduction of the feed near A or B unnecessarily results into a large number of plates in the rectifying or stripping section. The optimum location will be the one at which the liquid phase composition value crosses the feed composition value x_F.

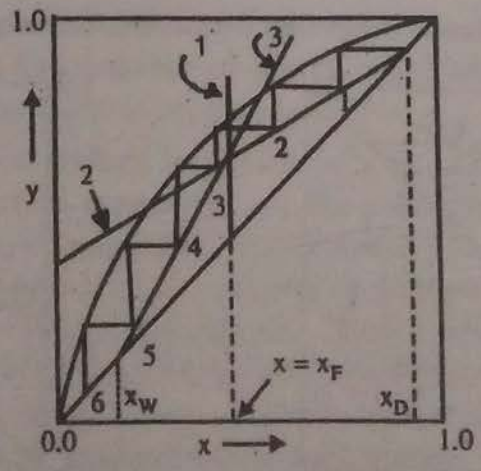

Fig. 1.18 : Feed plate location

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Referring Fig. 1.18, we have :

Theoretical plates including reboiler = n = 6.

Theoretical plates in column = n – 1 = 6 – 1 = 5

Feed plate - third from top.

Partial Condenser :

The total condenser is probably most commonly used on fractionating columns but in cases where the overhead product is desired as a vapour or where condensation of the entire vapour would be difficult, a partial condenser is used.

When the component to be removed (requiring large pressure at reasonable condenser temperatures) as a distillate has very low boiling point, then the complete condensation is difficult. In such cases, some part of the overhead vapour is condensed to provide a required reflux and the remainder of the overhead vapour is removed as a distillate.

With a partial condenser or dephlegmator, the distillate is removed in the vapour state. When sufficient time of contact between liquid reflux and vapour product is provided then these two streams leave the condenser in equilibrium with each other and thus the partial condenser acts as one theoretical stage for the separation, i.e. the partial condenser is equal to an additional stage (for enrichment) in the distillation unit. This is true normally, and so the partial condenser is represented by the triangle on the xy diagram.

Let x_C be the composition of the liquid reflux and y_D be the composition of the distillate withdrawn (vapour product).

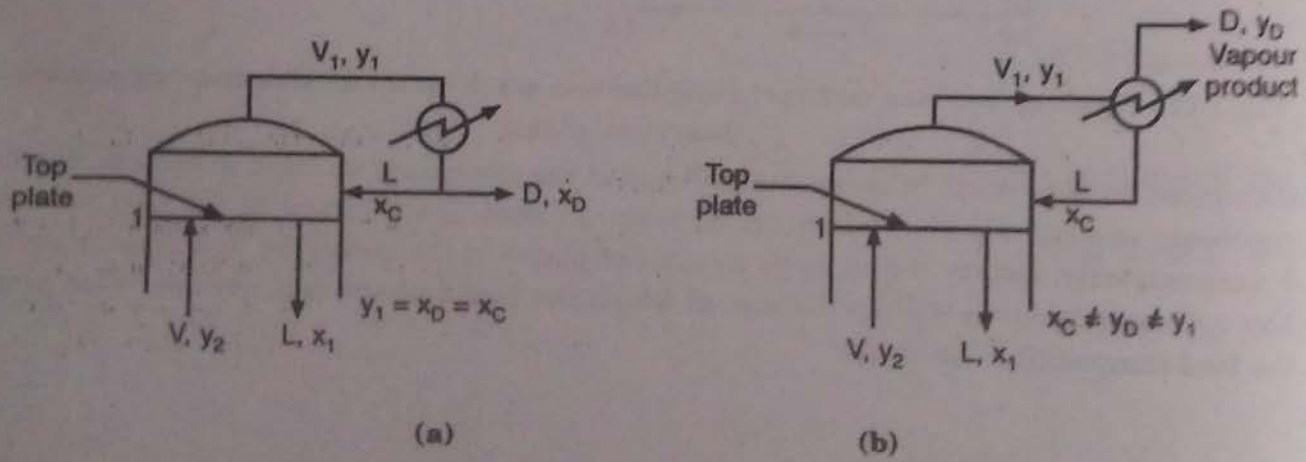

Fig. 1.19 : Material balance diagrams for (a) total condenser and (b) partial condenser

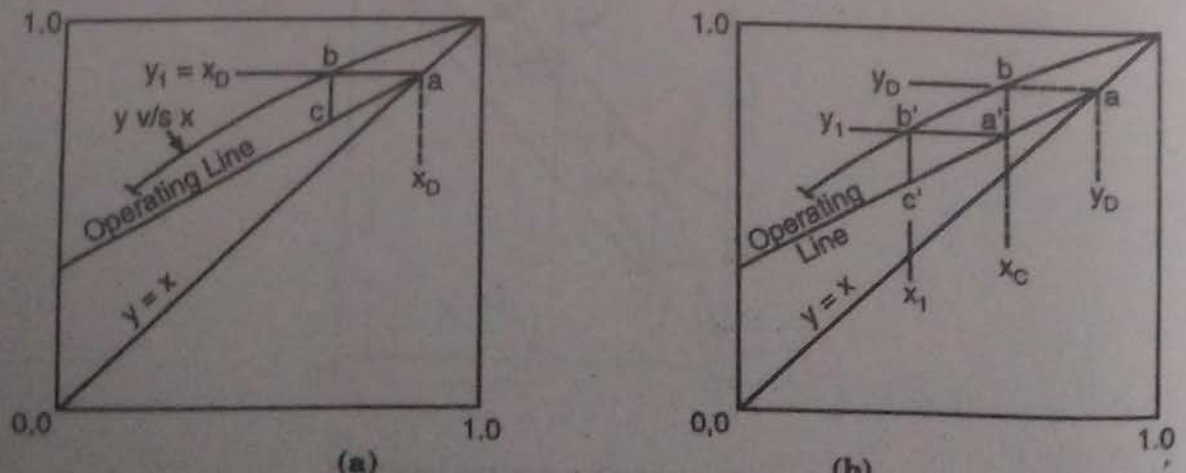

Fig. 1.20 : Graphical construction of top plate using : (a) total condenser, (b) partial condenser

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Vapour leaving the partial condenser has a composition y_D [Refer Fig. 1.19 (b)] which is in equilibrium with x_C, the composition of liquid reflux.

The operating line for the rectifying section with a total condenser is :

$$y_{n+1} = (R/R + 1)\, x_n + x_D/R + 1$$

The operating line for the rectifying section with a partial condenser is :

$$y_{n+1} = (R/R + 1)\, x_n + y_D/R + 1$$

only difference is x_D is replaced by y_D.

This operating line passes through the point (y_D, y_D) on the diagonal where y_D is the composition of the distillate withdrawn (in the vapour state).

As y_D is in equilibrium with x_C, the partial condenser is represented by the triangle aba'. As far as the column is concerned, the operating line ends at point a' with the coordinates (x_C, y_1) and the triangle a'b'c' represents the top plate in the column.

In case of partial condenser, we have to adopt the same procedure of construction of stages as that described previously for the total condenser [only difference is x_D is to be replaced by y_D – (composition of vapour product)] and count the number of stages between y_D and x_W. The total number of theoretical plates counted starting from y_D to x_W will include a reboiler as well as a partial condenser (each of them acting as one theoretical stage).

Number of theoretical plates counted between y_D and $x_W = n + 2$.

Number of theoretical plates required in the column $= n = n + 2 - 2$.

For example, suppose we counted 10 theoretical plates between $y_D - x_W$ (i.e. including the partial condenser and the reboiler) then

$$\text{Number of theoretical plates required (in the column)} = (n + 2) - 2$$

$$= 10 - 2 = 8$$

(as the reboiler and the partial condenser are both equivalent to one theoretical plate.)

Reflux Ratio :

1. Infinite reflux ratio or total reflux ratio, R_∞ :

The total reflux operation is of interest to know the minimum number of plates required (it sets a lower limit on the number of plates required) and is not a practical method of operation as at total reflux the product rate is zero.

A column operating under total reflux is shown in Fig. 1.21. In this case, enough material is charged to the distillation assembly and the column is operated under total reflux. During the operation of the column, the vapour issuing from the top is condensed and all the condensed stream is fed back to the column as reflux. Also all the liquid going to the reboiler is vaporised and is fed to the column. Thus, under the condition of total reflux $F = 0$, $D = 0$ and $W = 0$, therefore $V_{n+1} = L_n$ throughout the column. The slope (L/V) of both the operating lines becomes unity and hence both the operating lines coincide with the diagonal and a minimum number of stages (theoretical plates) are then required for a desired degree of separation $(x_D - x_W)$. The total reflux operation corresponds to a maximum reboiler heat supply and condenser cooling capacity for the separation.

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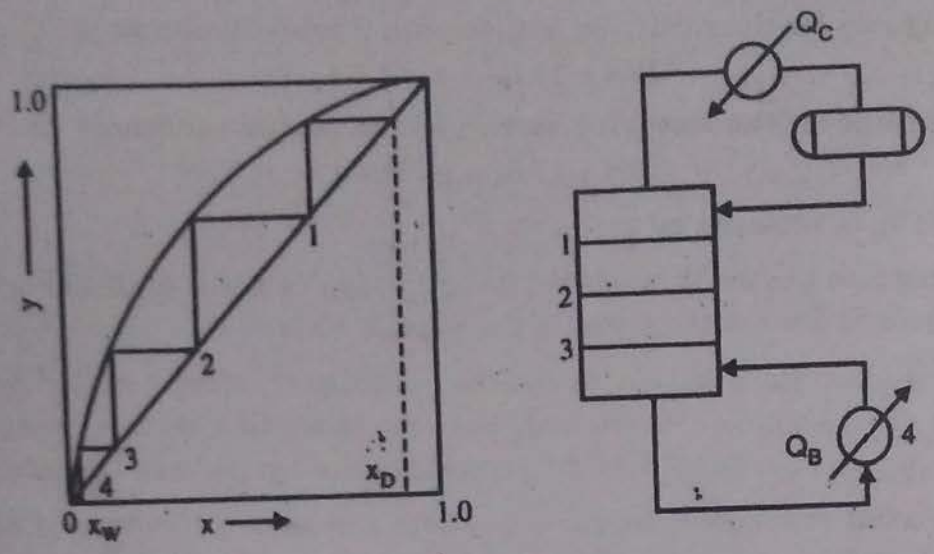

Fig. 1.21 : Operation under total reflux

2. Minimum Reflux Ratio, R_m :

At total reflux, the operating lines of both sections coincide with the diagonal and to effect a desired separation, the number of stages (theoretical plates) required are minimum. As the reflux ratio is reduced, the operating lines move towards the equilibrium curve along the feed line and the number of stages increase. Ultimately, further reduction in the reflux ratio results into a condition wherein the top operating line touches the equilibrium curve as shown in Fig. 1.22. The point of intersection of the operating lines lies on the equilibrium curve and at this point step becomes very close together, so that a zone of nearly constant composition is formed near the feed plate and an infinite number of plates are then required. This represents a condition of the minimum reflux. Thus, the minimum reflux ratio is *that reflux ratio at which an infinite number of plates are required for a desired separation*. At minimum reflux ratio, the requirement of heat supply to the reboiler and that of coolant supply to the condenser are minimum.

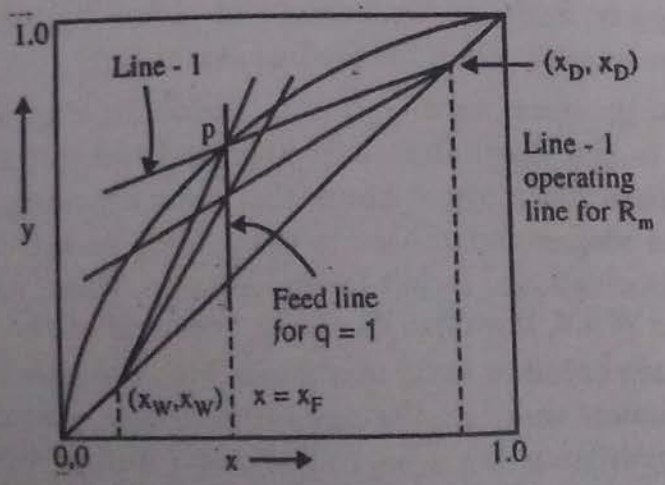

Fig. 1.22 : Minimum Reflux ratio

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Calculation of minimum reflux ratio (R_m) :

For liquid at its bubble point, the feed line is vertical (parallel to y-axis) and the point C lies on the equilibrium curve and has the co-ordinates (x_F, y_F).

Draw the equilibrium curve and the diagonal and draw the feed line, a vertical through point P (x_F, x_F) on the diagonal. The point of intersection of the feed line with the equilibrium curve is also a point of intersection of the operating lines at minimum reflux.

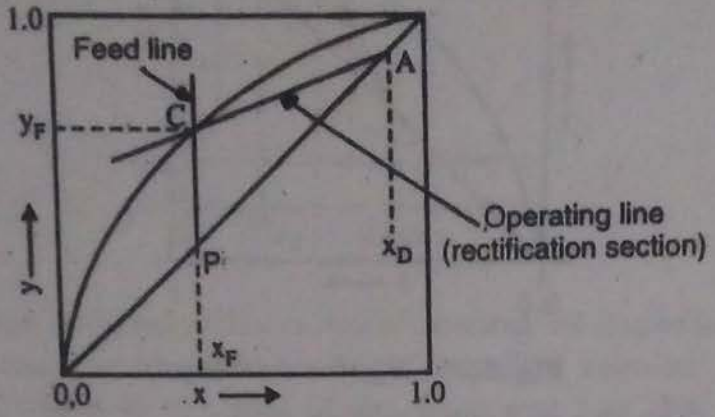

Fig. 1.23 : Method of obtaining R_m (saturated liquid feed)

Slope of the operating line AC is given as :

$$\frac{R_m}{R_m + 1} = \frac{x_D - y_F}{x_D - x_F} \quad \ldots (1.79)$$

$$R_m (x_D - x_F) = (R_m + 1)(x_D - y_F)$$

$$R_m x_D - R_m x_F = R_m x_D - R_m y_F + x_D - y_F$$

$$\therefore \quad R_m (y_F - x_F) = x_D - y_F$$

$$R_m = \frac{x_D - y_F}{y_F - x_F} \quad \ldots (1.80)$$

Similarly for feed as a saturated vapour, the feed line is horizontal.

Slope of the line AC is given as :

$$\frac{R_m}{R_m + 1} = \frac{x_D - y'}{x_D - x'} \quad \ldots (1.81)$$

$$\therefore \quad R_m = \frac{x_D - y'}{y' - x'}$$

$$= \frac{x_D - x_F}{x_F - x'} \quad (\text{as } y' = x_F) \quad \ldots (1.82)$$

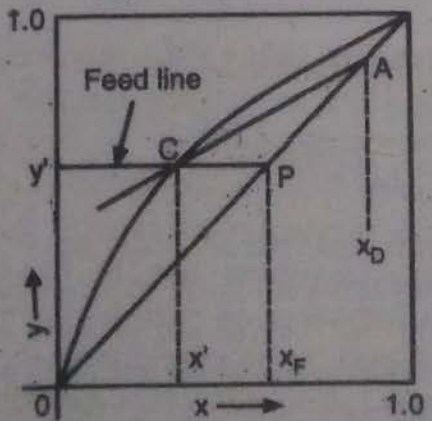

Fig. 1.24 : R_m for saturated vapour

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For feed as a cold liquid : i.e. for 'q' values greater than one :

$$R_m = \frac{x_D - y'}{y' - x'} \quad \ldots (1.83)$$

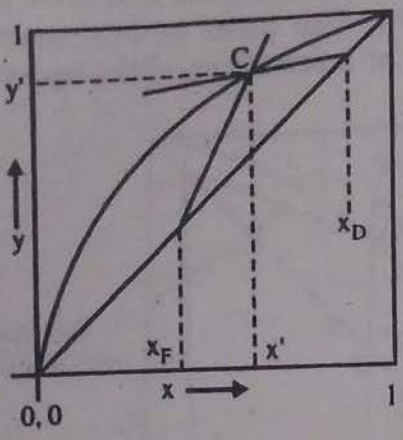

Fig. 1.25 : 'R_m' for cold liquid

For 'q' values less than one and greater than zero :

$$\text{Slope} = \frac{R_m}{R_m + 1} = \frac{x_D - y'}{x_D - x'}$$

$$\therefore \quad R_m = \frac{x_D - y'}{y' - x'} \quad \ldots (1.84)$$

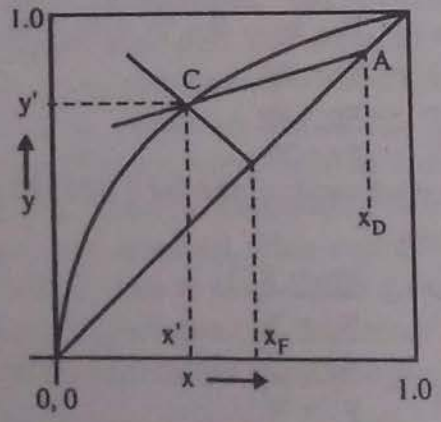

Fig. 1.26 : 'R_m' for feed-partially vapour

3. Optimum Reflux Ratio :

Any reflux ratio between infinite reflux ratio requiring a minimum number of plates and minimum reflux ratio requiring an infinite number of plates is a workable system which requires finite stages for a desired degree of separation. At minimum reflux ratio, as infinite number of plates are required, the fixed cost is also infinite while the cost of heat supply to the reboiler and the condenser coolant is minimum. As the reflux ratio is increased, the number of plates decreases and the fixed cost decreases at first, passes through a minimum and then increases as with higher reflux ratio the diameter of the column and sizes of reboiler and condenser increase. The operating cost increases continuously as the reflux ratio is increased as it is directly proportional to (R + 1). At total

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reflux, though the number of plates are minimum, the cost of heat supply to reboiler and

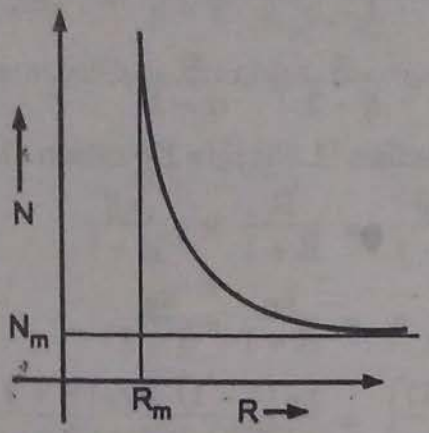

Fig. 1.27 : Relation between reflux ratio and number of stages for distillation

condenser coolant is maximum and also large capacity reboiler and condenser are needed. The total cost which is the sum of the fixed cost and the operating cost also decreases to a minimum and then increases with reflux ratio. The optimum reflux ratio occurs at a point where the sum of the fixed cost and the operating cost is minimum. The optimum reflux ratio is defined as that reflux ratio at which the sum of the fixed cost and operating cost is minimum. As a rough approximation, **the optimum reflux ratio is usually 1.1 to 1.5 times the minimum reflux ratio.**

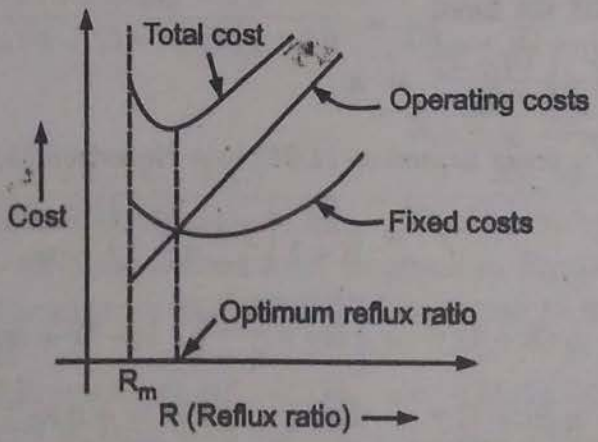

Fig. 1.28 : Optimum reflux ratio for distillation

Fenske - Underwood Equations :

Minimum Reflux Ratio (R_m) : The minimum reflux ratio, requiring infinite number of plates, can be obtained analytically when the relative volatility is taken as constant over the concentration range concerned ($x_D - x_W$). So let us obtain the required relationship with which we can calculate R_m.

For the normal equilibrium curve which is concave downward throughout its length, the operating line of the rectifying section, at minimum reflux, passes through the point of intersection of the q-line and the equilibrium curve.

The equation of the operating line for the rectifying section dropping the subscripts is :

$$y = \frac{R}{R+1} x + \frac{x_D}{R+1} \quad \ldots (1.85)$$

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The q-line equation is

$$y = \frac{-q}{1-q}x + \frac{x_F}{1-q}$$

$$y = \frac{q}{q-1}x - \frac{x_F}{q-1} \quad \ldots (1.86)$$

Putting value of y from Equation (1.85) into Equation (1.86),

$$\therefore \quad \frac{q}{q-1}x - \frac{x_F}{q-1} = \frac{R}{R+1}x + \frac{x_D}{R+1}$$

$$\frac{q}{q-1}x - \frac{R}{R+1}x = \frac{x_D}{R+1} + \frac{x_F}{q-1}$$

$$x\left[\frac{q(R+1) - R(q-1)}{(q-1)(R+1)}\right] = \frac{x_D(q-1) + x_F(R+1)}{(R+1)(q-1)}$$

$$x = \frac{x_D(q-1) + x_F(R+1)}{q(R+1) - R(q-1)}$$

$$x = \frac{x_D(q-1) + x_F(R+1)}{qR + q - Rq + R}$$

$$x = \frac{x_D(q-1) + x_F(R+1)}{(R+q)}$$

$$\therefore \quad x_a = x = \frac{x_D(q-1) + x_F(R+1)}{(R+q)}$$

From Equation (1.86), we have

$$\left[y + \frac{x_F}{q-1}\right]\frac{(q-1)}{q} = x \quad \ldots (1.87)$$

Putting the value of x from Equation (1.85) into Equation (1.87),

$$y = \frac{R}{R+1}\left[y + \frac{x_F}{q-1}\right]\frac{(q-1)}{q} + \frac{x_D}{(R+1)}$$

$$q(R+1)y = \left[Ry + \frac{Rx_F}{q-1}\right][q-1] + qx_D$$

$$q(R+1)y = Ry(q-1) + Rx_F + qx_D$$

$$qRy + qy - Ryq + Ry = Rx_F + qx_D$$

$$y(R+q) = Rx_F + qx_D$$

$$y = \frac{Rx_F + qx_D}{(R+q)}$$

$$\therefore \quad y_a = y = \frac{Rx_F + qx_D}{R+q}$$

So the coordinates of the point of intersection of the operating lines with the q-line are :

$$x_a = \frac{x_D(q-1) + x_F(R+1)}{(R+q)} \quad \text{and} \quad y_a = \frac{Rx_F + qx_D}{(R+q)}$$

So when this point of intersection of the operating line (of the rectification section) with the q-line lies on the equilibrium curve then R will be R_m and the coordinates represent the equilibrium values.

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The coordinates of the point of intersection (of feed line and operating line of rectification section) on the equilibrium curve (replacing R by R_m) : (x_a, y_a) are given as

$$x_a = \frac{x_D (q-1) + x_F (R_m + 1)}{(R_m + q)} \text{ and } y_a = \frac{R_m x_F + q x_D}{(R_m + q)}$$

As x_a and y_a represent the equilibrium values, they can be substituted in the definition of α.

We have, $$\alpha = \frac{y(1-x)}{x(1-y)} \quad \text{[from Equation (1.26)]}$$

$$\alpha = \frac{\frac{1}{x} - 1}{\frac{1}{y} - 1} \quad \ldots (1.88)$$

Put the values of x_a and y_a in Equation (1.88) as x_a (= x) and y_a (= y) are the coordinates of the point of intersection of operating line and q-line lying on the equilibrium curve.

$$\alpha = \frac{\frac{(R_m + q)}{x_D (q-1) + x_F (R_m + 1)} - 1}{\frac{(R_m + q)}{(R_m x_F + q x_D)} - 1}$$

$$\alpha = \frac{(R_m + q) - [x_D (q-1) + x_F (R_m + 1)] (R_m x_F + q x_D)}{[x_D (q-1) + x_F (R_m + 1)] \; [(R_m + q) - (R_m x_F + q x_D)]}$$

Rearranging, we get

$$\frac{\alpha [x_D (q-1) + x_F (R_m + 1)]}{(R_m + q) - [x_D (q-1) + x_F (R_m + 1)]} = \frac{R_m x_F + q x_D}{(R_m + q) - (R_m x_F + q x_D)}$$

$$= \frac{R_m x_F + q x_D}{R_m + q - R_m x_F - q x_D}$$

$$\frac{\alpha [x_D (q-1) + x_F (R_m + 1)]}{(R_m + q) - [x_D (q-1) + x_F (R_m + 1)]} = \frac{R_m x_F + q x_D}{R_m (1 - x_F) + q (1 - x_D)} \quad \ldots (1.89)$$

We have arranged the RHS in desired form as given in Equation (1.89).

Now rearrange the denominator of the LHS of Equation (1.89) in a suitable form and then put in Equation (1.89).

Denominator of LHS of Equation (1.89) $= (R_m + q) - [x_D (q-1) + x_F (R_m + 1)]$

$(R_m + q) - [x_D (q-1) + x_F (R_m + 1)] = R_m + q - x_D q + x_D - x_F R_m - x_F$

Add 1 and subtract 1.

$$= R_m + q - x_D q + x_D - x_F R_m - x_F + 1 - 1$$

Collecting the terms, we get

$$= R_m (1 - x_F) + (1 - x_F) + q (1 - x_D) - 1 (1 - x_D)$$

$$= (R_m + 1)(1 - x_F) + (q - 1)(1 - x_D)$$

Put the value of denominator of the LHS in Equation (1.89) to get a desired relationship.

$$\frac{\alpha [x_D (q-1) + x_F (R_m + 1)]}{(R_m + 1)(1 - x_F) + (q-1)(1 - x_D)} = \frac{R_m x_F + q x_D}{R_m (1 - x_F) + q (1 - x_D)}$$

$$\frac{R_m x_F + q x_D}{R_m (1 - x_F) + q (1 - x_D)} = \frac{\alpha [x_D (q-1) + x_F (R_m + 1)]}{(R_m + 1)(1 - x_F) + (q-1)(1 - x_D)} \quad \ldots (1.90)$$

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Equation (1.90) is the desired relationship for R_m and can be conveniently solved for any value of q.

With the help of the above equation, we can calculate R_m (minimum reflux) for a given x_F, x_D, α and feed condition (q).

Case I : R_m for q = 0, i.e., Feed is a saturated vapour at its dew point.

$$\frac{R_m x_F + q x_D}{R_m (1 - x_F) + q (1 - x_D)} = \frac{\alpha [x_D (q - 1) + x_F (R_m + 1)]}{(R_m + 1)(1 - x_F) + (q - 1)(1 - x_D)}$$

For saturated vapour feed, put q = 0 in the above equation and then the above relationship reduces to :

$$\frac{R_m x_F}{R_m (1 - x_F)} = \frac{\alpha [- x_D + x_F (R_m + 1)]}{(R_m + 1)(1 - x_F) - (1 - x_D)}$$

R_m/R_m on LHS of above equation will cancel and rearranging it, we get,

$$x_F (R_m + 1)(1 - x_F) - x_F (1 - x_D) = \alpha [- x_D + x_F (R_m + 1)] (1 - x_F)$$

Divide the LHS and RHS by $x_F (1 - x_F)$.

$$(R_m + 1) - \frac{(1 - x_D)}{(1 - x_F)} = \frac{\alpha}{x_F} [- x_D + x_F (R_m + 1)]$$

$$(R_m + 1) - \frac{(1 - x_D)}{(1 - x_F)} = \frac{- \alpha x_D}{x_F} + \alpha (R_m + 1)$$

Rearranging, we get

$$\frac{\alpha x_D}{x_F} - \frac{(1 - x_D)}{(1 - x_F)} = \alpha (R_m + 1) - (R_m + 1)$$

$$\therefore \quad \alpha (R_m + 1) - (R_m + 1) = \frac{\alpha x_D}{x_F} - \frac{(1 - x_D)}{(1 - x_F)}$$

$$(\alpha - 1)(R_m + 1) = \left[\frac{\alpha x_D}{x_F} - \frac{1 - x_D}{1 - x_F}\right]$$

$$R_m + 1 = \frac{1}{\alpha - 1}\left[\frac{\alpha x_D}{x_F} - \frac{1 - x_D}{1 - x_F}\right]$$

$$R_m = \frac{1}{\alpha - 1}\left[\frac{\alpha x_D}{x_F} - \frac{1 - x_D}{1 - x_F}\right] - 1 \qquad \ldots (1.91)$$

Equation (1.91) is the desired relationship for R_m when the feed is a saturated vapour.

Case II : For q = 1 : Feed is a liquid at its boiling point, i.e., feed is a saturated liquid.

We have :

$$\frac{R_m x_F + q x_D}{R_m (1 - x_F) + q (1 - x_D)} = \frac{\alpha [x_D (q - 1) + x_F (R_m + 1)]}{(R_m + 1)(1 - x_F) + (q - 1)(1 - x_D)}$$

For saturated liquid feed, i.e., liquid at its bubble point, put q = 1 and then the above relationship reduces to

$$\frac{R_m x_F + x_D}{R_m (1 - x_F) + (1 - x_D)} = \frac{\alpha x_F (R_m + 1)}{(R_m + 1)(1 - x_F)}$$

$$R_m x_F (R_m + 1)(1 - x_F) + x_D (R_m + 1)(1 - x_F) = \alpha x_F (R_m + 1)[R_m (1 - x_F) + (1 - x_D)]$$

$$R_m x_F (R_m + 1)(1 - x_F) + x_D (R_m + 1)(1 - x_F) = \alpha x_F (R_m + 1) R_m (1 - x_F)$$

$$+ \alpha x_F (R_m + 1)(1 - x_D)$$

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Divide each term of the RHS and LHS of the above Equation by $(R_m + 1)(1 - x_F)$.

$$R_m x_F + x_D = \alpha x_F R_m + \frac{\alpha x_F (1 - x_D)}{1 - x_F}$$

$$x_D - \frac{\alpha x_F (1 - x_D)}{1 - x_F} = \alpha x_F R_m - R_m x_F$$

$$\alpha x_F R_m - R_m x_F = x_D - \frac{\alpha x_F (1 - x_D)}{1 - x_F}$$

Divide each term of the LHS and RHS by x_F.

$$\alpha R_m - R_m = \frac{x_D}{x_F} - \frac{\alpha (1 - x_D)}{(1 - x_F)}$$

$$(\alpha - 1) R_m = \left[\frac{x_D}{x_F} - \frac{\alpha (1 - x_D)}{(1 - x_F)}\right]$$

$$R_m = \frac{1}{(\alpha - 1)} \left[\frac{x_D}{x_F} - \frac{\alpha (1 - x_D)}{(1 - x_F)}\right] \quad \ldots (1.92)$$

Equation (1.92) is the desired relationship for R_m for a saturated liquid feed (i.e., q = 1).

Alternative Approach :

Now, we will obtain the same relationships with the help of slope of the operating line for a minimum reflux and using the relationship : $y = \alpha x/[1 + (\alpha - 1) x]$

Case I : Feed is a liquid at its bubble point (saturated liquid feed).

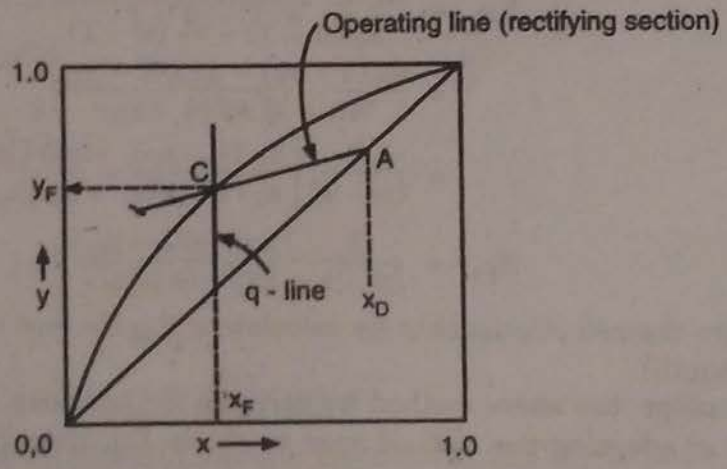

Fig. 1.29 : Method of obtaining R_m (saturated liquid feed)

The slope of the operating line AC is

$$\frac{R_m}{R_m + 1} = \frac{x_D - y_F}{x_D - x_F} \quad \ldots (1.93)$$

$$R_m (x_D - x_F) = (R_m + 1)(x_D - y_F)$$

$$R_m x_D - R_m x_F = R_m x_D - R_m y_F + x_D - y_F$$

$$R_m (y_F - x_F) = x_D - y_F$$

$$R_m = \frac{x_D - y_F}{y_F - x_F} \quad \ldots (1.94)$$

y_F and x_F are the coordinates of the point C i.e. the point of intersection of the operating line and the q-line lying on the equilibrium curve for minimum reflux conditions. So y_F and x_F are equilibrium values as the point C is on the equilibrium curve.

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Equilibrium relationship between x and y is given by

$$y = \alpha x/1 + (\alpha - 1)\, x$$

$$\therefore \quad y_F = \frac{\alpha x_F}{1 + (\alpha - 1)\, x_F} \qquad \ldots (1.95)$$

Put the value of y_F from Equation (1.95) into Equation (1.94).

$$R_m = \frac{x_D - \alpha x_F/[1 + (\alpha - 1)\, x_F]}{\frac{\alpha x_F}{1 + (\alpha - 1)\, x_F} - x_F}$$

$$R_m = \frac{x_D\,[1 + (\alpha - 1)\, x_F] - \alpha x_F}{\alpha x_F - x_F\,[1 + (\alpha - 1)\, x_F]}$$

$$R_m = \frac{x_D\,(1 + \alpha x_F - x_F) - \alpha x_F}{\alpha x_F - x_F\,[1 + \alpha x_F - x_F]}$$

$$R_m = \frac{x_D + \alpha x_D\, x_F - x_D\, x_F - \alpha x_F}{\alpha x_F - x_F\,(1 + \alpha x_F - x_F)}$$

Collecting the terms in the numerator,

$$R_m = \frac{x_D\,(1 - x_F) - \alpha x_F\,(1 - x_D)}{\alpha x_F - x_F - \alpha x_F^2 + x_F^2}$$

Collecting the terms in the denominator,

$$R_m = \frac{x_D\,(1 - x_F) - \alpha x_F\,(1 - x_D)}{x_F\,(\alpha - 1) - x_F^2\,(\alpha - 1)}$$

$$= \frac{x_D\,(1 - x_F) - \alpha x_F\,(1 - x_D)}{(\alpha - 1)\, x_F\,(1 - x_F)}$$

$$= \frac{1}{(\alpha - 1)}\left[\frac{x_D\,(1 - x_F)}{x_F\,(1 - x_F)} - \frac{\alpha x_F\,(1 - x_D)}{x_F\,(1 - x_F)}\right]$$

$$R_m = \frac{1}{(\alpha - 1)}\left[\frac{x_D}{x_F} - \frac{\alpha(1 - x_D)}{(1 - x_F)}\right] \qquad \ldots (1.96)$$

This is the desired relationship for calculating R_m for feed - a liquid at its bubble point (saturated liquid).

You can adopt the above method for deriving the equation for R_m for saturated liquid feed instead of adopting the method used to obtain Equation (1.92). Equations (1.92) and (1.96) are identical.

Case II : Feed is a saturated vapour at its dew point (i.e. for feed condition of q = 0).

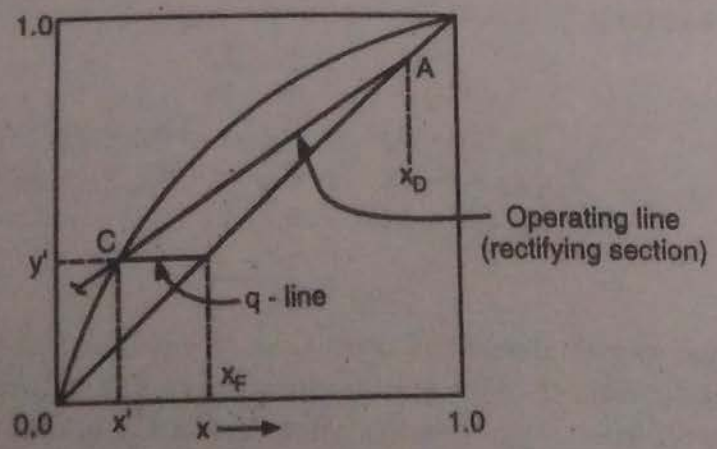

Fig. 1.30 : Method of obtaining R_m (saturated vapour feed)

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The slope of operating line AC is

$$\frac{R_m}{R_m + 1} = \frac{x_D - y'}{x_D - x'}$$

Solving, we get

$$R_m = \frac{x_D - y'}{y' - x'}$$

But $y' = x_F$

$$\therefore \quad R_m = \frac{x_D - x_F}{x_F - x'} \qquad \ldots (1.97)$$

The point C is on the equilibrium curve.

The point C is the point of intersection of the q-line and the operating line lying on the equilibrium curve for minimum reflux.

So y' $(= x_F)$ and x' are the equilibrium values of vapour and liquid phase compositions.

We have to substitute x' in terms of α and y' in the above equation.

$$y = \frac{\alpha x}{1 + (\alpha - 1)\, x}$$

$$y' = \frac{\alpha x'}{1 + (\alpha - 1)\, x'}$$

$$y' + \alpha x'y' - x'y' = \alpha x'$$

$$y' = \alpha x' - \alpha x'y' + x'y'$$

$$x' (\alpha - \alpha y' + y') = y'$$

$$x' [\alpha (1 - y') + y'] = y'$$

$$\therefore \quad x' = \frac{y'}{[y' + \alpha (1 - y')]}$$

but we have, $y' = x_F$

$$\therefore \quad x' = \frac{x_F}{x_F + \alpha (1 - x_F)} \qquad \ldots (1.98)$$

Put the value of x' from Equation (1.98) into Equation (1.97).

$$R_m = \frac{x_D - x_F}{x_F - x_F/[x_F + \alpha (1 - x_F)]}$$

$$= \frac{(x_D - x_F)\, [x_F + \alpha (1 - x_F)]}{x_F\, [x_F + \alpha (1 - x_F)] - x_F}$$

$$= \frac{(x_D - x_F)\, [x_F + \alpha - \alpha x_F]}{x_F\, [x_F + \alpha - \alpha x_F] - x_F}$$

$$R_m = \frac{(x_D - x_F)\, [x_F + \alpha - \alpha x_F]}{x_F^2 + \alpha x_F - \alpha x_F^2 - x_F}$$

Rearrange the denominator of the above equation.

$$R_m = \frac{(x_D - x_F)\, [x_F + \alpha - \alpha x_F]}{\alpha x_F (1 - x_F) - x_F (1 - x_F)}$$

$$R_m = \frac{(x_D - x_F)\, [x_F + \alpha - \alpha x_F]}{x_F (1 - x_F)\, (\alpha - 1)} \qquad \ldots (1.98\ a)$$

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Similarly, rearrange the numerator of above equation :

$$\text{Numerator} = (x_D - x_F)[x_F + \alpha - \alpha x_F]$$
$$= x_D x_F + \alpha x_D - \alpha x_D x_F - x_F^2 - \alpha x_F + \alpha x_F^2$$
$$= x_F(x_D - x_F) + \alpha x_D(1 - x_F) - \alpha x_F(1 - x_F)$$

Putting the value of numerator in Equation (1.98 a)

$$R_m = \frac{x_F(x_D - x_F) + \alpha x_D(1 - x_F) - \alpha x_F(1 - x_F)}{x_F(1 - x_F)(\alpha - 1)}$$

$$R_m = \frac{1}{(\alpha - 1)}\left[\frac{x_F(x_D - x_F)}{x_F(1 - x_F)} + \frac{\alpha x_D(1 - x_F)}{x_F(1 - x_F)} - \frac{\alpha x_F(1 - x_F)}{x_F(1 - x_F)}\right]$$

$$= \frac{1}{\alpha - 1}\left[\frac{x_D - x_F}{1 - x_F} + \frac{\alpha x_D}{x_F} - \alpha\right]$$

Add 1 and subtract 1 on the RHS in the bracket of the above equation

$$= \frac{1}{(\alpha - 1)}\left[\frac{x_D - x_F}{1 - x_F} + \frac{\alpha x_D}{x_F} - \alpha + 1 - 1\right]$$

$$= \frac{1}{(\alpha - 1)}\left[\frac{x_D - x_F}{1 - x_F} + \frac{\alpha x_D}{x_F} - (\alpha - 1) - 1\right]$$

$$= \frac{1}{(\alpha - 1)}\left[\frac{x_D - x_F}{1 - x_F} + \frac{\alpha x_D}{x_F} - 1\right] - \left(\frac{\alpha - 1}{\alpha - 1}\right)$$

$$= \frac{1}{(\alpha - 1)}\left[\frac{\alpha x_D}{x_F} + \frac{x_D - x_F}{1 - x_F} - 1\right] - 1$$

$$= \frac{1}{(\alpha - 1)}\left[\frac{\alpha x_D}{x_F} + \frac{x_D - x_F - 1 + x_F}{1 - x_F}\right] - 1$$

$$= \frac{1}{(\alpha - 1)}\left[\frac{\alpha x_D}{x_F} + \frac{x_D - 1}{1 - x_F}\right] - 1, \quad x_D - 1 = -(1 - x_D)$$

$$\therefore \quad R_m = \frac{1}{(\alpha - 1)}\left[\frac{\alpha x_D}{x_F} - \frac{(1 - x_D)}{(1 - x_F)}\right] - 1 \qquad \ldots (1.99)$$

Equation (1.99) is the desired relationship for obtaining R_m from the knowledge of α, x_F and x_D for saturated vapour feed.

The Number of Plates at Total Reflux by Fenske's Method :

In cases where the relative volatility is taken as constant over the range of temperature encountered in a column, Fenske derived an analytical expression for calculating the number of plates required at total reflux for achieving a desired degree of separation.

For a binary system comprising of A and B with A as the more volatile component, the relative volatility is given by,

$$\alpha = \frac{y_A/x_A}{y_B/x_B}$$

but $x_B = 1 - x_A$ and $y_B = 1 - y_A$

$$\therefore \quad \alpha = \frac{y_A/(1 - y_A)}{x_A/(1 - x_A)}$$

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$$\frac{y_A}{1-y_A} = \alpha \frac{x_A}{1-x_A}$$

$$\frac{y}{1-y} = \alpha \frac{x}{1-x} \quad \ldots (1.100)$$

x and y are the equilibrium compositions of liquid and vapour phases in terms of molefraction of more volatile component (A) respectively.

Equation (1.100) for plate n + 1 can be written as

$$\frac{y_{n+1}}{1-y_{n+1}} = \alpha_{n+1} \frac{x_{n+1}}{1-x_{n+1}} \quad \ldots (1.101)$$

We have, the operating line for rectifying section :

$$y_{n+1} = \frac{L_n}{V_{n+1}} x_n + \frac{x_D D}{V_{n+1}}, \; V_{n+1} = L_n + D$$

At total reflux, D = 0 and L = V, so we get,

$$y_{n+1} = x_n$$

So replacing y_{n+1} by x_n, Equation (1.101) becomes

$$\frac{x_n}{1-x_n} = \alpha_{n+1} \frac{x_{n+1}}{1-x_{n+1}} \quad \ldots (1.102)$$

At the top of the column if a total condenser is used, the composition of the vapour leaving the top plate and entering a condenser and that of the distillate removed from it will be the same.

$\therefore$ $y_1 = x_D$, y is the composition of the vapour leaving the plate - 1 (i.e. the top plate)

Equation (1.101) becomes

$$\frac{x_D}{1-x_D} = \alpha_1 \frac{x_1}{1-x_2} \quad \ldots (1.103)$$

For plate - 2, Equation (1.102) becomes

$$\frac{x_1}{1-x_1} = \alpha_2 \frac{x_2}{1-x_2} \quad \ldots (1.104)$$

For plate - 3, Equation (1.102) becomes

$$\frac{x_2}{1-x_2} = \alpha_3 \frac{x_3}{1-x_3} \quad \ldots (1.105)$$

For plate - n, Equation (1.102) becomes

$$\frac{x_{n-1}}{1-x_n} = \alpha_n \frac{x_n}{1-x_n} \quad \ldots (1.106)$$

Combining Equation (1.103) and all Equations upto plate-n, we get

$$\frac{x_D}{1-x_D} = \alpha_1 \cdot \frac{x_1}{1-x_1} = \alpha_1 \alpha_2 \left(\frac{x_2}{1-x_2}\right)$$

$\therefore$ $$\frac{x_D}{1-x_D} = \alpha_1 \alpha_2 \alpha_3 \ldots \alpha_n \left(\frac{x_n}{1-x_n}\right)$$

$$\frac{x_D}{1-x_D} = \alpha_1 \alpha_2 \alpha_3 \ldots \alpha_{n-1} \alpha_n \left(\frac{x_n}{1-x_n}\right)$$

If some average relative volatility $\alpha_{avg.}$ is used throughout the column, then

$$\alpha_{avg.} = \alpha_1 = \alpha_2 = \ldots = \alpha_n$$

$\therefore$ $$\frac{x_D}{1-x_D} = (\alpha_{avg.})^n \frac{x_n}{1-x_n} \quad \ldots (1.107)$$

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To reach the bottom product W of composition x_W, n_{min} (i.e. n_m) plates and a reboiler are required. The reboiler itself represents an equilibrium stage (reboiler acts as one equilibrium stage). Equation (1.107) then gives

$$\frac{x_D}{1 - x_D} = (\alpha_{avg.})^{n_m + 1} \cdot \frac{x_W}{1 - x_W}$$

n_m is the total minimum number of theoretical plates required.

$$\ln [x_D/(1 - x_D)] = \ln [(\alpha_{avg.})^{n_m + 1} \cdot x_W/(1 - x_W)]$$

$$\ln [x_D/(1 - x_D)] = (n_m + 1) \ln (\alpha_{avg.}) + \ln [x_W/(1 - x_W)]$$

$$(n_m + 1) \ln (\alpha_{avg.}) = \ln [x_D/(1 - x_D)] - \ln [x_W/(1 - x_W)]$$

$$= \ln [[x_D/(1 - x_D)]/[x_W/(1 - x_W)]]$$

$$(n_m + 1) \ln (\alpha_{avg.}) = \ln \left[\frac{x_D (1 - x_W)}{x_W (1 - x_D)}\right]$$

$$n_m + 1 = \frac{\ln \left[\frac{x_D (1 - x_W)}{x_W (1 - x_D)}\right]}{\ln (\alpha_{avg.})} \quad \ldots (1.108)$$

$$n_m = \frac{\ln [x_D (1 - x_W)/x_W (1 - x_D)]}{\ln (\alpha_{avg.})} - 1 \quad \ldots (1.109)$$

Equation (1.109) is known as the Fenske's equation. This Equation gives the total minimum number of theoretical plates required to produce products of compositions x_D and x_W. n_m – total minimum number of theoretical plates required at total reflux thus can be calculated from the knowledge of terminal compositions x_D and x_W. For a moderate variation of α from the top to the bottom of the column, a geometric mean of α at the top and the bottom products i.e. $\sqrt{\alpha_1 \alpha_W}$ is used in place of α_{avg}.

Side Streams :

Side streams from the fractionating column are generally removed in case of multi-component systems to obtain the products of intermediate composition (i.e. petroleum products). These are taken out from intermediate locations along the column height.

To get the idea regarding a construction of theoretical stages in such situations, let us consider the fractionation of a binary system with one side stream in the rectifying section shown in Fig. 1.31.

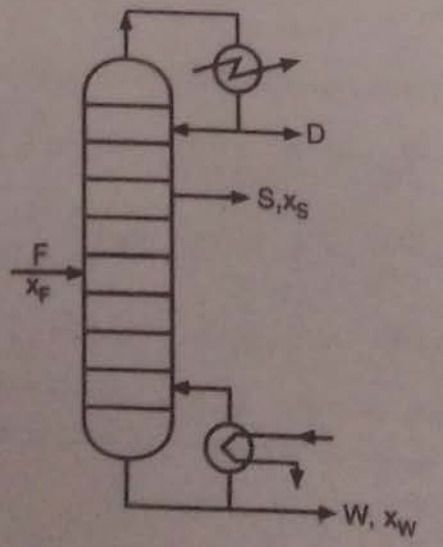

Fig. 1.31 : Column with side stream

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Let S be the molar flow rate of the side stream removed and x_s be the composition of the side stream with respect to the more volatile composition (A).

Column has three sections/column is divided into three sections :

(i) rectifying section above side stream

(ii) rectifying section below side stream

(iii) stripping section.

(1) Assuming constant molar overflow, the operating line for the rectifying section above the side stream is given by :

$$y_{n+1} = \frac{L_n}{V_{n+1}} x_n + \frac{D x_D}{V_{n+1}}$$

$$y_{n+1} = \frac{L}{V} x_n + \frac{D x_D}{V}$$

$$V = L + D \qquad \therefore \; L/V = R/R+1$$

V is the vapour flow up the column and L is the liquid flow down the column in this portion of the rectifying section.

(2) Let L_s be the liquid flow down the column below the side stream in the rectifying section.

Let V_s be the vapour flow below the side stream in the rectifying section.

The material balance for any plate in the rectifying section between the feed plate and the plate from which the side stream is taken out (i.e. the rectifying section above the feed plate and the below side stream) is :

$$V_s = L_s + S + D \qquad \ldots (1.110)$$

For the component A, it becomes :

$$V_s \cdot y_{n+1} = L_s x_n + S x_s + D x_D$$

$$y_{n+1} = \frac{L_s}{V_s} x_n + \frac{S x_s + D x_D}{V_s} \qquad \ldots (1.111)$$

This is the equation of the operating line applicable for a portion of the rectifying section between feed and side stream.

Let us obtain the point of intersection of the above given operating line and the diagonal $y = x$.

Dropping subscripts n, n + 1, the operating line equation is

$$y = \frac{L_s}{V_s} x + \frac{S x_s + D x_D}{V_s}$$

Diagonal :

$$y = x$$

$$x = \frac{L_s}{V_s} x + \frac{S x_s + D x_D}{V_s}$$

$$(V_s - L_s) x = S x_s + D x_D$$

$$x = \frac{S x_s + D x_D}{V_s - L_s}$$

We have :

$$V_s = L_s + S + D$$

$\therefore$

$$V_s - L_s = S + D$$

$\therefore$

$$x = y = \frac{S x_s + D x_D}{S + D}$$

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This is the required point of intersection of the operating line with the diagonal.

Let us find the point of intersection of two operating lines of the rectifying section.

We have : $$y = \frac{L}{V} x + \frac{D x_D}{V}$$

and $$y = \frac{L_s}{V_s} x + \frac{S x_s + D x_D}{V_s}$$

$$\frac{L}{V} x + \frac{D x_D}{V} = \frac{L_s}{V_s} x + \frac{S x_s + D x_D}{V_s}$$

But $$V_s = V$$

$$\therefore \quad L x + D x_D = L_s x + S x_s + D x_D$$

$$(L - L_s) x = S x_s$$

But $$L_s = L - S$$

$$(L - L + S) x = S x_s$$

$$x = x_s$$

So these two operating lines intersect at $x = x_s$.

(3) The operating line for the stripping section is given in the usual manner (Line-3).

$$y_{m+1} = \frac{L_m}{V_{m+1}} x_m - \frac{W x_W}{V_{m+1}}$$

$$y_{m+1} = \frac{L'}{V'} x_m - \frac{W x_W}{V'} \quad \ldots (1.112)$$

L' is the liquid flow down the column in the stripping section and V' is the vapour flow up the column in the stripping section.

The side stream is normally removed as liquid.

$$\therefore \quad L_n = L_s + S$$

$$L_s = L_n - S$$

(i) $L_s = L - S$, where L is liquid reflux to the column, $L = RD$.

(ii) $V_s = V$, where $V = L + D$

(iii) $L' = L_s + qF$

(if feed is a liquid at its bubble point, $L' = L_s + F$)

$$V_s = V = V' + (1 - q) F$$

$$V' = V - (1 - q) F$$

So we have to plot three operating lines on the xy diagram.

(i) The operating line for the portion of the rectifying section above the side stream.
As usual this line passes through the point (x_D, x_D) on the diagonal and has a slope of $R/R + 1 = L/V$ and an intercept of $x_D/R + 1$ on y-axis.
Draw this line through the point (x_D, x_D) on the diagonal with the intercept of $x_D/R + 1$ on y-axis.

(ii) The operating line for the portion of the rectifying section upto the side stream (i.e. between feed plate and the side stream).
This line passes through the point $[(S x_s + D x_D)/(S + D), (S x_s + D x_D)/(S + D)]$ on the diagonal and has a slope of L_s/V_s. This line intersects the top operating line, of slope $R/R + 1$, at $x = x_s$.

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So draw this line through the point $(S\ x_s + D\ x_D)/(S + D)$ on the diagonal, intersecting the top operating line at $x = x_s$ (or with a slope L_s/V_s).

(iii) The operating line for the stripping section is drawn in the usual manner, i.e., through the point (x_W, x_W) on the diagonal and with a slope equal to L'/V'.

This line passes through the point of intersection of the q-line and the operating line (between feed and side stream) of slope L_s/V_s.

Construct ideal stages as per the method previously described between the equilibrium curve and an appropriate operating line.

Effect of side streams : Removal of a side stream decreases the liquid rate below the side stream and so the side stream increases the number of plates required.

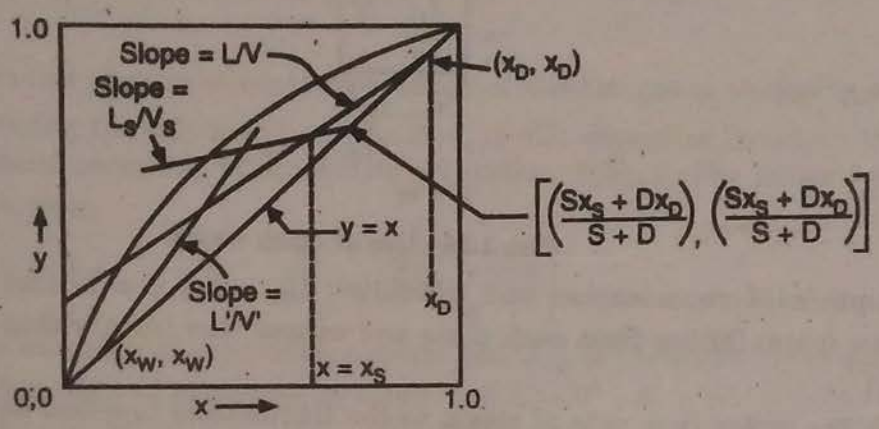

Fig. 1.32 : Effect of side stream

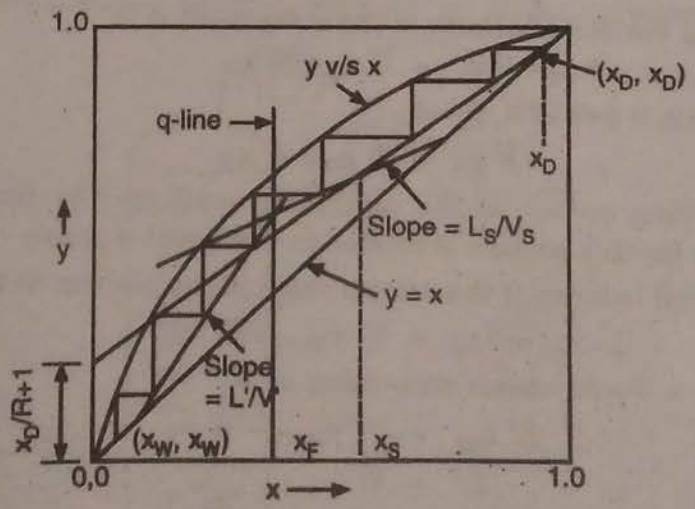

Fig. 1.33 : Construction of ideal stages - with side stream (feed-saturated liquid)

Use of Open Steam :

In the fractionation of an aqueous solution wherein the non-aqueous component is removed as the distillate and water is removed as the bottom product, the heat requirement of the fractionation may be provided by introducing steam directly at the bottom of the fractionating column, i.e., underneath the bottom plate (open steam). This procedure thus

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eliminates the need of a reboiler (heating surface) normally used. But with this arrangement, for a given reflux ratio and distillate composition, more plates will be required in the column. Even though the plates required are more, the cost of plates is less than the cost of the reboiler and its maintenance.

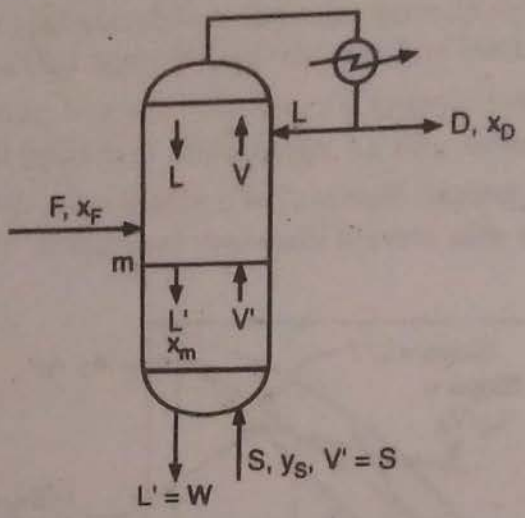

Fig. 1.34 : Use of open steam

For equimolal vaporisation and overflow : $L_m = L_{m-1} = L'$ and $V'_{m+1} = V_m = V'$ [liquid flow (rate) falling from each plate and vapour flow (rate) rising from each plate are constant.]

If S is the molar flow rate of steam to the fractionator (assume saturated at a column pressure) then the overall material balance is

$$F + S = D + W$$

The overall material balance of the component A is

$$F x_F + S y_s = D x_D + W x_W$$

As the steam is free of A, $y_s = 0$

$$\therefore \quad F x_F = D x_D + W x_W \qquad \ldots (1.113)$$

The rectifying section of the column is unaffected by the use of open steam, so the operating line for this section is located in the usual manner.

The material balance of the component A below plate m in the stripping section is

$$L' x_m + S y_s = V' y_{m+1} + W x_W$$

as $y_s = 0$ – for steam containing no A.

$$L' x_m = V' y_{m+1} + W x_W$$

We have : $L' = W$

$$\therefore \quad L' x_m = V' y_{m+1} + L' x_W$$

$$L' (x_m - x_W) = V' y_{m+1}$$

$$\therefore \quad \frac{L'}{V'} = \frac{L'}{S} = \frac{y_{m+1}}{x_m - x_W}, \text{ as } V' = S \qquad \ldots (1.114)$$

So the operating line of the stripping section passes through the point ($x = x_W$, $y = 0$). As this point is on x-axis, a graphical construction must be continued upto the x-axis of the x-y diagram. The slope of the operating line is L'/V' ($= L'/S$).

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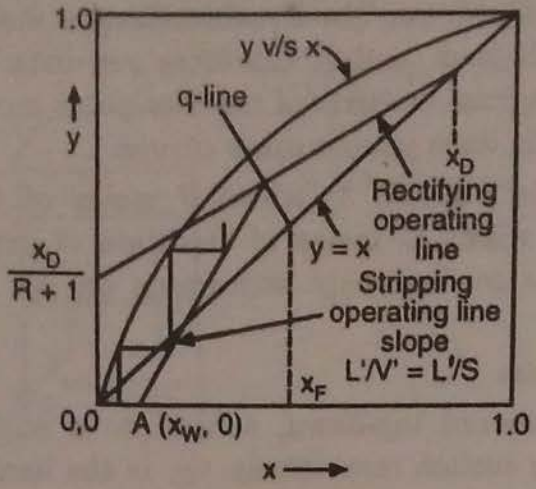

Fig. 1.35 : Use of open steam (construction of ideal stages in stripping section)

If steam entering the column is superheated, it will vapourise liquid on the plate at the bottom to an extent necessary to bring it to saturation. If λ_m is the molar latent heat at a column pressure, then

$$V' = S'\left[1 + \frac{H_{VS'} - H_{VS}}{\lambda_m}\right] \quad \ldots (1.115)$$

where $H_{VS'}$ is the enthalpy of superheated steam and H_{VS} is the enthalpy of saturated steam. S' is the molar flow rate of superheated steam and $L' = V' - S' + W$.

3. Ponchon – Savarit Method :

When molar latent heat is not constant and there is a substantial heat of mixing, then for establishing the relationship between number of plates, product composition and

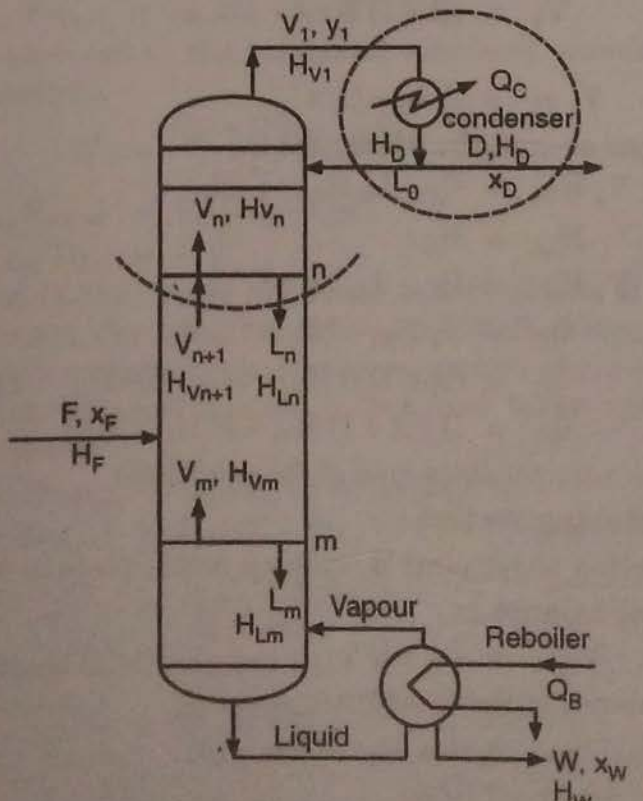

Fig. 1.36 : Fractionating column (Material and enthalpy balance)

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liquid/vapour ratio, we have to adopt the method developed by Ponchon and Savarit. Determination of the number of plates by this method is applicable for both constant and varying molar overflow situations. The Ponchon-Savarit method is based on the use of an enthalpy-composition diagram and it therefore requires detailed enthalpy data. The enthalpy-composition diagram consists of the dew-point curve (H v/s y), the bubble-point curve (H v/s x) and the tie lines joining these curves.

A continuous distillation unit to which F moles of feed of composition x_F is fed continuously and from which D moles of distillate of composition x_D and W moles of bottom product x_W are continuously withdrawn on per unit time basis is shown in Fig. 1.36.

Overall material balance :

The plates are numbered top-down, subscripts n and m indicating the rectifying section and the stripping section respectively. Q_C is the heat removed in a condenser and Q_B is the heat added to a reboiler.

The overall material balance over the column is :

$$F x_F = D x_D + W x_W \quad \ldots (1.116)$$

The overall energy balance over the column is :

$$F H_F = D H_D + Q_C + W H_W - Q_B \quad \ldots (1.117)$$

(Assuming negligible heat losses)

So heat to be added to the reboiler is :

$$Q_B = D H_D + Q_C + W H_W - F H_F \quad \ldots (1.118)$$

Equation (1.118) gives the reboiler heat i.e. heat to be added at the reboiler.

The material balance over the condenser is

$$V_1 = L_o + D$$

$$V_1 = (R + 1) D \quad \ldots (1.118\ a)$$

For component A :

$$V_1 y_1 = D x_D + L_o x$$

The enthalpy balance around the condenser is :

$$V_1 H_{V1} = Q_C + L_o H_{L_o} + D H_D$$

but $$H_{L_o} = H_D$$

$$V_1 H_{V1} = Q_C + L_o H_D + D H_D$$

$$Q_C = V_1 H_{V1} - (L_o + D) H_D$$

$$Q_C = (R + 1) D H_{V1} - (L_o + D) H_D, \quad \text{as } L_o/D = R$$

$$Q_C = D [(R + 1) H_{V1} - R H_D - H_D] \quad \ldots (1.119)$$

Equation (1.119) provides the heat load of the condenser.

(i) Rectifying/enriching section :

For the complete section above plate n, where n is any plate in this section.

The overall material balance is :

$$V_{n+1} = L_n + D$$

and the material balance of the component A is :

$$V_{n+1} y_{n+1} = L_n x_n + D x_D$$

$$V_{n+1} y_{n+1} - L_n x_n = D x_D \quad \ldots (1.120)$$

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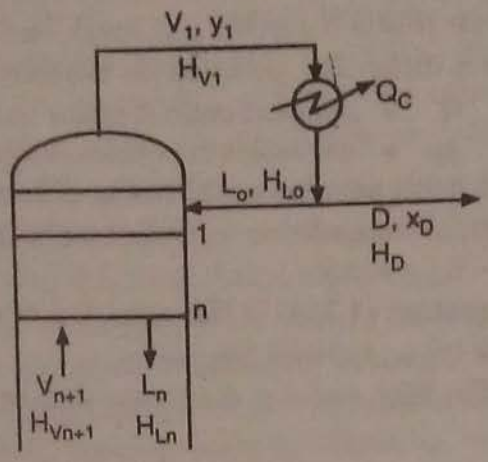

Fig. 1.37 : Rectifying section

The LHS of Equation (1.120) is the difference in the rate of flow of component A, up–down, i.e., net flow of A upward. For a given distillation, as the RHS of Equation (1.120) is constant, it follows that the net rate of flow of component A upward is constant irrespective of plate number in this section and is always equal to A which is permanently withdrawn from the top.

We have $D = V_{n+1} - L_n$

$$\therefore \quad V_{n+1}\, y_{n+1} = L_n\, x_n + (V_{n+1} - L_n)\, x_D$$

$$V_{n+1} y_{n+1} = L_n\, x_n + V_{n+1}\, x_D - L_n\, x_D$$

$$V_{n+1}(x_D - y_{n+1}) = L_n\,(x_D - x_n)$$

$$\frac{L_n}{V_{n+1}} = \frac{x_D - y_{n+1}}{x_D - x_n} \qquad \ldots (1.121)$$

Similarly, the enthalpy balance with negligible heat loss is

$$V_{n+1} H_{Vn+1} = L_n H_n + Q_C + D H_D$$

Let Q' be the heat removed in the condenser and heat associated with the distillate removed per mole of distillate.

$$\therefore \quad Q' = \frac{Q_C + D H_D}{D} = \frac{Q_C}{D} + H_D$$

$$V_{n+1} H_{Vn+1} = L_n H_{Ln} + DQ'$$

$$\therefore \quad V_n H_{Vn+1} - L_n H_{Ln} = DQ' \qquad \ldots (1.122)$$

The LSH of Equation (1.122) is the difference in rate of flow of heat, up–down, i.e., the net rate of heat flow upward. For given conditions, the RHS of Equation (1.122) is constant, so the net rate of heat flow upward is constant irrespective of plate number in this section. The net rate of heat flow upward is equal to the heat taken out permanently with the distillate and at the condenser.

$$V_{n+1} H_{Vn+1} = L_n H_{Ln} + (V_{n+1} - L_n)\, Q'$$

$$V_{n+1}(Q' - H_{Vn+1}) = L_n\,(Q' - H_{Ln})$$

$$\frac{L_n}{V_{n+1}} = \frac{Q' - H_{Vn+1}}{Q' - H_{Ln}} \qquad \ldots (1.123)$$

So combining Equations (1.121) and (1.123), we get

$$\frac{L_n}{V_{n+1}} = \frac{x_D - y_{n+1}}{x_D - x_n} = \frac{Q' - H_{Vn+1}}{Q' - H_{Ln}} \qquad \ldots (1.124)$$

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So Equation (1.124) on the enthalpy-composition diagram, i.e., on the Hxy diagram is a straight line passing through points $V_{n+1}(H_{V\,n+1}, y_{n+1})$, L_n (H_{Ln}, x_n) and $\Delta_D(Q', x_D)$.

$\Delta_D(Q', x_D)$ is called as a difference point as its coordinates represent the difference in rates of flow :

Q' = net heat out/net moles out

x_D = net moles A out/net moles out.

Δ_D is a fictitious stream with amount equal to the difference between vapour and liquid streams between two plates, of composition x_D and of enthalpy Q'.

$$\therefore \quad V_{n+1} - L_n = \Delta_D \qquad \ldots (1.125)$$

On the xy diagram, Equation (1.124) is the equation of a straight line of slope L_n/V_{n+1} and passing through points (y_{n+1}, x_n) and (x_D, x_D).

Fig. 1.38 shows both the Hxy and x-y diagrams and are drawn for the case of a total condenser.

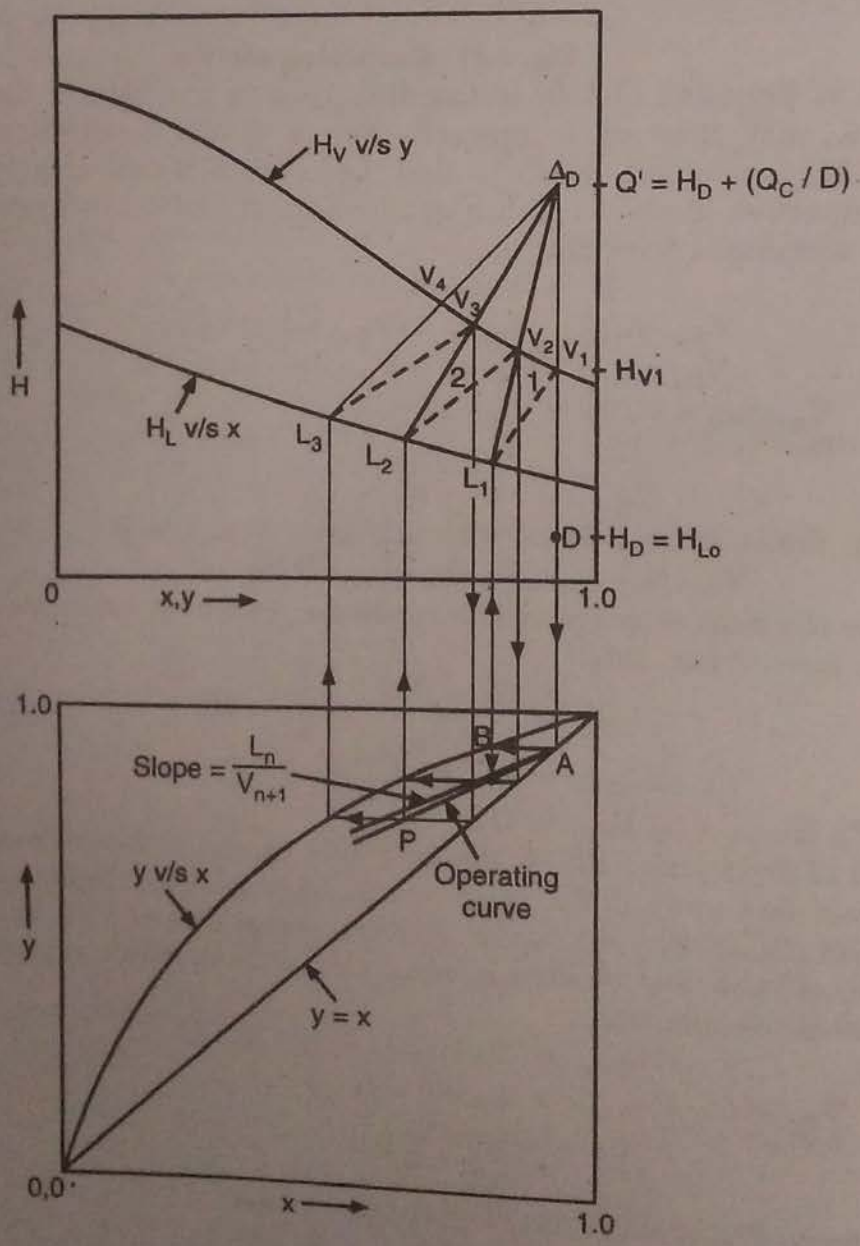

Fig. 1.38 : Rectifying section – A case of total condenser and reflux below bubble point

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In case of a total condenser, the distillate D and reflux L_0 are having the same composition and enthalpy content, i.e. on the Hxy diagram they are having the same coordinates (x_D, H_D). So in Fig. 1.38, they are plotted at point D. The point D indicates that both the distillate and reflux are below the bubble point (i.e. they are subcooled liquids). When both D and L_0 are at the bubble point (i.e. if only the latent heat is removed in the condenser) then the point D will be on the saturated liquid curve (H_L v/s x).

Locate point Δ_D on the Hxy diagram with its coordinates (x_D, Q'). Extend a line through points D, V_1 upward and on this line mark Δ_D at which H = Q'. The points Δ_D, D and V_1 are on the same line as they are having the same composition (x_D).

When the saturated vapour V_1 issuing from plate-1 to the condenser is condensed completely then it has the same composition as that of D and L_0. The saturated vapour V_1 is shown on the saturated vapour curve (i.e. H_V v/s y) with its coordinates (x_D, H_{V1} – enthalpy content of vapour). The liquid L_1 leaving plate-1, which is in equilibrium with V_1, is located on the bubble point curve/saturated liquid curve at the end of a tie line-1. For drawing a tie line; from V_1, drop a vertical on the xy diagram to cut the diagonal at point A then proceed horizontally towards the equilibrium curve to cut it at B and then draw a vertical through B to cut H_L v/s x curve at L_1. Join V_1L_1. V_1 L_1 is the required tie line.

As Equation (1.124) is applicable to any plate in the rectifying section, V_2 can be located on the saturated vapour curve by drawing a line from L_1 to Δ_D. The three quantities V_{n+1}, L_n and Δ_D will always lie on the straight line through Δ_D. The tie line through V_2 locates point L_2 on the saturated liquid curve. The point P is produced as shown in Fig. 1.38 and all such points produce the operating curve that passes through $y = x = x_D$. The difference point Δ_D is used in this way for further construction till the feed plate is reached.

We have,

$$V_{n+1} = L_n + D$$

and

$$\frac{L_n}{V_{n+1}} = \frac{Q' - H_{V\,n+1}}{Q' - H_{Ln}} = \frac{x_D - y_{n+1}}{x_D - x_n}$$

$$\frac{L_n}{L_n + D} = \frac{Q' - H_{V\,n+1}}{Q' - H_{Ln}}$$

$$\frac{L_n + D}{L_n} = \frac{Q' - H_{Ln}}{Q' - H_{V\,n+1}}$$

$$1 + \frac{D}{L_n} = \frac{Q' - H_{Ln}}{Q' - H_{V\,n+1}}$$

$$\frac{D}{L_n} = \frac{Q' - H_{Ln}}{Q' - H_{V\,n+1}} - 1$$

$$= \frac{Q' - H_{Ln} - Q' + H_{V\,n+1}}{Q' - H_{V\,n+1}}$$

$$\frac{D}{L_n} = \frac{H_{V\,n+1} - H_{Ln}}{Q' - H_{V\,n+1}}$$

$$\frac{L_n}{D} = \frac{Q' - H_{V\,n+1}}{H_{V\,n+1} - H_{Ln}} \quad \ldots (1.126)$$

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When the above equation is applied at the top plate, we get the external reflux ratio.

$$R = \frac{L_o}{D} = \frac{Q' - H_{V1}}{H_{V1} - H_{Lo}} = \frac{\text{line length } \Delta_D\ V_1}{\text{line length } V_1\ L_o}$$

$$= \frac{\text{line length } \Delta_D\ V_1}{\text{line length } V_1\ D} \quad \ldots (1.127)$$

For a given reflux ratio, the line-length ratio given by Equation (1.127) can be used to locate the difference point Δ_D and the y-coordinate of Δ_D can then be used to obtain the heat load of the condenser.

Partial condenser :

When a partial condenser is used, a distillate is withdrawn in the vapour state (as a saturated vapour) and the condensed liquid provides the reflux. Only some part of the overhead vapour leaving the top plate is condensed to provide the reflux and the remainder of it is removed as the distillate.

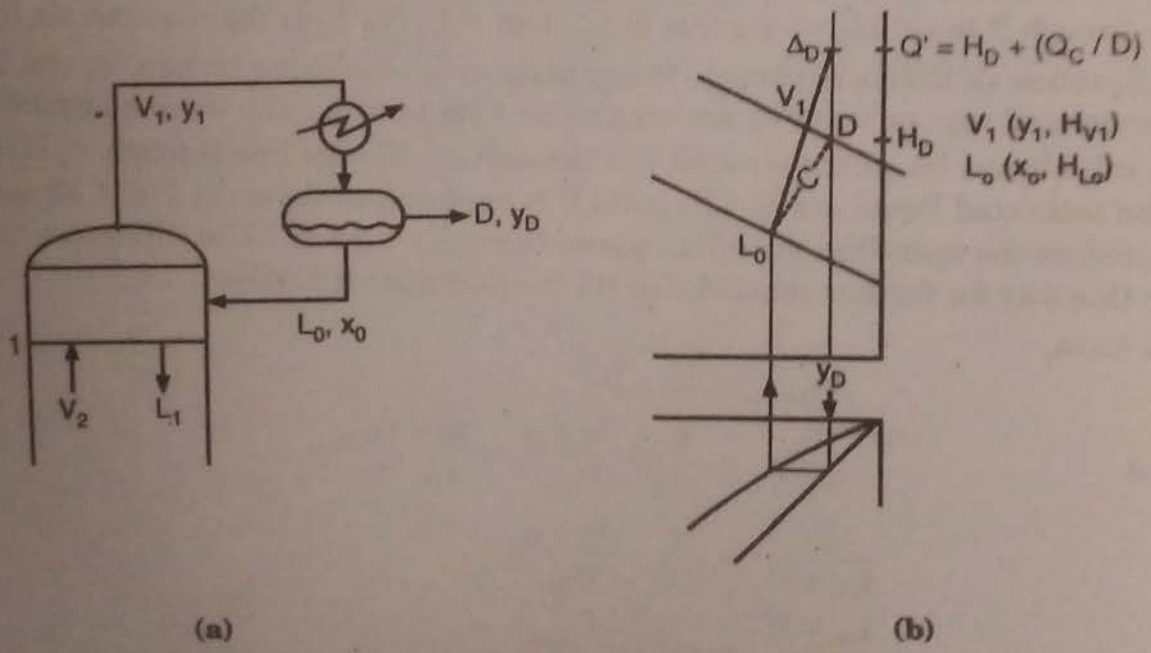

Fig. 1.39 : Partial condenser (or dephegmator)

In the case of a partial condenser, Δ_D is plotted at an abscissa y_D, where y_D is the composition of the vapour product/distillate withdrawn. Normally, the vapour leaving the condenser as the product is in equilibrium with the liquid reflux, the liquid reflux L_o will be at the end of the tie line C. V_1 is located by the construction line $L_o\ \Delta_D$. As each of reboiler and partial condenser acts as one theoretical stage, the number of theoretical plates required in the column will be $(n + 2) - 2$ [i.e. the number of tie lines (between y_D and x_W) – 2].

The reflux ratio (external reflux ratio) $= R = L_o/D = \dfrac{\text{line length } \Delta_D\ V_1}{\text{line length } V_1\ L_o}$.

The Stripping Section :

For the complete section below plate m, where m is any plate in the stripping section, the overall material balance is

$$L'_m = V'_{m+1} + W \quad \ldots (1.128)$$

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Material balance of component A is :

$$L'_m x_m = V'_{m+1} y_{m+1} + W x_W \quad \ldots (1.129)$$

$$L'_m x_m - V'_{m+1} y_{m+1} = W x_W \quad \ldots (1.130)$$

The LHS of Equation (1.130) is the difference in rate of flow of component A, down–up i.e. the net rate of flow of component A downward. As the RHS of Equation (1.130) is constant for a given distillation condition, the difference in rate of flow is independent of plate number so it is the same for any plate in this section and is equal to the rate of removal of A as the bottom product.

An enthalpy balance over the same portion of the column is

$$L'_m H_{Lm} + Q_B = V'_{m+1} H_{V\,m+1} + W H_W \quad \ldots (1.131)$$

Let Q" be the net flow of heat out outward at the bottom per mole of residue/bottom product.

$$\therefore \quad Q'' = \frac{WH_W - Q_B}{W} = H_W - Q_B/W \quad \ldots (1.132)$$

With Equations (1.131) and (1.132), we get

$$L'_m H_{Lm} - V'_{m+1} H_{V\,m+1} = WQ'' \quad \ldots (1.133)$$

The LHS of Equation (1.133) is the difference in rate of heat flow, down-up. This Equation is applicable for any plate in the column and the LHS is equal to the net rate of heat flow out at the bottom. Let us eliminate W from Equations (1.129) and (1.130) :

$$L'_m = V'_{m+1} + W$$

$$L'_m x_m = V'_{m+1} y_{m+1} + W x_W$$

$$L'_m x_m = V'_{m+1} y_{m+1} + (L'_m - V'_{m+1}) x_W$$

$$L'_m x_m = V'_{m+1} y_{m+1} + L'_m x_W - V'_{m+1} x_W$$

$$\therefore \quad \frac{L'_m}{V'_{m+1}} = \frac{y_{m+1} - x_W}{x_m - x_W} \quad \ldots (1.134)$$

We have, $L'_m = V'_{m+1} + W$

$$L'_m H_{Lm} - V'_{m+1} H_{Vm+1} = WQ''$$

Eliminating W between these two equations gives

$$L'_m H_{Lm} - V'_m H_{V\,m+1} = (L'_m - V'_{m+1}) Q''$$

$$L'_m H_{Lm} - V'_m H_{V\,m+1} = L'_m Q'' - V'_{m+1} Q''$$

$$\therefore \quad \frac{L'_m}{V'_{m+1}} = \frac{H_{V\,m+1} - Q''}{H_{Lm} - Q''} \quad \ldots (1.135)$$

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Combining Equations (1.134) and (1.135) gives

$$\frac{L'_m}{V'_{m+1}} = \frac{H_{V\,m+1} - Q''}{H_{Lm} - Q''} = \frac{y_{m+1} - x_W}{x_m - x_W} \quad \ldots (1.136)$$

Equation (1.136) is a straight line on the Hxy diagram passing through points $V_{m+1}(H_{V\,m+1}, y_{m+1})$, L_m, (H_{Lm}, x_m) and $\Delta_W (Q'', x_W)$.

Δ_W is a difference point with coordinates (Q'', x_W).

Q'' = net heat out/net moles out, at the bottom

x_W = moles A out/net moles out, at the bottom

Δ_W is a fictitious stream which is equal to the net flow outward (W) of properties (Q'', x_W) (i.e. of composition x_W and of enthalpy Q'').

$$L'_m - V'_{m+1} = \Delta_W \quad \ldots (1.137)$$

On the xy diagram, Equation (1.136) is a straight line passing through points (y_{m+1}, x_m) and (x_W, x_W) with a slope of L'_m/V'_{m+1}.

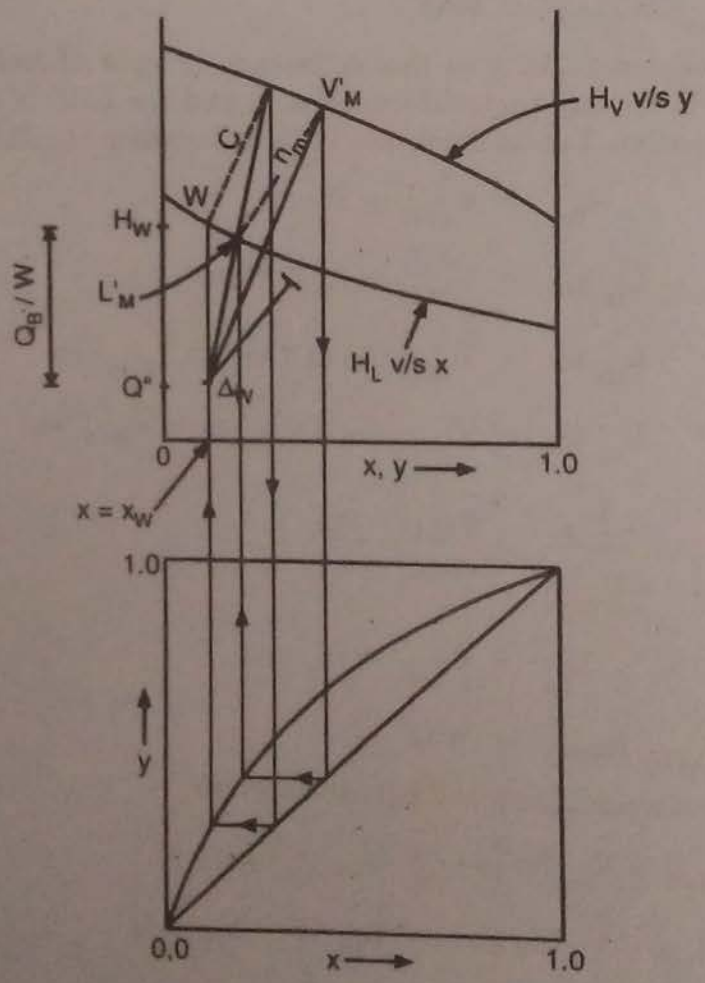

Fig. 1.40 : Stripping section (with bottom product-saturated liquid)

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Equation (1.136) is applicable to all plates in the stripping section. On the Hxy diagram, locate point W on the saturated liquid curve at (x_W, H_W) and point Δ_W at (x_W, Q''). W and Δ_W are located on the same line as they have the same composition x_W. (Refer Fig. 1.40). All the operating lines represented by Equation (1.136) pass through Δ_W (the difference point). The line from V'_{m+1} (representing the vapour leaving the reboiler and entering the plate at the bottom of the column) to Δ_W intersects the saturated liquid curve at point L'_m, the liquid leaving the bottom plate. The vapour V'_M leaving the bottom plate is in equilibrium with the liquid L'_M and is at the end of the tie line n_M. The procedure of construction of equilibrium stages similar to that described for the rectifying section is applied here. Note that Fig. 1.40 is drawn for the case where the vapour leaving the reboiler is in equilibrium with the residue leaving the reboiler so that the reboiler itself provides one theoretical stage (represented by a tie line C). The number of ideal plates are determined by alternating construction lines to Δ_W and tie lines, each tie line represents an equilibrium stage.

The complete fractionator :

Overall material balance over the fractionater is

$$F = D + W \qquad \ldots (1.138)$$

Overall material balance of the component A is

$$F x_F = D x_D + W x_W \qquad \ldots (1.139)$$

Overall enthalpy balance for negligible heat losses over the fractionator is

$$Q_B = D H_D + W H_W + Q_C - F H_F$$

We have,

$$Q' = (Q_C/D) + H_D$$

$$Q'' = H_W - (Q_B/W)$$

$$Q_B = (H_W - Q'') W$$

and

$$Q_C = (Q' - H_D) D$$

$$\therefore \quad (H_W - Q'') W = D H_D + W H_W + (Q' - H_D) D - F H_F$$

$$W H_W - Q''W = D H_D + W H_W + Q'D - H_D D - F H_F$$

$$\therefore \quad F H_F = DQ' + WQ'' \qquad \ldots (1.140)$$

Eliminating F between Equations (1.138) and (1.139) gives

$$x_F (D + W) = D x_D + W x_W$$

$$x_F D + x_F W = D x_D + W x_W$$

$$W (x_F - x_W) = D (x_D - x_F)$$

$$D/W = \frac{x_F - x_W}{x_D - x_F} \qquad \ldots (1.141)$$

Eliminating F between Equations (1.138) and (1.140) gives

$$(D + W) H_F = DQ' + WQ''$$

$$D H_F + W H_F = DQ' + WQ''$$

$$W (H_F - Q'') = D (Q' - H_F)$$

$$\frac{D}{W} = \frac{H_F - Q''}{Q' - H_F} \qquad \ldots (1.142)$$

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Combining Equations (1.141) and (1.142), we get

$$D/W = \frac{x_F - x_W}{x_D - x_F} = \frac{H_F - Q''}{Q' - H_F} \quad \ldots (1.143)$$

On the Hxy diagram, Equation (1.143) is a straight line passing through points $\Delta_D (Q', x_D)$, F (H_F, x_F), and $\Delta_W (Q'', x_W)$.

$$\therefore \quad F = \Delta_D + \Delta_W \quad \ldots (1.144)$$

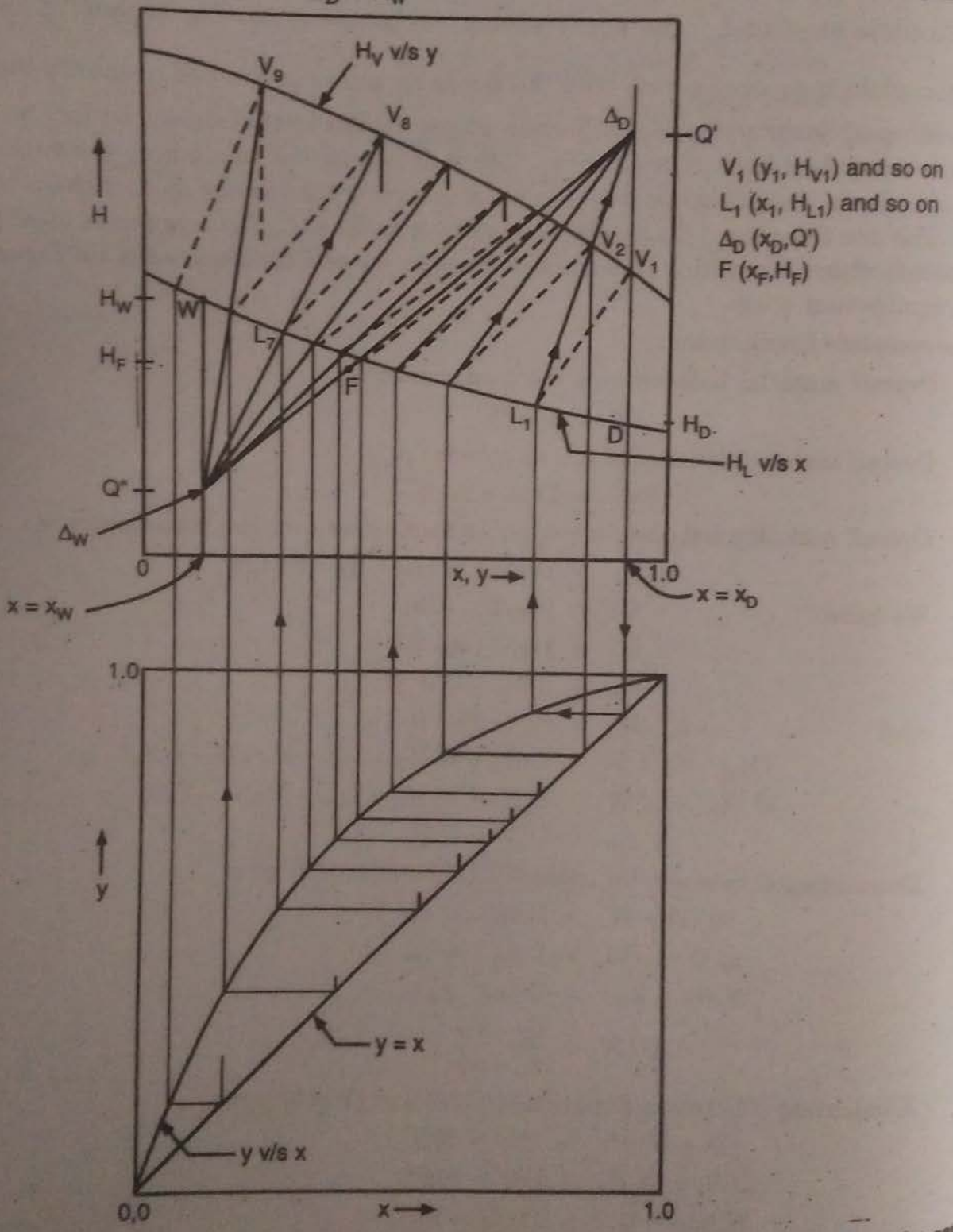

Fig. 1.41 : Complete fractionator, feed-cold liquid (below bubble point) and total condenser

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In Fig. 1.41, the location of point F (x_F, H_F) indicates that the feed is a liquid below its bubble point (cold feed). In the case of a saturated liquid, the feed point F will lie on the bubble-point curve/saturated liquid curve and in the case of a saturated vapour feed (feed-saturated vapour), F will lie on the saturated vapour curve/dew-point curve. In any case, note that Δ_D (x_D, Q'), Δ_W (x_W, Q") and F (x_F, H_F) must lie on a single straight line.

Stepwise Procedure for Determination of Number of ideal plates using Ponchon-Savarit Method :

(i) Plot the Hxy diagram and the xy diagram as shown in Fig. 1.42.

(ii) Locate the point F on the Hxy diagram. Coordinates of F are (x_F, H_F). F represents the condition of a feed. In Fig. 1.42, F is located on the saturated liquid curve which indicates that the feed is a saturated liquid.

(iii) Locate abscissas $x = x_D$ and $x = x_W$, by drawing verticals, through them, representing the top and bottom compositions.

(iv) Locate Δ_D vertically on vertical line through $x = x_D$, i.e. locate Δ_D on the ordinate at $x = x_D$. The coordinates of Δ_D are (x_D, Q').

∴ Locate Δ_D at $H = Q'$ on the ordinate at $x = x_D$.

(v) We know that Δ_D, F and Δ_W lie on the same straight line. Join Δ_D – F and extend this line to intersect the vertical line $x = x_W$ in the desired point Δ_W.

Locate Δ_W at which the extension of line Δ_D F cuts the ordinate at $x = x_W$.

(vi) Locate V_1 on the saturated vapour curve at which the ordinate through $x = x_D$ (i.e. vertical line $x = x_D$) intersects the curve, i.e., locate V_1 at the intersection of the vertical line $x = x_D$ with the saturated vapour curve.

V_1 represents the condition of vapour leaving the top plate and has coordinates (y_1, H_{V1}).

(vii) Now, we have to make use of the xy diagram, i.e. the equilibrium curve to draw tie lines L_1V_1, L_2V_2, etc.

(viii) We have to locate L_1 (x_1, H_{L1}) on the saturated liquid curve. Through $x = x_D$ on the Hxy diagram, drop a vertical to cut the diagonal (y = x) on the xy diagram at point A. Through A, proceed horizontally to get point B on the equilibrium curve and then through B, draw a vertical on the Hxy diagram to cut the saturated liquid curve at L_1. Join V_1L_1 which is a tie line representing an equilibrium stage (1).

Similar procedure is to be adopted for getting L_2, L_3 ... and drawing tie lines V_2L_2 , V_3L_3 .... (L_1 represents the condition of liquid on top plate.)

(ix) L_1, V_2 and Δ_D will lie on the same straight line. So join Δ_D – L_1 and mark V_2 (y_2, H_{V2}) at which line Δ_D – L_1 intersects the saturated vapour curve.

(V_2 represents the condition of vapour on the second plate.)

The same procedure is adopted for locating V_3, V_4 ... on the saturated vapour curve (as L_n, V_{n+1} and Δ_D lie on the same straight line.)

(x) Through V_2, drop a vertical on the xy diagram to cut the diagonal, from there move horizontally to the equilibrium curve and from that point on the equilibrium curve, draw a vertical to cut the saturated liquid curve at L_2. Join L_2 - V_2 which is a tie line representing the second ideal plate in the column from the top.

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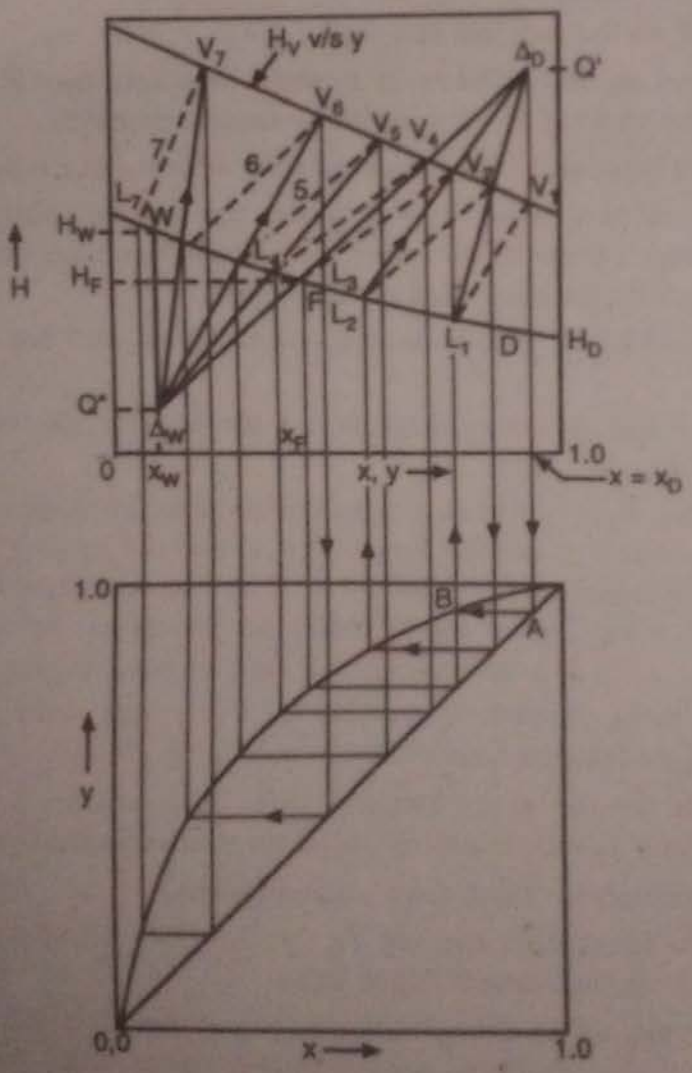
(xi) L_2, V_3 and Δ_D will lie on the same straight line. Join $L_2 - \Delta_D$ to cut the saturated vapour curve at V_3.

(xii) Repeat the above mentioned procedure for further construction of the number of theoretical plates till the feed plate is reached.
Use Δ_D for construction till the feed plate is reached.
The tie line which crosses the line $\Delta_D - F - \Delta_W$ represents a feed plate.

(xiii) Make use of Δ_W for the further construction work when any such tie line constructed crosses the line joining $\Delta_D - F - \Delta_W$.

(xiv) Suppose as an example (not true in Fig. 1.42) a tie line $V_7 - L_7$ crosses the line through points Δ_W, F and Δ_D, the plate 7th from the top is the feed plate. Afterwards join Δ_W to L_7 and extend it to cut the saturated vapour curve at V_8 (Δ_W, V_8 and L_7 lie on the same straight line.)
In Fig. 1.43 a tie line V_3L_3 crosses the line through points Δ_D, F and Δ_W, so the plate 3rd from the top is the feed plate. Afterwards L_3 is joined to Δ_W and $\Delta_W L_3$ is extended to meet the saturated vapour curve at V_4.



$L_1 (x_1, H_{L1})$ and so on

$V_1 (y_1, H_{V1})$ and so on

$Q' = H_D + (Q_C/D)$

$Q'' = H_W - (Q_R/W)$

Fig. 1.42 : Determination of number of plates using enthalpy-composition diagram (feed at its bubble point, F on H_L v/s x curve, feed plate 3rd from top, total plates n + 1 = 7 ∴ n = 6)

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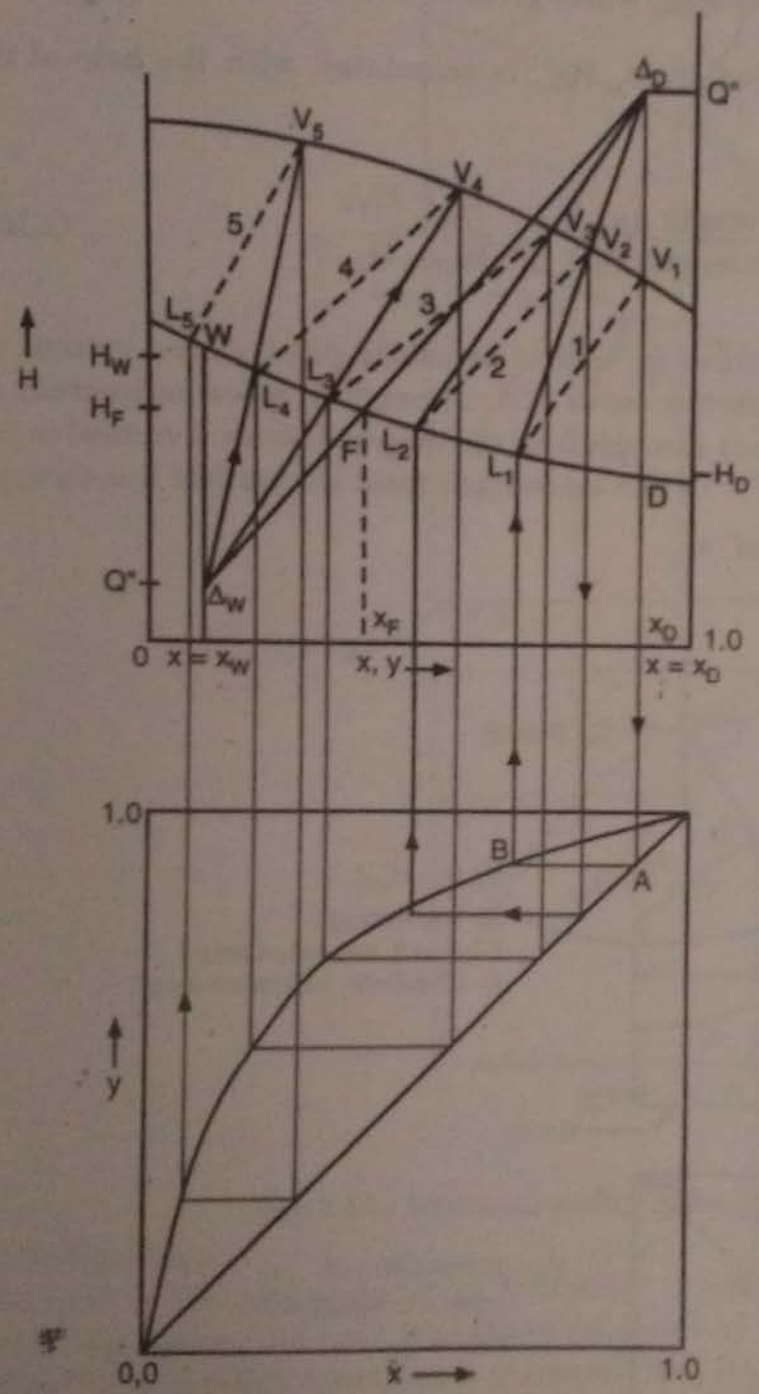
(xv) For the stripping section, adopt the same procedure of construction of the equilibrium stages (tie lines) as is given for the rectifying section and continue the construction work using Δ_W till Δ_W (x_W, Q") is exactly reached or covered. Count the number of tie lines. The number of tie lines counted from x_D to x_W represent the number of ideal plates including reboiler (i.e. n + 1), so the number of ideal plates required in the column will be n.



L_1 (x_1, H_{L1}) and so on

V_1 (y_1, H_{V1}) and so on

$Q' = H_D + (Q_C/D)$

$Q'' = H_W - (Q_B/W)$

Fig. 1.43 : Determination of number of theoretical plates using enthalpy-composition diagram (F, W, D at saturated conditions, feed plate 3rd from top, total plates including reboiler = 5)

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Minimum Reflux Ratio :

At minimum reflux ratio, requiring an infinite number of plates to achieve the desired degree of separation, both the heat requirement of the reboiler and the cooling water requirement of the condenser are minimum.

For the lowest possible heat load of the reboiler and the condenser cooling load, on the Hxy diagram, the tie line passes through F (x_F, H_F). The tie line passing through F intersects a vertical line through $x = x_D$ at Δ_{Dm}, which gives the value of ordinate Q'_m. Once Q'_m is determined, the minimum reflux ratio, R_m, is calculated with the help of the following equation :

$$R_m = \frac{\text{line length } \Delta_{Dm} V_1}{\text{line length } V_1 D} = \frac{Q'_m - H_{V1}}{H_{V1} - H_D} \quad \ldots (1.145)$$

Locating tie line through F :

Through F (on the saturated liquid curve for feed at its bubble point) drop a vertical on the xy diagram which will cut the equilibrium curve at A. Through A, move horizontally to the diagonal y = x at point B (y_F) and then through B on the diagonal, draw a vertical to cut the saturated vapour curve at V_F. Join F – V_F and extend it to meet the vertical line through $x = x_D$ at Δ_{Dm} and read the ordinate Q'_m of Δ_{Dm}.

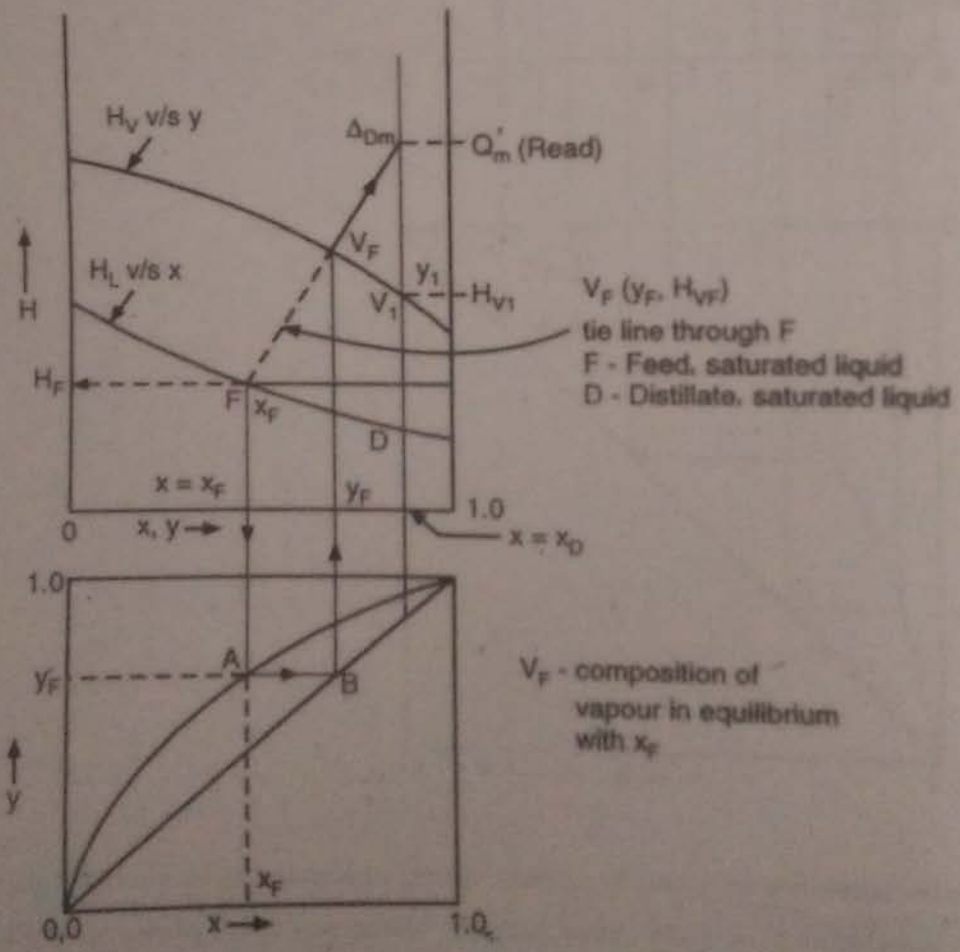

Fig. 1.44 : Minimum reflux ratio (feed-saturated liquid) (locating Δ_{Dm})

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Take any value of x on H_L v/s x, go to the equilibrium curve to get y, proceed vertically on H_V v/s y and see that F, x and y lying on the same straight line or not. Repeat the procedure till they lie on the same line. Here F, x' on H_L v/s x and y' on H_V v/s y lie on the same straight line.

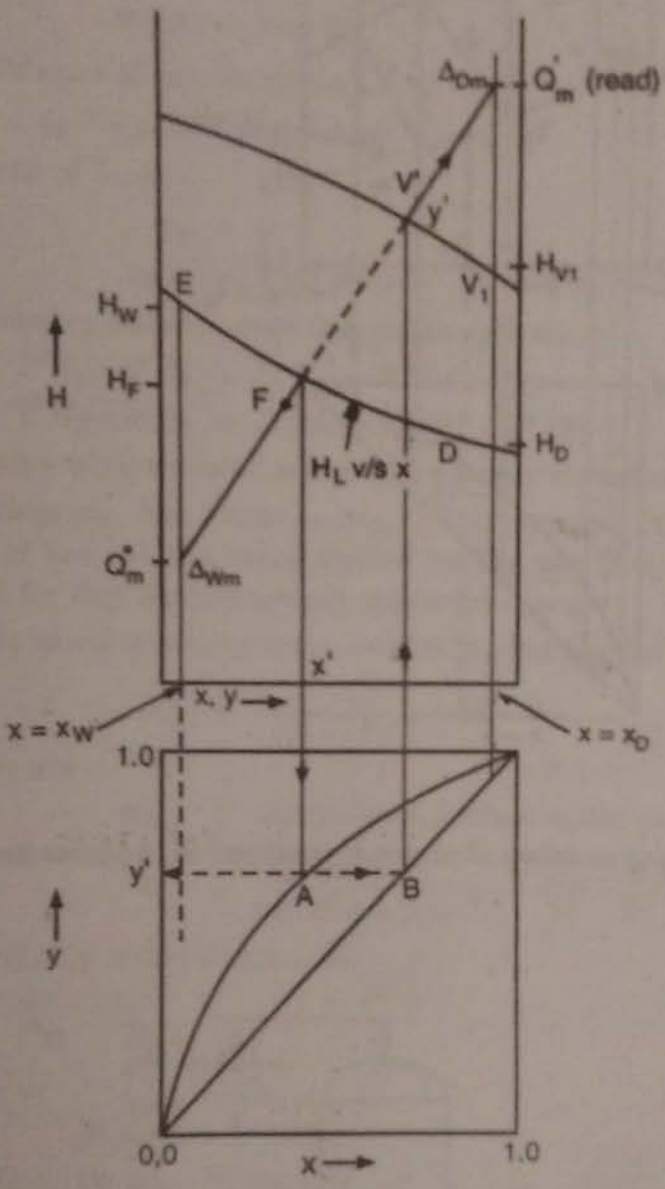

Fig. 1.45 : Minimum reflux ratio (feed-cold liquid)

Total Reflux Ratio :

At total reflux ratio, requiring minimum number of plates to achieve the desired degree of separation, the head load of the reboiler and the cooling load of the condenser are infinite. The total reflux operation is of interest only because that it sets a lower limit on the number of plates required and is not a practical method of operation as at total reflux the product rate is zero.

At total reflux, Δ_D and Δ_W points are at infinity (∞) and consequently the lines joining points L_n (x_n, H_{Ln}) and Δ_D as well as points (x_m, H_{Lm}) and Δ_W are all vertical.

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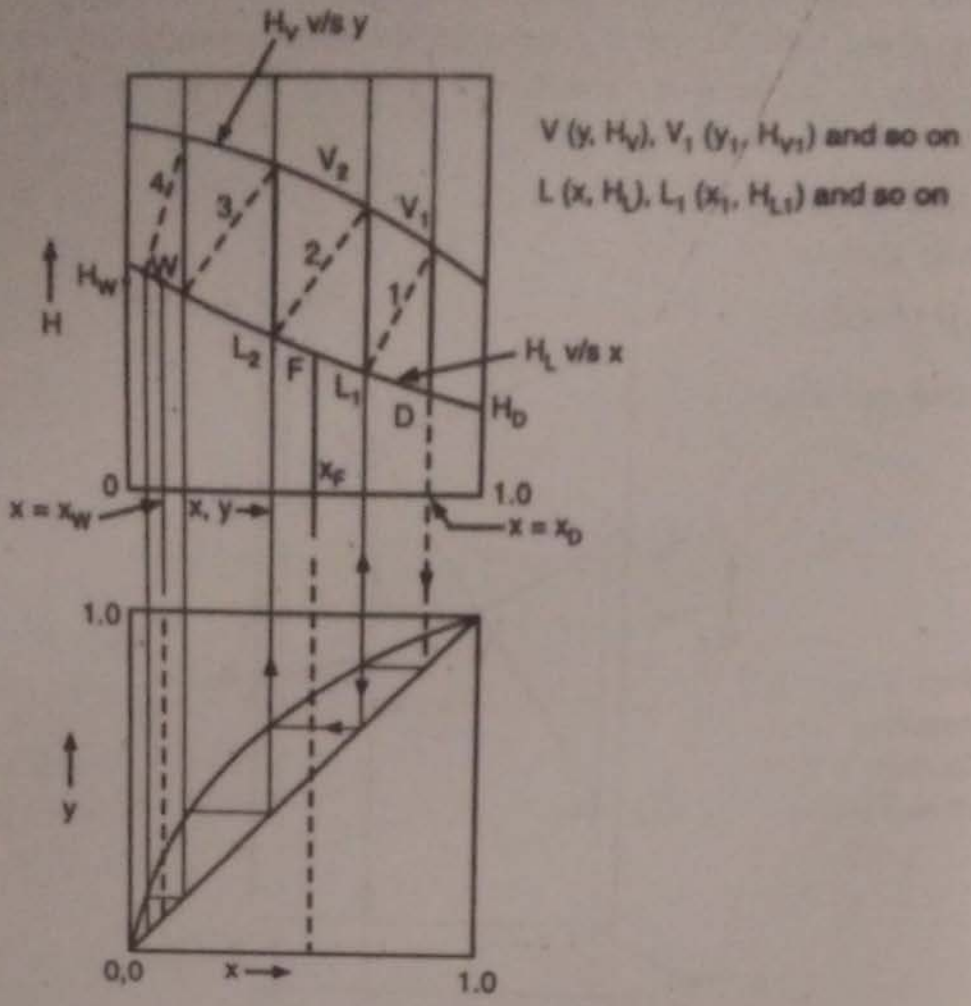

Fig. 1.46 : Obtaining number of plates at total reflux (4 plates including reboiler)

Use of Open Steam :

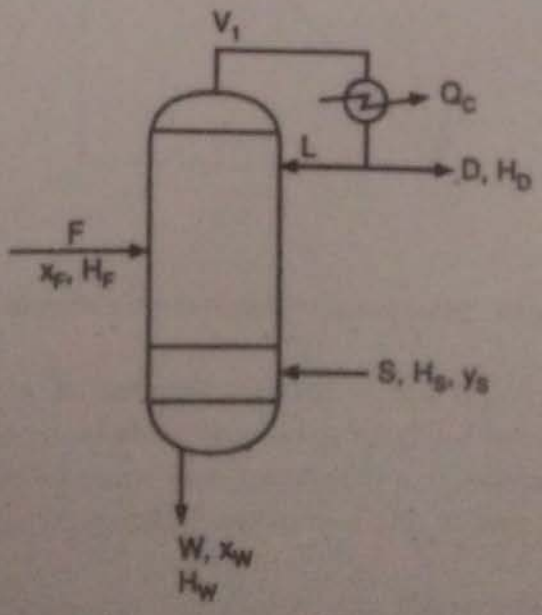

Fig. 1.47 : Use of open steam

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In the fractionation of a water solution to obtain the more volatile non-aqueous component as the distillate and where water is removed as the bottom product, the heat required can be provided by introduction of steam directly to the bottom of the column, thus eliminating the need of a reboiler-normally used.

The material balance over the entire column is :

$$F + S = D + W \qquad \ldots (1.146)$$

The material balance of the component A is

$$x_F F + y_S S = x_D D + x_W W \qquad \ldots (1.147)$$

The steam is free of A, so

$$y_S = 0$$

$$\therefore \quad x_F F = x_D D + x_W W$$

The energy/enthalpy balance over the entire column is

$$F H_F + S H_S = D H_D + W H_W + Q_C$$

$$F H_F + S H_S = D (H_D + Q_C/D) + W H_W \qquad \ldots (1.148)$$

H_S is the enthalpy of steam used and S is the molar rate of steam introduced.

On the Hxy diagram, the point $\Delta_D (x_D, Q')$ is located in the usual manner as the rectifying section of the column is unaffected by the use of open steam and all equations previously derived for this section are all applicable here.

Here Δ_W has its usual meaning which is nothing but the net flow outward.

$$\Delta_W = L'_m - V'_{m+1} = W - S \qquad \ldots (1.149)$$

and its coordinates are :

$$x_{\Delta_W} = \text{net moles A out/net moles out} = W x_W/(W - S)$$

$$Q'' = \frac{\text{net heat out}}{\text{net moles out}} = \frac{W H_W - S H_S}{W - S}$$

where H_S is the enthalpy of the steam used.

We have : $L'_m - V'_{m+1} = \Delta_W$

$$L'_m x_m - V'_{m+1} y_{m+1} = \Delta_W x_W$$

$$L'_m H_{Lm} - V'_{m+1} H_{V\,m+1} = \Delta_W \cdot Q''$$

$$\frac{L'_m}{V'_{m+1}} = \frac{y_{m+1} - x_{\Delta_W}}{x_m - x_{\Delta_W}} = \frac{H_{V\,m+1} - Q''}{H_{Lm} - Q''} \qquad \ldots (1.150)$$

m may be any plate in the stripping section.

A graphical construction for the number of theoretical plates required will be the same as that previously described for the section above the feed plate (rectifying section). From the section below the feed plate (stripping section), the only difference is that a point Δ_W with coordinates x_{Δ_W}, Q" is used for determining point $(y_{m+1}, H_{V\,m+1})$ corresponding to the point (x_m, H_{Lm}).

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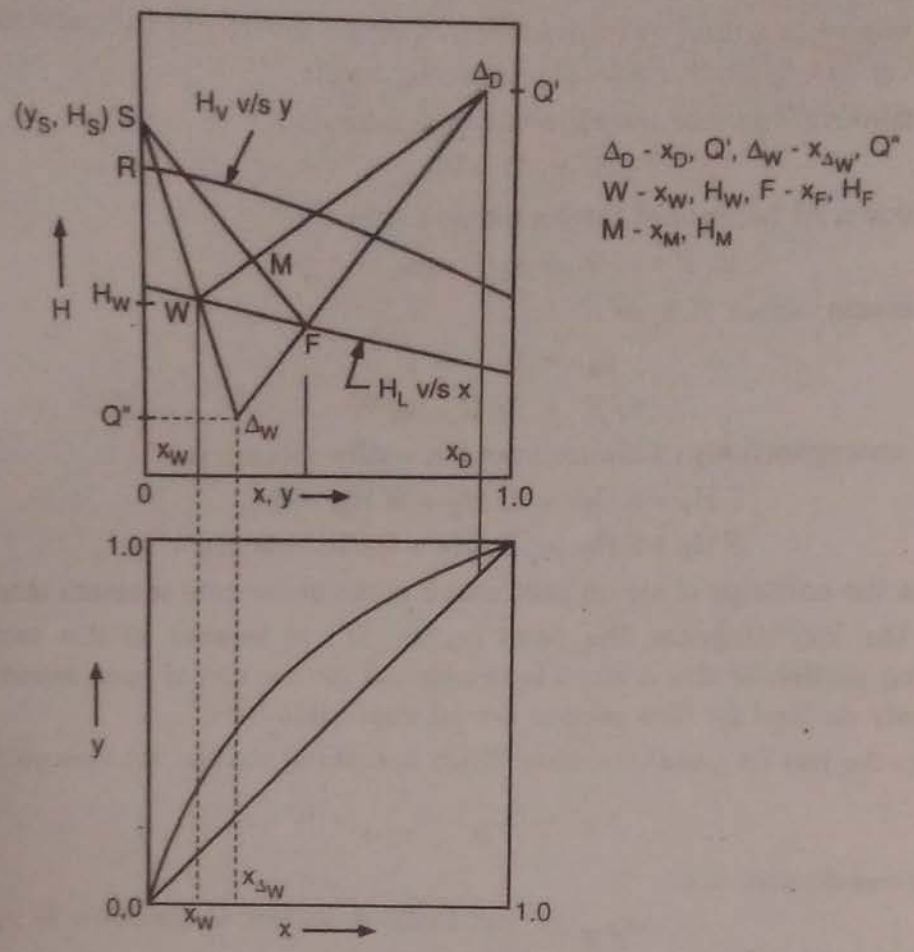

Fig. 1.48 : Use of open steam
(ideal stage construction not shown.
Determination of steam required by locating M is shown)

Determination of amount of steam used :

The amount of steam used can be determined by means of Equations (1.146), (1.147) and (1.148).

Let

$$M = D + S$$

$$M\,x_M = D\,x_D + S\,y_S \quad \ldots (1.151)$$

$$M\,H_M = D\,(H_D + (Q_C/D) + W\,H_W \quad \ldots (1.152)$$

So the point M (x_M, H_W) must lie on a straight line joining the points W (x_W, H_W) and Δ_D (x_D, Q').

From Equations (1.146), (1.147) and (1.148), it is clear that the point M must also lie on the straight line through points (y_S, H_S) and F (x_F, H_F). These points are known. In Fig. 1.48, the steam used is shown slightly superheated (point S). S will be at R for the saturated steam at a given pressure.

The intersection of the two lines S – F and W – Δ_D determines the point M (x_M, H_M). Furthermore,

$$\frac{S}{F} = \frac{H_M - H_F}{H_S - H_M} = \frac{x_M - x_F}{y_S - x_M} \quad \ldots (1.153)$$

All quantities in Equation (1.153) can be known except S, so that S can be determined.

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Batch Distillation :

Batch distillation is used extensively in small scale production units where the same piece of equipment is to be used for many different mixtures and where small quantities of the liquid mixtures are to be handled. It is useful when more than one product is to be obtained (i.e. different quality products) and when the liquid mixtures to be separated are high in solid content, tar, etc. as it keeps the solids separated (in reboiler/still) which are removed at the termination of the process. A column may be mounted on the top of reboiler (if column is not large) or may be supported independently, with connections for vapour and liquid streams as shown in Fig. 1.49.

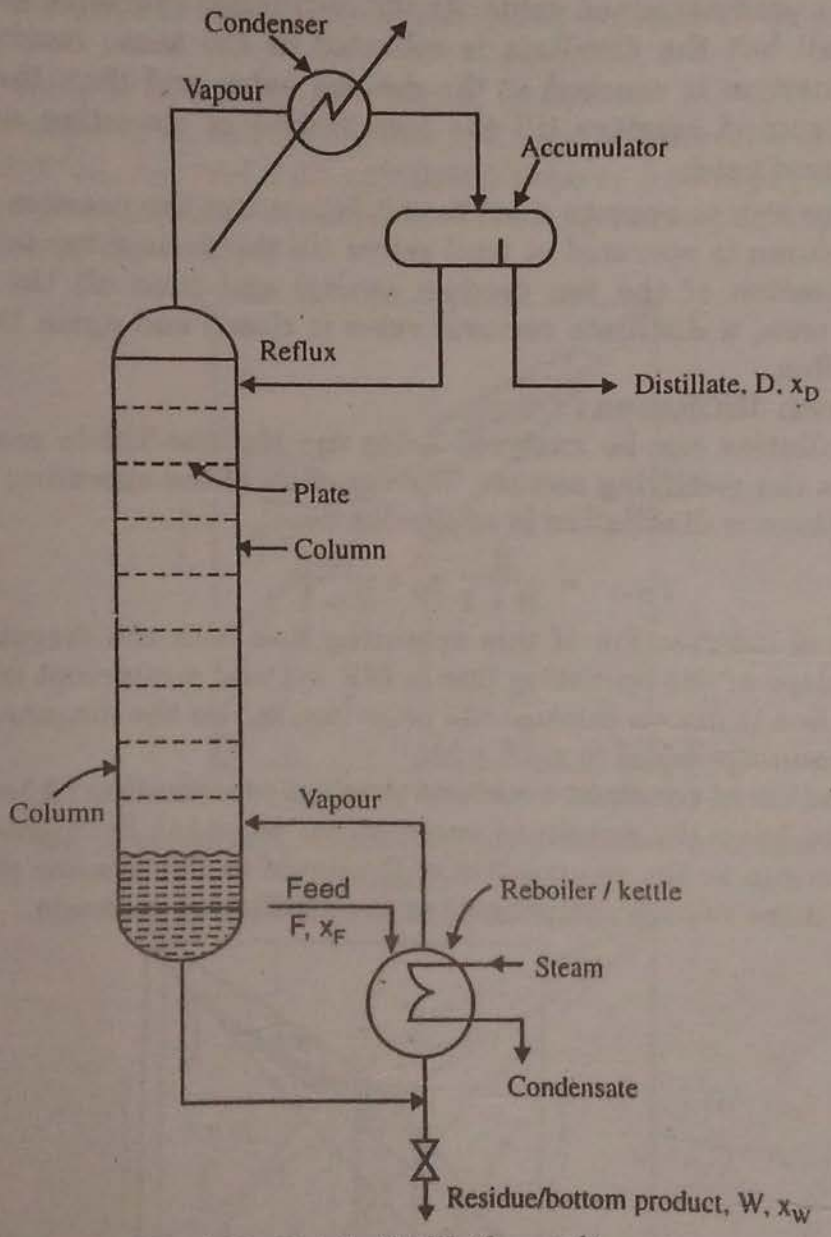

Fig. 1.49 : Batch distillation unit

In batch distillation, a specific quantity of a liquid mixture is charged to a reboiler/still, heating is applied, vapours generated flow upward through a fractionating column and a part of the liquid from a condenser runs down the column as reflux. The entire fractionating column acts as the rectifying/enriching section. As the distillate will be rich in the more volatile component, the liquid in the reboiler becomes steadily weaker in the more volatile component as the operation proceeds and hence the purity of the product will steadily fall.

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Batch distillation operation may be carried by varying the reflux ratio so as to get a constant overhead composition. In this case, initially a column is operated under total reflux and then some value of the reflux ratio is adjusted. But as the distillation proceeds, the top product quality may fall. Thus, to keep the top product quality to be constant, the reflux ratio is increased. Reflux ratio is continually increased till it reaches a maximum value and then it is reduced and the cut is taken into a separate receiver. It may be charged to the next batch.

Another method of operating a batch distillation unit is to operate a column under a constant reflux ratio. The column is operated under total reflux initially and then the reflux is set to a predetermined value. As the distillation proceeds, the top products quality will steadily fall but the distillate is collected in the same receiver until the average distillate composition is reached at the desired value and then the overhead product is collected in a second receiver till the termination of operation and the same may be charged to the next batch.

Still another way to operate a batch unit is to adopt the practice of a cycling procedure wherein the column is operated at total reflux till the desired top temperature is achieved (which is indication of the top product purity) and then all the overhead product is withdrawn at once, a distillate removal valve is closed and again the column is operated under total reflux.

Analysis of batch distillation :

Batch distillation can be analysed using the McCabe-Thiele method. Here the entire column acts as the rectifying section. The equation of the operating line for the rectifying section of continuous distillation is applicable here.

$$y_{n+1} = \frac{R}{R+1} x_n + \frac{x_D}{R+1} \quad \ldots (1.154)$$

The point of intersection of this operating line with the diagonal has coordinates of x_D, x_D. The slope of the operating line is R/R + 1 and y-intercept is x_D/R + 1. On the x-y diagram, this line is drawn through the point (x_D, x_D) on the diagonal with a slope equal to R/R + 1 (or y-intercept equal to x_D/R + 1).

(i) **Operation at constant overhead product composition :** A batch distillation unit may be operated to keep the overhead composition constant by increasing the reflux ratio throughout the run as the composition of the liquid in the reboiler changes (decreases with respect to the more volatile component) as the distillation proceeds.

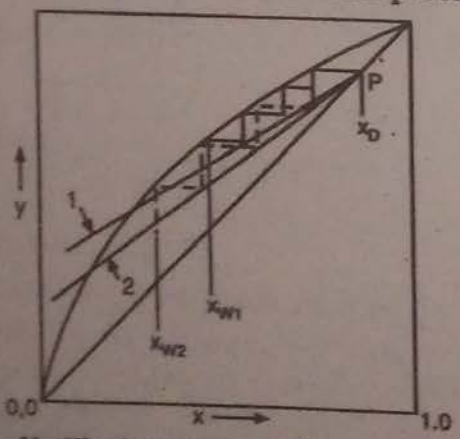

Fig. 1.50 : Batch distillation with constant overhead composition

In this case for every reflux ratio there is one corresponding operating line and all such operating lines of different slopes (as R is different) will radiate from the point P (x_D, x_D) on the diagonal.

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A typical diagram is shown in Fig. 1.50 for a batch distillation unit with four ideal stages including reboiler. The operating line-1 is for an initial reflux ratio R_1 and the operating line-2 is for a reflux ratio R_2 after certain charge has been removed.

To determine the reflux ratio for a constant overhead composition x_D and a given x_W and the number of ideal stages, we have to adopt a trial and error procedure. Draw the operating line randomly, construct the stages and see that whether the construction of the last stage ends exactly at x_W or not. If not, then draw a new operating line and repeat the procedure till we can construct the given number of stages, with the last stage exactly ending at the given x_W. If we construct the given number of stages, and end exactly at x_W then measure either the slope or read the intercept and then obtain R, as slope = R/R + 1 and intercept = x_D/R + 1.

For any desired value of R, x_W may be obtained as follows :

For any desired value of R, calculate either the slope or intercept, draw the operating line [through the point (x_D, x_D) and with calculated slope or intercept], mark off the steps corresponding to the given number of stages and read x_W [corresponding to the last stage giving the composition of the residue].

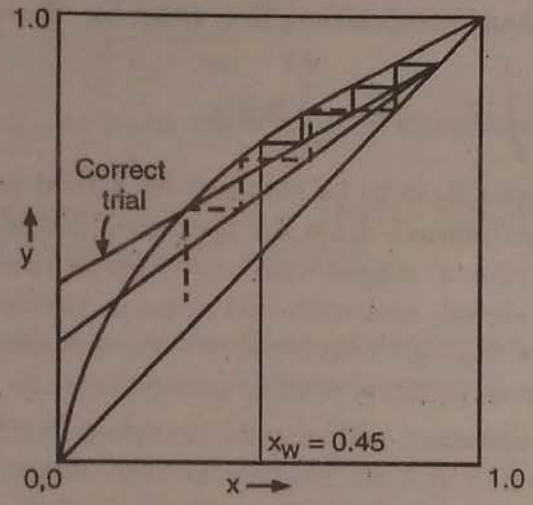

Fig. 1.51 : Finding R for given x_W and constant x_D, given number of stages = 4 including the reboiler

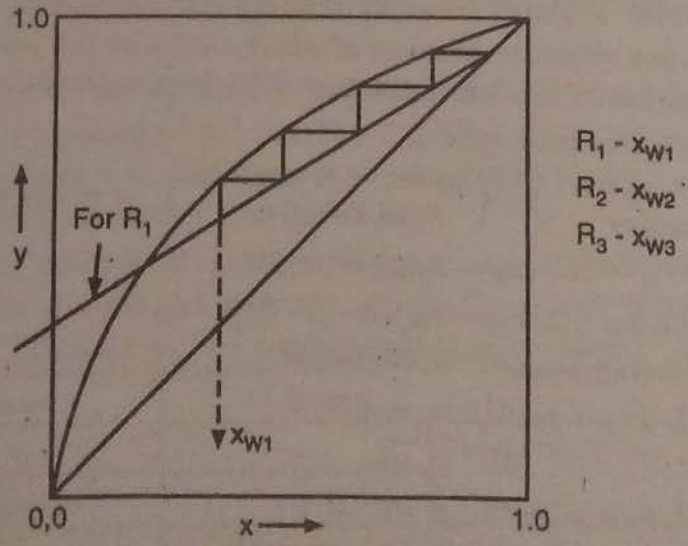

Fig. 1.52 : Finding x_W values at various R values for constant x_D and given number of stages = 4

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The amount of distillate (D) and remaining charge (W) can be calculated from the following equations :

$$\frac{D}{F} = \frac{x_F - x_W}{x_D - x_W} \quad \ldots (1.155)$$

and

$$\frac{W}{F} = \frac{x_D - x_F}{x_D - x_W} \quad \ldots (1.156)$$

where, x_F – molefraction of the more volatile component in the initial charge

x_D – overhead product composition in terms of molefraction of the more volatile component

x_W – molefraction of the more volatile component in the charge left in the reboiler (residue)

F – kmol of charge to the unit.

For R_1, R_2 ... read x_{W1}, x_{W2} ... and get D using Equation (1.155).

For a constant overhead composition, R is given by $R = dL/dD$

$$\therefore \int_0^L dL = L = \int_{R_1}^{R_f} R\, dD.$$

The initial reflux ratio R_1 is to be obtained by a trial and error procedure. For a given x_D, x_W and n (number of plates), draw the operating line through the point (x_D, x_D) on the diagonal and construct the 'n' stages with construction exactly ending in $x_W = x_F$, and then measure a slope or intercept, and with this slope or intercept, calculate R_1. Then assume suitable values of R, i.e. R_2, R_3 ... R_f, construct the n stages and read the corresponding x_W and calculate D. Construct a plot of R v/s D and evaluate L.

(ii) Operation at constant reflux ratio (varying overhead composition) : If the batch distillation unit is operated at a constant reflux ratio R, the composition of the more volatile component in the overhead product will continuously fall.

Consider that at any time t there are L moles of a liquid mixture of composition x_W in the reboiler. Over a small interval of time dt, assume that dD moles of distillate of composition x_D are obtained because of which moles of the mixture change from L to L–dL and the composition of the liquid mixture falls from x_W to $x_W - dx_W$.

Over a short time interval :

$$\begin{pmatrix}\text{moles of A at } t = t \\ \text{in reboiler}\end{pmatrix} - \begin{pmatrix}\text{moles of A at } t + dt \\ \text{in reboiler}\end{pmatrix} = \text{moles of A obtained as distillate}$$

$$L x_W - (L - dL)(x_W - d x_W) = x_D\, dD$$

$$L x_W - L x_W + L\, dx_W + x_W\, dL - dL\, dx_W = x_D\, dD$$

dLdW being small it is neglected.

$$\therefore \quad L\, dx_W + x_W\, dL = x_D\, dD$$

but $$dD = dL$$

$$\therefore \quad L\, dx_W + x_W\, dL = x_D\, dL$$

$$L\, dx_W = (x_D - x_W)\, dL$$

$$\therefore \quad \frac{dL}{L} = \frac{dx_W}{x_D - x_W}$$

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Integrating between limits :

Initial : t = 0, at the start of distillation

$L = F$ = moles of the liquid mixture charged to the reboiler with x_F molefraction of A

$\therefore \quad x_W = x_F$

Final : t = t, at the end of distillation :

$L = W$ = moles of the residue/charge left in the reboiler with x_W molefraction of A

$$x_W = x_W$$

$$\int_F^W \frac{dL}{L} = \int_{x_F}^{x_W} \frac{dx_W}{x_D - x_W}$$

Reversing the limit and integrating, we get

$$\ln \frac{F}{W} = \int_{x_W}^{x_F} \frac{dx_W}{x_D - x_W} \qquad \ldots (1.157)$$

The value of integral on the right hand side of Equation (1.157) can be evaluated graphically.

In this case, we can obtain the values of residue left after distillation, distillate obtained and composition of the composited distillate.

Determination of W, D, etc. :

In this case, you will be provided with the quantity of the initial charge F, the composition of charge x_F, the number of stages required, the composition of charge left/residue x_W and the reflux ratio R.

From a given value of R, calculate the slope of the operating line. As R is fixed, all the operating lines originating from different x_D values will have the same slope (lines are parallel to each other).

Draw the operating lines of constant slope starting from different values of x_D on the diagonal.

Step-off the given number of stages to get the values of x_W.

We have to draw the operating lines in such a way that after stepping off the number of stages, the value of x_W obtained should cover the composition of the initial charge (x_F) and the composition of the final residue or charge left (x_W).

In brief, by plotting a number of operating lines of constant slopes (parallel operating lines) from various values of x_D on the diagonal, obtain the values of x_W by stepping-off the given number of stages. Tabulate x_D and corresponding x_W values.

x_W	x_{W1}	–	–	–
x_D	x_{D1}	–	–	–
$x_D - x_W$	$x_{D1} - x_{W1}$	–	–	–
$1/(x_D - x_W)$	$1/(x_{D1} - x_{W1})$	–	–	–

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Construct a plot of $1/(x_D - x_W)$ v/s x_W and measure the area under the curve between initial composition of the charge (i.e. x_F) and the composition of the residue left after distillation is complete (final residue composition, i.e. x_W). The area under the curve gives the value of integral on the RHS of Equation (1.157).

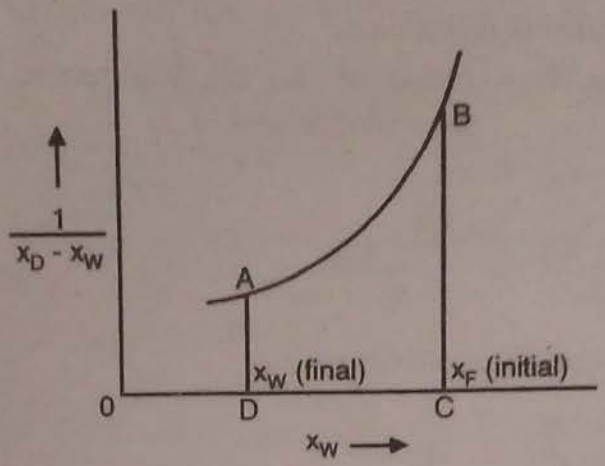

Fig. 1.53 : Graphical integration of RHS of Equation (1.157)

$$\text{RHS of Equation (1.157)} = \int_{x_F}^{x_W} \frac{dx_W}{x_D - x_W} = \text{Area ABCD} \times \text{scale x-axis} \times \text{scale y-axis.}$$

$$\text{With ln } F/W = \int_{x_W}^{x_F} \frac{dx_W}{x_D - x_W}\text{, get W, moles of residue left.}$$

We have : $F = D + W$ $\therefore$ get D, moles of distillate

We have : $x_F F = W\, x_W + D\, x_{D\,(avg)}$, get $x_{D\,(avg)}$ – composited/average composition of the distillate (composition of composited distillate).

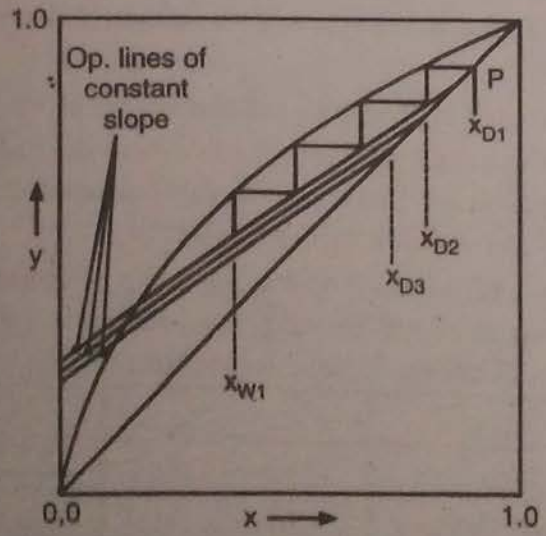

Fig. 1.54 : Construction of parallel operating lines (lines of same slope) and obtaining values of x_W and x_D for constant reflux ratio (Number of stages = 4)

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Azeotropes :

An azeotrope is a liquid mixture with an equilibrium vapour of the same composition as the liquid [the composition of the vapour (generated from the liquid) is the same as that of the liquid mixture]. As the dew point and the bubble point are identical (same) at the azeotropic composition, the azeotropes are called constant boiling mixtures (As the liquid and vapour compositions are the same, the boiling point of the liquid mixture remains constant as the vapour is generated from it.) Close boiling components showing small deviations from ideality (Raoult's law) may form an azeotrope. An azeotrope exhibits a maximum or a minimum boiling point relative to boiling points of the pure liquids and thus are classified as maximum boiling and minimum boiling azeotropes. A minimum boiling azeotrope will boil at a temperature lower than the boiling points of the pure components and a maximum boiling azeotrope will boil at higher temperature than the boiling points of pure components. The temperature–composition diagrams and equilibrium diagrams for the mixtures forming an azeotrope at constant pressure are shown in Fig. 1.55. Here we see that the equilibrium curve crosses the diagonal - an indication of the existence of an azeotrope, i.e., y = x and the mixture will not be separated by straight forward/ordinary distillation.

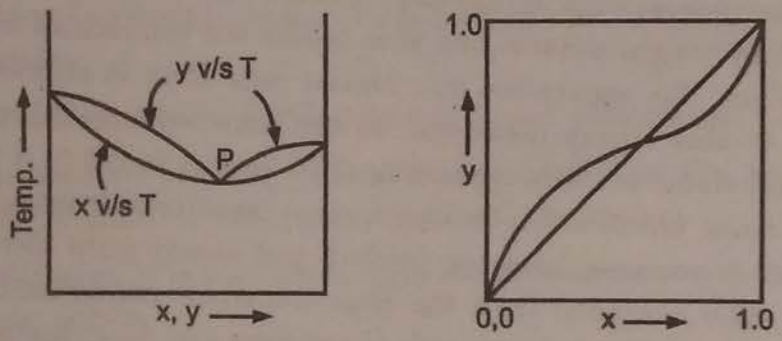

Fig. 1.55 : Minimum boiling azeotrope system carbon disulfide – acetone at one atmosphere pressure

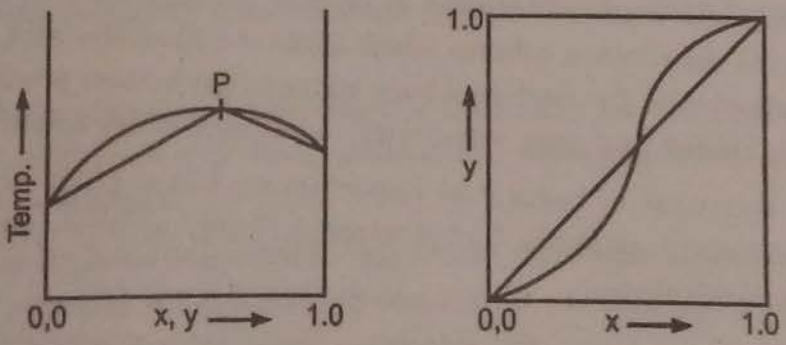

Fig. 1.56 : Maximum boiling azeotrope system chloroform – acetone at one atmosphere pressure

Refer to Fig. 1.55. For all the mixtures of compositions less than P, an equilibrium vapour is richer in the more volatile component than the liquid, while for all the mixtures of compositions greater than P, the equilibrium vapour is less rich in the more volatile component than the liquid. The mixture of composition P gives the vapour of composition

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identical with the liquid. (At the point P, the vapour and liquid are identical in composition and separation past that point by simple/ordinary distillation technique is not possible).

In such cases, complete separation by ordinary fractionation may be impossible. The constituents of a binary azeotrope are separated completely by - (1) adding a third component to the binary mixture and (2) changing the system pressure.

The third component added (which is relatively volatile) to a binary azeotrope usually form a low boiling azeotrope with one or both of the feed components and this low boiling azeotrope is usually withdrawn as the overhead product/distillate. *The third component added to break the binary azeotrope to effect its separation into pure components* is called as the **entrainer/azeotrope breaker**.

Azeotropic distillation is a distillation technique wherein a third component is added to a binary azeotrope (feed) to effect the complete separation by forming an azeotrope with one or both of the feed components and removing the azeotrope thus formed usually as the overhead product.

A binary azeotrope system containing ethanol and water forms a minimum boiling azeotrope at 96% by weight ethanol and thus blocks the production of pure components by ordinary distillation. The separation into almost pure form is effected by adding benzene as an entrainer to this binary azeotrope. In the azeotropic distillation technique of this system, an alcohol-water mixture containing 96% alcohol is fed to a first column, benzene used as an entrainer forms a low boiling ternary azeotrope (containing benzene, alcohol and water) which is removed as a top product and nearly pure alcohol is taken out as a bottom product. The overhead from the first column is condensed, phase separation is achieved in a decanter. The benzene rich phase from the decanter is returned to the first distillation column and the water rich phase is sent to a second distillation column. The overhead from the second column containing benzene in a major proportion is fed back to the first column while aqueous solution of alcohol is withdrawn as a bottom product. It is then fed to a third distillation column which gives the alcohol-water azeotrope as overhead product (to be mixed with the feed) and pure water as the bottom product. Thus, the net effect is to obtain pure alcohol and pure water from the alcohol-water azeotrope.

Azeotropic mixtures of industrial importance : water-nitric acid, water-hydrochloric acid and water-many alcohols (ethyl alcohol, butyl alcohol, propyl alcohol, isobutyl alcohol, isopropyl alcohol).

Choice of entrainer :

(i) It should preferably form a low-boiling azeotrope with one of the constituents of the binary azeotrope which is desired to separate.

(ii) The new azeotrope formed must be of sufficient volatility in order to make it readily separable from the remaining constituent.

(iii) It should be cheap, readily available, non-corrosive, chemically stable, non-toxic and it should have low freezing point and low heat of vaporisation.

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Extractive distillation refers to the distillation process wherein a high-boiling solvent is added to alter the relative volatilities of the constituents of a feed mixture.

It is necessary to alter the relative volatility of the original constituents because of similarities of the vapour pressures of the constituents or the presence of an azeotrope.

The solvent selected is less volatile than either of the two constituents (of a binary mixture) and so as to maintain a high concentration of the solvent throughout the column it is usually introduced a few stages below the top stage.

An example of extractive distillation is the separation of toluene (b.p. = 110.8°C) from iso-octane (b.p. = 99.3°C) using phenol as an extractive solvent. In the presence of phenol, the relative volatility of iso-octane increases.

Steam Distillation :

Steam distillation is a distillation process with open steam (live steam), i.e., it is a process in which the steam is in direct contact with the distilling system. It can be carried out batchwise or continuously.

Steam distillation is possible under the following conditions (necessary conditions to use steam distillation) :

(i) When the material to be distilled does not react with steam.

(ii) When the desired product - the volatile material or the nonvolatile residue is practically immiscible with water (insoluble in water).

With steam distillation it is possible to separate substances at a temperature lower than their normal boiling points.

Steam distillation is commonly used :

(i) To separate high boiling substances from nonvolatile impurities.

(ii) To separate relatively small amount of volatile impurity from a large amount of material.

(iii) Where the material to be distilled is thermally unstable - decomposition of the material might occur if direct distillation were employed - i.e., in case of heat sensitive materials.

(iv) To remove high boiling volatile impurities from still higher boiling substances - e.g., the removal of high boiling impurities causing undesirable odour or taste from still higher boiling edible vegetable oils.

(v) Where the material cannot be distilled by indirect steam heating even under low pressure because of the high boiling temperature - where vaporisation temperatures cannot be reached by steam heat.

Typical examples of steam distillation include concentration and purification of essentials oils for perfumery, deodourisation of fats and oils and purification of long chain fatty acids.

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Steam distillation is especially employed in cases where the substances involved cannot withstand the temperatures of distillation and decompose (i.e., for heat sensitive materials). Substances of this kind can be separated by reducing the partial pressure of the volatile component. This can be done by making use of an inert vapour that decreases the temperature of distillation. The inert vapour used should be practically immiscible with the components to be distilled. Steam is generally used for this purpose and the operation is called as steam distillation. Steam is widely used as : it is immiscible with many organic compounds, it provides required heat for vaporisation and it is readily available at low cost.

In steam distillation, the liquid is distilled by admitting steam directly into the liquid in a distillation still (the steam directly admitted in a pool of the liquid to be distilled is called as live steam). The steam carries with it vapours of the volatile liquid which is then condensed and the liquid is separated from water by gravity while the non-volatile material remains behind in the still.

The process of steam distillation may make use of superheated steam which provide sufficient heat to vaporise the desired component without self-condensing or it may make use of saturated steam which provide sufficient heat to vaporise the desired component by partly condensing.

In steam distillation, the liquid is distilled by blowing steam into a distillation still. The steam carries with it vapours of volatile liquid, which is then condensed to separate the liquid from water.

Suppose we want to steam distill toluene containing small amount of high boiling impurity at atmospheric pressure. For steam distillation of an impure toluene, saturated steam is blown into a distillation still containing some impure toluene. Suppose that the liquid in the still is heated only by the condensation of the steam. This results in the production of a water layer in the still. As the temperature rises the vapour pressures of both the toluene and water layer rise and when the sum of the two vapour pressures equals 760 mmHg, the mixture begins to distill. At this point, the vapour pressure of toluene is p_A° mmHg and that of water is $760 - p_A^\circ$ mmHg. It is found that the mixture begins to distill at 84°C. The vapour mixture passing over contains toluene with a partial pressure of 350 mmHg and water with a partial pressure of 410 mmHg. The ratio of toluene to water in the vapour by volume is 350/410 and by weight it is $350 \times 92/410 \times 18$ (Mol. wt. of toluene is 92 and that of water is 18).

When a water phase/layer is present and the effect of a non-volatile material on the vaporisation is neglected then for any volatile material containing a high boiling impurity during steam distillation we have three phases and two components (phases - two liquid phases - e.g., toluene and water, and one vapour phase. Components - e.g. toluene and water in the case of steam distillation of an impure toluene).

Thus, according to the phase rule, $F = C - P + 2$, the value of F, the degrees of freedom, is one, i.e., there is only one variable that can be varied independently and setting the

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temperature or pressure fixes the system. Therefore, at a given pressure of distillation, the temperature as well as the composition of the system is fixed and thus the system is completely defined.

The condensed water and toluene being immiscible do not affect each other properties. Hence, the partial pressure of each component is equal to the vapour pressure (each component exerts a partial pressure equal to its vapour pressure at the boiling point of the mixture). Thus, if the pressure is atmospheric, temperature adjusts itself so that the sum of the partial pressures of the two components equals one atmosphere. The temperature so reached is less than the boiling points of the pure components. Therefore, if the pressure is atmospheric (1 atm = 101.325 kPa) then it is possible to steam distill the material at a temperature lower than 100°C (the boiling point of water at 1 atm) thus blocking/avoiding destructive temperatures (temperatures causing the decomposition of the material to be distilled).

If P is the total pressure of the system, p_A is the partial pressure of A vapour (= vapour pressure of liquid A) and p_B is the partial pressure of water vapour (= vapour pressure of water), then

$$P = p_A + p_B$$

For atmospheric distillation :

$$p_A + p_B = 1 \text{ atm} = 760 \text{ mmHg}$$

The steam requirement for the process can be determined as follows :

Let y_A and y_B represent the mole fractions of liquid (desired) and water (steam) respectively in the vapour phase, then

$$p_A = y_A \cdot P \quad \text{and} \quad p_B = y_B \cdot P$$

$$\frac{y_A}{y_B} = \frac{p_A}{p_B} = \frac{p_A}{(P - p_B)}$$

$$y_A = N_A/(N_A + N_B) \quad \text{and} \quad y_B = N_B/(N_A + N_B)$$

where N_A and N_B are the moles of liquid and water respectively.

$$\therefore \quad \frac{N_A}{N_B} = \frac{p_A}{(P - p_A)}$$

We have : $N_B = W_B/M_B$ and $N_A = W_A/M_A$

$$\frac{W_A/M_A}{W_B/M_B} = \frac{p_A}{(P - p_A)}$$

$$\frac{W_A}{W_B} = \frac{p_A M_A}{(P - p_A) M_B}$$

or

$$\frac{W_B}{W_A} = \frac{(P - p_A) M_B}{p_A M_A} = \frac{(P - p_A^\circ) M_B}{p_A^\circ M_A} \quad \text{(Note that : } p_A = p_A^\circ\text{)} \quad \ldots (1.158)$$

where W_B and W_A are the kg of water and liquid respectively, and M_B and M_A are the molecular weights of water and liquid respectively.

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In order to obtain the steam requirement in steam distillation process, the values of $(P - p_A^\circ)$ and p_A° at the distillation temperature are substituted and the ratio W_B/W_A is computed. This is the amount of open steam condensed per unit weight of liquid condensed.

The actual steam is generally higher than that obtained by using Equation (1.158). This is because of (i) steam may be used externally to provide for the sensible heat and latent heat of liquid, (ii) actual state of equilibrium is difficult to achieve owing to inadequate contact between steam and liquid.

In equation (1.158) the vapour pressure of the liquid was assumed to be equal to the partial pressure of the liquid in the vapour phase. However, in actual practice these two are different because of the vaporisation efficiency.

The vaporisation efficiency is defined as :

$$\eta_v = p_A/p_A^\circ$$

The vaporisation efficiency is rarely unity.

Therefore, the steam requirement for atmospheric steam distillation should be computed from the following equation :

$$\frac{W_B}{W_A} = \left(\frac{101.325 - \eta_v p_A}{\eta_v p_A}\right)\frac{M_B}{M_A} \quad \ldots (1.159)$$

Distillation Towers/Columns :

Distillation (fractionation) is the most widely used mass transfer operation in the chemical industry. In case of this operation, the gas-liquid contacting devices for mass transfer are : plate columns and packed columns.

A column is also known as a tower. It is essentially a tall vertical cylindrical shell, with number of nozzles. The column consists of a series of plates or trays or variety of packings (usually random packings).

Based upon the contacting devices used for mass transfer to take place, the towers/columns are divided into two types namely, plate or tray columns and packed columns.

Plate Columns :

The process of distillation with rectification makes use of either a plate (tray) column or packed column, for intimate phase contacting, the former being more common. In the plate columns, the process of mass transfer is stage wise and in packed towers/columns, the process of mass transfer is continuous.

Each plate in a vertical column represents a stage as on each plate, phases (gas and liquid) are brought into intimate contact, mass is transferred from one phase to another (to and from both the phases) and finally phases are separated and leave the plate/tray in equilibrium (ideal case). A plate column is a vertical column divided into sections by a number of horizontal plates. In this column the liquid enters the plate from the top, flows over the plate and then from plate to plate via downcomers. A certain height of a liquid pool is maintained on each plate with the help of an outlet weir. Gas/vapour enters in the column from bottom inlet and flows upward through the opening in the plate, it bubbles through the pool of liquid on the plate, disengages from liquid and passes to the plate above. In this way, vapour and liquid are brought into intimate contact on the every plate.

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Various plates used for contacting of phases are :

1. **Bubble cap plate/tray**
2. **Sieve plate and**
3. **Valve plate.**

In plate columns, the minimum plate spacing is 300 mm. For columns of diameter 1.5 m or more, spacing of 450 or 600 mm is used for easier tray access. In plate columns, plates are provided with **downcomer** and **weirs**. The function of the downcomer is to provide a passage for the downward flow of the liquid from a plate above to the plate below. The downcomer area for the each plate is limited to 10 % of the total area of the plate. Pipe downcomer, segmental downcomer and chord type downcomer are various types of downcomers that can be used. A depth of liquid on the plate required for gas contacting is maintained by an overflow (outlet) weir. An inlet weir helps to distribute liquid as it enters the plate from the downcomer and prevents impingement of liquid on a contacting device.

1. Bubble cap plate (tray) :

In case of bubble cap tray, gas/vapour flows through a riser, reverses flow under cap,

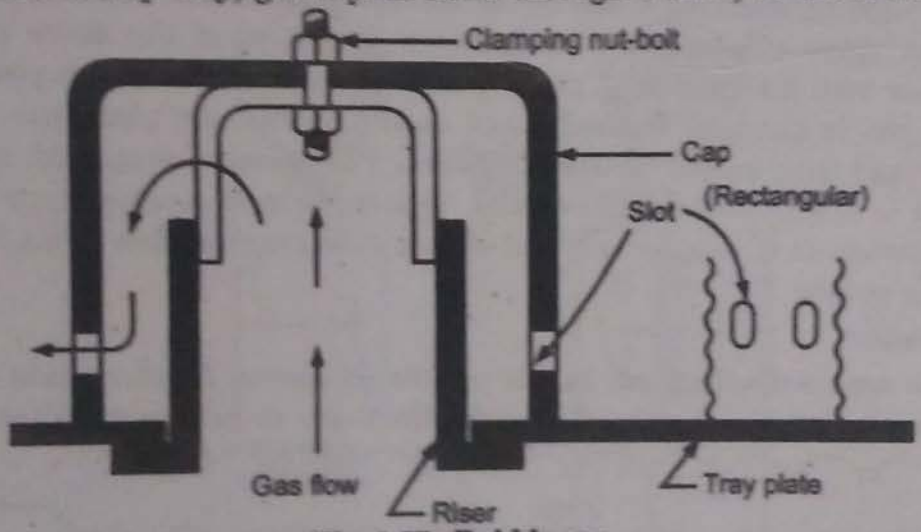

Fig. 1.57 : Bubble cap

moves downward through the annular space between the riser and the cap, and finally it bubbles into a pool of liquid through a series of slots (openings) provided along the periphery of the cap at the bottom. The slots may be circular, rectangular, or trapezoidal in shape.

Bubble cap disperses the gas phase into the liquid as the fine bubbles, it prevents the liquid drainage through the gas passage at low gas rates. It directs the gas flow first horizontally and then vertically upward through the pool of liquid in the form of bubbles. The bubble cap columns can be operated with high and constant stage efficiencies over a wide range of the vapour and liquid flow rates. Bubble cap plates are now rarely used for new installations because of (i) their relatively high cost and (ii) high pressure drop.

2. Sieve plate :

Very common plates in use in recent years are the perforated plates because of their (i) simplicity and (ii) low cost. A sieve plate is the perforated plate employing the perforations for dispersing gas into liquid on the plate. It is a metal sheet having hundreds

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of round perforations in it of the size ranging between 3 to 12 mm (5 mm is a common size). The total area of holes ranges from 5 to 15 % of the plate area. In the sieve plate column, liquid flows across the plate and gas passes vertically upward through the holes, in the form of bubbles through the liquid on the plate. Gas flow through the

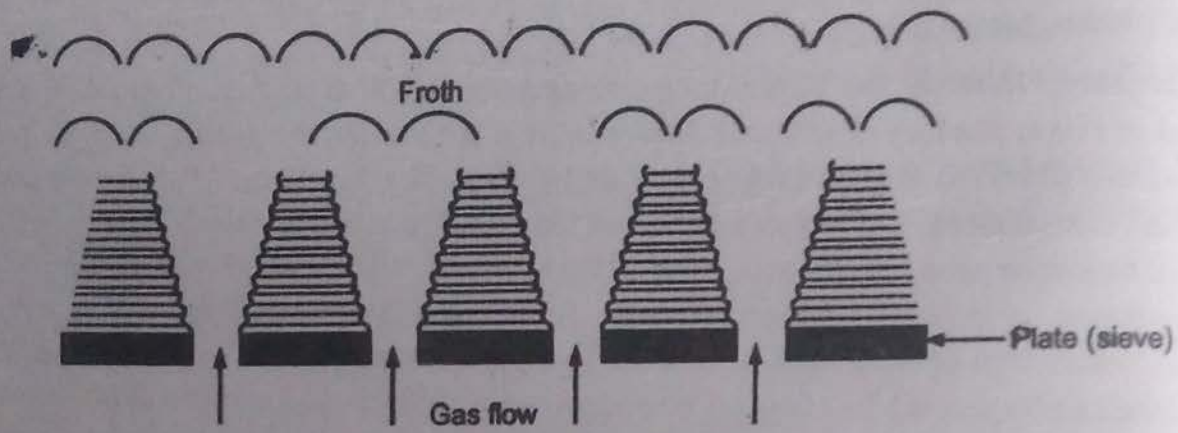

Fig. 1.58 : Sieve plate disperser

perforations prevents the down-flow of liquid through perforation in the plate but at low gas flow rates **weeping** (draining of the liquid through perforations in plate) is severe which in turn reduces stage efficiencies, and this is a limitation of the sieve plate. Gas-liquid contact is poor with the sieve plate as compared to that with bubble cap plate because in the sieve plate gas is directed vertically and in the bubble cap plate gas is first directed horizontally and then moves vertically upward. The pressure drop with sieve plate is low as compared to it with bubble cap columns. This is the most common type of plate (or tray) contacting device. It is usually operated over a wide range of flow rates between weeping and flooding points.

3. Valve plate :

It is the recent development in the perforated plates. A valve plate is nothing but a perforated metal sheet wherein the perforations are covered with liftable caps (valves).

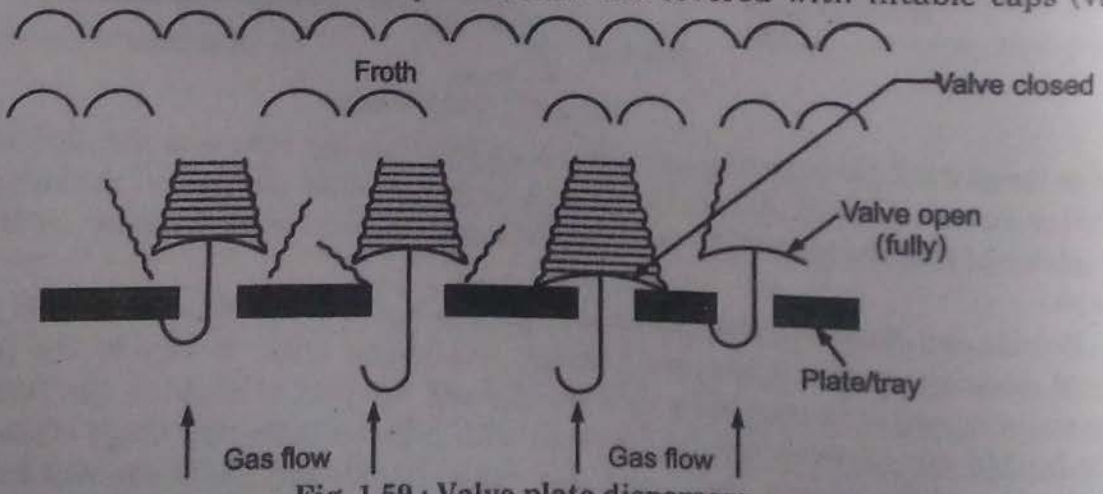

Fig. 1.59 : Valve plate dispersers

The valves are metal discs and are held in the plate by means of legs which restrict the upward motion of caps. The valve provides variable orifices for vapour flow i.e. opening in the plate will be small at the low vapour rate and large at high vapour/gas rate. Thus, weeping will be greatly reduced with this plate at low gas rates. Valve plates can be operated over a wide range of phase flow rates. Gas-liquid contact with the valve plate is

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much better than it with sieve plate as liftable caps direct the gas horizontally into the liquid on the plate while in sieve plate gas passes vertically upward through the liquid. The cost of the valve plate is in between that of bubble cap and sieve plate, approximately twenty per cent higher than that of the sieve plate.

Liquid flow patterns over plate/tray :

The type of liquid flow pattern over a plate depends upon liquid to gas flow ratio.

Various flow patterns are :

1. **Cross flow :** It is most frequently used. Provides a good length of the liquid path. Here the liquid flows across a plate and then falls to the plate below through a passage at the other side of the plate.

2. **Reverse flow :** Provides a very long liquid path especially suitable for low liquid to gas ratios.

3. **Split flow :** Especially suitable for handling high liquid to gas ratios. Here the liquid flowing across the plate is splitted into at least two paths.

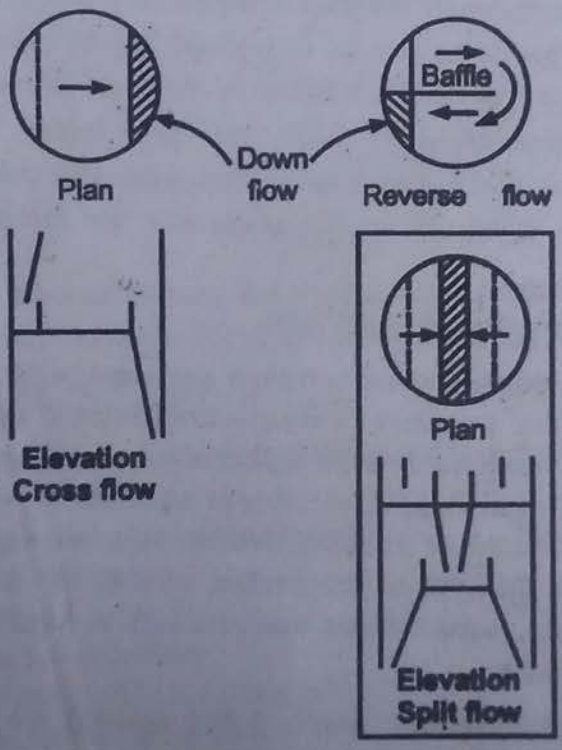

Fig. 1.60 : Arrangement for liquid flow over a plate

Downcomer and weir :

A downcomer is the passage for liquid flow from a plate above to the plate below which may be in the form of circular pipes or segments of the tower/column set aside (for liquid flow) by vertical plates (partitions). The discharge end of the downcomer is generally projected far enough into the liquid on the plate so as to avoid vapour (gas) from rising up the downcomer to short-circuit the plate above. For this purpose, inlet weir may be used. A exit weir or outlet weir is used for maintaining a certain depth of liquid (liquid pool) on each plate needed for the gas contacting. Straight rectangular weirs with length equal to 60 to 80% of the tower diameter are frequently used.

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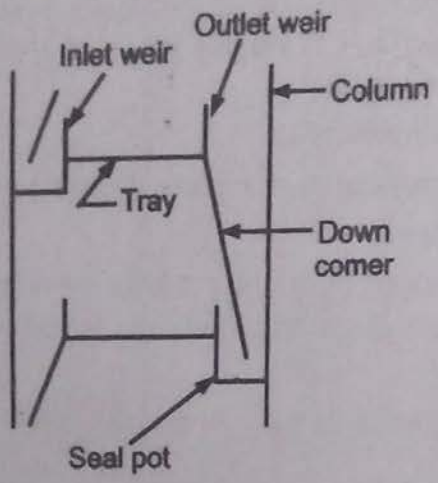

Fig. 1.61 : Downcomer - weir seal plot

Entrainment :

As gas/vapour bubbles through the pool of liquid on a plate, a large amount of turbulence is set up and the liquid particles can be entrained by the rising gas. *Carry over of liquid particles by the rising gas from a plate below to the plate above* is known as **entrainment** and is much more pronounced at high gas flow rates. Entrainment reduces the concentration change per plate and hence, decreases the plate efficiency. Tray spacing, vapour flow rate, depth of liquid on the plate are the factors on which the amount of entrainment depends.

PACKED COLUMNS FOR DISTILLATION :

For distillation operation, packed columns are especially useful whenever we have to carry out operation at low pressure (vacuum distillation) and whenever we are dealing with heat sensitive materials. Packed columns are cheaper than plates for column diameter less than 600 mm. A packed column consists of a cylindrical shell filled with some sort of packings (rings or saddles) which offer a large interfacial area for mass transfer. A common arrangement of the packed column for distillation operation is shown in Fig. 1.62, wherein the vapour flows upward and the refluxed liquid flows downward giving a continuous contact.

The ability of a given packing to effect the desired mass transfer between gas and liquid phases (i.e. to effect the desired degree of separation) is usually expressed as the height equivalent to one theoretical plate (HETP). In plate columns wherein a process of enrichment is stage wise, the vapour leaving the plate is richer with respect to more volatile component than the vapour entering the plate by one equilibrium stage. In packed columns, the same enrichment of the vapour will occur in a certain height of packing which is termed as the height equivalent to one theoretical plate or height equivalent to a theoretical plate. Thus, in packed columns, one equilibrium step is represented by a certain height of a packed bed and the required height of packing for the desired degree of separation is given by : HETP × number of ideal plates required. HETP is defined as the height of packing (packed bed) giving the enrichment of vapour same as that can be obtained by one theoretical plate (in stage-wise operation).

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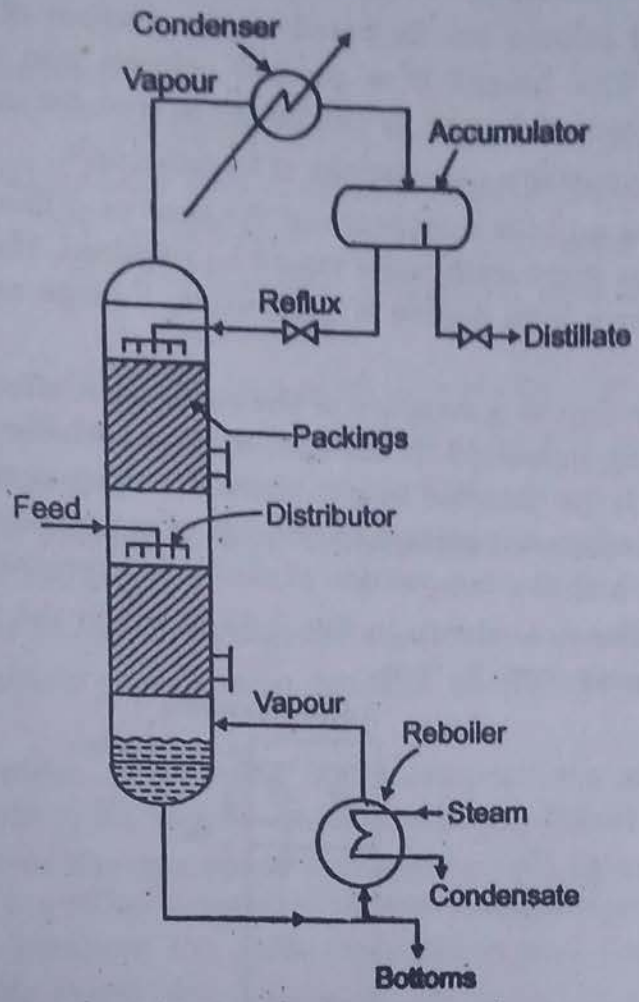

Fig. 1.62 : Packed column for continuous distillation

HETP can be estimated with the help of the following empirical equation such as

$$HETP = k_1 \cdot G^{k_2} \cdot D_t^{k_3} \cdot Z^{1/3} \cdot \alpha \cdot \mu_L/\rho_L \quad \ldots (1.160)$$

where k_1, k_2 and k_3 are empirical constants for packing and are function of type and size of packing.

- G – Superfacial gas mass velocity
- D_t – Tower diameter
- Z – Height of packing
- α – Relative volatility

μ_L and ρ_L are viscosity and density of liquid respectively.

HETP is an experimental quantity, a characteristic of each packing and depends on the type and size of the packing, flow rates of each fluid and concentration.

Method of Transfer Units :

Fractionation by means of a packed column involves continuous countercurrent contact of the vapour and liquid whereas fractionation by means of a plate column involves stagewise contact of the vapour and liquid. So in the packed column there is a continuous change in the composition of the components and in the plate column there is a discontinuous change in the composition of components (effected at each plate).

The theoretical unit of separation in a packed column is the transfer unit and not the theoretical plate (as it in a plate column). The efficiency of a packing consequently can be measured as the height of packing to give a separation of one transfer unit (height of transfer unit).

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The height of a packed column can be based on the number of transfer units and the height of transfer unit. The height of a packed column can easily be obtained by multiplying the number of transfer units by the height of transfer unit.

Z = Number of transfer units × height of transfer unit.

The number of transfer units is a measure of the ease or difficulty of separation. More difficult the separation, the more such units would be required. Hence, if very high purity products are desired or for a high degree of separation, a large number of transfer units will be required.

The height of transfer unit is a measure of the separation effectiveness of a particular packing for chemicals being processed. If the rate of mass transfer is high and the surface area for mass transfer is large then the height of transfer unit is small. For determination of the height of transfer unit, we require a knowledge of mass transfer coefficients in units consistent with those in which the composition of phases is expressed.

Consider a packed column as shown in Fig. 1.63 in which the composition of the more volatile component increases from y_b to y_t.

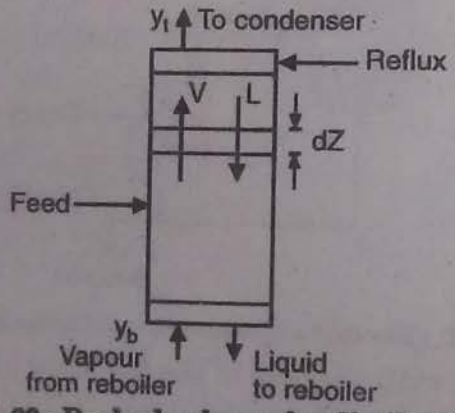

Fig. 1.63 : Packed column for distillation

Let A be the empty column cross-sectional area, Z be the height of packed section of a column, 'a' be the active interfercial surface per unit volume of packing (interfacial area per unit volume of packing).

Consider a differential element of height dZ, through which two phases pass countercurrently, wherein a differential change in vapour composition, dy, takes place. At the same time, a differential change in liquid composition is dx.

For steady-state operation, the material balance of the more volatile component A over the differential height dZ assuming constant molar overflow i.e. neglecting change in molar flow rate in any particular section is

$$V\,dy = L\,dx \qquad \ldots (1.161)$$

where V is the vapour flow rate in the enriching section of the column in moles/time and L is the molar flow rate of the liquid in mole/time. V is constant in the entire enriching section. It will be also constant in the stripping section but of a different magnitude depending upon the feed conditions.

Each term of Equation (1.161) is the indication of moles of A diffusing i.e. the rate of addition or depletion of A to the vapour phase or from the liquid phase and is a statement of the rate of transfer of A across the transfer area in dZ. This must be equal to the total driving force divided by the total resistance.

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So the rate of change of the component A within the phase must be equal to the rate of transfer to the phase.

Thus, for V-phase (vapour phase) :

$$V\,dy = k_G\,(y_i - y)\,dS = K_G\,(y^* - y)\,dS \quad \ldots (1.162)$$

where dS is the interfacial transfer area associated with the differential column height dZ.

dV = differential volume of packing in height dZ = AdZ

dS = differential interfacial area for mass transfer = aAdZ in height dZ

$$V\,dy = k_G\,(y_i - y)\,aA\,dZ = K_G\,(y^* - y)\,aA\,dZ$$

$$V\,dy = k_G\,a\,(y_i - y)\,A\,dZ = K_G\,a\,(y^* - y)\,A\,dZ \quad \ldots (1.163)$$

where $(y_i - y)$ is the driving force over the gas film

$(y^* - y)$ is the overall driving force for the gas phase/vapour phase - (overall driving force in terms of vapour composition).

K_G is the overall gas phase mass transfer coefficient in mol/(time)·(unit driving force)

$K_G \cdot a$ is the volumetric overall mass transfer coefficient in mol/(volume) · (time) · (unit driving force).

In distillation operation, vapour and liquid compositions are expressed in terms of molefraction, so the units of $K_G \cdot a$ will be mol/(volume) · (time) · (molefraction).

Use of $K_G \cdot a$ requires the gas phase composition (y^*) in equilibrium with the bulk liquid composition and use of $k_G \cdot a$ requires the gas phase composition (y_i) at the interface. It is possible only to measure the bulk composition and determination of interface composition is ordinarily impossible.

$$\therefore \quad V\,dy = K_G\,a\,(y^* - y)\,A\,dZ \quad \ldots (1.164)$$

$$(V/A)\,dy = K_G\,a\,(y^* - y)\,dZ$$

$$dZ = \frac{V/A}{K_G\,a}\left[\frac{dy}{y^* - y}\right]$$

Integrating between limits,

$$\int_0^Z dZ = \int_{y_b}^{y_t} \frac{V/A}{K_G\,a}\left(\frac{dy}{y^* - y}\right)$$

The molar flow rates (V and L) are assumed to be constant in the enriching section. The overall mass transfer coefficient is a function of these phase flow rates. If the physical properties do not vary appreciable, K_G is also constant.

$$\therefore \quad \int_0^Z dZ = \frac{V/A}{K_G\,a}\int_{y_b}^{y_t} \frac{dy}{y^* - y}$$

$$Z = \frac{V/A}{K_G\,a}\int_{y_b}^{y_t} \frac{dy}{y^* - y} \quad \ldots (1.165)$$

In Equation (1.165), the integral term represents the integrated ratio of the change in vapour composition to the available driving force (in terms of vapour composition) tending

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to bring this about. This term is defined by Chilton and Colburn as the number of transfer units (N_t). In our present case, N_t is $N_{t_{OG}}$.

$$N_{t_{OG}} = \int_{y_b}^{y_t} \frac{dy}{y^* - y} \quad \ldots (1.166)$$

$N_{t_{OG}}$ is the number of overall gas transfer units (i.e. number of transfer units based on gas phase).

The quantity outside the integral of Equation (1.165) is called the height of a transfer unit (H_t). In present case, H_t is $H_{t_{OG}}$.

$$H_{t_{OG}} = \frac{V/A}{K_G a} \quad \ldots (1.167)$$

Thus, Equation (1.165) can be written as

$$Z = H_{t_{OG}} \cdot N_{t_{OG}} \quad \ldots (1.168)$$

A similar series of equations can be written in terms of composition of the more volatile component in liquid phase. The column height can be evaluated using changes in composition of A in liquid phase and corresponding rate.

$$V\,dy = K_G a (y^* - y) A\,dZ = L\,dx = K_L a (x - x^*) A\,dZ \quad \ldots (1.169)$$

$$L\,dx = K_L a (x - x^*) A\,dZ \quad \ldots (1.170)$$

where $x - x^*$ is the overall driving force in terms of liquid phase composition.

$K_L a$ is the overall volumetric mass transfer coefficient for liquid phase in mol /(time) · (volume) · (unit driving force).

K_L is the overall liquid phase mass transfer coefficient in mol/(time) · (area) · (unit driving force).

So for liquid phase, we can write,

$$N_{tOL} = \int_{x_b}^{x_t} \frac{dx}{x - x^*} \quad \ldots (1.171)$$

N_{tOL} is the number of overall liquid phase transfer units (number of transfer units based on liquid phase).

$$H_{tOL} = \frac{L/A}{K_L a} \quad \ldots (1.172)$$

H_{tOL} is the height of an overall liquid transfer unit.

H_{tOG} and H_{tOL} are having dimension of length while N_{tOL} and N_{tOG} are dimensionless.

Plate Efficiency :

By using either the McCabe-Thiele method or the Ponchon-Savarit method, we obtain the number of ideal/theoretical/equilibrium stages required for achieving the desired degree of separation. In case of an ideal stage, the vapour leaving a plate is in equilibrium with the liquid which leaves it. In actual practice, complete equilibrium is never reached on any plate. This is due to short contact time and incomplete mixing on any plate. The vapour leaving an actual plate is usually weaker in the more volatile component than the vapour in equilibrium with the liquid leaving it. Thus, for achieving the desired degree of separation in actual practice, we require more number of plates in a column than the ideal

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plates. To apply either of the methods to an actual case, it is necessary to convert the number of ideal plates to the actual plates. The factor that is used for converting the ideal plates into the actual plates is the plate efficiency. The reboiler delivers vapour directly from the boiling liquid and is thus considered as a perfect plate.

Types of plate efficiency :

(i) Overall efficiency/overall plate efficiency – pertains to the entire column.

(ii) Murphree efficiency – pertains to any single plate.

(iii) Murphree local/point efficiency – pertains to a specific location on a given plate.

Overall efficiency is the ratio of the number of ideal plates (theoretical plates) required to effect a given degree separation to the number of actual plates required to effect the same separation.

$$\eta_o = \frac{\text{theoretical/ideal plates required}}{\text{actual plates required}} \times 100$$

For example, if we need six theoretical plates to effect the desired separation and the overall efficiency is 60% then the actual number of plates required are 6/0.60 = 10. The overall efficiency may vary from 30 to 100 percent.

The Murphree plate efficiency :

It is the efficiency of any single plate based on the vapour phase composition.

For the n^{th} plate in the column, the Murphree efficiency is given by

$$\eta_M = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}}$$

where, y_n is the average composition of the vapour leaving plate.

y_{n+1} is the average composition of the vapour entering plate

y_n^* is the composition of vapour in equilibrium with liquid leaving plate.

It applies to any individual plate in a column and is defined as the actual change in the average vapour composition achieved by a given plate divided by the change in the average vapour composition if the vapour leaving the plate were in equilibrium with the liquid leaving the plate. This efficiency is less convenient to use as it involves plate to plate compositions rather than the terminal compositions.

Point/Local efficiency : It pertains to a particular place on the plate surface (applies to single location on a given plate). It is defined by

$$\eta' = \frac{y_{n,\,local} - y_{n+1,\,local}}{y_{local}^* - y_{n+1,\,local}}$$

where $y_{n,\,local}$ is the composition of vapour leaving specific location on plate n

$y_{n+1,\,local}$ is the composition of vapour entering plate n at the same location.

y_{local}^* is the composition of vapour in equilibrium with the liquid at the same location (i.e. in equilibrium with x_{local}).

To obtain satisfactory plate efficiencies, it is highly necessary that the plates must operate properly. Adequate and intimate contact between vapour and liquid is essential. Each of the unfavourable phenomena such as entrainment, excessive foaming, weeping, short-circuiting and flooding lower the plate efficiency.

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SOLVED EXAMPLES

Ex. 1.1 : A mixture containing benzene and toluene with 40% benzene and 60% toluene is to be separated in a fractionating column to give product containing 96% benzene and bottom product containing 95% toluene. Feed is a mixture of two-third vapour and one-third liquid. Find out the number of theoretical stages required if reflux ratio of 1.5 times the minimum is used.

Data : Relative volatility $\alpha_{AB} = 2.5$.

Solution : Basis : Feed containing 40% benzene and 60% toluene by weight.

Molecular weight of benzene (C_6H_6) = $6\times12+6\times1 = 78$

Molecular weight of toluene ($C_6H_5CH_3$) = $7\times12+8\times1 = 92$

In 100 kg of feed there are 40 kg of benzene and 60 kg of toluene.

$$\text{Molefraction of benzene in feed} = x_F = \frac{40/78}{40/78 + 60/92} = 0.44$$

Distillate contains 96% benzene by weight.

$$\text{Molefraction of benzene in distillate} = x_D = \frac{96/78}{96/78 + 4/92} = 0.966$$

Bottom product contains 95% toluene and 5% benzene by weight.

$$\text{Molefraction of benzene in bottom product} = x_W = \frac{5/78}{5/78 + 95/92} = 0.058$$

$$\alpha_{AB} = 2.5$$

Generate x-y data with the help of equation :

$$y = \frac{\alpha x}{1 + (\alpha - 1)x}$$

Take x = 0, 0.1, 0.2 ... 1 and evaluate the corresponding y values.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.22	0.385	0.52	0.625	0.714	0.79	0.85	0.91	0.96	1.0

Plot the equilibrium curve.

Procedure to determine the minimum reflux ratio (R_m) :

q is the fraction of the feed that is liquid. Feed is two-third vapour and one-third liquid, so 1 mole of feed results in 1/3rd mole of saturated liquid in the stripping section.

$$\therefore \quad q = 1/3 = 0.333$$

$$\text{or } q = 1 - f, \text{ where f is fraction of feed that is vaporised}$$

$$= 1 - 2/3 = 1 - 0.667 = 0.333$$

$$\text{Slope of feed line} = \frac{-q}{1-q} = \frac{-0.333}{1-0.333} = -0.5$$

$$\text{Slope} = \tan\theta = -0.5$$

$$\therefore \quad \theta = -26.56^\circ$$

$$\text{Intercept on y-axis} = x_F/(1-q) = 0.44/(1-0.333) = 0.66$$

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Draw the feed line through the point Q (0.44, 0.44) on the diagonal with a slope equal to -0.5 ($\theta = -26.56^\circ$) or with an intercept on y-axis equal to 0.66 to cut the equilibrium curve at point P. Read y' and x' as the coordinates of the point P.

From graph, $y' = 0.51$ and $x' = 0.295$.

Minimum reflux ratio for feed partially vaporised is given by

$$R_m = \frac{x_D - y'}{y' - x'}$$

$$R_m = \frac{0.966 - 0.51}{0.51 - 0.295} = 2.12$$

Operating reflux ratio = $R = 1.5\, R_m = 1.5 \times 2.12 = 3.18$

Operating line of the rectification section :

This line passes through the point (0.966, 0.966) on the diagonal and its intercept on y-axis is

$$= x_D/(R + 1) = 0.966/(3.18 + 1) = 0.231$$

Draw this line through M (0.966, 0.966) on the diagonal with an intercept of 0.231 on y-axis.

Draw the operating line of the stripping section through the point (x_W, x_W) = N (0.058, 0.058) on the diagonal and passing through the point of intersection of q-line, i.e., feed line and operating line of the rectification section. Do step-by-step construction between the equilibrium curve and the operating lines starting from the point M (0.966, 0.966) and reaching exactly or covering the point N (0.058, 0.058) and count the plates.

Number of theoretical plates including reboiler = n = 11.

Number of theoretical plates required in the column for the separation = n – 1 = 11 – 1 = 10 ... **Ans.**

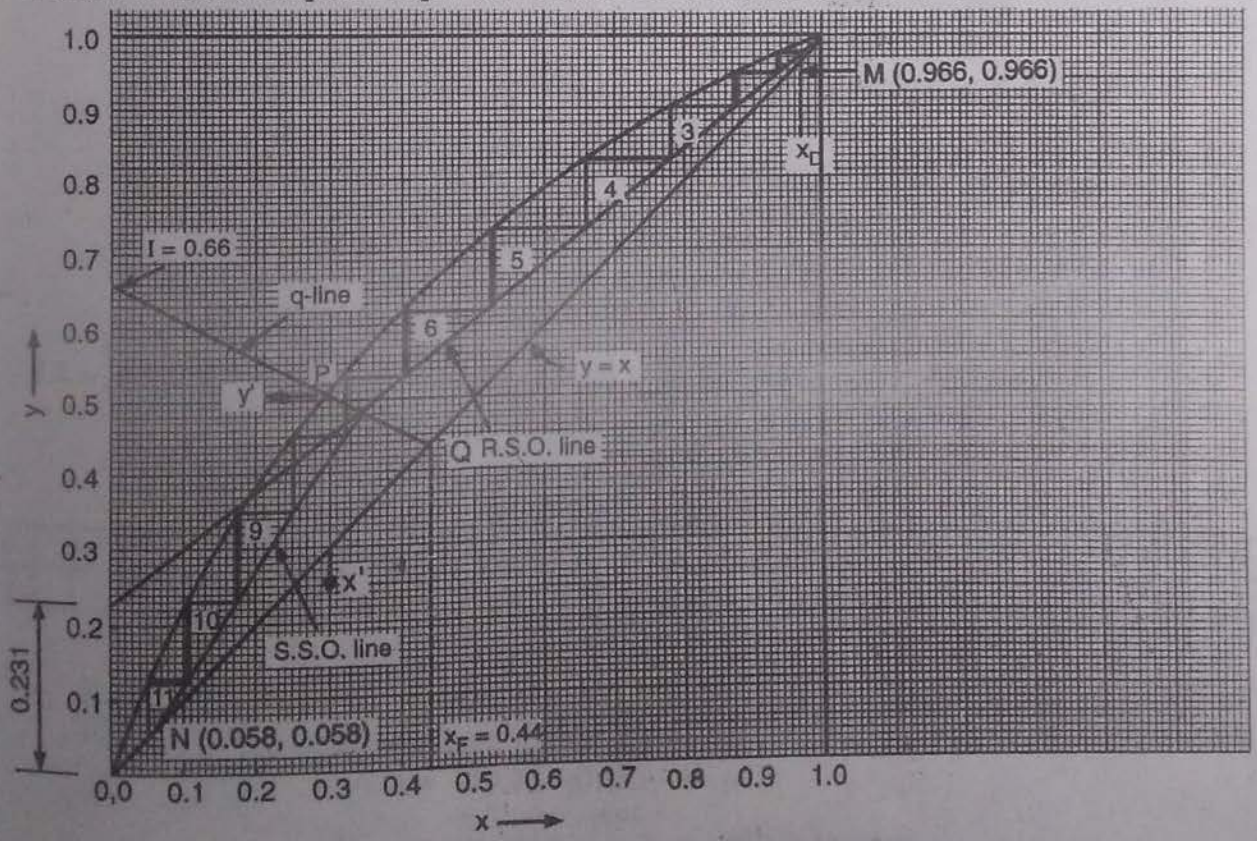

Fig. E 1.1 : Graphical construction of ideal stages

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Ex. 1.2 : A mixture of 35 mole % A and 65 mole % B is to be separated in distillation column. The concentration of A in the distillate is 93 mole % and 96 mole % of all A is in distillate. The feed is half vapour and reflux ratio is 4 : 1. The relative volatility (α_{AB}) is 2.5. How many equilibrium stages are required in each section of column ? What could be the minimum reflux ratio for such operation ?

Solution : Basis : 100 kmol of feed containing A and B.

$$F = D + W$$

$$100 = D + W \qquad W = 100 - D$$

$$x_F = \frac{35}{100} = 0.35, \; x_D = \frac{93}{100} = 0.93$$

$$\text{A in feed} = 0.35 \times 100 = 35 \text{ kmol}$$

96 mole % of A is in distillate

$$\therefore \quad \text{A in distillate} = 0.96 \times 35 = 33.6 \text{ kmol}$$

Distillate contains 93 mole % A.

$$\text{Mole \% A} = \frac{\text{A in distillate}}{\text{distillate}} \times 100$$

$$\text{distillate} = \frac{\text{A in distillate}}{(\text{Mole \% A}/100)}$$

$$\therefore \quad \text{Amount of distillate} = \frac{33.6}{0.93} = 36.13 \text{ kmol}$$

$$\text{Bottom product} = \text{Feed} - \text{distillate}$$

$$\text{Bottom product} = W = 100 - 36.13 = 63.87 \text{ kmol}$$

$$\text{A in bottom product} = 35 - 33.6 = 1.4 \text{ kmol}$$

$$\text{Mole \% A in bottom product} = \frac{1.4}{63.87} \times 100 = 2.2$$

Check :

$$\text{Mole \% B in bottom product} = 100 - 2.2 = 97.8$$

$$\text{B in distillate} = \text{Distillate} - \text{A in distillate}$$

$$= 36.13 - 33.6 = 2.53 \text{ kmol}$$

$$\text{B in feed} = \text{B in distillate} + \text{B in bottom product}$$

$$0.65 \times 100 = 65 = 2.53 + 0.978 \times 63.87 = 64.995 \approx 65$$

$D = 36.13$ kmol, $W = 63.87$ kmol, $x_F = 0.35$, $x_D = 0.93$, $x_W = \frac{2.2}{100} = 0.022$, $\alpha_{AB} = 2.5$.

$$y = \frac{\alpha x}{1 + (\alpha - 1)\, x}$$

Take x = 0, 0.1, 0.2 ... 1.0 and evaluate y and tabulate the data.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.217	0.385	0.517	0.625	0.714	0.789	0.854	0.91	0.957	1.0

(i) Plot the equilibrium data and diagonal.

(ii) Plot the q-line.

Feed is half vapour (given), so feed is half liquid.

$$\therefore \quad q = \text{fraction of feed that is liquid}$$

$$q = 0.5, \; (q = 0.5/1 = 0.5)$$

$$\text{slope of q-line} = \frac{-q}{1-q} = \frac{-0.5}{1-0.5} = -1.0$$

$$\text{slope} = \tan\alpha = -1.0 \qquad \therefore \quad \alpha = -45^\circ$$

$$x_F = 0.35$$

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Draw the q-line through the point Q (0.35, 0.35) on the diagonal and with $\alpha = -45°$ (measured from LHS).

(iii) Draw the operating line of the rectifying section through the point P having the coordinates (0.93, 0.93) on the diagonal and with an intercept on y-axis equal to 0.186.

$$\text{Intercept} = x_D/(R+1), \quad R = 4$$
$$= 0.93/(4+1) = 0.186$$

(iv) Draw the operating line for the stripping section through the point M (0.02, 0.02) on the diagonal and passing through the point of intersection of the rectification line and q-line.

(v) Construct the stages starting from x_D (0.93) and covering x_W (0.02).

While constructing triangles representing stages/plates, use the rectification section operating line upto $x_F = 0.35$ and then use the operating line of the stripping section upto $x_W = 0.02$.

(vi) Count the theoretical stages :

Number of theoretical stages including reboiler = n = 10.

Number of theoretical plates required = n – 1 = 10 – 1 = 9 ... Ans.

Number of theoretical plates in rectification section = 4 ... Ans.

Number of theoretical plates in stripping section = 5 ... Ans.

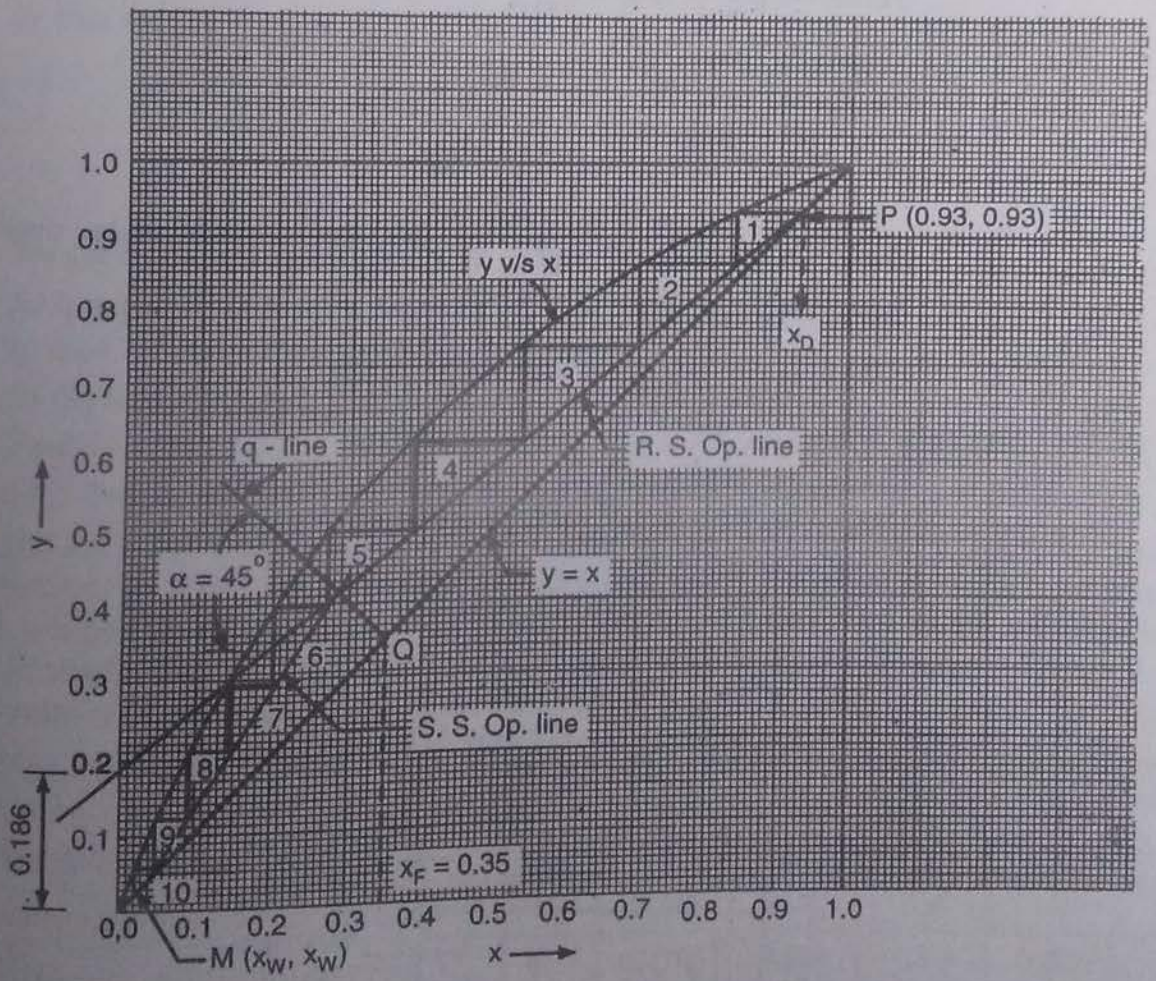

Fig. E 1.2 : Graphical construction of ideal plates

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Ex. 1.3 : A binary distillation column is operating under conditions specified below :

Feed rate = 350 kmol/h, overhead product rate = 150 kmol/h

Molefraction of more volatile component in overhead product = 0.97

Bottom product = 0.02

Bottom product rate = 200 kmol/h

Reflux ratio = 3.5

In the stripping section, it was found that the molefraction of the more volatile component in the vapour leaving a plate is 0.33 while its molefraction in the liquid coming to the same plate is 0.25. Assuming constant molar overflow, determine whether the feed is vapour, liquid or partially vaporised.

Solution : Basis : 350 kmol/h of feed to column.

$$D = 150 \text{ kmol/h}, \quad x_D = 0.97$$

$$W = 200 \text{ kmol/h}, \quad x_W = 0.02$$

$$R = 3.5$$

In the stripping section, we have :

$$y_{m+1} = \frac{L_m}{V_{m+1}} x_m - \frac{W x_W}{V_{m+1}}$$

$$L_m = V_{m+1} + W$$

$$V_{m+1} = L_m - W$$

$$y_{m+1} = \frac{L_m}{L_m - W} x_m - \frac{W x_W}{L_m - W}$$

$$y_{m+1} = \frac{L'}{L' - W} x_m - \frac{W}{L' - W} \cdot x_W, \text{ for constant molar overflow}$$

We have, $y_{m+1} = 0.33$ and $x_m = 0.25$

$$0.33 = \frac{L'}{(L' - 200)} \times 0.25 - \frac{200 \times 0.02}{(L' - 200)}$$

$$0.33 (L' - 200) = 0.25 L' - 4$$

$$\therefore \quad L' = 825 \text{ kmol/h}$$

$V_{m+1} = V'$, $L' = V' + W$, for stripping section

$$V' = L' - W$$

$$V' = 825 - 200 = 625 \text{ kmol/h}$$

$$F = 350 \text{ kmol/h}$$

$$L/D = R \quad \therefore L = RD$$

$$L = 3.5 \times 150 = 525 \text{ kmol/h in rectifying section}$$

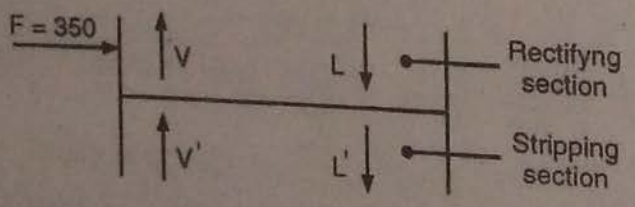

Fig. E 1.3

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L = 525 and L' = 825, so remaining (825 – 525) liquid should come from feed.

$$L' = L + qF \text{ – in stripping section}$$
$$825 = 525 + q(350)$$
$$q = 0.8572$$

Check :

$$L' = L + \text{part of feed which is liquid.}$$
$$\text{Part of feed which is liquid} = L' - L = 825 - 525 = 300 \text{ kmol/h}$$
$$\text{Also, part of feed which is liquid} = qF = 0.8572\,(350) = 300 \text{ kmol/h}$$
$$\text{Part of feed which is vapour} = (1 - q)\,F$$
$$= (1 - 0.8572) \times 350 = 50 \text{ kmol/h}$$
$$\text{Also, part of feed which is vapour} = \text{feed} - \text{part of feed that is liquid}$$
$$= 350 - 300 = 50 \text{ kmol/h}$$
$$= 50 \text{ kmol/h}$$

In the rectifying section :

$$V = V' + \text{part of feed that is vapour}$$
$$= 625 + 50 = 675 \text{ kmol/h} \quad \ldots (1)$$

In this section, we have,

$$V = L + D$$
$$= RD + D = (R + 1)\,D$$
$$= (3.5 + 1) \times 150$$
$$= 675 \text{ kmol/h} \quad \ldots (2)$$

We get the same results by (1) and (2).

As q is less than one (q = 0.8572), feed is partially vaporised. ... **Ans.**

In feed, we have : liquid + vapour.

In the feed mixture :

Feed in a liquid form : 85.72% and feed in a vapour form : 14.28%.

i.e., feed is (300/350 i.e.) $6/7^{th}$ liquid and (50/350 i.e.) $1/7^{th}$ vapour. ... **Ans.**

Ex. 1.4 : 100 kmol/h of a feed containing 40 mole % hexane and 60 mole % heptane is to be distilled in a column consisting of a still pot and a plate and a partial condenser. The feed, a liquid mixture at its bubble point, is fed to a reboiler from which a residue is drawn continuously. The reflux is returned to a plate. The distillate contains 80 mole % hexane. The ratio of liquid reflux flow to distillate is 2.

(i) Using the McCabe-Thiele method, calculate the distillate flow rate in moles/h and the bottom composition.

Equilibrium data :

Molefraction hexane in liquid, x	0	0.045	0.192	0.4	0.69	1
Molefraction hexane in vapour, y	0	0.1775	0.538	0.78	0.932	1

(ii) If the relative volatility $\alpha = 6$, do the calculation analytically.

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Solution : Basis : (i) 100 kmol/h of feed to the distillation unit.

Molefraction of hexane in feed = $x_F = \frac{40}{100} = 0.4$.

Molefraction of hexane in distillate product = $y_D = x_D = \frac{80}{100} = 0.80$.

Molefraction of hexane in bottom = x_W = ?

$$\frac{\text{Liquid reflux}}{\text{Distillate}} = \frac{L}{D} = 2, \quad R = 2$$

We have,

$$V = L + D$$

$$V y_{n+1} = L x_n + D y_D$$

$$y_{n+1} = \frac{L}{V} x_n + \frac{D}{V} y_D$$

$$\frac{L}{V} = \frac{L}{L+D} = \frac{L/D}{L/D+1} = \frac{2}{2+1} = \frac{2}{3}$$

$$D/V = D/L + D = \frac{1}{(L/D+1)} = \frac{1}{(2+1)} = \frac{1}{3}$$

$$L/D = 2 \text{ if } D = 1 \text{ then } L = 2.$$

Partial condenser is considered to act as one theoretical stage (if sufficient contact time between vapour product and liquid reflux is provided then vapour and liquid streams will leave in equilibrium, y_D is in equilibrium with x_o).

$$y_D = 0.8$$

So mark the point $y_D = x_D = 0.8$ on the y = x line, i.e., diagonal.

Conditions in the condenser are fixed and as x_o is in equilibrium with y_D, the point (x_o, y_D) is located. Draw the operating line through the point $y_D = x_D = 0.8$, on the y = x line, with a slope equal to L/V or an intercept equal to $D \cdot y_D/V$.

Slope of the operating line = L/V = 2/3.

Intercept of the operating line on the y-axis = $D\, y_D/V = y_D/R + 1 = 0.8/2 + 1 = 0.267$.

$$\text{Slope} = \frac{2}{3} = 0.66667$$

$$\tan\theta = 0.66667 \qquad \therefore \ \theta = 33.69^o$$

Draw a line through $y_D = 0.8$ on the diagonal making an angle of 33.69° to the line-through y_D and parallel to x-axis.

OR mark intercept = 0.267 on y-axis and joint it with $y_D = 0.8$ on the diagonal to get the operating line.

Partial condenser and reboiler each corresponds to one theoretical stage, so including a plate there are totally three theoretical stages (3 ideal plates) so step-off three theoretical stages by constructing the triangles between the equilibrium curve and the operating line and read x_W corresponding to the third stage.

From Fig. E 1.4 (a), $x_W = 0.135$

Composition of bottom product = $x_W = 0.135$ ... **Ans. (i)**

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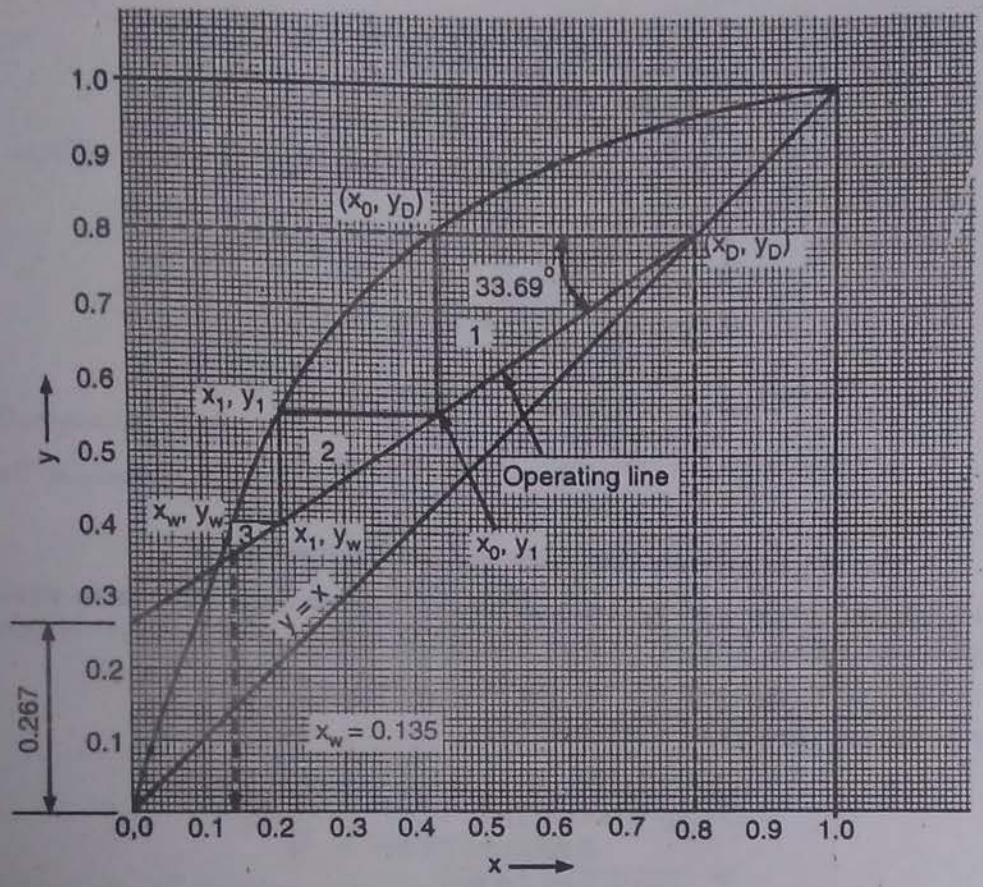

Fig. E 1.4 (a) : Determination of x_W by construction of stages

Overall material balance :

$$F = W + D$$

Material balance of hexane (overall)

$$x_F F = W x_W + D y_D$$

$$W = F - D = 100 - D$$

$$0.40 \times 100 = (100 - D) \times 0.135 + D \times 0.80$$

$$D = 39.85 \text{ kmol/h}$$

Distillate flow rate = 39.85 kmol/h ... Ans. (i)

(ii) Determination of x_W and D analytically for $\alpha = 6.0$.

Given : $\alpha = 6$

The x-y equilibrium relationship is given as

$$y = \frac{\alpha x}{1 + \alpha (1 - x)}$$

Rearranging it, we get,

$$x = \frac{y}{y + \alpha (1 - y)}$$

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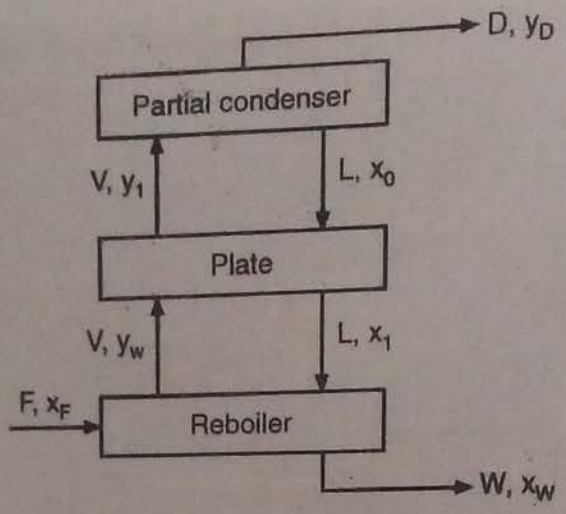

Fig. E 1.4 (b) : Reboiler, plate and partial condenser

We have in a partial condenser, y_D (composition of vapour distillate) is in equilibrium with x_0 (composition of reflux liquid).

We have $y_D = 0.8$

We will determine x_0 (in equilibrium with y_D).

$$x_0 = \frac{y_D}{y_D + \alpha (1 - y_D)}$$

$$= \frac{0.80}{0.80 + 6 (1 - 0.80)}$$

$$= 0.40$$

x_0, y_1 is on operating line, so using x_0, we will determine y_1.

Material balance about the partial condenser :

$$V y_{n+1} = L X_n + D y_D$$

$$V y_1 = L x_0 + D y_D$$

$$y_1 = \frac{L}{V} x_0 + \frac{D}{V} y_D$$

$$= \frac{R}{R+1} x_0 + \frac{1}{R+1} y_D$$

$$= \frac{2}{3} (0.40) + \frac{1}{3} (0.80)$$

$$= 0.533$$

x_1 is in equilibrium with y_1 and is determined as follows :

$$x_1 = \frac{y_1}{y_1 + \alpha (1 - y_1)}$$

$$= \frac{0.533}{0.533 + 6 (1 - 0.533)}$$

$$= 0.16$$

Using x_1, we will determine y_W by doing a material balance about the plate and condenser together.

$$V y_W = L x_1 + D y_D$$

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$$y_W = \frac{L}{V} x_1 + \frac{D}{V} y_D$$

$$= \frac{2}{3} (0.16) + \frac{1}{3} (0.80)$$

$$= 0.373$$

x_W is in equilibrium with y_W and is determined as follows :

$$x_W = \frac{y_W}{y_W + \alpha (1 - y_W)}$$

$$= \frac{0.373}{0.373 + 6 (1 - 0.373)}$$

$$= 0.090$$

Composition of bottom = 0.090 **... Ans. (ii)**

Overall material balance :

$$F = W + D$$

$$W = F - D = 100 - D$$

Material balance of hexane over the entire unit :

$$x_F F = D y_D + W x_W$$

$$x_F F = D y_D + (100 - D) x_W$$

$$0.40 \times 100 = D (0.80) + (100 - D) \times 0.09$$

$$D = 43.67 \text{ kmol/h}$$

Molar flow rate of distillate = D = 43.67 kmol/h **... Ans. (ii)**

Ex. 1.5 : A liquid mixture of benzene-toluene is to be distilled in a tower at 101.3 kPa pressure. The feed of 100 kmol/h is liquid and it contains 45 mole % benzene and 55 mole % toluene and enters at 327.6 K (54.6 °C). A distillate containing 95 mole % benzene and 5 mole % toluene and bottoms containing 10 mole % benzene and 90 mole % toluene are to be obtained. The reflux ratio is 4 : 1. The average heat capacity of the feed is 159 kJ/(kmol.K) and the average latent heat is 32099 kJ/kmol. Calculate the kmol/h of distillate and bottom product and the number of plates required.

Data : B.P. of benzene = 353 K and B.P. of toluene = 383.8 K

x	1	0.78	0.581	0.411	0.258	0.130	0
y	1	0.90	0.777	0.632	0.450	0.261	0

Solution : Basis : 100 kmol/h of feed

$$F = 100 \text{ kmol/h}$$

$$x_F = 45/100 = 0.45$$

$$x_D = 95/100 = 0.95, \quad x_W = 10/100 = 0.1$$

$$100 = D + W \qquad \therefore \; W = 100 - D$$

and $$0.45 \times 100 = 0.95 D + 0.1 W$$

$$45 = 0.95 D + 0.1 (100 - D)$$

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$\therefore$ $D = 41.2$ kmol/h

$W = 100 - 41.2 = 58.8$ kmol/h

Molar flow rate of distillate = 41.2 kmol/h ... Ans.

Molar flow rate of Residue/Bottom product = 58.8 kmol/h ... Ans.

(ii) Determination of number of theoretical plates required :

Feed is liquid and it enters at 327.6 K

Average C_p of feed = 159 kJ/(kmol.K)

Boiling point of benzene = 353 K

Boiling point of toluene = 383.8 K

Molefraction of benzene in feed = x_F = 0.45

Molefraction of toluene in feed = 1 – 0.45 = 0.55

Mean boiling point of feed = T

$$T = \sum_{i=1}^{n} (\text{molefraction} \times \text{boiling point})$$

$$T = 0.45 \times 353 + 0.55 \times 383.8 = 369.94 \text{ K}$$

Average latent heat = $\lambda_{avg.}$ = 32099 kJ/kmol.

Temperature of feed = T_f = 327.6 K

$$q = \frac{\text{Heat to convert 1 mole of feed to saturated vapour}}{\text{Molar latent heat of vaporisation}}$$

$$= \frac{C_p (T - T_f) + \lambda_{avg}}{\lambda_{avg}}$$

$$q = \frac{159 (369.94 - 327.6) + 32099}{32099}$$

$$= 1.2097 \approx 1.21$$

$$\text{Slope of q-line} = \frac{-q}{1-q} = \frac{-1.21}{1-1.21} = 5.762$$

$$\text{Slope} = \tan\alpha = 5.762$$

$\therefore$ $\alpha = \tan^{-1}(5.762) = 80.15^\circ$

(i) Draw the equilibrium curve and diagonal.

(ii) Draw the operating line of rectifying section through the point P (0.95, 0.95) on the diagonal and with an intercept on the y-axis of $x_D/(R + 1) = 0.95/(4 + 1) = 0.19$.

(iii) Draw the q-line through the point (0.45, 0.45) on the diagonal and with $\alpha = 80.15^\circ$.

(iv) Draw the operating line of the stripping section through the point having coordinates (0.1, 0.1) on the diagonal and passing through the point of intersection of q-line and top operating line.

(v) Construct the stages between the equilibrium curve and the operating lines required for the separation starting from the point (0.95, 0.95) and covering the point (0.1, 0.1) and count them.

Number of theoretical plates including reboiler = n = 8

Number of theoretical plates required = n – 1 = 8 – 1 = 7 ... Ans.

Feed plate : 5th from top. ... Ans.

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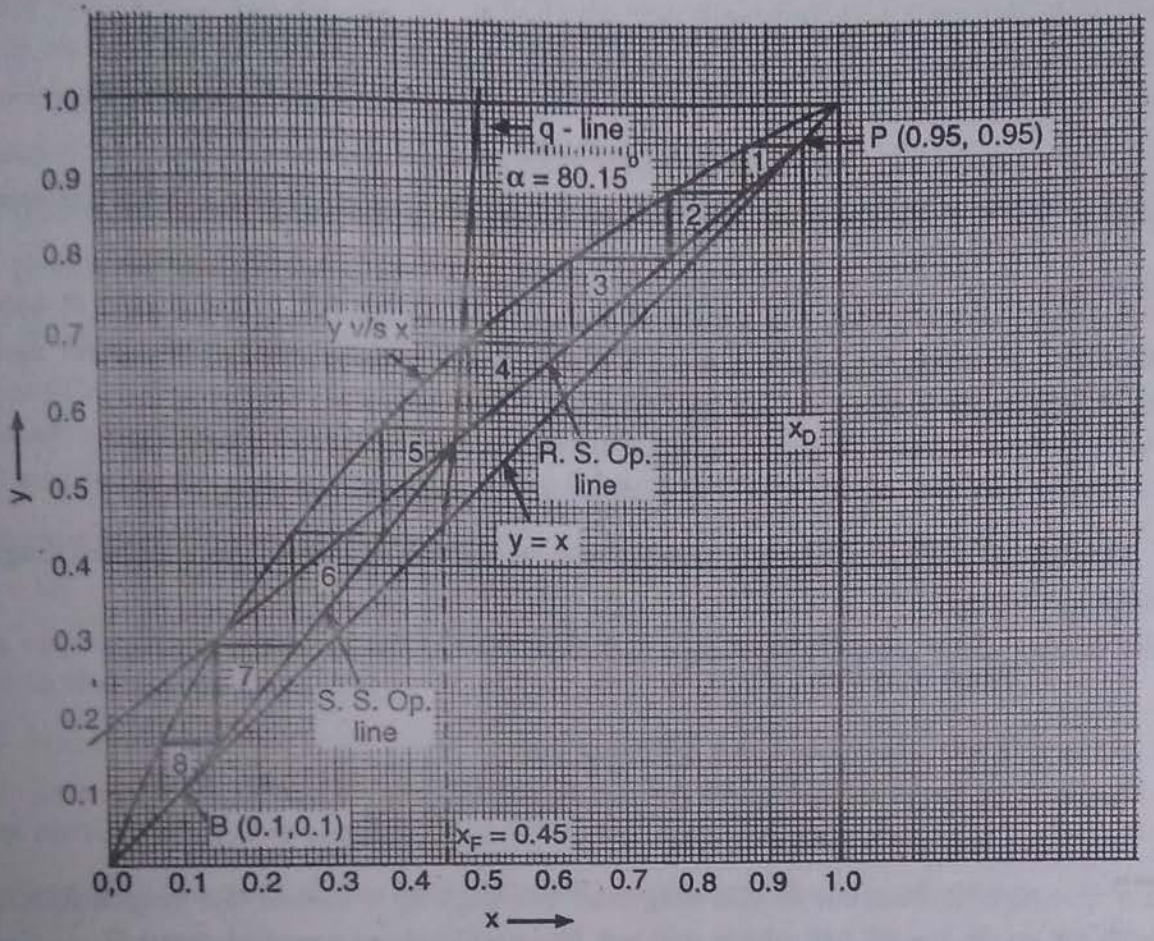

Fig. E 1.5 : Construcion of ideal stages

Ex. 1.6 : The feed consists of methanol and water containing 80 mole % methanol and 20 mole % water. The overhead product is to contain 99 mole % methanol and the bottom product contains 0.5 mole % methanol. The feed is cold and for each mole of feed, 0.2 moles of vapour is condensed on the feed plate. The reflux ratio is 1.35 and the reflux is at its bubble point.

Calculate :

(i) the minimum reflux ratio

(ii) number of theoretical plates using total condenser and

(iii) number of theoretical plates using a reboiler and partial condenser operating with reflux in equilibrium with a vapour going to the final condenser.

Data :

x	0	0.10	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
y	0	0.42	0.58	0.67	0.73	0.78	0.83	0.87	0.92	0.96	1

Solution : Basis : 100 kmol of feed

F = 100 kmol

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$$x_F = 80/100 = 0.8$$

$$x_D = 99/100 = 0.99$$

$$x_W = 0.5/100 = 0.005$$

q is the number of moles of saturated liquid resulting in stripping section for each mole of feed.

As for each mole of feed, 0.2 moles of vapour is condensed, total moles of saturated liquid resulting in stripping section are 1.2 moles per 1 mole of feed. The larger moles are resulting in stripping section as the feed is cold.

$$q = \frac{1 + 0.2}{1} = 1.2$$

(i) Minimum reflux ratio : The point of intersection of the q-line with the diagonal is (x_F, x_F with $x_F = 0.80$).

$$\text{Slope of q-line} = \frac{-q}{1-q}$$

$$= \frac{-1.2}{1-1.2} = 6$$

$$\tan\alpha = 6 \quad \therefore \alpha = 80.54^\circ$$

Draw the equilibrium curve and diagonal. Through $x_F = 0.8$ on the diagonal, draw the q-line with an angle $\alpha = 80.54^\circ$ which will cut the equilibrium curve at point P.

Join PQ (not shown in Fig. E 1.6 (a)) and the slope of PQ will give a value of R_m.

The coordinates of the point P are :

$$y' = 0.927 \quad \text{and} \quad x' = 0.82$$

For the cold feed, R_m is given as

$$R_m = \frac{x_D - y'}{y' - x'}$$

$$R_m = \frac{0.99 - 0.927}{0.927 - 0.82} = 0.588 \approx 0.59$$

The minimum reflux ratio = $R_m = 0.59$ ... Ans. (i)

(ii) Number of theoretical plates with total condenser.

$$x_D = 0.99, \quad x_W = 0.005$$

Locate point Q (0.99, 0.99) on the diagonal.

$$R = 1.35 \text{ (given)}$$

Intercept on y-axis of operating line (rectifying section) is :

$$\text{Intercept} = x_D/(R+1) = 0.99/(1.35+1) = 0.42$$

Draw the top operating line (operating line for rectification section) through the point Q (0.99, 0.99) on the diagonal and with an intercept on the y-axis equal to 0.42.

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Locate the point (x_W, x_W, x_W = 0.005) on the diagonal and through this draw the operating line of the stripping section which passes through the point of intersection of q-line and top operating line.

Construct the stages starting from x_D and reaching exactly or covering x_W (0.005).

Number of theoretical plates required including reboiler = n = 9 ... **Ans. (ii)**

Number of theoretical plates required = n – 1 = 9 – 1 = 8 ... **Ans. (i)**

(iii) Number of theoretical plates – using partial condenser :

The method of constructing the number of plates in the case of a partial condenser (followed by a final condenser) is the same as that given in (ii). We have to start with point (x_D, x_D) and so on. So a separate graphical procedure is not required, but the difference is :

Composition of the vapour leaving the partial condenser is y_D which is the same as x_D, $y_D = x_D$.

The vapour leaving the partial condenser y_D is in equilibrium with liquid which is refluxed to the column from it (x_c).

The operating line ends at point b and triangle abc represents the top plate in the column. As vapour leaving the partial condenser y_D is in equilibrium with liquid reflux (x_c), the partial condenser is equivalent to one theoretical plate/stage.

Number of theoretical plates using partial condenser including reboiler = n = 9.

Number of theoretical plates required in the column (using partial condenser) = n – 2 = 9 – 2 = 7 ...**Ans. (iii)**

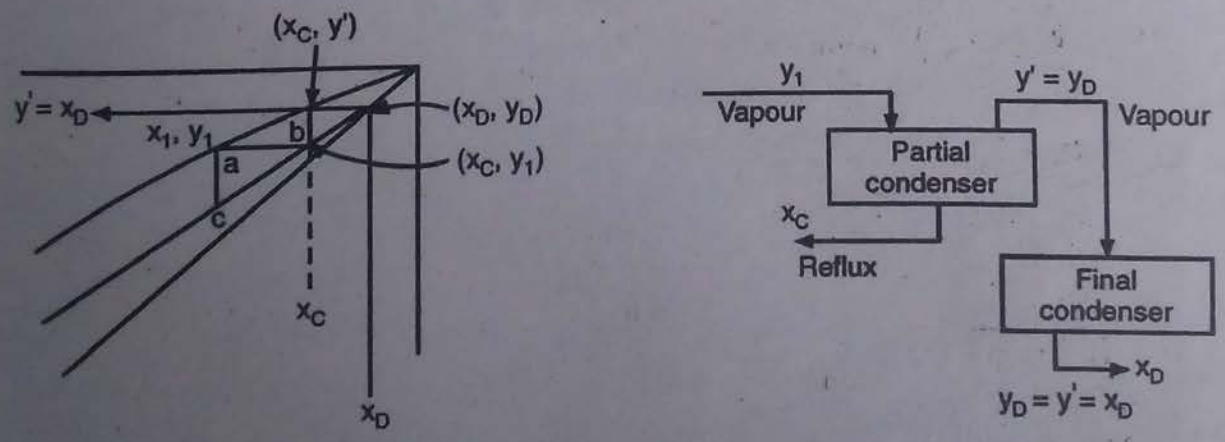

Fig. E 1.6 (a) : Partial condenser

[Number of theoretical plates required in the column = n – 2, 2 = 1 + 1 : one for reboiler and one for partial condenser as reboiler and partial condenser are each equivalent to one theoretical stage. 2 is subtracted from n (total theoretical plates) to get theoretical plates in the column.]

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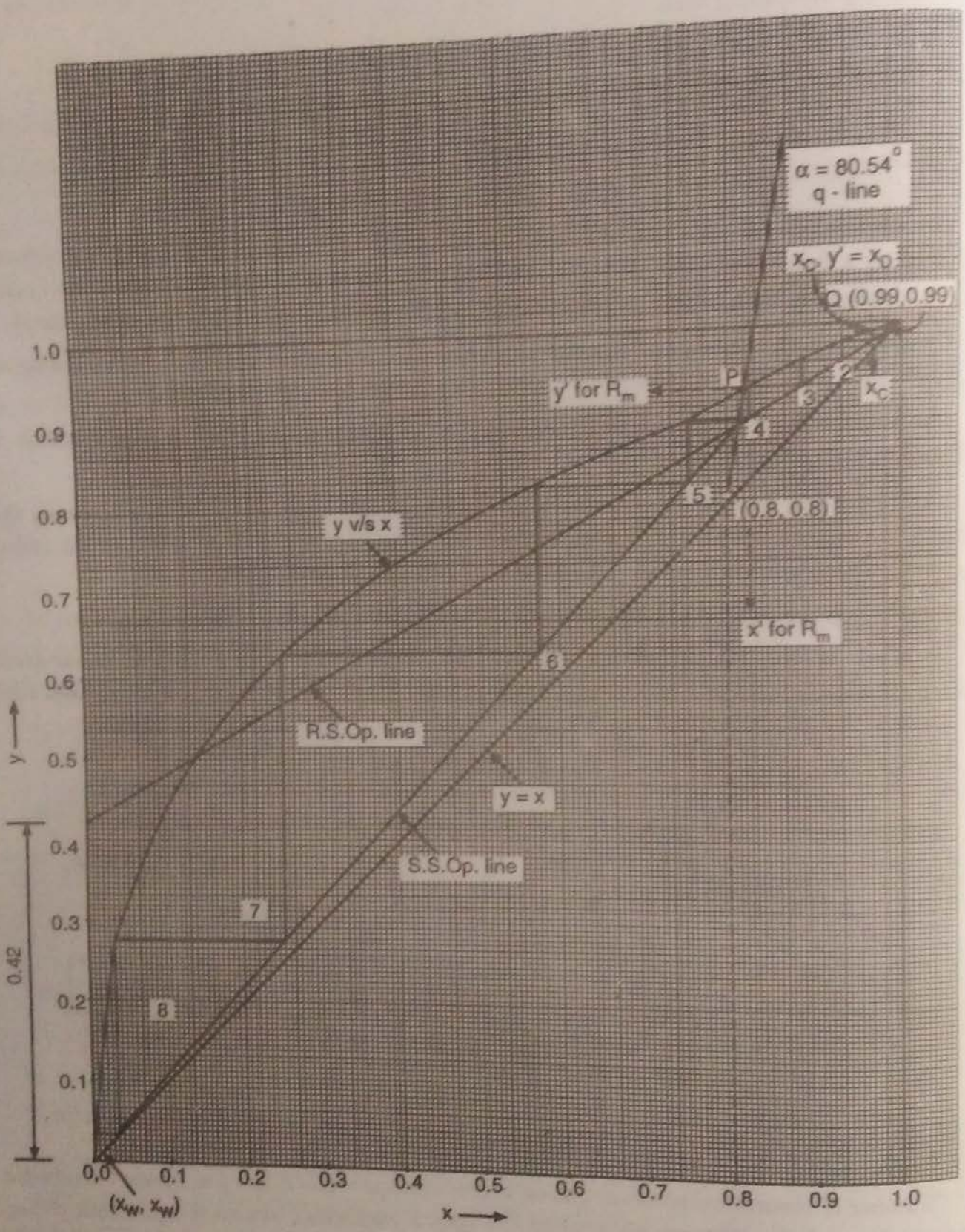

Fig. E 1.6 (b) : Determination of number of ideal plates

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Ex. 1.7 : Feed containing 0.16 molefraction ethyl alcohol and 0.84 molefraction water is continuously distilled in a fractionating column to yield a product containing 0.77 molefraction alcohol and a waste containing 0.02 molefraction alcohol. It is proposed to withdraw 25% of the alcohol in the entering stream as a side stream with a molefraction of 0.5 alcohol.

Determine the number of theoretical plates required and the plate from which the side stream should be withdrawn.

Data : (i) Reflux ratio is 2, (ii) Feed is liquid at its boiling point, (iii) Equilibrium data are :

x	0.	0.05	0.10	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.89	1
y	0	0.325	0.423	0.538	0.577	0.62	0.65	0.705	0.76	0.825	0.89	1

Solution : Basis : 100 kmol of feed to column.

Alcohol in feed = $0.16 \times 100 = 16$ kmol

Alcohol entering the column = 16 kmol

25% of the alcohol in the entering stream is withdrawn as a side stream containing 0.5 molefraction alcohol in it.

Alcohol in side stream = $0.25 \times 16 = 4$ kmol

Side stream removed = S = 4/0.5 = 8 kmol

Overall material balance :

$$F = D + W + S$$
$$100 = D + W + 8$$
$$D + W = 92, \quad W = 92 - D$$

Alcohol balance :

$$x_F F = x_D D + x_W W + x_S S$$
$$0.16 \times 100 = 0.77 D + 0.02 W + 0.5 \times 8$$
$$16 = 0.77 D + 0.02 W + 8$$
$$12 = 0.77 D + 0.02 W$$
$$12 = 0.77 D + 0.02 (92 - D)$$

$\therefore$ $D = 13.55$ kmol and $W = 78.45$ kmol

In the top section of the column :

$$L_n/D = R = 2$$
$$L_n = 2D = 2 \times 13.55 = 27.10 \text{ kmol}$$
$$V_{n+1} = L_n + D = 27.10 + 13.55 = 40.65 \text{ kmol}$$

The slope of operating line is always L/V.

Slope of the operating line = L_n/V_{n+1}

= $27.10/40.65 = 0.67$

This operating line of slope 0.67 is to be used from x_D to x_S, i.e., from 0.77 to 0.5 while constructing the stages.

In the section of column from the side stream to the feed plate :

$$V_s = V_{n+1} = 40.65 \text{ kmol}$$
$$L_n = L_s + S$$
$$L_s = L_n - S = 27.10 - 8 = 19.10 \text{ kmol}$$

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The slope of the operating line for the section of the column between side stream and feed plate is

$$\text{Slope} = L_s/V_s = 19.10/40.65 = 0.47$$

In the bottom section of the column below the feed plate :

As the feed is at its boiling point, q = 1

$$L_m = L_s + qF = 19.10 + 1 \times 100 = 119.10 \text{ kmol}$$

$$L_m = V_{m+1} + W$$

$$V_{m+1} = L_m - W$$

$$= 119.10 - 78.45 = 40.65 \text{ kmol}$$

The slope of the operating line for this section (stripping section) is

$$L_m/V_{m+1} = 119.10/40.65 = 2.93$$

Refer Fig. E 1.7.

(i) Draw the equilibrium curve, y v/s x, diagonal y = x, q-line which is a vertical line at $x_F = 0.16$ (as q = 1, it is parallel to y-axis).

(ii) Draw the top operating line through the point P (x_D, x_D), i.e., P (0.77, 0.77) on the diagonal with a slope equal to 0.67 or with an intercept on y-axis equal to $x_D/(R + 1)$.

$$\text{slope} = \tan\alpha = 0.67 \quad \therefore \alpha = 34^\circ$$

$$\text{Intercept} = \frac{x_D}{R+1} = \frac{0.77}{2+1} = 0.2566 \approx 0.257$$

This line is to be used from x_D = 0.77 to x_S = 0.5 while constructing stages.

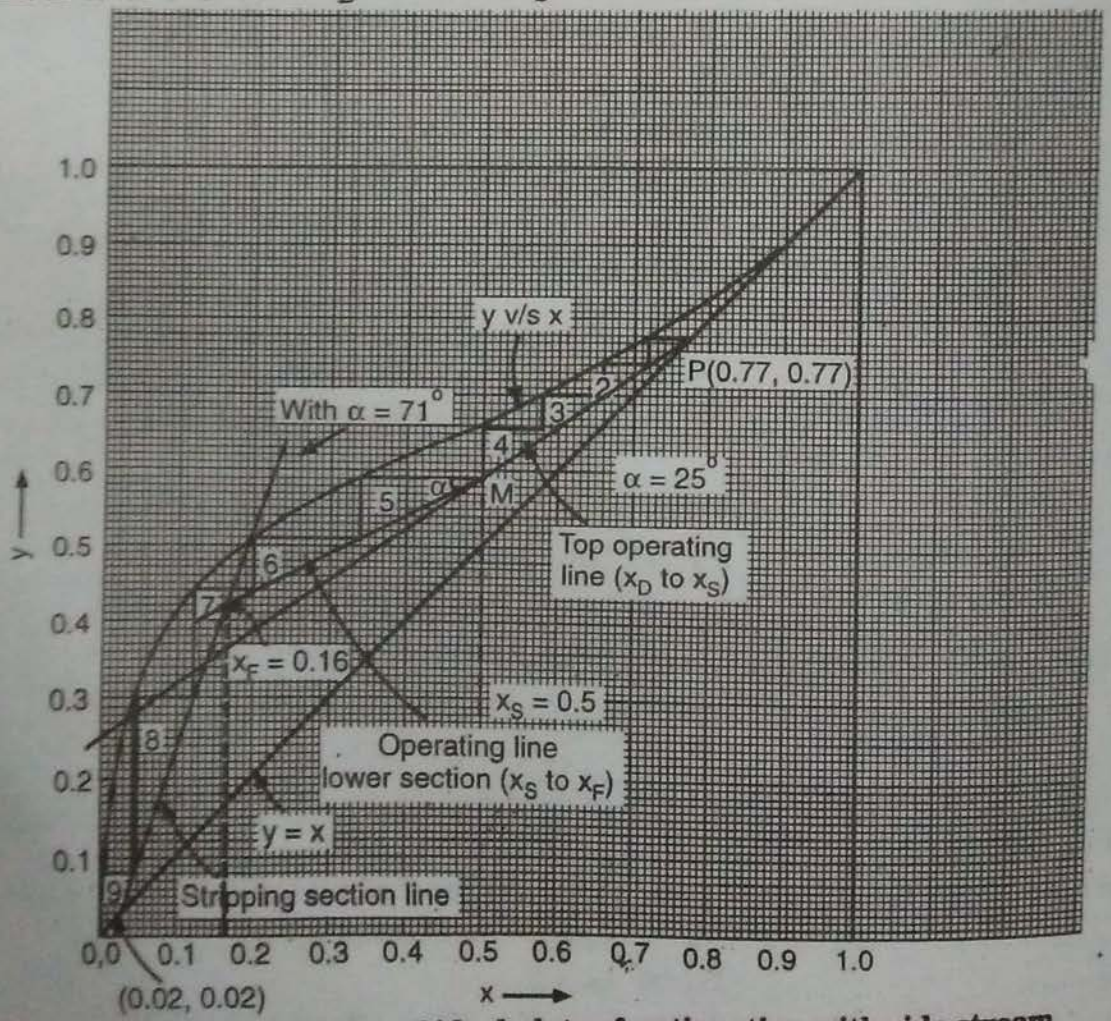

Fig. E 1.7 : Construction of ideal plates-fractionation with side stream

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(iii) Draw the operating line of the lower part of rectifying section (portion above the feed plate upto the side stream) through $x_s = 0.5$ on the previous operating line with a slope equal to 0.47.

$$\text{slope} = \tan\alpha = 0.47 \quad \therefore \quad \alpha = 25^\circ$$

[Draw the operating line for this section of the column through the point, at which a vertical through $x_S = 0.5$ intersect the top operating line, with angle of 25°, i.e., through the point M at an angle of 25° to the horizontal through M.]

Use this line for constructing stages from $x = 0.5$ to $x = x_F = 0.16$.

(iv) Draw the operating line for the stripping section through the point (x_W, x_W), i.e., (0.02, 0.02) on the diagonal with slope equal to 2.93.

$$\text{slope} = 2.93 = \tan\alpha \quad \therefore \quad \alpha = 71^\circ$$

Use this line for constructing stages upto $x_W = 0.02$.

From plot : Number of theoretical plates including reboiler = n = 9.

Number of theoretical plates required = n – 1 = 9 – 1 = 8 ... **Ans.**

Side stream should be withdrawn from 4th plate from the top. ... **Ans.**

Ex. 1.8 : A mixture of 35 mole % A and 65 mole % B is to be separated in a distillation column. The concentration of A in the distillate is 92 mole % and 96% of component A is recovered in distillate. The feed is all vapour. The reflux ratio is 4 and relative volatility is 2.4. How many equilibrium stages are required in each section of column ? What is the minimum reflux ratio ?

Solution : Basis : 100 kmol of feed

$$\text{Moles of A in feed} = 0.35 \times 100 = 35 \text{ kmol}$$

$$\text{Moles of A in distillate} = 0.96 \times 35 = 33.6 \text{ kmol}$$

$$\text{Moles of A in bottom product} = 35 - 33.6 = 1.4 \text{ kmol}$$

$$\text{Moles of B in feed} = 0.65 \times 100 = 65 \text{ kmol}$$

$$\text{Mole \% A in distillate} = 92$$

$$\therefore \quad 92 = \frac{\text{Moles of A}}{\text{Moles of distillate}} \times 100$$

$$\text{Moles of distillate} = \frac{33.6}{92} \times 100$$

$$= 36.52 \text{ kmol}$$

$$F = D + W$$

$$W = F - D = 100 - 36.52 = 63.48 \text{ kmol}$$

$$\text{Mole \% A in bottom product} = \frac{1.4}{63.48} \times 100$$

$$= 2.2$$

Molefraction of A in feed $= x_F = 35/100 = 0.35$

Molefraction of A in distillate $= x_D = 92/100 = 0.92$

Molefraction of A in bottom product $= x_W = 2.2/100 = 0.022$

$$\alpha = 2.4$$

Generate x-y data using

$$y = \frac{\alpha x}{1 + (\alpha - 1)x}$$

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Take x = 0, 0.1, 0.2 ... 1.0 and evaluate the corresponding values of y.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.21	0.375	0.507	0.615	0.706	0.783	0.848	0.906	0.956	1.0

Feed is all saturated vapour (given).

$\therefore$ q = 0 and so the q-line is parallel to x-axis through the point (x_F, x_F) on the diagonal.

$$R = 4 \text{ (given)}$$

(i) Draw the equilibrium curve.

(ii) Draw the q-line parallel to the x-axis through the point Q (0.35, 0.35) on the diagonal.

(iii) Draw the operating line of the rectification section through the point A (0.92, 0.92) on the diagonal having an intercept of 0.184 on y-axis.

$$\text{Intercept on y-axis} = x_D/(R + 1)$$
$$= 0.92/(4 + 1) = 0.184$$

(iv) Draw the operating line of the stripping section through the point B (0.022, 0.022) on the diagonal and passing through the point of intersection of q-line and top operating line.

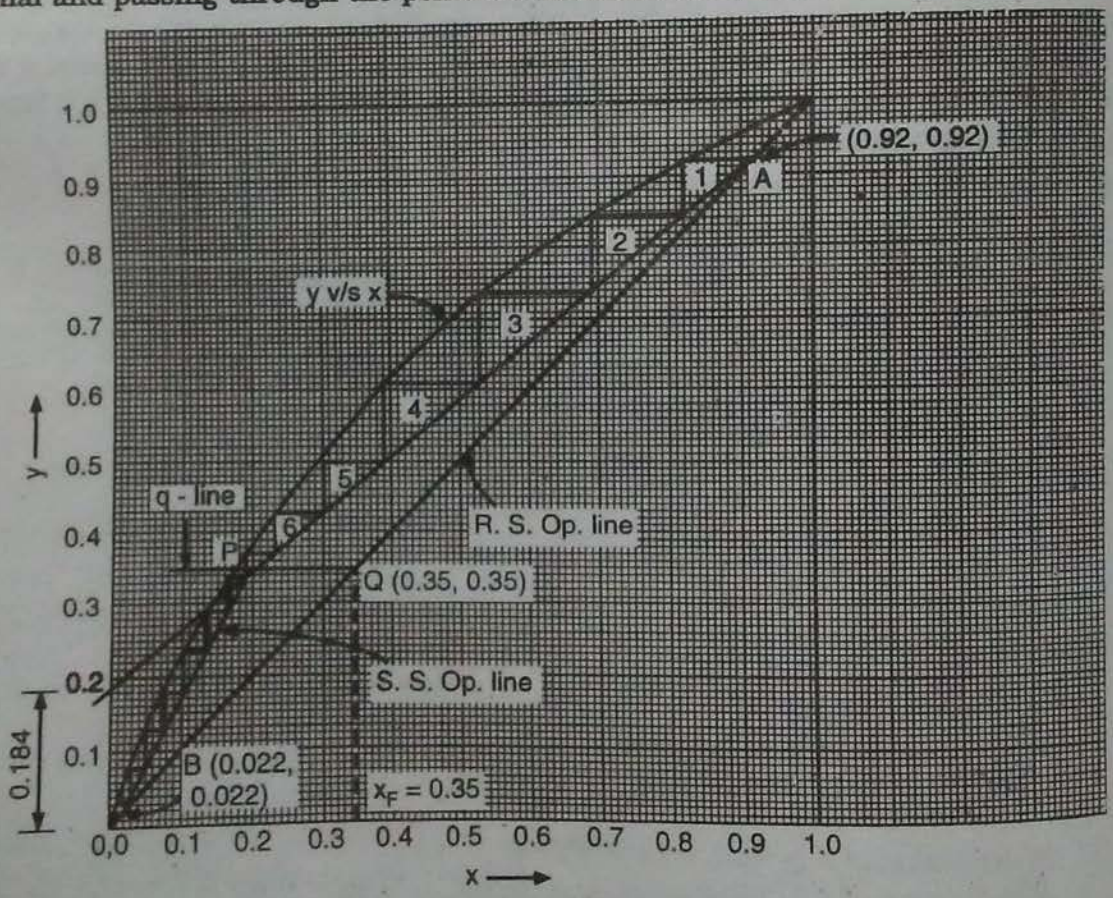

Fig. E 1.8 : Determination of ideal stages and R_m

(v) Construct the triangles representing ideal stages/plates, starting from the point (x_D = 0.92, x_D = 0.92) and reaching exactly or covering the point (x_W = 0.022, x_W = 0.022), between the equilibrium curve and the operating lines. Use the top operating line (for the construction) upto x_F = 0.35 and then use the operating line for the stripping section upto x_W = 0.022.

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Number of theoretical plates including reboiler = n = 14

Number of theoretical plates required = n – 1 = 14 – 1 = 13.

Number of theoretical plates in the rectification section = 6.

Number of theoretical plates in the stripping section = 7. ... **Ans.**

Minimum reflux ratio for feed-saturated vapour is given by

$$R_m = \frac{x_D - x_F}{x_F - x'}$$

where $x_D = 0.92$, $x_F = 0.35$

x' from graph - x coordinate of the point P = 0.183

$$R_m = \frac{0.92 - 0.35}{0.85 - 0.183} = 3.41$$ ... **Ans.**

Ex. 1.9 : A continuous fractionating column is to be designed to separate 14000 kg/h of a mixture of 40% benzene and 60% toluene at 1 atm into an overhead product containing 97% benzene and a bottom product containing 98% toluene. These percentages are by weight. The feed is available at 293 K (20 °C) and 1 atm. A reflux ratio of three times the minimum is used.

Determine : (a) The quantities of overhead and bottoms in kmol/h.

(b) The number of real/actual plates required and the optimum position of feed plate.

Given :

(i) Benzene and toluene form an ideal system with a relative volatility of 2.5.

(ii) The molal latent heats of benzene and toluene are 30800 and 33300 kJ/kmol respectively.

(iii) The bubble point of feed at 1 atm is 368 K (95 °C) and its specific heat is 1.85 kJ/(kg.K).

(iv) Overall efficiency of 60% can be assumed.

Solution : Basis : 14000 kg/h of feed to column.

Molecular weight of benzene [C_6H_6] = 78

Molecular weight of toluene [$C_6H_5CH_3$] = 92

Feed contains 40% benzene and 60% toluene by weight.

In 100 kg feed : 40 kg benzene and 60 kg toluene.

$$\text{Molefraction of benzene in feed} = x_F = \frac{40/78}{40/78 + 60/92}$$

$$= 0.44$$

$$\text{Molefraction of toluene in feed} = 1 - 0.44 = 0.56$$

$$\text{Average molecular weight of feed} = M_{avg}$$

$$M_{avg} = \sum M_i x_i$$

$$= 78\,(0.44) + 92\,(0.56) = 85.84 \text{ kg/kmol}$$

$$\text{Molar flow rate of feed to column} = F = \frac{14000}{85.84}$$

$$= 163.09 \text{ kmol/h} \approx 163 \text{ kmol/h}$$

Benzene in overhead product = 97% by weight

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$$\text{Molefraction of benzene in distillate} = \frac{97/78}{97/78 + 3/92}$$

$$x_D = 0.9744$$

Toluene in bottoms = 98% by weight

$\therefore$ Benzene in bottoms = 100 – 98 = 2% by weight

$$\text{Molefraction of benzene in bottoms} = \frac{2/78}{2/78 + 98/92}$$

$$x_W = 0.0235$$

Overall material balance :

$$F = D + W$$

$$163 = D + W \quad \therefore \quad W = 163 - D$$

Material balance of benzene :

$$x_F F = x_D D + x_W W$$

$$0.44\,(163) = 0.9744\,(D) + 0.0235\,(W)$$

$$71.72 = 0.9744\,D + 0.0235\,(163 - D)$$

$$\therefore \quad D = 71.395 = 71.4 \text{ kmol/h}$$

$$W = 163 - 71.4 = 91.6 \text{ kmol/h}$$

Molar flow rate of overhead product = 71.4 kmol/h.

Molar flow rate of bottoms = 91.6 kmol/h. ... Ans.

Molar latent heat of benzene = 30800 kJ/kmol

Molar latent heat of toluene = 33300 kJ/kmol

Molar latent heat of feed = λ

$$\lambda = 0.44\,(30800) + 0.56\,(33300)$$

$$= 32200 \text{ kJ/kmol}$$

Specific heat of feed = 1.85 kJ/(kg.K)

$M_{avg.}$ of feed = 85.84 kg/kmol

Molar specific heat of feed = C_p = 1.85×85.84

$$= 158.8 \text{ kJ/(kmol.K)}$$

Feed temperature = T_f = 293 K

Bubble point of feed = T_b = 368 K

Calculation of q for cold feed at 293 K :

$$q = \frac{\text{heat to convert 1 mole of feed to saturated vapour}}{\text{molar latent heat of feed}}$$

$$q = \frac{C_p\,(T_b - T_f) + \lambda}{\lambda} \quad \text{(for 1 mole feed)}$$

$$= \frac{158.8\,(368 - 298) + 32200}{32200}$$

$$= 1.3698 \approx 1.37$$

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$$\text{Slope of q-line (feed line)} = \frac{-q}{1-q} = \frac{-1.37}{1-1.37} = 3.7$$

Slope = $\tan\alpha = 3.7$ $\therefore$ $\alpha = 74.87^\circ = 75^\circ$

Relative volatility = 2.5

$$y = \frac{\alpha x}{1 + (\alpha - 1)x}$$

Generate x-y data using the above equation.

Take x = 0, 0.1, 0.2 ... 1.0, evaluate the corresponding y values and tabulate.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.22	0.385	0.52	0.625	0.714	0.79	0.854	0.91	0.96	1.0

(i) Plot the equilibrium curve.

(ii) Draw the q-line through the point (0.44, 0.44) on the diagonal and with an angle of 75° which will cut the equilibrium curve at point P. Read the coordinates of the point P.

The minimum reflux is given by the slope of line AP (not shown in Fig. E 1.9 (a)).

R_m for cold liquid is given by

$$R_m = \frac{x_D - y'}{y' - x'}$$

From the plot, the coordinates of the point P are $y' = 0.73$, $x' = 0.515$ and we have, $x_D = 0.9744$.

$$\therefore \quad R_m = \frac{0.9744 - 0.73}{0.73 - 0.515} = 1.1367$$

Operating reflux ratio $= R = 3\,R_m$ (given)

$$R = 3 \times 1.1367 = 3.41$$

Intercept of the rectification section operating line (the top operating line) on y-axis is

$$\text{Intercept} = x_D/(R+1) = 0.9744/(3.41+1) = 0.221$$

(iii) Draw the operating line of the rectifying section through the point (0.9744, 0.9744) on the diagonal and with an intercept of 0.221 on y-axis.

(iv) Draw the stripping section operating line through the point (0.0235, 0.0235) on the diagonal and passing through the point of intersection of the q-line and the top operating line.

(v) Construct the triangles (representing theoretical plates) between the equilibrium curve and the top operating line starting from the point (0.9744, 0.9744) on the diagonal upto $x_P = 0.44$ and then do the construction between the equilibrium curve and the operating line of the stripping section reaching exactly or covering the point (0.0235, 0.0235) on the diagonal. Count the triangles, i.e., theoretical plates.

Number of theoretical plates including reboiler = n = 11.

Number of theoretical plates required in the column = n – 1 = 11 – 1 = 10.

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Overall plate efficiency = 60%

Real/Actual plates required = 10/0.60

= 16.67 ≈ 17 plates ... Ans.

$$\left[\% \text{ efficiency} = \frac{\text{theoretical plates}}{\text{actual plates}} \times 100\right]$$

There are chances of variation of theoretical plates upto 5% due to errors while plotting and should be taken into account.

Enlarged scales are used for both x and y and steps are constructed.

From the Fig. E 1.9 (a),

$y' = 0.7266 \approx 0.73$, $x' = 0.515$

and number of theoretical plates required = n – 1 = 11 – 1 = 10.

Actual plates = 10/0.6 = 16.67 = 17 plates.

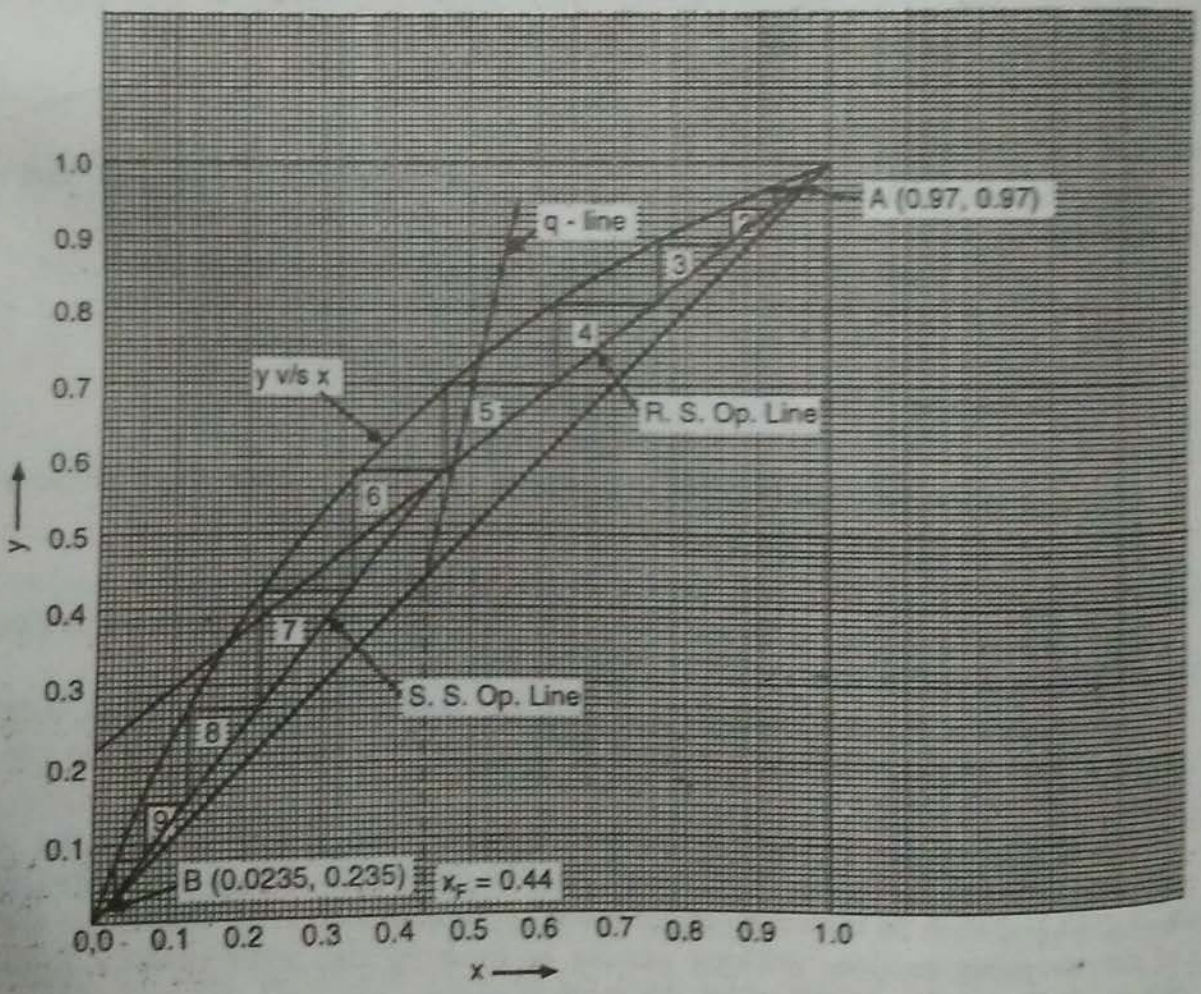

Fig. E 1.9 (a) : Construction of ideal stages

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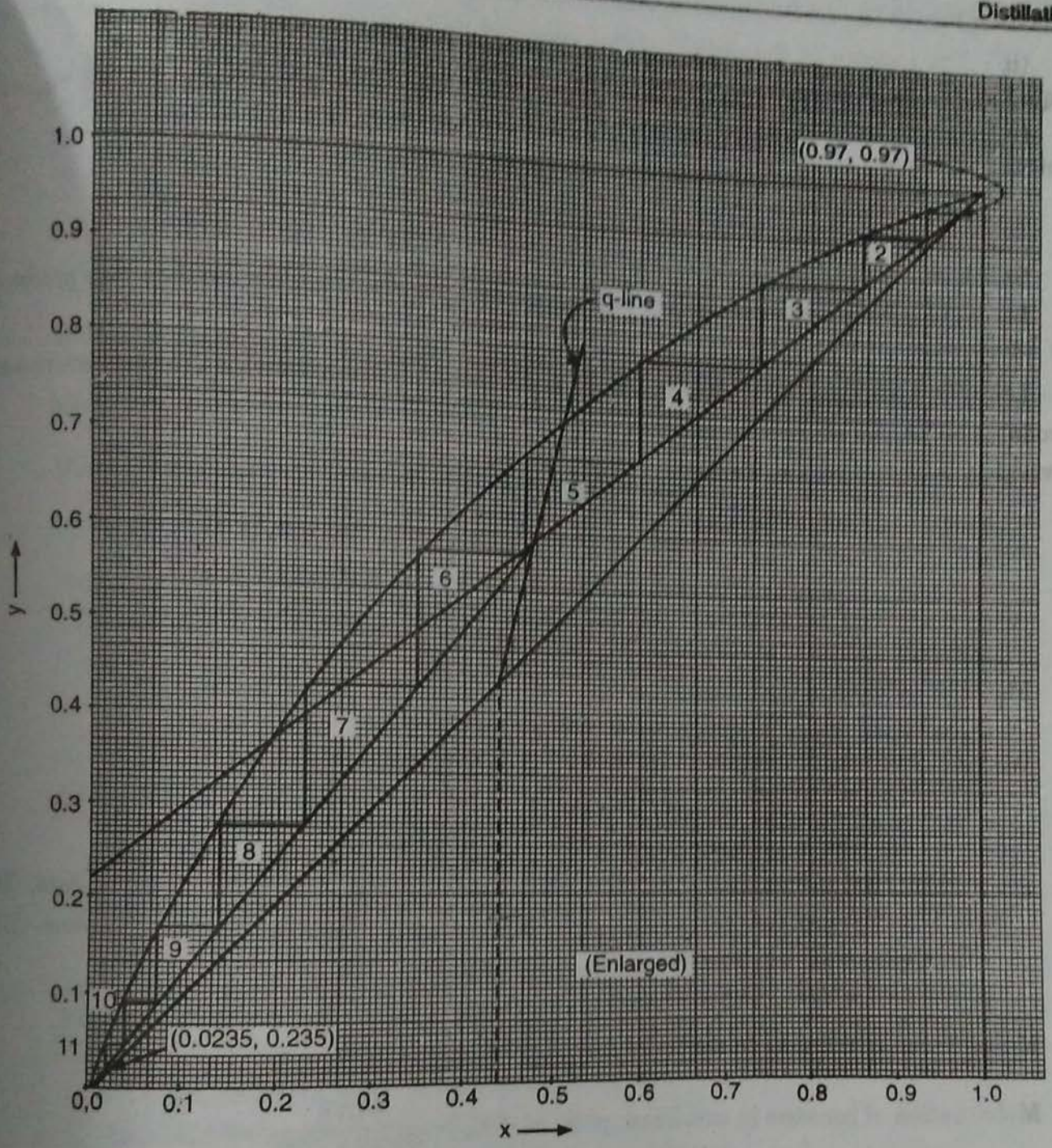

Fig. E 1.9 (b) : Construction of ideal stages (Enlarged scale)

Ex. 1.10 : A continuous fractionating column is used to handle 2.5 kg/s of feed containing 40% benzene and 60% toluene. The overhead product contains 97% benzene and bottom product contains 98% toluene. A reflux ratio of 3.5 kmol/kmol is used. Molar latent heat of benzene and toluene may be taken as 30000 kJ/kmol. Take relative volatility = 2.5. Calculate :

(i) The mass flow rates of overhead and bottom product.

(ii) Steam requirement. The steam is available at 240 kPa pressure and latent heat of steam at 240 kPa is 2186 kJ/kg.

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(iii) The number of theoretical plates required and position of feed plate if the feed is liquid at 295 K and its specific heat is 1.84 kJ/(kg.K).

(iv) The diameter of column if column operates at atmospheric pressure and vapour velocity is 1 m/s.

Boiling point of benzene = 353.3 K and

Boiling point of toluene = 383.8 K.

Solution : Basis : 2.5 kg/s of feed to column.

Let F', D' and W' be the mass flow rate of the feed, overhead product and bottom product in kg/s respectively.

Let x'_F, x'_D and x'_W be the mass fraction (weight fraction) of benzene in the feed, overhead product and bottom product respectively.

Overall material balance :

$$F' = D' + W'$$

$$2.5 = D' + W' \qquad \therefore \; W' = 2.5 - D'$$

$$x'_F = 40/100 = 0.40, \quad x'_D = 97.100 = 0.97$$

$$x'_W = (100 - 98)/100 = 0.02$$

Material balance of benzene :

$$0.4\,(2.5) = 0.97\,D' + 0.02\,W'$$

We have : $W' = 2.5 - D'$

$$1 = 0.97\,D' + 0.02\,(2.5 - D')$$

$$D' = 1 \text{ kg/s}$$

$$W' = 2.5 - 1 = 1.5 \text{ kg/s}$$

Mass flow rate of distillate = 1 kg/s ... Ans. (i)

Mass flow rate of bottom product = 1.5 kg/s ... Ans. (ii)

Molecular weight of benzene = 78

Molecular weight of toluene = 92.

$$\text{Molefraction of benzene in feed} = x_F = \frac{40/78}{40/78 + 60/92} = 0.44$$

$$\text{Molefraction of benzene in overhead product} = x_D = \frac{97/78}{97/78 + 3/92} = 0.974$$

$$\text{Molefraction of benzene in bottom product} = x_W = \frac{2/78}{98/92 + 2/78} = 0.0235$$

Average molecular weight of feed = $M_{avg.}$

$$M_{avg.} = 0.44\,(78) + 0.56\,(92) = 85.84 \text{ kg/kmol}$$

Mass flow rate of feed = 2.5 kg/s

Molar flow rate of feed = 2.5/85.84 = 0.0291 kmol/s

Overall material balance :

$$F = D + W$$

$$0.0291 = D + W, \qquad \therefore \; W = 0.0291 - D$$

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Material balance of benzene :

$$x_F F = x_D D + x_W W$$
$$0.44\,(0.0291) = 0.974\,D + 0.0235\,W$$
$$0.44\,(0.0291) = 0.974\,D + 0.0235\,(0.0291 - D)$$
$$\therefore \quad D = 0.01275 \text{ kmol/s and } W = 0.01635 \text{ kmol/s}$$
$$R = 3.5 \text{ (given)}$$
$$R = L_n/D$$
$$L_n = RD = 3.5 \times 0.01275 = 0.0446 \text{ kmol/s}$$
$$V_{n+1} = L_n + D = 0.0446 + 0.01275 = 0.05735 \text{ kmol/s}$$

Boiling point of benzene = 353.3 K and that of toluene = 383.8 K.

Mean/average boiling point of feed = $\sum_{i=1}^{n}$ (mole fraction × boiling point)

Mean boiling point of feed = $T_b = 0.44\,(353.3) + 0.56\,(383.8) = 370.38$ K

Temperature of feed = $T_f = 295$ K

Specific heat of feed = 1.84 kJ/(kg.K)

Molar specific heat of feed = $1.84 \times 85.84 = 157.95$ kJ/(kmol.K)

λ for benzene and toluene = 30000 kJ/kmol

$$q = \frac{C_p\,(T_b - T_f) + \lambda}{\lambda}$$
$$= \frac{157.95\,(370.38 - 295) + 30000}{30000} = 1.4$$

As the feed is cold, $q = 1.4$

$$L_m = L_n + qF \qquad \text{(below the feed plate)}$$
$$L_m = 0.0446 + 1.4\,(0.0291) = 0.08534 \text{ kmol/s}$$
$$L_m = V_{m+1} + W$$
$$V_{m+1} = L_m - W$$
$$= 0.08534 - 0.01635 = 0.06899 = 0.069 \text{ kmol/s}$$

Boilup rate/vapour rate at the bottom of the column = V_{m+1}

Vapour rate = 0.069 kmol/s

Molar latent heat = 30000 kJ/kmol

Heat load of the reboiler = 0.069×30000

= 2070 kJ/s

Steam at 240 kPa is used to meet the heat load of the reboiler. λ_s of steam = 2186 kJ/kg.

$$\dot{m}_s \lambda_s = \text{Heat load of the reboiler}$$
$$\dot{m}_s\,(2186) = 2070$$
$$\dot{m}_s = 0.9469 \approx 0.95 \text{ kg/s}$$

Steam rate required/steam consumption = 0.95 kg/s ... **Ans. (ii)**

Generate the equilibrium data.

$$\alpha = 2.5$$
$$y = \frac{\alpha x}{1 + (\alpha - 1)\,x}$$

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Take x = 0, 0.1, 0.2 ... 1.0 and evaluate the corresponding values of y and tabulate.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.22	0.385	0.52	0.625	0.714	0.79	0.854	0.91	0.96	1.0

(i) Plot the equilibrium curve.

(ii) Slope of q-line :

$$q = 1.4$$

$$\text{slope} = \frac{-q}{1-q} = \frac{-1.4}{1-1.4} = 3.5$$

$$\text{slope} = \tan\theta = 3.5 \quad \therefore \quad \theta = 74^\circ$$

Draw the q-line through the point (0.44, 0.44) on the diagonal with an angle of 74°.

(iii) Intercept of rectification section operating line is :

$$\text{Intercept} = x_D/(R+1)$$
$$= 0.974/(3.5+1) = 0.216$$

Draw this operating line through the point (0.974, 0.974) on the diagonal with an intercept of 0.216 on y-axis.

(iv) Draw the operating line of stripping section through the point (0.0235, 0.0235) on the diagonal and passing through the point of intersection of q-line and top operating line.

(v) Construct the stages.

Number of theoretical plates including reboiler = n = 11.

Number of theoretical plates required = n – 1 = 11 – 1 = 10 ... **Ans.** (iii)

Position of feed plate : 5th from top. ... **Ans.** (ii)

Vapour rate at the top of the column = V_{n+1} = 0.05735 kmol/s.

At the top, benzene is almost pure, so we can assume that the temperature at the top is 353.3 K at 1 atm pressure.

1 kmol of gas at 1 atm pressure and 273 K temperature occupies volume of 22.4 m^3.

$$\text{Column pressure} = 1 \text{ atm}$$

$$P_1V_1 = n_1RT_1$$

$$P_2V_2 = n_2RT_2$$

$$\frac{P_1V_1}{P_2V_2} = \frac{n_1}{n_2}\frac{T_1}{T_2}$$

$$V_1 = \frac{V_2}{n_2} \cdot n_1 \cdot \frac{T_1}{T_2} \cdot \frac{P_1}{P_2}$$

$$V_2 = 22.4 \text{ m}^3$$

$$n = 1 \text{ kmol}$$

$$T_2 = 273 \text{ K}$$

$$P_1 = P_2 = 1 \text{ atm}$$

$$T_1 = 353.3 \text{ K}$$

$$n_1 = 0.05735 \text{ kmol}$$

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Volumetric flow rate of vapour at top $= \frac{22.4}{1} \times 0.05735 \times \frac{353.3}{273} \times \frac{1}{1} = 1.6625 \text{ m}^3/\text{s}.$

Vapour velocity = 1 m/s

Volumetric flow rate = cross-sectional area (A) × velocity

$$A = \frac{1.6625}{1} = 1.6625 \text{ m}^2$$

$$A = \frac{\pi}{4} D_t^2 = 1.6625$$

$$D_t = 1.455 \approx 1.46 \text{ m (based on vapour flow at the top).}$$

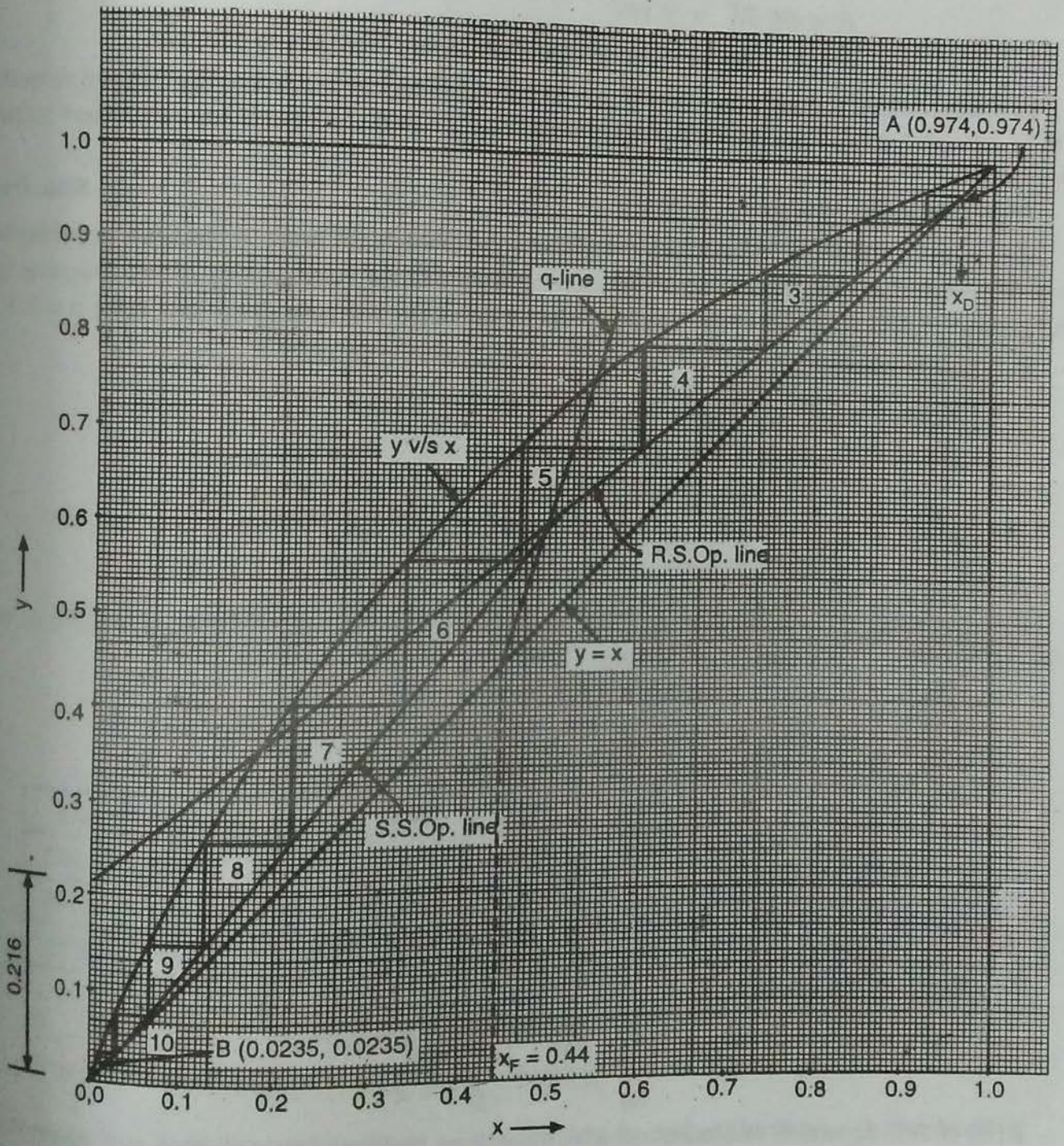

Fig. E 1.10 : Determination of theoretical plates

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Temperature at bottom of column is assumed to be 383.8 K.

$$V_{m+1} = 0.069 \text{ kmol/s}$$

Vapour rate at the bottom = 0.069 kmol/s

$$\text{Volumetric flow rate at the bottom} = \frac{22.4}{1} \times 0.069 \times \frac{383.8}{273} \times \frac{1}{1}$$

$$= 2.173 \text{ m}^3/\text{s}$$

Vapour velocity = 1 m/s

Volumetric flow rate = Area × Velocity

$$2.173 = A \times 1 \quad \therefore \quad A = 2.173 \text{ m}^2$$

$$A = \pi/4 \; D_t^2 = 2.173$$

$$D_t = 1.66 \text{ m (based on vapour flow at the bottom)}$$

In order to limit the vapour velocity through the column to 1 m/s (i.e. the vapour velocity not to exceed 1 m/s), the column diameter has to be 1.66 m (has to be the larger value obtained).

Diameter of the column = 1.66 m ... **Ans. (iv)**

Ex. 1.11 : 100 moles of benzene and toluene containing 50 mole % benzene is subjected to a differential distillation at atmospheric pressure till the composition of the benzene in the residue is 33% by mole. Calculate the total moles of the mixture distilled. Take $\alpha = 2.4$.

Solution : Basis : 100 kmol of feed.

Molefraction of benzene in feed $= x_F = 50/100 = 0.50$

Molefraction of benzene in residue $= x_W = 33/100 = 0.33$

$$\alpha = 2.4$$

We have, $$y = \frac{\alpha x}{1 + (\alpha - 1)\, x}$$

Generate the equilibrium data.

For this example, we require equilibrium data starting from $x_F = 0.5$ to $x_W = 0.33$, so we will generate the data between $x = 0.3$ to $x = 0.6$.

Take $x = 0.3$

$$y = \frac{2.4 \times 0.3}{1 + (2.4 - 1)\, 0.3} = 0.507$$

x	0.3	0.35	0.4	0.45	0.5	0.55	0.60	0.2
y	0.507	0.564	0.615	0.662	0.706	0.746	0.783	0.375
(y – x)	0.207	0.214	0.215	0.212	0.206	0.196	0.183	0.175
1/(y – x)	4.83	4.67	4.65	4.72	4.85	5.1	5.46	5.71

The Rayleigh equation is

$$\ln (F/W) = \int_{x_W}^{x_F} \frac{dx}{y - x} = \int_{0.33}^{0.5} \frac{dx}{y - x}$$, y is in equilibrium with x.

RHS of this equation is evaluated graphically by plotting $1/(y - x)$ v/s x and measuring the area under the curve between the limits $x = x_F = 0.5$ and $x = x_W = 0.33$.

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From Fig. E 1.11 :

Area under the curve between x = 0.50 and x = 0.33 is 788 mm².

Area under the curve = 7.88 cm²

$$\ln \frac{F}{W} = \int_{0.33}^{0.5} \frac{dx}{y - x}$$

= Area × (x-axis scale) × (y-axis scale)

$$= 7.88 \times \frac{0.1}{1} \times \frac{1}{1} = 0.788$$

$$\therefore \quad \frac{F}{W} = e^{0.788}$$

$$\frac{F}{W} = 2.199 = 2.2$$

$$W = F/2.2$$

but $F = 100$ kmol

$\therefore$ $W = 100/2.2 = 45.45$ kmol

Overall material balance :

$$F = D + W$$

$$D = F - W = 100 - 45.45 = 54.55 \text{ kmol}$$

Moles of mixture distilled = 54.55 kmol ... Ans.

(i.e., 54.55 mole % of feed is distilled).

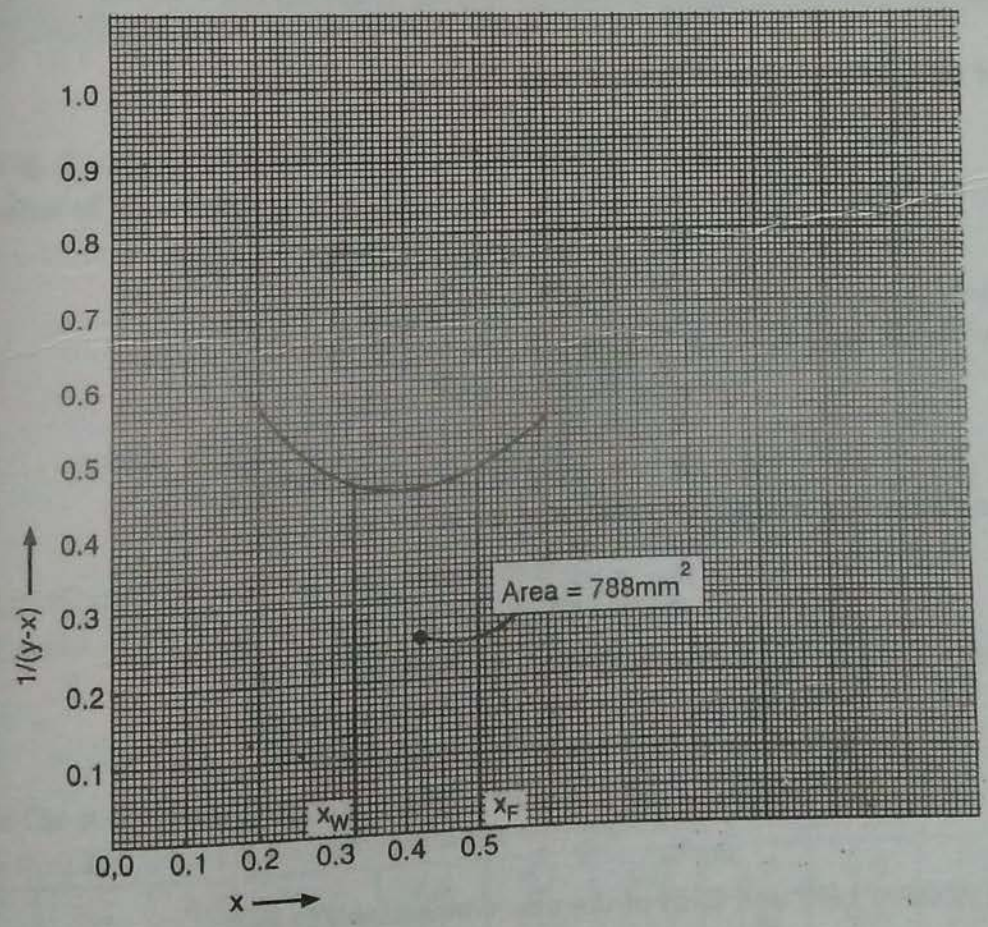

Fig. E 1.11 : Graphical evaluation of integral of the Rayleigh Equation

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Ex. 1.12 : A binary distillation column is operating at the following conditions :

Feed rate = 150 kmol/h, all vapour.

Distillate rate = 100 kmol/h with 98 mole % volatile component.

Reboiler steam = 4500 kg/h.

Latent heat of steam used = 2200 J/g.

Latent heat of column liquid = 30000 J/mol.

(i) Calculate the operating reflux ratio.

(ii) If for the above column liquid, the relative volatility is 2.2 and feed contains 50 mole % of more volatile component, how many stages are present in the rectification section ?

Solution : Basis : 150 kmol/h of feed to column.

Let, V – molar flow rate of vapour in rectifying section, kmol/h

V' – molar flow rate of vapour in stripping section, kmol/h

F – feed to the column

q – fraction of feed that is liquid.

The flow rate of vapour in the rectifying section is given by

$$V = V' + (1 - q) F$$

$$\therefore \quad V' = V - (1 - q) F$$

Flow rate of vapour in the stripping section, i.e., from the reboiler :

$$V' = V - (1 - q) F$$

For rectifying section at the top, we have :

$$V = L + D$$

$$V/D = L/D + 1$$

$$V = (R + 1) D, \text{ where } R = L/D$$

$$\therefore \quad V' = (R + 1) D - (1 - q) F$$

Given : $F = 150$ kmol/h

Feed is all vapour, so $q = 0$.

$$D = 100 \text{ kmol/h}$$

$$V' = (R + 1) 100 - (1 - 0) 150$$

$$V' = 100 (R + 1) - 150 \text{ kmol/h}$$

The steam required at the reboiler is given by

$$\dot{m}_s = \frac{V' \lambda}{\lambda_s}$$

where $\dot{m}_s$ is the steam consumption, kg/h

V' is the vapour rate from the reboiler, kmol/h

λ_s is the latent heat of steam used, kJ/kg

λ is the latent heat of liquid mixture, kJ/kmol

$$\dot{m}_s \lambda_s = V' \lambda$$

The units of LHS and RHS of the above equation are kJ/h.

$$\dot{m}_s \text{ (kg/h) } \lambda_s \text{ (kJ/kg)} = V' \text{ (kmol/h) } \lambda \text{ (kJ/kmol)}$$

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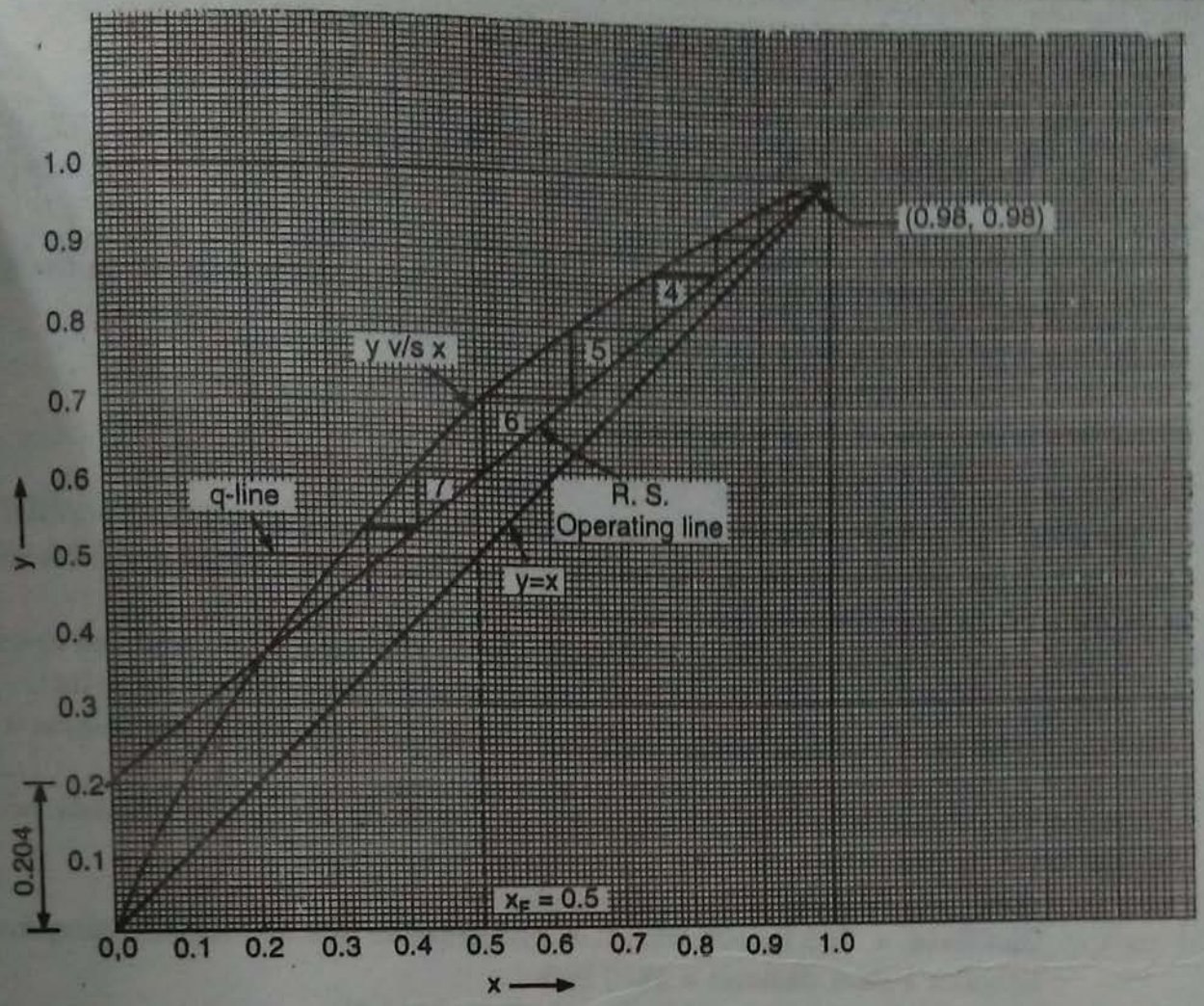

Fig. E 1.12 : Determination of ideal stages in the rectification section

Putting value of V', we get,

$$\dot{m}_s \lambda_s = [100 (R + 1) - 150] \lambda$$

$$\dot{m}_s = 4500 \text{ kg/h}$$

$$\lambda_s = 2200 \text{ J/g} = 2200 \text{ kJ/kg}$$

$$\lambda = 30000 \text{ J/mol} = 30000 \text{ kJ/kmol}$$

$$4500 \times 2200 = [100 (R + 1) - 150] \times 30000$$

$$330 = 100 (R + 1) - 150$$

$$\therefore \quad R = 3.8$$

Operating reflux ratio = R = 3.8 kmol/kmol ... **Ans. (i)**

Determination of number of stages in rectifying section :

$$\alpha = 2.2 \text{ given}$$

$$y = \frac{\alpha x}{1 + (\alpha - 1) x}$$

Generate the x – y data.

Take x = 0, 0.1, 0.2 ... 1.0 and calculate the corresponding y values.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.2	0.355	0.485	0.59	0.69	0.77	0.84	0.9	0.95	1.0

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$$x_D = 0.98, \quad x_F = 0.5$$
$$q = 0$$

As q = 0, the feed line (q-line) is parallel to x-axis through the point (0.5, 0.5) on the diagonal.

$$x_D = 0.98$$

So the operating line of the rectifying section has the point (0.98, 0.98) on the diagonal.

Intercept of the operating line on y-axis.

$$\text{Intercept} = x_D/(R + 1)$$
$$= 0.98/3.8 + 1$$
$$= 0.204$$

So draw the q-line through the point (0.5, 0.5) on the diagonal and parallel to x-axis (as q = 0). Draw the operating line for rectifying section through the point (0.98, 0.98) on the diagonal and having intercept on y-axis = 0.204.

Construct the equilibrium stages between the equilibrium curve and the operating line till we reach exactly or just cross the feed line. Count the stages above the q-line.

Number of theoretical stages (plates) required in the rectifying section = 7.

... **Ans.** (ii)

Ex. 1.13 : The following information is available from the records of a binary fractionating column :

Feed = 180 kmol/h and 60% vaporised.

Distillate = 100 kmol/h with 0.98 molefraction of more volatile component.

Reboiler steam demand = 4200 kg/h.

Latent heat of steam used in the reboiler = 2200 J/g.

Latent heat of column liquid = 3×10^4 J/mol.

Calculate : (i) The operating reflux ratio.

(ii) The composition of vapour entering that plate from which the liquid overflow contains 0.7 molefraction of the more volatile component.

Solution : Basis : 180 kmol/h of feed to column.

V – molar flow rate of vapour in rectifying section

V' – molar flow rate of vapour in stripping section

F – feed rate to the column

q – fraction of feed that is liquid.

The flow rate of vapour in the rectifying section is given by,

$$V = V' + (1 - q) F$$
$$\therefore \quad V' = V - (1 - q) F$$

We have, $V = L + D$, for rectifying section at the top

$$V/D = L/D + 1$$
$$\therefore \quad V = (R + 1) D$$
$$\therefore \quad V' = (R + 1) D - (1 - q) F$$

The feed is 60% vaporised (given).

i.e., feed is 40% liquid + 60% vapour, so q = fraction of the feed that is liquid = 0.40.

$$D = 100 \text{ kmol/h}, \quad F = 180 \text{ kmol/h}$$
$$V' = (R + 1) 100 - (1 - 0.4) 180$$
$$V' = 100 (R + 1) - 108 \text{ kmol/h}$$

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The steam required at the reboiler is given by,

$$\dot{m}_s = \frac{V'\lambda}{\lambda_s}$$

where $\dot{m}_s$ is the steam consumption, kg/h

V' is the vapour rate from the reboiler, kmol/h

λ_s is the latent heat of steam, kJ/kg

λ is the latent heat of liquid mixture, kJ/kmol.

$$\dot{m}_s \lambda_s = V'\lambda$$

The units of RHS and LHS are kJ/h.

$$\dot{m}_S \text{ (kg/h)} \; \lambda_S \text{ (kJ/kg)} = V' \text{ (kmol/h)} \; \lambda \text{ (kJ/kmol)}$$

Putting the value of V', we get,

$$\dot{m}_s \lambda_s = [100 (R + 1) - 108] \lambda$$

$\dot{m}_s = 4200$ kg/h

$\lambda_s = 2200$ J/g $= 2200$ kJ/kg

$\lambda = 3 \times 10^4$ J/mol $= 3 \times 10^4$ kJ/kmol

$$4200 \times 2200 = [100 (R + 1) - 108] \times 3 \times 10^4$$

$$\therefore \quad R = 3.16$$

Operating reflux ratio = 3.16 ... Ans. (i)

OR

Vapour flow rate from reboiler (in stripping section) = V'

$$V' = \frac{\dot{m}_s \lambda_s}{\lambda} = \frac{4200 \times 2200}{3 \times 10^4} = 308 \text{ kmol/h}$$

Feed is 60% vaporised.

i.e. Feed is 60% vapour and 40% liquid.

Feed as a vapour $= 0.6 \times 180 = 108$ kmol/h

Vapour flow rate in rectifying section = V' + feed as a vapour

$V = 308 + 108$

$V = 416$ kmol/h

At the top of rectifying section, we have

$V = L + D$

$\therefore \quad L = V - D$

$= 416 - 100 = 316$ kmol/h (reflux liquid)

$$\text{Operating reflux ratio} = \frac{L}{D} = \frac{316}{100} = 3.16 \text{ kmol/kmol}$$ ... Ans. (i)

Feed as a liquid $= 0.4 \times 180 = 72$ kmol/h

Liquid reflux to the column = 316 kmol/h

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Liquid flow in stripping section $= L' = 316 + 72$

$$L' = L + qF = 316 + 72 = 388 \text{ kmol/h}$$

In the stripping section, we have :

$$L' = V' + W$$

$$W = L' - V'$$

$$W = 388 - 308 = 80 \text{ kmol/h} \quad \ldots (1)$$

We have, $F = D + W \quad \ldots (2)$

$$W = F - D = 180 - 100 = 80 \text{ kmol/h}$$

W from (1) and (2) is same.

So the calculations done are correct.

(ii) Calculation of vapour composition :

Composition of liquid that overflows, say plate-n $= x_n = 0.7$

Composition of vapour stream entering plate-n $= y_{n+1}$.

The operating line for the rectifying section is :

$$y_{n+1} = \frac{R}{R+1} x_n + \frac{x_D}{R+1}$$

We have, $x_n = 0.7, \quad R = 3.16, \quad x_D = 0.98$

$$y_{n+1} = \frac{3.16}{(3.16+1)} (0.7) + \frac{0.98}{(3.16+1)} = 0.767 \simeq 0.77$$

Composition of vapour entering that plate from which liquid overflow contains 0.7 molefraction of more volatile component $= 0.77$ molefraction of more volatile component

... **Ans.** (ii)

Ex. 1.14 : A fractionating column operating at 1 atm pressure is fed with saturated liquid feed containing 40 mole % ethanol and 60 mole % water. The column produces saturated liquid overhead product containing 80 mole % ethanol and a saturated liquid bottom product containing 20 mole % ethanol. The reflux ratio is 2 kmol/kmol.

(i) Determine the number of theoretical stages required for given separation and feed plate location.

(ii) The heat load of the reboiler in kJ/kmol of bottom product.

(iii) The heat load in the condenser in kJ/kmol of overhead product.

Data : Relative volatility = 2.2

Latent heat of ethanol = 90523 kJ/kmol

Latent heat of water = 39718 kJ/kmol

Solution : Basis : 1 kmol of feed.

Molefraction of ethanol in feed $= x_F = 40/100 = 0.40$

$x_D = 80/100 = 0.80$ (overhead composition)

$x_W = 20/100 = 0.20$ (bottom product composition)

Overall material balance :

We have, $F = D + W$

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Material balance of ethanol :

$$F x_F = D x_D + W x_W$$
$$F x_F = D x_D + (F - D) x_W$$
$$1 \times 0.4 = D \times 0.8 + (1 - D) \times 0.20$$
$$\therefore \quad D = 0.3333 \text{ kmol}$$
$$W = 1 - 0.3333 = 0.6667 \text{ kmol}$$

Moles of overhead product = 0.3333 kmol

Moles of bottom product = 0.6667 kmol

Feed is a saturated liquid, so q = 1.0.

V – vapour flow in rectifying section

V' – vapour flow in stripping section

$$V = V' + (1 - q) F$$

The vapour flow in the rectifying section is given by,

$$V = V' + (1 - q) F$$

For saturated liquid feed, q = 1.0.

$$V = V' + (1 - 1) F$$
$$\therefore \quad V = V'$$

At top of the column, we have,

$$V = L + D$$
$$V/D = L/D + 1$$
$$V = (R + 1) D$$

R = 2 and D = 0.3333 kmol

$$V = (2 + 1) \times 0.3333$$
$$= 1 \text{ kmol}$$

Vapour flow to condenser = V = 1 kmol

Vapour flow from reboiler = V' = V = 1 kmol

Overhead product contains 80% ethanol and 20% water (molar basis)

λ_{water} = 39718 kJ/kmol

$\lambda_{ethanol}$ = 90523 kJ/kmol

Mean latent heat of condensation of overhead product vapour = $\sum_{i=1}^{n} x_i \lambda_i$

$$= 0.2 (39718) + 0.8 (90523)$$
$$= 80362 \text{ kJ/kmol}$$

Overhead product is a saturated liquid, so only the latent heat is removed in the condenser.

Heat load in condenser = $V \lambda_D$.

where, V = vapour flow to condenser = 1 kmol

λ_D = mean latent heat of overhead vapour = 80362 kJ/kmol

Heat load in the condenser = Heat removed in the condenser

$$= V \lambda_D$$
$$= 1 \times 80362 = 80362 \text{ kJ}$$

Overhead product/distillate = 0.3333 kmol.

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Required : Heat load in condenser is to be expressed as kJ per kmol of overhead product.

Heat load in condenser per kmol of overhead product $= \frac{80362}{0.3333} = 241110$ kJ ... Ans. (iii)

Vapour flow from reboiler $= V' = 1$ kmol.

Bottom product contains 20 mole % ethanol and 80 mole % water.

Mean latent heat of vaporisation of bottom liquid $\lambda_B = \sum_{i=1}^{n} x_i \lambda_i = 0.2\,(90523) + 0.80\,(39718)$

$\lambda_B = 49879$ kJ/kmol

Heat load of the reboiler $= V' \lambda_B$

$= 1 \times 49879$

$= 49879$ kJ

Moles of bottom product $= 0.6667$ kmol

Heat load of the reboiler per 1 kmol bottom product $= \frac{49879}{0.6667} = 74815$ kJ ... Ans. (ii)

Mean latent heat of liquid/vapour $= \sum_{i=1}^{n} x_i \lambda_i$

x_i is the molefraction of i^{th} component.

λ_i is the latent heat of i^{th} component.

Determination of number of theoretical stages :

$x_F = 0.40$, $x_D = 0.80$, $x_W = 0.20$

Feed is a saturated liquid $\therefore$ $q = 1$

Relative volatility $= 2.2$

$$y = \frac{\alpha x}{1 + (\alpha - 1)\, x}$$

Take $x = 0, 0.1, 0.2 \ldots 1.0$ and evaluate the corresponding y values.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.2	0.355	0.485	0.59	0.69	0.77	0.84	0.9	0.95	1.0

Draw the q-line through the point (0.4, 0.4) on the diagonal and parallel to y-axis as $q = 1$. Locate the point (x_D, x_D), i.e., the point (0.8, 0.8) on the diagonal as well as the point (x_W, x_W), i.e., point (0.2, 0.2) on the diagonal.

Intercept of the operating line of rectifying section on y-axis :

Intercept $= x_D/R + 1$

$= 0.8/2 + 1 = 0.267$

Draw the operating line for the rectifying section through the point (0.8, 0.8) on the diagonal and having intercept of 0.267 on the y-axis. Draw the operating line for the stripping section through the point (0.2, 0.2) on the diagonal and passing through the point of intersection of the q-line and the operating line of rectifying section.

Theoretical stages are constructed between the equilibrium curve and the operating lines starting from $x_D = 0.8$ and reaching exactly or covering $x_W = 0.2$.

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From plot :

Number of theoretical stages including reboiler = n = 6.

Number of theoretical stages/plates required = n – 1 = 6 – 1 = 5

Feed plate location = 4th from top. ... **Ans. (i)**

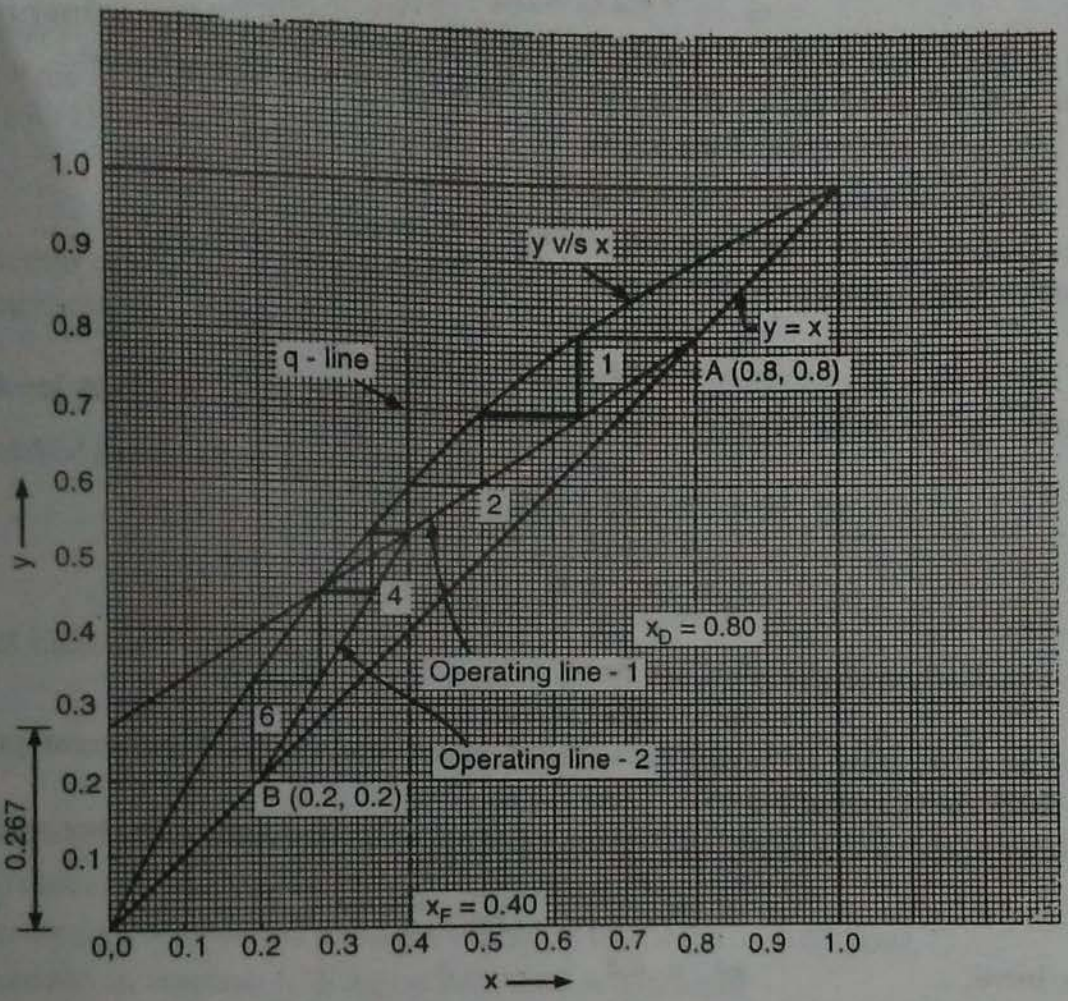

Fig. E 1.14 : Determination of the number of ideal plates

Ex. 1.15 : A feed containing 40 mole % benzene and 60 mole % toluene is to be distilled at atmospheric pressure to recover 95% of the original benzene.

Calculate : (i) the mole % of the mixture which should be distilled and (ii) the composition of the distillate obtained if the distillated is carried out by

(a) simple equilibrium distillation/flash distillation,

(b) differential distillation collecting all the distillate together.

Take relative volatility = 2.5.

Solution : Basis : 1 kmol of feed.

(a) **Flash distillation :**

Molefraction of benzene in feed = x_F = 40/100 = 0.4.

Moles of benzene in feed = 0.4 × 1 = 0.4 kmol

Benzene recovered = 95% of its original value.

Benzene recovered (in vapour) = 0.95 × 0.4 = 0.38 kmol

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If f is the molar fraction of feed converted to vapour (fractional vaporisation) then the material balance of benzene is

$$x_F F = f F y + (1 - f) F x$$

$$\therefore \quad x_F = f y + (1 - f) x$$

where y is the composition of vapour in equilibrium with liquid (residue) in the flash separator. x is the composition of the residue (liquid) in the flash separator.

Benzene in the feed = 0.4 kmol.

Benzene in the vapour (recovered) = 0.38 kmol

$\therefore$ Benzene in the residue = 0.4 – 0.38 = 0.02 kmol

Benzene in the vapour recovered = $f \cdot y \cdot F$ = 0.38 kmol

but $F = 1$ kmol

$\therefore$ $fy = 0.38$ kmol

Benzene in the liquid residue in the flash separator = $(1 - f) x \cdot F$ = 0.02 kmol

but $F = 1$ kmol

$\therefore$ $(1 - f) x = 0.02$ kmol

$$x = 0.02/(1 - f)$$

or $$f = \frac{x - 0.02}{x}$$

For equilibrium distillation, we have

$$x_F = fy + (1 - f) x \qquad \ldots (A)$$

To find f, we have to assume a value of x and adopt a trial-and-error procedure till LHS = RHS for the equation (A).

y value in the above equation (A) is in equilibrium with x and to be obtained from :

$$y = \frac{\alpha x}{1 + (\alpha - 1) x}, \quad \alpha = 2.5$$

We have, $x_F = 0.4$

$\therefore$ $fy + (1 - f) x = 0.4$ ... (B)

Trial-1 : Take $x = 0.30$, $f = \frac{0.3 - 0.02}{0.3} = 0.9333$, $y = \frac{2.5 \times 0.3}{1 + (2.5 - 1) 0.3} = 0.5172$

$$fy + (1 - f) x = 0.9333 (0.5172) + (1 - 0.9333) \times 0.3$$

$$= 0.5027 \neq 0.40 \text{ [see Equation (B)]}$$

x	$f = \frac{x - 0.02}{x}$	$y = \frac{\alpha x}{1 + (\alpha - 1) x}$	$fy + (1 - f) x$ (has to be 0.40)
0.30	0.9333	0.5172	0.5027
0.20	0.90	0.385	0.3665
0.22	0.9091	0.4135	0.3959
0.23	0.9130	0.4275	0.4103
0.215	0.9070	0.4064	0.3885
0.222	0.91	0.4163	0.3988
0.223	**0.9103**	**0.4177**	**0.4002 (Desired)**

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∴ From the trial and error procedure performed above :

$x = 0.223, \quad y = 0.4177$

$f = 0.9103$

As $f = 0.9103$

which is the fraction of feed that is converted to vapour.

Mole % of the feed-liquid mixture distilled i.e., mole % of feed vaporised $= 100\,f$

$= 100 \times 0.9103$

$= 91.03$ ... **Ans. (a)**

Molefraction of benzene in the vapour, i.e., in the distillate from the separator = 0.4177

Vapour composition/distillate composition :

Mole % benzene $= 0.4177 \times 100 = 41.77$

Mole % toluene $= 100 - 41.77 = 58.33$ ... **Ans. (a)**

(b) Differential distillation :

$F = 1$ kmol of feed

$F = D + W$

$W = F - D = 1 - D$ kmol

Material balance of benzene :

$x_F F = D\, y_{D\,avg} + W\, x_W$

Let $y_{D\,avg} = x_D$ for convenience

$x_F F = D\, x_D + W\, x_W$

Moles of benzene in feed $= 0.4 \times 1 = 0.4$ kmol

% recovery of benzene = 95

Benzene in composited distillate $= D\, x_D = 0.95 \times 0.4 = 0.38$ kmol

Benzene in residue $= W\, x_W = 0.4 - 0.38 = 0.02$ kmol

$\alpha = 2.5$ (given)

For constant relative volatility for simple distillation, we have :

$$\ln \frac{F\,x_F}{W\,x_W} = \alpha \ln \frac{F\,(1 - x_F)}{W\,(1 - x_W)}$$

$$\log \frac{F\,x_F}{W\,x_W} = \alpha \log \frac{F\,(1 - x_F)}{W\,(1 - x_W)}$$

$$\log \frac{1 \times 0.4}{0.02} = 2.5 \log \frac{F\,(1 - x_F)}{W\,(1 - x_W)}$$

$$\log \frac{F\,(1 - x_F)}{W\,(1 - x_W)} = \frac{\log (0.4/0.02)}{2.5}$$

$$= 0.5204$$

$$\log F\,(1 - x_F) - \log W\,(1 - x_W) = 0.5204$$

$$\log W\,(1 - x_W) = \log 1 \times (1 - 0.40) - 0.5204$$

$$= -0.2218 - 0.5204 = -0.7422$$

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We have, $W x_W = 0.02 \quad \therefore W = 0.02/x_W$

$$\log\left[\frac{0.02}{x_W}(1 - x_W)\right] = -0.7422$$

$$\log\left[\frac{0.02 - 0.02\,x_W}{x_W}\right] = -0.7422$$

$$\frac{0.02 - 0.02\,x_W}{x_W} = 10^{-0.7422}$$

$$\frac{0.02 - 0.02\,x_W}{x_W} = 0.181$$

$$0.181\,x_W = 0.02 - 0.02\,x_W$$

$$x_W = 0.0995 \approx 0.10$$

We have, $W x_W = 0.02$

$$W = 0.02/x_W$$

$$= 0.02/0.10 = 0.2 \text{ kmol}$$

$$F = D + W = 1 \text{ kmol}$$

$$D = F - W = 1 - 0.2 = 0.8 \text{ kmol}$$

As F = 1 kmol and D = 0.8 kmol.

The mole % original feed mixture distilled = $\frac{0.8}{1} \times 100 = 80$ ... Ans. (b)

We have, $D x_D = 0.38$ kmol

$$D = 0.8 \text{ kmol}$$

$$y_{D\,avg} = x_D = 0.38/0.8 = 0.475$$

Molefraction of benzene in composited distillate obtained = 0.475

Molefraction of toluene in composited distillate obtained = 1 – 0.475 = 0.525

Composition of composited distillate :

benzene = 47.5 mole %

toluene = 52.5 mole % ... Ans. (b)

Ex. 1.16 : A feed containing 50 mole % benzene and 50 mole % toluene is to be distilled in a fractionating column to produce a distillate containing 90 mole % benzene and bottoms containing 90 mole % toluene. Feed rate to the column is 10000 kg/day and feed is at its bubble point. The operating reflux ratio is 1.5 times the minimum and the overall plate efficiency is 0.75. Determine the actual plates required, the diameter and height of the column.

Data : $\alpha = 2.28$, column operates at 1 atm pressure.

boiling point of benzene = 353.3 K

boiling point of toluene = 383.8 K

Vapour velocity at top = 0.565 m/s

Vapour velocity at the bottom = 0.562 m/s, plate spacing = 450 mm.

Assume that the top temperature to be equal to the boiling point of benzene and the bottom temperature to be equal to the boiling point of toluene.

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Solution : Basis = 10000 kg/day of feed to the column.

Molefraction of benzene in feed = 50/100 = 0.50

Molecular weight of benzene = 78

Molecular weight of toluene = 92

M_{avg} = Average molecular weight of feed

= 0.5 × 78 + 0.5 × 92 = 85

Molar flow rate of feed to the column = 10000/85 = 117.65 kmol/day

$F = 117.65/24$

= 4.9 kmol/h

Molefraction of benzene in overhead product = x_D

$x_D = 90/100 = 0.90$

Mole % toluene in bottoms = 90

Mole % benzene in bottoms = 100 – 90 = 10

Molefraction of benzene in bottoms = $x_W = 10/100 = 0.10$.

Overall material balance :

$$F = D + W$$

Material balance of benzene :

$$F x_F = D x_D + W x_W$$

$$F x_F = D x_D + (F - D) x_W$$

$$4.9 \times 0.5 = D \times 0.90 + (4.9 - D) \times 0.10$$

$\therefore$ $D = 2.45$ kmol/h

$W = 2.45$ kmol/h

$\alpha = 2.28$

$$y = \frac{\alpha x}{1 + (\alpha - 1) x}$$

Generate x – y data. Take x = 0, 0.1, 0.2 ... 1.0 and evaluate the corresponding y values.

$x = 0.10$

$$y = \frac{2.28 \times 0.10}{1 + (2.28 - 1) \times 0.10} = 0.202$$

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.202	0.363	0.494	0.603	0.695	0.774	0.842	0.901	0.954	1.0

Plot the x - y data, i.e., the equilibrium curve.

Feed is at its bubble point, so q = 1.

The q-line is parallel to y-axis through the point (0.5, 0.5) on the diagonal. It will cut the equilibrium curve at point P. Read y_F, the y coordinate of the point P.

$y_F = 0.695$ (this value also given in data corresponding to $x_F = 0.50$).

For feed at its bubble point, R_m is given by

$$R_m = \frac{x_D - y_F}{y_F - x_F}$$

$$= \frac{0.90 - 0.695}{0.695 - 0.50} = 1.051$$

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Given : $R = 1.5\ R_m$

Operating reflux ratio $R = 1.5 \times 1.051 = 1.5765 \approx 1.58$

Intercept of the operating line of the rectifying section on y-axis :

$$\text{Intercept} = x_D/(R + 1)$$
$$= 0.90/(1.58 + 1) = 0.35$$

Draw the operating line of the rectifying section through the point (0.90, 0.90) on the diagonal and with an intercept of 0.35 on y-axis.

Draw the operating line of the stripping section through the point (0.1, 0.1) on the diagonal and passing through the point of intersection of the q-line with the rectifying section operating line.

Construct the theoretical stages between the equilibrium curve and the operating lines starting from $x_D = 0.9$ and reaching exactly or covering $x_W = 0.10$.

Number of theoretical plates including reboiler = n = 11.

Number of theoretical plates required = n – 1 = 11 – 1 = 10.

Overall plate efficiency = 75%.

$$\text{Actual plates required} = \frac{\text{Theoretical plates}}{0.75}$$
$$= 10/0.75$$
$$= 13.33 = 14$$

Actual plates required = 14 ... Ans.

Determination of the column height :

Plate spacing = 450 mm = 0.45 m

Number of plates = 14

So 14 plates require 13 number of spacings to be provided.

A spacing of 0.45 m is also to be provided (i) above the top plate and (ii) below the bottom plate.

$$\text{Column height} = 13 \times 0.45 + 1 \times 0.45 + 1 \times 0.45$$
$$= 6.75 \text{ m}$$ ... Ans.

Determination of the column diameter :

(i) Velocity of vapour at the top = 0.565 m/s

Vapour flow at the top = V = (R + 1) D

$$= (1.58 + 1) \times 2.45$$
$$= 6.321 \text{ kmol/h}$$
$$= 1.756 \times 10^{-3} \text{ kmol/s}$$

Top temperature = boiling point of benzene = 353.3 K

Volumetric vapour flow = nRT/P

$$= \frac{1.756 \times 10^{-3} \times 8.31451 \times 353.3}{101.325}$$
$$= 0.05091 \text{ m}^3/\text{s}$$

[Column pressure = 1 atm = 101.325 kPa, R = 8.31451 m^3.kPa/(kmol.K) and T = 353.3 K]

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Volumetric vapour flow rate = cross-sectional area × vapour velocity

Cross-sectional area = 0.05091/0.565 = 0.09011 m^2

$\pi/4\ D^2$ = 0.09011

$\therefore$ D = 0.338 m = 338 mm

(ii) Velocity of the vapour at the bottom = 0.562 m/s

$V' = V - (1 - q)\ F$ but $q = 1$

$\therefore$ $V' = V = 6.321$ kmol/h $= 1.756 \times 10^{-3}$ kmol/s

Vapour flow at the bottom = $V' = 1.756 \times 10^{-3}$ kmol/s

Bottom temperature = boiling point of toluene

= 383.8 K

Volumetric vapour flow at bottom = nRT/P

$$= \frac{1.756 \times 10^{-3} \times 8.31451 \times 383.8}{101.325}$$

= 0.0553 m^3/s

Volumetric vapour flow rate = cross-sectional area × vapour velocity

Volumetric vapour flow rate = $\pi/4\ D^2 \times 0.562 = 0.0553$

$\therefore$ D = 0.354 m = 354 mm

Higher value of diameter should be recommended.

$\therefore$ Diameter of the column = 354 mm ... **Ans.**

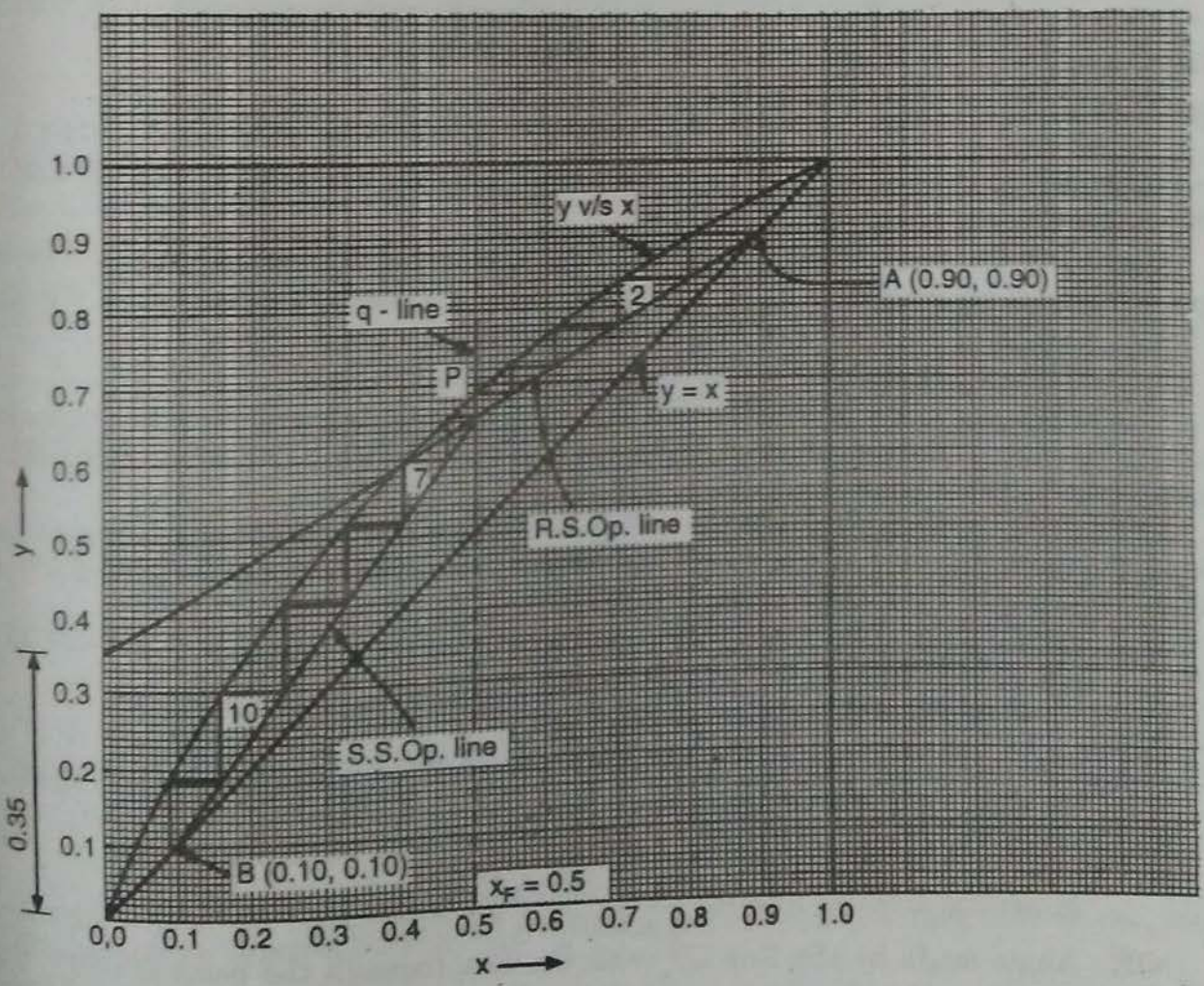

Fig. E 1.16 : Determination of ideal plates

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Ex. 1.17 : The data of molefraction of carbon disulphide in the liquid and in the vapour evolved from the mixture during the distillation of a carbon disulphide-carbon tetrachloride mixture is as follows :

x	0	0.20	0.40	0.60	0.80	1.0
y	0	0.445	0.65	0.795	0.91	1.0

Determine graphically the number of theoretical plates required.

Data : Reflux ratio = 3, purity of product = 97% (mole basis)

Percentage of carbon disulphide in waste liquor = 5% (mole basis)

Carbon disulphide is more volatile component.

Slope of the stripping section line = 1.4.

Determine the minimum slope of the rectifying line in this case.

Solution : Purity of product, i.e., distillate = 97% by mole

∴ Molefraction of carbon disulphide in the distillate = x_D = 0.97.

% of carbon disulphide in the waste liquor = 5% (mole basis)

∴ x_W = 5/100 = 0.05

R = reflux ratio = 3

Intercept of the operating line of rectifying section = $x_D/R + 1 = 0.97/3 + 1 = 0.2425$

Plot the equilibrium data.

Draw the operating line of the rectifying section through the point Q (0.97, 0.97) on the diagonal and with an intercept on y-axis equal to 0.2425.

Slope of the stripping section line = 1.4.

$$\tan \alpha = 1.4 \qquad \therefore \quad \alpha = 54.46^\circ \approx 55^\circ$$

Draw the operating line of the stripping section through the point M (0.05, 0.05) on the diagonal with an angle of 55° to the base.

Construct the equilibrium/theoretical stages between the equilibrium curve and the operating lines starting from x_D = 0.97 and reaching exactly or covering x_W = 0.05.

Number of theoretical plates including reboiler = n = 9

Number of theoretical plates required = n – 1 = 9 – 1 = 8 ... **Ans.**

The minimum slope of the operating line of the rectifying section corresponds to an infinite number of theoretical stages. The slope of the operating line of the stripping section is assumed to remain constant.

Extend the stripping section operating line to cut the equilibrium curve at point P. Draw the operating line of the rectifying section through the point Q (0.97, 0.97) and the point P (for a minimum slope of the line QP).

Measure the slope of the line QP or read the coordinates of the point P.

$$\text{Slope of operating line of rectifying section} = \text{Slope of line QP} = \frac{x_D - y'}{x_D - x'}$$

We have, x_D = 0.97

x', y' are the coordinates of point P.

$$x' = 0.53 \quad \text{and} \quad y' = 0.75$$

$$\text{Minimum slope of line QP} = \frac{0.97 - 0.75}{0.97 - 0.53} = 0.50 \qquad \text{... Ans.}$$

OR : Angle made by the line QP with the line through the point Q (0.97, 0.97) and parallel to x-axis (not shown) measured in anticlockwise direction = 26.5°.

$$\text{Slope} = \tan(26.5^\circ) = 0.4986 \approx 0.50.$$

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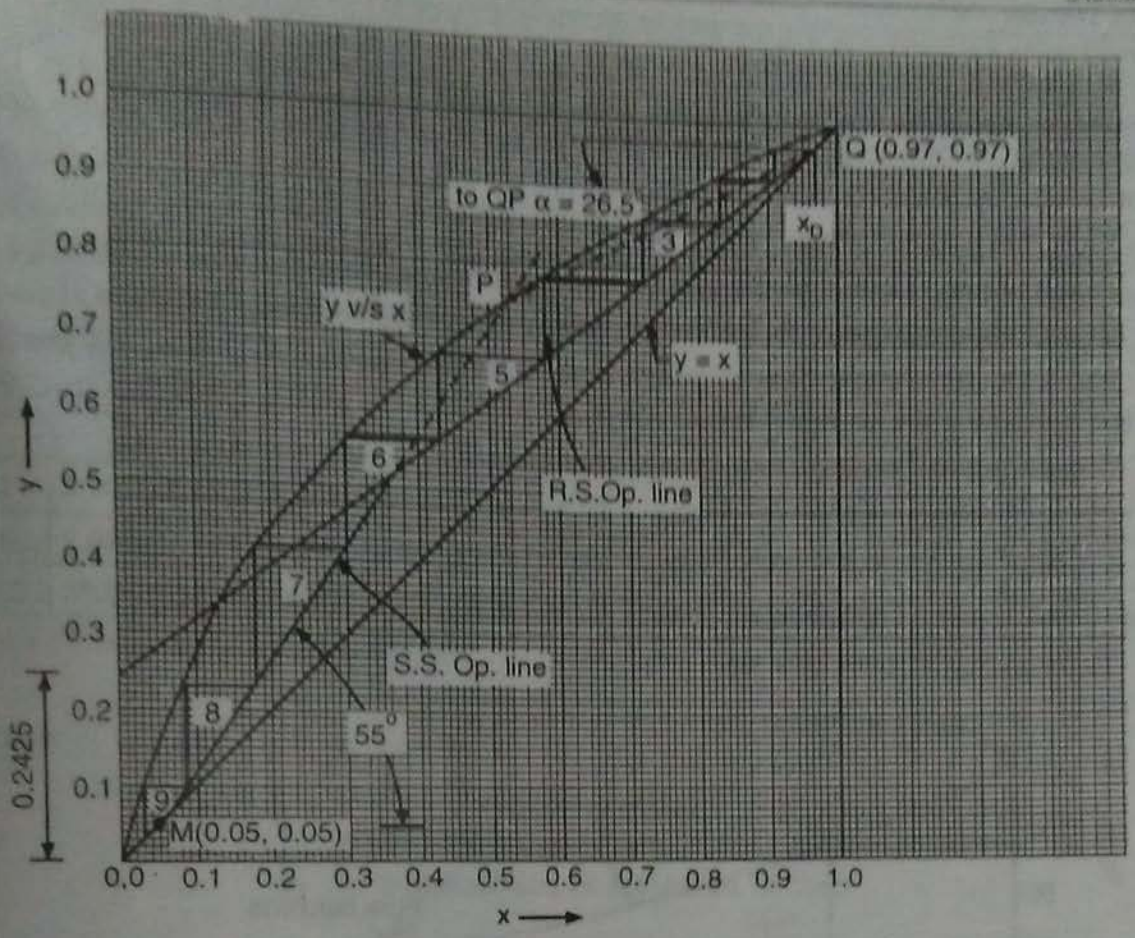

Fig. E 1.17 : Determination of ideal stages and minimum slope of operating line of rectifying section

Ex. 1.18 (A) : A liquid mixture containing 40 mole % benzene and 60 mole % toluene is subjected to flash distillation at a separator pressure of 101.325 kPa to vaporise 50 mole % of feed. What will be the equilibrium composition of vapour and liquid ?

Data :

x	0	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0.	0.13	0.21	0.375	0.5	0.6	0.7	0.77	0.83	0.9	0.95	1.0

Solution : Basis : Feed containing 40 mole % benzene

x_F = molefraction of benzene in feed

= mole % benzene / 100

$$x_F = \frac{40}{100} = 0.4$$

50 mole % of the feed is vaporised

f = molefraction of feed that is vaporised $= \frac{50}{100} = 0.5$

Slope of the operating line for flash distillation $= -\frac{(1-f)}{f}$

$$\text{Slope} = \frac{-(1-0.5)}{0.5} = -1.0, \tan\theta = -1.0 \quad \therefore \theta = -45^\circ$$

Draw the equilibrium diagram with the help of data given.

The point of intersection of the operating line and the diagonal is (x_F, x_F), i.e., (0.4, 0.4).

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Mark point (0.4, 0.4) on the diagonal and draw the operating line through it with a slope equal to – 1.0 ($\theta = -45°$) which will cut the equilibrium curve at point say P. Read the coordinates of the point P-the equilibrium liquid phase and vapour phase compositions from x-axis and y-axis respectively.

From the graph – [Refer to Fig. E 1.18 (A + B)]

Equilibrium : Liquid phase composition = 0.3 molefraction of benzene

Equilibrium : Vapour phase composition = 0.5 molefraction of benzene ... **Ans.**

Ex. 1.18 (B) : In the case of problem given in exercise 1.18 (A), find out the temperature in the separator for an equilibrium stage.

Temperature - composition (liquid phase) data :

T, °C	80	82.9	85	87	90.5	93.3	96.1	99	101.6	104.5	107.2	110	110.1
T, K	353	355.9	358	360	363.5	366.3	369.1	372	374.6	377.5	380.2	383	383.1
x	1.0	0.9	0.77	0.66	0.55	0.46	0.37	0.29	0.21	0.14	0.075	0.013	0

Refer to Fig. E 1.18 (A + B).

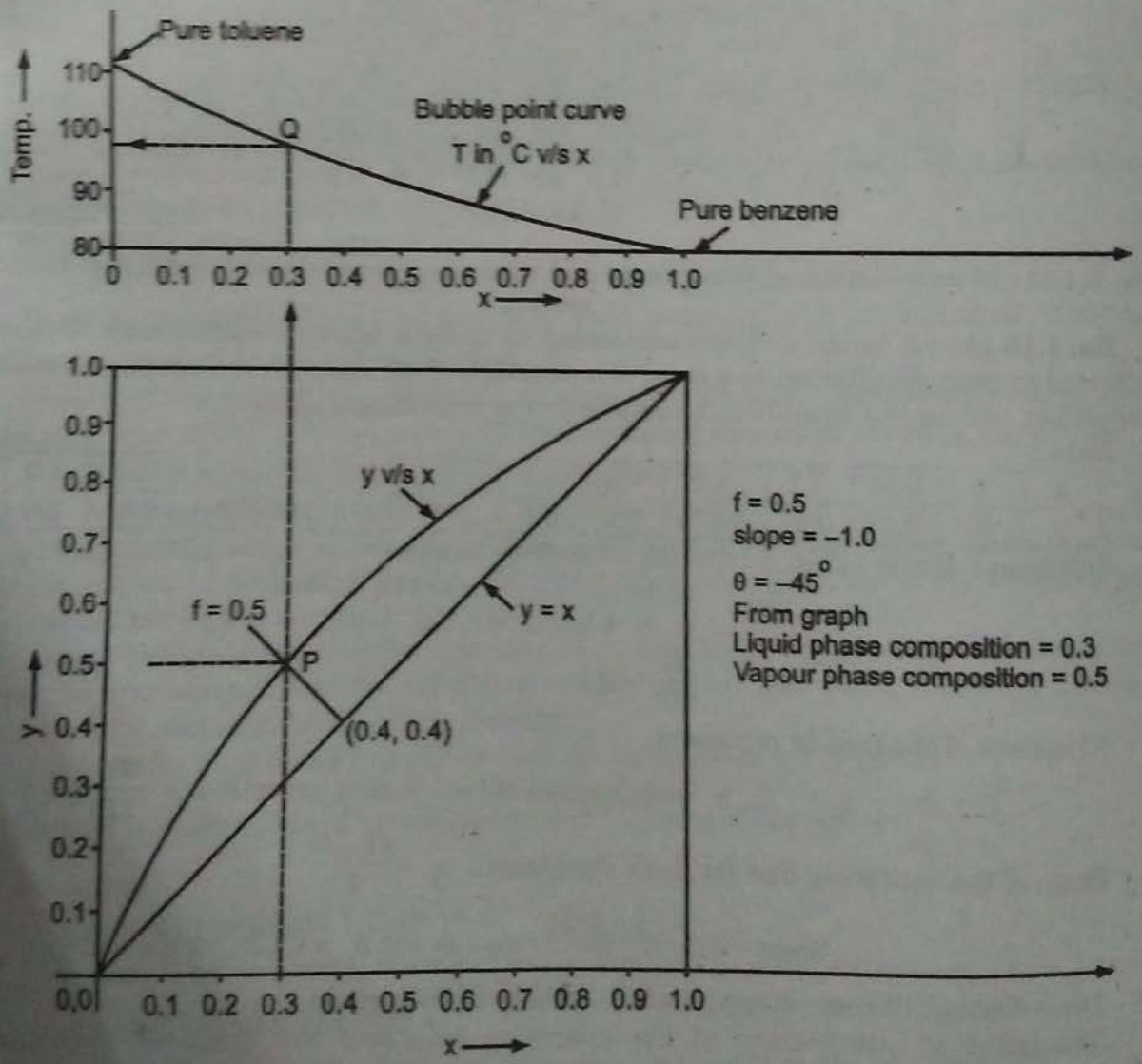

Fig. E 1.18 (A + B) : Determination of equilibrium compositions and temperature in separator

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Plot a graph of temperature (bubble point) v/s liquid phase composition, equilibrium diagram and draw the operating line through the point (0.4, 0.4) on the diagonal with a slope equal to minus one (at $\theta = -45^0$) which will cut the equilibrium curve at the point P. Through the point P, draw a vertical to cut bubble point curve at the point Q and then read temperature corresponding to the point Q on y-axis.

Temperature in the separator (from the graph) = 371 K (98 °C) ... Ans.

Ex. 1.19 : Liquid mixture containing 40 mole % methanol and 60 mole % water is fed to the differential distillation at atmospheric pressure with 60 mole % of the liquid is distilled. Find the composition of the composited distillate and the residue.

Equilibrium data :

x	0.05	0.1	0.2	0.3	0.4	0.5
y	0.27	0.42	0.57	0.66	0.73	0.78

Solution : Basis : 100 kmol of feed

Let F, D, W be the kmol of feed, distillate and residue respectively.

Given : 60 mole % of feed is distilled

$\therefore$ $D = 0.60 \times 100 = 60$ kmol

We know : $F = D + W$

$100 = 60 + W$

$\therefore$ $W = 40$ kmol

x_F = molefraction of methanol in feed

$$x_F = \frac{40}{100} = 0.4$$

x_W = molefraction of methanol in residue left in still

The Rayleigh equation :

$$\ln\left(\frac{F}{W}\right) = \int_{x_W}^{x_F} \frac{dx}{y - x}$$

The RHS of this equation is evaluated graphically by plotting a graph of 1/(y–x) v/s x and measuring the area under the curve between the limits : $x = x_F$ and $x = x_W$.

$$\text{LHS of equation} = \ln\left(\frac{F}{W}\right) = \ln\left(\frac{100}{40}\right) = 0.916$$

Plot a graph of 1/(y – x) v/s x and measure the area under the curve from $x_F = 0.50$ till area equals 0.916 and then integration is stopped, and the corresponding value of x represents x_W (composition of residue). Prepare the table as shown below from the data provided :

x	0.05	0.1	0.2	0.3	0.4	0.5
y	0.27	0.42	0.57	0.66	0.73	0.78
1/(y – x)	4.54	3.12	2.7	2.8	3.0	3.6

From Fig. E 1.19, $x_W = 0.075$ molefraction of methanol in residue.

$y_{Davg.}$ – composited distillate composition.

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Methanol material balance :

$$x_F F = y_{D\,avg.} D + x_W W$$

$$0.40 \times 100 = y_{D\,avg.} (60) + 0.075 \times 40$$

$$y_{D\,avg.} = 0.62$$

Scales : Refer Fig. E 1.19.

x-axis : 20 mm = 0.1 units

y-axis : 20 mm = 1.0 units

$$\ln \frac{F}{W} = \int_{x_F}^{x_W} \frac{dx}{y - x}$$

$$0.916 = \text{Area} \times (\text{x-axis scale}) \times (\text{y-axis scale})$$

$$0.916 = \text{Area} \left(\frac{0.1}{2}\right)\left(\frac{1}{2}\right)$$

$$\text{Area} = 36.64 \text{ cm}^2$$

Measure the area under the curve from

x_F = 0.4 till we get area = 36.64 cm^2

and read x_W on x-axis.

∴ From Fig. E 1.19, x_W = 0.075 molefraction of methanol in residue.

x_W = 0.075

Composition of residue = x_W = 0.075

Composition of composited distillate = $y_{D_{avg}}$ = 0.62 ... Ans.

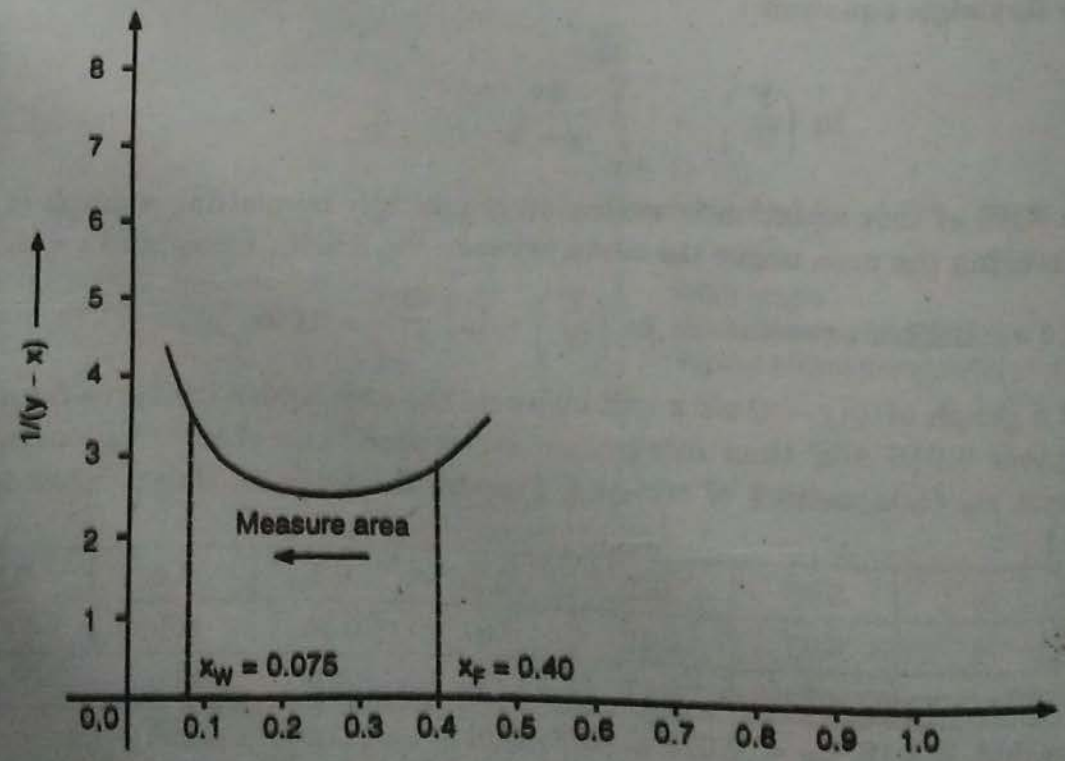

Fig. E 1.19 : Determination of x_W-simple distillation

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Ex. 1.20 : Feed containing 40 mole % benzene and 60 mole % toluene is to be distilled in a fractionating column to get top product (distillate) containing 90 mole % benzene and bottoms containing not more than 10 mole % benzene. The reflux ratio of 3 kmol per kmol of product is used. Assuming feed enters the column at its bubble point (saturated liquid), find the number of theoretical plates required and position of the feed plate by **Lewis-Sorel method.**

Equilibrium data :

x	0	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
y	0	0.13	0.21	0.375	0.5	0.6	0.7	0.77	0.83	0.9	0.95	1

Solution : Basis : 100 kmol of feed to column

$$F = D + W \text{ (overall material balance)}$$

$$F = 100 = D + W \quad \ldots (A)$$

$$\text{Molefraction of benzene in feed} = \frac{\text{mole \% benzene in feed}}{100}$$

$$x_F = 40/100 = 0.4$$

Similarly, $x_D = 90/100 = 0.90$ and $x_W = 10/100 = 0.10$

Material balance of benzene :

$$x_F F = x_D D + x_W W$$

$$0.4 \times 100 = 0.9\,D + 0.1\,W$$

$$40 = 0.9\,D + 0.1\,W$$

From equation (A), $D = 100 - W$

$$\therefore \quad 40 = 0.9\,(100 - W) + 0.1\,W$$

Solving, we get $W = 62.5$ kmol

and $D = 37.5$ kmol

$$R = \text{Reflux ratio} = 3 = \frac{L}{D}$$

but $L = L_n$

$$\therefore \quad L_n = 3\,D$$

$$L_n = 3\,(37.5)$$

$$= 112.5 \text{ kmol}$$

$$L_n + D = 112.5 + 37.5 = 150 \text{ kmol}$$

Rectifying section operating line :

$$x_{n+1} = \frac{L_n}{L_n + D}\, x_n + \frac{D\,x_D}{L_n + D}$$

$$\therefore \quad y_{n+1} = \frac{112.5\,x_n}{150} + \frac{37.5 \times 0.9}{150}$$

$$y_{n+1} = 0.75\,x_n + 0.225 \quad \ldots (B)$$

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As the feed is at its bubble point, $q = 1$

$$L_m = L_n + qF$$

$$\therefore \quad L_m = L_n + F$$

$$= 112.5 + 100 = 212.5 \text{ kmol}$$

and $$L_m - W = 212.5 - 62.5 = 150 \text{ kmol}$$

The operating line of the stripping section is

$$y_{m+1} = \frac{L_m}{L_m - W} x_m - \frac{W}{L_m - W} x_W$$

$$y_{m+1} = \frac{212.5}{150} x_m - \frac{62.5}{150} \times 0.1$$

$$y_{m+1} = 1.415\, x_m - 0.042 \quad \ldots (C)$$

Distillate composition $= x_D = 0.9$

$$\therefore \quad y_n = x_D = 0.9$$

Obtain the value of x_n corresponding to y_n (x_n in equilibrium with y_n)

The value of x_n corresponding to y_n from the equilibrium diagram (Fig. E 1.20) is 0.79

$$y_{n+1} = 0.75\, x_n + 0.225$$

$$= 0.75 \times 0.79 + 0.225$$

$$= 0.818$$

Obtain the value of x_{n+1} from the equilibrium diagram corresponding to y_{n+1}.

$$\therefore \quad x_{n+1} = 0.644$$

$$y_{n+2} = 0.75\, x_{n+1} + 0.225$$

$$y_{n+2} = 0.75 \times 0.644 + 0.225 = 0.708$$

x_{n+2} from the equilibrium diagram corresponding to y_{n+2} is 0.492

$$y_{n+3} = 0.75\, x_{n+2} + 0.225$$

$$y_{n+3} = 0.75 \times 0.492 + 0.225 = 0.594$$

x_{n+3} from the equilibrium diagram corresponding to y_{n+3} is 0.382

x_{n+3} is near to x_F (0.40), so that feed should be introduced on plate n + 3 from the top (4^{th} plate).

Stripping section : Now perform the calculation using the equation of the operating line of the stripping section.

For the stripping section, the operating line is :

$$y_{m+1} = 1.415\, x_m - 0.042$$

For the sake of convenience (in this portion of column also) retain n for m.

$$\therefore \quad y_{n+1} = 1.415\, x_n - 0.042$$

$$y_{n+4} = 1.415\, x_{n+3} - 0.042$$

$$y_{n+4} = 1.415 \times 0.382 - 0.042$$

$$= 0.498$$

x_{n+4} from the equilibrium diagram corresponding to y_{n+4} is 0.298

$$y_{n+5} = 1.415 \times 0.298 - 0.042 = 0.379$$

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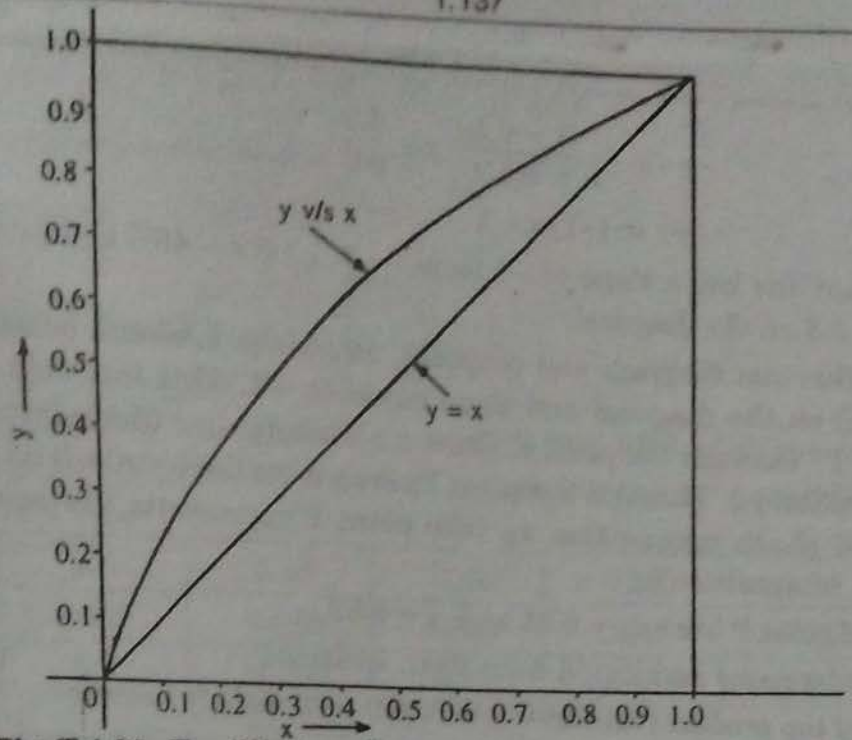

Fig. E 1.20 : Equilibrium diagram for benzene - toluene system

x_{n+5} from the equilibrium diagram corresponding to y_{n+5} is 0.208

$$y_{n+6} = 1.4165\, x_{n+5} - 0.042$$

$$y_{n+6} = 1.4165 \times 0.208 - 0.042 = 0.252$$

x_{n+6} from the equilibrium diagram corresponding to $y_{n+6} = 0.252$ is equal to 0.120

$$\therefore \quad y_{n+7} = 1.415 \times 0.120 - 0.042 = 0.127$$

x_{n+7} from the equilibrium diagram corresponding to y_{n+7} is equal to 0.048.

As we have crossed x_W at $x_{n+7} = 0.048$, stop the calculations.

[As $x_{n+6} = 0.12$ is greater than $x_W = 0.1$, one more step has to be incorporated so that the liquid phase composition obtained at any case should not be more than x_W.]

Number of ideal plates including reboiler = 7 + 1 = 8

Hence, there will be 7 ideal plates in the column. **... Ans.**

Ex. 1.21 : A feed of 50 mole % hexane and 50 mole % octane is fed to a pipe still through a pressure reducing valve and then into a flash disengaging chamber. The vapour and liquid leaving the chamber are assumed to be in equilibrium. If the fraction of the feed converted to vapour is 0.5, find the composition of the top and bottom products. Equilibrium data for system is given below.

Molefraction of hexane in liquid (x)	1	0.69	0.4	0.192	0.045	0
Molefraction of hexane in vapour (y)	1	0.932	0.78	0.538	0.1775	0

Solution : Basis : 1 mol of feed.

Material balance line for flash distillation is –

$$y = -\frac{1-f}{f}\, x + \frac{x_F}{f}$$

$$x_F = \frac{50}{100} = 0.5$$

f = fraction of feed converted to vapour = 0.5

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$$\therefore \quad y = -\frac{(1-0.5)}{0.5}x + \frac{0.5}{0.5}$$

$$y = (-1)x + 1$$

Material balance line has a slope of – 1 ($\tan\theta = 1 \quad \therefore \quad \theta = -45^{o}$) and passes through the point $x = y = x_F = 0.5$ on the diagonal.

Draw the equilibrium diagram and diagonal. Draw the material balance line through the point (0.5, 0.5) on the diagonal and slope equal to – 1. This line cuts the equilibrium curve at point say P. Through the point P, draw a horizontal line (dotted) to cut y-axis to get the vapour composition y_D. Through the point P, drop a vertical (dotted) on x-axis to get the equilibrium liquid phase composition x_W (the point P represents the equilibrium vapour and liquid phase composition).

Coordinates of point P are : $x_W = 0.31$ and $y = 0.69$.

Composition of vapour and liquid from flash chamber :

Composition of top product (vapour) = $0.69 \times 100 = 69$ mole % hexane.

Composition of bottom product (liquid) = $0.31 \times 100 = 31$ mole % hexane.

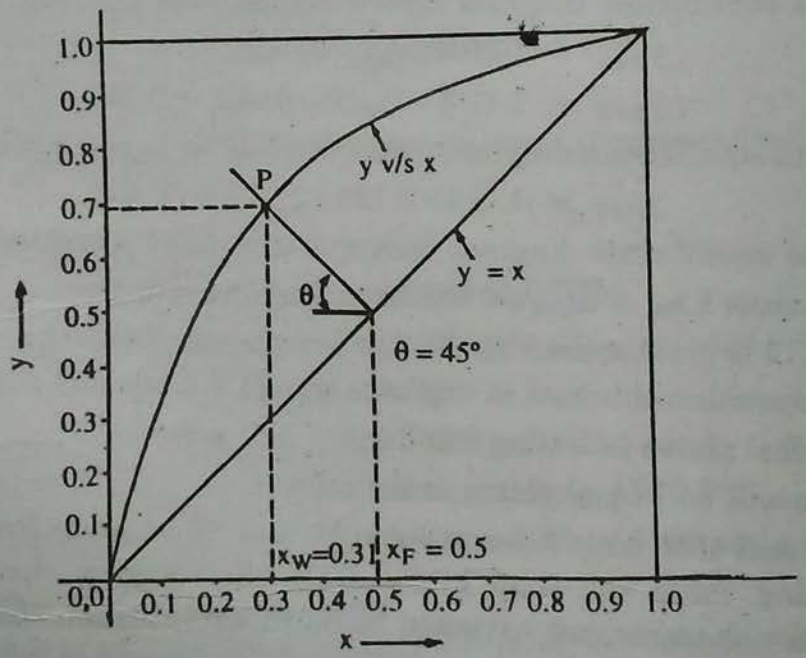

Fig. E 1.21 : Determination of compositions – flash distillation

Ex. 1.22 : 100 kmol of mixture containing 50 mole % n-heptane (more volatile) and 50 mole % n-octane is subjected to a differential distillation at atmospheric pressure; with 60 mole % of liquid distilled. Compute the composition of the composited distillate and the residue using Rayleigh equation.

Equilibrium data :

x	0.5	0.46	0.42	0.38	0.34	0.32
y	0.689	0.648	0.608	0.567	0.523	0.497

Solution : Basis : 100 kmol of feed

$$F = 100 \text{ kmol}$$

60 mole % of feed is distilled

$$\therefore \quad D = 0.60 \times 100 = 60 \text{ kmol}$$

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$\therefore$ $W = F - D = 100 - 60 = 40$ kmol

Molefraction of heptane in feed = $x_F = \frac{50}{100} = 0.50$

Rayleigh equation :

$$\ln F/W = \int_{x_W}^{x_F} \frac{dx}{y - x}$$

LHS = ln F/W = ln (100/40) = 0.916. The RHS of this equation is evaluated graphically by plotting 1/(y – x) v/s x and measuring the area under the curve between limits $x = x_F$ and $x = x_W$.

x	0.5	0.46	0.42	0.38	0.34	0.32
y	0.689	0.648	0.608	0.567	0.523	0.497
1/(y – x)	5.3	5.32	5.32	5.35	5.465	5.65

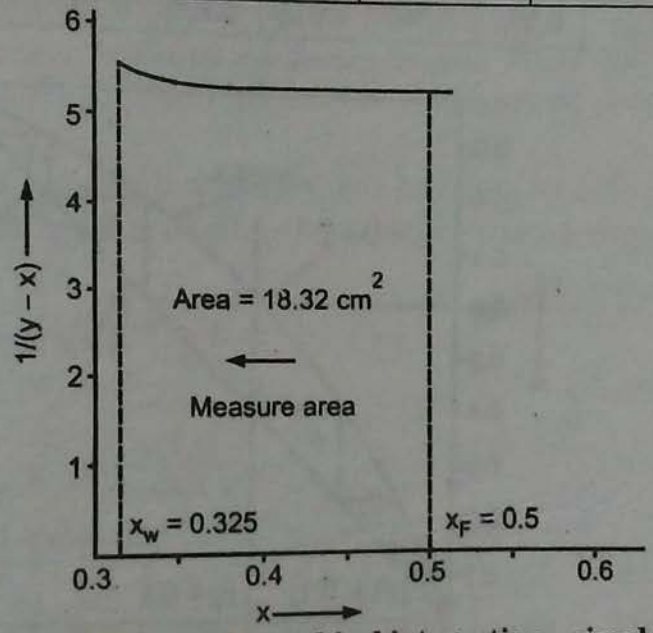

Fig. E 1.22 : Determination of x_W by graphical integration – simple distillation

0.916 = Area under the curve × Scale x-axis × Scale y-axis

$$0.916 = \text{Area} \times \frac{0.1}{2} \times \frac{1}{1}$$

Area = 18.32 cm²

Measure the area under the curve from $x_F = 0.5$ till area equals 18.32 cm² and read the corresponding value of x_W from x-axis.

From Fig. E 1.22, $x_W = 0.325$

Material balance of heptane :

$$F x_F = D y_{D\,avg} + W x_W$$

$$100 \times 0.5 = 60\, y_{D\,avg} + 40 \times 0.325$$

$$y_{D\,avg.} = 0.6167$$

Bottom product / Residue composition = 0.325 molefraction of n-heptane

Composited distillate composition = 0.6167 molefraction of n-heptane.

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Ex. 1.23 : A mixture of benzene and toluene containing 40 mole % benzene is to be separated to give a product of 90 mole % of benzene at a top and a bottom product with not more than 10 mole % benzene. Using an average value of 2.4 for the volatility of benzene relative to toluene, calculate the number of theoretical plates required at total reflux.

Also calculate the minimum reflux ratio, if the feed is liquid at its bubble point.

Solution :

Relative volatility $= \alpha = 2.4$

$$y = \frac{\alpha x}{1 + (\alpha - 1) x} = \frac{2.4 x}{1 + 1.4 x}$$

For x = 0, 0.1, 0.2 ... 1.0, obtain corresponding values of y and plot the equilibrium diagram.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.21	0.375	0.507	0.615	0.706	0.783	0.850	0.906	0.95	1.0

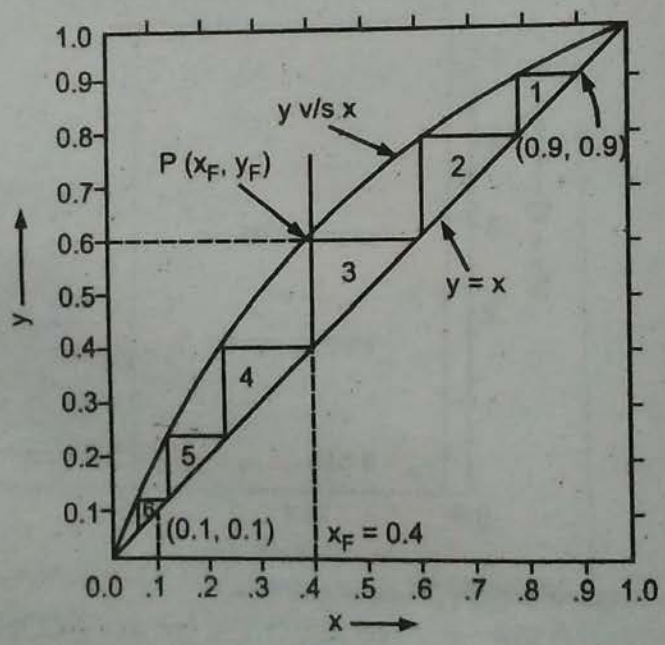

Fig. E 1.23 : Determination of No. of plates at total reflux and R_m

Molefraction of benzene in feed $= x_F = \frac{40}{100} = 0.40$

Molefraction of benzene in distillate $= x_D = \frac{90}{100} = 0.90$

Molefraction of benzene in bottom product $= x_W = \frac{10}{100} = 0.10$

At total reflux both the operating lines will coincide with the diagonal.

At total reflux, for finding the number of ideal plates, start constructing the triangles between the diagonal and the equilibrium curve from the point (0.9, 0.9) till we reach or cover the point (0.1, 0.1) on the diagonal.

Number of plates including reboiler $= n = 6$

Number of plates in column $= n - 1 = 6 - 1 = 5$

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Feed is a liquid at its bubble point. So draw the feed line through the point (0.4, 0.4) on diagonal and parallel to y-axis.

The feed line cuts the equilibrium curve at point P (x_F, y_F).

Read the coordinates of the point P

$$x_F = 0.4, \quad y_F = 0.615$$

The minimum reflux ratio (R_m) for a saturated liquid feed is given by

$$R_m = \frac{x_D - y_F}{y_F - x_F}$$

$$= \frac{0.90 - 0.615}{0.615 - 0.40} = 1.325 \quad \text{... Ans.}$$

Ex. 1.24 : 1000 kmol/h of an ethanol–propanol mixture containing 65 mole percent ethanol is to be separated in a continuous plate column operating at 101.325 kPa total pressure. The desired terminal compositions in terms of molefraction of ethanol are –

$$x_D = 0.92 \text{ and } x_W = 0.07$$

The feed is saturated vapour and total condenser is used. When the reflux flow rate is four times the amount of top product, find the number of theoretical plates required for the separation.

Relative volatility of ethanol-propanol system may be taken as 2.10.

Solution : Basis : 1000 kmol/h of ethanol-propanol mixture containing 65 mole % ethanol.

$$\alpha = 2.10$$

$$y = \frac{\alpha x}{1 + (\alpha - 1) x}$$

$$y = \frac{2.1 x}{1 + 1.1 x}$$

Take x = 0, 0.1, 0.2 ... 1.0 and find the corresponding values of y.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.19	0.34	0.47	0.58	0.67	0.76	0.83	0.89	0.95	1.0

$$x_F = \frac{65}{100} = 0.65$$

$$x_D = 0.92, \quad x_W = 0.07$$

Operating line of rectifying section has the point Q (0.92, 0.92) on diagonal.

$$\text{Intercept of this line} = \frac{x_D}{R + 1}$$

$$\text{L = reflux flow rate} = 4D$$

$$\therefore \quad L/D = R = 4$$

$$\text{Intercept} = \frac{0.92}{4 + 1} = 0.184$$

Draw this line on the equilibrium diagram. The feed is a saturated vapour (i.e., q = 0) therefore the feed line is parallel to x-axis and the point of intersection of it with the diagonal is (0.65, 0.65). Draw the feed line through the point (0.65, 0.65) on the diagonal.

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Operating line of the stripping section passes through the point of intersection of feed line and operating line of the rectifying section. Draw this line through the point P (0.07, 0.07) on the diagonal.

Starting from the point Q (0.92, 0.92), construct the triangles which represent ideal plates between the equilibrium curve and the operating lines till we cover exactly or cross the point P (0.07, 0.07) on the diagonal.

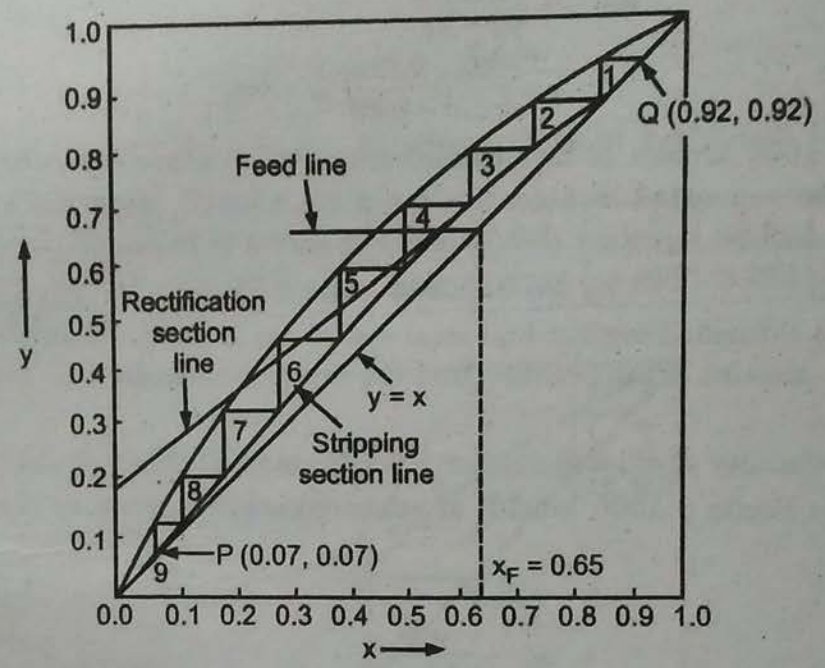

Fig. E 1.24 : Construction of ideal stages

From Fig. E 1.24,

Number of theoretical plates including reboiler = n = 9

Number of theoretical plates in column = n – 1 = 9 – 1 = 8

Location of feed plate = 4^{th} from top. ... Ans.

Ex. 1.25 : A methanol-water solution contains 50% by weight methanol. It is to be continuously rectified at atmospheric pressure (101.325 kPa) to obtain a distillate containing 95% methanol and residue containing 5% methanol (by weight). The feed enters the column at its bubble point. It is proposed to operate column at a reflux ratio of 1.5. Find graphically the number of theoretical plates required to effect a given separation and position of feed plate. Also determine R_m graphically and obtain the relation between R_m and R.

The equilibrium data are :

x	0	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.27	0.42	0.57	0.66	0.73	0.78	0.83	0.87	0.93	0.96	1.0

Solution :

Feed contains 50% methanol by weight

Distillate contains 95% methanol by weight

Residue contains 5% methanol by weight

Molecular weight of methanol = 32

Molecular weight of water = 18

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1 kg of solution contains 0.5 kg of methanol and 0.5 kg of water.

$$\therefore \quad x_F = \text{molefraction of methanol in feed} = \frac{0.50/32}{0.50/32 + 0.50/18} = 0.36$$

$$x_D = \text{molefraction of methanol in distillate} = \frac{0.95/32}{0.95/32 + 0.05/18} = 0.914$$

$$x_W = \text{molefraction of methanol in residue} = \frac{0.05/32}{0.05/32 + 0.95/18} = 0.03$$

1. Construct the x-y plot.

2. Draw the feed line. As the feed is at its bubble point (q = 1.0), so the feed line is parallel to y-axis through the point ($x_F = 0.36$, $x_F = 0.36$) on the diagonal.

The operating line of the rectifying section has an intercept on y-axis which is given by :

$$\text{Intercept} = \frac{x_D}{R+1} = \frac{0.914}{1.5+1} = 0.366$$

3. Draw the operating line of rectification section on the equilibrium diagram. It passes through the point ($x_D = 0.914$, $x_D = 0.914$) on the diagonal and has an intercept on y-axis of 0.366. Draw the operating line of the stripping section through the point ($x_W = 0.03$, $x_W = 0.03$) on the diagonal and passing through the point of intersection of feed line and top operating line (of rectification section). Construct the stages by usual method and count them.

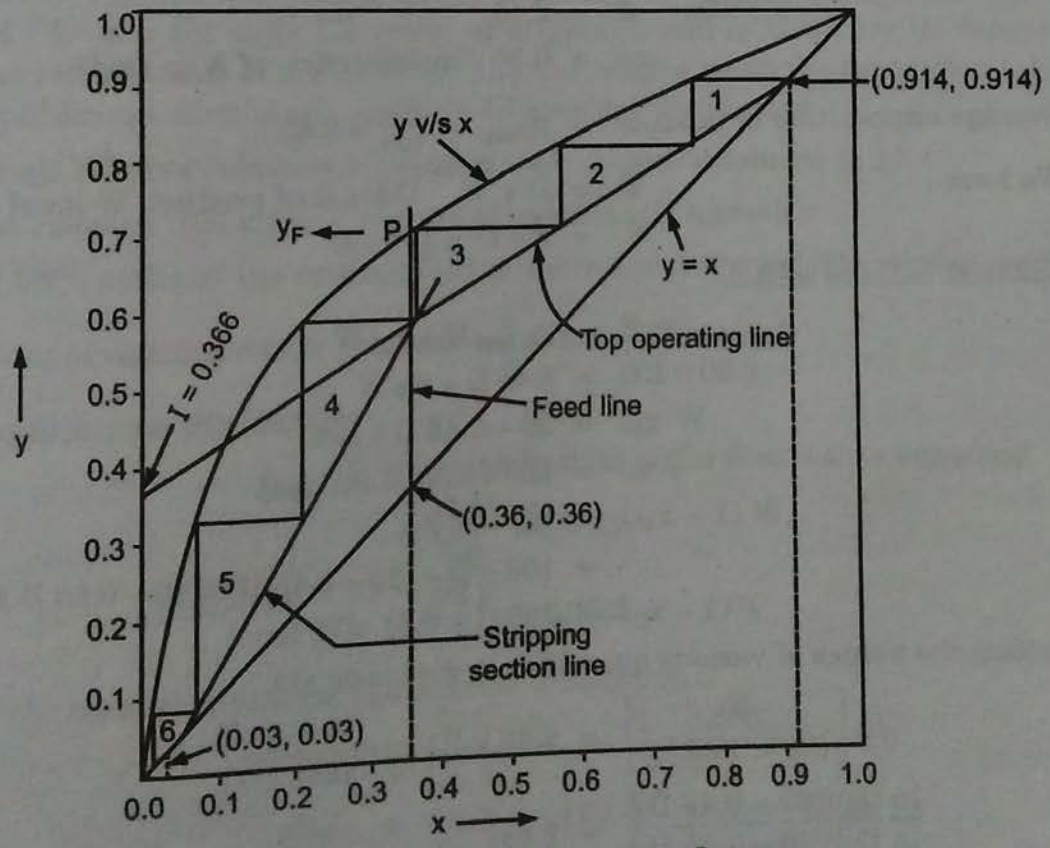

Fig. E 1.25 : Construction of ideal stages

From E 1.25, total plates including reboiler = n = 6

Number of plates in column = n – 1 = 6 – 1 = 5 ... **Ans.**

Feed plate = 4th from top

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Determination of R_m : q-line cuts the equilibrium curve at the point P.

Read y-coordinate of the point P

$$y_P = 0.70$$

The minimum reflux ratio is given by

$$R_m = x_D - y_P/(y_P - x_F)$$

$$= (0.914 - 0.70)/(0.70 - 0.36) = 0.63 \quad \text{... Ans.}$$

$$R_m = 0.63 \quad \text{and} \quad R = 1.5$$

Let, $R = Z R_m \qquad 1.5 = Z(0.63) \qquad \therefore Z = 2.38$

The relationship between R and R_m is

$$R = 2.38 R_m \quad \text{... Ans.}$$

Ex. 1.26 : A 30 mole % of component A in a mixture of A and B is to be distilled in a batch operation to obtain a product having an average composition of 45 mole % A. Calculate the amount of residue per 100 kmol of charge when $\alpha_{AB} = 2.45$.

Solution : Basis : 100 kmol of feed containing 30 mole % A.

This is a case of simple batch distillation without reflux i.e. simple distillation – with constant relative volatility.

For constant relative volatility, we have :

$$\ln \frac{F\, x_F}{W\, x_W} = \alpha \ln \frac{F\,(1 - x_F)}{W\,(1 - x_W)} \quad \ldots (1)$$

$$\alpha = 2.45$$

$$x_F = 0.30 \text{ (molefraction of A in feed)}$$

Average composition of product $= y_{D\,avg.} = \frac{45}{100} = 0.45.$

We have : $F = D + W$, D-kmol of product, W-kmol of residue

$\therefore \quad W = F - D = 100 - D$

Material balance of A :

$$x_F F = y_{D\,avg.} D + x_W W$$

$$0.30 \times 100 = 0.45\, D + x_W W$$

$\therefore \quad W x_W = 30 - 0.45\, D$ kmol

$$F x_F = 100 \times 0.30 = 30 \text{ kmol}$$

$$W(1 - x_W) = W - W x_W$$

$$= 100 - D - (30 - 0.45\, D) = 70 - 0.55\, D \text{ kmol}$$

$$F(1 - x_F) = 100(1 - 0.3) = 70 \text{ kmol}$$

Putting the values of various quantities in Equation (1),

$$\ln\left[\frac{30}{(30 - 0.45\, D)}\right] = 2.45 \ln\left[\frac{70}{(70 - 0.55\, D)}\right]$$

$$\frac{\ln [30/(30 - 0.45\, D)]}{\ln [70/(70 - 0.55\, D)]} = 2.45$$

$$\ln\left[\frac{30\,(70 - 0.55\, D)}{70\,(30 - 0.45\, D)}\right] = 2.45$$

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$$\frac{30\,(70 - 0.55\,D)}{70\,(30 - 0.45\,D)} = e^{2.45} = 11.59$$

$$30\,(70 - 0.55\,D) = 70\,(30 - 0.45\,D) \times 11.59$$

$$210 - 1.65\,D = 2433.9 - 36.51\,D$$

$$34.86\,D = 2223.9$$

$$\therefore \quad D = 63.795 \approx 63.8 \text{ kmol}$$

$$W = F - D = 100 - 63.8 = 36.2 \text{ kmol}$$

Amount of residue obtained per 100 kmol of feed = 36.2 kmol ... **Ans.**

We have : $W\,x_W = 30 - 0.45\,D$

$$= 30 - 0.45 \times 63.8$$

$$= 1.29$$

$$x_W = 1.29/W$$

$$= 1.29/36.2 = 0.0356$$

Composition of the residue left = $x_W = 0.0356$ ... **Ans.**

Ex. 1.27 : A mixture containing 70 mole % benzene and 30 mole % toluene is to be vaporised at 760 mm Hg until 1/3 moles of original liquid mixture are in vapour phase. Calculate the compositions of the distillate and the residue when the separation takes place under (i) equilibrium distillation and (ii) differential distillation.

The average relative volatility of benzene with respect to toluene is 2.5.

Solution : Basis : 100 kmol of mixture of benzene and toluene.

Given : 1/3rd moles of the original feed mixture are converted into vapour product.

Amount of vapour product produced $= \frac{1}{3} \times 100 = 33.33$ kmol.

(i)Equilibrium distillation :

f = molar fraction of the feed that is vaporised

$$= \frac{33.33}{100} = 0.3333$$

or $$f = \frac{1/3\,F}{F} = 0.3333$$

Slope of the operating line for equilibrium distillation :

$$\text{slope} = \frac{-(1-f)}{f} = \frac{-(1-0.3333)}{0.3333} = -2.0$$

$$\text{slope} = \tan\theta = -2.0 \qquad \therefore \ \theta = -63.43^\circ$$

Molefraction of benzene in feed $= x_F = \frac{70}{100} = 0.70$

Relative volatility $= \alpha = 2.5$

$$y = \frac{\alpha x}{1 + (\alpha - 1)\,x}$$

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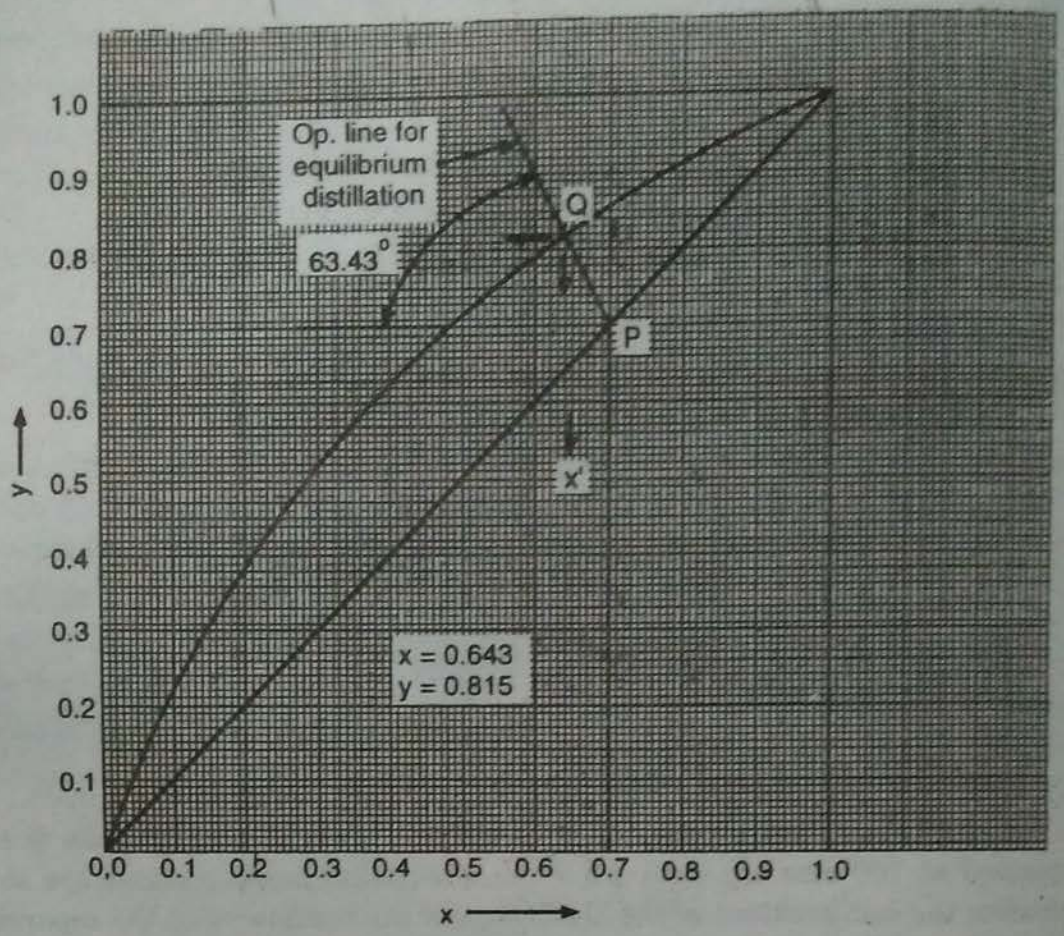

Fig. E 1.27 (a) : Flash/Equilibrium distillation

Generate the equilibrium data.

x	0	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1.0
y	0	0.22	0.385	0.52	0.625	0.71	0.79	0.85	0.91	0.96	1.0

Plot the equilibrium data.

The point of intersection of the operating line for equilibrium distillation and the diagonal is (x_F, x_F).

Draw the operating line through the point P (0.70, 0.70) on the diagonal with slope equal to -2.0 ($\theta = -63.43^\circ$). This line will cut the equilibrium curve at point Q. Read the coordinates of the point Q to get the equilibrium composition of liquid and vapour phase.

From the plot :

Liquid phase composition/Residue composition = 0.643.

Vapour phase composition/Distillate composition = 0.815 ... Ans.

(Equilibrium compositions in terms of molefraction of benzene).

(ii) Differential distillation :

Amount of vapour product obtained $= \frac{1}{3} \times 100 = 33.33$ kmol

i.e. amount of distillate obtained = D = 33.33 kmol.

We have, $F = D + W$

$W = F - D = 100 - 33.33 = 66.67$ kmol

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For differential distillation :

$$\ln (F/W) = \int_{x_W}^{x_F} \frac{dx}{y - x}$$

$$\ln (F/W) = \int_{x_W}^{0.70} \frac{dx}{y - x}, \text{ as } x_F = 0.7$$

$$\int_{x_W}^{0.70} \frac{dx}{y - x} = \ln (100/66.67) = 0.4054 \approx 0.405 \quad \ldots (1)$$

We can determine x_W (residue composition) using Equation (1).

Plot $1/(y - x)$ v/s x.

The area under the curve between $x = x_F = 0.70$ and x_W gives the value of the integral of Equation (1).

x	0.1	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.65	0.67
y	0.22	0.385	0.52	0.625	0.71	0.79	0.85	0.91	0.823	0.835
$1/(y - x)$	8.33	5.40	4.54	4.44	4.76	5.26	6.67	9.10	5.78	6.06

Area under the curve × scale y-axis × scale x-axis = 0.405.

$$\therefore \quad \text{Area} \times \frac{1}{1} \times \frac{0.10}{1} = 0.405$$

$$\therefore \quad \text{Area} = 4.05 \text{ cm}^2 = 405 \text{ mm}^2$$

Measure the area under the curve starting from $x = 0.70$ till it is equal to 405 mm² and then read the corresponding value of x_W.

From E 1.27 (b) :

The area between $x = 0.70$ and $x = 0.63$ is 416 mm²

So the area between $x = 0.70$ and $x = 0.632$ is 405 mm².

$$\therefore \quad x_W = 0.632$$

Composition of the residue = 0.632

Material balance of benzene :

$$F x_F = W x_W + D y_{D \text{ avg.}}$$

$F = 100$ kmol, $D = 33.33$ kmol, $W = 66.67$ kmol.

$$100 \times 0.7 = 66.67 \times 0.632 + 33.33 \, y_{D \text{ avg.}}$$

$$\therefore \quad y_{D \text{ avg.}} = 0.836$$

With differential distillation :

Composition of residue = $x_W = 0.632$

Composition of composited distillate = $y_{D \text{ avg.}} = 0.836$ ... Ans.

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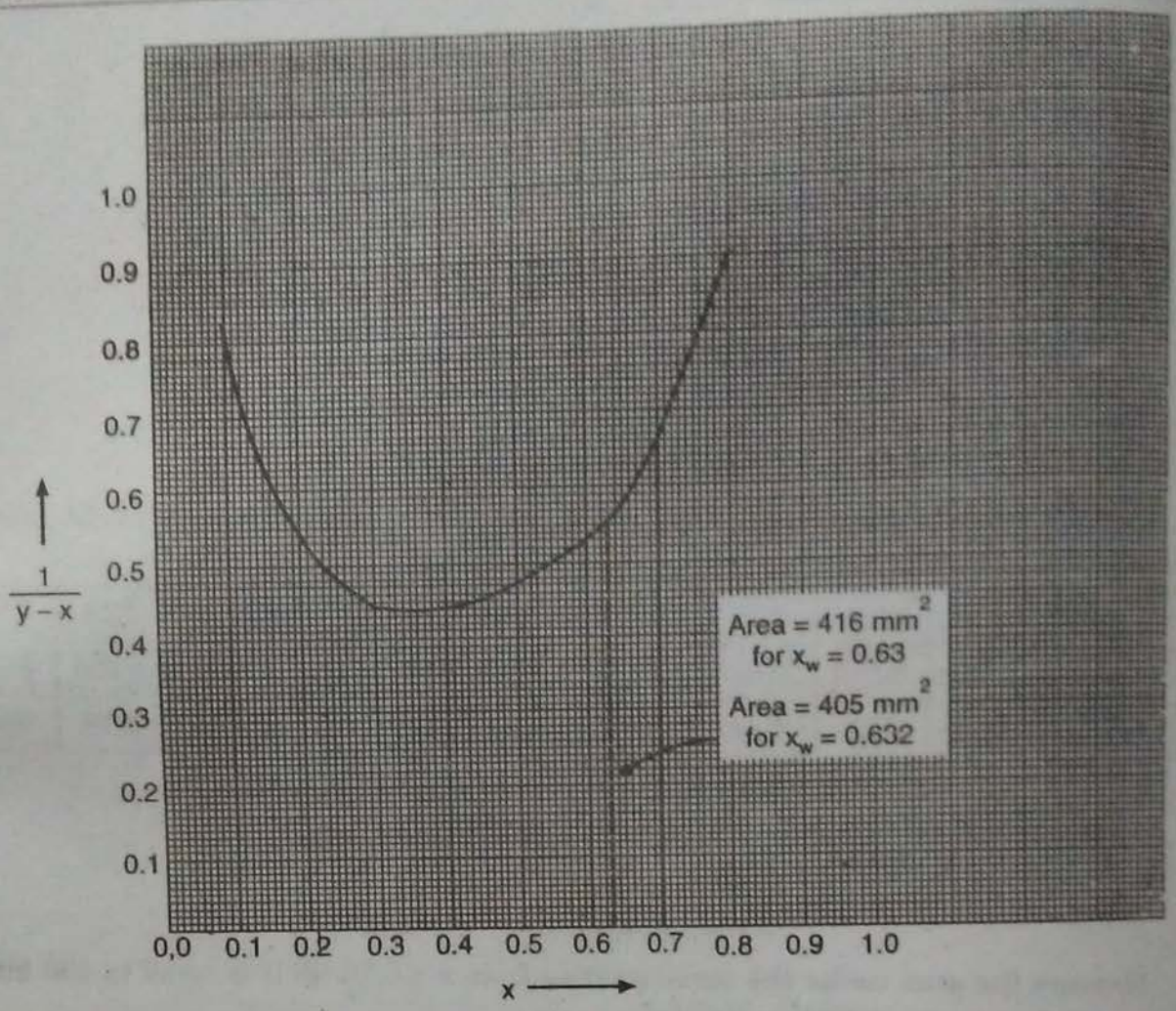

Fig. E 1.27 (b) : Determination of x_W (differential distillation)

Ex. 1.28 : A binary mixture containing 40 mole % light component is to be separated into 5 mole % bottom product and 90 mole % distillate product. The feed flow rate is 100 kmol/h and feed is 50% vapour and 50% liquid as it enters the column. The reflux ratio is 2.625. The VLE for the column can be approximated as a constant relative volatility of 2.5. Assume theoretical trays and a partial reboiler, total condenser, and saturated liquid reflux. What is the optimum feed tray location and how many theoretical trays are required ? If overall plate efficiency is 60%, how many actual trays are required ?

Solution : Basis : 100 kmol/h of binary mixture containing 40 mole % light component (A) and 60 mole % heavy component.

Molefraction of A in feed = $x_F = 40/100 = 0.40$

A in bottom product = 5 mole %

Molefraction of A in bottom product = $x_W = 5/100 = 0.05$

Molefraction of A in distillate = $x_D = 90/100 = 0.90$

Relative volatility = $\alpha_{AB} = \alpha = 2.5$.

$$y = \frac{\alpha x}{1 + (\alpha - 1)\, x}$$

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Generate the equilibrium data and tabulate.

x	0	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1.0
y	0	0.22	0.384	0.52	0.625	0.714	0.79	0.85	0.91	0.96	1.0

Plot the x - y data and diagonal y = x.

Given : Feed is 50% vapour and 50% liquid.

$$q = \text{fraction of feed that is liquid}$$

$$\therefore \quad q = 0.50$$

$$\text{slope of q-line} = \frac{-q}{1-q} = \frac{-0.5}{1-0.5} = -1.0$$

$$\text{slope} = -1.0, \quad \tan\theta = -1.0, \quad \theta = -45^\circ$$

(i) Plot the q-line through the point P (0.40, 0.40) on the diagonal and with slope of – 1.0 (at an angle of – 45°).

(ii) Operating line of rectifying section :

$$\text{Reflux ratio} = R = 2.625$$

$$\text{Intercept of this line on y-axis} = x_D/(R+1)$$

$$= 0.90/(2.625 + 1) = 0.248$$

This operating line has point A (0.90, 0.90) on the diagonal, so plot this line passing through the point A (0.90, 0.90) on the diagonal and with intercept equal to 0.248.

(iii) The operating line of the stripping section intersects the diagonal at the point (0.05, 0.05). So plot this line through the point B (0.05, 0.05) and passing through the point of intersection of q-line and operating line of the rectification section.

(iv) Construct the ideal trays/stages starting from the point A (0.90, 0.90) and reaching or covering the point B (0.05, 0.05).

(v) Count the number of ideal stages or trays.

Here it is given that the partial reboiler is used, it indicates that it will not give enrichment equivalent to one ideal stage, so whatever the ideal stages we have constructed from A to B are the ideal stages/trays that are there in the column. This reboiler provides partial vaporisation (mixture issuing from the reboiler comprises of both liquid and vapour). This is a situation that occurs with a thermosyphon reboiler. So the safest practice is not to assume such a reboiler equivalent to a theoretical stage and to provide a stage in the column itself.

Number of ideal trays required in column = n

From Fig. E 1.28 :

Number of ideal stages required for desired separation = n = 10.

As the partial reboiler is used (not acting as a plate),

Number of ideal trays required in column = n = 10 ... **Ans.**

Plate efficiency = 65%

Actual number of trays required = 10/0.60 = 16.67 ≈ 17 ... **Ans.**

Feed plate location = 5th from top. ... **Ans.**

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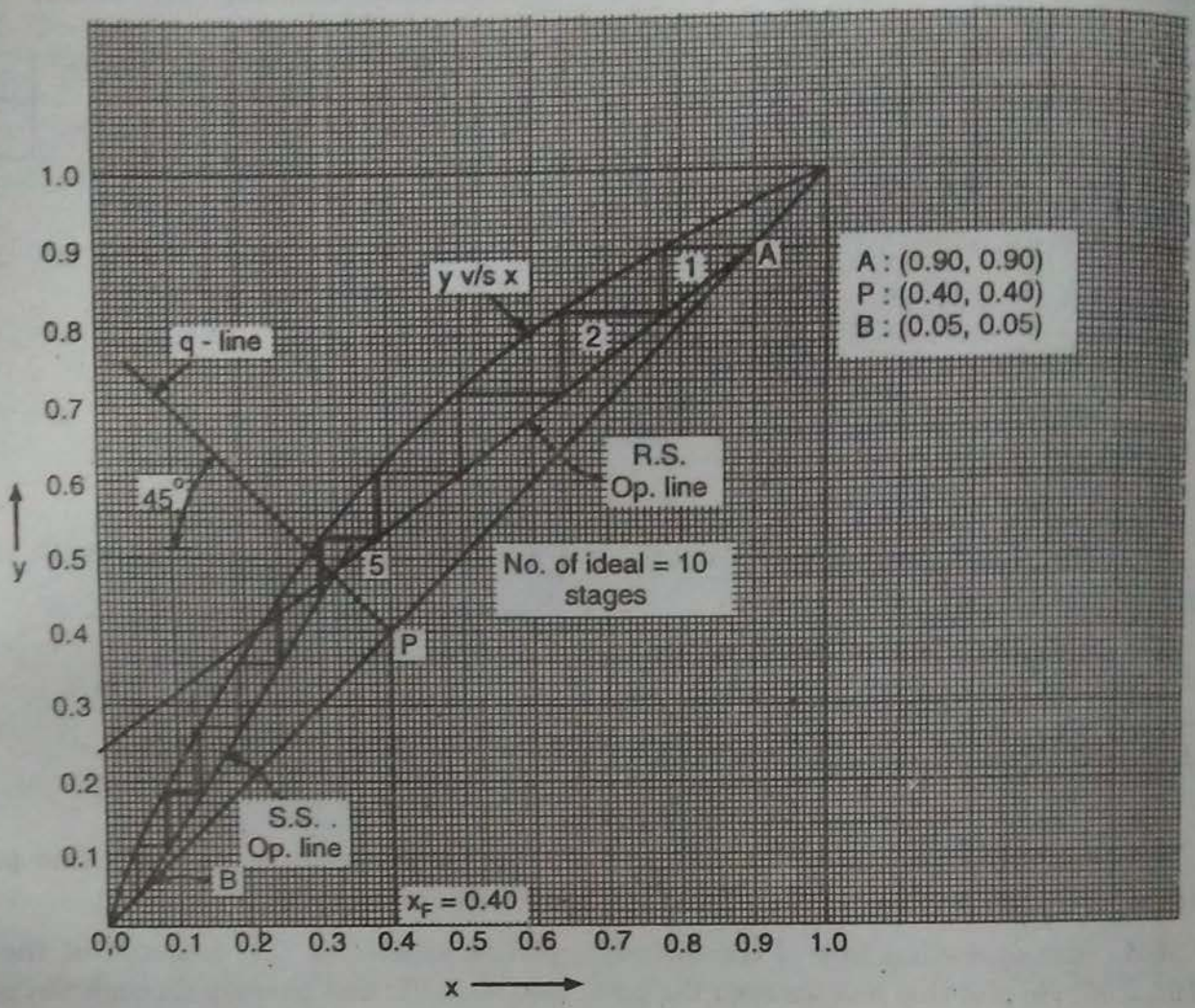

Fig. E 1.28 : Determination of number of ideal plates (partial reboiler and total condenser)

Ex. 1.29 : A binary mixture of methanol (30% by weight) and ethanol (70% by weight) is to be separated by fractionation to obtain each of 95% purity by weight. Calculate the minimum reflux and the number of ideal plates at total reflux if the feed is 30% vaporised. Also calculate the actual number of plates if a reflux ratio of twice the minimum is used when the overall plate efficiency is 60%.

The system follows Raoult's law. The following vapour pressure data are available.

Temperature, °C	64	67	70	73	76	78
Methanol, mm Hg	760	820	920	1020	1150	1260
Ethanol, mm Hg	420	430	470	610	690	760

Solution : Basis : A binary mixture of methanol and ethanol.

Molecular weight of methanol (CH_3OH) = 32

Molecular weight of ethanol (C_2H_5OH) = 46

Feed contains 30% methanol. ∴ Weight fraction methanol = 0.30.

Methanol is more volatile component.

$$\text{Molefraction of methanol in feed} = x_F = \frac{0.30/32}{0.30/32 + 0.70/46}$$

$$= 0.38$$

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Weight % methanol in distillate = 95%

Weight fraction methanol in distillate = 95/100 = 0.95

$$\text{Molefraction of methanol in distillate} = x_D = \frac{0.95/32}{0.95/32 + 0.05/46}$$

$$= 0.96$$

Weight % ethanol in bottom product = 95%

Weight % methanol in bottom product = 100 – 95 = 5%

Weight fraction methanol in bottom product = 5/100 = 0.05

$$\text{Molefraction of methanol in bottom product} = x_W = \frac{0.05/32}{0.05/32 + 0.95/46}$$

$$= 0.07$$

Relative volatility, $\alpha_{AB} = \alpha = p_A^\circ / p_B^\circ$

At 64 °C, $p_A^\circ = 760$ mm Hg and $p_B^\circ = 420$ mm Hg.

$$\therefore \quad \alpha = 760/420 = 1.809$$

T, °C	64	67	70	73	76	78
α	1.809	1.744	1.735	1.672	1.667	1.658

$$\text{Average value of } \alpha = \frac{1.809 + 1.744 + 1.735 + 1.672 + 1.667 + 1.658}{6}$$

$$= 1.714$$

We have, $y = \alpha x / 1 + (\alpha - 1) x$

Generate the equilibrium data : Take x = 0, 0.10, 0.20 ... 1.0 and evaluate corresponding y values.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.16	0.30	0.42	0.53	0.63	0.72	0.80	0.873	0.94	1.0

(i) At total reflux, both the operating lines coincide with the diagonal, so construct the ideal stages starting from the point (0.96, 0.96) on the diagonal and ending or covering the point (0.07, 0.07).

Refer Fig. E 1.29 (a).

Number of ideal stages including reboiler = n = 11.

Number of ideal plates required in column at total reflux = 11 – 1 = 10 ...Ans.

(ii) Calculation of minimum flux :

Feed 30 % vaporised, f = fraction of feed vaporised = 0.30.

$$q = 1 - f$$

$$= 1 - 0.30 = 0.70$$

Feed is 30% vaporised and so 70% liquid.

$$q = \text{fraction of feed that is liquid}$$

$$q = 0.70 \ (= 70/100)$$

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slope of q-line/feed line $= -(q/1-q)$

$$\text{slope} = -\left(\frac{0.7}{1-0.7}\right) = -2.33$$

slope $= \tan\theta = -2.33$, $\theta = -66.80^{o} \approx 67^{o}$

Draw the feed line through the point (0.38, 0.38) on the diagonal with a slope equal to – 2.33 (with an angle – 67°)

or with intercept on y-axis $= \dfrac{x_F}{1-q}$

$= 0.38/(1-0.7)$

$= 1.267$

which will cut the equilibrium curve at point P.

Through the point A (0.96, 0.96) on the diagonal, draw the operating line A – P of the rectification section and read the coordinates x', y' of point P (the line A – P is not drawn as the figure may become crowdy).

Minimum reflux ratio (for partially vaporised feed) $= R_m = \dfrac{x_D - y'}{y' - x'}$

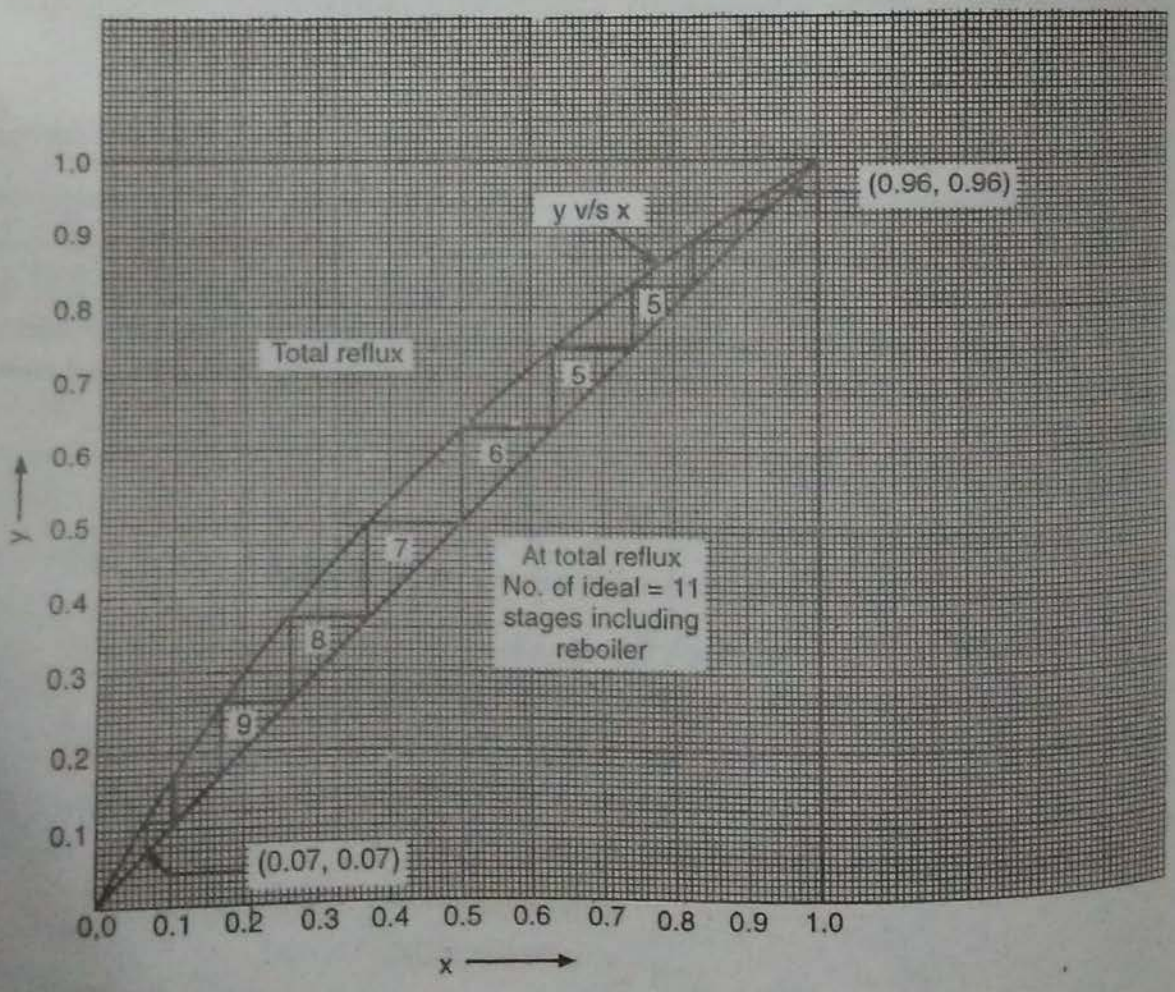

Fig. E 1.29 (a) : Determination of number of ideal plates at total reflux

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From Fig. E 1.29 (b) :

Coordinates of P are : $x' = 0.345$, $y' = 0.465$

We have, $x_D = 0.96$

$$R_m = \frac{0.96 - 0.465}{0.465 - 0.345} = 4.125$$

Minimum reflux ratio = 4.125 ... Ans.

Given : Operating reflux ratio = $2 R_m$

$R = 2 \times 4.125 = 8.25$

The intercept of the operating line of the rectification section on y-axis is

intercept = $x_D/(R + 1) = 0.96/(8.25 + 1) = 0.104$

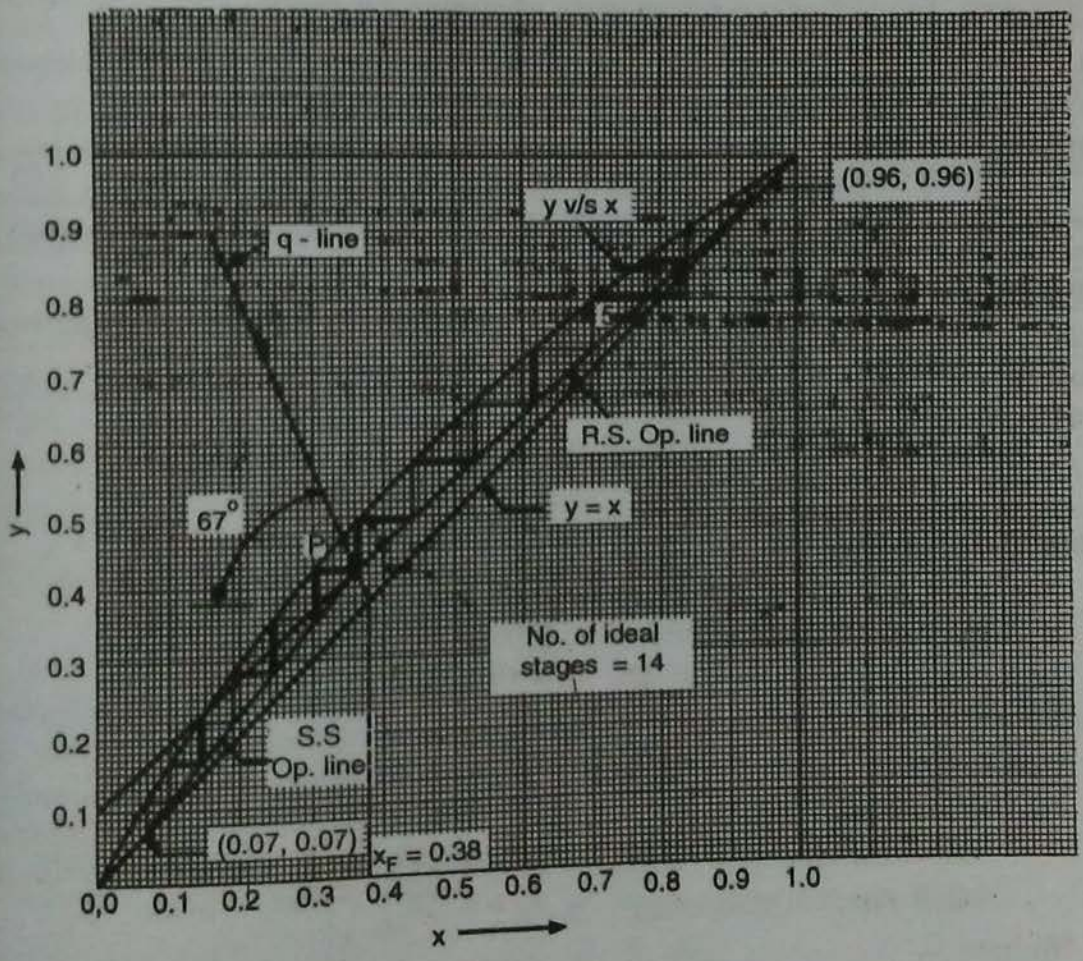

Fig. E 1.29 (b) : Determination of ideal stages/plates at $R = 2 R_m$

(i) The operating line of the rectification section is plotted through the point (0.96, 0.96) on the diagonal and with y-axis intercept = 0.104.

(ii) The operating line of the stripping section is plotted through the point (0.07, 0.07) on the diagonal and passing the point of intersection of q-line/feed line and operating line of the rectification section.

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(iii) Construct the stages starting from the point (0.96, 0.96) till the point (0.07, 0.07) is reached or covered.

From Fig. E 1.29 (b) :

Number of ideal plates/stages including reboiler = n + 1 = 14.

Number of ideal plates required in the column = n = 14 – 1 = 13.

Overall plate efficiency = 60%

Actual number of plates required in the column for the separation

= 13/0.60 = 21.67 ≈ 22 ... Ans.

Ex. 1.30 : A charge containing 55 mole % ethyl alcohol and 45 mole % water is distilled with a constant reflux ratio R of 2.1. What will be the average composition of the distillate and the heat required per kmol of product if the distillation is stopped when the composition in the still has fallen to 0.105 molefraction of alcohol ? A batch distillation column has four ideal plates.

The average latent heat of alcohol-water mixture is 4000 kJ/kmol.

Equilibrium data :

x	0	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.89
y	0	0.43	0.526	0.577	0.615	0.655	0.70	0.754	0.82	0.89

Solution : Basis : 100 kmol of charge.

This is a case of batch distillation with constant reflux ratio (resulting in continuous change in overhead composition with time of distillation).

The operating line is

$$y_{n+1} = \frac{R}{R+1} x_n + \frac{x_D}{R+1}$$

with point on diagonal (x_D, x_D) and slope = R/R + 1 and intercept = x_D/R + 1.

Reflux ratio = R = 2.1

Slope of all operating lines = R/R + 1 = 2.1/3.1 = 0.6774

Slope = $\tan\theta$ = 0.6774 $\therefore$ $\theta = 34°$

Number of ideal plates in the column = 4

Number of ideal stages including the reboiler = 4 + 1 = 5.

Initial charge to the unit = F = 100 kmol

Feed/charge composition = x_F = 55/100 = 0.55

Final residue composition = x_W = 0.105

We have :

$$\ln F/W = \int_{x_W}^{x_F} \frac{dx_W}{x_D - x_W}$$

$$\ln F/W = \int_{0.105}^{0.55} \frac{dx_W}{x_D - x_W}$$

The integral of the above equation is to be evaluated graphically to find out W.

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Material balance of ethyl alcohol :

$$x_F F = W x_W + D x_{D\,avg.}$$

$x_{D\,avg.}$ is the average composition of distillate.

Now to evaluate the integral, we have to generate data of x_D and corresponding x_W by plotting a number of operating lines of constant slope = 0.6774 to cover $x_W = x_F = 0.55$ (initial composition) and $x_W = 0.105$ (final composition).

So draw operating lines, parallel lines, of slope 0.6774 (at an angle of 34°) through point on the diagonal, read x_D, construct five stages, read corresponding x as x_W, and tabulate.

OR : Assume x_D (to cover both feed composition $x_F = 0.55$ and final residue composition $x_W = 0.105$), find intercept $(x_D/R + 1)$, plot operating line, construct five stages and read x_W. In this case, also lines are parallel to each other. This procedure is adopted.

For $x_D = 0.80$

$$\text{intercept} = x_D/(R+1) = 0.78/(2.1 + 1) = 0.2516$$

Plot the operating line through the point (0.78, 0.78) on the diagonal and with y-axis intercept equal to 0.2516.

Construct the five stages between the equilibrium curve and the operating line and read x_W. $\therefore x_{W1} = 0.64$ (construction is not shown).

Take $x_D = 0.78$, 0.75 and 0.72, adopt the same procedure to find x_W (construction of stages is shown for $x_D = 0.75$).

As the operating lines (lines parallel to each other) are very close to each other, lines are plotted on two equilibrium diagrams for clarity.

x_D	$x_D/(R+1)$	x_W	$x_D - x_W$	$1/(x_D - x_W)$
0.80	0.2581	0.64	0.16	6.25
0.78	0.2516	0.42	0.36	2.8
0.75	0.2419	0.13	0.62	1.61
0.72	0.232	0.05	0.67	1.49

$$\ln (F/W) = \int_{0.105}^{0.55} \frac{d x_W}{x_D - x_W}$$

Evaluate the integral graphically.

For evaluating integral graphically, plot $1/(x_D - x_W)$ against x_W and measure the area under the curve between the limits $x_W = 0.105$ (final residue composition) and $x_W = x_F = 0.55$ (composition of the charge).

From Fig. E 1.30 (c) :

Area under the curve = 1104 mm² = 11.04 cm²

$$\ln (F/W) = \text{Area} \times \text{scale y-axis} \times \text{scale x-axis}$$

$$= 11.04 \times \frac{1}{1} \times \frac{0.1}{1}$$

$$= 1.104$$

$$F/W = e^{1.104} = 3.0162$$

$$W = F/3.0162, \quad F = 100 \text{ kmol}$$

$\therefore$

$$= 100/3.0162$$

$$= 33.15 \text{ kmol}$$

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Overall material balance : $F = D + W$

$D = F - W = 100 - 33.15 = 66.85$ kmol

Material balance of ethyl alcohol :

$F x_F = W x_W + D x_{D\ avg.}$

$100 \times 0.55 = 33.15 \times 0.105 + 66.85 (x_{D\ avg.})$

$x_{D\ avg.} = 0.77066 \approx 0.771$

Average composition of distillate (composition of composited distillate) = 0.771 ... **Ans.**

$R = L/D$

$\therefore L = RD = 2.1 \times 66.85 = 140.385$ kmol

Reflux = 140.385 kmol, Distillate = 66.85 kmol

Latent heat of alcohol-water mixture = $\lambda = 4000$ kJ/kmol

Total heat required (during distillation) = Heat to provide reflux + Heat to provide distillate

$= L\lambda + D\lambda = \lambda (L + D)$

$= 4000 (140.385 + 66.85) = 828940$ kJ

Total heat required per kmol of product (distillate) = 828940/distillate

= 828940/66.85 = 12400 kJ = 12.4 MJ ... **Ans.**

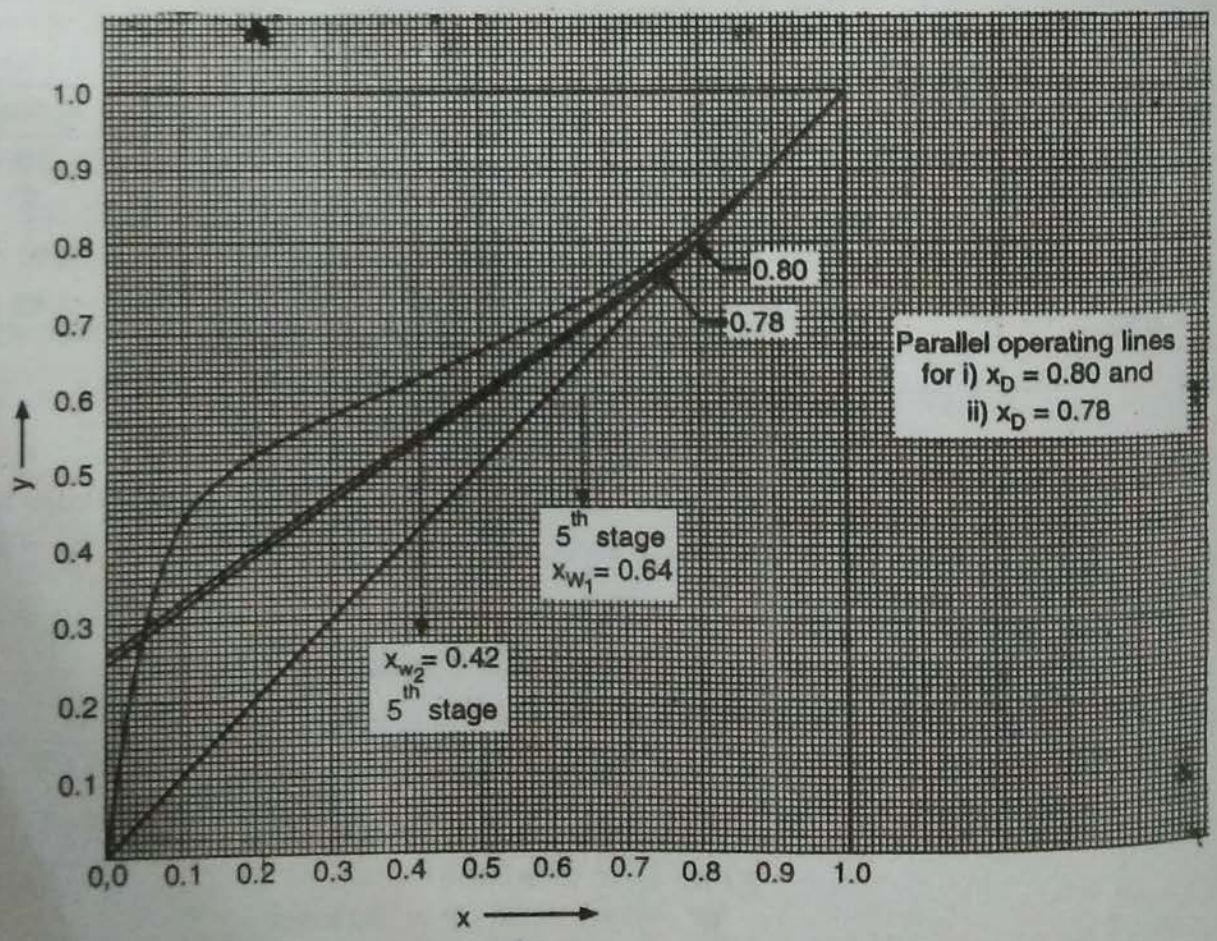

Fig. E 1.30 (a) : Batch distillation with constant R

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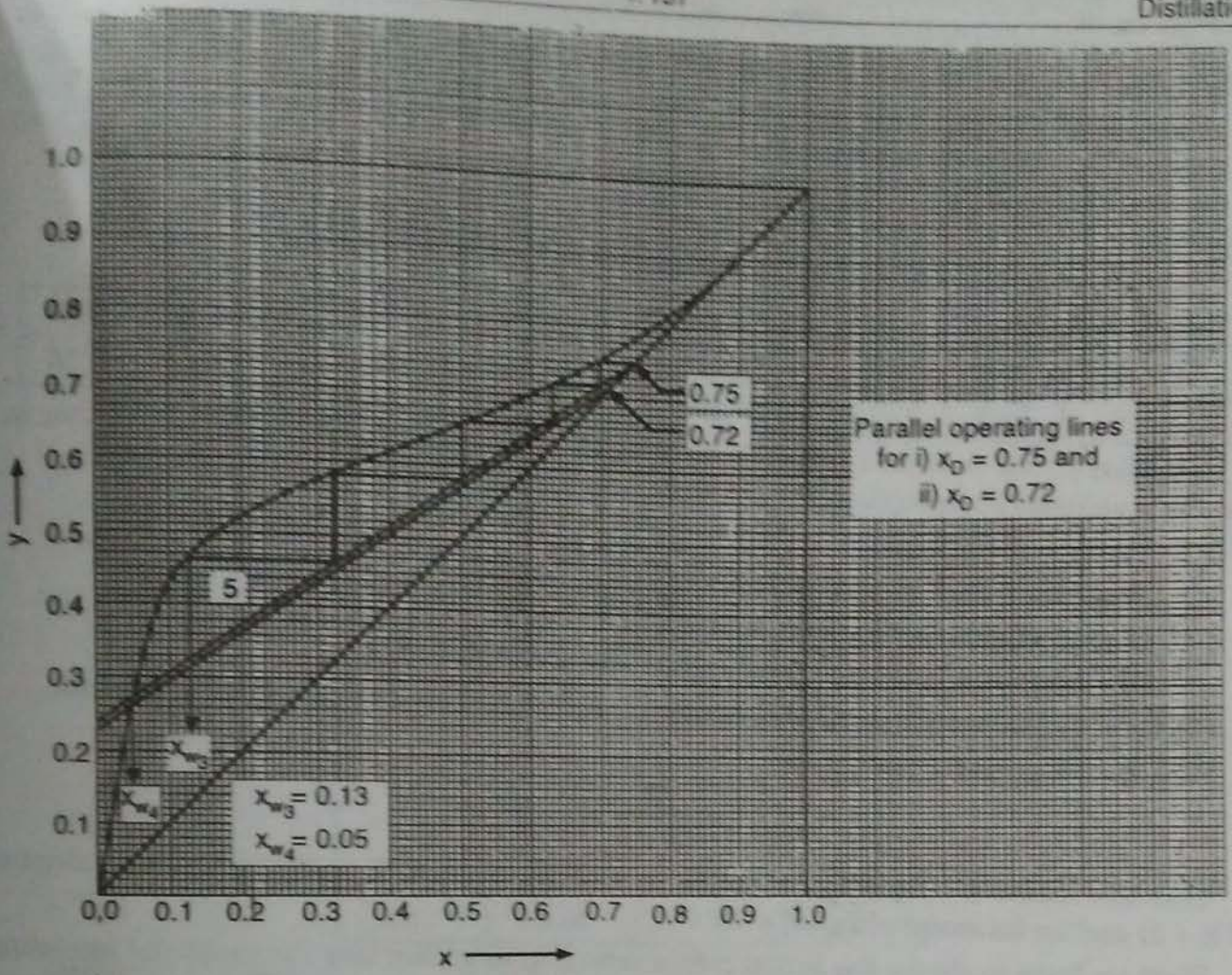

Fig. E 1.30 (b) : Batch distillation with constant R

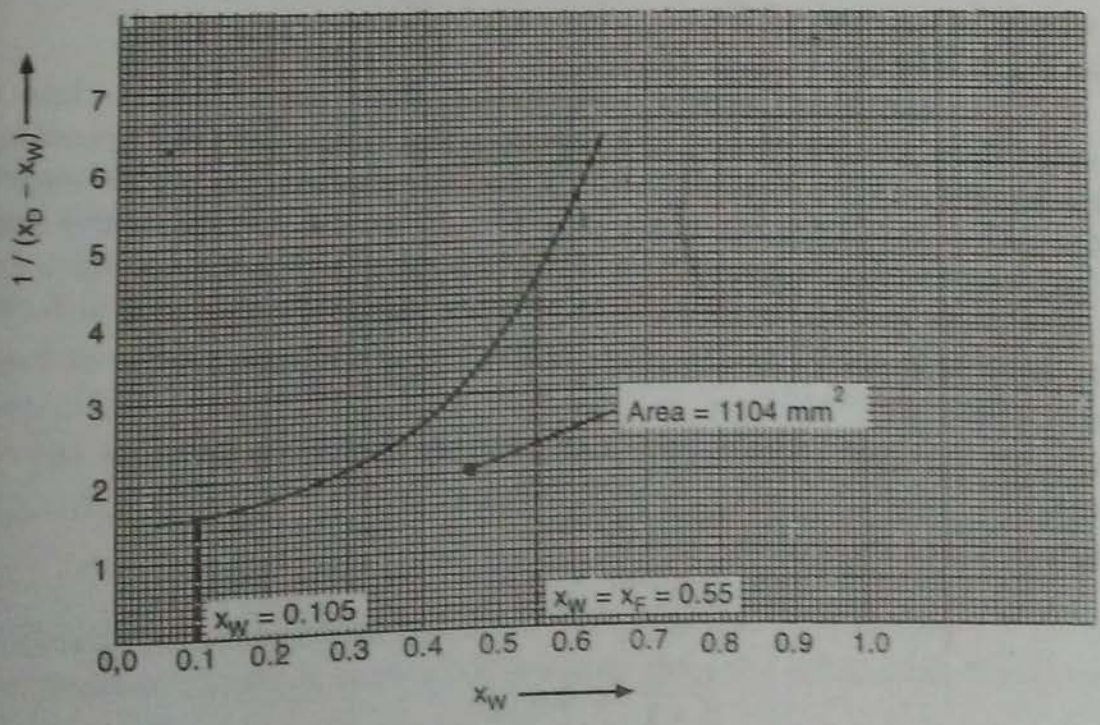

Fig. E 1.30 (c) : Batch distillation - evaluation of integral

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Ex. 1.31 : A charge containing 55 mole % ethyl alcohol and 45 mole % water is distilled to give a top product containing 75 mole % ethyl alcohol. The column has four ideal plates and the distillation has stopped when the reflux ratio has to be increased beyond 4.0. Find the amount of distillate obtained per 100 kmol of charge and heat required per mol of product.

The average latent heat of alcohol-water mixture is 4000 kJ/kmol.

Equilibrium data :

x	0	0.10	0.20	0.30	0.40	0.50	0.6	0.7	0.8	0.89
y	0	0.43	0.526	0.577	0.615	0.655	0.7	0.754	0.82	0.89

Solution : This is a case of batch distillation with varying reflux ratio (resulting in constant overhead composition).

Number of ideal plates in column = 4.

Number of ideal stages including reboiler = 4 + 1 = 5.

Overhead composition (wrt MVC) = x_D = 75/100 = 0.75

Feed/charge composition (wrt MVC) = x_F = 55/100 = 0.55.

We have to find out the distillate obtained and the heat required per kmol of product.

The operating line equation is

$$y_{n+1} = \frac{R}{R+1} x_n + \frac{x_D}{R+1}$$

This line intersects the diagonal at the point P (x_D, x_D) [i.e., P (0.75, 0.75)], has a slope of R/(R + 1) and an intercept of x_D/(R + 1).

First we have to choose the initial reflux ratio (R_1) in such a way that with the operating line corresponding to R_1, we can construct five ideal stages starting from x_D (0.75) and ending exactly at $x_{W1} = x_F = 0.55$.

Refer Fig. E 1.30 (a).

For this you can draw random lines through the point P such as line PQ (other construction lines are not shown) and construct the stages between operating line and equilibrium curve and see whether we are ending the construction of fifth stage exactly at $x_{W1} = x_F = 0.55$ or not. If not, repeat the procedure till our construction of five stages exactly ends at $x_{W1} = x_F = 0.55$. The construction of stages is not shown on the x-y plot to avoid it to be crowdy as we have to draw operating lines for other values of reflux ratio also.

The operating line PQ : exactly gives us five stages and construction ending exactly at $x_{W1} = x_F = 0.55$.

Remember that such cases require high accuracy in plotting the curve and constructing the stages.

Intercept of line PQ on y-axis = 0.40.

$$\frac{x_D}{R+1} = 0.4$$

$$\frac{0.75}{R+1} = 0.4$$

$$R_1 = R = 0.875 \text{ (initial reflux ratio)}$$

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Now assume various values for R till R = 4 (given), find either slope or y-axis intercept. Plot the operating lines, construct five stages, read x_W corresponding to the fifth stage and tabulate.

For

$$R_2 = R = 1.5$$

$$\text{intercept} = x_D/R + 1$$

$$= 0.75/2.5 = 0.30$$

Draw the operating line through the point P and with intercept on y-axis = 0.30. Construct five stages and read the corresponding x_{W2}, construction not shown.

$$x_{W2} = 0.35$$

$$\text{For } R_3 = R = 2$$

$$\text{intercept} = x_D/R + 1 = 0.75/3 = 0.25$$

and

$$x_{W3} = 0.19$$

$$\text{For } R_4 = R = 4$$

$$\text{intercept} = x_D/R + 1 = 0.75/5 = 0.15$$

$$x_{W4} = 0.06$$

R	x_W	x_D	D
0.875	0.55	0.75	0
1.5	0.35	0.75	50
2	0.19	0.75	64.3
4	0.06	0.75	71

Evaluate D using

$$\frac{D}{F} = \frac{x_F - x_W}{x_D - x_W}$$

$$D = F\left(\frac{x_F - x_W}{x_D - x_W}\right)$$

where $F = 100$ kmol (charge) and $x_D = 0.75$

For $x_{W1} = x_F = 0.55$, $D = 0$

For $x_{W2} = 0.35$, $D = 100\left(\frac{0.55 - 0.35}{0.75 - 0.35}\right) = 50$ kmol

For $x_{W3} = 0.19$, $D = 100\left(\frac{0.55 - 0.19}{0.75 - 0.19}\right) = 64.3$ kmol

For $x_{W4} = 0.06$, $D = 100\left(\frac{0.55 - 0.06}{0.75 - 0.06}\right) = 71.0$ kmol

Distillate obtained = D = 71 kmol ... **Ans.**

If the reflux ratio is assumed to be adjusted continuously so as to keep the top product of constant composition (constant quality) then at any time, the reflux ratio R is given by

$$R = \frac{dL}{dD}$$

$$dL = R\, dD$$

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$$\int_0^L dL = \int_{R_1}^{R_4} R\,dD$$

$$L = \int_{R_1}^{R_4} R\,dD = \int_{0.875}^{4} R\,dD$$

Plot D v/s R and the area above the curve - bounded by the curve and y-axis gives the value of the integral of right hand side.

From Fig. E 1.31 (b).

Area above the curve = 960 mm^2 = 9.6 cm^2

$$L = \int_{0.875}^{4} R\,dD$$

= 9.60 × scale y-axis × scale x-axis

$$= 9.60 \times \frac{10}{1} \times \frac{1}{1}$$

= 96 kmol

To provide reflux dL, the heat to be removed in the condenser is λ dL.

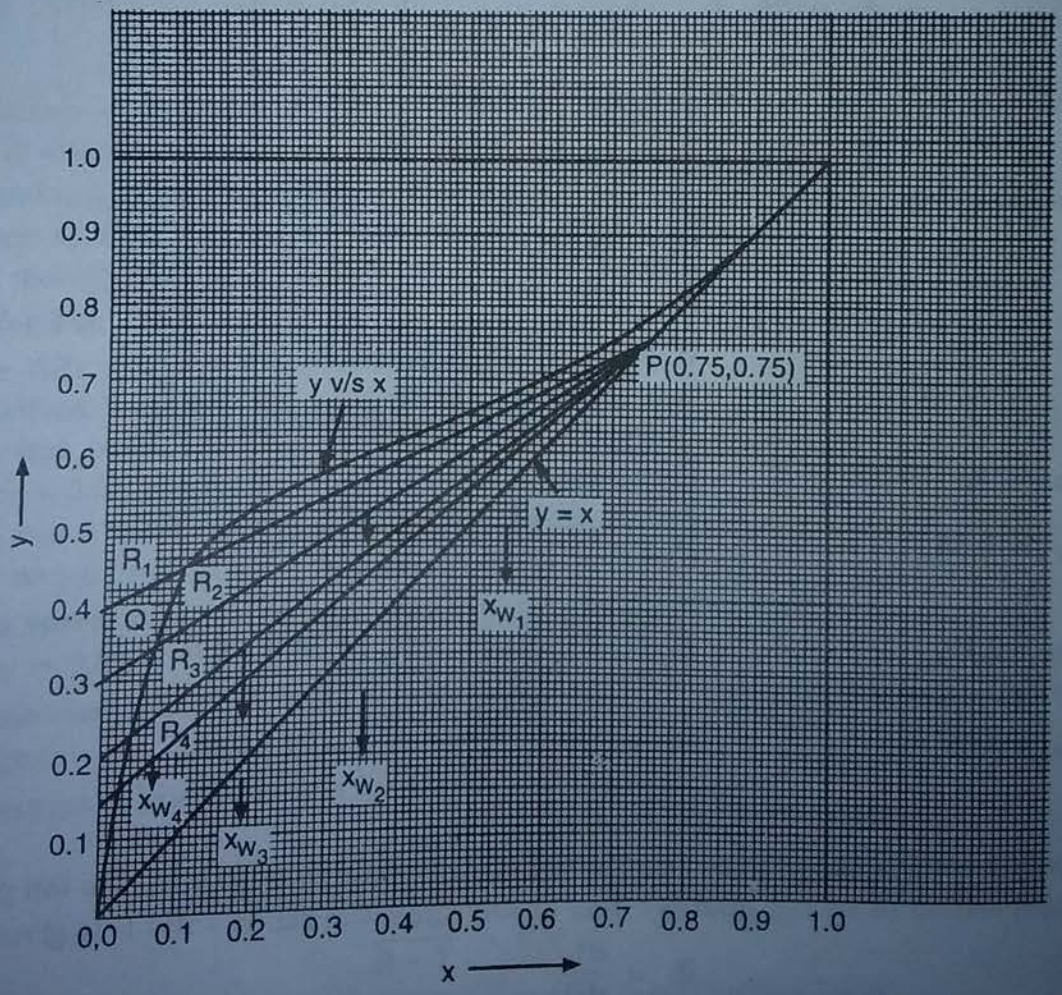

Fig. E 1.31 (a) : Batch distillation-constant x_D determination of x_W for various R values

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Therefore, heat to be supplied to the reboiler to provide this reflux during total distillation operation is

$$Q_1 = \lambda \int_0^L dL = \lambda \int_{R_1}^{R_4} R\, dD$$

λ = 4000 kJ/mol (given)

$Q_1 = \lambda L = 4000 \times 96 = 384000$ kJ

Heat to be supplied to the reboiler to provide total distillate $= Q_2 = \lambda \cdot D$

$= 4000 \times 71$

$= 284000$ kJ

Total heat to be supplied (provided) to the reboiler $= Q_1 + Q_2$

$= 384000 + 284000$

$= 668000$ kJ

$\therefore$ Heat to be provided/supplied to the reboiler per kmol of product $= \frac{668000}{71} = 9408$ kJ

≈ 9.4 MJ ... Ans.

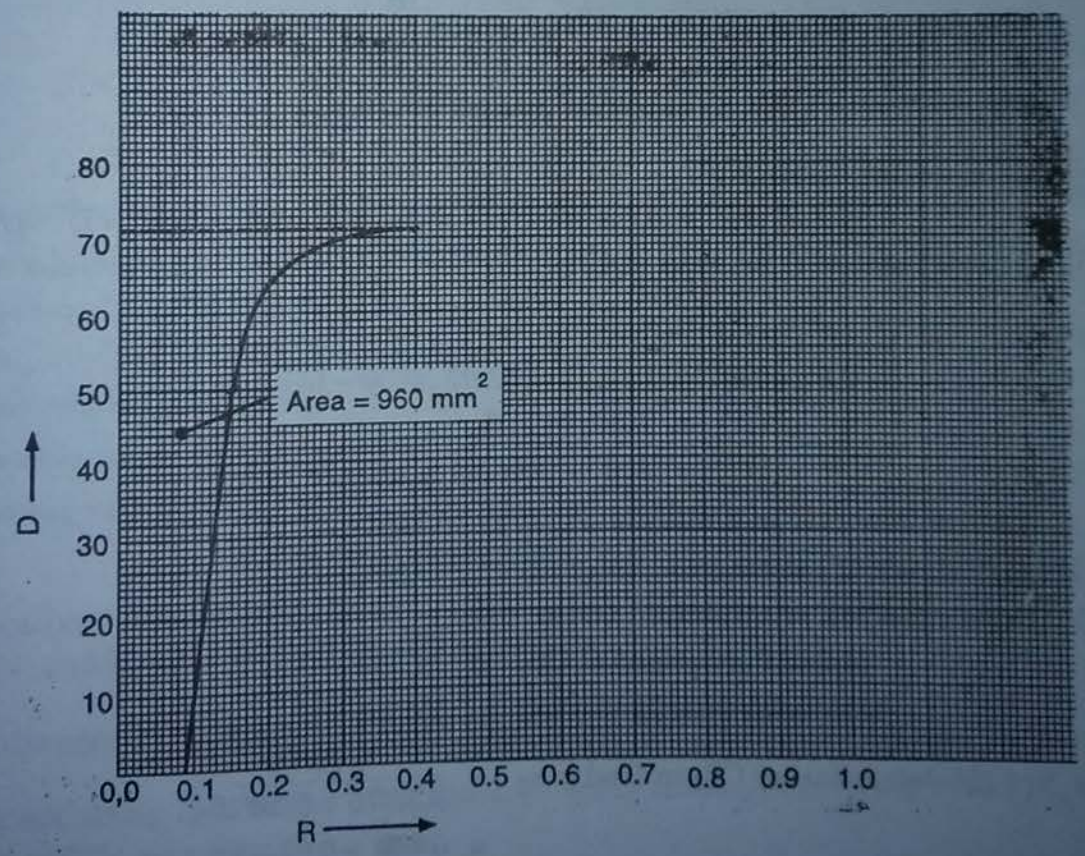

Fig. E 1.31 (b) : Graphical integration for determination of L

Ex. 1.32 : A solution containing CCl_4 and CS_2 in equal masses is to be continuously fractionated at the rate of 4000 kg/h. The overhead product is to contain 95 weight % CS_2 and the residue 0.5 weight % CS_2. The feed will be 30 mole % vaporised before it enters the

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column. The total condenser is used, and reflux is returned to the column at bubble point. Equilibrium data (x, y = molefraction of CS_2) are as follows :

T, °C	76.7	74.9	70.3	63.8	59.3	55.3	50.4	48.5	46.3
x	0	0.02960	0.1106	0.2585	0.3908	0.5318	0.7574	0.8604	1.0
y^*	0	0.0823	0.2660	0.4950	0.6340	0.7470	0.8780	0.9320	1.0

(i) Determine the product rates.

(ii) Determine the minimum reflux ratio.

(iii) Determine the number of theoretical plates if the reflux ratio is twice the minimum.

(iv) Determine the minimum number of plates graphically and by using Fenske's equation.

Solution : Basis : 4000 kg/h of feed to the column containing CS_2 and CCl_4.

$$F' = 4000 \text{ kg/h}$$

Let D' and W' be the mass flow rates of distillate and bottom product respectively.

As the feed to the column contains equal amounts of CS_2 and CCl_4 on the weight basis, we have 50 weight % CS_2 and 50 weight % CCl_4 in feed.

Weight fraction of CS_2 in feed $= x'_F = \frac{50}{100} = 0.50.$

Weight fraction of CS_2 in distillate $= x'_D = \frac{95}{100} = 0.95.$

Weight fraction of CS_2 in bottom product $= x'_W = \frac{0.5}{100} = 0.005.$

Overall material balance :

$$F' = D' + W', \qquad \therefore W' = F' - D' = 4000 - D', \text{ kg/h}$$

(i) Overall material balance of CS_2 :

$$x'_F F' = x'_D D' + x'_W W'$$

$$0.5 \times 4000 = 0.95\, D' + 0.005\,(4000 - D')$$

$$\therefore \quad D' = 2095.24 \text{ kg/h}$$

$$W' = 4000 - 2095.24 = 1904.76 \text{ kg/h}$$

Mass flow rate of distillate = 2095.24 kg/h

Mass flow rate of bottom product = 1904.76 kg/h ... Ans. (i)

(ii) Determination of minimum reflux ratio :

Molecular weight of CS_2 = 76

Molecular weight of CCl_4 = 154

$$\text{Molefraction of } CS_2 \text{ in feed} = x_F = \frac{0.50/76}{0.50/76 + 0.50/154}$$

$$= 0.669 \simeq 0.67$$

$$\text{Molefraction of } CS_2 \text{ in distillate} = x_D = \frac{0.95/76}{0.95/76 + 0.05/154}$$

$$= 0.975$$

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Molefraction of CS_2 in bottom product = $x_W = \dfrac{0.005/76}{0.005/76 + 0.995/154}$

$= 0.01$

The point of intersection of q-line and diagonal is P (x_F, x_F) i.e. P (0.67, 0.67).

Refer Fig. E 1.32 (a).

Given : Feed is 30 mole % vaporised (so 70% liquid and 30% vapour).

q = fraction of feed that is liquid

$q = 1 - 0.30 = 0.7$

or $q = 70/100 = 0.70$

$$\text{slope of q-line} = \frac{-q}{1-q} = \frac{-0.7}{1-0.7} = -2.3333$$

$$\text{slope} = \tan\theta = -2.3333 \quad \therefore \quad \theta = -66.8^\circ = -67^\circ$$

So through the point P (0.67, 0.67) on the diagonal, draw the q-line with a slope equal to – 2.3333, i.e., at an angle of – 67°, to cut the equilibrium curve at point Q. Read the coordinates of the point Q.

From Fig. E 1.32 (a), $y' = 0.80$ and $x' = 0.615$

Minimum reflux ratio for a partially vaporised feed is given by

$$R_m = \frac{x_D - y'}{y' - x'}$$

$$= \frac{0.975 - 0.80}{0.80 - 0.615} = 0.946$$

Minimum reflux ratio = $R_m = 0.946$ ... **Ans. (ii)**

[For determining R_m, we have to draw the operating line AQ for the rectification section. The slope of this line is $R_m/(R_m + 1) = x_D - y'/(x_D - x')$ which on solving for R_m gives $R_m = (x_D - y')/(y' - x')$. Line AQ is not shown in Fig. E 1.32 (a) for the sake of clarity of other construction work.]

(iii) Determination of number of ideal plates :

Operating reflux ratio = $R = 2\,R_m$

$R = 2 \times 0.946 = 1.892$

The operating line of rectifying section intersects the diagonal at the point A (0.975, 0.975) and has a slope of $R/(R + 1)$ and intercept of $x_D/(R + 1)$.

$$\text{Intercept on y-axis} = \frac{x_D}{R+1} = \frac{0.975}{1.892 + 1} = 0.337 \approx 0.34$$

Draw the operating line of the rectification section through the point A (0.975, 0.975) with intercept of 0.34 on y-axis.

The point of intersection of the operating line of the stripping section with the diagonal is the point B (0.01, 0.01). Draw this operating line through the point B (0.01, 0.01) and passing through the point of intersection of the top operating line and the q-line.

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Construct the ideal stages starting from A and reaching exactly or covering B and count them. Construction of the last stage (stage 12) is not clearly seen in the diagram as three corners of the triangle representing the stage are very close to each other.

Number of ideal stages/plates including reboiler = n + 1 = 12 ... Ans.

Number of ideal plates required (in column) = n = 12 – 1 = 11 ... Ans. (iii)

(iv) Determination of minimum number of plates :

Minimum number of ideal plates results at infinite reflux ratio.

At total reflux ratio, both the operating lines coincide with the diagonal.

So to find the minimum number of ideal plates, construct the ideal stages between the equilibrium curve and the diagonal starting from the point A (0.975, 0.975) and ending exactly or covering the point B (0.01, 0.01).

Minimum no. of ideal plates including reboiler (graphically) = n_m +1 =9 ... Ans. (iv)

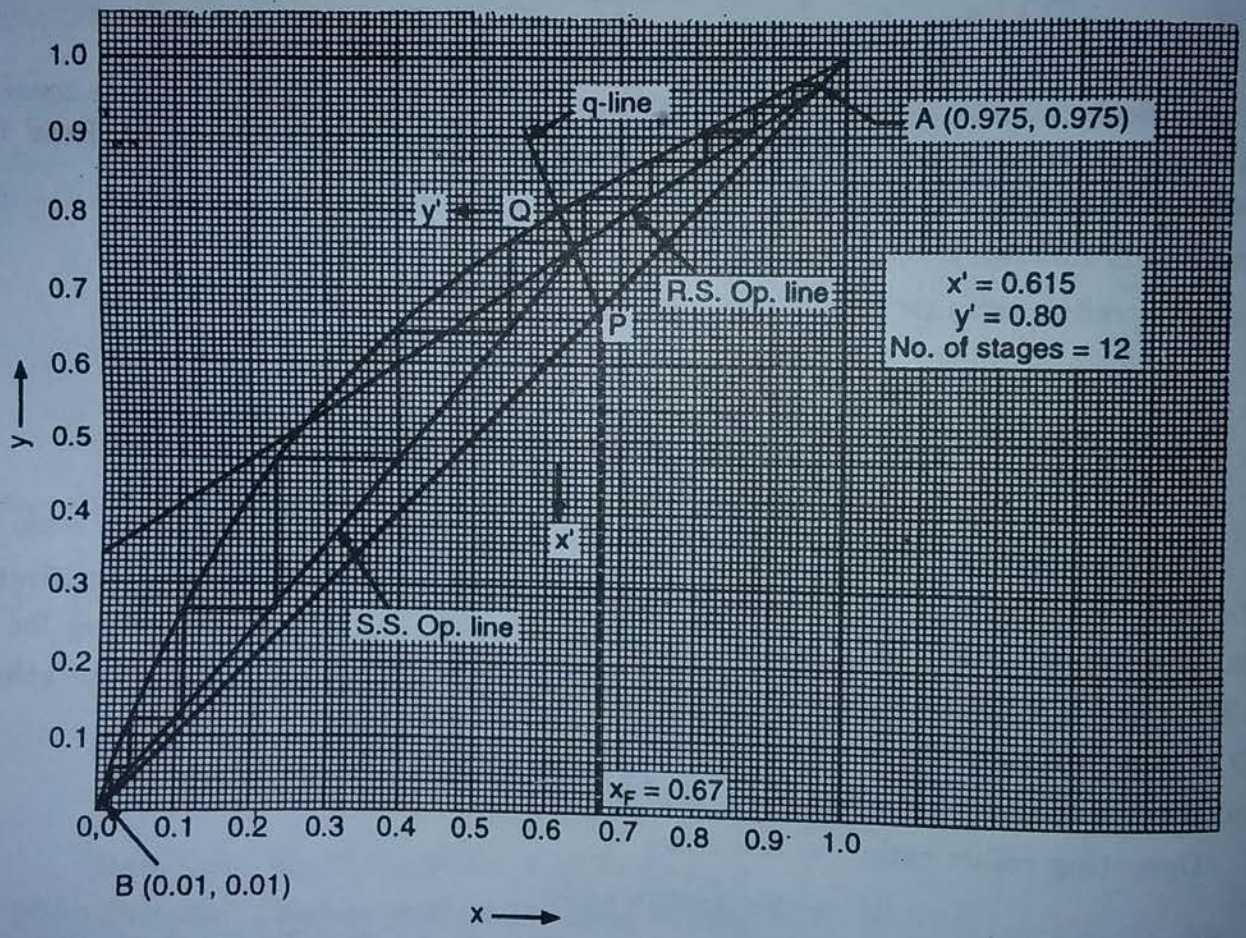

Fig. E 1.32 (a) : Determination of ideal plates (at R = 2 R_m) and R_m

(v) Minimum number of ideal plates using the Fenske's equation :

For this we have to obtain the average value of relativity from the equilibrium data.

$$\alpha = \frac{y^* (1 - x)}{x (1 - y^*)}$$

Find α for each pair of $x - y^*$.

x	0.0296	0.1106	0.2585	0.3908	0.5318	0.7574	0.8604
y^*	0.0823	0.2660	0.4950	0.6340	0.7470	0.8780	0.9320
α	2.94	2.914	2.812	2.7	2.60	2.305	2.224

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The average value of relative volatility is :

$$\alpha_{avg.} = \frac{2.94 + 2.914 + 2.812 + 2.7 + 2.6 + 2.305 + 2.224}{7}$$

$$= 2.64$$

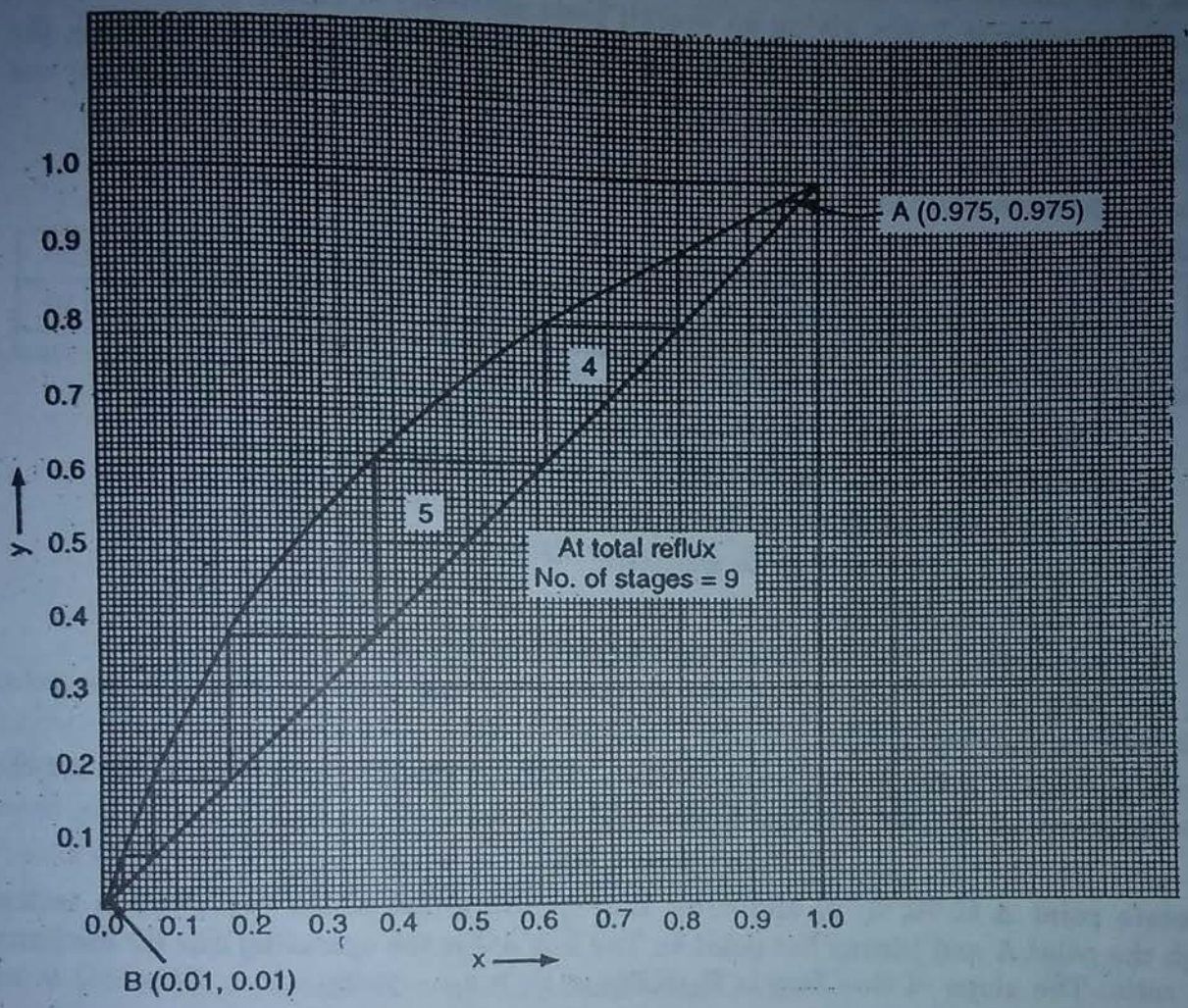

Fig. 1.32 (b) : Determination of the minimum number of ideal plates

As per Fenske's method, the minimum number of ideal stages including reboiler (i.e., to produce products x_D and x_W) are given by :

$$n_m + 1 = \frac{\ln\left[\left(\frac{x_D}{1 - x_D}\right)\left(\frac{1 - x_W}{x_W}\right)\right]}{\ln(\alpha_{avg.})}$$

$$n_m + 1 = \frac{\ln\left[\left(\frac{0.975}{1 - 0.975}\right)\left(\frac{1 - 0.01}{0.01}\right)\right]}{\ln(2.64)}$$

$$= 8.51$$

Minimum number of ideal plates including reboiler (graphically) = $n_m + 1 = 9$

Minimum number of ideal plates including reboiler (by Fenske's method) = $n_m + 1 = 8.51$

... **Ans.**

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The minimum number of ideal plates obtained graphically and by using Fenske's equation are almost same.

Ex. 1.33 : It is desired to concentrate a mixture of ethanol and water from 40 mole % to 70 mole % ethanol. A fractionating column 1.2 m in diameter and having 10 plates is available. It is known that the minimum superficial vapour velocity in the column at atmospheric pressure is 1 m/s, giving an overall plate efficiency of 50%.

Assuming that the mixture is fed to the column as a boiling liquid and using the operating reflux ratio of twice the minimum value, determine the feed plate location and distillate mass flow rate.

Assume top temperature to be 353 K.

Equilibrium data :

x	0	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.89
y	0	0.43	0.526	0.577	0.615	0.655	0.7	0.754	0.82	0.89

Solution :

Molefraction of ethanol in feed = x_F = 40/100 = 0.40

Molefraction of ethanol in distillate = x_D = 70/100 = 0.70

Actual plates in the column = 10

Overall plate efficiency = 50%

Number of ideal plates in the column = $0.50 \times 10 = 5$

(i) Determination of the minimum reflux ratio. Plot the x-y data.

As the feed is a boiling liquid, the q-line is vertical passing through the point P (0.40, 0.40) on the diagonal.

Draw a vertical through the point P (0.40, 0.40) on the diagonal which will cut the equilibrium curve at Q. Read the coordinates of point Q.

$$x_F = 0.40, \quad y_F = 0.615$$

[Locate point A (0.70, 0.70) and draw the operating line of the rectification section through the point A and joining the point Q. The line AQ is the operating line for minimum reflux ratio. The slope of this line is $R_m/(R_m + 1) = x_D - y_F/x_D - x_F$. (line AQ is not shown)]. Solving for R_m, we get :

$$R_m = \frac{x_D - y_F}{y_F - x_F} = \frac{0.70 - 0.615}{0.615 - 0.40} = 0.395$$

Operating reflux ratio = $R = 2 \times 0.395 = 0.79$

The operating line of rectification section has the point P (0.70, 0.70) on the diagonal.

The intercept of this line (on y-axis) is

$$\text{Intercept} = \frac{x_D}{R + 1} = \frac{0.70}{0.79 + 1} = 0.391$$

Draw the operating line of the rectification section through point P (0.70, 0.70) on the diagonal and with y-axis intercept = 0.391.

Construct the stages till the point $x_F = 0.4$ is covered and count the ideal stages, the triangle within which the q-line falls represents the feed plate.

Feed plate location : 4^{th} from the top. ... Ans.

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Distillate contains 70 mole % ethyl alcohol

Molefraction of ethyl alcohol in distillate = 0.70

Molefraction of water in distillate = 1 – 0.70 = 0.30

Average molecular weight of product/distilate = $0.70 \times 46 + 0.30 \times 18 = 37.6$ kg/kmol

Vapour velocity based on empty column cross-section = 1.0 m/s

Cross-sectional area of column = $\pi/4\ (1.2)^2$

= 1.13 m^2

Volumetric vapour flow rate at the top = 1.0×1.13

= 1.13 m^3/s

Top temperature = T = 353 K

Pressure = P = 1 atm = 101.325 kPa

Assume that vapour follows ideal gas law.

$PV' = n'RT$

n' = molar flow rate of vapour at top, kmol/s

V' = volumetric flow rate of vapour at top, m^3/s

R = 8.31451 kPa · m^3/(kmol.K)

$n' = PV'/RT$

$= 101.325 \times 1.13/8.31451 \times 353$

= 0.039 kmol/s

For constant molar overflow - molar vapour flow rate and molar liquid flow rate above feed plate each are constant throughout the top section.

Molar flow rate of vapour at top = V = 0.039 kmol/s.

Slope of the operating line for the rectification section :

$$\frac{L}{V} = \frac{R}{R+1} = \frac{0.79}{1+0.79} = 0.4413$$

$L = 0.4413\ V = 0.4413 \times 0.039 = 0.0172$ kmol/s

At $R = 0.79$

$L/D = 0.79$

$D = L/0.79 = 0.0172/0.79 = 0.022$ kmol/s

or $R = L/D$

$D = L/R$

We have, $V = L + D \quad \therefore L = V - D$

$$D = \frac{V - D}{R}$$

$DR = V - D$

$D(1 + R) = V$

$\therefore D = V/(1 + R)$

$= 0.039/(1 + 0.79) = 0.022$ kmol/s

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Molar flow rate of distillate = 0.022 kmol/s ... Ans.

Average molecular weight of distillate = 37.6 kg/kmol

Mass flow rate of distillate = 0.022×37.6

= 0.827 kg/s ... Ans.

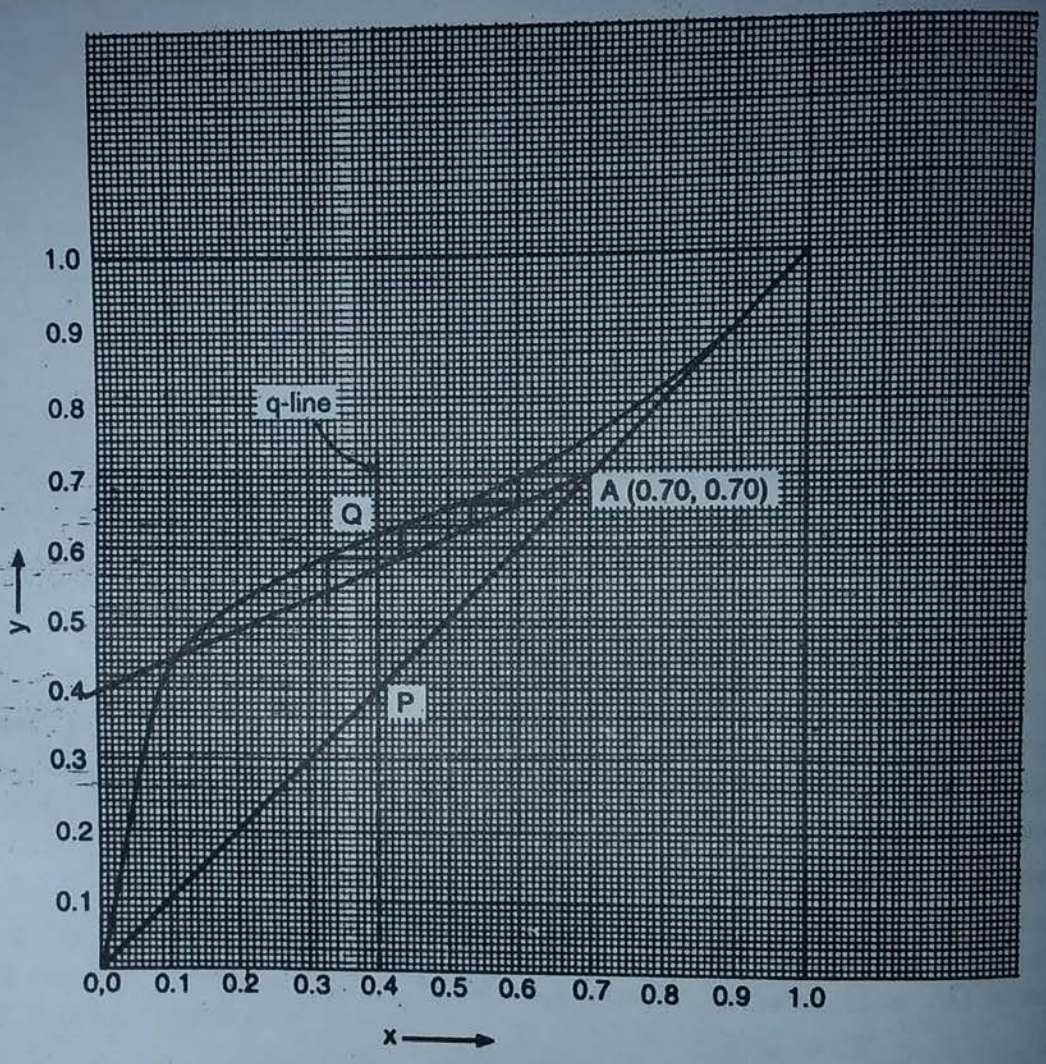

Fig. E 1.33 : Determination of R_m and feed plate location

Ex. 1.34 : A feed mixture containing 42 mole % heptane and 58 mole % ethylbenzene is to be fractionated at 101.325 kPa to produce a distillate containing 97 mole % heptane and the residue containing 99 mole % ethylbenzene.

Calculate : (i) The minimum reflux ratio.

(ii) The number of theoretical stages for R = 2.5.

(iii) The minimum number of theoretical stages at total reflux.

(iv) The heat load on condenser and reboiler in kJ/h for a feed rate of 5000 kg/h.

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Vapour-liquid liquid equilibria and enthalpy data for heptane-ethylbenzene at 101.325 kPa are :

T °C	x mole %	y mole %	H_L kJ/kmol	H_V kJ/kmol
136.2	0	0	24787	60795
129.5	8	23.2	23853	59468
122.9	18.5	42.8	23112	58283
119.7	25.1	51.4	22777	58074
116	33.5	60.8	22442	56985
110.8	48.7	72.9	22024	56315
106.2	65.1	83.4	21647	54891
103	78.8	90.4	21605	54180
100.2	91.4	96.3	21454	53527
98.5	100	100	21354	53070

Chemical formula of heptane = C_7H_{16}

Chemical formula of ethylbenzene = $C_6H_5C_2H_5$

Solution : Basis : 5000 kg/h of feed mixture.

Heptane is more volatile component.

Molefraction of heptane in feed = $x_F = 42/100 = 0.42$

Mole % ethylbenzene in residue = 99%.

Mole % heptane in residue = 100 – 99 = 1 %

Molefraction of heptane in residue = $x_W = 1/100 = 0.01$

Mole % heptane in distillate = 97.

Molefraction of heptane in distillate = $x_D = 97/100 = 0.97$

Molecular weight of heptane = $C_7H_{16} = 100$

Molecular weight of ethylbenzene = $C_6H_5C_2H_5 = 106$

Average molecular weight of feed = $M_{avg.} = \Sigma m_i x_i$

$$M_{avg.} = 0.42 \times 100 + 0.58 \times 106 = 103.48 \text{ kg/kmol}$$

Molar flow rate of feed = F = 5000/103.48

$$= 48.318 \approx 48.32 \text{ kmol/h}$$

Overall material balance :

$$F = D + W$$

$$W = F - D = 48.32 - D, \text{ kmol/h}$$

Material balance of heptane :

$$x_F F = x_D D + x_W W$$

$$x_F F = x_D D + x_W (48.32 - D)$$

$$0.42 \times 48.32 = 0.97 D + 0.01 (48.32 - D)$$

$$D = 20.6366 \approx 20.64 \text{ kmol/h}$$

$$W = 48.32 - 20.64 = 27.68 \text{ kmol/h}$$

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(a) Determination of minimum reflux ratio (R_m) :

Feed – saturated liquid with $x_F = 0.42$.

Reflux and distillate – saturated liquid with $x_D = 0.97$.

Vapour leaving the column from top - saturated vapour with $y_1 = x_D = 0.97$.

(i) Plot Hxy diagram : H_L v/s x and H_V v/s y.

(ii) Plot x-y diagram [Fig. E 1.34 (b)].

(iii) From x-y diagram read/obtain y_F -composition of vapour in equilibrium with feed composition ($x_F = 0.42$). From x-y plot [Fig. E 1.34 (b)], for $x_F = 0.42$, $y_F = 0.68$.

(iv) On the Hxy diagram, draw a vertical at $x = x_D = 0.97$ as shown in Fig. E 1.34 (a) or E 1.34 (c). The points L_0 and D will be at same location on Hxy diagram – as being saturated liquid. Mark L_0 at which the vertical at $x = x_D = 0.97$ cuts the H_L v/s x curve.

As the vapour at the top is a saturated vapour, mark V_1 at which a vertical through $x = x_D = 0.97$ cuts the H_V v/s y curve.

(v) The feed is a saturated liquid with $x_F = 0.42$. So mark the point F on H_L v/s x curve at which a vertical through $x = x_F = 0.42$ cuts the H_L v/s x curve.

(vi) Mark point E at on H_V v/s y curve at $y_F = 0.68$.

(vii) **For minimum reflux,** the tie line passes through the point F. The tie line joining equilibrium vapour and liquid phase composition passes through the point F.

(viii) So draw a tie line FE and extend it to cut the vertical at $x = x_D = 0.97$ at point $\Delta_{D\,min}$.

(ix) Read the ordinates of the points $\Delta_{D\,min}$, V_1 and L_0.

From Fig. E 1.34 (a) or (c), y-coordinate of $\Delta_{D\,min} = Q'_m = 94000$ kJ/kmol.

In Fig. E 1.34 (c), as both the H_L v/s x and H_V v/s y are almost flat over the region $x = y = 0.97$ and $x = y = 1.0$, it is difficult to read exact values of H_{V1} and H_{L0} when we take a small scale for y-coordinate. If we take a large scale for y-coordinate then the point $\Delta_{D\,min}$ will go out of graph paper (as well as the points Δ_D and Δ_W required for construction of stages).

So using interpolation : Calculation of H_{V1} :

For $y = 0.963$ $H_V = 53527$

For $y = 0.97$ $H_{V1} = ?$

For $y = 1.0$ $H_V = 53070$

$$\frac{0.97 - 0.963}{1 - 0.97} = \frac{H_{V1} - 53527}{53070 - H_{V1}}$$

$$\therefore \quad H_{V1} = 53440 \text{ kJ/kmol} \qquad \ldots (1)$$

Extra : From the Hxy diagram drawn with an enlarged scale for y-coordinate, we can read H_{V1}.

From Fig. E 1.34 (a).

For $y_1 = x_D = 0.97$

$$H_{V1} = \frac{160.3}{30} \times 10000 = 53433 \text{ kcal/kmol} \qquad \ldots (2)$$

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So the value of H_{V1} from (1) and (2) is almost same.

Calculation of H_{L0} by interpolation :

For	$x = 0.914$	H_L	$= 21454$
For	$x = 0.97$	H_{L_0}	$= ?$
For	$x = 1.0$	H_L	$= 21354$

$$\frac{0.97 - 0.914}{1 - 0.97} = \frac{H_{L0} - 21454}{21354 - H_{L0}}$$

$$\therefore \quad H_{L_0} = 21389 \text{ kJ/kmol}$$

Extra : From the Hxy diagram with an enlarged scale for y-coordinate [Fig. E 1.34 (a)], reading an exact value of H_{L0} is still difficult.

From Fig. E 1.34 (c) or (a) : Read y-coordinate of $\Delta_{D\ min}$.

$$Q'_m = 94000 \text{ kJ/kmol}$$

From Fig. E 1.34 (c) :

$$Q'_m = 94 \text{ mm} \times \frac{10000}{10 \text{ mm}} = 94000 \text{ kJ/kmol}$$

Fig. E 1.34 (a) is drawn only for the sake of clarity as far as the points L_0, V_1, W, F are concerned.

$$R_m = \frac{Q'_m - H_{V1}}{H_{V1} - H_{L0}}$$

$Q'_m = 94000$ kJ/kmol, $H_{L0} = 21389$ kJ/kmol

$H_{V1} = 53440$ kJ/kmol

$$R_m = \frac{94000 - 53440}{53440 - 21389}$$

$$= 1.265$$

We will refer to Fig. E 1.34 (c) :

$\therefore$ Minimum reflux ratio $= R_m = 1.265$ ... Ans. (i)

Given : Reflux ratio $= R = 2.5$

$$R = \frac{L_0}{D} = \frac{Q' - H_{V1}}{H_{V1} - H_{L0}}$$

$H_{V1} = 53440$ kJ/kmol, $H_{L0} = 21389$ kJ/kmol

$$2.5 = \frac{Q' - 53440}{53440 - 21389}$$

$$Q' = 133567.5 \text{ kJ/kmol}$$

$$Q' = H_D + (Q_C/D) = 133567.5 \text{ kJ/kmol}$$

The coordinates of the point Δ_D are x_D, Q'.

The coordinates of the point Δ_W are x_W, Q''.

The coordinates of the point F are x_F, H_F.

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(iii) Determination of number of stages : [Refer Fig. E 1.34 (c)]

(i) Through $x = x_D = 0.97$, draw a vertical and locate L_0 on H_L v/s x curve, V_1 on H_V v/s y curve at which this vertical cuts the respective curves.

(ii) Locate $F(x_F, H_F)$ on H_L v/s x curve at $x = x_F = 0.42$.

(iii) Locate point Δ_D on the vertical through $x = x_D = 0.97$ at Q' (y-coordinate) = 133567.5 kJ/kmol as x_D and Q' are the coordinates of Δ_D.

(iv) Draw a vertical through $x = x_W = 0.01$ on the x-axis. Locate W on H_L v/s x curve at $x = x_W = 0.01$.

(v) The points Δ_D, F and Δ_W are lying on the same straight line. Join $\Delta_D - F$ and extend to cut the vertical at $x = x_W = 0.01$ at point Δ_W. So we have obtained the point Δ_W.

(vi) Construct the tie lines starting from the point V_1 ($y_1 = x_D = 0.97$) and reaching exactly or covering $x_W = 0.01$ (i.e., the point W). For this we have to make use of the equilibrium curve.

(vii) Construction of a tie line V_1L_1 : For $y_1 = x_D = 0.97$, read the value of x_1 from the equilibrium curve (the x-coordinate of the point P on the equilibrium curve). For $y_1 = 0.97$, $x_1 = 0.93$.

(viii) Mark point L_1 on the H_L v/s x curve at $x = x_1 = 0.93$.

(ix) Join V_1L_1 which is the required tie line.

(x) L_1, V_2 and Δ_D lie on the same straight line. So $L_1\,\Delta_D$ which will cut the H_V v/s y curve at V_2. Read y_2 corresponding to V_2 i.e. x-coordinate of the point V_2. $y_2 = 0.94$.

(xi) Read x_2 corresponding to $y_2 = 0.94$ from the equilibrium curve (as we read x_1 for y_1). For $y_2 = 0.94$, $x_2 = 0.87$.

(xii) Mark point L_2 on H_L v/s x curve at $x = x_2 = 0.87$.

(xiii) The points L_2, V_3, Δ_D lie on the same straight line. Join L_2, Δ_D and mark V_3 (y_3, H_{V3}) at which line $L_2\,\Delta_D$ intersects the H_V v/s y curve.

(xiv) Adopt the above mention procedure for constructing further tie lines.

(xv) Use the point Δ_D till we cross the point F (till we are short of the point F). Once we cross the point F then make use of the point Δ_W for construction of tie lines (i.e., stages) in the stripping section till we exactly reach or cover the point W.

(xvi) Each tie line represents an ideal stage. So count the tie lines.

(xvii) After crossing the point F, join Δ_W to L and extend the line $\Delta_W L$ to get V on H_V v/s y curve. For example, join Δ_W to L_7 and extend line $\Delta_W\,L_7$ to cut H_V v/s y curve at V_8 (as Δ_W, L_7 and V_8 lie on the same straight line).

From Fig. E 1.34 (c) :

Number of theoretical plates including reboiler = n = 11.

Number of theoretical plates required = $n - 1 = 11 - 1 = 10$ ... Ans. (ii)

(ii) Determination of number of stages at total reflux :

At total reflux, D = 0 and W = 0.

$(Q_C/D) + H_D = Q'$ is at ∞. Similarly, Q'' is at $-\infty$.

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At $R = \infty$, the Δ points are at ∞ on the Hxy diagram. So that the line joining the points L_1V_2, L_2V_3 ... are parallel to y-axis as Δ points are at ∞.

Construction of number of stages is shown in Fig. E 1.34 (d) in which the x-y plot is drawn directly below the Hxy diagram for convenience.

Number of theoretical plates including reboiler = 7

Number of theoretical plates required = 7 – 1 = 6 ... **Ans. (ii)**

[If the x-y diagram is drawn elsewhere then : For $y_1 = x_D = 0.97$, read x_1 from the equilibrium curve. Mark L_1 at $x = x_1$ on the H_L v/s x curve. Join V_1L_1 – required tie line. Through L_1 draw the line parallel to y-axis to cut H_V v/s y curve at V_2. Read y_2 of V_2. For y_2 read x_2 from the equilibrium curve and ... . Each tie line represents one stage. Do the construction starting from x_D and reaching exactly or covering x_W.]

We have,

$$Q' = H_D + Q_C/D = 133567.5 \text{ kJ/kmol}$$

$$H_D = H_{L0} = 21389 \text{ kJ/kmol (interpolated value)}$$

$$\frac{Q_C}{D} = Q' - H_D$$

$$= 133567.5 - 21389 = 112178.5 \text{ kJ/kmol}$$

$$D = 20.64 \text{ kmol/h}$$

Q_C = Heat load on the condenser

$$Q_C = (Q' - H_D)\,D \quad \text{... (1)}$$

$$= 112178.5 \times 20.64$$

$$= 2315364 \text{ kJ/h}$$

Heat load on the condenser = 2315364 kJ/h (= 643 kW) ... **Ans. (iv)**

Extra : We have, $Q_C = D\,[(R + 1)\,H_{V1} - RH_{L0} - H_D]$ ... (2)

but $H_{L0} = H_D$ in our case

$$Q_C = D\,[(R + 1)\,H_{V1} - RH_D - H_D]$$

$$= D\,[(R + 1)\,H_{V1} - H_D\,(R + 1)]$$

$$= D\,[(R + 1)\,(H_{V1} - H_D)]$$

We have, $D = 20.64$ kmol/h, $R = 2.5$

$H_D = 21389$ kJ/kmol, $H_{V1} = 53440$ kJ/kmol (interpolated values)

$$Q_C = 20.64\,[(2.5 + 1)\,(53440 - 21389)]$$

$$= 2315364 \text{ kJ/h}$$

Q_C by (1) and (2) is same.

y-coordinate of the point Δ_W is Q''.

From Fig. E 1.34 (c) :

$$Q'' = \left(61.5 \text{ mm} \times \frac{10000}{10 \text{ mm}}\right) = -61500 \text{ kJ/kmol}$$

$$Q'' = H_W - \frac{Q_B}{W}$$

$$Q_B = W\,(H_W - Q'')$$

Q_B is the heat load on the reboiler.

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From Fig. E 1.34 (c), it is very difficult to measure correctly/read correctly the y-coordinate of the point W.

So let us obtain H_W (y-coordinate of the point W) by interpolation.

For $x = 0$ $H_L = 24787$ kJ/kmol

$x = 0.01$ $H_L = H_W$

$x = 0.08$ $H_L = 23853$ kJ/kmol

$$\frac{0.01 - 0}{0.08 - 0.01} = \frac{H_W - 24787}{23853 - H_W}$$

$$\therefore \quad H_W = 24670 \text{ kJ/kmol}$$

Extra : We can correctly read the y-coordinate of the point W from Fig. E 1.34 (a) with enlarged scale for y-axis.

$$H_W = 74 \text{ mm} \times \frac{10000}{30 \text{ mm}} = 24667 \text{ kJ/kmol}$$

H_W by interpolation = 24670 kJ/kmol

H_W from Fig. E 1.34 (a) = 24667 kJ/kmol (almost same)

We have, $W = 27.68$ kmol/h

$$Q_B = W (H_W - Q'') \quad \text{... (1)}$$

$$= 27.68 [24670 - (-61500)]$$

$$= 2385186 \text{ kJ/h}$$

Heat load on the reboiler (Heat input to the reboiler) = 2385186 kJ/kmol ... **Ans. (iv)**

Extra : H_F from Fig. E 1.34 (a)

$$H_F = 66.5 \text{ mm} \times \frac{10000}{30 \text{ mm}} = 22167 \text{ kJ/kmol}$$

H_F by interpolation :

$$\frac{0.42 - 0.335}{0.487 - 0.42} = \frac{H_F - 22442}{22024 - H_F}$$

$$\therefore \quad H_F = 22208 \text{ kJ/kmol}$$

So the values of H_F are near about same.

Use a value of H_F as 22208 kJ/kmol

We have, $Q_B = D H_D + W H_W + Q_C - F H_F$ ... (2)

$$= 20.64 (21389) + 27.68 (24670) + 2315364 - 48.32 \times 22208$$

$$= 2366608 \text{ kJ/h}$$

Heat load of the reboiler = 2366608 kJ/h ... by (2)

Heat load of the reboiler = 2385186 kJ/h ... by (1)

% variation in the result obtained $= \dfrac{2385186 - 2366608}{2385186} \times 100 = 0.779 \approx 0.78$

So very slight variation is there in the calculation of Q_B by (1) and (2).

Extra : H_D = enthalpy of distillate, H_W = enthalpy of bottom product

H_F = enthalpy of feed, Q_C = heat load of the condenser

So in examination to accommodate the Hxy diagram completely on graph paper, use small scale for y-coordinate. Plot x-axis centrally of graph paper, draw separate x-y diagram and proceed. With this – the curves are flat and determination of y-coordinate correctly is very difficult, so adopt the practice of interpolation for the determination of H_W, H_F etc.

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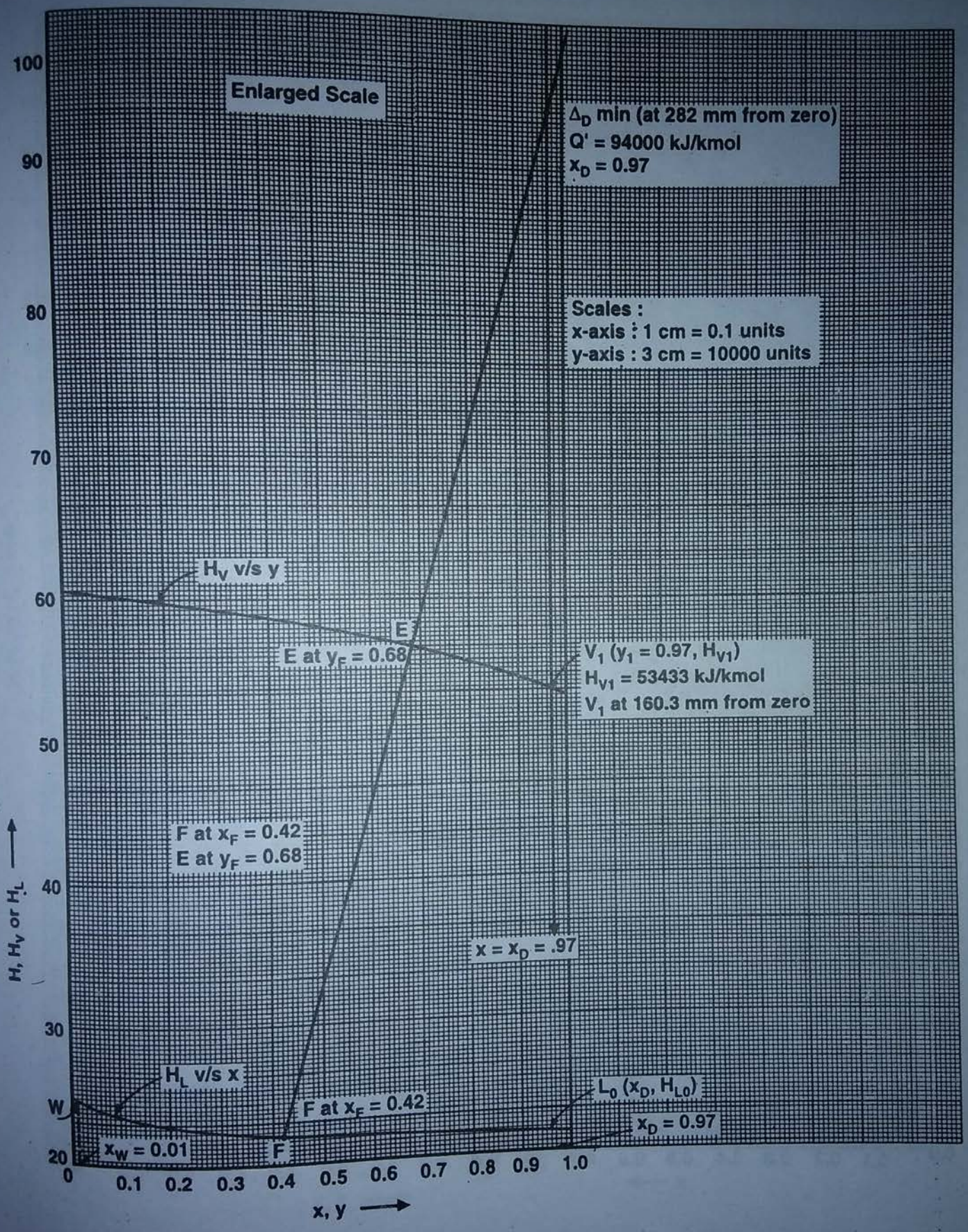

Fig. E 1.34 (a) : Hxy diagram

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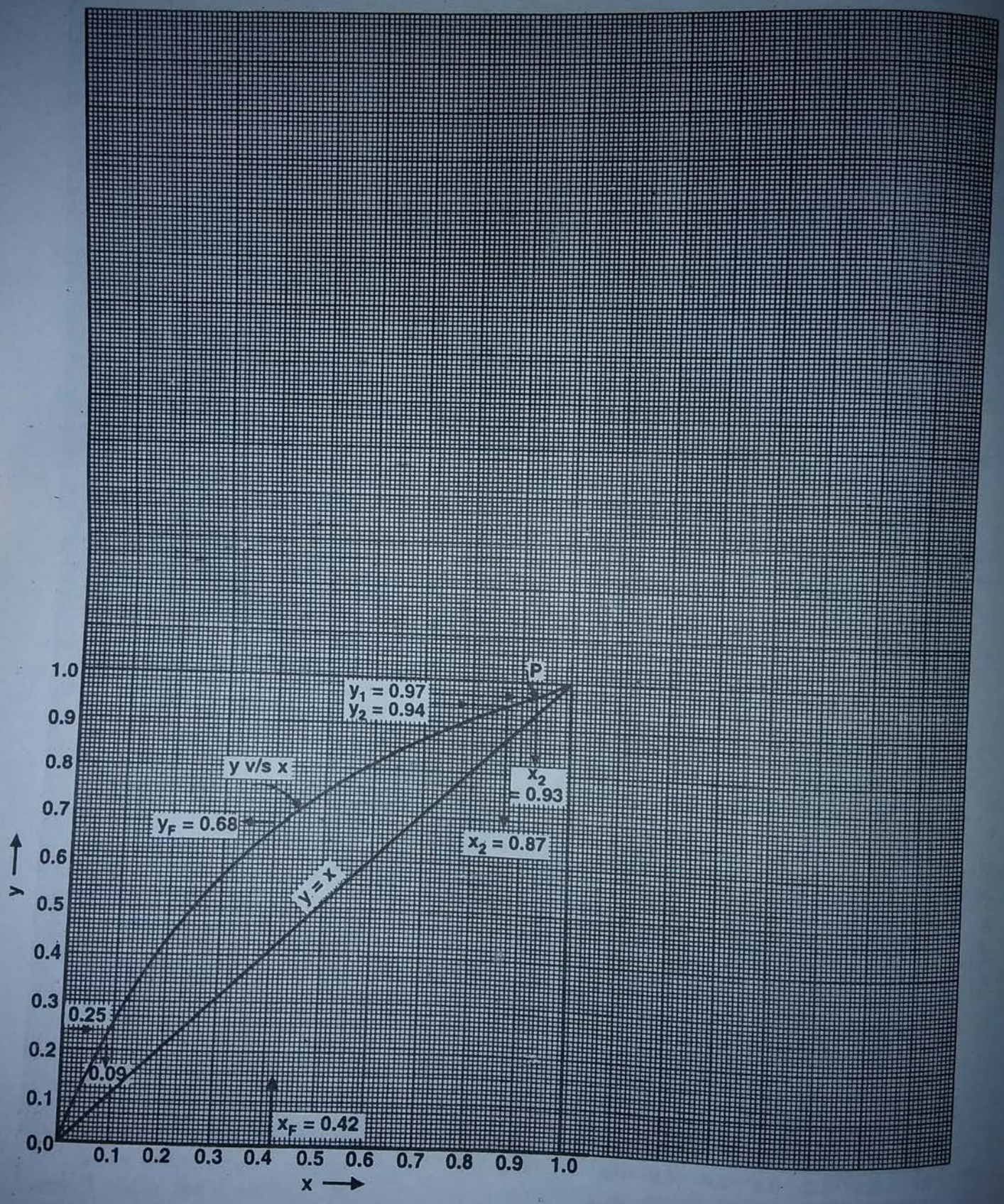

Fig. E 1.34 (b) : Equilibrium curve (for getting x for corresponding y)

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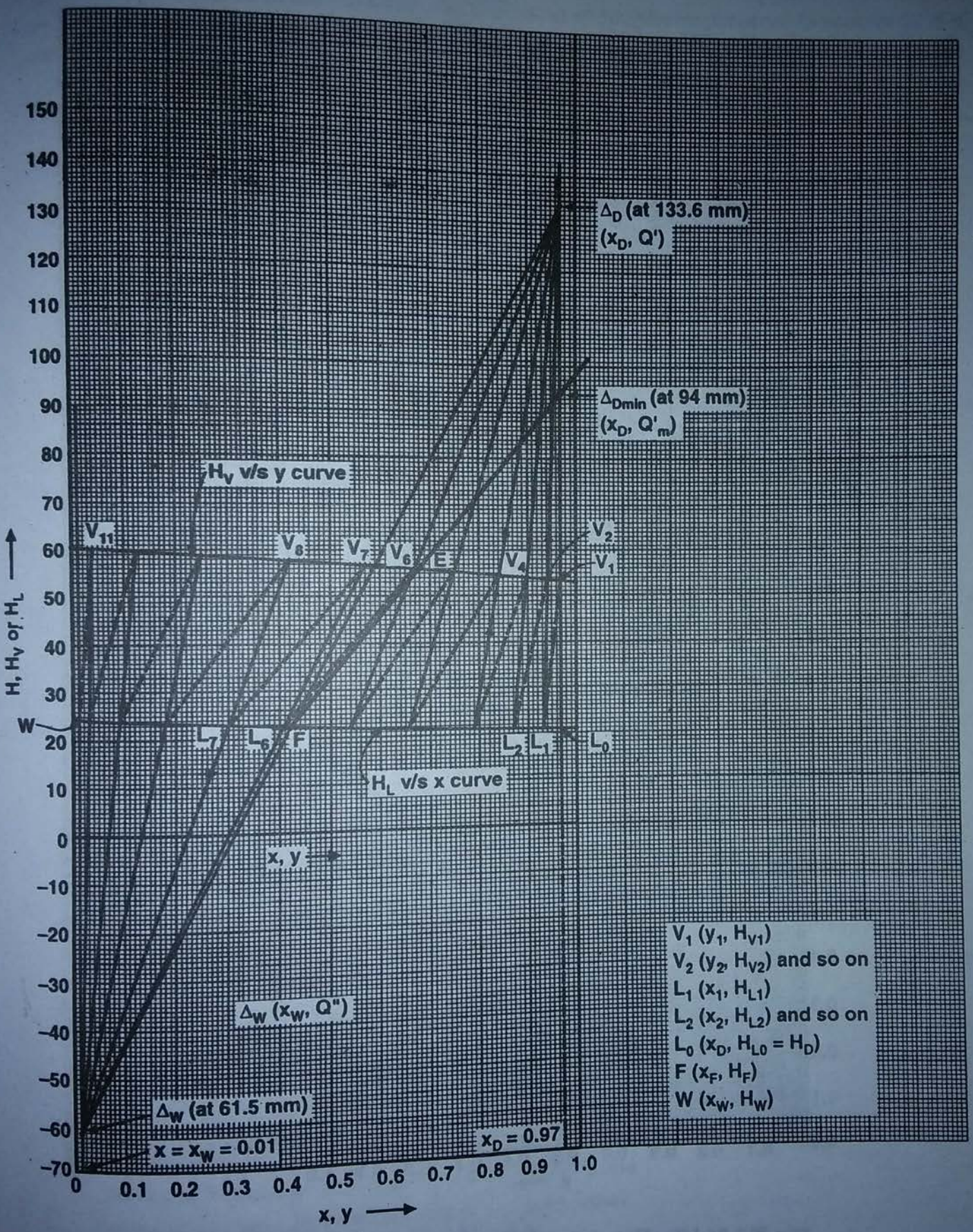

Fig. E 1.34 (c) : Construction of the ideal stages on Hxy diagram

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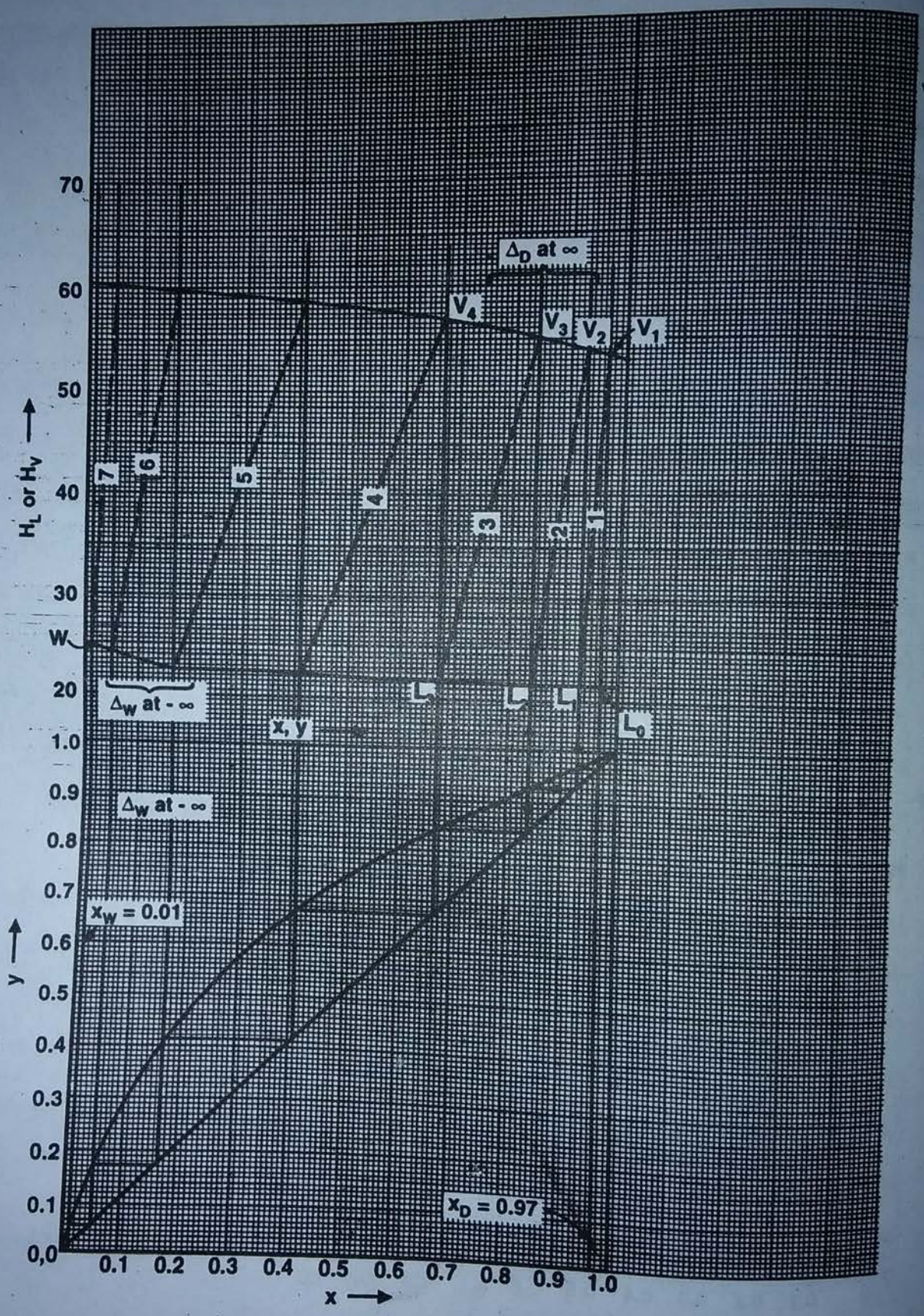

Fig. E 1.34 (d) : Determination of number of stages at total reflux

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Ex. 1.35 : A batch of crude pentane containing 15 mole % butane and 85 mole % pentane is subjected to simple batch distillation at atmospheric pressure to remove 90% of the butane. What would be the composition of remaining liquid ? Also determine the quantity of pentane removed per 1 kmol of feed.

Data : Average value of relativity of butane with respect to pentane is 3.5.

Solution : Basis : 1 kmol of feed

$$F = 1 \text{ kmol}$$

It is a simple distillation technique that is adopted here.

For the case of constant relative volatility, we have,

$$\ln \frac{F\,x_F}{W\,x_W} = \alpha \ln \frac{F\,(1 - x_F)}{W\,(1 - x_W)}$$

x_F = molefraction of butane in feed/charge = 15/100 = 0.15

Butane removed = 90%

Butane in feed = $0.15 \times 1 = 0.15$ kmol

Butane removed = $0.90 \times 0.15 = 0.135$ kmol

Butane in residual liquid left = $0.15 - 0.135 = 0.015$ kmol

Pentane in feed = 85%

Pentane in feed = $0.85 \times 1 = 0.85$ kmol

Butane in feed = $F\,x_F = 0.15$ kmol

Pentane in feed = $F\,(1 - x_F) = 0.85$ kmol

Butane in residual liquid = $W\,x_W = 0.015$ kmol

Pentane in residual liquid = $W\,(1 - x_W)$ kmol

Relative volatility = $\alpha = 3.5$

$$\ln\left(\frac{0.15}{0.015}\right) = 3.5 \ln\left(\frac{0.85}{W\,(1 - x_W)}\right)$$

$$2.3025 = 3.5 \ln [0.85/W\,(1 - x_W)]$$

$$W\,(1 - x_W) = 0.44 \text{ kmol}$$

∴ Amount of pentane in residual liquid = $W\,(1 - x_W) = 0.44$ kmol

Amount of pentane removed = Amount of pentane in charge – Amount of pentane in residual liquid

Amount of pentane removed = $0.85 - 0.44 = 0.41$ kmol ... **Ans.**

Amount of residual liquid left = (butane + pentane) left

$= W\,x_W + W\,(1 - x_W)$

$= 0.015 + 0.44$

$= 0.455$ kmol

Composition of liquid remaining :

Moles of butane in residual liquid = 0.015 kmol

Moles of pentane in residual liquid = 0.44 kmol

Moles of liquid remaining /residual liquid = $0.015 + 0.44 = 0.455$ kmol

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Molefraction of butane in residual liquid = 0.015/0.455 = 0.03296

x_W = 0.033

Mole % butane in liquid left in still (residual liquid) = 0.033 × 100 = 3.3

Mole % pentane in liquid left in still (residual liquid) = 100 – 3.3 = 96.7 ∴ Ans.

Ex. 1.36 : In case of steam distillation of benzaldehyde at atmospheric pressure with condensed steam as the only source of heat, estimate the steam requirement per kg of benzaldehyde (associated with a very small amount of non-volatile impurity).

Data : Benzaldehyde charged to kettle = 100 kg

Temperature of benzaldehyde charged = 305 K (32 °C)

Steam pressure : 0.7 kgf/cm².g (178 kPa)

Vaporisation efficiency = 90%

Latent heat of benzaldehyde = 356 kJ/kg

Specific heat of benzaldehyde = 1.79 kJ/(kg.K)

Molecular weight of benzaldehyde = 106

Enthalpy of steam at 1 atm = H = 2676 kJ/kg

Enthalpy of steam at 0.7 kgf/cm².g pressure = 2700 kJ/kg

Temperature of steam at 0.7 kgf/cm².g = 388 K (115 °C)

Enthalpy of steam (condensed) at the temperature of distillation = 410 kJ/kg

Vapour pressure data for benzaldehyde and water at various temperatures are as follows :

Temperature, °C	62	75	90	99.6
Benzaldehyde, mmHg	10	20	40	60
Water, mmHg	169.8	289.1	527.8	749.2

Solution : Basis : 100 kg of benzaldehyde charged. As the benzaldehyde and water are completely insoluble (the vapour pressure of either component cannot be influenced by the pressure of other), each exerts its true vapour pressure at the prevailing temperature, and when the sum of the vapour pressures equals the total pressure of 1 atm (760 mmHg), the mixture boils.

To determine the temperature of steam distillation :

From the given data, tabulate the vapour pressure of benzaldehyde and total pressure at various values of temperature.

Temperature, T, °C	62	75	90	99.6
Benzaldehyde, p_A^o, mmHg	10	20	40	60
Water, p_B^o, mmHg	163.8	289.1	527.8	749.2
Total pressure, P, mmHg	173.8	309.1	567.8	809.2

$$P = p_A^o + p_B^o$$

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Total pressure of distillation = 760 mmHg (given).

Plot the vapour pressure of benzaldehyde and total pressure along x-axis and temperature along y-axis.

Refer Fig. E 1.36.

From P = 760 mmHg on the x-axis, draw a vertical to cut the T v/s P curve and then proceed horizontally to read the corresponding steam distillation temperature from the y-axis.

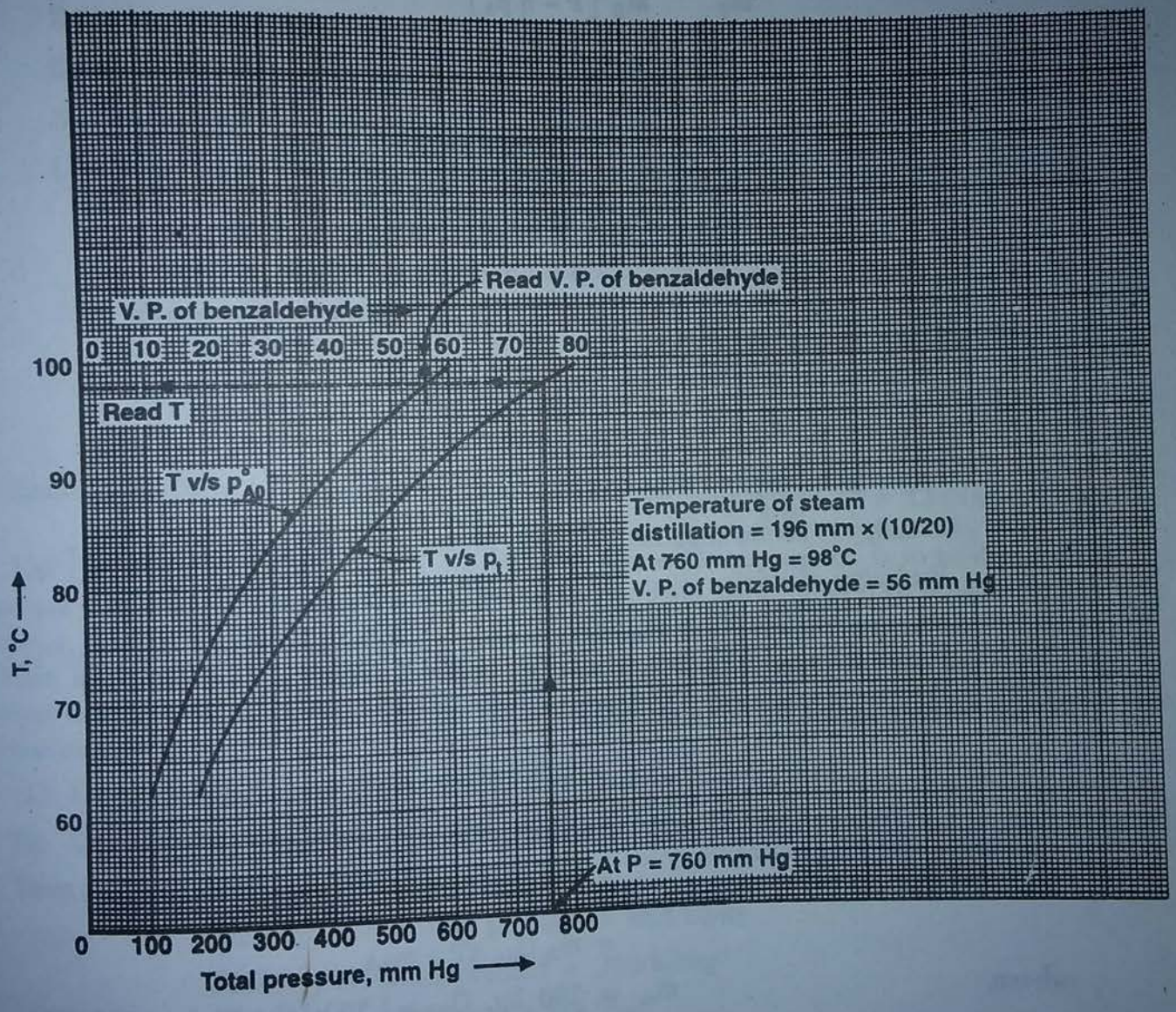

Fig. E 1.36

Temperature of steam distillation (temperature at which the mixture boils) = 98 °C.

So long as the two phases are present, the mixture will boil at this temperature and produce a vapour of constant composition.

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The horizontal line also cuts the T v/s v.p. of benzaldehyde curve. Read the corresponding vapour pressure of benzaldehyde.

Vapour pressure of benzaldehyde at 98 °C = 56 mmHg.

Here steam is used as a carrier and is in direct contact with the charge. Some of the steam gets totally condensed at 98 °C and other will go along with benzaldehyde vapour in the vapour stream (steam - which partly donates heat). The steam supplies the necessary heat and also acts as a carrier.

The mass relationship of benzaldehyde and steam (carrier) in the vapour is given by

$$\frac{m_A}{m_B} = \frac{M_A}{M_B}\left[\frac{\eta\, p_A}{P - \eta\, p_A}\right]$$

where A – benzaldehyde

B – water

m_A – mass of benzaldehyde = 100 kg

m_B – mass of carrier steam

M_A = 106, M_B = 18

η – vaporisation efficiency = 0.90

P = 760 mmHg, $p_A = p_A^o$ = 56 mmHg

$$\frac{100}{m_B} = \frac{18}{106}\left[\frac{0.90 \times 56}{760 - 0.90 \times 56}\right]$$

$\therefore$ m_B = 239.10 kg

Amount of carrier steam = 239.10 kg

Enthalpy of steam supplied = H_1 = 2700 kJ/kg – steam at 0.7 kgf/cm^2.g

Enthalpy of carrier steam = H_2 = 2676 kJ/kg – steam at 760 mmHg

So the difference $H_2 - H_1$ is the enthalpy removed from the carrier steam and utilised for supplying heat for steam distillation.

Enthalpy obtained from carrier steam = $m_B (H_2 - H_1)$

= 239.1 (2700 – 2676)

= 5738.4 kJ

Heat required for benzaldehyde to vaporise = sensible heat + latent heat

= $m_A C_{PA} \Delta T + m_A \lambda$

where, m_A = 100 kg, C_{PA} = 1.79 kJ/(kg.K), λ = 356 kJ/kg

$\Delta T = T - T_f = 98 - 32 = 66$ °C

T = Temperature of distillation

Heat required for benzaldehyde to vaporise = $(100 \times 1.79 \times 66) + (100 \times 356)$ = 47414 kJ

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Energy balance :

$$\begin{pmatrix}\text{Heat required to}\\ \text{vaporise benzaldehyde}\end{pmatrix} = \begin{pmatrix}\text{Heat from}\\ \text{carrier steam}\end{pmatrix} + \begin{pmatrix}\text{Heat from}\\ \text{condensed}\\ \text{steam}\end{pmatrix} - \begin{pmatrix}\text{Heat with}\\ \text{condensed}\\ \text{steam}\\ \text{at 98 °C}\end{pmatrix}$$

$$\begin{pmatrix}\text{Heat from}\\ \text{condensed}\\ \text{steam}\end{pmatrix} - \begin{pmatrix}\text{Heat with}\\ \text{condensed}\\ \text{steam}\end{pmatrix} = 47414 - 5738.4 = 41675.6 \text{ kJ}$$

Let m_C be the condensed steam.

Enthalpy of condensed steam at 98 °C = 410 kJ/kg

Heat with the condensed steam = $m_C \times 410 = 410\, m_C$ kJ

Enthalpy of steam at 0.7 kgf/cm^2.g = 2700 kJ/kg

Latent heat from the condensed steam used for distillation = $m_C \times 2700 = 2700\, m_C$

$\therefore$ $2700\, m_C - 410\, m_C = 41675.6$

$\therefore$ $m_C = 18.2$ kg

$\therefore$ Steam requirement for 100 kg charge = condensed steam (heating steam) + carrier steam

= 239.10 + 18.2

= 257.3 kg

Amount of steam (of pressure 0.7 kgf/cm^2.g) required per 1 kg of benzaldehyde $= \frac{257.3}{100} \times 1 = 2.573$ kg ... **Ans.**

Ex. 1.37 : A distillation column operating at atmospheric pressure is to be designed to separate 30000 kg/h of a solution of benzene and toluene, containing 0.40 weight fraction benzene, into an overhead product containing 0.97 weight fraction benzene and a bottom product containing 0.98 weight fraction toluene. A reflux ratio of 3.5 kg reflux per kg product is to be used. The feed is liquid at its boiling point and reflux is to be returned to the column at 100 °C.

(a) Determine the quantity of top and bottom product.

(b) Calculate the condenser duty and rate of heat input to the reboiler.

Data : Enthalpy of feed = 73.5 kJ/kg

Enthalpy of overhead product = 28.7 kJ/kg

Enthalpy of bottom product = 86.9 kJ/kg

Enthalpy at plate one = 232 kJ/kg

Solution : Basis : 30000 kg/h of feed to the column.

Let F', D', and W' be the mass flow rate of feed, overhead product and bottom product respectively.

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Let x_F', x_D' and x_W' be the weight fraction of benzene in feed, overhead and bottom product respectively.

$$x_F' = 0.40, \quad x_D' = 0.97$$

Weight fraction of toluene in bottom product = 0.98

$$\therefore \quad x_W' = 1 - 0.98 = 0.02$$

Overall material balance :

$$F' = D' + W'$$

$$30000 = D' + W', \quad W' = 30000 - D'$$

Material balance of benzene :

$$F' x_F' = D' x_D' + W' x_W'$$

$$0.4\,(30000) = D'\,(0.97) + W'\,(0.02)$$

$$12000 = 0.97\,D' + 0.02\,W'$$

$$12000 = 0.97\,D' + 0.02\,(30000 - D')$$

$$D' = 12000 \text{ kg/h}$$

$$W' = 30000 - 12000 = 18000 \text{ kg/h}$$

Mass flow rate of top (overhead) product = 12000 kg/h

Mass flow rate of bottom product = 18000 kg/h ...Ans. (a)

Reflux ratio = $R = L'/D' = 3.5$ kg/kg

The condenser duty is given by

$$Q_C = D' [(R + 1) H_{V1}' - R H_L' - H_D']$$

where H_{V1}' = enthalpy at plate one = enthalpy of vapour leaving the column = 232 kJ/kg

$H_D' = H_L'$ = enthalpy of overhead product = 28.7 kJ/kg

[enthalpy of reflux liquid = enthalpy of overhead product]

$R = 3.5$ kg/kg

$D' = 12000$ kg/h

$$Q_C = 12000\,[(3.5 + 1) \times 232 - 3.5 \times 28.7 - 28.7]$$

$$= 10978200 \text{ kJ/h} = 3049.5 \text{ kW}$$

... Ans. (b)

The heat input to the reboiler (Reboiler heat) is given by

$$Q_B = D' H_D' + W' H_W' + Q_C + Q_L - F' H_F'$$

where, Q_L = sum of heat losses = 0.0

$D' = 12000$ kg/h, $H_D' = 28.7$ kJ/kg, $W' = 18000$ kg/h,

$F' = 30000$ kg/h, $H_F' = 73.5$ kJ/kg, $Q_C = 10978200$ kJ/h

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$$H'_W = 86.9 \text{ kJ/kg}$$

$$Q_B = (12000 \times 28.7) + (18000 \times 86.9) + 10978200 + 0 - (30000 \times 73.5)$$
$$= 10681800 \text{ kJ/h}$$

Heat input to the reboiler = 10681800 kJ/h **... Ans. (b)**

[Here notations with prime are used and they indicate that the data as well as quantities are in weight units.]

Ex. 1.38 : Given a binary system for which the vapour-liquid equilibrium data may be presented by the following equation :

$$y = \frac{\alpha x}{1 + (\alpha - 1)x}$$

Prove that the equilibrium diagram in weight fraction unit coincides with the equilibrium diagram in molefraction unit.

Solution : For a binary system, vapour-liquid equilibrium data may be represented by the following equation :

$$y = \frac{\alpha x}{1 + (\alpha - 1)x} \quad \ldots (1)$$

where, y is the molefraction of more volatile component in vapour phase

x is the molefraction of more volatile component in liquid phase

α is the relative volatility of A (more volatile component) with respect to B.

The equilibrium data, i.e., the composition of vapour phase (y) in equilibrium with the composition of liquid phase (x) is generated with the help of Equation (1) by taking x = 0, 0.1, 0.2, ... 1.0 and evaluating the corresponding y values (α should be known for the system under consideration).

x – 0, 0.1, 0.2 ... 1

y – 1.0

and then y is plotted against x to get the equilibrium diagram.

Equation (1) can be rearranged as :

$$\alpha = \frac{y(1-x)}{(1-y)x}$$

$$\alpha = \frac{y/(1-y)}{x/(1-x)} = \frac{y(1-x)}{(1-y)x} \quad \ldots (2)$$

We have, $y = \frac{\alpha x}{1 + (\alpha - 1)x}$

$$y[1 + (\alpha - 1)x] = \alpha x$$

$$y + \alpha xy - xy = \alpha x$$

$$y(1-x) = \alpha x(1-y)$$

$$\therefore \quad \alpha = \frac{y(1-x)}{(1-y)x} = \frac{y/(1-y)}{x/(1-x)}$$

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So the vapour-liquid equilibrium data in molefraction is given by Equation (1) or rearranged Equation (1) i.e. Equation (2).

Now we will express the composition of phases in terms of weight fraction of more volatile component (A).

Let y' be the weight fraction of A in vapour phase then $1 - y'$ be the weight fraction of B in vapour phase [as for binary system of A and B, $\Sigma y_i' = 1$.]

Similarly, let x' be the weight fraction of A in liquid phase and so $1 - x'$ be the weight fraction of B in liquid phase. The phases are in equilibrium with each other, so the compositions in weight fraction are the equilibrium compositions.

For 1 mol of vapour phase :

$$\text{Moles of A} = y \times 1 = y \text{ mol}, \quad y \text{ is molefraction of A}$$

$$\text{Moles of B} = (1 - y) \times 1 = (1 - y) \text{ mol}$$

Let M_A and M_B be the molecular weights of A and B respectively.

$$\text{Mass of A in vapour phase} = yM_A \text{ g}$$

$$\text{Mass of B in vapour phase} = (1 - y) M_B \text{ g}$$

Weight fraction of A in vapour phase is given by :

$$y' = \frac{\text{mass of A}}{\text{mass of A + mass of B}}$$

$$y' = \frac{yM_A}{yM_A + (1 - y) M_B}$$

$$y'yM_A + y'M_B - yy'M_B = yM_A$$

$$y'M_B (1 - y) = yM_A (1 - y')$$

$$\frac{1 - y}{y} = \frac{(1 - y') M_A}{y'M_B}$$

$$\frac{y}{1 - y} = \frac{y'M_B}{(1 - y') M_A} \quad \ldots (3)$$

For 1 mol of liquid phase :

$$\text{Moles of A} = x \times 1 = x \text{ mol}, \quad x \text{ is molefraction of A}$$

$$\text{Moles of B} = (1 - x) \times 1 = (1 - x) \text{ mol}$$

$$\text{Mass of A in liquid phase} = xM_A \text{ g}$$

$$\text{Mass of B in liquid phase} = (1 - x) M_B \text{ g}$$

Weight fraction of A in liquid phase is given by :

$$x' = \frac{\text{mass of A}}{\text{mass of A + mass of B}}$$

$$= \frac{xM_A}{xM_A + (1 - x) M_B}$$

$$x'xM_A + x'M_B + xx'M_B = xM_A$$

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$$x'M_B(1-x) = xM_A(1-x')$$

$$\frac{x}{1-x} = \frac{x'M_B}{(1-x')M_A} \quad ...(4)$$

Put the value of $y/(1-y)$ from Equation (3) and that of $x/(1-x)$ from Equation (4) into Equation (2),

$$y = \alpha x/[1+(\alpha-1)x]$$

i.e.

$$\alpha = \frac{y(1-x)}{(1-y)x} = \frac{y/(1-y)}{x/(1-x)}$$

$$\alpha = \frac{y'M_B/(1-y')M_A}{x'M_B/(1-x')M_A}$$

$$\alpha = \frac{y'M_B}{(1-y')M_A} \cdot \frac{(1-x')M_A}{x'M_B}$$

$$\alpha = \frac{y'}{(1-y')} \frac{(1-x')}{x'} = \frac{y'/(1-y')}{x'/(1-x')} \quad ...(5)$$

Which on rearranging yields

$$y' = \frac{\alpha x'}{1+(\alpha-1)x'} \quad ...(6)$$

It is clear that as Equations (2) and (5) or Equations (1) and (6) are identical, the equilibrium diagram in weight fraction units coincides with the equilibrium diagram in molefraction units.

Ex. 1.39 : Pertinent data on a binary system : heptane-ethylbenzene are as follows :

x	0	0.08	0.185	0.251	0.333	0.487	0.651	0.788	0.914	1.0
y	0	0.233	0.428	0.514	0.608	0.729	0.834	0.904	0.963	1.0

A feed mixture composed of 42 mole % heptane and 58 mole % ethylbenzene is to be fractionated to yield an overhead product of 97 mole % heptane and a residue of 99 mole % ethylbenzene.

(i) Using reflux ratio of 1.5, determine the number of equilibrium stages needed for saturated liquid feed and a bubble point reflux by the McCabe-Thiele method.

(ii) Determine the minimum reflux ratio.

(iii) Determine the number of equilibrium stages at total reflux.

Solution : Basis : Feed containing 42 mole % heptane and 58 mole % ethylbenzene at its bubble point.

Molefraction of heptane in feed :

$$x_F = 42/100 = 0.42$$

Molefraction of heptane in distillate :

$$x_D = 97/100 = 0.97$$

Molefraction of heptane in residue :

$$x_W = (100-99)/100 = 0.01$$

Reflux ratio $= R = 1.5$

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Graphical procedure for determination of number of equilibrium stages :

Refer Fig. E 1.39 (a).

(i) Plot the equilibrium diagram.

(ii) Draw the feed line/q-line. For saturated liquid feed, q = 1 so the feed line is parallel to y-axis. Through the point L (0.42, 0.42) on the diagonal, draw a line parallel to y-axis.

(iii) The point of intersection of operating line of the rectification section with the diagonal has the coordinates : 0.97, 0.97.

$$\text{Intercept on y-axis of this line} = \frac{x_D}{R+1} = \frac{0.97}{1.5+1} = 0.388 \approx 0.39.$$

So draw the operating line of rectifying section through the point P (0.97, 0.97) on the diagonal with intercept equal to 0.39.

(iv) The point of intersection of operating line of the stripping section with the diagonal has coordinates : 0.01, 0.01. So draw this line through the point Q (0.01, 0.01) on the diagonal and passing through the point of intersection of top operating line and feed line.

(v) Construct the stages between the equilibrium curve and the operating lines starting from the point P and ending exactly or covering the point Q.

Number of equilibrium stages including reboiler = n = 13 ... Ans. (i)

Number of equilibrium plates in column = n – 1 = 13 – 1 = 12 ...Ans. (i)

Refer Fig. E 1.39 (b).

Determination of the number of ideal stages at total reflux :

Draw the equilibrium diagram. Mark points P (0.97, 0.97) and Q (0.01, 0.01).

Construct the equilibrium stages between equilibrium curve and diagonal starting from the point P and reaching exactly or covering the point Q and count them.

Number of equilibrium stages including reboiler (at total reflux) = n = 7 ... Ans. (iii)

Number of ideal plates in the column (at total reflux) = 7 – 1 = 6 ... Ans. (iii)

Determination of R_m (for saturated liquid feed) :

At R_m, the rectification line and the feed line meet each other at a certain point on the equilibrium curve. In our case, point R (line PR) is not shown. The coordinates of the point R are :

$$y_F = 0.68, \quad x_F = 0.42$$

For saturated liquid feed, R_m is given by,

$$R_m = \frac{x_D - y_F}{y_F - x_F} = \frac{0.97 - 0.68}{0,68 - 0.42} = 1.115$$

Minimum reflux ratio (R_m) = 1.115 ... Ans. (ii)

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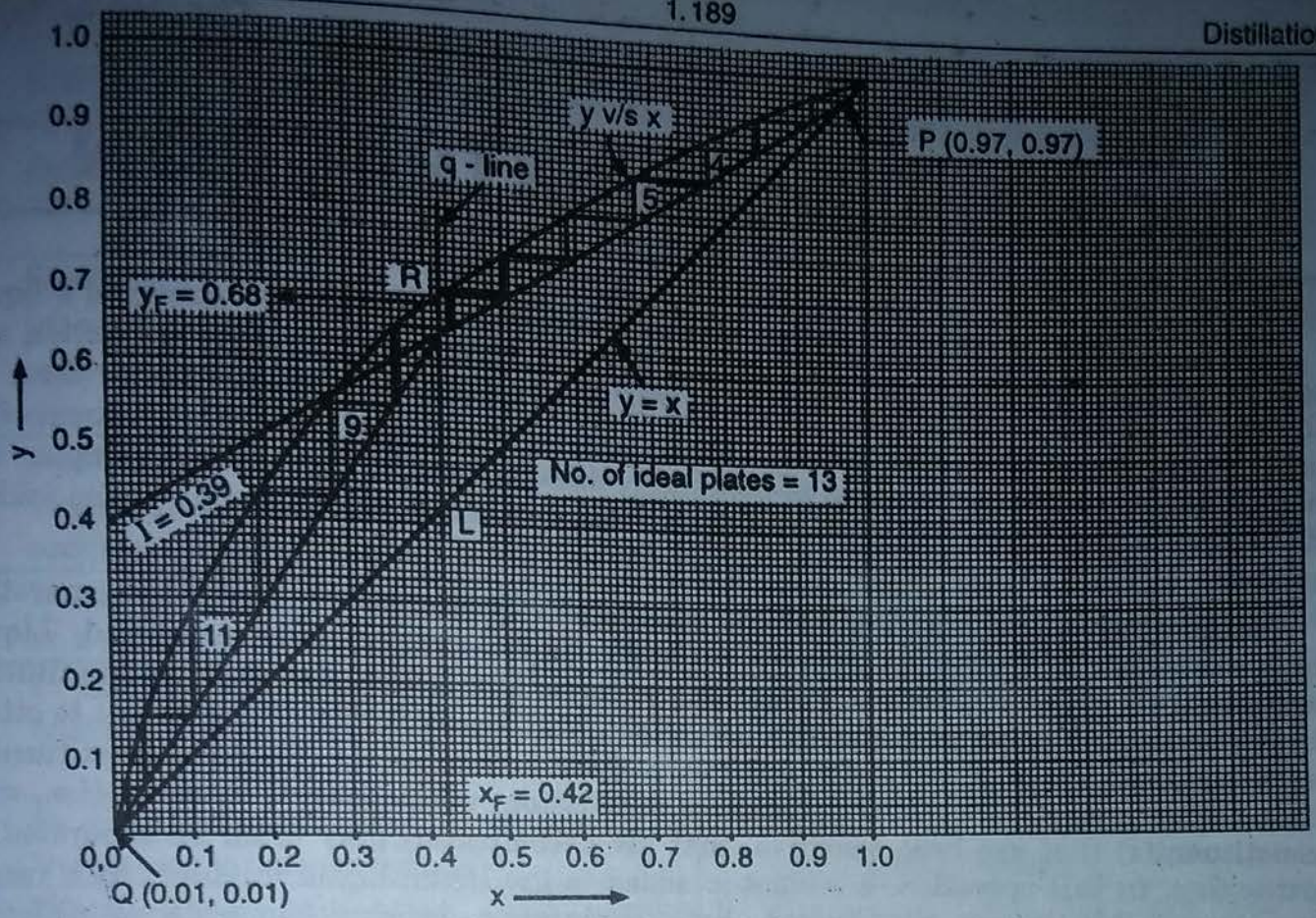

Fig. E 1.39 (a)

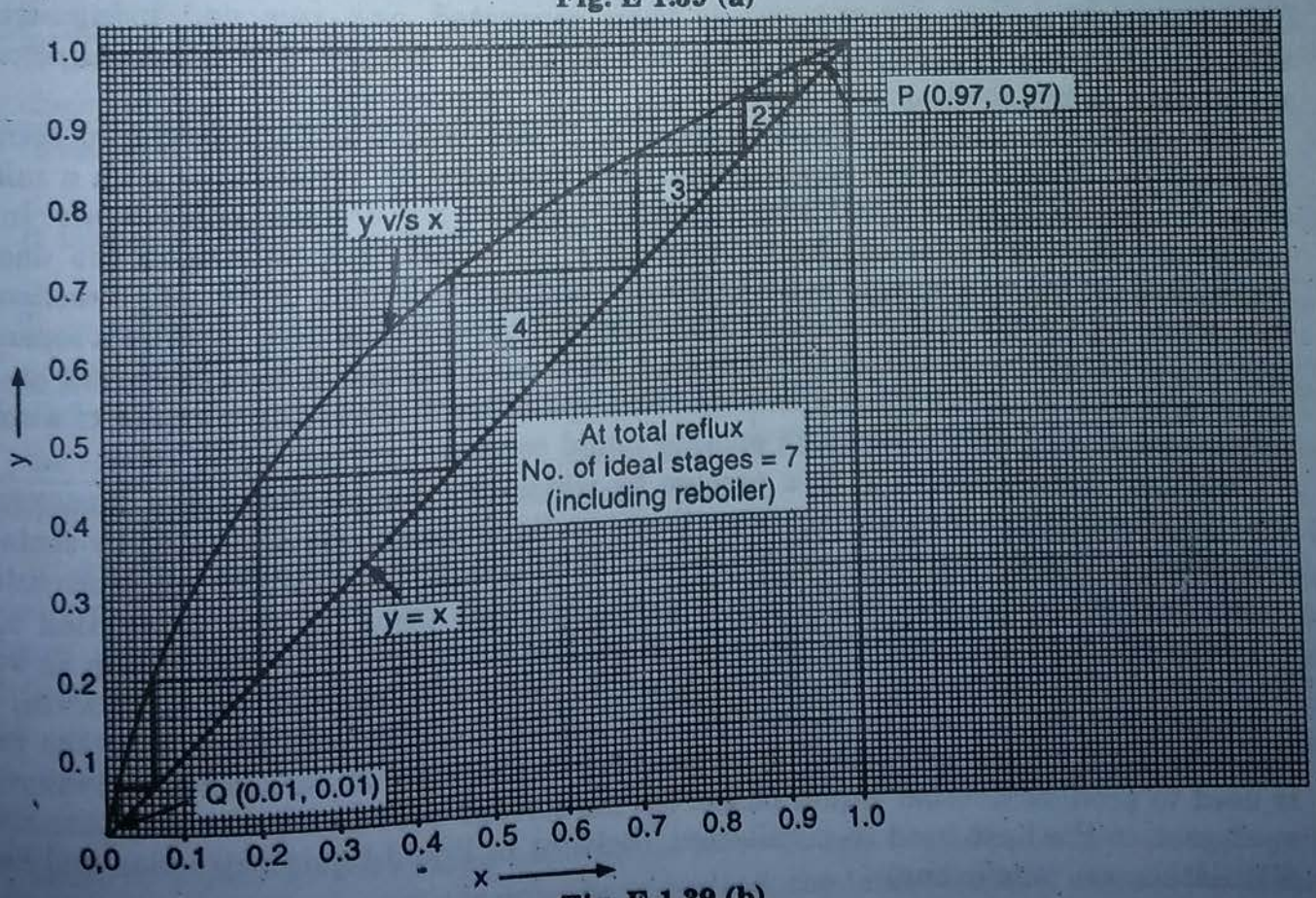

Fig. E 1.39 (b)

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LIQUID-LIQUID EXTRACTION

The techniques of removing one component from a solid or liquid by means of a liquid solvent fall into two categories, the first one is called as leaching or solid extraction and the second as liquid extraction. **Leaching** is used to dissolve soluble matter from its mixture with an insoluble solid whereas **liquid extraction** is used to separate two miscible liquids by the use of solvent which preferentially dissolves one of them. In this chapter, our discussion is restricted to liquid extraction – the extraction of the liquid phase with a solvent.

When a mixture of liquids is not easily separable by distillation, i.e. when separation by distillation is ineffective or very difficult, liquid extraction is employed. Liquid mixtures : (i) with constituents having very close boiling points, or (ii) with constituents having too high boiling points for the heating medium available (i.e. it is difficult to attain such temperatures easily with the heating medium available), or (iii) with constituent(s) that cannot withstand the temperature of distillation, even under a vacuum (i.e., with constituent(s) that are heat sensitive, that get decomposed) may often be separated by extraction. In this operation, a solvent is added to the liquid-liquid mixture. As a result, two immiscible layers are formed, both containing varying amounts of different components. The isolated layers are then separated and removed using density difference. For the recovery of the solvent for reuse, extraction is followed often by distillation or evaporation.

Liquid-liquid extraction (liquid extraction or solvent extraction) refers to *an operation in which the components of a liquid mixture are separated by contacting it with a suitable insoluble liquid solvent, which preferentially dissolves one or more components.* In this operation, the separation of the components of a solution depends upon the unequal distribution of the components between the two immiscible liquids. In liquid extraction, the feed solution to be handled represents one phase and solvent to be used to effect a separation represents the second phase. In this operation, the two immiscible phases in contact are both liquid and so is a liquid-liquid operation. This operation involves the transfer of a soluble component of a liquid mixture into another liquid solvent.

Solvent Extraction : It is a process by which two liquids forming a homogeneous mixture (the mixture may contain more than two liquids also) are separated by contacting with another liquid into which one of the two liquids dissolves/separates out preferentially.

Both distillation and liquid-liquid extraction are used for the separation of the constituents of a liquid mixture. In order to achieve a separation by distillation as well as by liquid-liquid extraction, it is necessary to have two phases. In distillation, it is necessary to have liquid and vapour phases, and the heat is used to produce the vapour phase. In liquid-liquid extraction, it is necessary to have two liquid phases and the solvent is used to produce another liquid phase. So the solvent used in liquid-liquid extraction is analogous to the heat used in distillation. (Solvent in liquid-liquid extraction and heat in distillation are analogous).

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Distillation and extraction both are used for the separation of the constituents of any liquid mixture based on the relative merits/demerits of the individual methods.

Difference between distillation and extraction :

Distillation	Extraction
1. Distillation/fractionation is an operation in which constituents of the liquid mixture are separated by using thermal energy.	1. Extraction is an operation in which constituents of the liquid mixture are separated by using insoluble liquid solvent.
2. Distillation utilises differences in vapour pressure or volatility of the components to effect a separation.	2. Extraction utilises differences in solubility of the components to effect a separation.
3. In distillation, relative volatility is used as a measure of the degree of separation.	3. In extraction, selectivity is used as a measure of the degree of separation.
4. In distillation, a new phase is created by addition of heat.	4. In extraction, a new insoluble liquid phase is created by addition of solvent to the original mixture.
5. In distillation, mixing and separation of the phases is easy and rapid.	5. In extraction, the phases are hard to mix and harder to separate.
6. Distillation/fractionation gives almost pure products.	6. Extraction itself does not give pure products and needs further processing.
7. It does not offer more flexibility in the choice of operating conditions.	7. It does offer more flexibility in the choice of operating conditions.
8. It requires thermal energy.	8. It requires mechanical energy for mixing and separation.
9. It needs heating and cooling provisions.	9. It does not need heating and cooling provisions.
10. It is a primary choice for separation of components of liquid mixture.	10. It is a secondary choice for separation of components of liquid mixture.
11. Extractive distillation is the extraction of the vapour phase with solvent.	11. Liquid extraction is the extraction of liquid phase with solvent.

Field of application of Liquid Extraction :

Whenever separation by both distillation and extraction is possible, the choice is usually distillation, irrespective of heating and cooling requirements. In extraction, the solvent should be recovered for reuse and hence extraction is usually followed by distillation for the recovery of the solvent. This combined operation is more complicated and more expensive than ordinary distillation. But whenever separation of the components of a liquid mixture is very difficult or ineffective by distillation, extraction can be thought of as an alternate process (alternative) to it (distillation).

Thus, (i) whenever very large amounts of latent heats are required (as in cases with very dilute solutions where water must be vaporised, the latent heats of organic liquids are substantially lower than that of water), (ii) whenever we are dealing with liquid mixtures forming azeotropes / close boiling mixtures and (iii) whenever we are dealing with substances that are thermally unstable/heat sensitive, separation by distillation is

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expensive, ineffective and difficult. In such cases, extraction is attractive and preferred. For example : (i) separation of acetic acid from dilute solution with water is done more economically by extraction followed by distillation than distillation alone because of high latent heat of vaporisation of water (distillation would be possible but extraction reduces substantially the amount of water to be vaporised). (ii) Long chain fatty acids can be removed from vegetable oils economically by extraction with liquid propane than high vacuum distillation technique. (iii) Separation of petroleum products that have about the same boiling range. Separation of aromatics from lube oil fractions using furfural as a solvent and (iv) Recovery of penicillin from the fermentation broth using butyl acetate.

A typical liquid-liquid extraction operation utilising differences in solubilities of the components of a liquid mixture may be described as consisting of the following steps :

1. Contacting the feed with a solvent.
2. Separation of the resulting phases, and
3. Removal / Recovery of solvent(s) from each phase.

In liquid-liquid extraction operation, the liquid mixture to be treated and a suitable insoluble solvent are contacted intimately, the constituents of the liquid mixture are distributed between the phases resulting into some degree of separation (which can be improved by a multistage contact), and then the phases are separated from one another based on the density difference. For example, acetone may be preferentially extracted from a solution in water with the help of chloroform and the resulting chloroform phase is found to contain a large part of acetone, but little water.

In extraction operations, the solution which is to be extracted is called the **feed** and the liquid with which the feed is contacted for the extraction purpose is called the **solvent**. The residual liquid solution (solvent lean) from which the solute is removed is called the **raffinate** (also termed as the raffinate phase/layer) and the solvent-rich product of the operation, containing the extracted solute is called the **extract** (also termed as the extract phase/layer). The extract phase contains the desired product in a larger proportion.

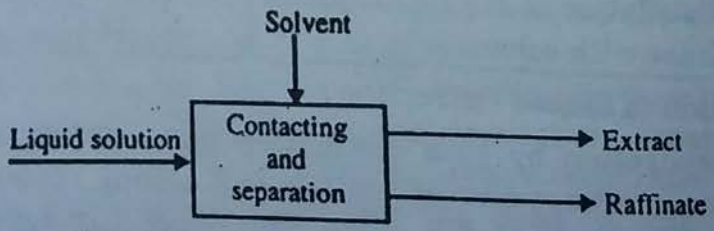

Fig. 2.1 : Block diagram for Liquid extraction

Thus, if we contact a solution of acetic acid in water with a solvent such as ethylacetate then two phases/layers will result. The extract (ester layer/organic layer) phase will contain most of the acetic acid in ethyl acetate with some water, while the raffinate (aqueous layer) phase will contain weaker acetic acid solution with a small amount of ethyl acetate. The amount of water in the extract and ethyl acetate in the raffinate depends upon their solubilities into one another.

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Distribution coefficient :

In dilute solutions at equilibrium, *the ratio of the concentrations of the solute in the two phases* is called as the distribution coefficient or distribution constant K. Thus,

$$K = C_E / C_R$$

where C_E and C_R are the concentrations of the solute in the extract and the raffinate phases, respectively.

The distribution coefficient can also be given in terms of weight fraction of the solute in the two phases in contact at equilibrium.

$$K' = y^*/x$$

where y^* is the weight fraction of the solute in the extract and x is the weight fraction of the solute in the raffinate.

Notation Adopted :

1. A is the feed solvent, B is the extracting solvent (A and B are pure and substantially insoluble liquids) and C is the solute that will distribute between two phases.

   Feed solutions (F) to be separated by extraction are comprised of A and C.

2. The above mentioned letters will be used to indicate the quantities and the locations of mixtures on the phase diagrams to follow. For batch operations, quantities are measured in mass (or weight) and that for continuous operations in mass/time. Therefore,

   E = mass or mass/time of solution E, i.e., an extract or extract phase, shown on a phase diagram by point E

   R = mass or mass/time of solution R, i.e., a raffinate or raffinate phase shown on a phase diagram by point R.

   B = mass or mass/time of solvent B

3. x = weight fraction of C in the A-rich phase, i.e., in the raffinate phase.

   y = weight fraction of C in the B-rich phase, i.e., in the extract phase.

   In case of immiscible liquids/insoluble liquids :

$$X = \frac{x}{(1-x)} = \frac{\text{mass C}}{\text{mass non-C in the raffinate phase}} = \frac{\text{mass C}}{\text{mass A}}$$

$$Y = \frac{y}{(1-y)} = \frac{\text{mass C}}{\text{mass non-C in the extract phase}} = \frac{\text{mass C}}{\text{mass B}}$$

Ternary Systems :

The addition of a new solvent to a binary liquid mixture of the solute in a solvent may result in the formation of several types of mixture/may yield different types of mixture :

(i) The solvent may be completely immiscible with the feed solvent. This is the ideal case.

(ii) The solvent may be partially miscible with the feed solvent, forming one pair of partially miscible liquid. If A is the feed solvent, C is the solute and B is the extracting solvent, then C dissolves in A and B completely while A and B dissolve only to a limited extent in each other. This is the most common type.

(iii) A homogeneous solution may be formed, then the solvent selected is not suitable and should be rejected.

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(iv) The solvent may lead to the formation of two or three partially miscible liquids. A and C are completely soluble, while A and B, and B and C dissolve only to a limited extent in each other (two pairs partially miscible). It is observed occasionally and not desired.

The most commonly encountered possibility (after addition of a solvent to a feed solution) is that given by type (ii). In liquid extraction, the equilibrium data (i.e., equilibrium compositions of the phases) can be presented by using triangular coordinates (e.g., equilateral triangular coordinates) as well as by using rectangular coordinates. In case of type (i) possibility, rectangular coordinates are commonly used to show the equilibrium relationship between the phases wherein the composition of the solute in one phase is plotted against the composition of the solute in another phase in equilibrium with it. The equilibrium data in case of type (ii) and type (iv) possibilities are often presented by using equilateral triangular coordinates.

Equilateral Triangular Coordinates/Equilateral Triangular Diagram : The equilateral triangular diagrams are extensively used for graphical presentation of the compositions of ternary systems/mixture. In the case of an equilateral triangle, we know that the sum of the perpendicular distances from any point inside the triangle to its three sides is equal to the altitude of the triangle, so we can assume here that the altitude represents 100 percent composition and the perpendicular distances to the three sides represent the percentages or fractions of the three components.

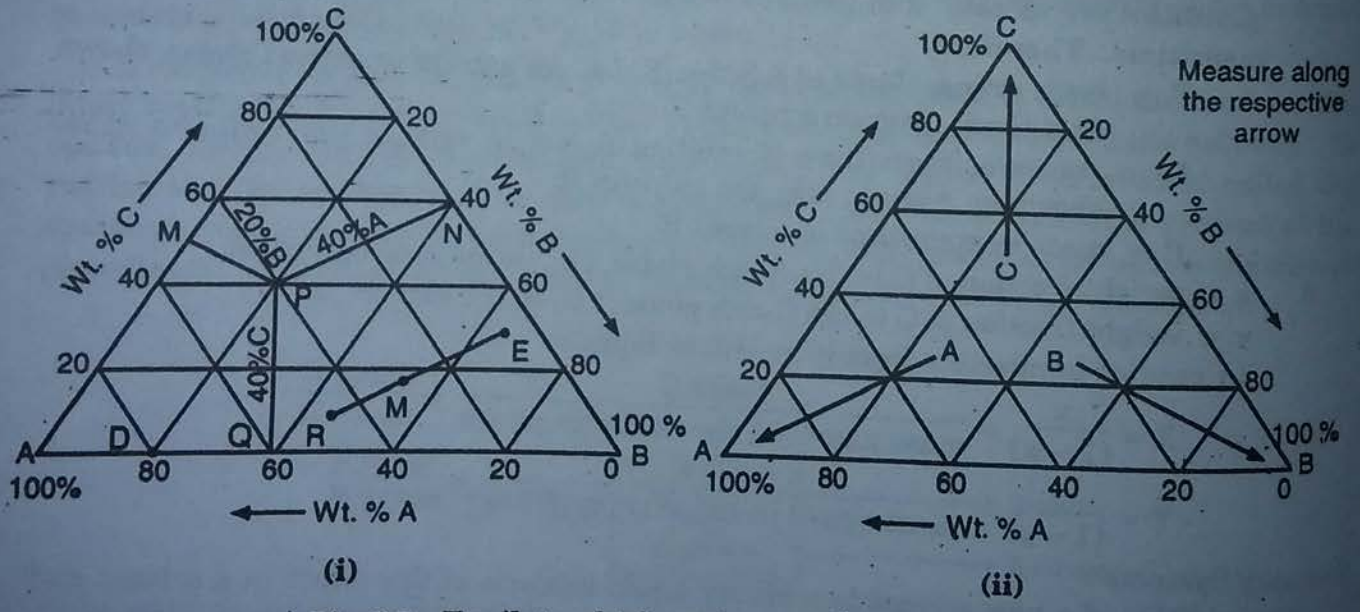

Fig. 2.2 : Equilateral triangular coordinates/diagram

Here we are considering a ternary system comprising of components A, C and B, so in Fig. 2.2, the apexes of the triangle are marked with 100 % A, 100 % C and 100 % B (each apex representing the pure component). Any point on a side of the triangle represents the locus of all possible mixtures of two out of any three components, i.e., it represents a binary mixture and any point inside the triangle represents a mixture of all three components. Therefore, a side AC represents the locus of all possible mixtures of A and C only, i.e., it represents a binary mixture of A and C. Similarly, a side AB represents a binary mixture of A and B, and a side BC, a binary mixture of C and B. For example, point D marked on the side AB

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represents a binary mixture containing 80 % A and 20 % B. (The corners of the triangle labelled A, B and C correspond to the pure components A, B and C respectively. The side of the triangle opposite to the corner labelled A implies the absence of A. Therefore, the horizontal lines across the triangle show increasing percentage of A from zero at the base to 100% at the apex).

Any point within the triangle represents a mixture consisting of all three components. If the two components are marked with their compositions then the third one is fixed because of the property of a triangle. (The property of an equilateral triangle is that the sum of the three perpendicular distances from any point within the triangle to the sides of the triangle is always equal to the height of the triangle or altitude of the triangle.) The mixture indicated by any point P consists of the three components A, B and C in the ratio of the perpendiculars PN, PM, PQ. The perpendicular distance from point P to the base AB, i.e., PQ represents the composition of C in the mixture at P. Similarly, the perpendicular distances PM (to the base AC) and PN (to the base BC) represent the compositions of B and A in the mixture at P respectively. So $x_p = 0.40$.

If R kg of a mixture at point R is added to E kg of a mixture at point E (Fig. 2.3), then the new mixture resulted (M kg) is shown on the straight line RE at point M, such that :

$$\frac{R}{E} = \frac{\text{line ME}}{\text{line RM}} = \frac{x_E - x_M}{x_M - x_R} \quad \ldots (2.1)$$

where x_R is the weight fraction of C in R
x_E is the weight fraction of C in E
x_M is the weight fraction of C in M

Now, we will establish Equation (2.1).

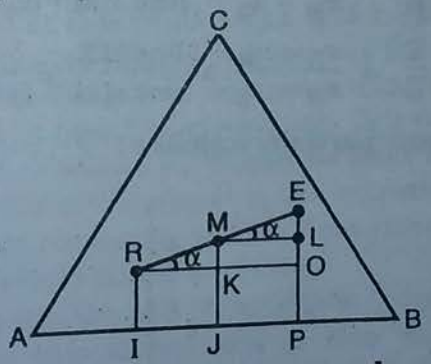

Fig. 2.3 : The mixture rule

Fig. 2.3 shows that the addition of R kg of mixture at R to E kg of mixture at E results in the formation of M kg of new mixture at M.

Line RI represents the weight fraction of C in R = x_R
Line MJ represents the weight fraction of C in M = x_M
Line EP represents the weight fraction of C in E = x_E

Overall material balance :

$$R + E = M$$

Material balance of component C :

$$R \text{ (line RI)} + E \text{ (line EP)} = M \text{ (line MJ)}$$

$$R x_R + E x_E = M x_M$$

$$R x_R + E x_E = (R + E) x_M$$

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Similarly, $R(x_M - x_R) = E(x_E - x_M)$

$$\frac{R}{E} = \frac{x_E - x_M}{x_M - x_R}$$

$$\frac{R}{E} = \frac{x_E - x_M}{x_M - x_R} = \frac{\text{line EP} - \text{line MJ}}{\text{line MJ} - \text{line RI}}$$

But line EP – line MJ = line EL

and line MJ – line RI = line MK = line LO

$$\frac{R}{E} = \frac{x_E - x_M}{x_M - x_R} = \frac{\text{line EL}}{\text{line LO}}$$

In the triangle EML, $EL = ME \cos \alpha$ i.e., line EL = line ME·cos α

In the triangle ERO, $EO = RE \cos \alpha$

but $EO = EL + LO$

$EL + LO = RE \cos \alpha$

but $EL = ME \cos \alpha$

$\therefore$ $LO = RE \cos \alpha - ME \cos \alpha = (RE - ME) \cos \alpha = RM \cos \alpha$

(as RE = RM + ME)

i.e., line LO = line RM · cos α

Putting the values of line EL and line LO,

$\therefore$

$$\frac{R}{E} = \frac{x_E - x_M}{x_M - x_R} = \frac{\text{line ME} \cdot \cos \alpha}{\text{line RM} \cdot \cos \alpha}$$

$$\frac{R}{E} = \frac{x_E - x_M}{x_M - x_R} = \frac{\text{line ME}}{\text{line RM}} = \frac{\text{length ME}}{\text{length RM}}$$

Systems of three liquids – One pair partially miscible :

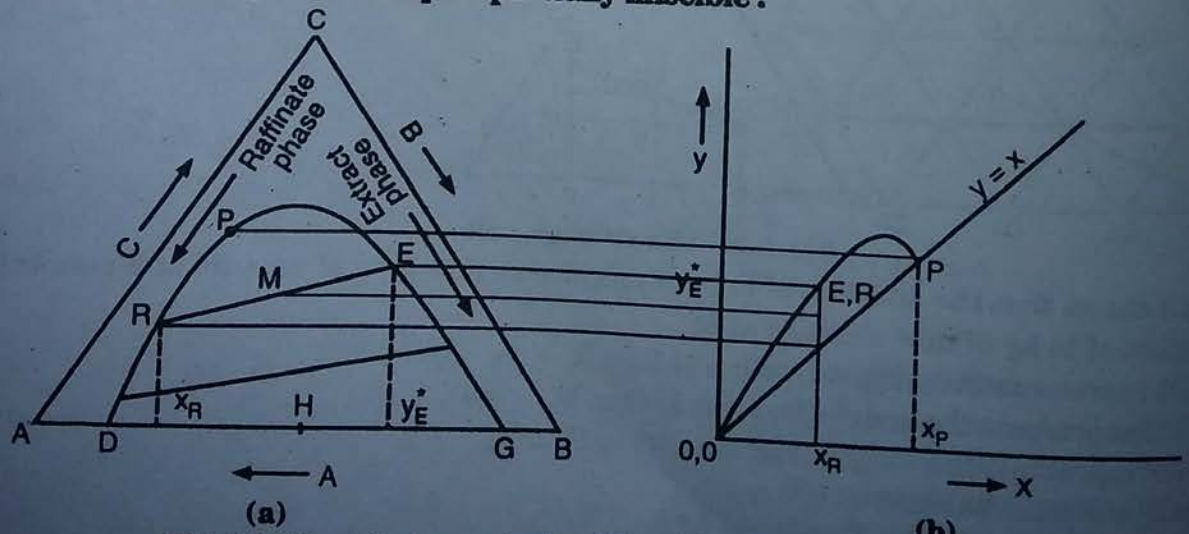

Fig. 2.4 : Equilibrium relationship of distribution of acetone (C) between water (A) and MIK (B) (System of three liquids, A and B partially miscible)

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Consider a system C (acetone), A (water) and B (methyl isobutyl ketone, MIK) at 298 K (25°) wherein acetone is the solute, water is the diluent (feed solvent) and methyl isobutyl ketone is the solvent used to extract acetone, i.e., extracting solvent. Acetone and water are completely miscible, and acetone and MIK are completely miscible while water and MIK are partially miscible with each other. Water (A) and MIK (B) dissolve to a limited extent in each other to give rise to the saturated liquid solutions (A - rich) at D and (B - rich) at G. Apex C represents 100 % acetone, and apexes A and B represent 100 % water and 100 % MIK respectively. Along a side AB the concentration of C is zero and same is true for A and B along BC and AC respectively. Every mixture of MIK and water (binary mixture) of composition lying between points D and G say at H forms two liquid phases (layers) of compositions at D and G and the relative amounts of phases depend upon the position of H according to the mixture rule Equation (2.1). When acetone is added to the two phase mixture of MIK and water, acetone gets distributed between the phases (layers) and compositions of the phases follow the raffinate phase and extract phase solubility curves. The curved line GEP indicates the composition of saturated MIK layer-extract phase and the curved line DRP indicates the composition of saturated water layer-raffinate phase. The area under the binodal solubility curve represented by the curved line DRPEG represents a two phase region that will split into two phases/layers in equilibrium with each other. Any mixture outside this curve will be a homogeneous solution of one liquid phase. The equilibrium compositions of two phases will lie on this curve. Any mixture underneath the curve at point M will split into two saturated liquid phases of equilibrium compositions indicated by points R and E. The line joining the equilibrium compositions of the two phases (raffinate and extract), i.e., line RE is called as a tie line. The tie line is the one which connects together the two phases in equilibrium with each other. The points of tie lines must be found by experiment. The point P on the binodal solubility curve represents a single phase which does not split into two phases (i.e., at this point, the two phases become identical) and at this point A-rich and B-rich curves merge; so the point P corresponds to a tie line of zero length and is known as the plait point (the compositions of the two phases become equal at this point). At a given temperature and pressure, the point P is fixed. The plait point must be found by experiment. It is not necessary that the plait point be at the peak of the binodal solubility curve and tie lines be parallel. In the system under consideration wherein C is the solute and B is the extraction solvent, the separation will increase as the plait point approaches A and tie lines increase in slope. All points outside the binodal curve represent single phase mixtures.

[The curve DRPEG separates the two phase region (lying under this curve) from the one phase region (lying outside this curve) and is also called the phase envelope/phase boundary. Compositions of phases must lie on the phase envelope. At the plait point F, the tie line reduces to a point where the extract and raffinate phases are identical].

From Fig. 2.4, it is clear that % of C in E (extract) is greater than that in R (raffinate) and so, in this case, the distribution of C favours the B-rich phase (extract phase). This is shown on the distribution diagram wherein the composition of C in the extract phase is plotted against the composition of C in the raffinate phase (the compositions expressed in terms of weight fractions). In Fig. 2.4 (b), the points E, R lie above the diagonal $y = x$ indicating that y^* is always greater than x.

So, in this case the distribution coefficient y^*/x is greater than unity. When the concentrations of C at the ends of tie lines are plotted against each other, we get the distribution curve shown in Fig. 2.4 (b).

For ternary systems of the type described above, the mutual solubility of A and B increases with increase in temperature and above the critical solution temperature they get

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dissolved completely. So, liquid extraction operations that are depending upon the formation of insoluble liquid phases must be carried at temperatures below such a temperature.

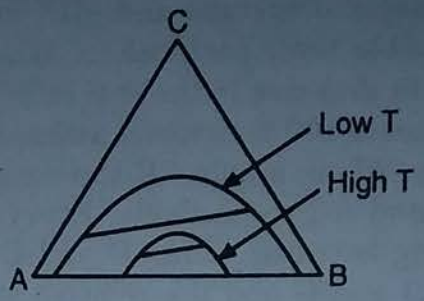

Fig. 2.5 : Effect of temperature

Except at very high pressures, the effect of pressure on the liquid equilibrium is very very small and thus can be neglected.

Systems of three liquids – Two pairs partially miscible :

An example of this type of ternary system is the system chlorobenzene (A) - methyl ethyl ketone (C) - water (B). Here A and C are completely miscible in one another, while the pairs A – B and B – C are partially miscible in one other, i.e., they show only limited solubility. The equilibrium relationship in such a case is as shown in Fig. 2.6.

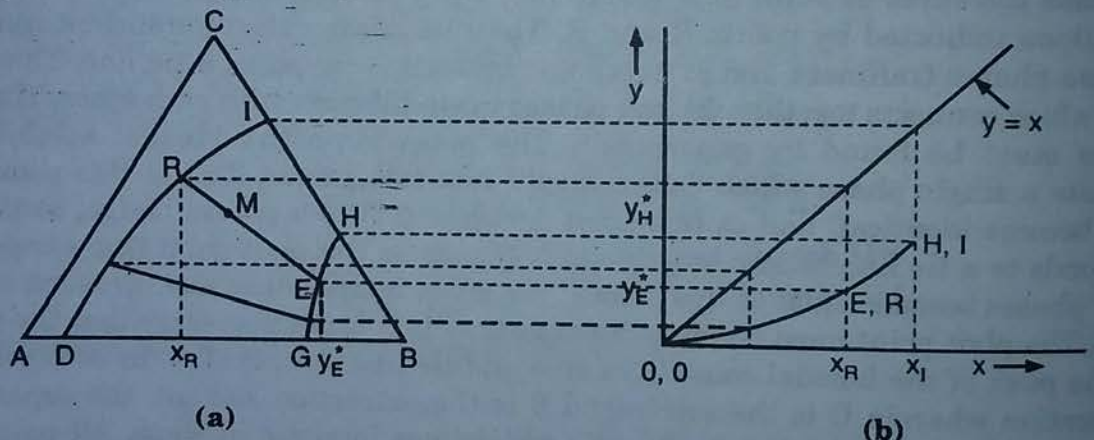

Fig. 2.6 : Systems of three liquids, A – B and B – C are partially miscible

In Fig. 2.6 (a), points D and G represent the mutual solubilities of A and B and points I and H represent the mutual solubilities of B and C. Curved lines DRI (A - rich / raffinate) and GEH (B - rich) are the ternary solubility curves. The mixture such as M within these curves forms two liquid phases in equilibrium represented by points R and E and RE is a tie line. Fig. 2.6 (b) shows the corresponding distribution curve. Any mixture outside the band between these solubility curves forms a homogeneous single phase solution.

Rectangular Coordinates :

Data for ternary systems (three component systems) are frequently presented on equilateral triangular diagrams. However, for engineering calculations, a right-angled triangle/a right-angled triangular diagram is more convenient for presenting the data, since ordinary rectangular-coordinate graph paper may be used in constructing the diagram.

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In case of equilateral triangles, the coordinate scales are always the same. In order to expand one concentration scale relative to the other, rectangular coordinates are used.

Fig. 2.7 shows the use of rectangular coordinates for presenting the equilibrium data wherein concentrations of B are plotted as abscissas and concentrations of C as ordinates. One can use unequal scale to expand the plot as desired. Equation (2.1) is applicable to all mixtures on Fig. 2.7 (a), irrespective of any inequality of the scales.

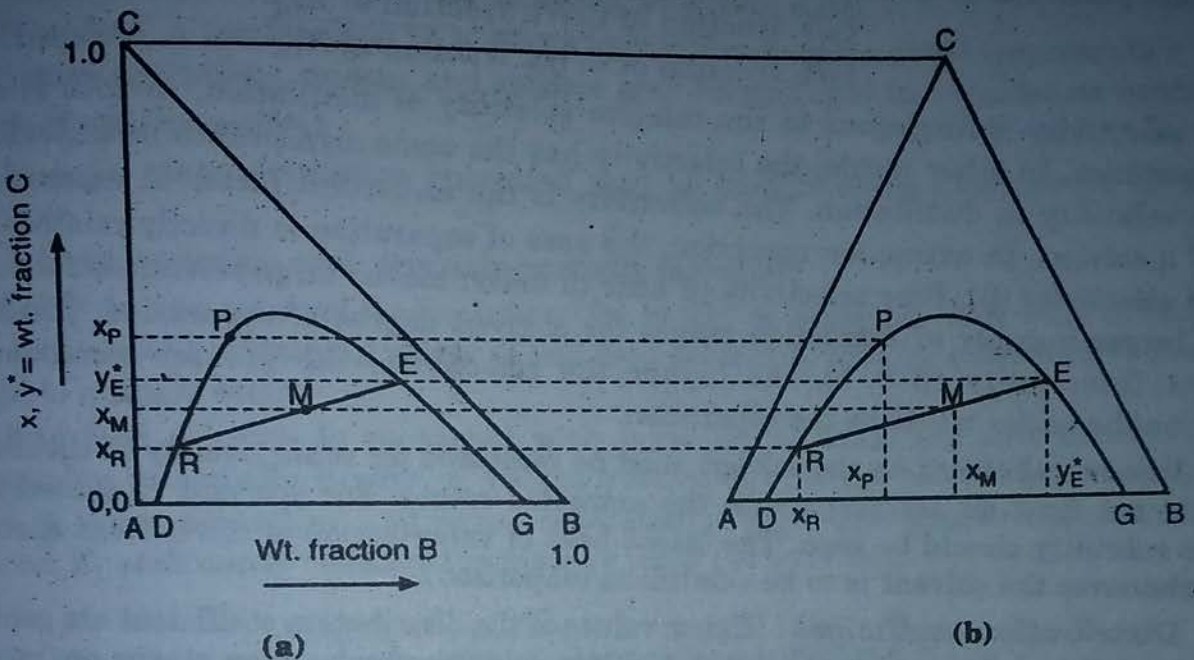

Fig. 2.7 : Rectangular coordinates (For system with one pair partially miscible/soluble)

One of the important factors in selecting a solvent for extraction is the selectivity of the solvent and is given by

$$\beta = \frac{y_E^* \cdot (\text{wt. fraction of A in R})}{x_R \cdot (\text{wt. fraction of A in E})}$$

The selectivity has the same significance in extraction as the relative volatility has in distillation. So it is a measure of the separation by extraction. When the selectivity is equal to one, separation by extraction is not possible. For all practical operations, the selectivity should be greater than one. The selectivity in extraction and the relative volatility in distillation are analogous. Poor selectivity (near unity) means large solvent to feed ratios and so a large number of extraction stages will be required for a specified separation.

Selection of Solvent for Extraction :

Any particular liquid to be used as a solvent for extraction will not possess all the properties considered desirable for extraction and hence, some compromise is usually necessary in selecting the best solvent out of various possible liquid solvents available.

While selecting a solvent for extraction, the qualities such as selectivity, recoverability, distribution coefficient, density, etc. should be given consideration.

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1. Selectivity : The *ratio of concentration ratio of solute to feed solvent in the extract phase to that in the raffinate phase* is called the selectivity or separation factor. It is a measure of the effectiveness of solvent for separating the constituents of a feed. The selectivity should be greater than one for all useful extraction operations and if it is equal to one, separation by extraction is not possible.

If B is the solvent, feed contains C and A, where C is the solute, and E and R are the equilibrium phases then the selectivity, β of solvent B is given as –

$$\beta = \frac{[\text{Wt. fraction of C/Wt. fraction of A}]_E}{[\text{Wt. fraction of C/Wt. fraction of A}]_R}$$

The selectivity is analogous to the relative volatility of distillation, in both meaning and importance. In other words, the selectivity has the same significance in extraction as relative volatility in distillation. The selectivity is the important factor in assessing the value of a solvent. In extraction operation, the ease of separation is directly related to the value of selectivity (β). Poor selectivity (β near to unity) means large solvent to feed ratios and a larger number of extraction stages for a given degree of separation. β must be different from unity ($\beta > 1$), the higher the selectivity, the greater would be the separation/the easier would be the separation.

2. Recoverability : As the solvent must be recovered for reuse usually by distillation, it should not form an azeotrope with the extracted solute. For low cost of recovery, the relative volatility should be high. The latent heat of vaporisation of the solvent should be small whenever the solvent is to be volatilised (vaporised).

3. Distribution coefficient : Higher values of the distribution coefficient are generally desirable as then less amount of solvent and less number of extraction stages are required for a given extraction duty.

4. Capacity : It is the ability of the solvent to dissolve the extracted solute. In ternary systems with one pair partially miscible, solvent can dissolve an infinite amount of solute and this results in generally lower solvent requirements than for ternary systems with two pairs partially miscible.

5. Density : A difference in densities of the saturated liquid phases should be larger for physical separation of the phases.

6. Insolubility of solvent : The solvent insoluble in the original liquid solvent (feed solvent) should be preferred and it should have high solubility for the solute to be extracted as then small amounts of solvent are required.

7. Interfacial tension : The interfacial tension should be high for the coalescence of emulsions to occur more readily, as the same is of greater importance than the dispersion.

8. The solvent should be stable chemically and inert towards the other components and should not be corrosive towards the common materials of construction.

9. The solvent should be cheap.

10. The solvent should be non-toxic and non-flammable.

11. The solvent should have low viscosity, freezing point, vapour pressure for ease in handling and storage (low viscosity is desired for better mass transfer rates and settling of dispersions).

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Liquid-liquid extraction operations may be carried out by using : (1) stagewise equipment (may be a single stage or multistage-crosscurrent, concurrent or countercurrent) and (2) differential contact/continuous contact equipment.

Stagewise Contact :

In **stagewise contact, it is presumed that each stage has facilities (i) of contacting the feed solution and the extracting solvent,** i.e., contacting the insoluble liquids and **(ii) of separating the product streams. Each stage** involved in the extraction operation **represents a theoretical or equilibrium stage** so that the product streams, i.e., the extract **and raffinate leave the stage in equilibrium with each other.**

Therefore, a combination of a mixer and settler forms a stage (represents a stage). In **multistage operations, mixers and settlers** may be arranged in cascades as needed.

Single Stage Operation :

A single stage extraction operation may be performed in a batchwise or continuous manner.

(A) A ternary system with one pair partially miscible :

Let F be the mass of feed solution (for batch) or mass per unit time of feed solution (for continuous) containing solute C and feed solvent A with x_F mass/weight fraction of C is fed to a single stage.

The feed solution is contacted with mass S_1 (or mass per unit time) of a solvent containing principally B with y_s weight fraction of C. If the solvent is pure B, $y_S = 0$ and $S_1 = B$. After contacting and phase separation, we get the equilibrium raffinate R_1 and extract E_1 each one is measured in mass or mass per unit time.

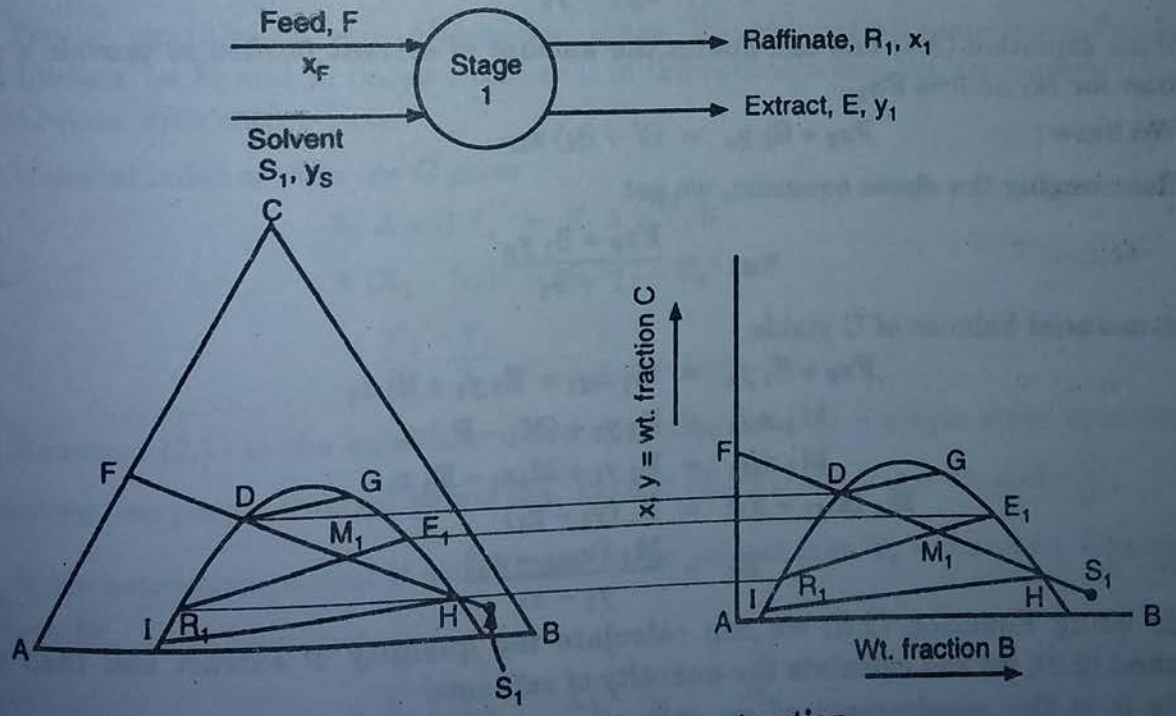

Fig. 2.8 : Single-stage extraction

Let x_1 be the weight fraction of C in the raffinate and y_1 be the weight fraction of C in the extract. As the effluent streams leave the stage in equilibrium, x_1 and y_1 represent the equilibrium values. If the solvent used for extraction is pure, $y_s = 0$ and so, S_1 will be plotted at the apex B.

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Adding solvent S to feed solution F produces a mixture M_1 which on settling splits into the two equilibrium phases, namely E_1 and R_1. These two phases are joined by the tie line passing through point M_1. (Fig. 2.8)

A total overall material balance yields :

$$F + S_1 = M_1 = E_1 + R_1 \quad \ldots(2.2)$$

From Equation (2.2), it is clear that F_1, S_1 and M_1 lie on a straight line as well as E_1, R_1 and M_1.

Point M_1 can be located on line FS_1 by the mixture rule, Equation (2.1) as :

$$\frac{F}{S_1} = \frac{\text{line length } M_1S_1}{\text{line length } FM_1}$$

One can also locate point M_1 by computing the concentration of C in the mixture M_1. Let x_{M1} is the weight fraction of C in the mixture M_1.

A material balance of C yields :

$$Fx_F + S_1y_s = M_1 x_{M1} \quad \ldots(2.3)$$

From Equation (2.3), x_{M1} can be computed.

Eliminating M_1 between Equations (2.2) and (2.3) gives

$$Fx_F + S_1 y_s = (F + S_1)\, x_{M1}$$

$$\therefore \quad \frac{S_1}{F} = \frac{x_F - x_{M1}}{x_{M1} - y_s} \quad \ldots(2.4)$$

With Equation (2.4) one can obtain the amount of solvent needed to provide a given location for M_1 on line FS_1.

We have : $Fx_F + S_1 y_s = (F + S_1)\, x_{M1}$

Rearranging the above equation, we get

$$x_{M1} = \frac{Fx_F + S_1 y_s}{F + S_1} \quad \ldots(2.5)$$

A material balance of C yields :

$$Fx_F + S_1 y_s = M_1 x_{M1} = E_1 y_1 + R_1 x_1$$

$$\therefore \quad M_1 x_{M1} = E_1 y_1 + (M_1 - E_1)\, x_1$$

$$M_1 x_{M1} = E_1 y_1 + M_1 x_1 - E_1 x_1$$

$$M_1 (x_{M1} - x_1) = E_1 (y_1 - x_1)$$

$$\therefore \quad E_1 = \frac{M_1 (x_{M1} - x_1)}{y_1 - x_1} \quad \ldots(2.6)$$

So using Equation (2.6) we can calculate the quantity of extract and then using Equation (2.2), we can calculate the quantity of raffinate.

As it is the requirement of an extraction operation that two insoluble/immiscible phases must form, point M_1 always lies within the heterogeneous area beneath the binodal solubility curve as shown in Fig. 2.8. Therefore, the minimum amount of solvent required is found out by locating M_1 at point D which provides an infinitesimal amount of extract at point G. In the same way, the maximum amount of solvent required is found out by locating M_1 at H which provides an infinitesimal amount of raffinate at point I.

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(B) Single stage operation - Immiscible solvents :

When the extraction solvent and feed solvent are insoluble (immiscible) and remain so over a concentration range of the solute involved then all A fed with the feed solution will remain in the raffinate (raffinate will contain only A and C) and all B in the solvent fed will remain in the extract (extract will contain only C and B).

In such cases, there is no need to use a triangular graph or rectangular graph described earlier. In this case, the solute concentrations are expressed on a C-free basis and the equilibrium solute concentrations are plotted as shown in Fig. 2.9.

Suppose that F mass or mass/time of feed solution containing A kg of A and C kg of C is contacted with a solvent containing B kg of B and after providing a sufficient contact time and applying settling, the phases are removed in equilibrium with each other.

Let

$$X_F = \frac{\text{mass C}}{\text{mass A}} = \frac{\text{kg C}}{\text{kg A}} \text{ in the feed solution}$$

$$X_1 = \frac{\text{mass C}}{\text{mass A}} \text{ in the raffinate}$$

$$Y_1 = \frac{\text{mass C}}{\text{mass B}} \text{ in the extract}$$

$$Y_S = \frac{\text{mass C}}{\text{mass B}} \text{ in the extraction solvent}$$

For an equilibrium stage/theoretical stage, the effluent streams leave the stage in equilibrium, so X_1 and Y_1 (mass ratios of C in the raffinate and extract respectively) are in equilibrium with each other.

Material balance of solute C gives

$$X_F A + B Y_S = X_1 A + Y_1 B$$

$$-A (X_1 - X_F) = B (Y_1 - Y_S)$$

$$\frac{Y_1 - Y_S}{X_1 - X_F} = -\frac{A}{B} \quad \ldots (2.7)$$

Equation (2.7) is the equation of the operating line for a single stage operation. The operating line passes through points (X_F, Y_S) and (X_1, Y_1) and has a slope of $-\frac{A}{B}$.

If the extraction solvent is pure (contains no solute) then $Y_S = 0$ and so, Equation (2.7) reduces to

$$\frac{Y_1 - 0}{X_1 - X_F} = \frac{Y_1}{X_1 - X_F} = -\frac{A}{B} \quad \ldots (2.8)$$

Equation (2.8) is the equation of the operating line for a single stage extraction when the solvent is pure B.

It is a straight line that passes through points $(X_F, 0)$ and (X_1, Y_1) and has a slope of $-\frac{A}{B}$.

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The operating line given by Equation (2.8) is plotted in Fig. 2.9.

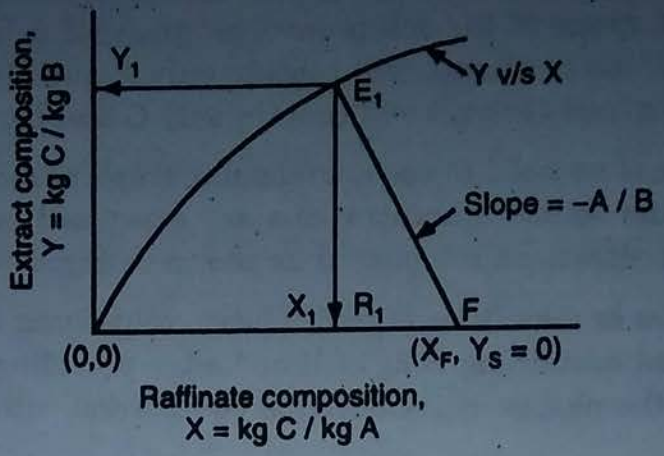

Fig. 2.9 : Single stage operation (insoluble solvents)

Multistage Cross-current Extraction : With Partially Miscible solvents :

It is simply an extension of single stage extraction. This can be performed in a batch or continuous manner. Fig. 2.10 shows a three-stage cross-current extraction operation. Here the raffinate from any stage acts as the feed for the next stage and as the raffinate moves from stage to stage it is contacted with a fresh solvent each time. From this type of arrangement, we get a single final raffinate whereas we get a separate extract from each stage and the extracts from all the stages can be combined to get the composited extract (E).

Fig. 2.10 indicates that different amounts of solvent of the same solute composition are used.

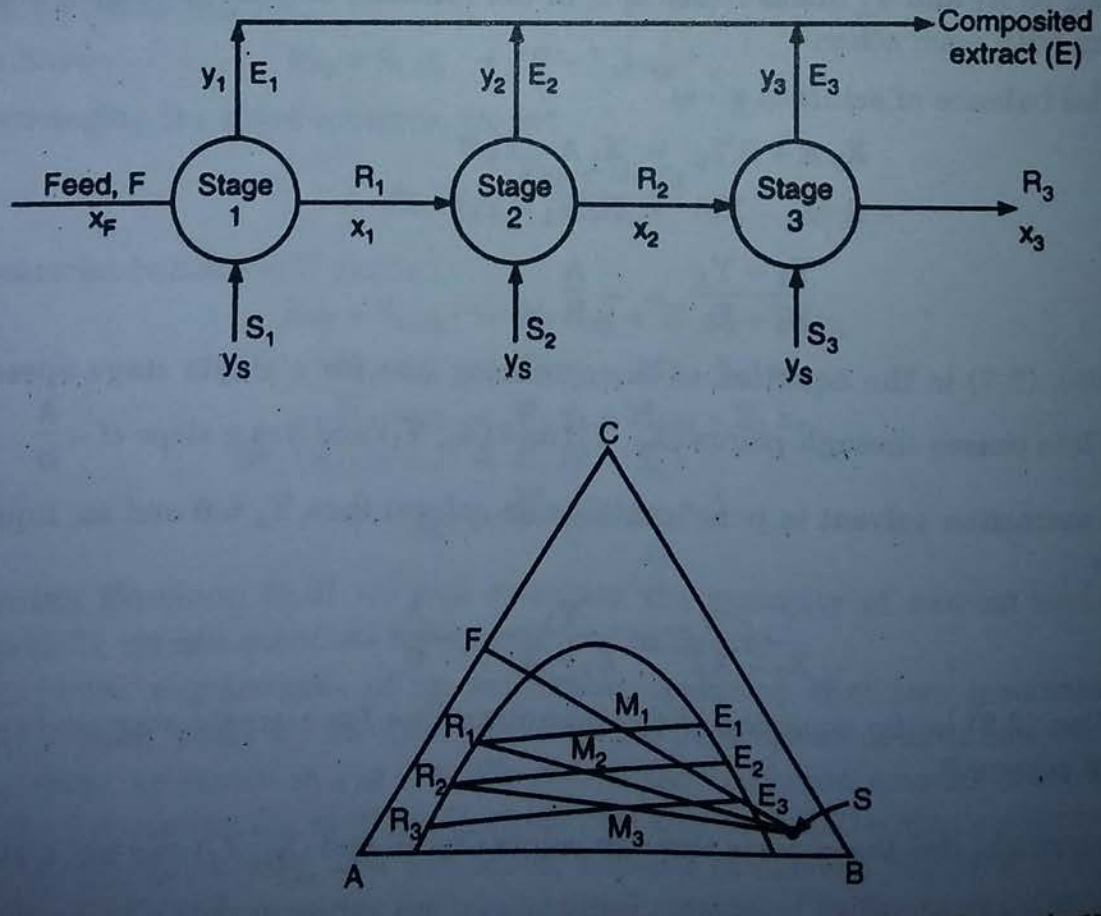

Fig. 2.10 : Multistage cross-current extraction operation (three-stage)

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For stage 1 :

The overall material balance is

$$F + S_1 = M_1 = E_1 + R_1$$

The material balance of solute C is

$$F x_F + S_1 y_s = M_1 x_{M1} = E_1 y_1 + R_1 x_1$$

$$F x_F + S_1 y_s = (F + S_1) x_{M1}$$

$$\therefore \quad \frac{S_1}{F} = \frac{x_F - x_{M1}}{x_{M1} - y_s} \qquad \ldots (2.9)$$

$$x_{M1} = \frac{F x_F + S_1 y_s}{M_1} = \frac{F x_F + S_1 y_s}{F + S_1} \qquad \ldots (2.10)$$

$$M_1 x_{M1} = E_1 y_1 + R_1 x_1$$

$$M_1 x_{M1} = E_1 y_1 + (M_1 - E_1) x_1$$

$$\therefore \quad E_1 = \frac{M_1 (x_{M1} - x_1)}{y_1 - x_1} \qquad \ldots (2.11)$$

For stage 2 :

The overall material balance is :

$$R_1 + S_2 = M_2 = E_2 + R_2$$

The material balance of solute C is :

$$R_1 x_1 + S_2 y_s = M_2 x_{M2} = E_2 y_2 + R_2 x_2$$

$$R_1 x_1 + S_2 y_s = M_2 x_{M2} = (R_1 + S_2) x_{M2}$$

$$\therefore \quad \frac{S_2}{R_1} = \frac{x_1 - x_{M2}}{x_{M2} - y_s} \qquad \ldots (2.12)$$

$$x_{M2} = \frac{R_1 x_1 + S_2 y_s}{M_2}$$

$$x_{M2} = \frac{R_1 x_1 + S_2 y_s}{R_1 + S_2} \qquad \ldots (2.13)$$

$$M_2 x_{M2} = E_2 y_2 + R_2 x_2$$

$$M_2 x_{M2} = E_2 y_2 + (M_2 - E_2) x_2$$

$$\therefore \quad E_2 = \frac{M_2 (x_{M2} - x_2)}{y_2 - x_2} \qquad \ldots (2.14)$$

On the same line we can write the above equations for the n^{th} stage as :

Overall material balance :

$$R_{n-1} + S_n = M_n = E_n + R_n$$

Material balance of solute C :

$$R_{n-1} x_{n-1} + S_n y_s = M_n \cdot x_{Mn} = E_n y_n + R_n x_n$$

$$\therefore \quad \frac{S_n}{R_{n-1}} = \frac{x_{n-1} - x_{Mn}}{x_{Mn} - y_s} \qquad \ldots (2.15)$$

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$$x_{Mn} = \frac{R_{n-1}\, x_{n-1} + S_n\, y_s}{R_{n-1} + S_n} \quad \ldots (2.16)$$

and

$$E_n = \frac{M_n\,(x_{Mn} - x_n)}{y_n - x_n} \quad \ldots (2.17)$$

We can use equal amount of solvent for each time.

The amount of solute extracted, i.e., amount of solute in the composited extract is

$$\text{Solute in the composited extract} = E_1 y_1 + E_2 y_2 + \ldots + E_n y_n$$

$$\text{Composited extract} = E_1 + E_2 + \ldots + E_n$$

Here one can also make use of rectangular coordinates (previously described).

The graphical construction for a multistage crosscurrent extraction on triangular coordinates is shown in Fig. 2.10. The amounts of solvents used for extraction S_1, S_2 and S_3 as are having the same composition, they are represented by point S.

Locate points F and S_1 and then locate M_1 on line FS such that $F/S_1 = M_1S/FM_1$. E_1, R_1 and M_1 also lie on the straight line. Locate E_1 and R_1 at the ends of a tie line through M_1. The tie line is located by a trial and error procedure. The coordinates of E_1 and R_1 will give compositions of the extract and raffinate from the stage - 1. The point R_1 now represents the feed to the stage - 2. R_1, S_2 and M_2 lie on the same line. Locate M_2 on line R_1S_2 (R_1 S) at x_{M2}. R_2 and E_2 are the ends of a tie line through M_2, so locate the tie line by a trial and error procedure. The coordinates of E_2 and R_2 will give compositions of the extract and raffinate leaving the stage-2. R_2 now represents the feed to the stage-3. Repeat the procedure described above and coordinates of E_3 and R_3 will give compositions of the extract and raffinate leaving the stage-3.

We can locate a tie line through M_1 by a trial and error procedure as given in Fig. 2.11. From the x - y diagram, project x and corresponding y on the triangular diagram at E_1 and R_1 and check whether the line joining E_1 R_1 passes through M_1 or not. If not, take another set of x and y till the line E_1 R_1 passes through M_1. In Fig. 2.11, a second trial is a correct trial.

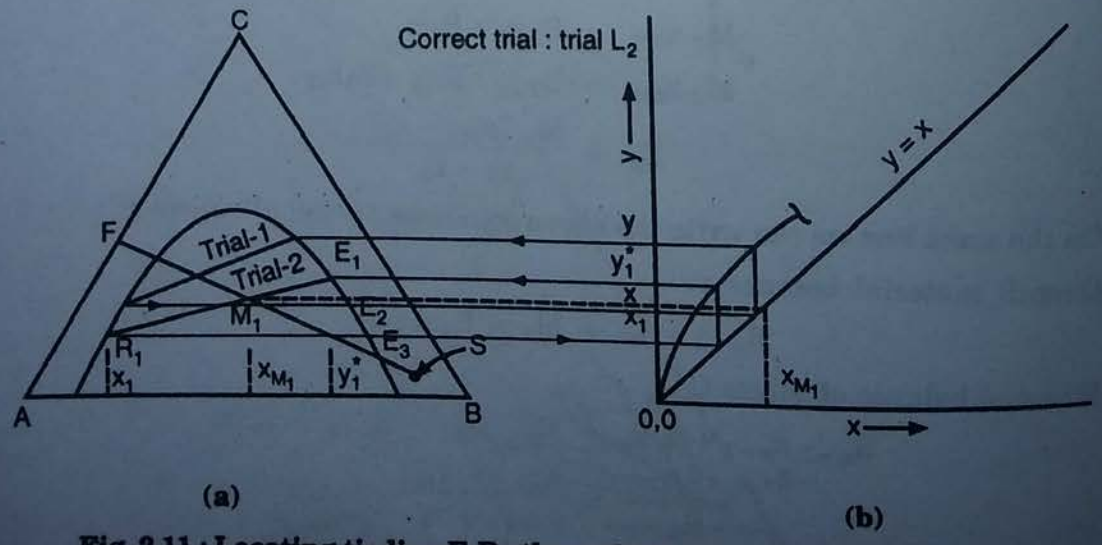

Fig. 2.11 : Locating tie line E_1R_1 through M_1 on triangular coordinates

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The same procedure is to be followed for further stages. We can adopt the same procedure when a ternary system is plotted on rectangular coordinates.

Multistage Crosscurrent Extraction - Immiscible Solvents :

Here we are considering a case where the feed solvent and extraction solvent are completely immiscible. In such cases, the raffinate from all stages will contain only A and C and the extract from each stage contain only B and C. Further, all A in the feed solution will be there in all raffinates and the extract from each stage contain all the solvent B fed to that stage. In such a case, we will express the concentration of C in all the streams in terms of a mass ratio (i.e., mass of C/mass of A or B) of solute C.

Case I : Consider that an equal amount of solvent is used in each stage and the solvent is pure B ($Y_s = 0$).

$$X_F = \frac{\text{mass C}}{\text{mass A}} \text{ in the feed solution}$$

$$X_1, X_2, \ldots = \frac{\text{mass C}}{\text{mass A}} \text{ in the raffinate from stage 1, 2, ...}$$

$$Y_1, Y_2, \ldots = \frac{\text{mass C}}{\text{mass B}} \text{ in the extract from stage 1, 2, ...}$$

B = Amount of pure B (solvent) added in each stage.

With this, for the stage - 1, the material balance of solute C is :

$$AX_F = AX_1 + BY_1$$

$$\frac{Y_1}{X_1 - X_F} = -\frac{A}{B} \quad \ldots (2.18)$$

Equation (2.18) is the operating line equation for stage-1, of slope – A/B, and passing through points $(X_F, 0)$ and (X_1, Y_1).

Now the raffinate from the stage - 1, i.e., R_1 with A and X_1 will be the feed to the stage-2 where it is again contacted with mass B of B.

Material balance of solute C over the stage - 2 :

$$AX_1 = AX_2 + BY_2$$

$$\therefore \quad \frac{Y_2}{X_2 - X_1} = -\frac{A}{B} \quad \ldots (2.19)$$

Equation (2.19) is the operating line equation for stage - 2, of slope – A/B, and passing through points $(X_1, 0)$ and (X_2, Y_2).

Similarly, for the stage - 3, we can write :

$$\frac{X_3}{X_3 - X_2} = -\frac{A}{B} \quad \ldots (2.20)$$

and for the n^{th} stage :

$$\frac{X_n}{X_n - X_{n-1}} = -\frac{A}{B} \quad \ldots (2.21)$$

As equal amounts of solvent are used for each stage, the slope of operating lines for all stages is the same (– A/B), so the operating lines in this case are parallel to each other on Y v/s X plot.

Now, we will do the graphical construction for a three-stage extraction.

(i) Plot Y v/s X.

(ii) The operating line for stage-1 passes through F $(X_F, 0)$ and E_1 (X_1, Y_1) on the equilibrium curve and has a slope of – A/B.

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So through (X_F, 0) draw a line of slope – A/B to cut the equilibrium curve at E_1.

As point E_1 is on the equilibrium curve, Y_1 and X_1 represent the equilibrium values.

This will give us the composition Y_1 of the extract and X_1 of the raffinate leaving the stage - 1.

(iii) Drop a vertical through E_1 to cut the x-axis at R_1 (X_1). The operating line for stage-2 passes through points (X_1, 0) and (X_2, Y_2) and has a slope of – A/B. So draw a line through X_1 on the x-axis of slope – A/B to cut the equilibrium curve at E_2. Point E_2 gives the compositions X_2 and Y_2 of the raffinate and the extract leaving the stage - 2.

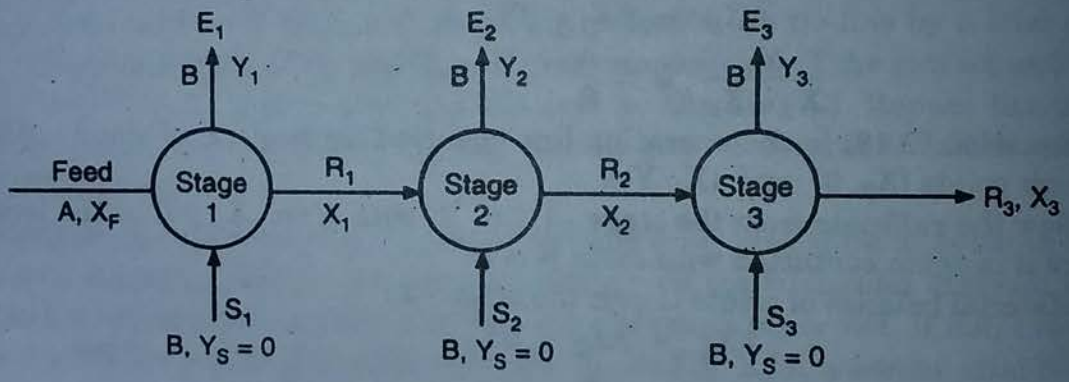
(iv) Adopt the above mentioned procedure to get the compositions X_3 and Y_3 of the final raffinate and extract leaving the stage-3.

When the equilibrium relationship is given by

$$Y = mX$$

then the equilibrium curve is represented by a straight line of slope m.

In such cases, for equal amounts of solvent pure B used for each stage in a multistage extraction, we can derive the equations for obtaining the composition of the final raffinate and the number of stages required.



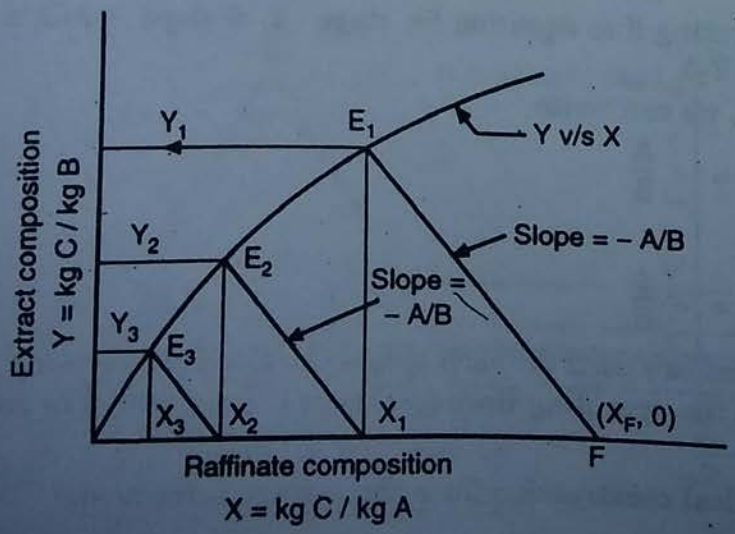


Fig. 2.12 : Multistage crosscurrent extraction - Immiscible solvents

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Material balance of solute over stage - 1 :

$$AX_F = AX_1 + BY_1$$

but $Y = mX$ i.e. $Y_1 = mX_1$

$$AX_F = AX_1 + Bm\,X_1 = (A + Bm)\,X_1$$

$$\therefore \quad X_1 = \left[\frac{A}{A + Bm}\right] X_F \qquad \ldots (2.22)$$

Further addition of a mass B of B to raffinate AX_1 gives us extract of composition Y_2 and raffinate of composition X_2 from the stage-2.

Material balance of solute over stage-2 :

$$AX_1 = AX_2 + BY_2$$

but $Y_2 = mX_2$

$$AX_1 = AX_2 + Bm\,X_2 = (A + Bm)\,X_2$$

$$\therefore \quad X_2 = \left(\frac{A}{A + Bm}\right) X_1 \qquad \ldots (2.23)$$

From Equations (2.22) and (2.23), we get

$$X_2 = \left[\frac{A}{A + Bm}\right]\left(\frac{A}{A + Bm}\right) X_F$$

$$X_2 = \left(\frac{A}{A + Bm}\right)^2 X_F \qquad \ldots (2.24)$$

So, Equation (2.24) for n stages becomes

$$X_n = \left[\frac{A}{A + Bm}\right]^n X_F \qquad \ldots (2.25)$$

We have,

$$X_n = \left[\frac{A}{A + Bm}\right]^n X_F$$

$$\left[\frac{A}{A + Bm}\right]^n = \frac{X_n}{X_F}$$

$$n \ln\left[\frac{A}{A + Bm}\right] = \ln\left(\frac{X_n}{X_F}\right)$$

$$\therefore \quad n = \frac{\ln\left(\frac{X_n}{X_F}\right)}{\ln\left[\frac{A}{A + Bm}\right]} \qquad \ldots (2.26)$$

So if A, B, X_F and n (number of stages) are known, we can calculate the composition of the final raffinate X_n leaving stage-n by using Equation (2.25).

If we know A, B, X_n and X_F, we can calculate the number of stages required to achieve a desired degree of separation by using Equation (2.26).

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Case II : Insoluble liquid - Unequal amounts of solvent, solvent contains some amount of solute (i.e. B ≠ constant throughout and $Y_s \neq 0$) :

The material balance of solute C over any stage-n is :

$$AX_{n-1} + B_n Y_s = AX_n + B_n Y_n \qquad \ldots (2.27)$$

Rearranging, we get

$$\frac{Y_n - Y_s}{X_n - X_{n-1}} = -\frac{A}{B_n} \qquad \ldots (2.28)$$

This is the equation of the operating line for stage-n. It is a straight line on X-Y coordinates of slope $-A/B_n$ and passes through points (X_{n-1}, Y_s) and (X_n, Y_n). The graphical construction for a three-stage extraction is shown in Fig. 2.13 where each operating line is drawn with a slope appropriate to that stage. Each of the operating lines drawn meets the equilibrium curve at a point whose coordinates are the equilibrium compositions of raffinate and extract from that stage. In this case, it is not possible to obtain the raffinate of concentration smaller than that in equilibrium with the entering solvent.

Equations (2.27) and (2.28) are also used with the concentrations of solute expressed as mass/volume, in which case X and Y are expressed in terms of mass solute/volume, and A and B are expressed in terms of volume (for batch) or volume per unit time (for continuous operations).

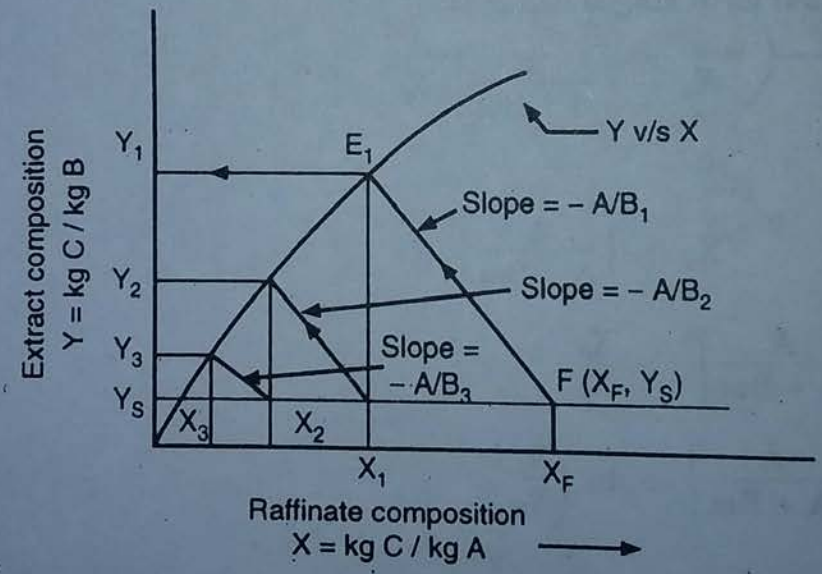

Fig. 2.13 : Multistage crosscurrent extraction (immiscible solvents) - unequal amounts of solvent used

In this case, if $Y_S = 0$ then point F $(X_F, 0)$ is on the x-axis. For unequal amounts of solvent, the operating lines are not parallel but for equal amounts, the operating lines are parallel to each other.

Continuous Multistage Countercurrent Extraction :

I. Systems of three liquids - one pair partially miscible : Fig. 2.14 is a schematic diagram of a typical continuous multistage countercurrent extraction operation. Here raffinate and extract streams flow from stage to stage in a countercurrent fashion.

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Fig. 2.14 shows a typical continuous multistage countercurrent extraction operation where each circle represents an extraction arrangement involving a mixer and a separator. Here, raffinate and extract streams flow from stage to stage in a countercurrent fashion. In this type of arrangement, the feed solution (F) is fed to the first stage which leaves it as raffinate R_1 which then passes through further stages and leaves from the n^{th} stage as final raffinate R_n. The fresh solvent enters the n^{th} stage and passes through the stages in the opposite direction and finally leaves the first stage as final extract E_1. So with this type of unit, we obtain two final products - (i) raffinate R_n and (ii) extract E_1.

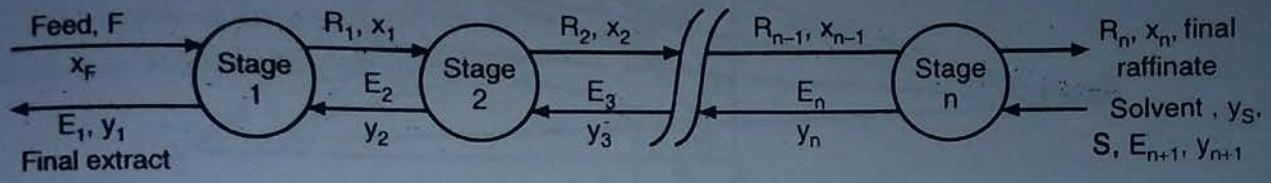

Fig. 2.14 : Multistage countercurrent extraction

Solvent stream to the unit = $S = E_{n+1}$, and $y_s = y_{n+1}$.

Material balance over stage-1 :

$$F + E_2 = R_1 + E_1 \quad \ldots (2.29)$$

$$\therefore \quad F - E_1 = R_1 - E_2 = \Delta_R \quad \ldots (2.30)$$

where Δ_R is the difference point and it represents the net flow outward at the stage - 1.

The material balance for the entire unit (over stages 1 to n) is

$$F + E_{n+1} = R_n + E_1 = M \quad \ldots (2.31)$$

but $E_{n+1} = S$

$$\therefore \quad F + S = R_n + E_1 = M \quad \ldots (2.32)$$

The material balance of solute C over the entire unit :

$$Fx_F + Sy_s = R_n x_n + E_1 y_1 = Mx_M \quad \ldots (2.33)$$

$$x_M = \frac{Fx_F + Sy_S}{M}$$

$$\therefore \quad x_M = \frac{Fx_F + Sy_S}{F + S} \quad \ldots (2.34)$$

Rearranging Equation (2.32), we get

$$F - E_1 = R_n - S = \Delta_R \quad \ldots (2.35)$$

where Δ_R is the difference point and it represents the total net flow outward at the stage-n i.e., the total net flow outward to the right of the entire unit (cascade).

Material balance over n^{th} stage :

$$R_{n-1} + S = E_n + R_n$$

$$\therefore \quad R_{n-1} - E_n = R_n - S = \Delta_R \quad \ldots (2.36)$$

The graphical procedure for determining the number of stages required to pass from an initial concentration of feed solution F to a final raffinate concentration R_n is shown in Fig. 2.15 using rectangular coordinates. The construction procedure for obtaining the number of stages on the equilateral triangle is identical with this.

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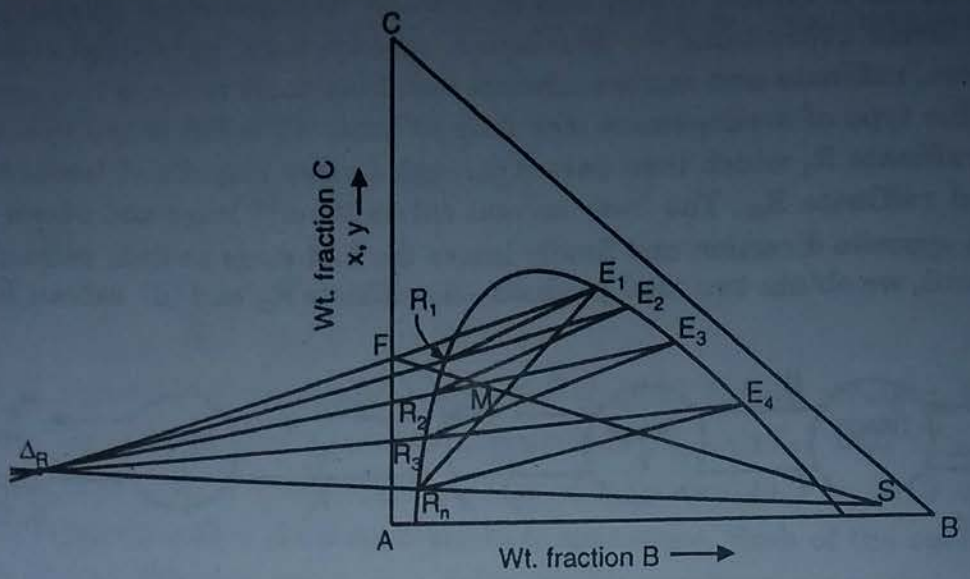

Fig. 2.15 : Graphical construction of number of stages, multistage countercurrent extraction system with one pair partially miscible

Equation (2.31) indicates that F, S and M lie on the same straight line. So join F – S and locate M on FS at x_M. Point F represents the composition of the feed solution and point S, the composition of the fresh solvent. Point M represents the composition of a mixture of F and S such that line length FM/line length MS = mass S / mass F. With this we can easily locate M. Equation (2.32) shows that points R_n, E_1 and M also lie on the same straight line. So the point of intersection of line $R_n - E_1$ and line F – S is the point M. Equation (2.35) indicates that the lines joining $F - E_1$ and R_n – S when extended further they will meet at point ΔR (the difference point).

Equation (2.36) indicates that the difference in flow rates between the raffinate leaving a stage R_{n-1} and the extract entering from the next stage E_n is constant throughout the cascade. So in general R_{n-1}, E_n and Δ_R lie on the same straight line.

Stepwise graphical construction : Refer Fig. 2.15.

(i) Locate points F and S. Locate M on line FS such that length FM/length MS = mass of S / mass of F or by calculating x_M.

(ii) Locate R_n on the solubility curve at x_n. Join R_n – M and extend to meet the solubility curve at E_1.

(iii) Join R_n – S and extend. Join E_1 – F and extend to meet the extended line R_n – S at Δ_R.

(iv) As R_1 and E_1 leave the stage - 1 in equilibrium, these must be the ends of a tie line. E_1 at y_1 is fixed, so from the tie line data find x_1 corresponding to y_1 and locate R_1 at x_1 on the solubility curve. (Prepare a plot of y v/s x by side of the rectangular graph showing the concentrations in the ternary system and x_1 corresponding to y_1 is projected on it from the x – y plot). Join $E_1 - R_1$, E_1R_1 is the required tie line.

(v) Equation (2.36) shows that R_1, E_2, Δ_R lie on the same straight line, so join $\Delta_R - R_1$ and extend it to intersect the curve at E_2 (of composition y_2).

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(vi) $E_2 - R_2$ is the tie line representing stage - 2. So find x_2 corresponding to y_2 from the tie line data and locate R_2 on the solubility curve at x_2 or get x_2 corresponding to y_2 from the x - y plot and locate R_2. Join $E_2 - R_2$, which is the required tie line.

(vii) R_2, E_3, Δ_R lie on the same straight line, join $\Delta_R - R_2$ and extend it to meet the solubility curve at E_3.

(viii) The above mentioned procedure of constructing the stages is repeated till a tie line drawn passes through or crosses point R_n. Each tie line so drawn represents an equilibrium or theoretical stage. Count the tie lines to obtain the number of stages required for passing from x_F to x_n.

Minimum Solvent Requirement :

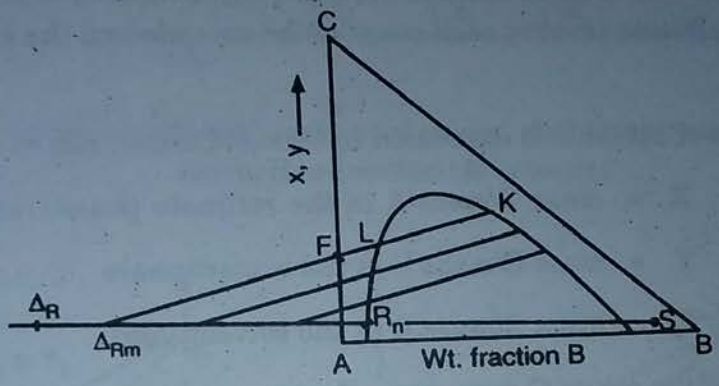

Fig. 2.16 (a) : Minimum solvent for countercurrent extraction

If a line from point Δ_R coincides with a tie line then infinite stages will be required and the position of point Δ_R in this case corresponds to the minimum amount of solvent required. The minimum amount of solvent required is determined by locating point Δ_{Rm}. For this we have to extend tie lines to intersect the extended $S - R_n$ line. If point Δ_R is on the LHS of the diagram then the point of intersection which is farthest from S is the point Δ_{Rm} (it is the point corresponding to the minimum solvent requirement). If point Δ_R is on the RHS of the diagram then the point of intersection (of a tie line with the extended $R_n - S$ line) which is nearest to S is the point Δ_{Rm}.

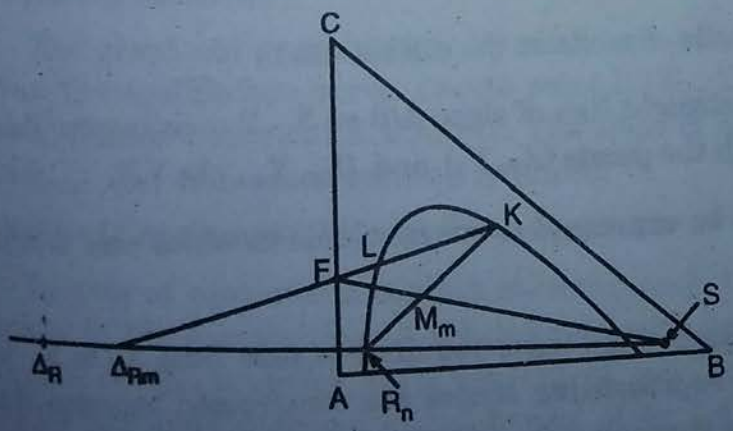

Fig. 2.16 (b) : Minimum solvent for countercurrent operation

Normally, the tie line, which when extended, passes through point F representing the feed, i.e., tie line K – L locates point Δ_{Rm} and thus gives the minimum amount of solvent required.

Join FS, K (E_{1m}) – R_n. The point of intersection of FS and E_{1m} – R_n is M_m. Read x_m and use Equation (2.34) for determining B_{min}.

When $x_M = x_m$, $S = B_{min}$ in Equation (2.34). With $y_S = 0$, $B_{min} = (F\, x_F / x_m) - F$.

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Multistage Countercurrent Extraction - Immiscible Solvents :

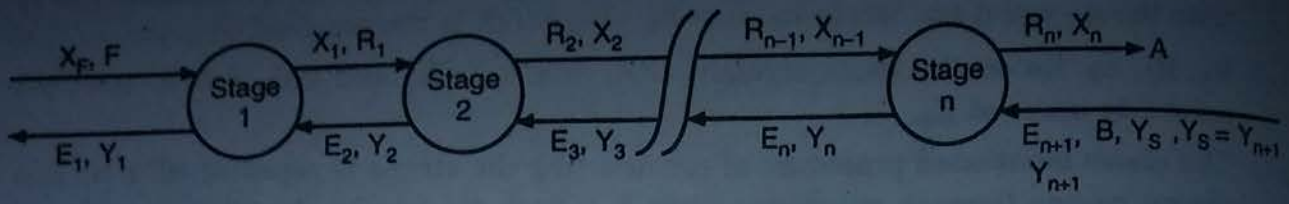

Fig. 2.17 : Multistage crosscurrent extraction-immiscible solvents

Consider that A and B are completely immiscible so that the solvent content of all extracts and the A content of all raffinates are constant (i.e., A kg of A in the feed solution will be there in the raffinate leaving each stage of the cascade and the same is true for B kg of B in the extract).

The concentration of solute C is expressed in terms of mass ratio as :

X = mass C/mass A in the raffinate phase (raffinate streams)

Y = mass C/mass B in the extract phase

Y_s = mass C/mass B in the solvent used

Material balance of the solute C over stage - 1 :

$$AX_F + BY_2 = AX_1 + BY_1 \qquad \ldots (2.37)$$

Material balance of the solute C over the entire unit :

$$AX_F + BY_{n+1} = AX_n + BY_1 \qquad \ldots (2.38)$$

$$A(X_F - X_n) = B(Y_1 - Y_{n+1})$$

$$\frac{Y_1 - Y_{n+1}}{X_F - X_n} = \frac{A}{B} \qquad \ldots (2.39)$$

or

$$Y_{n+1} = \frac{A}{B}(X_n - X_F) + Y_1 \qquad \ldots (2.40)$$

(Note here that $Y_{n+1} = Y_s$ = mass ratio of solute in the solvent used.)

Equation (2.39) is the equation of a straight line of slope A/B on X - Y coordinates. This is the operating line which passes through the points (X_F, Y_1) and $[X_n, Y_{n+1} (= Y_s)]$.

In equation (2.39), X and Y can also be expressed as mass/volume in which case A and B are expressed as volume/time.

The graphical construction of the number of stages required to pass from X_F to X_n is shown in Fig. 2.18 where the triangles representing stages are constructed between the equilibrium curve and the operating line starting from (X_F, Y_1) and covering (X_n, Y_{n+1}).

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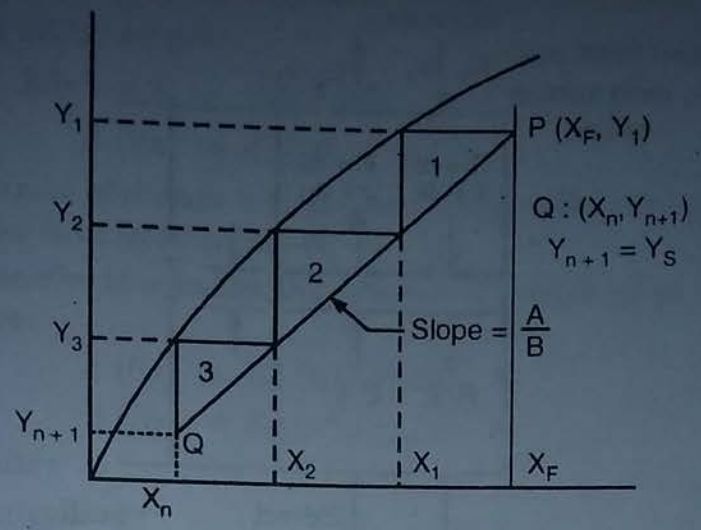

Fig. 2.18 : Construction of number of stages, multistage countercurrent extraction-immiscible solvents

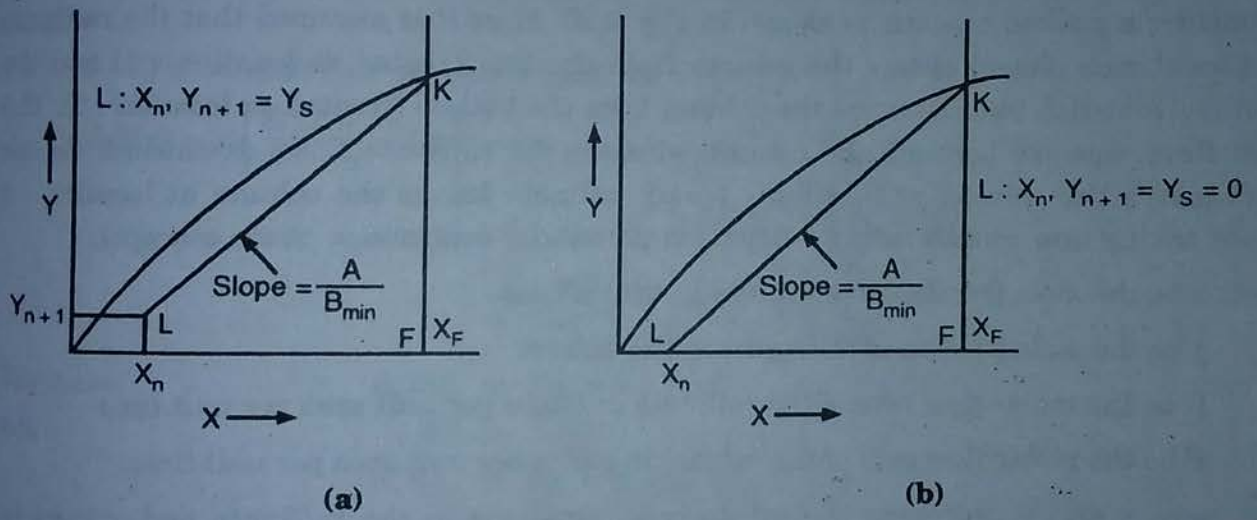

Fig. 2.19 : Minimum solvent, countercurrent extraction, immiscible solvents

Minimum Solvent :

The graphical construction for minimum solvent requirement is shown in Fig. 2.19. Draw the equilibrium curve. Locate point L (X_n, Y_{n+1}). Through X_F draw a vertical to cut the equilibrium curve at K. Join KL. The slope of KL is equal to A/B_{min}. With this slope find out B_{min}, i.e., minimum solvent required.

Design of Continuous Contact Columns :

In case of continuous contact columns such as packed columns there is a continuous change in the composition of either liquid with height of packing as it passes through the column. So the transfer units concept is used to design packed columns for extraction, i.e., for the determination of the height of an active portion of the extractor to effect the desired separation.

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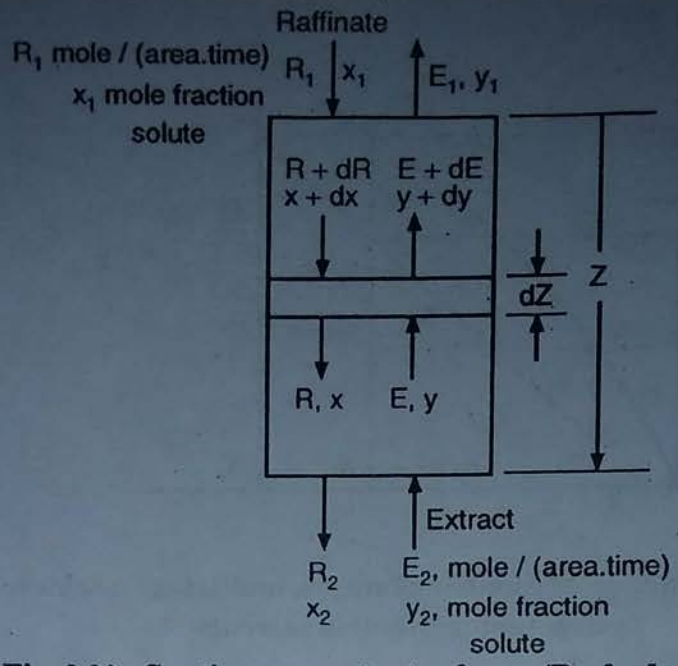

Fig. 2.20 : Continuous contact column (Packed column)

Consider a packed column as shown in Fig. 2.20. Here it is assumed that the raffinate (i.e., diluent-rich phase) enters the column from the top (treated as location - 1) and the extract (solvent-rich phase) enters the column from the bottom (treated as location - 2). The extract flows upward through the column whereas the raffinate flows downward. So the extract leaves the column at location - 1 and raffinate leaves the column at location - 2 (without taking into consideration - dispersed phase and continuous phase concept).

Let x be the mole fraction of the solute in the raffinate

y be the mole fraction of the solute in the extract

R be the molar flow rate of the raffinate in moles per unit area per unit time

E be the molar flow rate of the extract in moles per unit area per unit time.

So, here x and y represent the solute concentrations in the raffinate and extract in terms of molefraction.

S is the total interfacial surface per unit column cross-section.

$$S = \text{specific interfacial surface} \times \text{height}$$

$$S = \text{surface/volume of packing} \times \text{height}$$

$$S = aZ$$

$$\therefore \quad dS = adZ, \ m^2 \text{ of mass transfer surface } /m^2 \text{ of column cross-section}$$

Over a differential height dZ of the column, the solute balance is

$$(R + dR)(x + dx) + Ey = (E + dE)(y + dy) + Rx$$

$$Rx + Rdx + xdR + dRdx + Ey = Ey + Edy + ydE + dEdy + Rx$$

$$Rdx + xdR + dRdx = Edy + ydE + dEdy$$

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Neglecting dRdx and dEdy, we get

$$Rdx + xdR = Edy + ydE$$

$$\therefore \quad d(Rx) = d(Ey)$$

Over a height dZ, the rate of change of solute component within the phase must be equal to the rate of mass transfer to or from the phase.

The rate of mass transfer is equal to the product of overall mass transfer coefficient and overall driving force.

$$\therefore \quad d(Rx) = K_R (x - x^*)\, dS \qquad \ldots (2.41)$$

and $$d(Ey) = K_E (y^* - y)\, dS \qquad \ldots (2.42)$$

Number of Transfer Units :

(1) Raffinate Phase Controlling :

i.e., resistance to mass transfer resides in the raffinate phase.

The raffinate phase flow (R) and its composition (y) vary from one end of the column to the other. If the mutual solubility of diluent and solvent do not vary along the column and only solute is transferred then the term R (1 – x) representing diluent can be taken as constant.

As R (1 – x) is constant, we can write d(Rx) as

$$d(Rx) = d\left[\frac{R(1-x)}{(1-x)} \cdot x\right] = R(1-x) \cdot d\left[\frac{x}{1-x}\right]$$

$$= R(1-x) \frac{[(1-x)\,dx - x(-dx)]}{(1-x)^2}$$

$$= \frac{R(1-x)(dx)}{(1-x)^2} = \frac{R\,dx}{1-x}$$

We have, $$d(Rx) = K_R (x - x^*)\, dS$$

but $$dS = a\, dZ$$

$$d(Rx) = K_R\, a (x - x^*)\, dZ$$

We have, $$d(Rx) = \frac{R\,dx}{(1-x)}$$

$$\therefore \quad \frac{R\,dx}{1-x} = K_R\, a (x - x^*)\, dZ \qquad \ldots (2.43)$$

Multiplying both sides by mean value $(1 - x)_M$,

$$\frac{R(1-x)_M\,dx}{1-x} = K_R\, a (x - x^*)(1-x)_M\, dZ$$

$$\frac{(1-x)_M\,dx}{(1-x)(x-x^*)} = \frac{K_R\, a (1-x)_M\, dZ}{R}$$

$$\int_{x_2}^{x_1} \frac{(1-x)_M\,dx}{(1-x)(x-x^*)} = \frac{K_R\, a (1-x)_M}{R} \int_0^Z dZ$$

$$\int_{x_2}^{x_1} \frac{(1-x)_M\,dx}{(1-x)(x-x^*)} = \frac{K_R\, a (1-x)_M\, Z}{R}$$

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$$Z = \left[\int_{x_2}^{x_1} \frac{(1-x)_M \, dx}{(1-x)(x-x^*)}\right] \cdot \left[\frac{R}{K_R \, a \, (1-x)_M}\right]$$

$$Z = N_{tOR} \times H_{tOR} \quad \ldots (2.44)$$

Height of overall transfer unit based on the raffinate phase :

$$H_{tOR} = \frac{R}{K_R \, a \, (1-x)_M} \quad \ldots (2.45)$$

Number of overall transfer units based on the raffinate phase :

$$N_{tOR} = \int_{x_2}^{x_1} \frac{(1-x)_M \, dx}{(1-x)(x-x^*)} \quad \ldots (2.46)$$

Replace $(1-x)_M$ by the arithmetic mean.

$$(1-x)_M = \frac{[(1-x)+(1-x^*)]}{2}$$

$$= \int_{x_2}^{x_1} \frac{(1-x)+(1-x^*)}{2(1-x)(x-x^*)} \, dx = \int_{x_2}^{x_1} \frac{(2-x-x^*) \, dx}{2(1-x)(x-x^*)}$$

Add x and subtract x in the numerator.

$$= \int_{x_2}^{x_1} \frac{(2-x-x+x-x^*) \, dx}{2(1-x)(x-x^*)}$$

$$= \int_{x_2}^{x_1} \frac{2(1-x) \, dx}{2(1-x)(x-x^*)} + \int_{x_2}^{x_1} \frac{(x-x^*) \, dx}{2(1-x)(x-x^*)}$$

$$= \int_{x_2}^{x_1} \frac{dx}{(x-x^*)} + \frac{1}{2} \int_{x_2}^{x_1} \frac{dx}{(1-x)}$$

$$N_{tOR} = \int_{x_2}^{x_1} \frac{dx}{(x-x^*)} + \frac{1}{2} \ln \left[\frac{1-x_2}{1-x_1}\right] \quad \ldots (2.47)$$

where $(1-x)_M = \dfrac{(1-x^*)-(1-x)}{\ln [(1-x^*)/(1-x)]}$

(2) Extract Phase Controlling :

Derivation for N_{tOE} is similar to N_{tOR}, only thing is that we have to replace R by E and x by y.

$$Z = N_{tOE} \cdot H_{tOE} \quad \ldots (2.48)$$

$$H_{tOE} = \frac{E}{K_E \, a \, (1-y)_M} \quad \ldots (2.49)$$

and

$$N_{tOE} = \int_{y_2}^{y_1} \frac{dy}{(y^*-y)} + \frac{1}{2} \ln \left[\frac{1-y_1}{1-y_2}\right] \quad \ldots (2.50)$$

In Equations (2.45) to (2.50), the solute concentrations are expressed in terms of mole fractions.

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When x and y are expressed in terms of weight fractions then N_{tOE} and N_{tOR} expressions take the forms as given below :

$$N_{tOR} = \int_{x_2}^{x_1} \frac{dx}{(x - x^*)} + \frac{1}{2} \ln\left[\frac{1 - x_2}{1 - x_1}\right] + \frac{1}{2} \ln \frac{x_2 (r - 1) + 1}{x_1 (r - 1) + 1} \quad \ldots (2.51)$$

where r is the ratio of molecular weight of non-solute to molecular weight of solute.

Similarly,

$$N_{tOE} = \int_{y_2}^{y_1} \frac{dy}{y^* - y} + \frac{1}{2} \ln\left[\frac{1 - y_1}{1 - y_2}\right] + \frac{1}{2} \ln \frac{y_1 (r - 1) + 1}{y_2 (r - 1) + 1} \quad \ldots (2.52)$$

Dilute solutions :

For dilute solutions, the second term of the definition of N_{tOR} in Equation (2.47) becomes negligible and so can be discarded (only the integral term is important).

$$\therefore \quad N_{tOR} = \int_{x_2}^{x_1} \frac{dx}{x - x^*} \quad \ldots (2.53)$$

If the equilibrium curve and the operating line are straight over the concentration range involved, then we can use logarithmic averages of the terminal concentration differences for obtaining N_{tOR} and N_{tOE}.

$$\therefore \quad N_{tOR} = \frac{x_1 - x_2}{(x - x^*)_M} \quad \ldots (2.54)$$

and

$$N_{tOE} = \frac{y_1 - y_2}{(y^* - y)_M} \quad \ldots (2.55)$$

where $(x - x^*)_M$ = logarithmic average of the concentration differences at the ends of the column (raffinate phase)

$$= \frac{(x_1 - x_1^*) - (x_2 - x_2^*)}{\ln\left[\dfrac{x_1 - x_1^*}{x_2 - x_2^*}\right]}$$

and $(y^* - y)_M$ = logarithmic average of the concentration differences at the ends of the column (extract phase)

$$= \frac{(y_1^* - y_1) - (y_2^* - y_2)}{\ln\left[\dfrac{y_1^* - y_1}{y_2^* - y_2}\right]}$$

Equations (2.54) and (2.55) are applicable for a linear equilibrium curve given by :

$$y^* = mx + c$$

The solute transferred in terms of the mass transfer coefficients is given by

$$R (x_1 - x_2) = E (y_1 - y_2) = K_R \, a \, Z (x - x^*)_M = K_E \, a \, Z (y^* - y)_M \quad \ldots (2.56)$$

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Dilute solutions, Henry's law :

If in addition Henry's law applies then the equilibrium curve is a straight line passing through the origin (i.e., $m = y^*/x = x^*/y$, where x^* is the concentration in equilibrium with y and y^* is the concentration in equilibrium with x) and in such cases N_{tOE} and N_{tOR} are given by

$$N_{tOR} = \frac{\ln\left[\frac{x_1 - y_2/m}{x_2 - y_2/m}(1 - R/mE) + R/mE\right]}{1 - R/mE} \quad \ldots (2.57)$$

$$N_{tOE} = \frac{\ln\left[\frac{y_2 - mx_1}{y_1 - mx_1}(1 - mE/R) + mE/R\right]}{1 - mE/R} \quad \ldots (2.58)$$

Equations (2.54), (2.55), (2.57) and (2.58) given on a mole basis are equally applicable when solute concentrations are expressed on a weight basis (i.e., in terms of weight fractions), in which case we have to define m accordingly, and E and R have to be expressed as mass/(area)·(time).

When solute concentrations are expressed as c mol/volume, Equations (2.54), (2.55) and (2.56) are also applicable in which case we have to express the raffinate and extract phase flow rates as V volume/(area)·(time) and mass transfer coefficients as mol/(area)·(time)·Δc driving force.

Neglecting a change in the molar flow rate, the amount extracted in the height dZ is Edy.

$$\therefore \quad Edy = K_E (y^* - y)\, a\, dZ \quad \ldots (2.59)$$

$$\frac{dy}{y^* - y} = \frac{K_E a}{E} dZ$$

$$\int_{y_2}^{y_1} \frac{dy}{y^* - y} = \frac{K_E a}{E} \int_0^Z dZ$$

But $$\int_{y_2}^{y_1} \frac{dy}{y^* - y} = N_{tOE} = \frac{y_1 - y_2}{(y^* - y)_M}$$

$$\therefore \quad \frac{(y_1 - y_2)}{(y^* - y)_M} = \frac{K_E a}{E} \cdot Z$$

$$\therefore \quad E(y_1 - y_2) = K_E\, a\, Z\, (y^* - y)_M$$

Similar treatment is to be given for raffinate phase.

So we can write

$$R(x_1 - x_2) = E(y_1 - y_2) = K_E\, a\, Z\, (y^* - y)_M = K_R\, a\, Z\, (x - x^*)_M$$

K is overall mass transfer coefficient, mole/(area)·(time)·(mole fraction).

K_R is for raffinate phase and K_E is for extract phase.

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$(y^* - y)$ is overall driving force for extract phase and $(x - x^*)$ is overall driving force for raffinate phase both are in terms of mole fraction. 'a' is specific interfacial surface in m^2/m^3.

The number of transfer units is a measure of ease or difficulty of the separation (large number of transfer units are required for high degree of separation). The height of transfer unit is the measure of the separation effectiveness of a particular packing.

Extraction Equipments :

In liquid-liquid extraction operation, two phases must be brought into an intimate contact for mass transfer to occur and then the phases are separated. This operation may be carried out batchwise or continuously. Whenever we are dealing with simple systems and quantities to be handled are small, batchwise procedure is adopted and when large quantities are to be handled and several contacts are required, continuous operation becomes economical.

Most of the extraction equipments are operated continuously with either successive stage contacts or differential contacts.

Industrial extraction equipments are classified as :

1. Stage-type extractors and
2. Differential (continuous contact) extractors.

Mixer-settler (single and multistage), plate column, pulsed sieve plate column, etc. are examples of stage-type equipment/extractor.

Packed column, spray column, pulsed packed column, rotating disc contactors, etc. are examples of differential extractors.

In stage-type extractors, the two phases are allowed to mix together so as to reach equilibrium and then are separated before being passed countercurrent to each other. Advantages of stage-type contactors include simplicity of design, no axial mixing, high stage efficiency. These units are large and bulky owing to the requirement of a separator for the phase separation after each stage.

In differential extractors, the two phases are always in continuous contact. These are compact for a given throughput and require small floor space.

In case of plate column, packed column, spray column used for extraction operation, the phase interdispersion and countercurrent flow are produced by the force of gravity acting on the density difference between phases. In case of mixer settlers, rotating disc contactors, the phase interdispersion is produced by mechanical agitation and counter-current flow is still produced by gravity.

Mixer-Settlers :

A mixer-settler is a single-stage extraction device which comprises of a mixer for contacting the two liquid phases to effect mass transfer and a settler for mechanical separation of the phases. It is the most simple type of extractor and has many variations.

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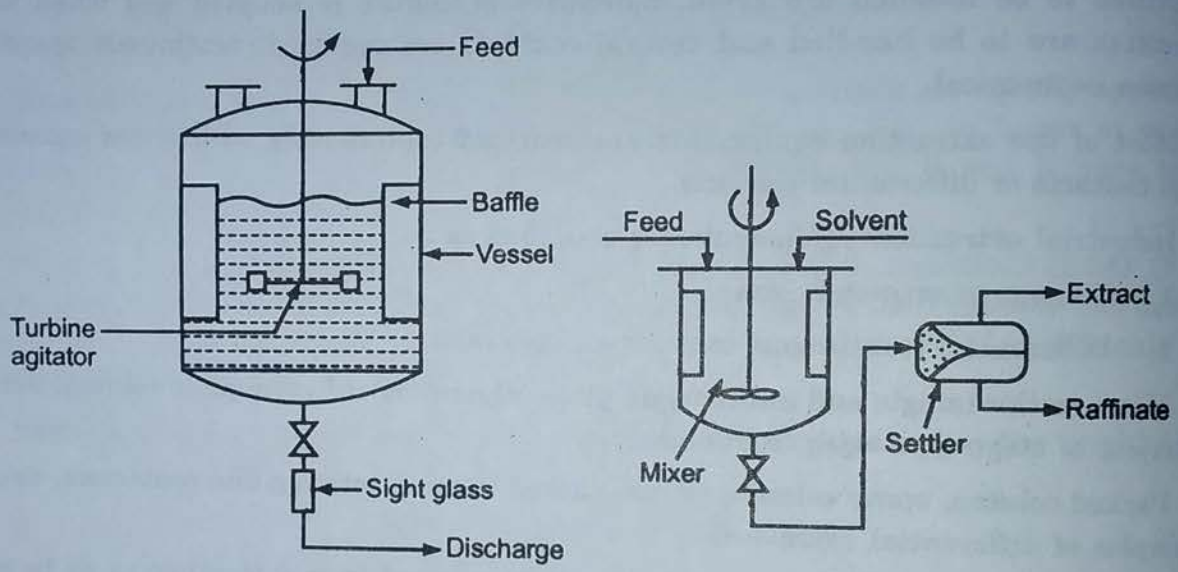
In case of extraction operations carried out a batchwise manner/fashion, the mixer and settler may be the same unit (i.e., mixing and settling is carried out in the same piece of equipment/in the same equipment) [See Fig. 2.21 (a)]. It consists of a vertical tank/vessel incorporating a turbine or propeller agitator. It is provided with charging nozzles at the top and a discharge connection provided with a sight glass at the bottom. The feed solution to be extracted is taken into the agitated vessel, required amount of the solvent is added, and the whole mass is agitated for a predetermined time. At the end of mixing cycle, agitation is stopped and settling is applied for the phase separation. Afterwards, the raffinate and extract phases are withdrawn from the bottom discharge connection into separate receivers.



(a) Mixer settler for batch operation (Mechanically agitated vessel)

(b) Continuous single-stage Mixer-settler

Fig. 2.21

For continuous extraction operation, the mixer and settler are separate units. [See Fig. 2.21 (b)]. The mixer is a small baffled-agitated tank provided with inlet-outlet connections and the settler is frequently a continuous gravity decanter (simple settler with a coalescer for emulsifying liquids).

In this extractor, the two phases are in continuous contact with each other in the mixer under thorough agitation before flowing to the settler for the phase separation.

One phase is usually dispersed into the other in the form of small droplets so as to produce large interfacial areas resulting in a faster extraction.

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When several contacts are required, the battery/train of mixer settlers operated in a

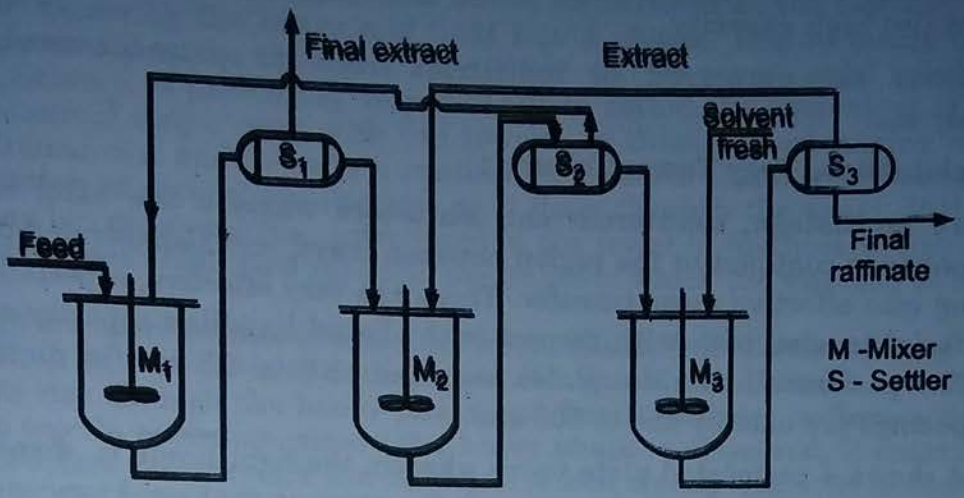

Fig. 2.22 : Mixer settler battery for countercurrent extraction

countercurrent fashion is used. Generally, liquids are pumped from one stage to the next, but if sufficient headroom is available gravity flow is arranged. A feed solution is introduced in the first mixer and a fresh solvent is fed to the last mixer so that the raffinate from each settler becomes the feed to the next settler, and the extract from the last settler acts as the solvent to the second last mixer, and so on. The final extract leaves the first settler while the final raffinate is taken out from the last settler.

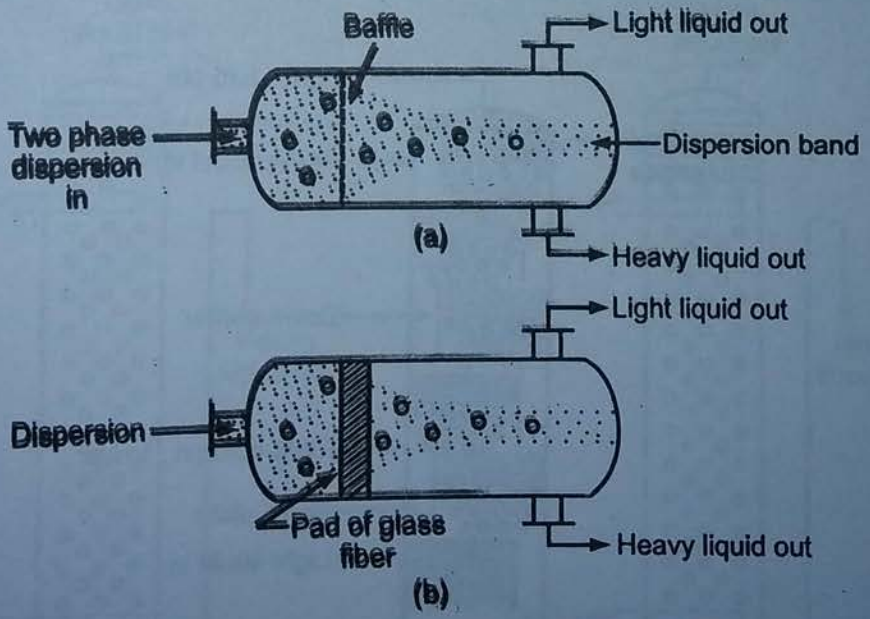

Fig. 2.23 : Gravity settlers (a) simple and (b) with coalescer

In case of liquids which emulsify easily and having nearly the same density, it is necessary to pass the mixer discharge through a pad of glass fibre to coalesce the droplets of the dispersed phase before gravity settling is feasible.

Since pumps and piping systems are involved in the installation of a train of mixer-settlers, initial investment on equipment is high.

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Advantages of mixer-settlers :

(i) High stage efficiency, (ii) good flexibility, (iii) capacity to handle liquids of high viscosity, and (iv) high capacity.

These units are employed in industries such as petrochemical, fertilizers, metallurgical, etc.

Perforated plate/Sieve Tray Tower/Column :

These are multistage, countercurrent contactors wherein the axial mixing of the continuous phase is confined to the region between trays, and redispersion occurs at each tray resulting into effective mass transfer. These are very effective, especially for systems of low interfacial tension both with respect to the liquid handling capacity and extraction efficiency. The perforations in the plates are about 1.5 to 4.5 mm in diameter and the plate/tray spacings are usually 150 to 600 mm.

Fig. 2.24 shows a perforated plate tower wherein the light liquid is dispersed. General arrangement of the plates and down comers is the same as that used for gas-liquid contact except that no weir is needed. Light liquid (dispersed phase) is introduced at the bottom, passes through the perforations in the plate in the form of fine droplets, which rise through the heavy continuous phase, coalesce into a layer beneath the plate, and again redisperses through the plate above. Heavy continuous phase liquid is introduced at the top, passes across each plate and flows downward from a plate above to the plate below via down comer. The principal interface is maintained at the top, light liquid is removed from the top and heavy liquid from the bottom. The heavy liquid can be dispersed in which case, the tower is turned so that the down comer of each plate becomes the upcomer.

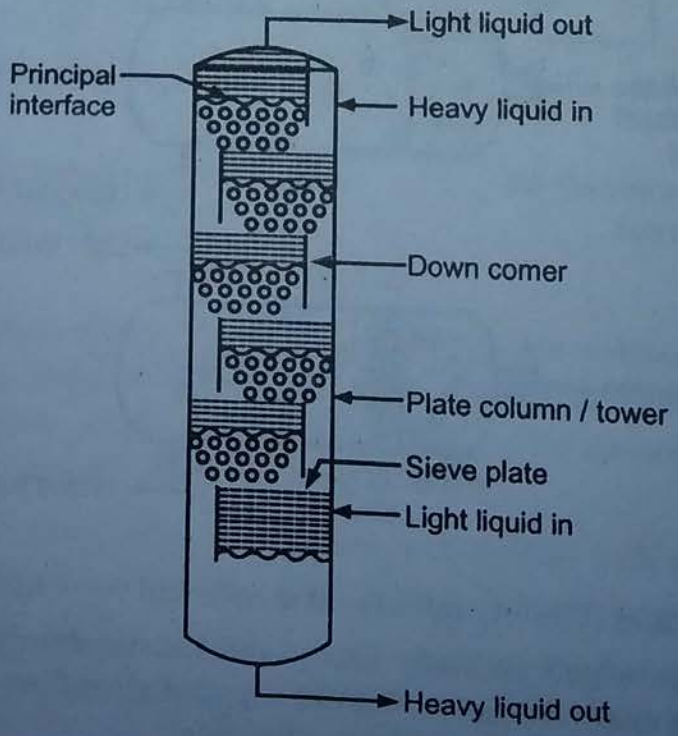

Fig. 2.24 : Perforated plate/tower for continuous countercurrent extraction (light-liquid dispersed)

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Spray Tower/Column :

Spray towers are the simplest of differential contactors. In its simplest form, it consists of an empty tower provided with inlet and outlet connections at the top and bottom, for introducing and removing the heavy and light liquid phases. Due to maximum freedom for a liquid movement, as the shell is empty, there will be severe axial mixing in these extractors and hence it is practically very difficult to obtain the equivalent of more than one or two theoretical stages even with tall towers. In these towers, mixing and settling proceeds simultaneously, and there is a continuous transfer of material between phases, and the composition of each phase changes as it flows through the tower. Either light or heavy phase may be dispersed (i.e., introduced in the tower in the form of fine drops), but generally the phase with high flow rate and the phase of which the viscosity is high should be dispersed.

Spray towers are operated into two distinct ways - Either light phase or heavy phase may be dispersed. When light phase is to be dispersed, it is introduced through a nozzle from the bottom, droplets rise through the heavy phase and finally coalesce to form a liquid-liquid interface at the top. In operation, where the heavy phase is dispersed, it enters the tower from the top, droplets fall through the light-phase and finally coalesce to form a liquid-liquid interface at the bottom of the tower and then leaves the tower in the form of stream through the bottom outlet. In this case, the light phase is introduced at the bottom, flows as a continuous phase and ultimately leaves the tower from the top. Fig. 2.25 shows both methods of operating spray towers.

Though the spray tower is simple to construct, its performance is poor because of considerable recirculation of the continuous phase (axial mixing) and little turbulence in it.

In spray towers, the droplets of dispersed phase rise or fall through the continuous phase under the influence of gravity, thus there is a limit for the amount of dispersed phase that can pass through the tower for any given flow rate of the continuous phase. Additional light

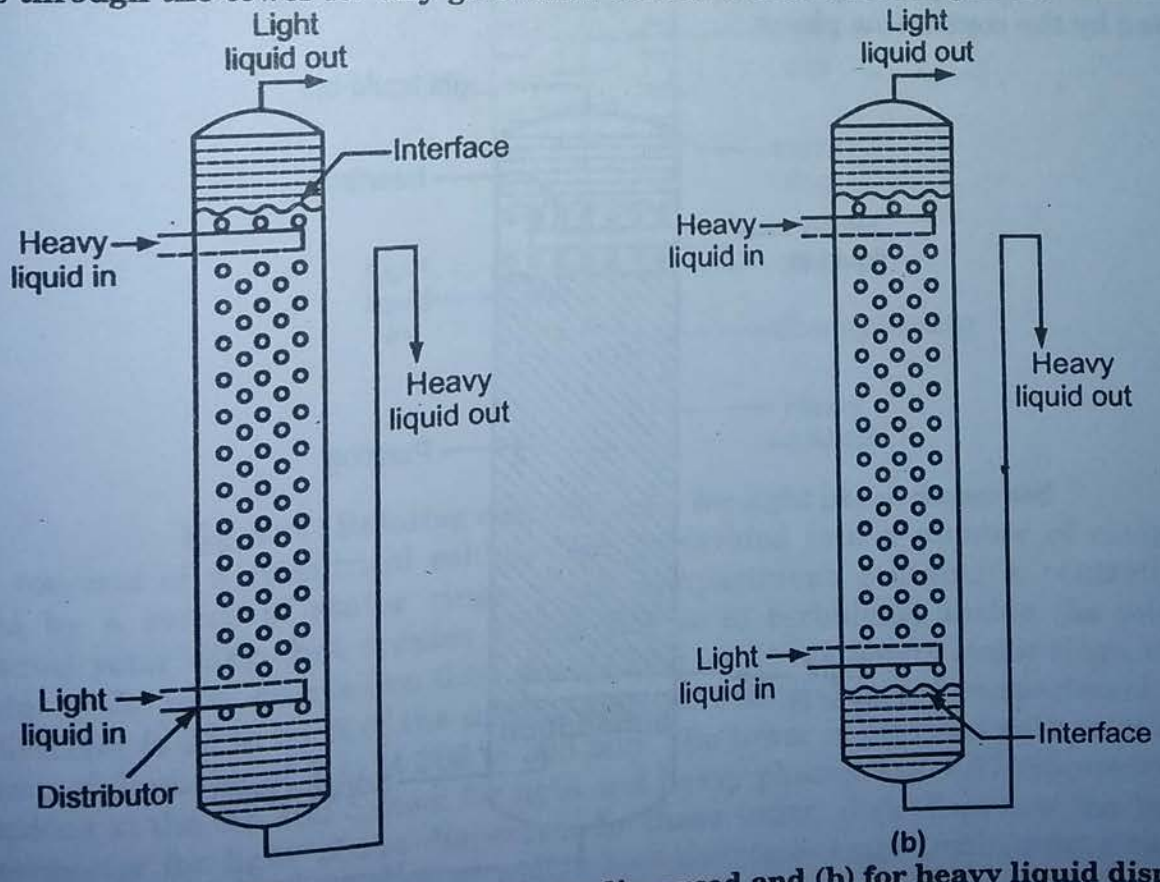

Fig. 2.25 : Spray towers (a) for light liquid dispersed and (b) for heavy liquid dispersed

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phase fed to the bottom of the tower (wherein the light liquid is dispersed) in excess of that flows upwards under the influence of gravity will be rejected through the bottom of the tower and the tower is then said to be flooded.

It is simple in construction and does not contain any internal structures like plate, packings, etc. so gives high throughputs per unit cross-sectional area.

Disadvantages of the spray column include – low efficiency due to axial mixing in the continuous phase and necessity for a tall tower to effect a given degree of separation.

Packed tower/Column :

Randomly packed towers used for gas-liquid contact are also adopted for liquid-liquid extraction. As packing provides large interfacial area for the phase contacting and causes the drop to coalesce and reform, the mass transfer rates in packed towers are very high than those obtained with spray towers as packings cut down the recirculation of the continuous phase. Packed towers are used in the petroleum industry. Packed towers are unsuitable to handle dirty liquids, suspensions or high viscosity liquids.

Packed tower shown in Fig. 2.26 is arranged for the light phase dispersed. It consists of a cylindrical shell filled with packings, packings rest on a support plate, and the shell is provided with inlet-outlet connections at the top and the bottom for introduction and withdrawal of liquid phases. The heavy liquid phase is fed from the top and the light liquid (dispersed through a distributor) is fed from the bottom. A large portion of the void space in packings is filled by the continuous phase which flows downward and remainder of the void space is filled with drops of light liquid which rise through the continuous phase and finally coalesce to form a liquid-liquid interface at top. The light liquid leaves the tower from the top and heavy liquid leaves from the bottom. There is a continuous transfer to material between phases, and the composition of each phase changes as it flows through the tower/column. Usually the packing material should be such that it should preferentially get wetted by the continuous phase.

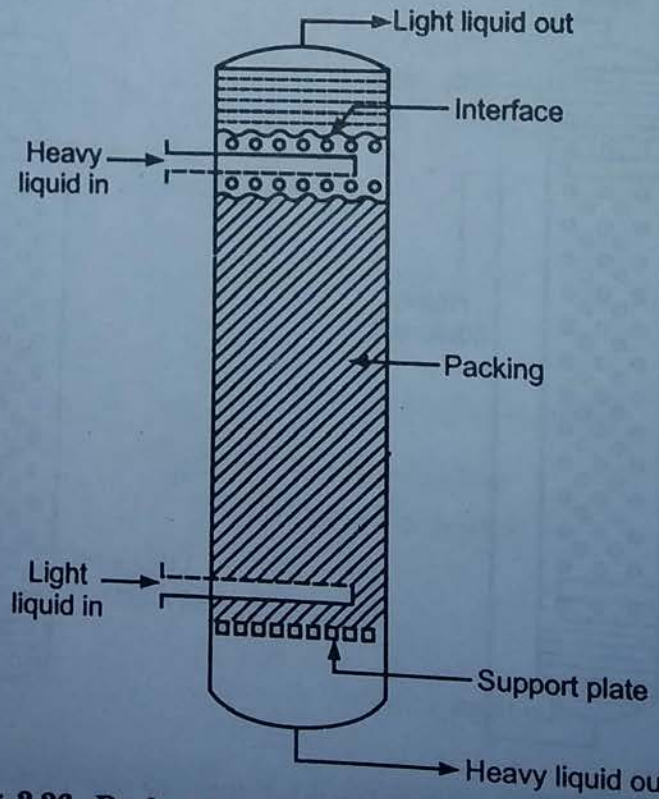

Fig. 2.26 : Packed tower for light phase dispersed

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Carbon and plastic packings get preferentially wetted by organic liquids and ceramic by aqueous solutions. The packing size should not be greater than one-eighth the tower diameter. The dispersed phase distributor is always embedded in the packing whenever the dispersed phase droplets do not wet the material of a packing support to avoid premature flooding (drops will have difficulty in entering the column).

Packed columns are easy to construct. They can be made to handle corrosive liquids at a reasonable cost. Their performance is better than spray column. The chief disadvantage of packed column is that the solids tend to collect in the packing and cause channeling.

Rotating Disk Contactor (R.D.C.) :

For systems of high interfacial tension, good dispersion is achieved by mechanical agitation of the liquids which in turn results in good mass transfer rates.

Rotating disk contactor is a mechanically agitated countercurrent extractor wherein agitation is brought with the help of rotating disks which usually runs at much higher speed than turbine type impellers.

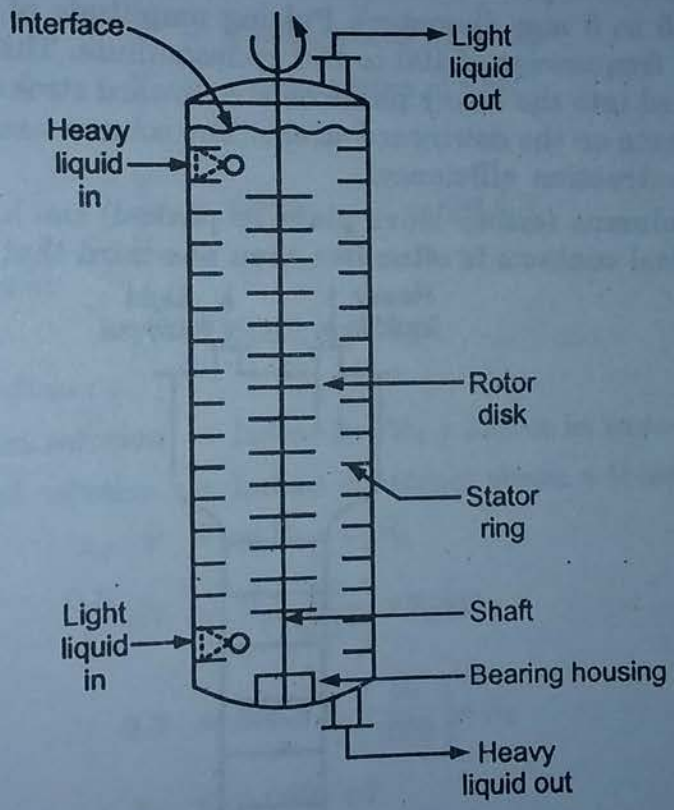

Fig. 2.27 : Rotating disc contactor for light phase dispersed

It consists of a cylindrical column that is divided into a number of compartments formed by a series of stator rings. Each compartment contains a centrally located horizontal rotor disk that creates a high degree of turbulence inside the column. The diameter of the rotor disk is less than the opening in the stationary stator rings, usually the disk diameter is 33 to 66 % of the column diameter. Recommended compartment height for a column of diameter of 2 m is 200 to 300 mm. The tower is provided with inlet and outlet connections at the top and bottom for light and heavy phases. Fig. 2.27 represents rotating-disk contactor for light phase dispersed. In these units, disk disperses the liquids and impels them outward towards tower wall, where the stator rings create quiet zones wherein the two phases can separate.

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It has reasonable capacity, low operating cost and high efficiency. With corrosive liquids, it is very difficult to maintain the internal moving parts (maintenance problem - a major disadvantage).

It is commonly used in the petroleum industry particularly for furfural extraction, separation of aromatics from aliphatics, etc. and pharmaceutical industries.

Pulse Column :

A pulse column may contain ordinary packings or sieve plates without down comers. Applications of oscillating pulse to the content of such columns increase the efficiency because the pulse increases both turbulence and interfacial areas.

Reciprocating pump is a common mechanical pulsing device which pulses the entire contents of the column at a frequent interval, so that a rapid reciprocating motion of relatively small amplitude is superimposed on usual flow of the liquid phases. Bellow or diaphrams made of steel teflon, etc. are also used as pulsing device.

A pulsed sieve plate column is a vertical column with a large number of sieve plates without down comers. The perforations in the plates of such column are smaller than non-pulsing column (1.5 to 3 mm diameter). Pulsing amplitude of 5 to 25 mm are generally recommended with frequencies of 100 to 260 cycles/minute. The pulsation causes the light liquid to be dispersed into the heavy phase on the upward stroke and heavy liquid phase to jet into the light phase on the downward stroke. The column has no moving parts, low axial mixing and high extraction efficiency.

In the pulse columns (either sieve plate or packed) the height required for a given number of theoretical contacts is often less than one-third that needed in the non-pulsing column.

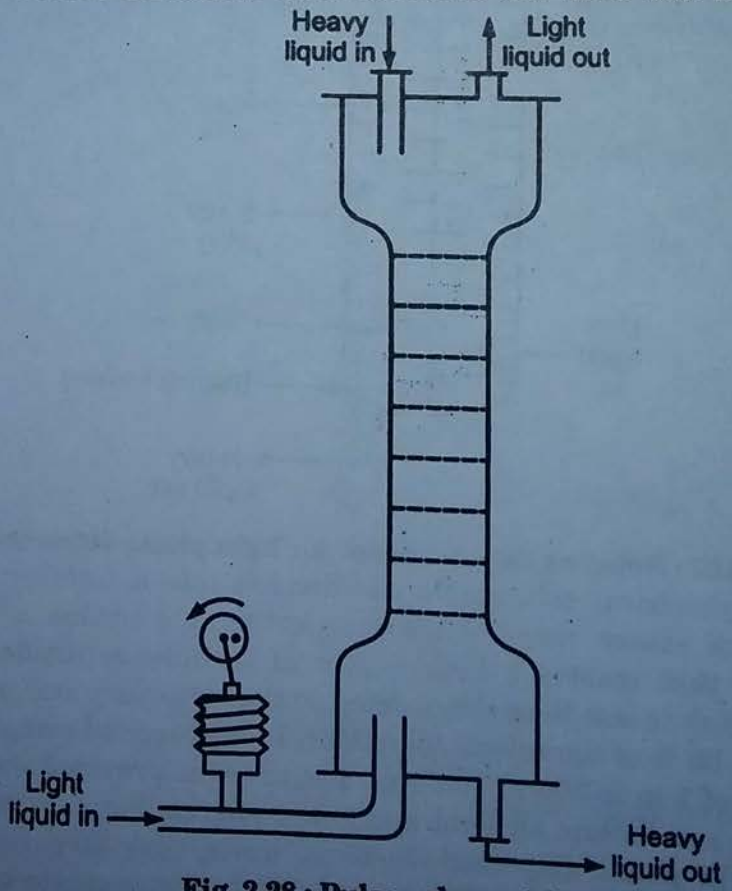

Fig. 2.28 : Pulse column (sieve plate)

Pulse columns are extensively used for processing radioactive solutions in atomic-energy work.

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SOLVED EXAMPLES

Ex. 2.1 : Carbon disulphide (CS_2) is used to extract iodine from its saturated aqueous solution. The distribution of iodine between carbon disulphide and water at equilibrium is given by

$$K = \frac{Y}{X} = \frac{\text{g iodine/1 } l \text{ of } CS_2}{\text{g iodine/1 } l \text{ of water}} = 588.2$$

(i) Calculate the concentration of iodine in the aqueous phase if 1 l of a saturated aqueous solution of CS_2 at 293 K (20°C) containing 0.3 grams of iodine per 1 litre of water is stirred with 50 ml of carbon disulphide.

(ii) Repeat for two ideal extractions using 25 ml of solvent each time. [1753] - 301 P. U.

Solution : Basis : 1 litre of saturated solution at 293 K

It contains 0.3 g of iodine.

(i) For one stage extraction using 50 ml of CS_2

X = g iodine/litre of water

Y = g iodine/litre of CS_2

At equilibrium,

$$\frac{Y}{X} = 588.2, \qquad Y = 588.2\,X$$

For single stage, we have

$$Y_1 = 588.2\,X_1$$

Material balance of iodine :

Iodine in saturated solution = Iodine in CS_2 + Iodine in water

Iodine in feed solution = Iodine in extract phase + Iodine in raffinate phase

$$x_F \cdot F = y_1 E_1 + x_1 R_1$$

$$\frac{0.3}{1} \times 1 = Y_1\left(\frac{50}{1000}\right) + X_1\,(1)$$

$$0.3 = 588.2\,X_1\left(\frac{50}{1000}\right) + X_1$$

$$X_1 = 0.0099 \text{ g/}l$$

Concentration of iodine in the aqueous phase for single stage extraction = 0.0099 g/l ...Ans. (i)

(ii) For two-stage extraction system using 25 ml of CS_2 for each stage :

Material balance of iodine for Stage-1 :

$$0.30 = 588.2\,X_1\left(\frac{25}{1000}\right) + X_1$$

$$X_1 = 0.0191 \text{ g/}l$$

Concentration of iodine in the solution (raffinate which is the feed to second stage) leaving the first stage = X_1 = 0.0191 g/l

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Material balance for stage-2 yields :

$$0.0191 \times 1 = 588.2\, X_2 \left(\frac{25}{1000}\right) + X_2$$

$$X_2 = 0.001268\ g/l$$

X_2 = Concentration of iodine in aqueous phase (raffinate) after stage-2

$$= 0.001268\ g/l \quad \text{... Ans (ii)}$$

Ex. 2.2 : A solution containing 5 % acetaldehyde and 95 % toluene is to be extracted with water in a five stage crosscurrent extraction unit to extract acetaldehyde. Toluene and water are essentially insoluble. If 25 kg of water each time are used per 100 kg of feed, calculate the amount of acetaldehyde extracted and the final concentration of the exit solution.

The equilibrium relationship is given as :

$$Y = 2.20\, X$$

where Y = kg acetaldehyde/kg water

X = kg acetaldehyde/kg toluene

Solution : Basis : 100 kg of feed solution containing 5 % acetaldehyde.

Amount of acetaldehyde in the feed solution $= \frac{5}{100} \times 100 = 5$ kg

Amount of toluene in the feed solution $= A = 100 - 5 = 95$ kg

kg of acetaldehyde/kg of toluene in the feed solution $= X_F = \frac{5}{95} = 0.05263$

Number of cross current stages $= n = 5$.

Amount of solvent (water) used for each stage $= B = 25$ kg

Equilibrium relationship is

$$Y = 2.2\, X$$

$$Y = mX \quad \text{(linear relationship)}$$

$\therefore$ $m = 2.2$

For n stages with equal quantity of solvent for each stage and for linear equilibrium relationship, we have

$$X_n = \left[\frac{A}{A + Bm}\right]^n \cdot X_F$$

where X_n is the concentration (in kg acetaldehyde/kg toluene) of the solute in the final raffinate from n^{th} stage.

$\therefore$

$$X_5 = \left[\frac{95}{95 + 25 \times 2.2}\right]^5 (0.05263)$$

$$X_5 = 0.005363$$

So, the final exit solution will contain 0.005363 kg of acetaldehyde per 1 kg of toluene.

$\therefore$ Final raffinate solution based on 1 kg toluene

$$= 0.005363 + 1 = 1.005363 \text{ kg}$$

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∴ Concentration of the exit toluene solution in terms of weight % acetaldehyde (solute) is

$$= \frac{0.005363}{1.005363} \times 100$$

$$= 0.533 \text{ \% by wt.} \quad \text{... Ans.}$$

Amount of acetaldehyde in the final exit solution

$$= \frac{0.005363}{1} \times 95$$

$$= 0.51 \text{ kg}$$

Amount of acetaldehyde removed from the feed solution = 5 – 0.51

(i.e., acetaldehyde extracted per 100 kg of feed) = 4.49 kg ... Ans.

Ex. 2.3 : A solution of nicotine in water containing 1 % nicotine is to be extracted with kerosene at 293 K (20°C). Water and kerosene are essentially insoluble. Assume the equilibrium relationship to be :

$$Y = 0.9X$$

where, Y = kg nicotine/kg kerosene

X = kg nicotine/kg water

(i) Determine the percentage extraction of nicotine if 100 kg of a feed solution is extracted with 150 kg of solvent (kerosene).

(ii) Repeat for three theoretical extractions using 50 kg solvent each time.

Solution : Basis : 100 kg of feed solution at 293 K.

The equilibrium relationship is

$$Y = 0.9X$$

(i) For single stage extraction,

$$Y_1 = 0.9X_1$$

where, Y_1 = kg nicotine/kg kerosene [Extract]

X_1 = kg nicotine/kg water [Raffinate]

X_0 = kg nicotine/kg water in feed solution

The feed solution contains 1 % nicotine by weight.

Amount of nicotine in feed = $0.01 \times 100 = 1$ kg

Amount of water in the feed solution (A) = 100 – 1 = 99 kg

$$X_F = \frac{1}{99} = 0.0101 \text{ kg nicotine/kg water}$$

B = kg of solvent = 150 kg of kerosene

As water and kerosene are immiscible solvents,

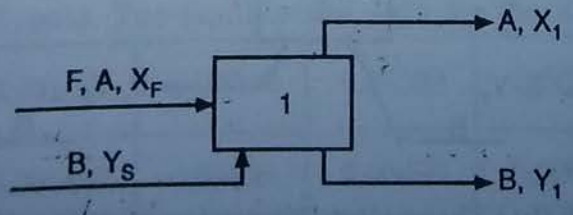

Fig. E2.3 (a) : Single stage extraction system

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The material balance of nicotine over a single stage extraction is :

kg nicotine in feed + kg nicotine in solvent used

= kg nicotine in extract + kg nicotine in raffinate

$$AX_F + BY_s = BY_1 + AX_1, \text{ but } Y_s = 0$$

$$\therefore \quad AX_F = BY_1 + AX_1$$

$$0.0101 \times 99 = 150\,Y_1 + 99\,X_1$$

We have $Y_1 = 0.9\,X_1$

$$0.0101 \times 99 = 150\,(0.9\,X_1) + 99\,X_1$$

$$\therefore \quad X_1 = 0.00427 \text{ kg nicotine/kg water}$$

Amount of nicotine removed from water

$$= (0.0101 - 0.00427) \times 99$$

$$= 0.5772 \text{ kg} \approx 0.58 \text{ kg}$$

$$\% \text{ extraction of nicotine} = \frac{0.58}{1} \times 100 = 58 \text{ (i.e., 58 \% of that in the feed)}$$

... Ans. (i)

(ii) For three theoretical extractions :

B = 50 kg for each time

For immiscible solvents for n stage, we have :

$$X_n = \left[\frac{A}{A + B_m}\right]^n X_F$$

For our case,

n = 3, A = 99 kg, B = 50 kg, m = 0.90, X_F = 0.0101 kg nicotine/kg water

The equilibrium relationship is

$Y = 0.9\,X$, is of the type :

$Y = mX$, m = 0.90

$$X_3 = \left[\frac{99}{99 + 50 \times 0.9}\right]^3 \times 0.0101$$

= 0.00328 kg nicotine/kg water

Amount of nicotine removed from water

$$= (0.0101 - 0.00328) \times 99 = 0.675 \text{ kg}$$

$$\% \text{ extraction of nicotine} = \frac{0.675}{1} \times 100 = 67.5$$

i.e., 67.5 % of nicotine in the feed is extracted using three-stages. ... Ans. (ii)

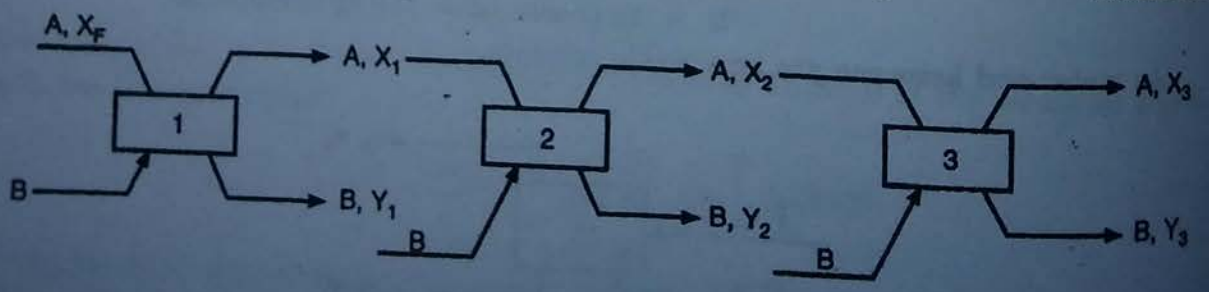

Fig. E2.3 (b) : Three-stage crosscurrent extraction system

B = Solvent for each stage = 50 kg

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Material balance of nicotine :

Stage - 1 :

$$AX_F = BY_1 + AX_1, \quad Y_1 = 0.90\,X_1$$
$$99 \times 0.0101 = 50\,(0.90\,X_1) + 99\,X_1$$
$$\therefore \quad X_1 = 0.00694 \text{ kg nicotine/kg water}$$

Stage - 2 :

$$AX_1 = BY_2 + AX_2, \quad Y_2 = 0.90\,X_2$$
$$99 \times 0.00694 = 50\,(0.90\,X_2) + 99\,X_2$$
$$\therefore \quad X_2 = 0.00477 \text{ kg nicotine/kg water}$$

Stage - 3 :

$$AX_2 = BY_3 + AX_3, \quad Y_3 = 0.90\,X_3$$
$$99 \times 0.00477 = 50\,(0.90\,X_3) + 99\,X_3$$
$$X_3 = 0.00328 \text{ kg nicotine/kg water}$$

So there will be 0.00328 kg nicotine per kg of water in raffinate (water phase) leaving the stage-3.

Nicotine in raffinate leaving stage-3 $= 0.00328 \times 99 = 0.325$ kg

Nicotine in feed solution $= 1$ kg

Nicotine extracted (removed) $= 1 - 0.325 = 0.675$ kg

% extraction of nicotine based on nicotine in feed $= \dfrac{0.675}{1} \times 100 = 67.5$

i.e., 67.5 % of nicotine in the feed is extracted using three stages with 50 kg solvent for each stage. **... Ans. (ii)**

Alternatively : Check :

Amount of nicotine extracted in kerosene

$$= BY_1 + BY_2 + BY_3$$
$$= B\,[Y_1 + Y_2 + Y_3]$$
$$= B\,[0.9\,X_1 + 0.9\,X_2 + 0.9\,X_3]$$
$$= 0.9B\,[X_1 + X_2 + X_3]$$
$$= 0.9 \times 50\,[0.00694 + 0.00477 + 0.00328]$$
$$= 0.67455 \text{ kg} \approx 0.675 \text{ kg}$$

% extraction of nicotine $= \dfrac{0.675}{1} \times 100 = 67.5$ **... Ans. (ii)**

Ex. 2.4 : 100 kg/h of a nicotine-water containing 10% by weight nicotine is extracted with 150 kg/h of kerosene containing 0.06% by weight nicotine in a countercurrent stage column. The nicotine content of exit water stream is 0.10 weight percent. Determine the number of theoretical stages required for this separation. Water and kerosene are essentially immiscible solvents. The equilibrium data for the above system is :

$X = \dfrac{\text{kg nicotine}}{\text{kg water}}$	0.001011	0.00246	0.00502	0.0075	0.00998	0.0204
$Y = \dfrac{\text{kg nicotine}}{\text{kg kerosene}}$	0.000807	0.00196	0.00456	0.00686	0.00913	0.0187

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Solution : Basis : 100 kg/h of nicotine-water solution and 150 kg/h of kerosene solvent as given.

Nicotine in the feed = 10% by weight = 0.010 weight fraction

Water in the feed = 1 – 0.01 = 0.99 weight fraction

Feed solvent (water) in the feed solution

$$= A = 100 \times 0.99 = 99 \text{ kg/h}$$

$$X_F = \text{kg nicotine/kg water in feed} = \frac{x_F}{1 - x_F}$$

$$X_F = \frac{0.01}{1 - 0.01} = 0.0101$$

Water and kerosene are immiscible solvents.

As water and kerosene are immiscible solvents, all water in feed will appear in raffinate phases and all kerosene in entering solvent stream will appear in extract phases.

Let n be the theoretical stages required for the said duty.

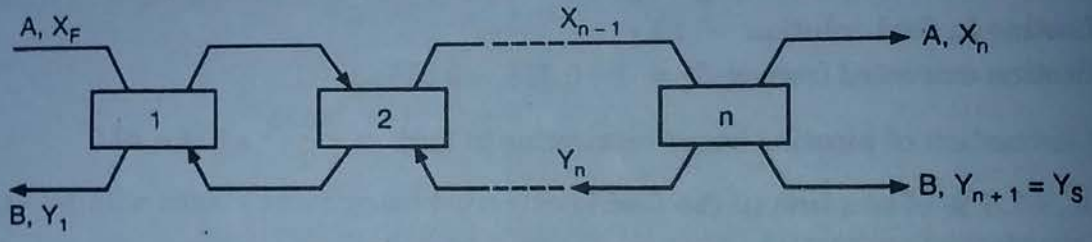

Fig. E2.4 (a) : Countercurrent n-stage extraction system

B = kg/h of kerosene to n^{th} stage (solute-free basis)

X_n = kg nicotine/kg water in water (raffinate) leaving n^{th} stage

Y_{n+1} = kg nicotine/kg kerosene in solvent kerosene entering n^{th} stage

X_F = kg nicotine/kg water in feed entering first stage

Y_1 = kg nicotine/kg kerosene in solvent (extract) leaving first stage

A = kg/h of water in feed as well as in raffinate

Nicotine in kerosene entering = 0.06% by weight = 0.0006 weight fraction

$$\therefore \quad Y_{n+1} = \frac{0.0006}{1 - 0.0006} = 0.0006004 \approx 0.0006$$

Nicotine in water (raffinate) leaving system = 0.10% by weight = 0.001 weight fraction

$$X_n = \frac{0.001}{1 - 0.001} = 0.001$$

B is the kg/h of kerosene solvent on solute-free basis.

But the solvent stream contains 0.0006 weight fraction of nicotine, so the pure solvent B to the n^{th} stage is :

$$B = 150\,(1 - 0.0006) = 149.91 \text{ kg/h}$$

Material balance of nicotine over the entire countercurrent extraction system :

Nicotine in (Feed + Solvent) = Nicotine in (Extract + Raffinate)

$$AX_F + BY_{n+1} = BY_1 + AX_n, \quad Y_{n+1} = Y_s$$

$$99 \times 0.0101 + 149.91 \times 0.0006 = 149.91\, Y_1 + 99 \times 0.001$$

$$Y_1 = 0.00661 \text{ kg nicotine/kg kerosene}$$

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The operating line represented by the above equation for this countercurrent extraction passes through the points M (X_F, Y_1) and L (X_n, Y_{n+1}).

Point M → (0.0101, 0.0066)

Point L → (0.001, 0.0006)

Plot : Y v/s X equilibrium data.

Plot : Operating line passing through the points L and M (the end points of the operating line) and step-off the number of stages required between points L and M.

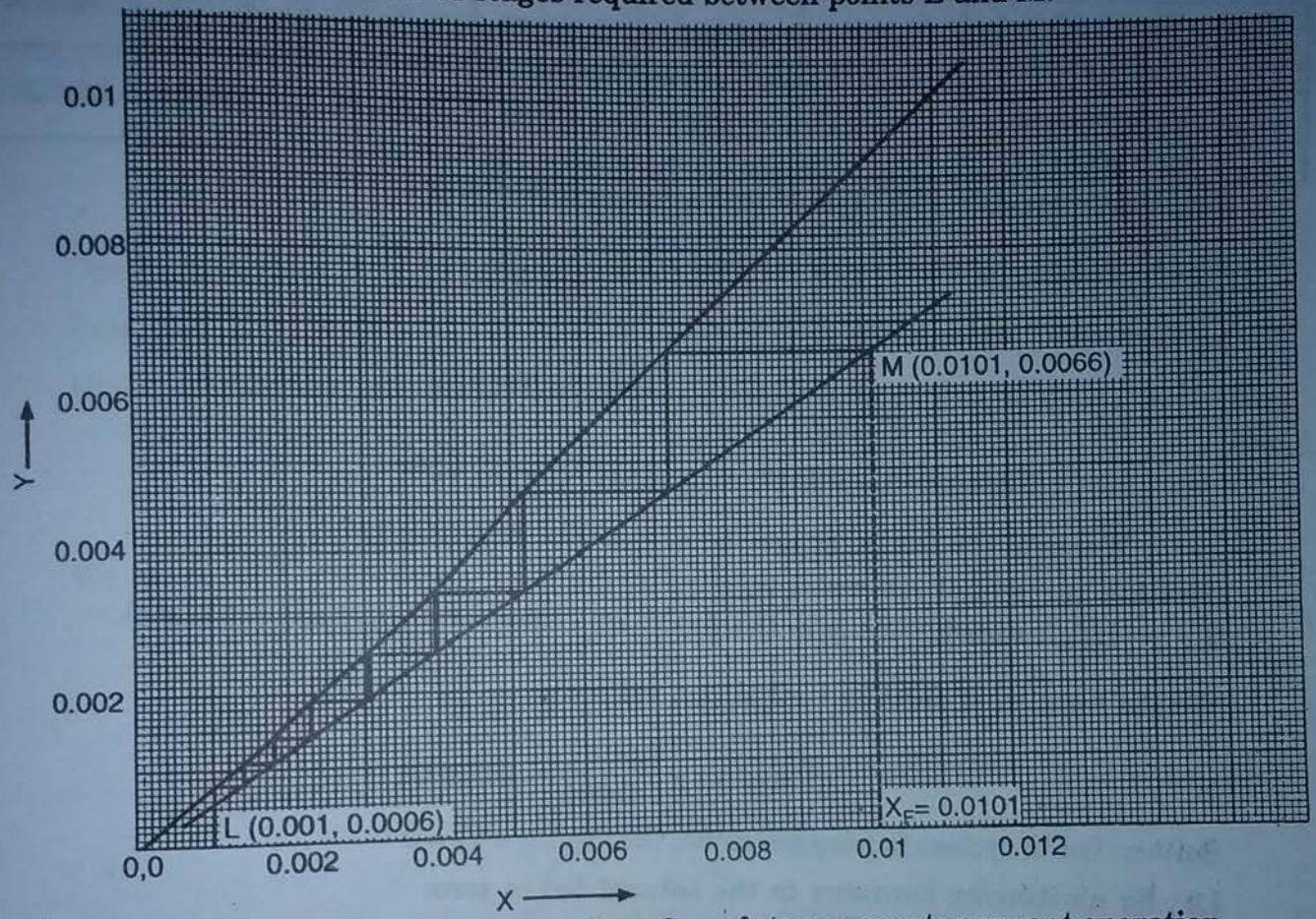

Fig. E 2.4 (b) : Determination of number of stages, countercurrent operation, insoluble solvents

$$A X_F + B Y_{n+1} = B Y_1 + A X_n$$

$$Y_{n+1} = \frac{A}{B}(X_n - X_F) + Y_1$$

This is the operating line for countercurrent extraction. It is a straight line with a slope equal to A/B and passing through the points (X_F, Y_1) and (X_n, Y_{n+1}).

The number of stages required to pass from X_F to X_n is found by drawing the steps (triangles) between the operating line and the equilibrium curve.

Y v/s X is plotted and then points M (0.0101, 0.0066) and L (0.001, 0.0006) are located. The operating line passing through these points is drawn [Refer Fig. E 2.4 (b)] and the number of stages are stepped-off to pass from X_F to X_n by constructing the triangles between the equilibrium curve and the operating line.

From Fig. E 2.4 (b), the number of stages required = n = 8 ...Ans.

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Ex. 2.5 : 1000 kg/h of a nicotine-water solution containing 1 % nicotine is to be extracted with kerosene to reduce the nicotine content to 0.1 %. Water and kerosene are immiscible solvents.

(i) What is the minimum solvent requirement, kg/h ?

(ii) If 1150 kg/h of solvent is used, how many theoretical stages are required ?

The equilibrium data are as follows :

$X = \frac{\text{kg nicotine}}{\text{kg water}}$	0	0.00101	0.00246	0.00502	0.00998	0.0204
$Y = \frac{\text{kg nicotine}}{\text{kg kerosene}}$	0	0.00807	0.00196	0.00456	0.00913	0.01870

Solution : Basis : 1000 kg/h of feed solution containing 1 % nicotine.

Weight fraction of nicotine in the feed solution $= \frac{1}{100} = 0.01$

kg nicotine/kg water in the feed solution $= X_F$

$$= \frac{0.01}{1 - 0.01} = 0.0101$$

Nicotine in the feed $= \frac{1}{100}(1000) = 10$ kg/h

Water in the feed solution $= A = 1000 - 10 = 990$ kg/h

Nicotine in the final exit water solution = 0.1 %

X_n = kg nicotine/kg water in the final exit water solution (final raffinate)

Weight fraction of nicotine in the final raffinate $= \frac{0.1}{100} = 0.001$

$$\therefore \quad X_n = \frac{0.001}{1 - 0.001} = 0.001001$$

Solvent fed to system is pure kerosene, (B) $Y_{n+1} = Y_s = 0$

i.e., kg nicotine/kg kerosene in the solvent fed is zero.

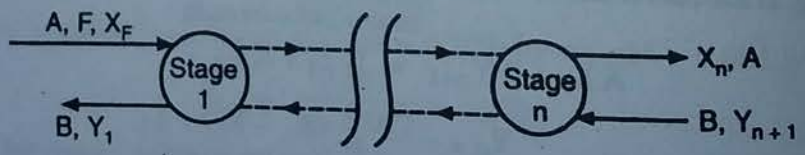

Fig. E2.5 (a) : Countercurrent extraction unit

Water and kerosene are essentially insoluble, so the solvent content of all extracts and A content of all raffinates are constant. All A will be in the raffinate and all B will be in the extract phase.

Overall nicotine balance :

$$AX_F + BY_{n+1} = AX_n + BY_1$$

$$Y_{n+1} = \frac{A}{B}(X_n - X_F) + Y_1$$

or

$$\frac{Y_1 - Y_{n+1}}{X_F - X_n} = \frac{A}{B}$$

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This is the equation of operating line, it is a straight line with a slope equal to A/B a it passes through the points (X_F, Y_1) and (X_n, Y_{n+1}).

$$A = 990 \text{ kg}, \quad B = 1150 \text{ kg}$$

$$\therefore \quad \text{Slope} = \frac{A}{B} = \frac{990}{1150} = 0.8608 = 0.861$$

$$X_n = 0.001001, \quad Y_{n+1} = Y_s = 0 \text{ (as pure solvent is use}$$

Draw the equilibrium curve and draw the operating line through L (0.001001, 0) with slope of 0.861 which will cut the vertical through X_F at M. Construct the stages betwe X_F and X_n and count them.

$$\text{Slope} = 0.861 = \tan \alpha$$

$$\therefore \quad \alpha = \tan^{-1}(0.861) = 40.73°$$

Draw a line through L at an angle α.

From Fig. E 2.5 (b),

Number of theoretical stages required to reduce nicotine content from 1 % to 0.1 %

$$= n = 8 \quad \text{... Ans. (}$$

$$Y_1 = 0.0078$$

We have, $$\frac{Y_1 - Y_{n+1}}{X_F - X_n} = \frac{A}{B}$$

$Y_{n+1} = 0$, $A = 990$ kg, $B = 1150$ kg.

$$X_F = 0.0101, \; X_n = 0.001001$$

$$Y_1 = Y_{n+1} + \frac{A}{B}(X_F - X_n)$$

$$Y_1 = 0 + \frac{990}{1150}(0.0101 - 0.001001)$$

$$Y_1 = 0.00783 = 0.0078$$

So the values of Y_1 are matching. You can plot operating line by first calculating Y_1 above and then locating Y_1 (from equilibrium curve) on vertical through X_F i.e. poin L and M and join LM.

Determination of the minimum solvent :

For determining the minimum solvent rate for the sake of clear understanding, on the equilibrium data is plotted separately. Refer Fig. E 2.5 (c).

Locate point L (X_n, Y_{n+1}), i.e., L (0.001001, 0) on the x-axis. Locate point X_F on t x-axis and through X_F draw a vertical to cut the equilibrium curve at K. Join LK a measure the slope.

For the minimum solvent rate and so for the infinite number of stages, the operati line starts at L and passes through K, the point of intersection of the vertical through X_F a the equilibrium curve. Draw line LK and measure the slope to get the minimum solve rate.

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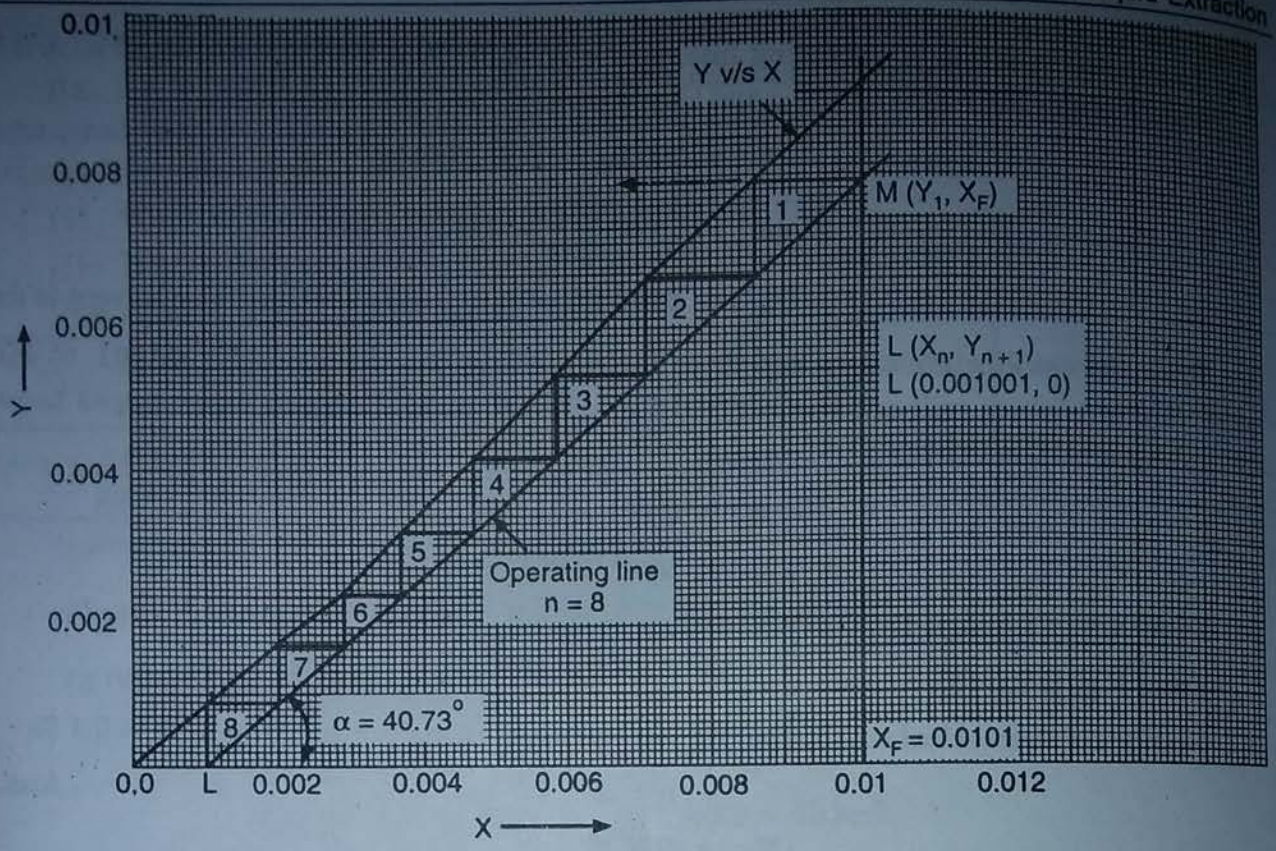

Fig. E 2.5 (b) : Determination of number of stages, countercurrent operation, insoluble solvents

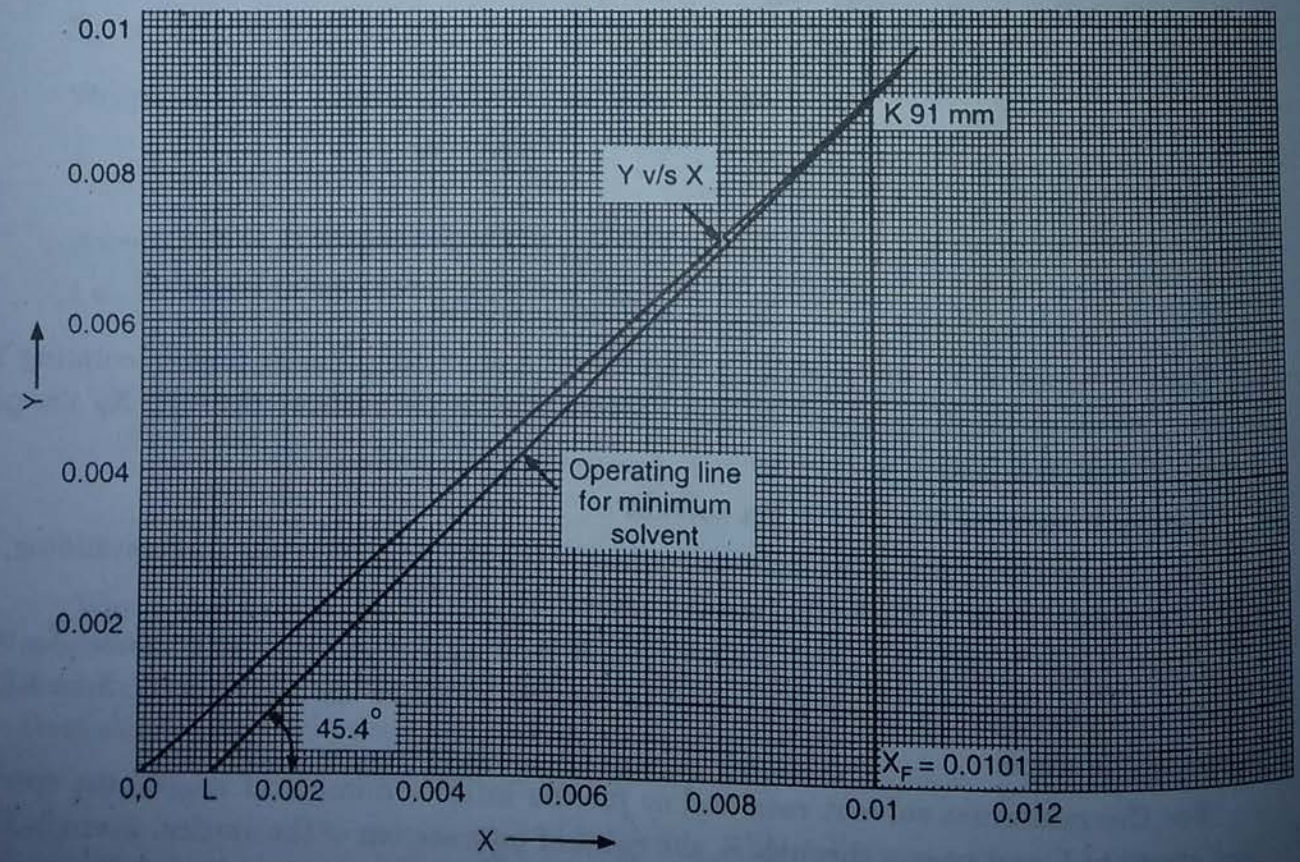

Fig. E 2.5 (c) : Determination of minimum solvent rate

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Slope of the operating line for minimum solvent rate :

$$= \frac{0.0092 - 0}{0.0101 - 0.001001} = 1.0111$$

$$\frac{A}{B} = 1.0111$$

$$B = \frac{990}{1.0111} = 979 \text{ kg/h}$$

OR Angle made by the operating line with x-axis = 45.4°

$$\text{Slope} = \tan \alpha$$

$$= \tan (45.4) = 1.014$$

$$\therefore \quad \frac{A}{B} = 1.014$$

$$B = \frac{A}{1.014} = \frac{990}{1.014} = 976.3 \text{ kg/h} \quad \text{... Ans. (i)}$$

Minimum solvent requirement = 976.3 kg/h ...Ans. (i)

Ex. 2.6 : For extraction of dioxane from water, benzene is used as an extraction solvent. At 298 K (25°C), the equilibrium distribution of dioxane between water and benzene is as given below :

Weight % dioxane in water	5.1	18.9	25.2
Weight % dioxane in benzene	5.2	22.5	32

At these concentrations, water and benzene are substantially insoluble.

1000 kg of solution containing 25 % dioxane is to be extracted with benzene to remove 95 % of the dioxane. The dioxane free benzene is used.

(i) Calculate the solvent requirement for single batch operation.

(ii) If the extraction were done in four-stage crosscurrent unit using 750 kg of pure benzene each time, determine the final raffinate concentration and the percentage removal of dioxane from feed solution.

Solution : Basis : 1000 kg of feed solution containing 25 % dioxane and 75 % water.

Weight % dioxane in water	5.1	18.9	25.2
Weight % dioxane in benzene	5.2	22.5	32

Convert this data in kg dioxane/kg liquid :

$$\text{Weight fraction dioxane in water} = \frac{5.1}{100} = 0.051$$

$$\frac{\text{kg dioxane}}{\text{kg water}} = X = \frac{0.051}{1 - 0.051} = 0.05374$$

$$\text{Weight fraction of dioxane in benzene} = \frac{5.2}{100} = 0.052$$

$$\frac{\text{kg dioxane}}{\text{kg benzene}} = Y = \frac{0.052}{1 - 0.052} = 0.00548$$

$\frac{\text{kg dioxane}}{\text{kg water}} = X$	0.05374	0.233	0.3369
$\frac{\text{kg dioxane}}{\text{kg benzene}} = Y$	0.0548	0.2903	0.4706

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Amount of dioxane in feed solution = $0.25 \times 1000 = 250$ kg

Amount of water in feed solution (A) = $1000 - 250 = 750$ kg

Dioxane removed = 95 %

Amount of dioxane removed from water solution = $0.95 \times 250 = 237.5$ kg

Amount of dioxane in exit water solution = $250 - 237.5 = 12.5$ kg

Amount of dioxane (removed) extracted = 237.5 kg

As water and benzene are essentially insoluble, all the water in feed solution will appear in raffinate phase (water solution) and all benzene will appear in extract phase.

(i) For single stage extraction :

kg dioxane/kg water in feed solution = $X_F = \dfrac{250}{750} = 0.3333$

kg dioxane/kg water in exit solution = $X_1 = \dfrac{12.5}{750} = 0.01667$

From the equilibrium curve, Y_1 corresponding to X_1 is

$$Y_1 = 0.017$$

Or In the viscinity of the origin-data : X = 0.05374 and Y = 0.0548 is very close so the Y v/s X plot-Y and X are almost appearing equidistant. But Y is greater than X (see the data) so to get Y_1 corresponding to $X_1 = 0.01667$ use :

$$Y_1 = \left[\frac{0.0548 - 0}{0.05374 - 0}\right] \times 0.01667 = 0.017$$

Material balance of dioxane :

$$AX_F = AX_1 + BY_1$$

$$750\,(0.3333) = 750\,(0.01667) + B \times 0.017$$

$$B = 13967 \text{ kg}$$

Solvent required for single stage = 13967 kg ... Ans. (i)

(ii) Four-stage crosscurrent operation :

Pure benzene is used, so $Y_s = 0$

Amount of solvent used for each stage = B = 750 kg

Material balance of dioxane for stage - 1 :

$$AX_F + BY_s = AX_1 + BY_1$$

but $Y_s = 0$

$$\therefore \quad AX_F = AX_1 + BY_1$$

$$\therefore \quad -\frac{A}{B} = \frac{Y_1}{X_1 - X_F}$$

This is the operating line for the first stage. It has a slope of – A/B and it passes through the points $(X_F, 0)$ and (X_1, Y_1).

The slope of the operating line is

$$\text{Slope} = -\frac{A}{B} = -\frac{750}{750} = -1.0$$

$$\text{Slope} = \tan\alpha = -1.0 \quad \therefore \quad \alpha = -45°$$

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As A = 750 kg for each stage and as same amount of solvent (750 kg) is used each time. The slope of all four operating line is same (the operating lines are parallel to each other).

Construction of four stages to obtain X_4 :

Locate point F (0.3333, 0) on the x-axis, representing the feed solution. Through the point F, draw the operating line at an angle of – 45° to the x-axis. This line will cut the equilibrium curve at the point P (X_1, Y_1).

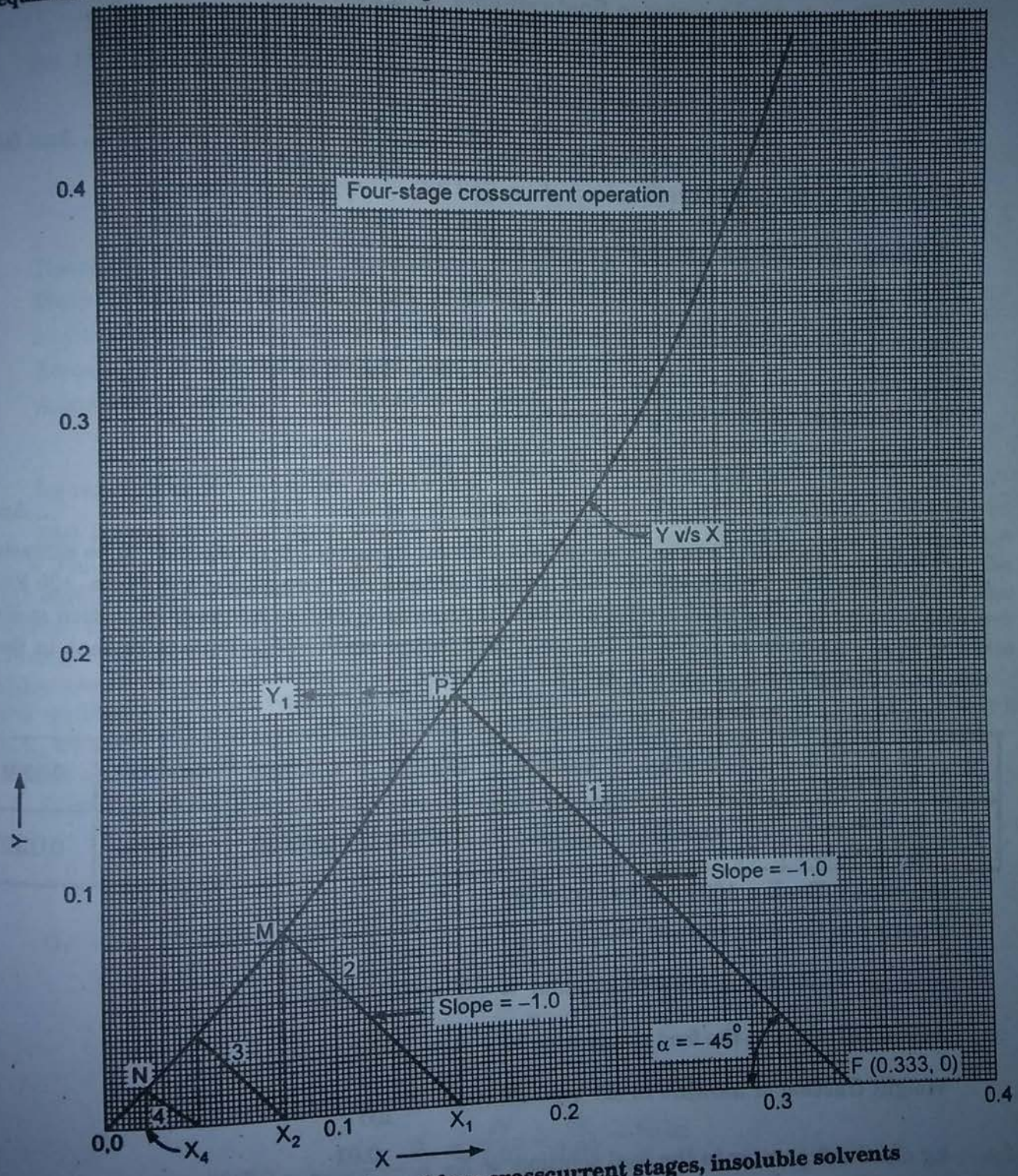

Fig. E 2.6 : Construction of four crosscurrent stages, insoluble solvents

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Through point P, drop a vertical to the x-axis at X_1. Through X_1, draw the operating line for stage-2, inclined at an angle of – 45° to the x-axis. This line will cut the equilibrium curve at the point M ($X_2 \cdot Y_2$). Repeat the procedure and draw the operating lines for stage-3 and -4. Read the value of X_4 from the x-axis (x-coordinate of the point N)

$$X_4 = 0.01875 \text{ kg dioxane/kg water}$$

∴ kg dioxane per kg water in final raffinate = 0.01875 (raffinate leaving stage-4)

i.e. 1 + 0.01875 = 1.01875 kg of final raffinate contains 0.0185 kg of dioxane

∴ Weight % of dioxane in final raffinate $= \frac{0.01875}{1.01875} \times 100$

$= 1.724\ \%$ ... Ans. (ii)

Amount of water in final raffinate = A = 750 kg

∴ Amount of dioxane in final raffinate $= \frac{0.01875}{1} \times 750$

$= 14.06$ kg

Amount of dioxane removed from feed solution

$= 250 - 14.08 = 235.94$ kg

% removal of dioxane from the feed solution

$= \frac{235.94}{250} \times 100$

$= 94.376 \approx 94.4\ \%$ ... Ans.

Ex. 2.7 : Nicotine (C) in a water (A) solution containing 1 % nicotine is to be extracted with kerosene (B) at 293 K (20°C). Water and kerosene are essentially insoluble. 100 kg of a feed solution is to be treated with kerosene in three-stage cross current extraction system using equal quantity of solvent each time. If nicotine removed is 66.3 % of that in feed, determine the quantity of solvent used for each stage.

Equilibrium data are as follows :

$X = \frac{\text{kg nicotine}}{\text{kg water}}$	0	0.00101	0.00246	0.00502	0.00751	0.00998	0.0204
$Y = \frac{\text{kg nicotine}}{\text{kg kerosene}}$	0	0.00807	0.00196	0.00456	0.00686	0.00913	0.0187

Solution : Basis : 100 kg of feed solution

Nicotine in the feed solution = 1 %

Amount of nicotine in the feed solution $= \frac{1}{100} \times 100 = 1$ kg

Amount of water in the feed solution = 100 – 1 = 99 kg

Weight fraction of nicotine in the feed solution $= \frac{1}{100} = 0.01$

kg nicotine/kg water in the feed solution $= X_F = \frac{0.01}{1 - 0.01} = 0.0101$

Water and kerosene are essentially insoluble, so all water will appear in the raffinate phase and all kerosene will appear in the extract phase each time.

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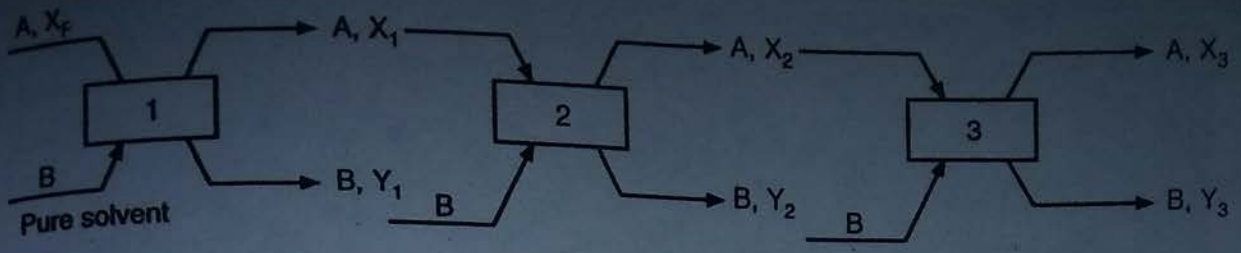

Fig. E 2.7 (a) : Three-stage crosscurrent extraction (insoluble solvents)

For 1st stage, nicotine balance is

$$AX_F + BY_s = AX_1 + BY_1, \quad Y_s = 0 \text{ (pure solvent)}$$

$$AX_F = AX_1 + BY_1$$

$$A(X_1 - X_F) = -BY_1$$

$$\frac{Y_1}{X_1 - X_F} = -\frac{A}{B}$$

The slope of operating line is equal to – A/B (This is valid for all the three stages).

Water in the final raffinate (water solution) = 99 kg

Nicotine extracted, i.e., removed from the water solution = 66.3 %

Amount of nicotine removed from the water solution = 0.663 (1) = 0.663 kg

Amount of nicotine in the exit water solution = Nicotine in feed – Nicotine extracted

$$= 1 - 0.663 = 0.337 \text{ kg}$$

kg nicotine/kg water in exit water solution $= X_3 = \frac{0.337}{99} = 0.003404 \approx 0.0034$

Plot the equilibrium curve [Fig. E 2.7 (b)]. Mark point F on the x-axis at $X_F = 0.0101$ as $Y_s = 0$ and point Q on the x-axis at $X_3 = 0.0034$. Draw a vertical from X_3 on the x-axis to cut the equilibrium curve. As there are three stages of extraction and as equal quantity of the solvent is used for each stage, the slope of three operating lines is the same and hence the lines are parallel to each other. We have to adopt a trial and error procedure to construct three parallel lines between X_F (F) and X_3. Draw a line through F, such as line FP and do the construction such that there will be three parallel lines and will exactly end in X_3 [i.e., we can construct exactly three lines starting from X_F and ending exactly at X_3. Only a correct trial is shown in Fig. E 2.7 (b)] and then measure the slope of any one of the lines.

From Fig. E 2.7 (b),

$$\text{Slope of line EP} = -\left(\frac{0.0063}{0.0101 - 0.0069}\right)$$

$$= -1.96875 = -1.97$$

Or measuring angle, $\theta = -63°$

$$\text{Slope} = \tan\theta = -1.963$$

$$\text{Take slope} = -1.97$$

$$\text{Slope} = -\frac{A}{B} = -1.97$$

$$\frac{A}{B} = 1.97$$

$$B = \frac{A}{1.97} = \frac{99}{1.97} = 50 \text{ kg} \quad \text{...Ans.}$$

Amount of solvent used for each stage = 50 kg

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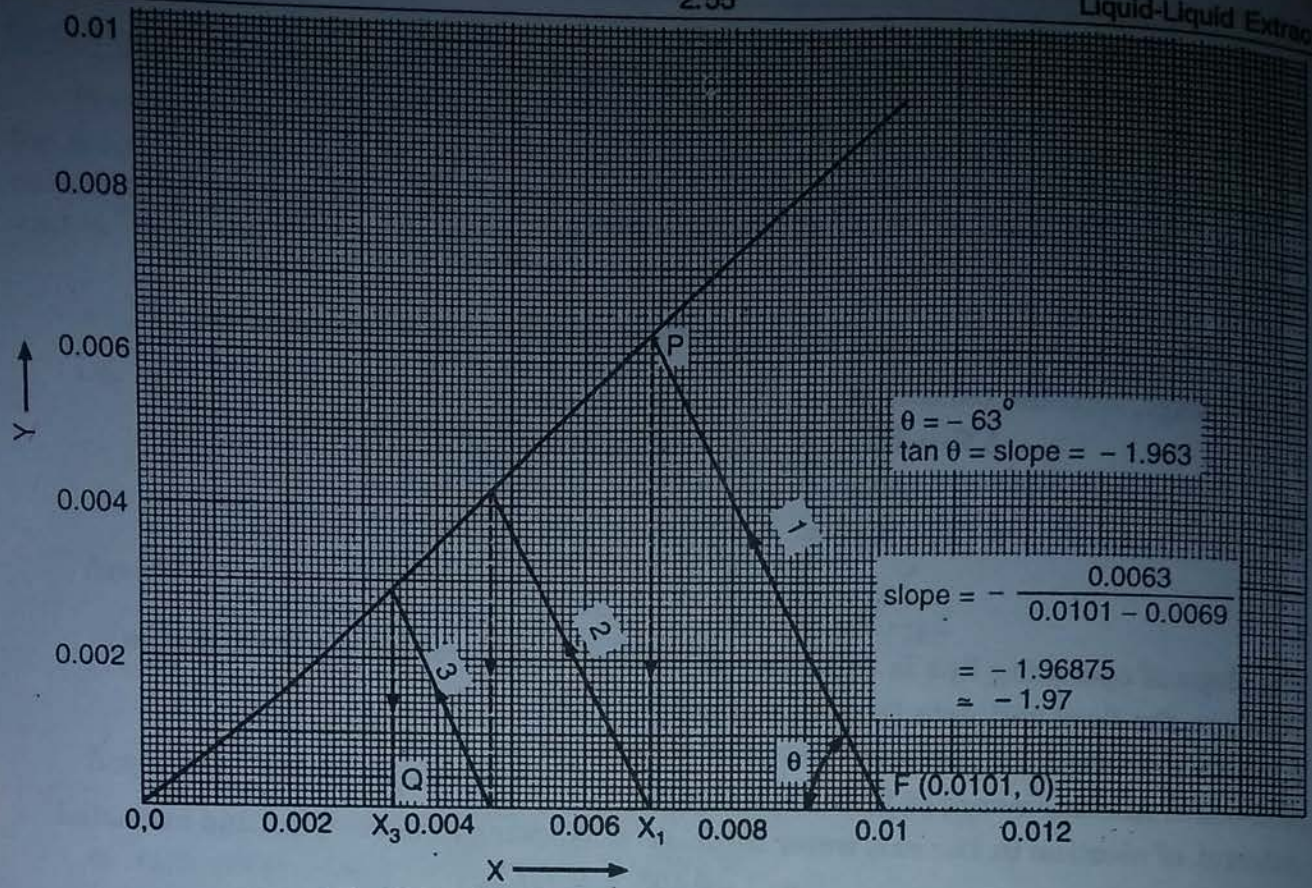

Fig. E 2.7 (b) : Determination of solvent requirement. Three-stage crosscurrent operation, insoluble/immiscible solvents

Ex. 2.8 : Nicotine-water solution containing 1 % nicotine by weight is to be extracted with kerosene at 293 K (20°C). Water and kerosene are essentially insoluble.

Determine : (i) the percentage extraction of nicotine if 100 kg of feed solution is extracted once with 150 kg of solvent, (ii) repeat for three theoretical (ideal) extractions using 50 kg of solvent each time.

The equilibrium data for extraction of nicotine expressed as kg nicotine per kg of liquid are as follows :

$X = \frac{\text{kg nicotine}}{\text{kg water}}$	0	0.00101	0.00246	0.00502	0.00751	0.00998	0.0204
$Y = \frac{\text{kg nicotine}}{\text{kg kerosene}}$	0	0.00807	0.00196	0.00456	0.00686	0.00913	0.0187

Solution : Basis : 100 kg of feed solution at 293 K

(i) Single stage extraction using 150 kg of pure kerosene (solvent).

Nicotine in the feed solution = 1 % by weight

Weight fraction of nicotine in the feed = 0.01

Weight fraction of water in the feed = 1 – 0.01 = 0.99

Nicotine in the feed solution = $\frac{1}{100} \times 100 = 1$ kg

Water in the feed solution (A) = $0.99 \times 100 = 99$ kg

kg nicotine/kg water in the feed solution = $X_F = \frac{0.01}{1 - 0.01} = 0.0101$

Amount of solvent (B) = 150 kg

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Water and kerosene are essentially insoluble, so water will appear all in raffinate and kerosene in extract.

Material balance of nicotine over the extraction system :

Nicotine in feed + Nicotine in solvent = Nicotine in extract + Nicotine in raffinate

$$AX_F + BY_s = AX_1 + BY_1, \text{ but } Y_s = 0$$

$$\therefore \quad AX_F = AX_1 + BY_1$$

$$BY_1 = -AX_1 + AX_F$$

$$Y_1 = -\frac{A}{B} X_1 + \frac{A}{B} X_F$$

$$Y_1 = -\frac{A}{B} [X_1 - X_F]$$

$$\frac{Y_1}{X_1 - X_F} = -\frac{A}{B}$$

This is the operating line with a slope equal to – A/B. Draw this line through point F [representing the feed solution with slope – A/B (Refer Fig. E 2.8)]. Let that it will cut the equilibrium curve at E_1 and point E_1 will give the composition Y_1 of the extract and X_1 of the raffinate.

$$A = 99 \text{ kg}, \; B = 150 \text{ kg}, \; A/B = 99/150 = 0.66$$

$$\text{Slope} = -\frac{A}{B} = -0.66$$

$$\text{Slope} = \tan\alpha = -0.66 \quad \therefore \quad \alpha = -33.42°$$

Draw a line through F (0.0101, 0) with an angle 33.42°, measuring the angle in the clockwise direction. This line cuts the equilibrium curve at E_1 which gives the composition of extract Y_1 and of raffinate X_1.

$$X_1 = 0.0043 \text{ kg nicotine/kg kerosene}$$

$$Y_1 = 0.0038 \text{ kg nicotine/kg kerosene}$$

$$\text{Nicotine removed from water} = A(X_F - X_1)$$

$$= 99 (0.0101 - 0.0043)$$

$$= 0.5742 \text{ kg}$$

$$\% \text{ extraction of nicotine} = \frac{0.5742}{1} \times 100 = 57.42 \quad \text{...Ans.}$$

(ii) For three-stage crosscurrent extraction using 50 kg of solvent each time.

We have, A = 99 kg, B = 50 kg for each time

$$\frac{A}{B} = \frac{99}{50} = 1.98$$

$$\text{Slope} = -\frac{A}{B} = -1.98 \quad \therefore \quad \alpha = \tan^{-1}(-1.98) = -63.2°$$

Draw the operating line through F (0.0101, 0) with an angle – 63.2° to cut the equilibrium curve at $E_{1\,(3)}$ which gives Y_1 and X_1. Draw a vertical through $E_{1\,(3)}$ to cut the x-axis in X_1. Through X_1, draw a line with a slope of – 1.98, i.e., at an angle of – 63.2° to cut the equilibrium curve at $E_{2\,(2)\,(3)}$ which gives Y_2 and X_2. Through X_2 on the x-axis, draw a line with $\alpha = -63.2°$ to cut the equilibrium curve at $E_{3\,(3)}$ which gives Y_3 and X_3.

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From Fig. E 2.8 :

$$X_3 = 0.0033 \text{ kg nicotine/kg water}$$

$$\text{Nicotine removed from water} = A (X_F - X_3)$$

$$= 99 (0.0101 - 0.0033) = 0.6732$$

$$\% \text{ extraction of nicotine} = \frac{0.6732}{1} \times 100 = 67.32$$

...Ans.

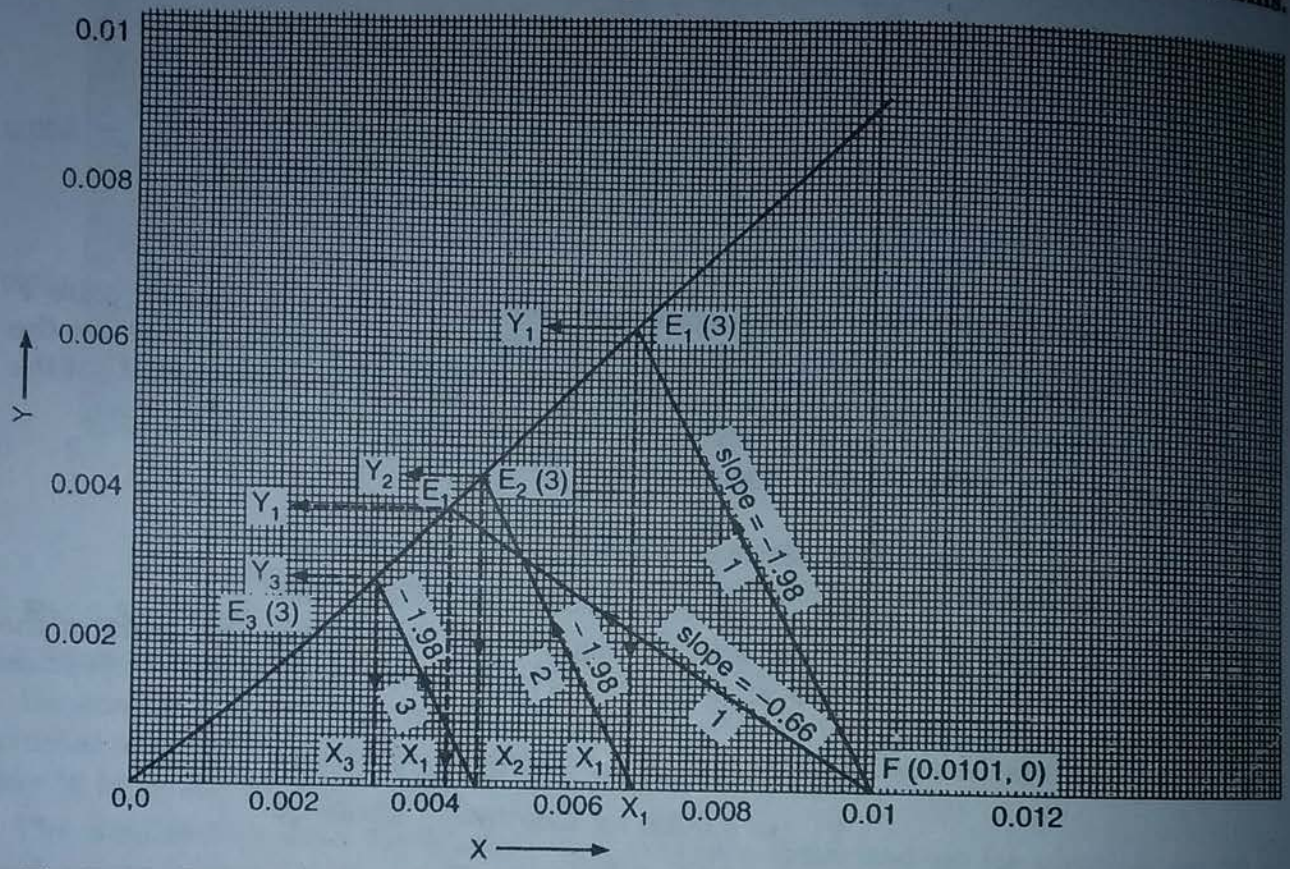

Fig. E 2.8 : Determination of extract and raffinate composition. Single and three-stage crosscurrent operation. Insoluble solvents

Extra (check for plot) :

From Fig. E 2.8 : $Y_1 = 0.0062, \quad Y_2 = 0.0042, \quad Y_3 = 0.0028$

$$\text{Nicotine totally extracted} = B [Y_1 + Y_2 + Y_3]$$

$$= 50 [0.0062 + 0.0042 + 0.0028]$$

$$= 0.66 \text{ kg}$$

$$\% \text{ extraction of nicotine} = \frac{0.66}{1} \times 100 = 66$$

...Ans.

$$\% \text{ error} = \frac{67.32 - 66}{67.32} \times 100 = 1.96 \text{ (tolerable)}$$

Ex. 2.9 : A 25 % solution of dioxane in water is to be continuously extracted at a rate of 1000 kg/h in countercurrent extraction system with benzene to remove 95 % of dioxane. The equilibrium distribution of dioxane between water and benzene is as follows :

Weight % dioxane in water	5.1	18.9	25.2
Weight % dioxane in benzene	5.2	22.5	32

At these concentrations, benzene and water are substantially insoluble.
Determine number of theoretical stages required if 900 kg of benzene is used per hour.

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Solution : Basis : 1000 kg/h of feed solution.

Weight % dioxane in water	5.1	18.9	25.2
Weight % dioxane in benzene	5.2	22.5	32

Convert this data in kg dioxane/kg liquid :

Weight % of dioxane in water = 5.1

$$\text{Weight fraction of dioxane in water} = \frac{5.1}{100} = 0.051$$

$$\frac{\text{kg dioxane}}{\text{kg water}} = X = \frac{0.051}{1 - 0.051} = 0.05374$$

$$\text{Weight fraction of dioxane in benzene} = \frac{5.2}{100} = 0.052$$

$$\frac{\text{kg dioxane}}{\text{kg benzene}} = Y = \frac{0.052}{1 - 0.052} = 0.00548$$

$\frac{\text{kg dioxane}}{\text{kg water}} = X$	0.05374	0.233	0.3369
$\frac{\text{kg dioxane}}{\text{kg benzene}} = Y$	0.0548	0.2903	0.4706

Dioxane in feed solution = $0.25 \times 1000 = 250$ kg/h

Water in feed solution = $A = 1000 - 250 = 750$ kg/h

Dioxane removed/extracted = 95 % of that in feed

Dioxane removed from the water solution

$= 0.95 \times 250 = 237.5$ kg/h

Dioxane in the final raffinate = $250 - 237.5 = 12.5$ kg/h

kg dioxane/kg water in the feed solution = $X_F = \frac{250}{750} = 0.3333$

kg dioxane/kg water in the final solution/raffinate = $X_n = \frac{12.5}{750} = 0.01667$

(All the water in the feed solution will appear in the raffinate phase at each stage.)

Water and benzene are substantially insoluble, so the solvent content of all extracts and A contents of all raffinates are constant. All A will appear in the raffinate phase and all B will appear in the extract phase.

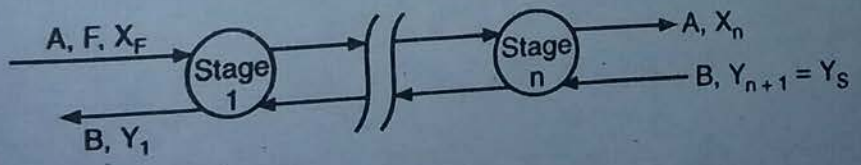

Fig. E 2.9 (a) : Countercurrent system

Overall dioxane balance :

$$AX_F + BY_{n+1} = AX_n + BY_1$$

$$Y_{n+1} = \frac{A}{B}(X_n - X_F) + Y_1$$

or

$$\frac{Y_1 - Y_{n+1}}{X_F - X_n} = \frac{A}{B}$$

This is the equation of operating line. It is a straight line with a slope equal to A/B and it passes through the points L (X_n, Y_{n+1}) and M (X_F, Y_1).

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$$A = 750 \text{ kg}, \quad B = 900 \text{ kg}$$

$$\text{Slope} = \frac{A}{B} = \frac{750}{900} = 0.833$$

$$\text{Slope} = \tan \alpha = 0.833$$

$$\therefore \quad \alpha = 39.8 \approx 40°$$

$X_n = 0.01667$, $Y_{n+1} = Y_s = 0$ as benzene used is pure

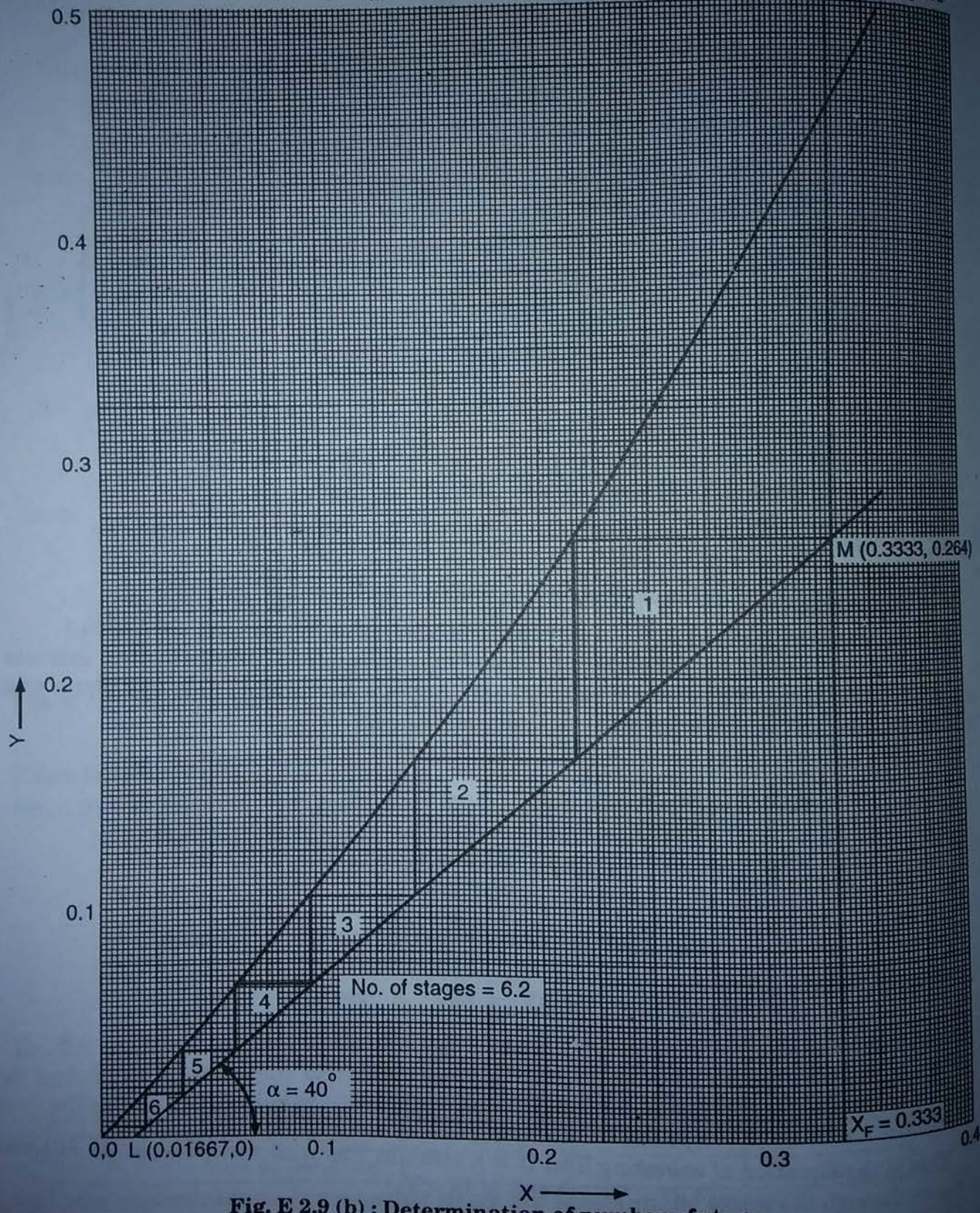

Fig. E 2.9 (b) : Determination of number of stages

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Draw the equilibrium curve Y v/s X and draw the operating line through L (0.01667, 0) on the x-axis with a slope of 0.833, i.e., with an angle $\alpha = 40°$ which will cut a vertical through X_F on the x-axis at M. Construct the stages between the equilibrium curve and the operating line starting from the point M (0.333, 0.264) and covering the point L (0.01367, 0), i.e., required for the desired separation – X_F and X_n and count them.

$$\frac{Y_1 - Y_{n+1}}{X_F - X_n} = \frac{A}{B}$$

$$\frac{Y_1 - 0}{0.3333 - 0.01667} = \frac{750}{900}$$

$$\therefore \quad Y_1 = 0.264$$

So locate the point M (0.3333, 0.264) on the vertical through X_F and join LM, this is a required operating line. Construct stages starting from the point M (0.333, 0.264) and ending or covering the point L (0.01667, 0) and count them. Refer Fig. E 2.9 (b).

Number of theoretical stages required = n = 6.2 ...Ans.

Amount of dioxane removed = 237.5 kg.

Ex. 2.10 : A 2000 kg of pyridine-water solution containing 50 % pyridine is to be extracted with an equal weight of pure chlorobenzene. The raffinate from the first extraction is to be re-extracted with a weight of solvent equal to the raffinate weight and so on.

Determine : (i) Theoretical stages required and (ii) Total quantity of solvent required to reduce the pyridine concentration to 2 %. Use rectangular co-ordinates.

Equilibrium tie line data in weight percent are :

Pyridine	Chlorobenzene	Water	Pyridine	Chlorobenzene	Water
0	99.95	0.05	0	0.08	99.92
11.05	88.28	0.07	5.02	0.16	94.82
18.95	79.90	1.15	11.05	0.24	88.71
24.10	74.28	1.62	18.90	0.38	80.72
28.60	69.15	2.25	25.5	0.58	73.92
31.55	65.58	2.87	36.15	1.85	62.05
35.05	61.00	3.95	44.95	4.18	50.87
40.60	53.00	6.40	53.20	8.90	37.90
49.00	37.80	13.20	49.00	37.80	13.20

Solution : Basis : 2000 kg of pyridine-water solution.

Feed solution contains 50 % pyridine.

$$x_F = \frac{50}{100} = 0.50$$

For first stage :

kg solvent required = kg feed solution (given)

Amount of solvent to first stage = 2000 kg

It is a crosscurrent extraction operation.

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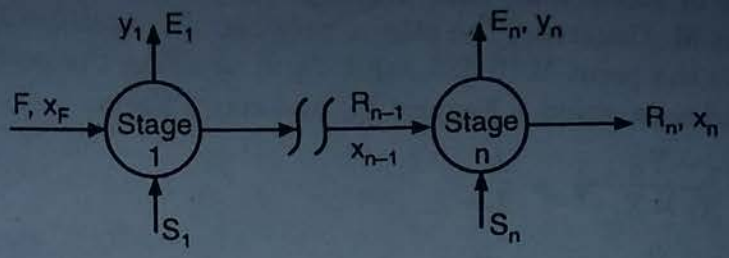

Fig. E 2.10 (a) : Crosscurrent extraction

The solvent used is pure, so $y_s = 0$ for all stages.

$$x_F = 0.5 \text{ and } x_n = \frac{2}{100} = 0.02$$

So we have to find the number of stages required between x_F and x_n.

Refer Fig. E 2.10 (b).

(i) Plot : (1) x, y v/s weight fraction of chlorobenzene and (2) y v/s x or x v/s y.

The horizontal rows of the data give the concentrations in equilibrium solutions.

(ii) Mark the point F on y-axis at $x_F = 0.5$ and the point S at weight fraction of B = 1.0 (pure B).

(iii) Join FS (S_1 = B). For pure solvent, the point S is at the B-apex.

(iv) Locate point M_1 on FS such that distance FM_1 = distance M_1S [as $F = S_1 = 2000$ kg]

Or $$M_1 = F + S_1 = 2000 + 2000 = 4000 \text{ kg}$$

Material balance of pyridine :

$$x_F F + S_1 y_s = x_{M1} M_1$$

$$0.5 \times 2000 + 2000\,(0) = x_{M1}\,(4000)$$

$$\therefore \quad x_{M1} = 0.25$$

So locate M_1 on FS such that M_1 is at $x_{M1} = 0.25$ (read from the y-axis)

Here solvent used is pure ($y_s = 0$), therefore, the point S (for S_1, S_2, ...) is at the B-apex.

Adopt a trial and error procedure to draw a tie line passing through M_1.

As the values of y are greater than x. For the values of y less than 0.3, the tie line slopes towards the origin of rectangular co-ordinates.

As $x_{M1} = 0.25$, y_1 (E_1) has to be more than 0.25 and less than 0.3 and x_1 (R_1) has to be less than 0.25.

Trial : Take E_1 ($y_1 = 0.28$) on rectangular co-ordinates – x, y v/s weight fraction of chlorobenzene plot, move horizontally towards y v/s x plot, get corresponding value of x_1 and from the diagonal y = x (corresponding to x_1) move horizontally to the previous plot and mark R_1 (x_1) and join $R_1 - E_1$ and see that the line R_1E_1 passes through M_1 or not. If it passes through M_1 then R_1E_1 is the required tie line and if not then take other value of E_1 (i.e., y_1) and repeat the procedure till we get a line R_1E_1 that passes through M_1.

For $y_1 = 0.26$ (point E_1), we get $x_1 = 0.22$ (point R_1) and tie line R_1E_1 exactly passes through M_1.

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So extract from stage - 1 contain 0.26 weight fraction of pyridine and raffinate 0.22 weight fraction of pyridine.

$$y_1 = 0.26 \quad \text{and} \quad x_1 = 0.22$$

E_1 is calculated by

$$E_1 = \frac{M_1 (x_{M1} - x_1)}{y_1 - x_1}$$

$$= \frac{4000 (0.25 - 0.22)}{0.26 - 0.22}$$

$$= 3000 \text{ kg}$$

$$M_1 = E_1 + R_1$$

$$R_1 = M_1 - E_1 = 4000 - 3000 = 1000 \text{ kg}$$

We have to adopt the above mentioned procedure till we cover $x = 0.02$.

For stage - 2 :

Amount of solvent = Amount of raffinate leaving first stage (given)

$$S_2 = R_1 = 1000 \text{ kg}$$

Join R_1S and locate x_{M2} on R_1S such that $R_1M_2 = M_2S$ as $R_1 : S_2 :: 1 : 1$

$(R_1/S_2 = 1000/1000 = 1)$

or $$R_1 x_1 + S_2 y_s = x_{M2} M_2$$

$$0.22 (1000) + 1000 (0) = x_{M2} (1000)$$

$$x_{M2} = 0.11$$

Mark M_2 (at $x_{M2} = 0.11$) on R_1S

Through M_2, draw a tie line R_2E_2 (adopting the above mentioned procedure for constructing the tie line).

So for the tie line R_2E_2 passing through M_2, we have $y_2 = 0.14$ and $x_2 = 0.07$

$$E_2 = \frac{M_2 (x_{M2} - x_2)}{y_2 - x_2}$$

$$E_2 = \frac{2000 (0.11 - 0.07)}{0.14 - 0.07}$$

$$= 1143 \text{ kg}$$

$$M_2 = R_2 + E_2$$

$$R_2 = M_2 - E_2 = 2000 - 1143 = 857 \text{ kg}$$

For stage - 3 :

$$R_2 = 857 \text{ kg}$$

Amount of solvent = Amount of raffinate entering the stage

$$= 857 \text{ kg}$$

$$M_3 = R_2 + S_3 = 857 + 857 = 1714 \text{ kg}$$

$$R_2 \cdot x_2 + S_3 y_s = M_3 \cdot x_{M3}$$

$$857 (0.07) + 857 (0) = 1714 \cdot x_{M3}$$

$$x_{M3} = 0.035$$

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Fig. E 2.10 (b) : Determination of theoretical stages, crosscurrent operation, one pair partially miscible

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Join R_3S and mark point M_3 on it at $x_{M3} = 0.035$ and draw tie line. R_3E_3 passing through M_3 (for constructing a tie line adopt the procedure given above).

For the tie line R_3E_3 passing through M_3, we have :

$$y_3 = 0.05 \text{ and } x_3 = 0.02$$

As $x_3 = 0.02$ – which is the final raffinate concentration 2 %, the procedure of construction should be stopped as we covered $x_n = 0.02$

$\therefore$ The number of theoretical stages = 3 ... **Ans. (i)**

Total solvent required $= S_1 + S_2 + S_3$

$= 2000 + 1000 + 857$

$= 3857$ kg ... **Ans. (ii)**

Ex. 2.11 : A 2000 kg of pyridine-water solution containing 50 % pyridine is to be extracted with an equal weight of pure chlorobenzene. The raffinate from the first extraction is to be re-extracted with a weight of solvent equal to the raffinate weight and so on.

Using equilateral triangular coordinates, determine : (i) theoretical stages required and (ii) total solvent required to reduce the pyridine concentration to 2 %.

Equilibrium tie line data in weight percent are :

Pyridine	Chlorobenzene	Water	Pyridine	Chlorobenzene	Water
0	99.95	0.05	0	0.08	99.92
11.05	88.28	0.67	5.02	0.16	94.82
18.95	79.90	1.15	11.05	0.24	88.71
24.10	74.28	1.62	18.90	0.38	80.72
28.60	69.15	2.25	25.50	0.58	73.92
31.55	65.58	2.87	36.10	1.85	62.05
35.05	61.00	3.95	44.95	4.18	50.87
40.60	53.00	6.40	53.20	8.90	37.90
49.00	37.80	13.2	49.00	37.80	13.20

Solution : Basis : 2000 kg of pyridine-water feed solution.

Pyridine in feed solution = 50 %

Pyridine in the final raffinate = 2 %

So we have to find out number of theoretical stages required for reducing the concentration of feed solution from 50 % to 2 % pyridine.

Weight fraction of pyridine in the feed $= x_F = \frac{50}{100} = 0.50$

Weight fraction of pyridine in the final raffinate $= x_n = \frac{2}{100} = 0.02.$

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This is a crosscurrent extraction operation.

Now, we will find the number of theoretical stages required for reducing the concentration of feed solution from 50 % to 2 %.

We have to go on constructing tie-lines till we reach or cover $x_n = 0.02$ (2 %) and number of such tie lines represent number of theoretical stages. Stages start with :

Stage - 1 : $F = 2000$ kg, $x_F = 0.5$

Amount of solvent = S_1 = Amount of feed (given)

$S_1 = 2000$ kg

Plot : (i) triangular diagram

(ii) x v/s y.

The horizontal rows of data give the concentrations in equilibrium solutions.

Refer Fig. E 2.11 (a) and (b).

Locate F at $x_F = 0.5$ on AC and S at the B apex (100 % B). Join FS.

S for S_1, S_2 will be located at B-apex as pure B is used as solvent.

Locate M_1 on FS such that $FM_1 = M_1S$ as $F = S_1 = 2000$ kg

i.e. $F/S_1 = M_1S/FM_1 = 1$

OR $M_1 = F + S_1 = 2000 + 2000 = 4000$ kg

$$F x_F + S_1 y_S = M_1 x_{M1}$$

$$2000\,(0.5) + 2000\,(0) = 4000\, x_{M1}$$

$$x_{M1} = 0.25$$

Mark M_1 on FS at $x_{M1} = 0.25$ (read along BC).

Adopt a trial and error procedure to draw a tie line through M_1 giving R_1 and E_1. For this plot x v/s y and make use of it for getting x in equilibrium with y. The x v/s y plot shows that for y less than 0.3, y is greater than x, so the tie lines will slope towards the apex A.

y_1 (point E_1) has to be greater than 0.25 and less than 0.3. For getting y_1 and corresponding x_1, we require three-four trials.

Take $y_1 = 0.28$, read the corresponding x_1 from the x v/s y plot. Mark y_1 and x_1 on the binodal solubility curve as E_1 and R_1, join R_1E_1 and see whether R_1E_1 passes through M_1 or not. If not take other value for y_1 and repeat the procedure till we get a line R_1E_1 passing through M_1 (it is not necessary to draw lines for each value of y_1 and x_1 on a triangular diagram).

For $y_1 = 0.26$ and x_1 (from y v/s x plot) = 0.22, we get a line R_1E_1 passing through M_1. R_1E_1 is the tie line.

This procedure is to be adopted for drawing the remaining tie lines till we cover $x_n = 0.02$

$$y_1 = 0.26 \text{ and } x_1 = 0.22$$

$$E_1 = \frac{M_1 (x_{M1} - x_1)}{y_1 - x_1} = \frac{4000\,(0.25 - 0.22)}{0.26 - 0.22}$$

$$= 3000 \text{ kg}$$

$$M_1 = R_1 + E_1$$

$$R_1 = M_1 - E_1 = 4000 - 3000 = 1000 \text{ kg}$$

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Stage - 2 : Feed to stage 2 : $R_1 = 1000$ kg

Given : Amount of solvent = S_2 = Amount of feed (raffinate from stage 1)

$= 1000$ kg

Join R_1S and locate M_2 on it such that $R_1/S_2 = M_2S/R_1M_2 = 1.0$

OR $R_1x_1 + S_2 y_s = M_2 x_{M2}$

$1000\,(0.22) + 1000\,(0) = 2000 \cdot x_{M2}$

$x_{M2} = 0.11$

Locate M_2 on R_1S at $x_{M2} = 0.11$

Repeat the procedure given above to obtain a tie line R_2E_2 passing through M_2.

y_2 has to be more than 0.11.

Take $y_2 = 0.15$, 0.14 and so on read x from x v/s y plot and so on.

By a trial and error procedure, we get a tie line R_2E_2 passing through M_2 for :

$y_2 = 0.14$ and $x_2 = 0.07$

$$E_2 = \frac{M_2\,(x_{M2} - x_2)}{y_2 - x_2}$$

$$= \frac{2000\,(0.11 - 0.07)}{0.14 - 0.07} = 1143 \text{ kg}$$

$M_2 = R_2 + E_2$

$R_2 = M_2 - E_2 = 2000 - 1143 = 857$ kg

Stage - 3 :

Given : Amount of solvent = Amount of raffinate entering the stage

$S_3 = 857$ kg

$M_3 = R_2 + S_3 = 857 + 857 = 1714$ kg

$R_2 x_2 + S_3 y_s = M_3 x_{M3}$

$857\,(0.07) + 857\,(0) = 1714\, x_{M3}$

$x_{M3} = 0.035$

Join R_3S, mark M_3 on it at $x_{M3} = 0.05$ and draw a tie line R_3E_3 passing through M_3.

For R_3E_3 passing through M_3, we have : $y_3 = 0.05$ and $x_3 = 0.02$

As $x_3 = 0.02$ – which is the final raffinate concentration of 2 %, further procedure is not required (as we covered $x_n = 0.02$).

$\therefore$ Number of theoretical stages required = 3.

Total solvent required = $2000 + 1000 + 857 = 3857$ kg ...Ans.

$$E_3 = \frac{M_3\,(x_{M3} - x_3)}{y_3 - x_3} = \frac{1714\,(0.035 - 0.02)}{0.05 - 0.02} = 857 \text{ kg}$$

$R_3 = 1714 - 857 = 857$ kg

Check : Pyridine in the feed = $0.50\,(2000) = 1000$ kg

Pyridine in the final raffinate = $0.02\,(857) = 17.14$ kg

Pyridine removed from the solution = $1000 - 17.14 = 982.86$ kg ... (1)

Pyridine extracted = $E_1 y_1 + E_2 y_2 + E_3 y_3$

$= 3000\,(0.26) + 1143\,(0.14) + 857\,(0.05)$

$= 982.85$ kg ... (2)

(1) = (2), so calculations are correct.

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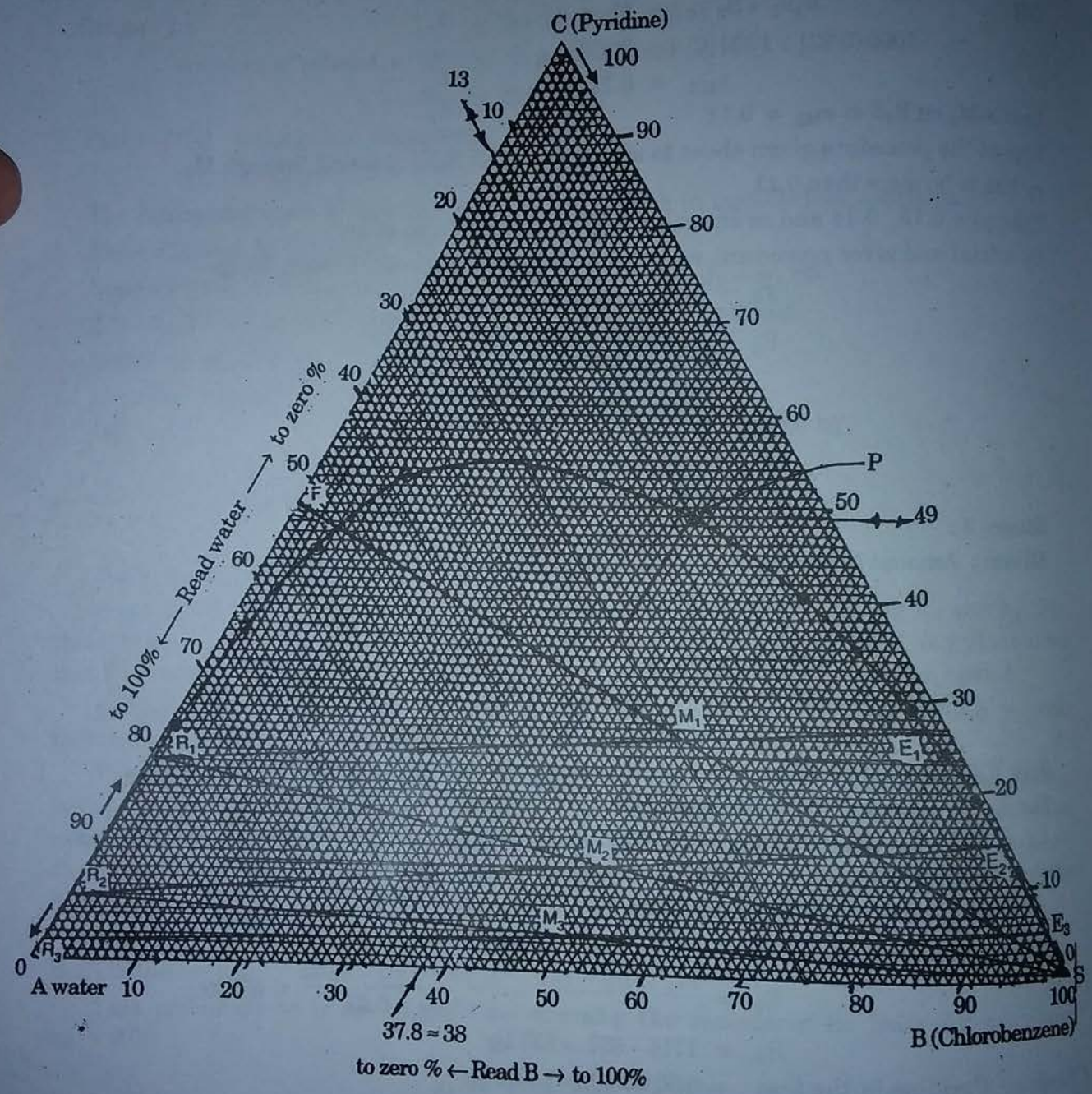

Fig. E 2.11 (a) : Pyridine – 49% + chlorobenzene – 37.8% + water – 13% by weight ⇒ Point P in the figure corresponds to the above equilibrium data

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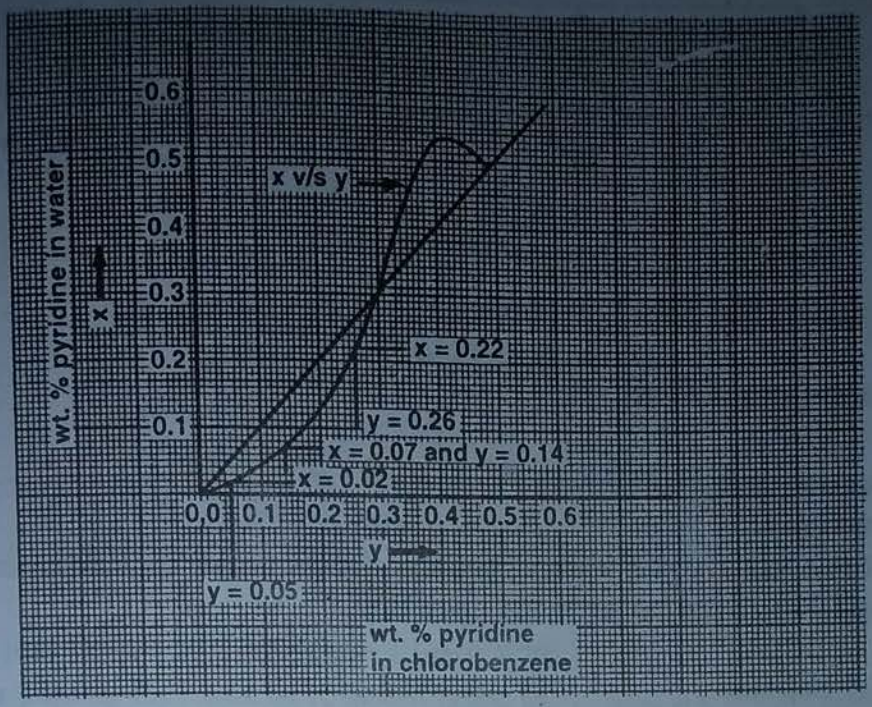

Fig. E 2.11 (b)

Ex. 2.12 : 100 kg of a solution of acetic acid (C) in water (A) containing 30 % acid by weight is to be extracted three times with isopropyl ether (B) at 293 K (20°C), using 40 kg of pure solvent in each stage (cross-current system). Determine (i) quantities and compositions of various streams, (ii) quantity of solvent that would be required if the same final raffinate concentration were to be obtained in single stage.

Equilibrium data at 293 K (20°C) are as follows :

Water layer			Isopropyl ether layer		
wt. % A.A., 100 x	Water	IPE	wt. % A.A., 100 y	Water	IPE
0.69	98.1	1.2	0.18	0.5	99.3
1.41	97.1	1.5	0.37	0.7	98.9
2.89	95.5	1.6	0.79	0.8	98.4
6.42	91.7	1.9	1.93	1.0	97.1
13.3	84.4	2.3	4.82	1.9	93.3
25.5	71.1	3.4	11.40	3.9	84.7
36.70	58.9	4.4	21.60	6.9	71.5
44.30	45.1	10.6	31.10	10.8	58.1
46.4	37.1	16.5	36.20	15.1	48.7

Solution : Basis : 100 kg of 30 % acetic acid solution.

F = 100 kg

x_F = **Weight fraction of acetic acid in feed** = $\frac{30}{100} = 0.30$

Pure solvent used for each stage = S = B = 40 kg

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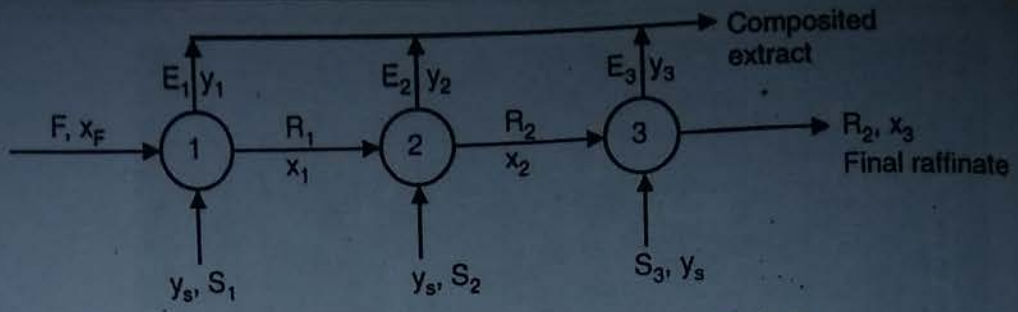

Fig. E 2.12 (a) : Crosscurrent operation

$$S_1 = S_2 = S_3 = S = B = 40 \text{ kg}$$

and $y_s = 0$ (as solvent is pure B)

Plot y v/s x

Plot x v/s wt. fraction IPE

Plot y v/s wt. fraction IPE

It is clear from the nature of plot y v/s x that the system under consideration is such that the tie lines will slope downward towards the apex B.

Stage - 1 :

Overall material balance :

$$F + S_1 = M_1 = E_1 + R_1$$

Material balance of acetic acid (C) :

$$F x_F + S_1 y_1 = M_1 x_{M1} = E_1 y_1 + R_1 x_1$$

$$E_1 = \frac{M_1 (x_{M1} - x_1)}{y_1 - x_1}$$

$$M_1 = F + S_1 = 100 + 40 = 140 \text{ kg}$$

Adding S_1 to F, produces a mixture M_1, which on settling splits into R_1 and E_1.

Material balance of acetic acid (C) :

$$100 (0.30) + 40 (0) = 140 (x_{M1})$$

$$\therefore \quad x_{M1} = 0.214$$

Refer Fig. E2.12 (b).

Adding S_1 to F, produces a mixture M_1, which splits on settling into E_1 and R_1.

(i) As the solvent used is pure B, the point S is located at the B-apex. Through the point S on x-axis at 100 % B (as B is pure), draw a line SF. Point F is at $x_F = 0.3$ on y-axis.

(ii) Mark point M_1 on line FS at $x_{M1} = 0.214$ (read x_{M1} from y-axis).

(iii) Now we have to draw a tie line R_1E_1. The tie line joining R_1E_1 passes through M_1. For drawing R_1E_1 we have to adopt a trial and error procedure.

(iv) x_1 has to be less than 0.3, moreover it has to be less than 0.285 at which FS line cuts x v/s wt. fraction of IPE plot. The tie line slopes downward towards the B apex as y is less than x for any value of x (as seen from a plot of y v/s x). Take x_1 (R_1 on x v/s wt. fraction IPE) = 0.28, 0.27, 0.26 ... etc., proceed towards the y v/s x curve horizontally, get the corresponding value of y_1 from the y v/s x curve and then proceed back to the original figure horizontally and mark a y_1 value at E_1 on the y v/s wt. fraction IPE. See that the line joining R_1E_1 passes through M_1 or not if does not passes through M_1, take a new value of x_1 (R_1) on the x v/s wt. fraction of IPE and repeat the same procedure till R_1E_1 passes through M_1 then corresponding x_1 and y_1 values represent the equilibrium values of wt. fraction of acetic acid in the raffinate and extract (maximum three trials are required).

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For a tie line R_1E_1 passing through M_1,

$$x_1 = 0.26 \text{ and } y_1 = 0.1175$$

Then
$$E_1 = \frac{M_1 (x_{M1} - x_1)}{y_1 - x_1}$$
$$= \frac{140 (0.214 - 0.26)}{0.1175 - 0.26}$$
$$= 45.2 \text{ kg}$$

We have
$$M_1 = E_1 + R_1$$
$$R_1 = M_1 - E_1 = 140 - 45.2 = 94.8 \text{ kg}$$

Stage - 2 :

$$M_2 = R_1 + S_2 = 94.8 + 40 = 134.8 \text{ kg}$$

and
$$R_1 x_1 + S_2 y_2 = M_2 x_{M2}$$
$$94.8 \times 0.26 + 40 (0) = 134.8 x_{M2}$$
$$\therefore \quad x_{M2} = 0.1828$$

x_2 has to be less than 0.26 and greater than 0.1828.

Draw a line joining R_1S and on it locate point M_2 at $x_{M2} = 0.1828$ (reading from y-axis). Adopt a trial and error procedure as described above till we get a tie line R_2E_2 passing through M_2 and read x_2 and y_2.

$$x_2 = 0.225 \text{ and } y_2 = 0.0975$$
$$E_2 = \frac{M_2 (x_{M2} - x_2)}{y_2 - x_2} = \frac{134.8 (0.1828 - 0.225)}{0.0975 - 0.225}$$
$$= 44.62 \text{ kg}$$

We have
$$M_2 = E_2 + R_2$$
$$R_2 = M_2 - E_2 = 134.8 - 44.62 = 90.18 \text{ kg}$$

Stage - 3 :

$$M_3 = R_2 + S_3 = 90.18 + 40 = 130.18 \text{ kg}$$
$$R_2 x_2 + S_3 y_s = M_3 x_{M3}$$
$$90.18 \times 0.225 + 40 (0) = 130.18 x_{M3}$$
$$x_{M3} = 0.156$$

Join R_2 S and locate on it M_3 at $x_{M3} = 0.156$ and adopting a similar procedure, draw a tie line $R_3 E_3$ passing through M_3 (x_3 has to be less than 0.225 and greater than 0.156) and read the values of x_3 and y_3.

$$x_3 = 0.195 \text{ and } y_3 = 0.08$$
$$E_3 = \frac{M_3 (x_{M3} - x_3)}{y_3 - x_3} = \frac{130.18 (0.156 - 0.195)}{0.08 - 0.195}$$
$$= 44.15 \text{ kg} \quad \text{...Ans.}$$

$$M_3 = R_3 + E_3$$
$$R_3 = M_3 - E_3 = 130.18 - 44.15 = 86.03 \text{ kg} \quad \text{... Ans.}$$

Acid content of final raffinate $= 0.195 \times 86.03 = 16.78 \text{ kg}$ ...Ans.

Composited extract $= E_1 + E_2 + E_3 = 45.2 + 44.62 + 44.15$
$= 133.77 \text{ kg}$ ... Ans.

Acid content of composited extract phase
$$= E_1 y_1 + E_2 y_2 + E_3 y_3$$
$$= 45.2 (0.1175) + 44.62 (0.0975) + 44.15 (0.08)$$
$$= 13.2 \text{ kg} \quad \text{... Ans.}$$

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(ii) For single stage extraction :

We have $x_3 = 0.195$ and $x_F = 0.195$.

For getting the final raffinate concentration of 0.195 in single stage, point M has to be the intersection of lines R_3E_3 and SF.

from Fig. E 2.12 (b), therefore corresponding x_M is :

$$x_M = 0.12$$

$$x_{M1} = x_M = 0.12 \text{ for single stage}$$

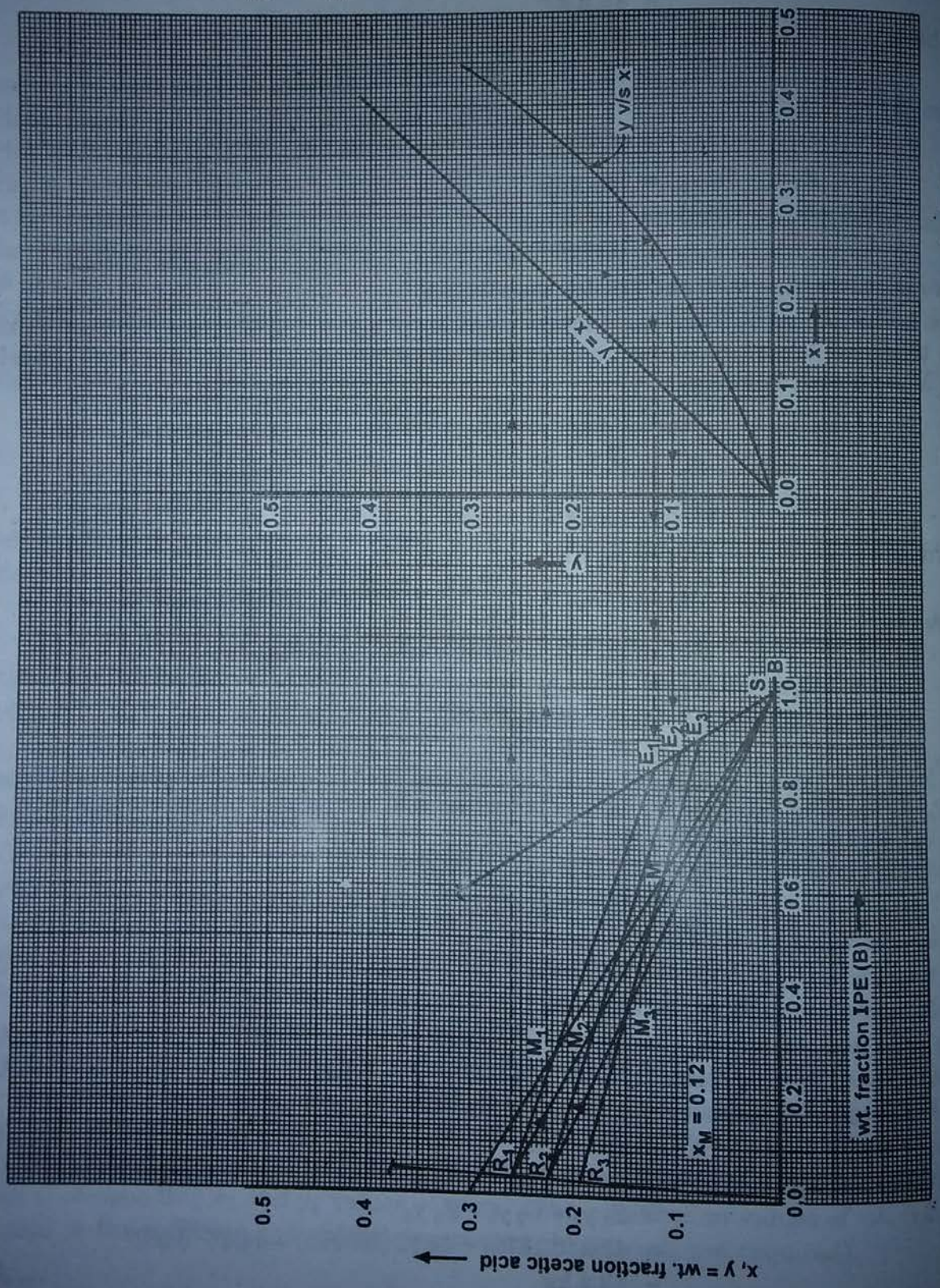

Fig. E 2.12 (b) : Three-stage crosscurrent operation of pair partially miscible

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The amount of solvent to provide a given location for M on the line FS can be computed by

$$\frac{S_1}{F} = \frac{x_F - x_{M1}}{x_{M1} - y_s}$$

$$S_1 = \frac{F\,(x_F - x_{M1})}{x_{M1} - y_s}$$

$$= \frac{100\,(0.30 - 0.12)}{0.12 - 0}$$

$$= 150 \text{ kg}$$

For three stage cross-current operation, total solvent required $= 3 \times 40 = 120$ kg

For the single stage operation, solvent requirement for the same final raffinate concentration = 150 kg ... Ans.

For a single stage :

$$M_1 x_{M1} = y_1 E_1 + x_1 R_1 = x_F F + S_1 y_s$$

$$M_1 x_{M1} = x_F F + S_1 y_s$$

$$M_1 (0.12) = 0.3 \times 100 + S_1 (0)$$

$$M_1 = 250 \text{ kg}$$

$$M_1 = F + S_1$$

$$S_1 = M_1 - F = 250 - 100 = 150 \text{ kg}.$$

Ex. 2.13 : 0.160 *l*/s of a solvent B is used to treat 0.40 *l*/s of a 10 % by weight solution of C in A, in three-stage countercurrent extraction system. Determine the composition of the final raffinate.

Using same amount of solvent equally distributed in three-stage crosscurrent extraction system, determine the concentration of final raffinate. A and B are essentially insoluble.

Data :

kg C/kg A = X	0.05	0.10	0.15
kg C/kg B = Y	0.069	0.159	0.258

density of C = 1200 kg/m^3

density of A = 1000 kg/m^3

density of B = 800 kg/m^3

Solution : Basis : 0.4 *l*/s of 10 % solution of C in A.

(i) Countercurrent (three-stage) extraction

Flow rate of B = 0.160 *l*/s

Density of B = 800 kg/m^3 = 0.80 kg/*l*

Mass flow rate of B = 0.160×0.80

= 0.128 kg/s

Density of A = 1000 kg/m^3 = 1 kg/*l*

Density of C = 1200 kg/m^3 = 1.2 kg/*l*

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Let x be *l*/s of C in total volumetric flow.

Volumetric flow of A = $0.4 - x$ *l*/s

Mass flow rate of C = $x \times 1.2 = 1.2\,x$ kg/s

Mass flow rate of A = $(0.4 - x) \times 1.0 = (0.4 - x)$ kg/s

Total mass flow rate of solution = $1.2\,x + (0.4 - x)$

$= (0.4 - 0.2\,x)$ kg/s

Weight % A in feed solution = 10

$$\frac{\text{Weight of A}}{\text{Total weight}} \times 100 = \text{Weight \% A}$$

$$\frac{1.2\,x}{(0.4 - 0.2\,x)} \times 100 = 10$$

$\therefore \quad x = 0.034$

Mass flow rate of A = $0.4 - 0.034$

$= 0.366$ kg/s

Mass flow rate of C = 1.2×0.034

$= 0.408 = 0.041$ kg/s

kg C/kg A in feed solution = $X_F = \dfrac{0.041}{0.366} = 0.112$

As A and B are essentially insoluble, so all A will appear in raffinate and all B will appear in extract.

For countercurrent extraction for insoluble liquids, we have :

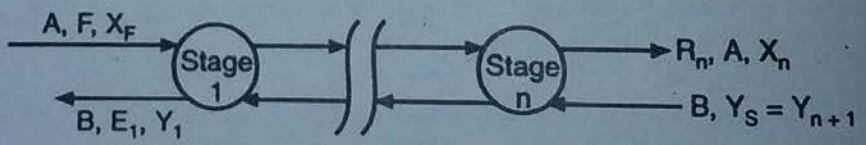

Fig. E 2.13 (a)

The overall material balance of component C over the entire system is

$$AX_F + BY_{n+1} = BY_1 + AX_n$$

$$AX_F + BY_s = BY_1 + AX_n, \qquad (\text{as } Y_{n+1} = Y_s)$$

$$\frac{A}{B} = \frac{Y_1 - Y_s}{X_F - X_n}$$

This is the equation of operating line which is a straight line with a slope equal to A/B. It passes through points M (X_F, Y_1) and P (X_n, Y_s).

Slope = A/B = 0.366/0.128 = 2.86

Slope = $\tan\alpha = 2.86 \quad \therefore \quad \alpha = \tan^{-1}(2.86) = 71°$

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Refer Fig. E 2.13 (b) and (c).

Draw the equilibrium curve Y v/s X.

Draw a vertical through $X_F = 0.112$ on the x-axis.

For our case, $Y_s = Y_{n+1} = 0$ (pure solvent B)

So the point P $(X_n, 0)$ lies on the x-axis.

For drawing the operating line through point P of a slope equal to 2.86 (i.e., inclined at an angle 71° to x-axis), we have to adopt a trial and error procedure.

The number of stages given are three. So the operating line is to be drawn by taking different values for X_n in such a way that we can construct exactly three stages between X_F and X_n.

Draw the operating line through point P with an angle 71° and mark its cutting point with the vertical through X_F as M and construct the stages between the equilibrium curve and the operating line. If we do not get three stages, then take a new value for X_n and repeat the procedure till we get a operating line (PM) with which we can construct exactly three stages. A trial and error procedure is shown in Fig. E 2.13 (c).

For n = 3 between M (X_F, Y_1) and P $(X_n, 0)$ read X_n from x-axis.

$$X_n = 0.055$$

$\therefore$ Composition of final raffinate $= X_n = X_3 = 0.055$ kg C/kg A ... **Ans.**

In final raffinate there are 0.055 kg of C per 1 kg of A.

So in 1.055 kg of raffinate there are 0.055 kg of C.

$$\text{Weight \% C in final raffinate} = \frac{0.055}{1.055} \times 100$$

$$= 5.21 \quad \text{... Ans.}$$

(ii) Crosscurrent operation - three stages

Total solvent to be used = Solvent used for countercurrent operation

$= 0.128$ kg/s

Given : Solvent is evenly distributed between the three stages.

$$\therefore \quad \text{Solvent used for each stage} = \frac{\text{Total solvent used}}{3}$$

$$= \frac{0.128}{3} = 0.0427 \text{ kg/s}$$

$$B_1 = B_2 = B_3 = B$$

For stage - 1 :

$B_1 = B = 0.0427$ kg/s and $X_F = 0.112$

As A and B are insoluble liquids, we have :

$$AX_F + B_1Y_s = AX_1 + BY_1$$

$$AX_F = AX_1 + BY_1, \quad (\text{as } Y_s = 0)$$

$$\frac{Y_1}{X_1 - X_F} = -\frac{A}{B}$$

This is the operating line equation which is a straight line through X_F on the x-axis of slope $-\frac{A}{B}$.

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As A and B are insoluble liquids (immiscible solvents) and $B_1 = B_2 = B_3 = 0.0427$ kg/s, the slope of all three operating lines is same (lines are parallel to each other).

The operating line - 1 passes through the point M (X_F, 0) on the x-axis and cuts the equilibrium curve at (Y_1, X_1).

$$\text{Slope} = -\frac{A}{B} = -\frac{0.366}{0.0427} = -8.57$$

$$\tan \alpha = -8.57 \qquad \therefore \ \alpha = \tan^{-1}(-8.57) = -83^\circ$$

So measure the angle in clockwise direction.

Refer Fig. E 2.13 (d).

Draw the equilibrium curve Y v/s X. Mark point M on the x-axis at $X_F = 0.112$ (as $Y_s = 0$) and through M draw the operating line with a slope of – 8.57, i.e., inclined at an angle – 83° which will cut the equilibrium curve at P (X_1, Y_1). Through P, drop a vertical on the x-axis and through the cutting point of the vertical with x-axis (i.e., X_1) draw the operating line-2 (representing stage 2), parallel to the operating line-1, which will cut the equilibrium curve at the point L (X_2, Y_2). Repeat the procedure for third stage and read X_3 on the x-axis.

$$X_3 = 0.0675$$

Composition of the final raffinate = 0.0675 kg C/kg A ... Ans.

Amount of the raffinate = Amount of A + Amount of C = (1 + 0.0675) kg

$$\text{Weight \% C in the final raffinate} = \left(\frac{0.0675}{1 + 0.0675}\right) 100 = 6.3 \quad \text{... Ans.}$$

In the graphical solution taking into account all possible errors, a variation in the answer obtained to the extent of 5 % should be permitted.

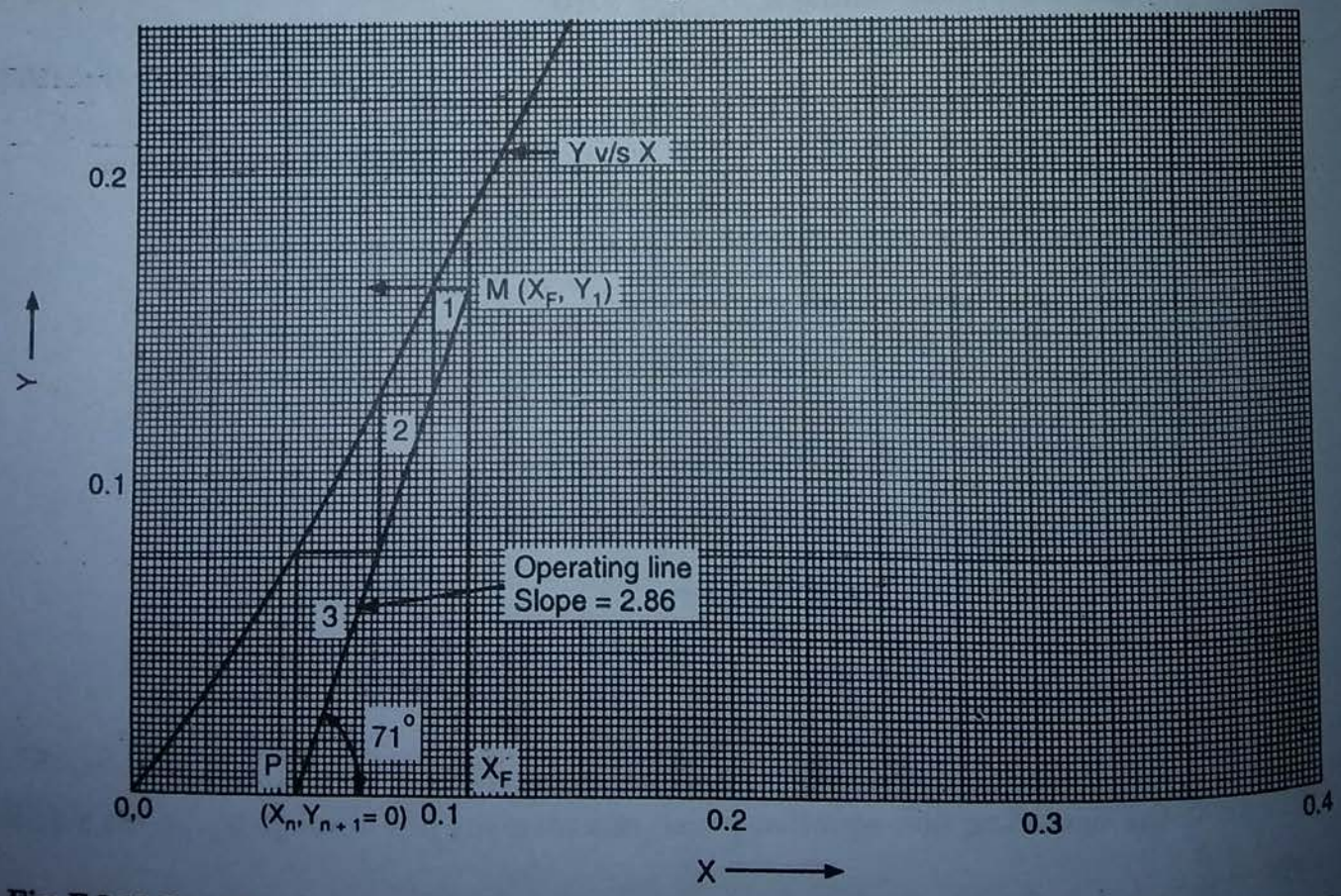

Fig. E 2.13 (b) : Three-stage countercurrent operation, determination of composition of final raffinate

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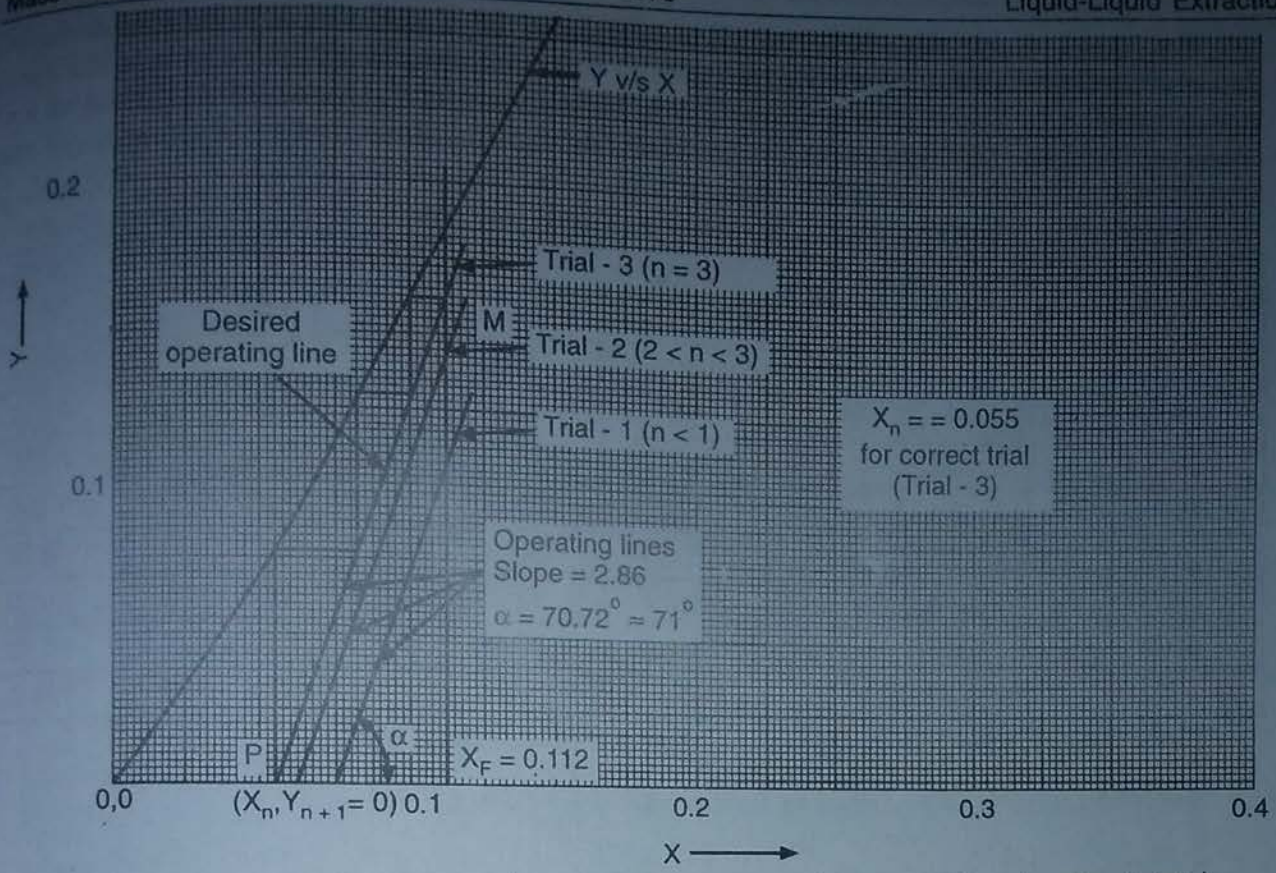

Fig. E 2.13 (c) : Three-stage countercurrent extraction operation showing trials

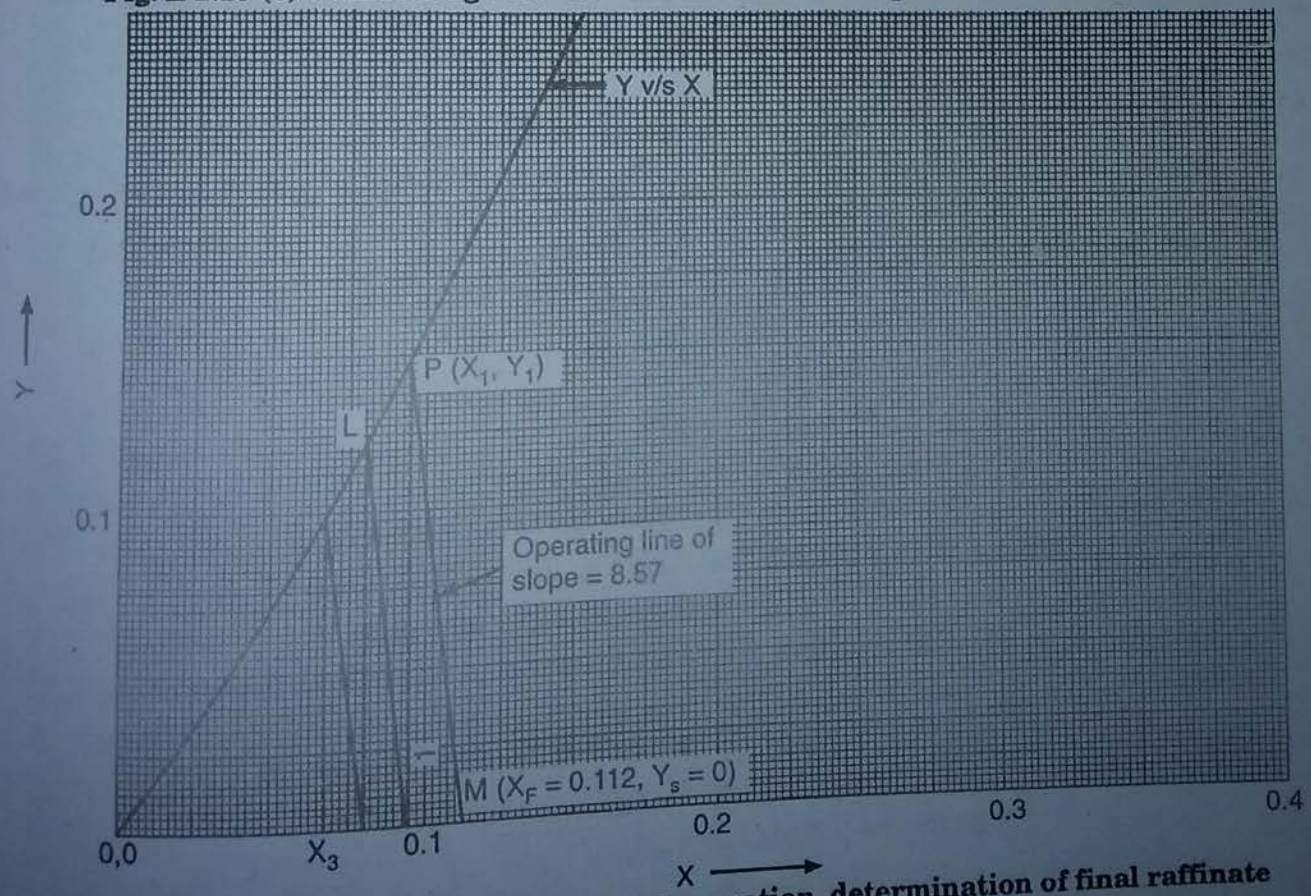

Fig. E 2.13 (d) : Three-stage crosscurrent operation, determination of final raffinate composition

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Ex. 2.14 : A pyridine-water solution containing 50 % pyridine is to be continuously and countercurrently extracted at the rate of 2.25 kg/s with chlorobenzene to reduce the pyridine concentration to 2 % in the final raffinate. Using rectangular co-ordinates - plot of x v/s y and x, y v/s wt. fraction of chlorobenzene, determine the number of theoretical stages required if 2.3 kg/s of solvent is used.

Equilibrium tie-line data in weight percent are :

Pyridine	Chlorobenzene	Water	Pyridine	Chlorobenzene	Water
0	99.95	0.05	0	0.08	99.92
11.05	88.28	0.67	5.02	0.16	94.82
18.95	79.90	1.15	11.05	0.24	88.71
24.10	74.28	1.62	18.90	0.38	80.72
28.60	69.15	2.25	25.50	0.58	73.92
31.55	65.58	2.87	36.10	1.85	62.05
35.05	61.0	3.95	44.95	4.18	50.87
40.60	53.0	6.40	53.20	8.90	37.90
49	37.8	13.2	49	37.8	13.2

Solution : Basis : 2.25 kg/s of pyridine-water solution feed.

Weight fraction of pyridine in feed = $x_F = \frac{50}{100} = 0.50$

Weight fraction of pyridine in final raffinate = $x_n = \frac{2}{100} = 0.02$

Solvent (B) flow rate = S = 2.3 kg/s

Refer Fig. E 2.14 (a) and (b).

Plot : x v/s y and x, y v/s weight fraction of chlorobenzene.

Mark point F on y-axis at $x_F = 0.50$. Mark the point S at the B-apex as the solvent is pure (i.e. at which wt. fraction of B = 1.0 as it is pure). Join FS.

Now locate M on FS such that :

$$\frac{S}{F} = \frac{\text{line MF}}{\text{line MS}}$$

$$\frac{\text{line MF}}{\text{line MS}} = \frac{2.3}{2.25} = 1.022$$

$$\text{line MF} = 1.022\ \text{MS}$$

We have : $FS = MF + MS = 112$ units (mm, measured)

$1.022\ MS + 1\ MS = 2.022\ MS = 112$, $\therefore$ MS = 55.40 units

and MF = 112 – 55.40 = 56.6 units

So locate M on FS such that line MF = 56.6 units and line MS = 55.4 units.

OR :

$$M = F + S = 2.25 + 2.3 = 4.55 \text{ kg/s}$$

$$F x_F + S y_S = M x_M$$

$$2.25\ (0.5) + 2.3\ (0) = 4.55\ x_M$$

$$\therefore \quad x_M = 0.247$$

Mark point M on FS at $x_M = 0.247$ (read from x-axis).

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Both the above mentioned procedures give the same location for point M on FS.

Locate R_n (raffinate from n^{th} stage) on the binodal solubility curve corresponding to 2 % pyridine (i.e., at $x_n = 0.02$). Join R_nM and project it to meet the equilibrium curve at E_1. Join FE_1, R_nS and project these lines to meet at Δ_R.

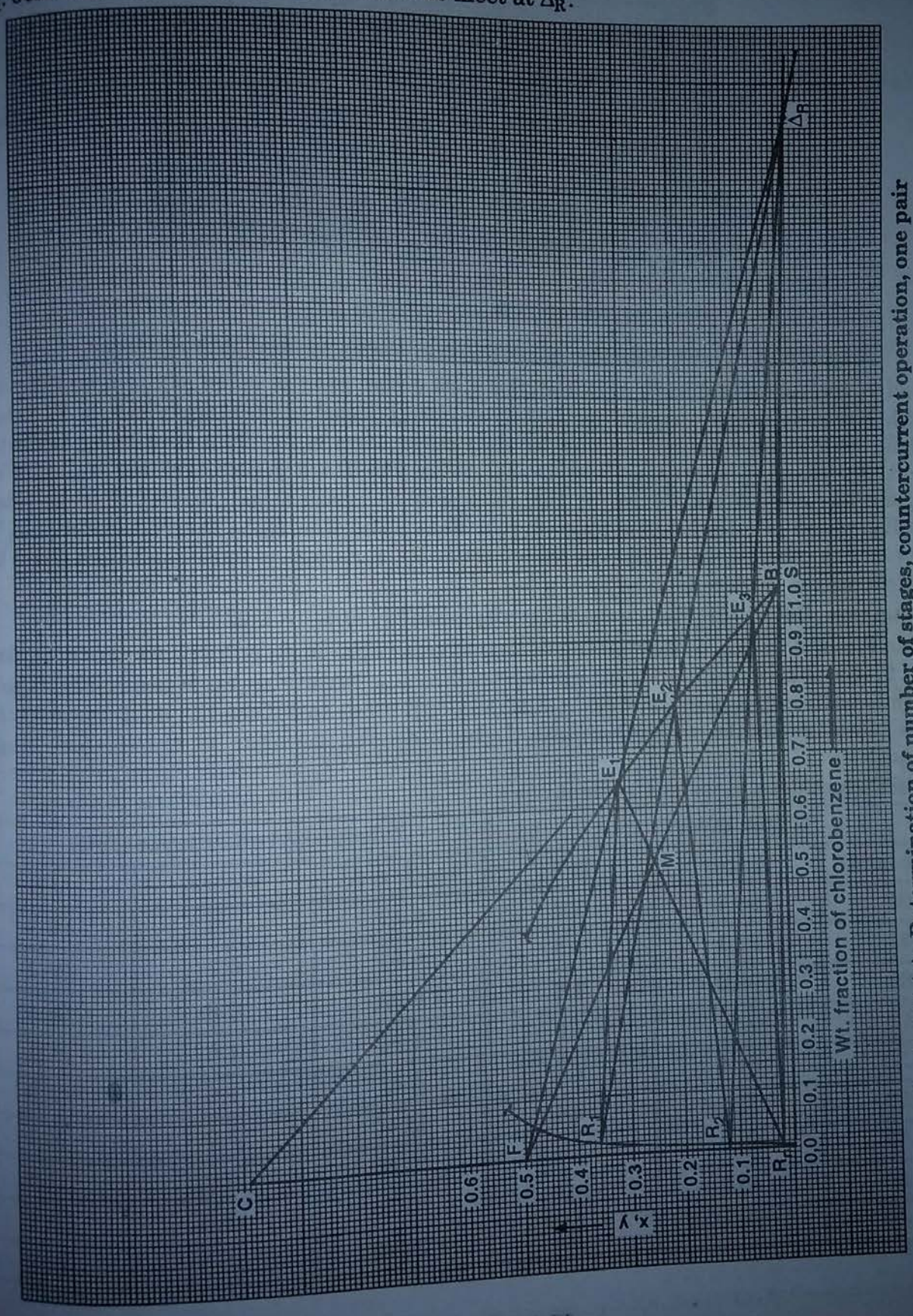

Fig. E 2.14 (a) : Determination of number of stages, countercurrent operation, one pair partially miscible

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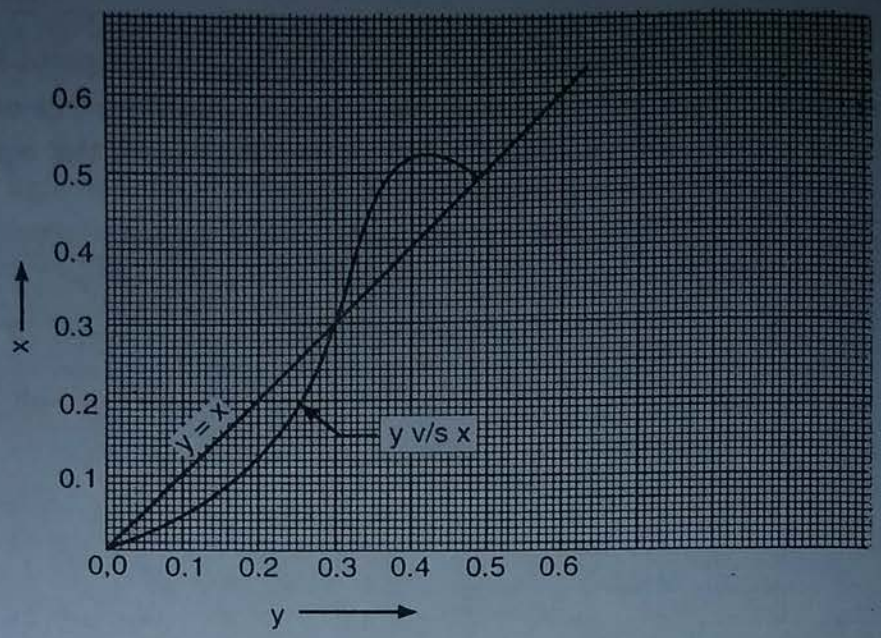

Fig. E 2.14 (b) : Equilibrium curve – required for construction of stages

Read y_1 corresponding to E_1 and from the x v/s y plot, get the value of x_1 (in equilibrium with y_1). Mark R_1 on the binodal solubility curve at location corresponding to x_1 value. Join R_1E_1 which is a required tie line for first stage. Join $R_1\Delta_R$. The line $R_1\Delta_R$ meets the curve at E_2. Read y_2 corresponding to E_2. From the x v/s y plot, get the value of x_2 corresponding to y_2. Mark R_2 on the binodal curve at location corresponding to x_2 value. Join R_2E_2, which is a tie line for stage-2, join $R_2\Delta_R$. The line $R_2\Delta_R$ meets the curve at E_3. The working is continued in this way till we cover R_n (i.e., $x_n = 0.02$) and then count the theoretical stages required.

In our case, the tie line R_3E_3 is such that R_3 is exactly at $x_n = 0.02$

From Fig. E 2.14 (a), number of theoretical stages required = n = 3 ... Ans.

Ex. 2.15 : 2.22 kg/s of an acetic acid-water solution containing 30 % acetic acid is to be countercurrently extracted with 5.55 kg/s of isopropyl ether to reduce the acid concentration to 2 % (final raffinate concentration). Determine the number of theoretical stages required.

Equilibrium data in weight percent are :

Water layer			Isopropyl ether layer (IPE)		
Acetic acid	Water	IPE	Acetic acid	Water	IPE
0.69	98.1	1.2	0.18	0.5	99.3
1.41	97.1	1.5	0.37	0.7	98.9
2.89	95.5	1.6	0.79	0.8	98.4
6.42	91.7	1.9	1.93	1.0	97.1
13.30	84.4	2.3	4.82	1.9	93.3
25.50	71.1	3.4	11.40	3.9	84.7
36.70	58.9	4.4	21.60	6.9	71.5
44.30	45.1	10.6	31.10	10.8	58.1
46.40	37.1	16.5	36.20	15.1	48.7

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Solution : Basis : 2.22 kg/s of 30 % acetic acid solution.

Solvent flow rate = 5.55 kg/s. Solvent is pure B, so $y_s = 0$.

Weight fraction of acetic acid in feed solution = $x_F = \frac{30}{100} = 0.30$.

Weight fraction of acetic acid in final raffinate = $x_n = \frac{2}{100} = 0.02$

We have to find out the number of stages required for reducing the acid concentration from $x_F = 0.3$ to $x_n = 0.02$ for countercurrent operation.

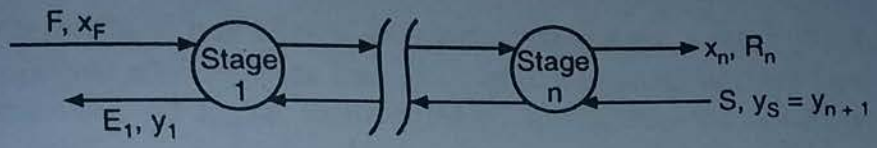

Fig. E 2.15 (a) : Crosscurrent extraction

For finding out number of stages, we will make use of rectangular coordinates.

Plot : y v/s x.

This is required for drawing tie lines, knowing y (E) read x from this graph to locate R at this x value on rectangular coordinates.

Plot : x, y v/s weight fraction of isopropyl ether (IPE) and draw a curve covering $x_F = 0.3$ only.

Refer Fig. E 2.15 (b) and (c).

Locate F on the y-axis at $x_F = 0.3$

Locate S on the x-axis at wt. fraction of IPE (B) = 1.0 (at the B-apex as it is pure – $y_s = 0$).

Join FS.

F = 2.22 kg/s, S = 5.55 kg/s, S/F = 5.55/2.22 = 2.5

Locate M on FS such that

$$\frac{\text{line FM}}{\text{line MS}} = \frac{S}{F} = 2.5$$

Total FS distance = 157 units (measured)

$$\frac{FM}{MS} = 2.5$$

$FM = 2.5\,MS$, $FS = FM + MS = 3.5\,MS$

$3.5\,MS = 157$ $\therefore$ MS = 44.86 units and FM = 112.14 units

So mark M on FS at a distance of 112.14 mm from F and read x_M from y-axis.

$x_M = 0.08625$

OR : Calculate x_M and locate M.

Material balance of acetic acid over entire unit :

$M = F + S$

$M = 2.22 + 5.55 = 7.77$ kg/s

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$$F x_F + S y_S = M x_M$$

$$2.22\,(0.30) + 5.55\,(0) = 7.77\, x_M$$

$$x_M = 2.22\,(0.3)/7.77 = 0.086$$

Both these procedures give the same location for M on FS.

Locate R_n on x v/s IPE (water rich curve) at $x_n = 0.02$

Join $R_n M$ and extend to meet IPE rich curve at E_1.

Join FE_1 and extend it to meet a line drawn through $R_n S$ at Δ_R. Lines FE_1 and $R_n S$ are extended to meet at point Δ_R.

At E_1, $y_1 = 0.10$, from y v/s x plot, read x_1, $x_1 = 0.23$ and locate R_1 on water rich curve. R_1E_1 is the required tie-line (stage-1). Join $R_1\Delta_R$ and so on and do the working till we cover $x_n = 0.02$ as per the procedure given in solution of Ex. 2.14. When number of stages required are more then this procedure makes the figure more complex in nature as there will be many lines most probably overlapping and crossing each other. So in such a case, a high accuracy is required to get the correct answer. It is not that we cannot adopt it, we can adopt this method but all that we have to do is to do the construction precisely to get the correct answer when stages are more.

So we will adopt the following alternate procedure for determining the required stages.

Procedure upto locating Δ_R is same as above.

After locating point Δ_R :

Draw a number of lines such as $RQ\Delta_R$ random lines - each cutting both water rich and IPE rich curves and passing through Δ_R and read the values of y_1 and corresponding x_1 for each line.

For line $FE_1\Delta_R$: $x = x_F = 0.3$ and $y_1 = 0.10$

For line $RQ\Delta_R$: $x = 0.145$ and $y = 0.0375$

Upto line $R_n S\Delta_R$: $x = x_n = 0.02$, $y = y_{n+1} = y_S = 0$

y_{n+1}	0	0.01	0.0375	0.05	0.075	0.08	0.10 = y_1
x_n	0.02	0.055	0.145	0.1755	0.2375	0.25	0.3 = x_F

Refer Fig. E 2.15 (d).

These data are plotted as the operating curve.

Plot : (i) tie-line data as the equilibrium curve (given as the data)

(ii) the operating curve y_{n+1} v/s x_n

and construct the stages between the equilibrium curve and the operating curve starting from the point (x_F, y_1), i.e., (0.3, 0.1) and ending or covering the point $(x_n, y_{n+1} = y_s)$, i.e., (0.02, 0) and count the stages.

Number of theoretical stages required = 8 ... Ans.

(As 7 stages are insufficient to cover x_n and to cover x_n, one more stage should be provided.)

For these plots, take larger scales for accuracy.

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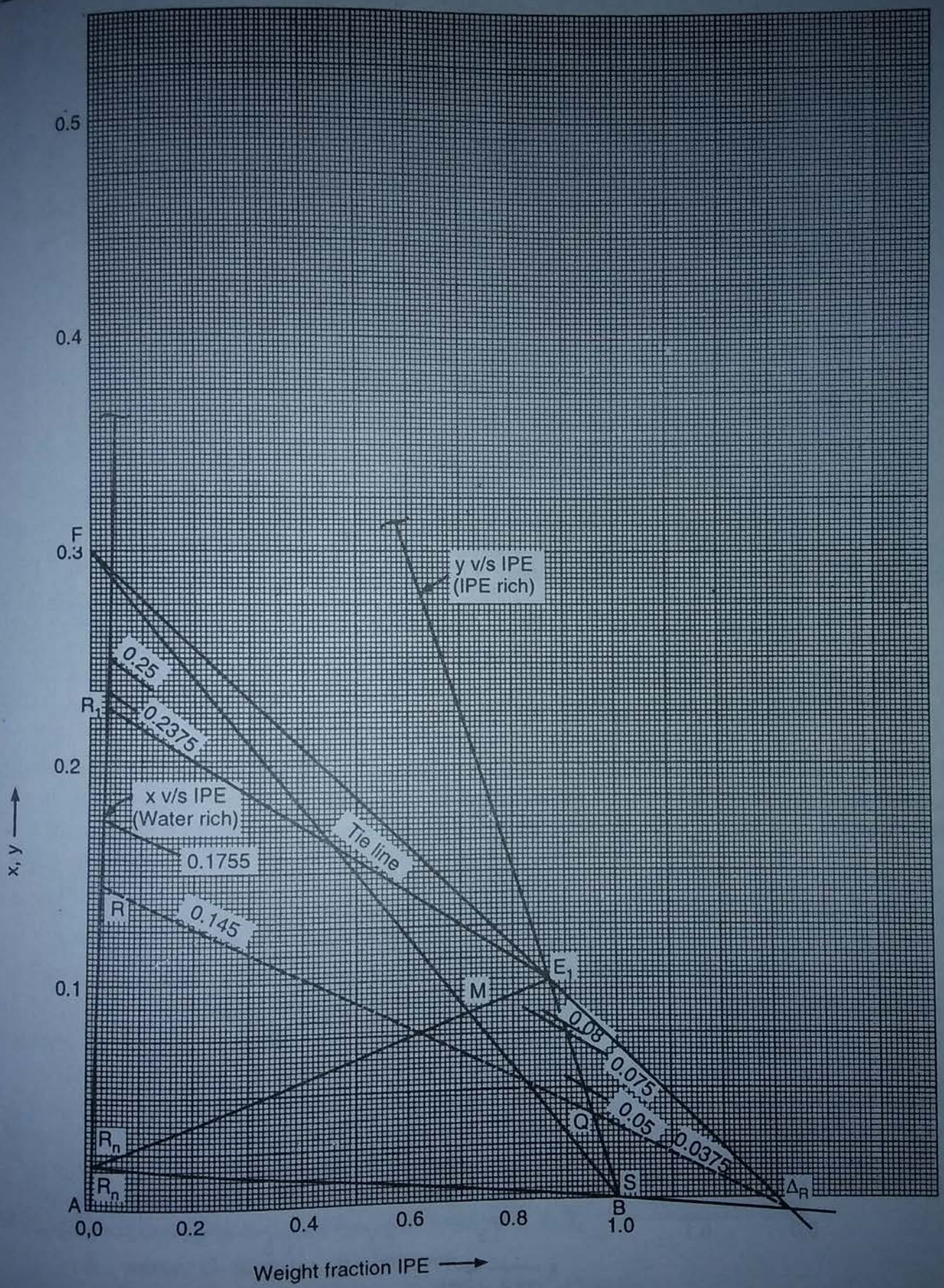

Fig. E 2.15 (b)

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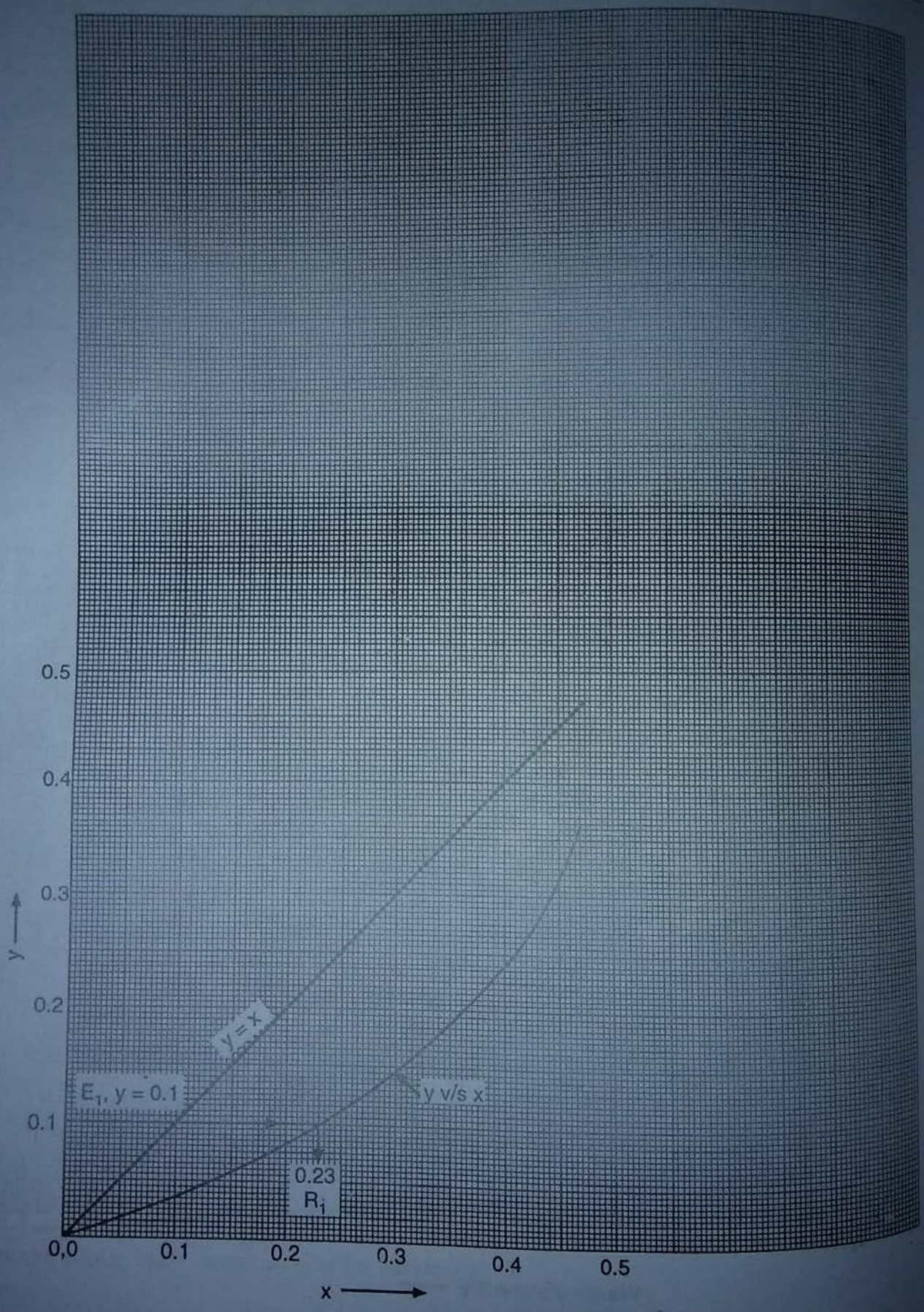

Fig. E 2.15 (c)

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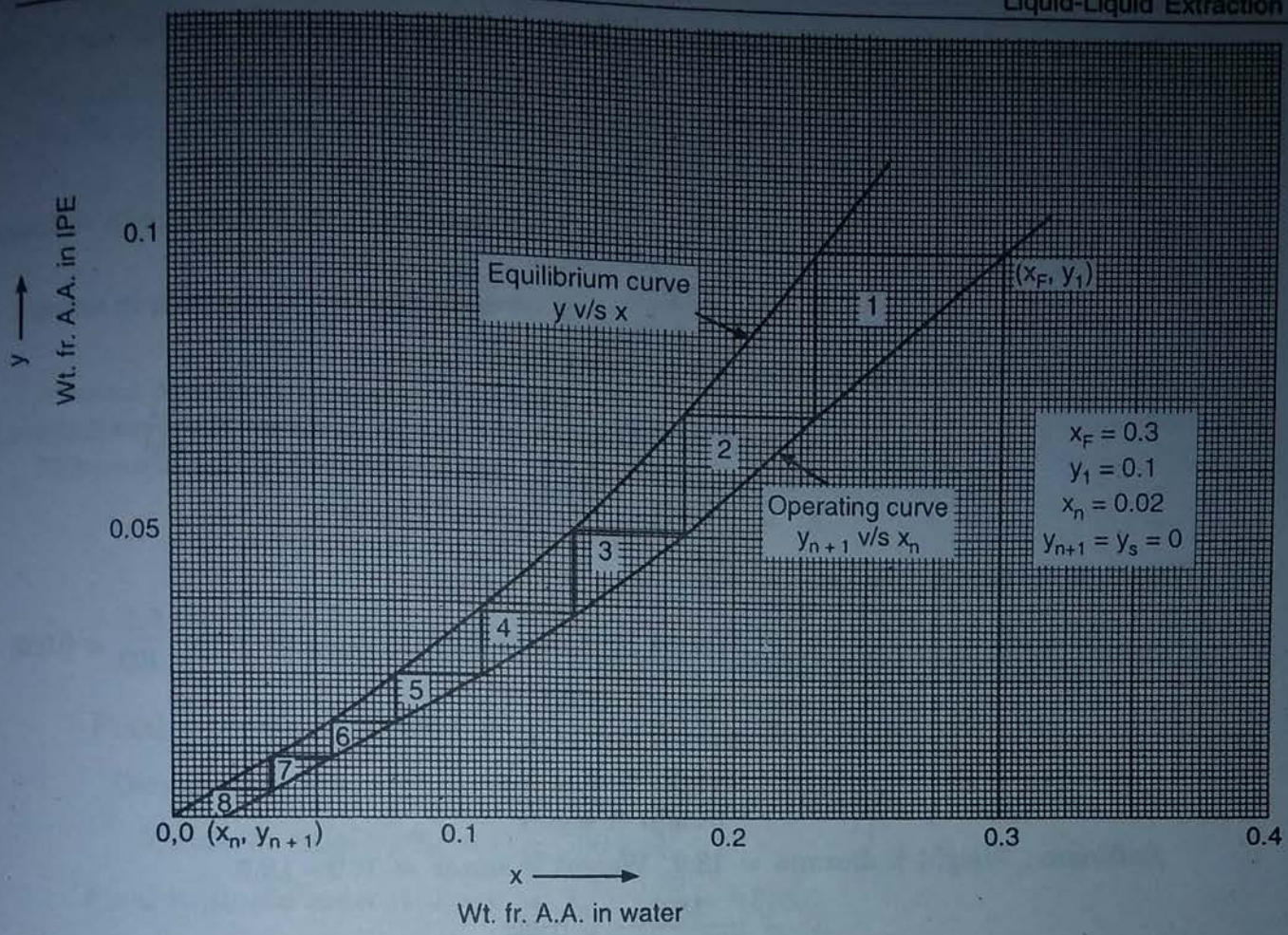

Fig. E 2.15 (d) : Determination of number of stages, countercurrent operation, one pair partially miscible

Ex. 2.16 : Water-dioxane solution forming a minimum boiling azeotrope cannot be separated by ordinary distillation techniques. Benzene forming no azeotrope with dioxane can be used as a extraction solvent. The equilibrium distribution of dioxane between water and benzene at 298 K (25°C) is as follows :

Weight % dioxane in water	5.1	18.9	25.2
Weight % dioxane in benzene	5.2	22.5	32

At these concentrations, benzene and water are essentially insoluble.

(i) 1000 kg of a 25 % dioxane and 75 % water solution is to be extracted with benzene to remove 95 % of the dioxane. The benzene is dioxane free.

Calculate the solvent requirement for a single stage batch operation.

(ii) If the extraction is done in countercurrent fashion with 1000 kg/h of feed solution containing 25 % of dioxane, what is the minimum solvent requirement in kg/h to remove 95 % of the dioxane ? Fresh benzene is used.

Solution : (i) Single stage operation :

Basis : 1000 kg of solution containing 25 % dioxane.

A - water, C- dioxane, B - benzene

Dioxane in feed solution = $0.25 \times 1000 = 250$ kg

Water in feed solution = $0.75 \times 1000 = 750$ kg

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Benzene and water are essentially insoluble. Raffinate will contain 750 kg of water and extract will contain all B.

X_F = kg dioxane/kg water in feed = 250/750 = 0.3333

Convert equilibrium data into X and Y.

$$X = \frac{x}{1-x} : \qquad x \text{ - weight fraction of dioxane in raffinate}$$

$$Y = \frac{y}{1-y} : \qquad y \text{ - weight fraction of dioxane in extract}$$

e.g. weight % dioxane in water = 5.1

$$x = \text{Weight fraction of dioxane in water} = \frac{5.1}{100} = 0.051$$

$$\therefore \quad X = \frac{x}{1-x} = \frac{0.051}{(1-0.051)} = 0.05374$$

Weight % dioxane in benzene = 5.2

$$y = \text{Weight fraction of dioxane in benzene} = \frac{5.2}{100} = 0.052$$

$$\therefore \quad Y = \frac{y}{1-y} = \frac{0.052}{1-0.052} = 0.05485$$

or $$X = \frac{\text{weight \% dioxane}}{\text{weight \% water}}$$

Raffinate : Weight % dioxane = 18.9, Weight % water = 100 – 18.9

$$\therefore \quad X = \frac{18.9}{100-18.9} = 0.233$$

Extract : Weight % dioxane = 22.5, Weight % benzene = 100 – 22.5

$$\therefore \quad Y = \frac{22.5}{100-22.5} = 0.2903$$

Equilibrium data :

X	0.05374	0.233	0.337
Y	0.05485	0.2903	0.4706

Plot Y v/s X.

Dioxane removed from solution = 95 %

So final raffinate will contain 5 % of the dioxane in feed.

Dioxane in feed = 250 kg

$$\text{Dioxane in raffinate} = \frac{5}{100} \times 250 = 12.5 \text{ kg}$$

$$X_1 = \frac{\text{kg dioxane}}{\text{kg water}} \text{ in raffinate} = \frac{12.5}{750} = 0.01667$$

For single stage, we have

$$BY_S + AX_F = AX_1 + BY_1$$

$$-A(X_1 - X_F) = B(Y_1 - Y_S)$$

$$\frac{-A}{B} = \frac{Y_1 - Y_S}{X_1 - X_F}$$

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For dioxane – free benzene, $Y_s = 0$

$$-\frac{A}{B} = \frac{Y_1}{X_1 - X_F}$$

X_1 and Y_1 are equilibrium values.

So for $X_1 = 0.01667$, read the value of Y_1 from the equilibrium curve.

$\therefore$ $Y_1 = 0.017$

$$-\frac{A}{B} = \frac{Y_1}{X_1 - X_F}$$

$$-\frac{750}{B} = \frac{0.017}{0.01667 - 0.333}$$

$$B = 13956 \text{ kg}$$

Amount of solvent required = 13956 kg ... **Ans. (i)**

Countercurrent Operation :

Minimum solvent requirement : Basis : 1000 kg/h of feed solution.

$$AX_F + BY_{n+1} = AX_n + BY_1$$

$$Y_{n+1} = Y_s$$

$$\frac{A}{B} = \frac{Y_1 - Y_s}{X_F - X_n}$$

Feed solution = 1000 kg/h, it contains 250 kg/h dioxane and 750 kg/h water

Final raffinate concentration = 5 % of the dioxane in feed

Dioxane in final raffinate $= \frac{5}{100} \times 250 = 12.5$ kg/h

Water in final raffinate = 750 kg/h

Final raffinate concentration $= X_n = \frac{12.5}{750} = 0.01667$

$Y_s = 0$ for fresh dioxane – free benzene

$X_F = \frac{250}{750} = 0.333$

The above operating line passes through the point Q (X_n, Y_s), i.e., Q (0.01667, 0) and the point P (X_F, Y_1), i.e., P (0.333, Y_1).

One end of the operating line, i.e., the point Q (0.01667, 0) is fixed. For the minimum solvent requirement, the operating line touches the equilibrium curve. The other end of the operating line should lie on a vertical through $X_F = 0.333$ as the point P is having coordinates : (0.333, Y_1). Draw a vertical through $X_F = 0.333$.

Now draw the operating line through the point Q (0.01667, 0) which will be a tangent to the equilibrium curve (so it touches the equilibrium curve).

This line will cut a vertical through $X_F = 0.333$ at point P. Read the y-coordinate of point P, which is Y_1.

Y_1 from graph = 0.38 – for minimum solvent

$$\frac{A}{B_{min}} = \frac{Y_{1\,(min)} - Y_s}{X_F - X_n}$$

$$\frac{A}{B_{min}} = \frac{0.38 - 0}{0.333 - 0.01667}$$

$$A = 750 \text{ kg/h}$$

$$B_{min} = \frac{750\,(0.333 - 0.01667)}{0.38} = 624.3 \text{ kg/h}$$

Minimum solvent requirement for 1000 kg/h solution = 624.3 kg/h ... **Ans. (ii)**

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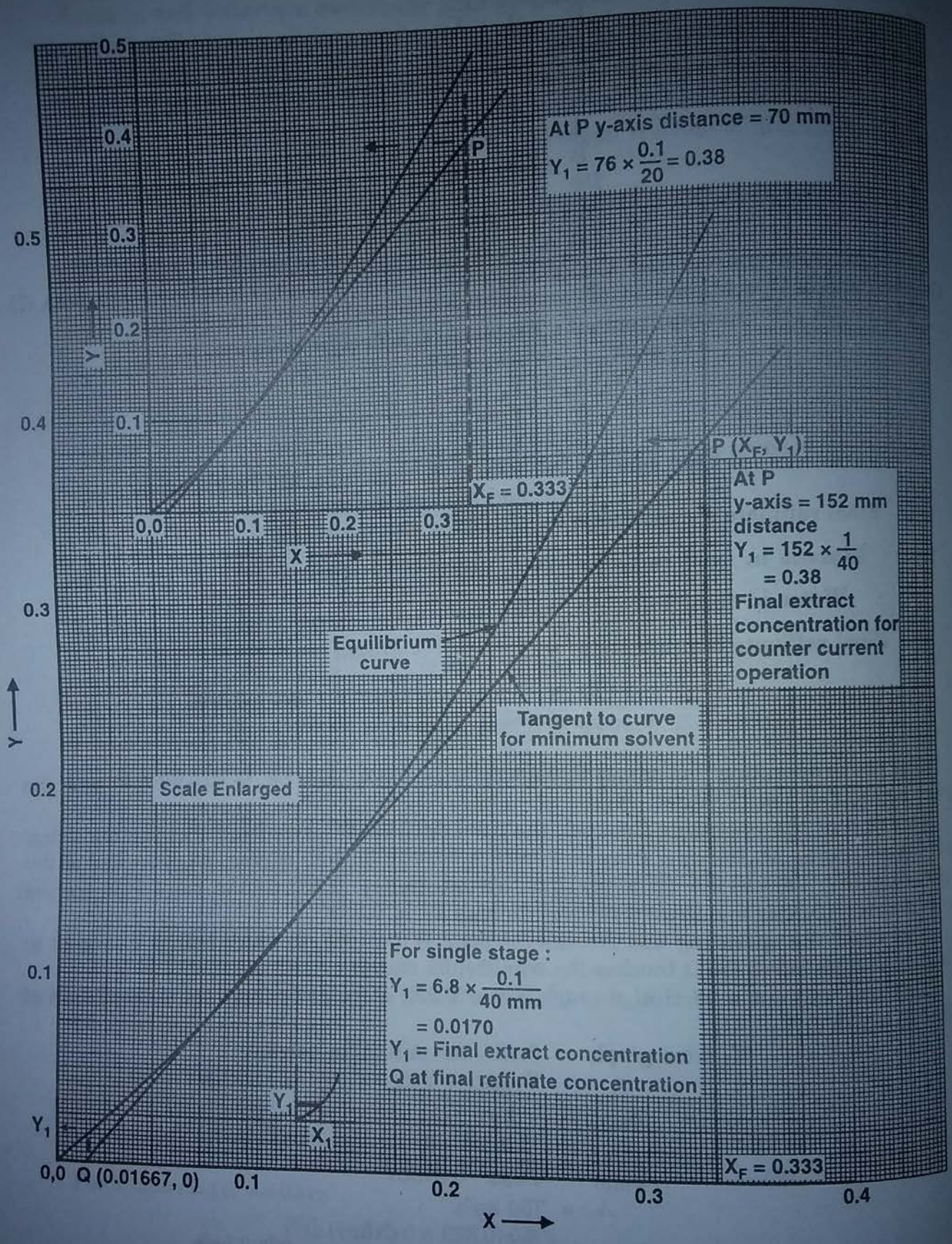

Fig. E 2.16 : Determination of minimum solvent requirement

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Ex. 2.17 : Picric acid is to be extracted from a dilute solution containing 0.1 mol picric acid per litre of solution using benzene as a solvent. 80 % of the original picric acid is to be recovered.

The equilibrium data for benzene-picric acid-water system at 298 K are as follows :

$C_S \times 10^2$	0.0932	0.225	1	2	5	10	18
$\frac{C_S}{C_A}$	2.23	1.45	1.705	0.505	0.32	0.24	0.187

where C_S and C_A are the equilibrium concentrations of picric acid in mol/l in benzene and aqueous phase respectively.

Determine the quantity of benzene required per 1 litre of aqueous solution for :

(i) single stage operation

(ii) three stage crosscurrent operation using equal quantities of fresh solvent in each stage.

Solution : Basis : 1 litre of dilute aqueous solution of picric acid.

Tabulate the data as C_S and C_A i.e. picric acid concentration in extract and raffinate respectively.

For example :

$$C_S \times 10^2 = 0.0932 \quad \therefore \quad C_S = 0.093 \times 10^{-2} \text{ mol/}l$$

$$\frac{C_S}{C_A} = 2.23 \quad \therefore \quad C_A = \frac{0.093 \times 10^{-2}}{2.23} = 4.17 \times 10^{-4} \text{ mol/}l$$

C_S, mol/l	0.093×10^{-2}	2.25×10^{-3}	0.01	0.02	0.05	0.10	0.18
C_A, mol/l	4.17×10^{-4}	1.55×10^{-3}	5.86×10^{-3}	0.0396	0.156	0.417	0.962

Plot C_S v/s C_A.

Given : 80 % of the picric acid is recovered in extract.

Picric acid in raffinate = 100 – 80 = 20 % of its original value.

Concentration of picric acid in feed solution = C_F = 0.10 mol/l

Amount of picric acid in feed solution = 0.1×1 = 0.1 mol

Picric acid in extract = 0.80×0.1 = 0.08 mol

Picric acid in raffinate = 0.20×0.1 = 0.02 mol

Concentration of picric acid in raffinate = $C_{A1} = \frac{0.02}{1} = 0.02$ mol/l

Concentration of picric acid in extract = C_{S1}, mol/l

For single stage, we have

$$BC_{S0} + AC_F = AC_{A1} + BC_{S1}$$

$$-\frac{A}{B} = \frac{C_{S1} - C_{S0}}{C_{A1} - C_F}$$

For fresh solvent used, $C_{S0} = 0$ (concentration of picric acid in incoming solvent benzene).

This operating line passes through the points $(C_F, 0)$ and (C_{A1}, C_{S1}).

As $C_{S0} = 0$, we get

$$-\frac{A}{B} = \frac{C_{S1}}{C_{A1} - C_F}$$

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C_{S1} is the equilibrium value of picric acid concentration in extract, C_{S1} and C_{A1} are the equilibrium values of picric acid concentration as the effluent streams leaving theoretical stage are in equilibrium. So read the value of C_{S1} from the plot corresponding to C_{A1} = 0.02 mol/*l*

From the plot : C_{S1} = 0.0155 mol/*l*

Amount of solvent (benzene) used = B litres

Picric acid in extract = 0.08 mol

$$\text{Concentration of picric acid in extract} = C_{S1} = \frac{\text{Moles of picric acid}}{\text{Volume of solvent}}$$

$$\text{Volume of solvent} = B = \frac{\text{Moles of picric acid}}{C_{S1}}$$

$$= \frac{0.08}{0.0155} = 5.16\,l$$

Benzene required per 1 *l* of solution = 5.16 *l* ... Ans. (i)

Crosscurrent operation :

Number of stages = 3, Feed solution = 1 *l*

Concentration of picric acid in feed solution = C_F = 0.1 mol/*l*

80 % of the picric acid is removed.

Picric acid in final raffinate = 20 % of its original value

Picric acid in final raffinate = 0.2×0.1 = 0.02 mol

$$\text{Concentration of picric acid in final raffinate} = \frac{0.02}{1} = 0.02 \text{ mol/}l$$

$\therefore$ C_{A3} = 0.02 mol/*l*

and C_{S3} = 0.0155 mol/*l* (equilibrium value)

For stage - 1 :

$$-\frac{A}{B} = \frac{C_{S1} - C_{S0}}{C_{A1} - C_F}$$

$$\text{Slope of operating line} = -\frac{A}{B}$$

For stage - 3 :

$$-\frac{A}{B} = \frac{C_{S3} - C_{S0}}{C_{A3} - C_{A2}}$$

As the equal amount of solvent is used in each stage, the operating lines will have the same slope (– A/B). We have to construct/draw three operating lines parallel to each other (as the slope is same) starting from point F (C_F, C_{S0}) and ending/covering exactly C_{A3}, the final raffinate concentration.

The operating line for the first stage passes through the point F (C_F, C_{S0}) and the operating line of the third stage passes through the point Q (C_{A3}, C_{S3}).

So locate point F (C_F, C_{S0}), i.e. F (0.10, 0) that represents the feed solution and draw the operating line through it which will cut equilibrium curve at P (C_{A1}, C_{S1}) and through C_{A1} on x-axis, draw the operating line parallel to the first one and so on and construct exactly three stages. For this, we have to adopt a trial and error procedure. Once we construct exactly three stages, measure the slope of any one of these operating lines.

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From the graph :

Slope of operating line FP :

$$\text{Slope} = -\frac{(0.0255 - 0)}{(0.1 - 0.067)} = -0.773$$

OR :

Measured angle α of the line PF with x-axis $= -57°$

We have :

$$\text{Slope} = -0.773$$

Considering scale, $\tan\alpha = -0.773 \times \frac{20/0.01}{10/0.01} = -1.546$

$$\alpha = \tan^{-1}(-1.546) = -57.1°$$

So the slope obtained is correct.

$$\text{Slope} = -\frac{A}{B} = -0.773$$

$$\frac{A}{B} = 0.773$$

$$B = \frac{A}{0.773}, \quad A = 1\,l$$

$$= \frac{1}{0.773} = 1.3\,l \text{ for one stage.}$$

Benzene required per stage $= 1.3\,l$

Benzene required for three stages $= 3 \times 1.3 = 3.9\,l$

Total benzene requirement for three – stage countercurrent operation $= 3.9\,l$ ... **Ans.(ii)**

Keep in mind that : Small error in plotting will create a large error in results. So we must accept the variation of ± 5 % in the results.

Extra : Check for plotting :

Feed solution $= 1\,l$

Concentration of picric acid in solution $= 0.10$ mol/l

Picric acid in feed solution $= 0.1 \times 1 = 0.1$ mol

Picric acid recovered $= 80$ % of that in feed

(1) Picric acid in extract $= 0.80 \times 0.1 = 0.08$ mol

Calculation of picric acid in extract graphically :

Read C_{S1}, C_{S2} and C_{S3} values indicating the concentrations of picric acid in extract from the first, second and third stage corresponding to the point at P, L and M on y-axis.

C_{S1} at P $= 0.0255$ mol/l

C_{S2} at L $= 0.02$ mol/l

C_{S3} at M $= 0.0155$ mol/l

Solvent used for each stage $= B = 1.3\,l$

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(2) Picric acid in composited extract (from graph)

$$= B [C_{S1} + C_{S2} + C_{S3}]$$
$$= 1.3 [0.0255 + 0.02 + 0.0155]$$
$$= 0.0793 \text{ mol}$$
$$(1) \approx (2)$$
$$\text{Error} = \frac{0.08 - 0.0793}{0.08} \times 100 = 0.875\ \%$$

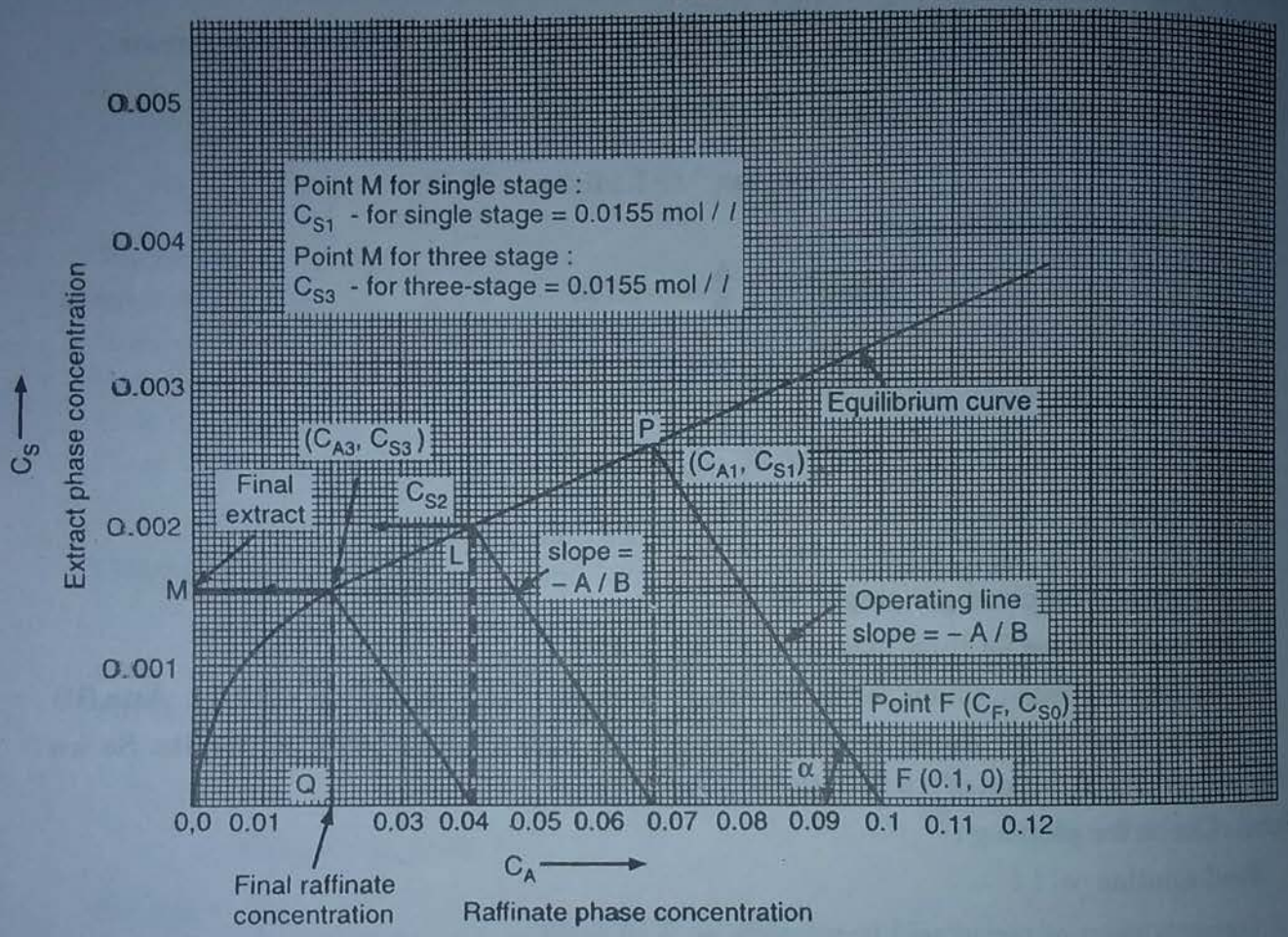

Fig. E 2.17 : Three-stage crosscurrent operation

Ex. 2.18 : Water-dioxane form a minimum boiling azeotrope at atmospheric pressure and thus cannot be separated by ordinary distillation methods. The benzene as it does not form an azeotrope with dioxane it can be used as a solvent for extraction of dioxane.

At 298 K (25°C), the equilibrium distribution of dioxane between water and benzene is as follows :

Weight % dioxane in water	5.1	18.9	25.2
Weight % dioxane in benzene	5.2	22.5	32

At these concentrations, benzene and water are substantially insoluble.

1000 kg of a 25 % dioxane and 75 % water solution is to be extracted with benzene to remove 95 % of the dioxane. The benzene used is dioxane free.

If the extractions were done with equal amounts of solvent in five crosscurrent stages, determine the quantity of total solvent required.

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Solution : Basis : 1000 kg of dioxane-water solution.

Benzene and water are substantially insoluble, so all the benzene fed will be in extract and all the water from feed solution will be in the raffinate.

So for insoluble liquids, convert the data into X and Y.

X = concentration of dioxane in raffinate in kg dioxane/kg water

Y = concentration of dioxane in extract in kg dioxane/kg benzene.

$$X = \frac{x}{1-x}, \qquad x = \text{weight fraction of dioxane in water phase}$$

(i) Weight % dioxane in water = 5.1 %

$$x = \frac{5.1}{100} = 0.051$$

$$X = \frac{0.051}{(1-0.051)} = 0.05374$$

$$Y = \frac{y}{1-y}, \quad y = \text{weight fraction of dioxane in benzene}$$

(ii) Weight % dioxane in benzene = 5.2

$$y = \frac{5.2}{100} = 0.052$$

$$Y = \frac{0.052}{1-0.052} = 0.05485$$

Similarly, convert the remaining data and tabulate.

X = kg dioxane/kg water	0.05374	0.233	0.337
Y = kg dioxane/kg benzene	0.05485	0.2903	0.4706

Amount of dioxane in feed solution = $0.25 \times 1000 = 250$ kg

Amount of water in feed solution = $1000 - 250 = 750$ kg

Dioxane to be removed = 95 % of its amount in feed solution

Dioxane remained in solution (in final raffinate) = 5 % of its amount in feed solution

Dioxane recovered/extracted = $0.95 \times 250 = 237.5$ kg

Dioxane in final raffinate = $0.05 \times 250 = 12.5$ kg

kg dioxane/kg water in feed solution = $X_F = \frac{250}{750} = 0.333$

kg dioxane/kg water in final raffinate = $X_5 = \frac{12.5}{750} = 0.01667$

There are five crosscurrent stages.

For stage - 1 :

$$AX_F + BY_S = AX_1 + BY_1$$

$$-\frac{A}{B} = \frac{Y_1 - Y_S}{X_1 - X_F} = \frac{Y_1 - 0}{X_1 - X_F}$$

$Y_S = 0$ as the benzene used is dioxane free.

For stage - 5 :

$$AX_4 + BY_S = AX_5 + BY_5, \quad Y_S = 0 \text{ (pure B)}$$

$$-\frac{A}{B} = \frac{Y_5 - Y_S}{X_5 - X_4} = \frac{Y_5 - 0}{X_5 - X_4}$$

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The operating line of first stage passes through the points F (X_F, Y_S) and P (X_1, Y_1) and has a slope of – A/B.

The operating line of fifth stage passes through the point (X_5, Y_5) which is a point on the equilibrium curve.

Locate point F (X_F, Y_S) i.e. F (0.333, 0).

Locate point Q at X_5 = 0.01667 on x-axis (final raffinate concentration).

Equal amounts of solvent are used in five crosscurrent stages.

As the amount of solvent used for each of the five crosscurrent stages is the same, the slope of all the five operating lines is same and is equal to – A/B (B is the solvent used in each stage).

For determination of the solvent requirement, we must know the slope of the operating line. For that we have to construct exactly five stages between X_F = 0.333 and X_5 = 0.01667. For constructing exactly five stages, we have to adopt a trial and error procedure.

Construction of stages should be started from the point F (0.333, 0) on the x-axis. Through the point F, draw randomly a line such as FP, this line cuts the equilibrium at point P (X_1, Y_1). Through P, drop a vertical on the x-axis to cut it at point L. Through L, draw a line (for second stage) parallel to the line FP and so on. All the operating lines should be parallel to each other. If we cannot exactly construct five stages between X_F and X_5, draw another line through point F and repeat the procedure till we can construct exactly five stages (starting from the point F, X_F = 0.333 and ending at the point Q, i.e., at X_5 = 0.01667).

With a third trial, we have constructed exactly five stages.

In such type problems, the accuracy in plotting and construction work is required.

From Fig. E 2.18 :

$$\text{Slope of line FP} = -\left(\frac{0.2-0}{0.333-0.17125}\right)$$

$$= -1.2365 \approx -1.24$$

$$\text{Slope} = -\frac{A}{B} = -1.24$$

$$B = \frac{A}{1.24} \qquad (A = 750 \text{ kg})$$

$$B = \frac{750}{1.24} = 605 \text{ kg}$$

Amount of solvent (benzene) required for each stage per 1000 kg of feed solution

= 605 kg

Total amount of solvent (benzene) required for five crosscurrent stages

= 5 × 605 = 3025 kg ... Ans.

Check for slope :

Measured angle of line FP = – 50.5°

Slope = – 1.24 (calculated above)

$\tan \alpha = -1.24$

$\therefore \quad \alpha = \tan^{-1}(-1.24) = -51.11°$

$$\text{Error} = \frac{-51.11 + 50.5}{-51.11} \times 100 = 1.2\ \% \text{ (acceptable)}$$

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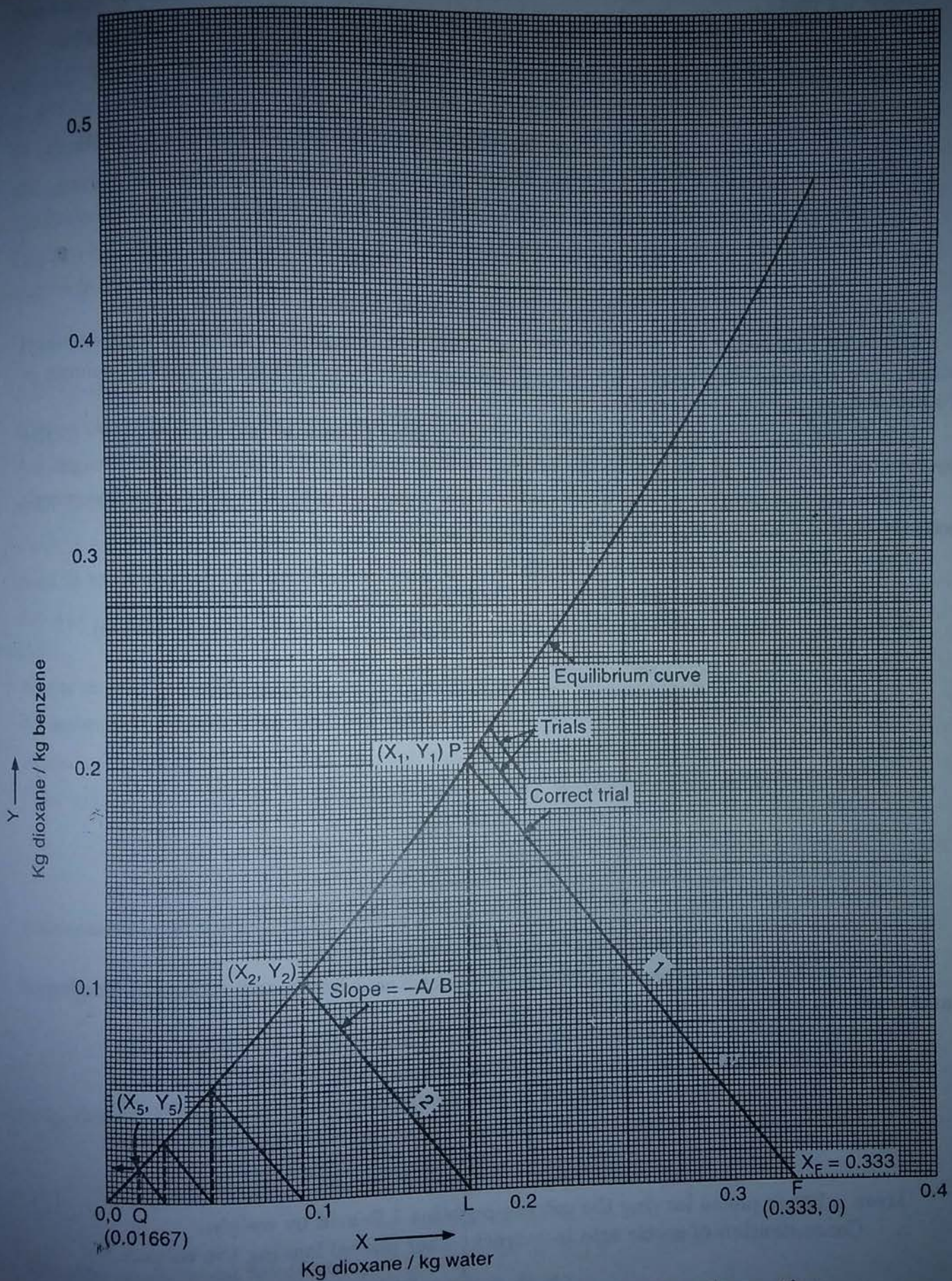

Fig. E 2.18 : Solvent requirement for five-stage crosscurrent operation

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Check for plot :

From graph : $Y_1 = 0.20$, $Y_2 = 0.09625$, $Y_3 = 0.05125$, $Y_4 = 0.03$, $Y_5 = 0.016875$

Dioxane (from plot) recovered in composited extract

$$= B(Y_1 + Y_2 + Y_3 + Y_4 + Y_5)$$
$$= 605\,[0.20 + 0.09625 + 0.03 + 0.016875 + 0.05125]$$
$$= 238.6 \text{ kg}$$

Dioxane recovered in extract = 237.5 kg

$$\% \text{ error} = \frac{238.6 - 237.5}{237.5} \times 100$$
$$= 0.463\ \% \text{ (acceptable)}$$

Ex. 2.19 : In order to extract acetic acid from dilute aqueous solution with isopropyl ether, two immiscible phases are passed countercurrently through a packed column of 75 mm diameter and 3 m length.

It is found that if 1800 kg/(m^2·h) of pure ether is used to extract 900 kg/(m^2· h) of 4 % acid by weight, then ether phase leaves the column with a concentration of 1 % acid by weight.

Calculate the number of overall transfer units based on raffinate phase and the overall extraction coefficient based on raffinate phase.

The equilibrium relationship is given by

weight % acid in ether phase = 0.30 (weight % acid in water phase)

Solution : Basis : 900 kg/(m^2·h) of aqueous solution of acetic acid to the column.

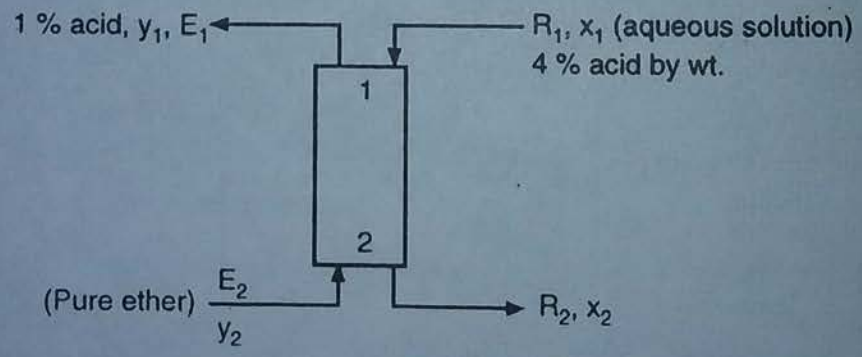

Fig. E 2.19 : Packed column for extraction

Here x and y refer to the concentrations of acetic acid expressed as weight fractions of acetic acid in raffinate and extract respectively.

Flow rate of raffinate (aqueous solution) to the column = R_1 = 900 kg/(m^2·h)

Flow rate of extract phase (pure ether) to the column = E_2 = 1800 kg/(m^2·h)

Given : Aqueous solution contains 4 % acid by weight.

∴ Concentration of acetic acid in raffinate (aqueous solution) to the column

$$= x_1 = \frac{4}{100} = 0.04$$

Given : Ether phase leaving the column contains 1 % acid by weight.

∴ Concentration of acetic acid in extract (ether phase) leaving the column

$$= y_1 = \frac{1}{100} = 0.01$$

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Pure ether is used as a solvent.

∴ Concentration of acetic acid in extract (solvent ether) entering the column = $y_2 = 0$

Given : Water and ether are immiscible, so all the ether will be in extract and all the water will be in raffinate.

So ether flow in extract will be equal to 1800 kg/(m²·h).

The extract leaving the column contains 1 % acetic acid, i.e., the extract leaving the column contains (100 – 1) = 99 % ether by weight.

i.e., 100 kg extract contains 99 kg of ether and 1 kg acid i.e. for 99 kg ether in extract, the extract is 100 kg.

∴ Flow rate of the extract leaving the column = $E_2 = \frac{100}{99} \times 1800 = 1818.2$ kg/(m²·h)

Acid in extract leaving the column = 0.01 (1818.2) = 18.18 kg/(m²·h)

Given : Raffinate (aqueous solution) entering the column contains 4 % acetic acid.

Acid in raffinate entering the column = 0.04 (900) = 36 kg/(m²·h)

Acid balance :

Acid in aqueous solution (R_1) = Acid in extract (E_1) + Acid in raffinate (R_2)

Acid in raffinate (R_2) leaving the column = 36 – 18.18 = 17.82 kg/(m²·h)

Aqueous solution/raffinate entering the column contains 100 – 4 = 96 % water by weight.

Water in raffinate (R_1) entering the column = 0.96 (900) = 864 kg/(m²·h)

As water and ether are immiscible, all the water in R_1 has to be there in R_2.

Water in raffinate (R_2) leaving the column = 864 kg/(m²·h)

Flow rate of the raffinate leaving the column = R_2 = (water + acid) in R_2

$= 864 + 17.82$

$= 881.82$ kg/(m²·h)

Concentration of acetic acid in raffinate (R_2) = $x_2 = \frac{17.82}{881.82} = 0.0202$

Average flow rate of the raffinate = $R = \frac{R_1 + R_2}{2}$

$= \frac{900 + 881.82}{2} = 890.91$ kg/(m²·h)

Average flow rate of the extract = $E = \frac{E_1 + E_2}{2}$

$= \frac{1800 + 1818.2}{2}$

$= 1809.1$ kg/(m²·h)

Here the equilibrium relationship is given as $y^* = 0.3\,x$

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For dilute solutions and for equilibrium relationship of the type $y^* = mx$ (equilibrium curve is a straight line passing through the origin), the number of transfer units based on the raffinate phase are given by the expression :

$$N_{tOR} = \frac{\ln\left[\frac{x_1 - y_2/m}{x_2 - y_2/m}(1 - R/mE) + R/mE\right]}{1 - R/mE}$$

We have : $R = 890.91$ kg/(m²·h), $E = 1809.1$ kg/(m²·h),

$m = 0.30$, $x_1 = 0.04$, $x_2 = 0.0202$, $y_2 = 0$.

$$N_{tOR} = \frac{\ln\left[\frac{0.04 - 0/0.3}{0.0202 - 0/0.3}(1 - 890.91/0.3 \times 1809.1) + 890.91/0.3 \times 1809.1\right]}{1 - 890.91/0.3 \times 1809.1}$$

$$= 1.5448 \approx 1.55$$

Number of transfer units based on the raffinate phase = 1.55 ... Ans. (i)

For dilute solutions the mass transfer rate in terms of mass transfer coefficient is given by :

$$R(x_1 - x_2) = K_R a Z (x - x^*)_M, \text{ based on raffinate phase}$$

where $$(x - x^*)_M = \frac{(x_1 - x_1^*) - (x_2 - x_2^*)}{\ln\left[\frac{(x_1 - x_1^*)}{(x_2 - x_2^*)}\right]}$$

Evaluate $(x - x^*)_M$.

$$x_1 = 0.04, \; x_2 = 0.0202$$

Given : The equilibrium relationship is $y^* = 0.3x$, so also $y = 0.3x^*$

$$0.3 = y^*/x = y/x^*$$

$$x^* = y/0.3$$

$$x_1^* = y_1/0.3$$

For $y_1 = 0.01$, $x_1^* = 0.01/0.3 = 0.03$

For $y_2 = 0$, $x_2^* = 0/0.3 = 0$

$$(x - x^*)_M = \frac{(0.04 - 0.033) - (0.0202 - 0)}{\ln\left[\frac{0.04 - 0.033}{0.0202 - 0}\right]}$$

$$= 0.01246$$

Z = height of column = 3 m

R = Average raffinate flow rate based on unit column cross-section = 890.91 kg/(m²·h)

$$K_R a = \frac{R(x_1 - x_2)}{Z(x - x^*)_M}$$

$$= \frac{890.91(0.04 - 0.0202)}{3 \times 0.01246}$$

$$= 471.91 \text{ kg/(m}^3\text{·h) (kg/kg)}$$ ... Ans. (ii)

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Ex. 2.20 : A continuous contact extraction column is used to extract a solute from a aqueous stream (F) using organic solvent (B). The distribution coefficient $(y^*/x) = 1.0$, where x and y are the mass fractions of the solute in the raffinate and extract phases respectively. The height of the transfer unit based on extract phase is 1 m. Assuming water and solvent are immiscible, find the height of the column required.

Data : (i) Flow rate of aqueous solution (F) = 100 kg/h

(ii) Flow rate of solvent (B) = 100 kg/h. The solvent used is pure.

(iii) Weight fraction of solute in aqueous solution = 0.10 and that in raffinate leaving the column is 0.01.

Solution : Basis : 100 kg/h of aqueous stream (solution) to the column.

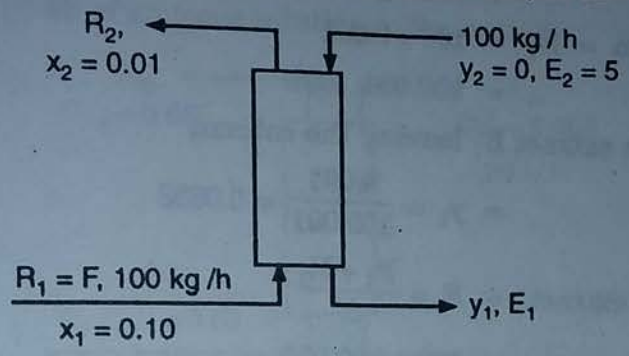

Fig. E 2.20 : Packed column for extraction

Here, the concentrations of solute in the raffinate and extract are expressed in terms of weight fractions x and y respectively.

Flow rate of the aqueous solution/raffinate entering the column = R_1 = 100 kg/h

Flow rate of the extract/solvent entering the column = E_2 = 100 kg/h

Concentration of the solute in the raffinate (aqueous solution) entering the column

$$= x_1 = 0.10$$

Concentration of the solute in the raffinate (R_2) leaving the column

$$= x_2 = 0.01$$

Pure solvent is used.

∴ Concentration of the solute in the extract (solvent), E_2, entering the column

$$= y_2 = 0$$

Solute in the raffinate/aqueous solution entering the column

$$= 0.10\,(100) = 10 \text{ kg/h}$$

Water in the raffinate/aqueous solution entering the column

$$= 100 - 10 = 90 \text{ kg/h}$$

Assuming that the solvent and water are immiscible, all the water in the aqueous solution (R_1) entering the column will be there in the raffinate (R_2) leaving the column.

Water in raffinate leaving the column (R_2) = 90 kg/h

Concentration of solute in raffinate (R_2) = x_1 = 0.01

∴ Concentration of water in the raffinate (R_2) = $1 - x_1 = 1 - 0.01 = 0.99$

(i.e., the 100 kg raffinate contains 99 kg water and 1 kg solute)

∴ Flow rate of the raffinate leaving the column = $R_2 = \dfrac{90}{0.99} = 90.91$ kg/h

Solute in the raffinate leaving the column = $0.01 \times 90.91 = 0.9091$ kg/h

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Solute balance :

Solute extracted by the solvent = Solute removed from aqueous solution

$= 10 - 0.9091$

$= 9.0909 \approx 9.091$ kg/h

Solute in extract (E_1) leaving the column = 9.091 kg/h.

Solvent being immiscible with water, all of it in input will appear in E_1.

Solvent in extract E_1 leaving the column = Solvent entering the column (E_2)

$= 100$ kg/h

Extract leaving the column = $E_1 = 100 + 9.091$

$= 109.091$ kg/h

Concentration of solute in extract E_1 leaving the column

$$= y_1 = \frac{9.091}{109.091} = 0.0833$$

$$\text{Average flow rate of extract} = E = \frac{E_1 + E_2}{2}$$

$$= \frac{100 + 109.091}{2} = 104.55 \text{ kg/h}$$

$$\text{Average flow rate of raffinate} = R = \frac{R_1 + R_2}{2}$$

$$= \frac{100 + 90.91}{2} = 95.455 \text{ kg/h}$$

The equilibrium relationship given is

$y^*/x = 1$, i.e., $y^* = x$, $\therefore m = 1.0$

For dilute solutions and for the equilibrium relationship of the type $y^* = mx$, the number of overall transfer units based on extract phase are given by the expression :

$$N_{tOE} = \frac{\ln\left[\frac{y_2 - mx_1}{y_1 - mx_1}(1 - mE/R) + mE/R\right]}{1 - mE/R}$$

Here $m = 1.0$, $y_1 = 0.0833$, $y_2 = 0$, $x_1 = 0.10$, $E = 104.55$ kg/h, $R = 95.455$ kg/h

$$N_{tOE} = \frac{\ln\left[\frac{0 - 1 \times 0.10}{0.0833 - 1 \times 0.1}\left(1 - \frac{1 \times 104.55}{95.455}\right) + \frac{1 \times 104.55}{95.455}\right]}{1 - \frac{1 \times 104.55}{95.455}}$$

$= 6.77$

Given : $H_{tOE} = 1$ m (height of the overall transfer unit based on the extract phase)

We have $Z = N_{tOE} \times H_{tOE}$

$Z = 6.77 \times 1 = 6.77$ m

$\therefore$ Height of the column (tower) required = 6.77 m $\approx$ 6.8 m ... Ans.

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Ex. 2.21 : An aqueous solution containing 0.690 kmol/m³ of solute is contacted with an organic solvent containing 0.004 kmol/m³ of solute. The exit raffinate stream from the column has a solute concentration of 0.685 kmol/m³. The organic phase flow rate is 20 *l*/h and that of aqueous solution is 30 *l*/h. The column height is 1.4 m and its cross-sectional area is 0.0045 m².

Determine : (i) the volumetric mass transfer coefficient and (ii) height of transfer unit, based on extract phase.

The equilibrium relationship is given as

$$C_{organic} = 0.025\, C_{aqueous}$$

Solution : Basis : 30 *l*/h of aqueous solution to the column.

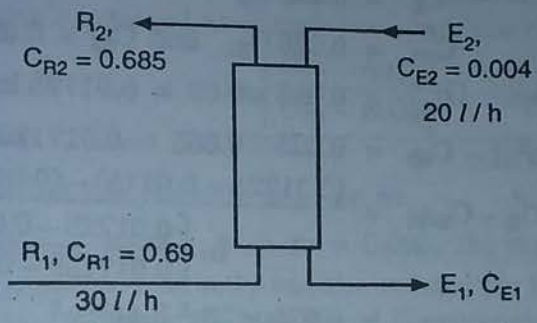

Fig. E 2.21 : Packed column for extraction

Assume that the water and organic solvent are immiscible.

Raffinate : aqueous phase, Extract = organic solvent phase

Volumetric flow rate of raffinate = $R_1 = R_2 = R = 30\ l/h = 0.03\ m^3/h$

Volumetric flow rate of extract = $E_2 = E_1 = E = 20\ l/h = 0.02\ m^3/h$

Here the concentrations of the solute in the raffinate and extract are expressed as C moles/volume, C_{R1} and C_{R2} in raffinate and C_{E1} and C_{E2} in extract.

Concentration of the solute in aqueous solution entering the column (R_1).

$$= C_{R1} = 0.690\ kmol/m^3$$

Concentration of the solute in raffinate leaving the column (R_2)

$$= C_{R2} = 0.685\ kmol/m^3.$$

Concentration of the solute in extract/Solvent entering the column (E_2)

$$= C_{E2}$$

$$C_{E2} = 0.004\ kmol/m^3$$

Concentration of solute in extract leaving the column (E_1) = C_{E1} kmol/m³

Material balance of solute :

Solute removed from raffinate = Solute added in extract

$$R\,(C_{R1} - C_{R2}) = E\,(C_{E1} - C_{E2})$$

$$0.03\,(0.690 - 0.685) = 0.02\,(C_{E1} - 0.004)$$

$$C_{E1} = 0.0115\ kmol/m^3$$

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For dilute solutions, the rate of mass transfer in terms of volumetric mass transfer coefficient is given by :

$$E\,(C_{E1} - C_{E2}) = K_E\,a\,Z\,(C_E^* - C_E)_M$$

where
$$(C_E^* - C_E)_M = \frac{(C_{E1}^* - C_{E1}) - (C_{E2}^* - C_{E2})}{\ln\left[\frac{C_{E1}^* - C_{E1}}{C_{E2}^* - C_{E2}}\right]}$$

$C_{E1} = 0.0115$ kmol/m^3, $C_{E2} = 0.004$ kmol/m^3

The equilibrium relationship is given by

$$C_{org}^* = 0.025\,C_{aqueous}$$

$$C_E^* = 0.025\,C_R$$

$$C_{E1}^* = 0.025\,C_{R1} \text{ and } C_{E2}^* = 0.025\,C_{R2}$$

For $C_{R1} = 0.69$ kmol/m^3, $C_{E1}^* = 0.025 \times 0.69 = 0.01725$ kmol/m^3

For $C_{R2} = 0.685$ kmol/m^3, $C_{E2}^* = 0.025 \times 0.685 = 0.017125$ kmol/m^3

$$(C_E^* - C_E)_M = \frac{(0.01725 - 0.0115) - (0.017125 - 0.004)}{\ln\left[\frac{0.01725 - 0.0115}{0.017125 - 0.004}\right]}$$

$$= 8.936 \times 10^{-3} \text{ kmol/m}^3$$

We have :

$$E\,(C_{E1} - C_{E2}) = K_E\,a\,Z\,(C_E^* - C_E)_M$$

In above equation

Note that : E is the volumetric flow rate of extract per unit cross-sectional area of column [volume/(area)·(time)] in m^3/(h·m^2)

C_E in kmol/m^3, Z in m

then units of K_E a are kmol/(m^3·h)·(kmol/m^3) driving force

We have E = 0.02 m^3/h

So express this E per unit area i.e. in m^2/h·m^2

$$E = \frac{0.02}{A} = \frac{0.02}{0.0045} = 4.444 \text{ m}^3/(\text{m}^2 \cdot \text{h})$$

$C_{E1} = 0.0115$ kmol/m^3, $C_{E2} = 0.004$ kmol/m^3

Z = 1.4 m, $(C_E^* - C_E)_M = 8.936 \times 10^{-3}$ kmol/m^3

$$K_E\,a = \frac{E\,(C_{E1} - C_{E2})}{Z\,(C_E^* - C_E)_M}$$

$$= \frac{4.444\,(0.0115 - 0.004)}{1.4 \times 8.936 \times 10^{-3}}$$

$$= 2.664 \text{ kmol/(m}^3\cdot\text{h)}\cdot\text{(kmol/m}^3\text{) driving force}$$

Volumetric mass transfer coefficient based on extract phase

$$= 2.664 \text{ kmol/(m}^3\cdot\text{h)}\cdot\text{(kmol/m}^3\text{) driving force} \quad \text{... Ans. (i)}$$

Height of transfer unit based on extract phase

$$= \frac{E}{K_E\,a} = \frac{4.444}{2.664} = 1.668 \simeq 1.67 \text{ m} \quad \text{... Ans. (ii)}$$

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OR :

We know that :

$$Z = N_{tOE} \times H_{tOE}$$

$$\therefore \quad H_{tOE} = \frac{Z}{N_{tOE}}$$

As Z is given (1.4 m), we can obtain the value of H_{tOE} knowing N_{tOE}.

For dilute solutions and for equilibrium relationship of the type $y^* = mx$, N_{tOE} is given by :

$$N_{tOE} = \frac{\ln\left[\frac{y_2 - mx_1}{y_1 - mx_1}(1 - mE/R) + mE/R\right]}{1 - mE/R}$$

In terms of concentrations expressed as C moles/volume, the above equation becomes

$$N_{tOE} = \frac{\ln\left[\frac{C_{E2} - mC_{R1}}{C_{E1} - mC_{R1}}(1 - mE/R) + mE/R\right]}{1 - mE/R}$$

In our case, the equilibrium relationship is given as

$C^*_{org.} = 0.025\, C_R$ i.e. $C^*_E = m\, C_R$ with $m = 0.025$, the above equation is applicable.

In the equation for N_{tOE}, one can directly use the volumetric flow rates of raffinate and extract phase i.e. R and E expressed as m^3/h or R and E expressed as $m^3/(m^2 \cdot h)$ i.e. on unit cross-section area as in equation there is a ratio of R/E.

We have : $C_{E2} = 0.0040$ kmol/m^3, $C_{E1} = 0.0115$ kmol/h, $m = 0.025$, $E = 0.025$ m^3/h,
$R = 0.03$ m^3/h, $C_{R1} = 0.690$ kmol/m^3

E and R are expressed as m^3/h.

$$N_{tOE} = \frac{\ln\left[\frac{0.004 - 0.025 \times 0.690}{0.0115 - 0.025 \times 0.690}\left(1 - \frac{0.025 \times 0.02}{0.03}\right) + \frac{0.025 \times 0.02}{0.03}\right]}{1 - \frac{0.025 \times 0.02}{0.03}}$$

$$= 0.83925$$

We know that :

$$Z = H_{tOE} \cdot N_{tOE}$$

$$H_{tOE} = \frac{Z}{N_{tOE}}, \quad Z = 1.4 \text{ m (given)}$$

$$= \frac{1.4}{0.83925} = 1.668 \approx 1.67 \text{ m}$$

Height of transfer unit based on extract phase = 1.67 m ... **Ans.**

We obtained the same answer irrespective of the method used. Adopt any one method.

Ex. 2.22 : An aqueous solution containing 1.5 kmol/m^3 of solute is contacted with an organic solvent containing 0.008 kmol/m^3 of solute. The aqueous phase leaving the column contains 1.3 kmol/m^3 of solute. The flow rates of aqueous phase and organic phase are 36 *l*/h and 90 *l*/h respectively. The column is 1 m tall and of 0.005 m^2 cross-sectional area. The equilibrium relationship is given as

$$C_{organic} = 0.3\, C_{aqueous}$$

Determine : (i) the overall mass transfer coefficient and
(ii) the height of transfer unit based on extract phase.

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Solution : Basis : 36 *l*/h of aqueous solution entering the column.

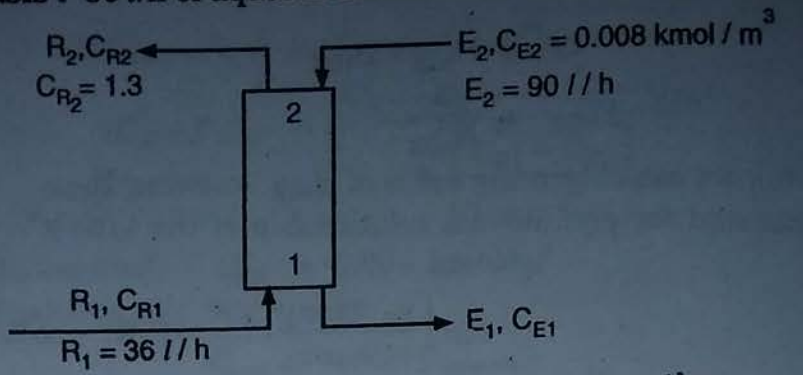

Fig. E 2.22 : Packed column for extraction

Here, the concentrations of the solute in raffinate and extract are expressed as C moles/volume, C_{R1} and C_{R2} for raffinate and C_{E1} and C_{E2} for extract.

Volumetric flow rate of raffinate = $R_1 = R_2 = R = 36\ l/h = 0.036\ m^3/h$

Volumetric flow rate of extract = $E_1 = E_2 = E = 90\ l/h = 0.09\ m^3/h$

Concentration of solute in raffinate i.e. aqueous solution entering the column (R_1)

$$= C_{R1} = 1.5\ kmol/m^3$$

Concentration of solute in raffinate leaving the column (R_2)

$$= C_{R2} = 1.3\ kmol/m^3$$

Concentration of solute in extract/solvent entering the column (E_2)

$$= C_{E2} = 0.008\ kmol/m^3$$

Concentration of the solute in extract leaving the column (E_1)

$$= C_{E1}\ kmol/m^3$$

Material balance of solute :

Solute removed from raffinate = Solute added in extract

$$R(C_{R1} - C_{R2}) = E(C_{E1} - C_{E2})$$

$$0.036(1.5 - 1.2) = 0.09(C_{E1} - 0.008)$$

$$\therefore \quad C_{E1} = 0.088\ kmol/m^3$$

For the dilute solutions, the rate of mass transfer in terms of volumetric mass transfer coefficient can be given as :

$$E(C_{E1} - C_{E2}) = K_E a Z (C_E^* - C_E)_M$$

In above equation, E is the volumetric flow rate of extract per unit cross-sectional area of column. Here E is expressed as volume/(area) (time) and has units of $m^3/(m^2 \cdot h)$.

$(C_E^* - C_E)_M$ = log mean concentration difference driving force

$$= \frac{(C_{E1}^* - C_{E1}) - (C_{E2}^* - C_{E2})}{\ln\left[\frac{C_{E1}^* - C_{E1}}{C_{E2}^* - C_{E2}}\right]}$$

$C_{E1} = 0.088\ kmol/m^3$, $C_{E2} = 0.008\ kmol/m^3$

The equilibrium relationship is

$$C_{org}^* = 0.3\ C_{aqueous}^*$$

$$C_E^* = 0.3\ C_R$$

With this obtain values of C_{E1}^* and C_{E2}^*.

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For $C_{R1} = 1.5$ kmol/m^3, $C^*_{E1} = 0.3\, C_{R1} = 0.3 \times 1.5 = 0.45$ kmol/m^3

For $C_{R2} = 1.3$ kmol/m^3, $C^*_{E2} = 0.3\, C_{R2} = 0.3 \times 1.3 = 0.39$ kmol/m^3

$$(C^*_E - C_E)_M = \frac{(0.45 - 0.088) - (0.39 - 0.008)}{\ln\left[\frac{0.45 - 0.088}{0.39 - 0.008}\right]}$$

$$= 0.372 \text{ kmol/m}^3$$

$E = 0.09$ m^3/h, $A = 0.005$ m^2

We have : $E\,(C_{E1} - C_{E2}) = K_E\, a\, Z\, (C^*_E - C_E)_M$

Here, E is volumetric flow rate of extract per unit cross-sectional area in m^3/(m^2·h)

C_{E1}, C_{E2}, $(C^*_E - C_E)_M$ - Concentrations in kmol/m^3

Z - height of column in m

K_E a - Volumetric overall mass transfer coefficient based on extract phase in kmol/(m^3·h) (kmol/m^3)

$E = 0.09$ m^3/h

Area $= 0.005$ m^2

E is expressed as per unit area :

$$E = \frac{0.09}{0.005} = 18 \text{ m}^3/(\text{m}^2 \cdot \text{h})$$

$C_{E1} = 0.088$ kmol/m^3, $C_{E2} = 0.008$ kmol/m^3

$Z = 1$ m

$$K_E\, a = \frac{E\,(C_{E1} - C_{E2})}{Z\,(C^*_E - C_E)_M}$$

$$= \frac{18\,(0.088 - 0.008)}{1 \times 0.372}$$

$= 3.871$ kmol/(m^3·h) (kmol/m^3) driving force

Volumetric overall mass transfer coefficient based on extract phase

$= K_E\, a = 3.871$ kmol/(m^3·h) (kmol/m^3) **... Ans. (i)**

Height of transfer unit based on extract phase

$$= H_{tOE} = \frac{E}{K_E\, a}$$

where E is volumetric flow rate of extract per unit area.

$$H_{tOE} = \frac{18}{3.871} = 4.649 \approx 4.65 \text{ m}$$ **... Ans. (ii)**

OR

Let us calculate H_{tOE} from :

$$Z = H_{tOE} \times N_{tOE}$$

where $Z = 1$ m, so obtain N_{tOE} to get H_{tOE}.

For dilute solutions and for equilibrium relationship of the type

$$y^* = mx, \; (m = y^*/x = x^*/y)$$

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(i.e., the equilibrium curve is a straight line through the origin), the number of transfer units based on extract phase are given by the expression :

$$N_{tOE} = \frac{\ln\left[\frac{y_2 - mx_1}{y_1 - mx_1}(1 - mE/R) + mE/R\right]}{1 - mE/R}$$

In terms of concentrations,

$$N_{tOE} = \frac{\ln\left[\frac{C_{E2} - mC_{R1}}{C_{E1} - mC_{R1}}(1 - mE/R) + mE/R\right]}{1 - mE/R}$$

where E and R are volumetric flow rates of extract per unit area in $m^3/(m^2 \cdot h)$.

$$E = \frac{0.09}{A} = \frac{0.09}{0.005} = 18\ m^3/(m^2 \cdot h)$$

$$R = \frac{0.036}{A} = \frac{0.036}{0.005} = 7.2\ m^3/(m^2 \cdot h)$$

$$m = 0.3,\quad C_{E2} = 0.008\ kmol/m^3$$

$$C_{R1} = 1.5\ kmol/m^3,\quad C_{E1} = 0.088\ kmol/m^3$$

$$N_{tOE} = \frac{\ln\left[\frac{0.008 - 0.3 \times 1.5}{0.088 - 0.3 \times 1.5}\left(1 - \frac{0.3 \times 18}{7.2}\right) + \frac{0.3 \times 18}{7.2}\right]}{1 - \frac{0.3 \times 18}{7.2}}$$

$$= 0.215$$

$$Z = H_{tOE} \times N_{tOE}$$

$$H_{tOE} = \frac{Z}{N_{tOE}} = \frac{1}{0.215} = 4.65\ m \quad \text{... Ans.}$$

∴ We obtained the same answer by the two different methods.

Ex. 2.23 : 1000 kg of a solution of acetic acid (C) and water (A) containing 30 % acid by weight is extracted 3 times with isopropyl ether (B). Each time 400 kg of pure ether is used. If the concentration of acid in A-phase is 4 times the concentration of acid in B-phase, (in wt. %), determine the final raffinate concentration. Also determine the quantity of solvent needed for single stage extraction. Assume that the ether and water are immiscible.

Solution : Basis : 1000 kg of solution containing 30 % acid by weight.

This is a three-stage crosscurrent operation.

As water and ether are to be treated as immiscible, all water will be there in raffinate and ether will be there in extract.

A - water, B - solvent ether, C - acetic acid.

For stage - 1 :

For immiscible liquids, the solute balance is :

$$AX_F + BY_S = AX_1 + BY_1$$

As the pure ether is used, $Y_S = 0$

$$\therefore \quad AX_F = AX_1 + BY_1$$

X_F = kg acid/kg water in feed solution

X_1 = kg acid/kg water in raffinate from stage - 1

Y_1 = kg acid/kg ether in extract from stage - 1

Acid in feed solution = C = 0.3×1000 = 300 kg

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Water in feed solution = A = $(1-0.3)\times 1000$ = 700 kg

Solvent used = B = 400 kg

Weight % acid in feed solution = 30

Weight fraction of acid in feed solution = $x_F = \frac{30}{100} = 0.30$

kg acid/kg water in feed solution = $X_F = \frac{x_F}{1-x_F}$

$$= \frac{0.30}{1-0.30} = 0.4286$$

The equilibrium relationship is given as :

Concentration of C in A-phase (raffinate) = 4 × concentration of C in B phase (extract)

i.e. $x = 4y$

$\therefore$ $y = 0.25x$

where x and y are the weight fractions of acetic acid in raffinate and extract phase respectively.

We have, X = kg acid/kg water

Y = kg acid/kg ether

$$X = \frac{x}{(1-x)}$$

and $$Y = \frac{y}{(1-y)}$$

$\therefore$ $$x = \frac{X}{1+X}$$

and $$y = \frac{Y}{1+Y}$$

Putting the values of x and y in the equilibrium relationship,

$$\frac{Y}{1+Y} = 0.25\,\frac{X}{1+X}$$

$$Y + XY = 0.25X + 0.25XY$$

$$Y + 0.7XY = 0.25X$$

$\therefore$ $$Y = \frac{0.25X}{(1+0.75X)} \quad \ldots (1)$$

We have : $AX_F = AX_1 + BY_1$

A = 700 kg, B = 400 kg, $X_F = 0.4286$

Express Y_1 in terms of X_1 with the help of equilibrium relationship :

$$Y_1 = \frac{0.25X_1}{(1+0.75X_1)}$$

Putting this value of Y_1 in equation (1)

$$700(0.4286) = 700X_1 + \frac{400\times 0.25X_1}{(1+0.75X_1)}$$

$$300 = 700X_1 + \frac{100X_1}{(1+0.75X_1)}$$

$$300(1+0.75X_1) = 700X_1(1+0.75X_1) + 100X_1$$

$$300 + 225X_1 = 700X_1 + 525X_1^2 + 100X_1$$

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$$525\,X_1^2 + 575\,X_1 - 300 = 0$$

$$1.75\,X_1^2 + 1.9167\,X_1 - 1 = 0$$

$$X_1 = \frac{-1.9167 \pm \sqrt{(1.9167)^2 - 4 \times 1.75 \times (-1)}}{2 \times 1.75}$$

$$X_1 = 0.3858 \text{ (Positive root)}$$

Acid content of raffinate leaving stage-1 is $X_1 = 0.3858$ kg/kg water

Raffinate from stage-1 is a feed to stage-2.

For stage-2 :

Material balance of acetic acid :

$$AX_1 + BY_s = AX_2 + AY_2$$

But $Y_s = 0$ as solvent is pure

$$\therefore \quad AX_1 = AX_2 + BY_2 \qquad \ldots(2)$$

X_1 is the acid content in kg/kg water in raffinate leaving stage-1

Y_2 is the acid content in kg/kg ether in extract leaving stage-2

Y_2 in terms of X_2 (from equilibrium relationship) is

$$Y_2 = \frac{0.25\,X_2}{(1 + 0.75\,X_2)}$$

$A = 700$ kg, $B = 400$ kg, $X_1 = 0.3858$

Putting Y_2 in terms of X_2 and the values of A, B etc. in Equation (2).

$$700\,(0.3858) = 700\,X_2 + \frac{400 \times 0.25\,X_2}{(1 + 0.75\,X_2)}$$

$$270\,(1 + 0.75\,X_2) = 700\,X_2\,(1 + 0.75\,X_2) + 100\,X_2$$

$$525\,X_2^2 + 597.5\,X_2 - 270 = 0$$

$$1.944\,X_2^2 + 2.213\,X_2 - 1 = 0$$

$$X_2 = \frac{-2.213 \pm \sqrt{(2.213)^2 - 4 \times 1.944\,(-1)}}{2 \times 1.944}$$

$$X_2 = 0.3464 \text{ (Positive root)}$$

The acid content of raffinate leaving stage-2 is $X_2 = 0.3464$ kg/kg water

Acid in raffinate leaving stage-2 in weight %

$$= \frac{0.3464}{1 + 0.3464} \times 100 = 25.73\,\%$$

Raffinate from stage-2 is the feed to the stage-3.

For stage-3 :

Material balance of acetic acid :

$$AX_2 + BY_s = AX_3 + BY_3$$

But $Y_s = 0$ as the solvent is pure

$$AX_2 = AX_3 + BY_3, \qquad \ldots(3)$$

X_3 is the acid content in kg acid/kg water in raffinate leaving stage-3 (i.e. final raffinate)

Y_3 is the acid content of extract (in kg acid/kg ether) leaving stage-3

Express Y_3 in terms of X_3.

$$Y_3 = \frac{0.25\,X_3}{(1 + 0.75\,X_3)}$$

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Putting the value of Y_3 in terms of X_3, A and B and X_2 in Equation (3)

$$700\,(0.3464) = 700\,X_3 + \frac{400 \times 0.25\,X_3}{(1 + 0.75\,X_3)}$$

$$242.48\,(1 + 0.75\,X_3) = 700\,X_3\,(1 + 0.75\,X_3) + 100\,X_3$$

$$525\,X_3^2 + 618.14\,X_3 - 242.48 = 0$$

$$2.165\,X_3^2 + 2.55\,X_3 - 1 = 0$$

$$X_3 = \frac{-2.55 \pm \sqrt{(2.55)^2 - 4\,(2.165) \times (-1)}}{2 \times 2.165}$$

Acid content of final raffinate = X_3 = 0.3104 kg/kg water

So the final raffinate contains 0.3104 kg acid with every 1 kg of water.

i.e., 1 + 0.3104 = 1.3104 kg raffinate contains 0.3104 kg acid and 1 kg water.

Concentration of acid in final raffinate expressed as weight fraction

$$= x_3 = \frac{0.3104}{1.3104} = 0.2368$$

Final raffinate concentration in terms of weight % acid

$$= 100\,x_3 = 0.2368 \times 100 = 23.68$$ ... Ans. (i)

(ii) The solvent requirement for reducing the acid content from 30 % to 23.68 % by weight using a single stage :

For a single stage, we have :

$$AX_F + BY_S = AX_1 + BY_1$$

$$AX_F = AX_1 + BY_1, \qquad Y_S = 0 \text{ as pure solvent is used}$$

Here $A = 700$ kg

$$X_F = \frac{0.3}{1 - 0.30} = 0.4286$$

$$X_1 = \frac{0.2368}{1 - 0.2368} = 0.3104 \text{ (which is } X_3 \text{ calculated above)}$$

We have $$Y = \frac{0.25\,X}{1 + 0.75\,X}$$

$$Y_1 = \frac{0.25\,X_1}{1 + 0.75\,X_1}$$

We have $X_1 = 0.3104$ (raffinate phase acid content in kg/kg water in single stage)

$\therefore$ $$Y_1 = \frac{0.25 \times 0.3104}{1 + 0.75 \times 0.3104} = 0.063$$ (extract-phase acid content in kg/kg ether in single stage)

$\therefore$ $$700 \times 0.4268 = 700 \times 0.3104 + B\,(0.063)$$

$$B = 1313 \text{ kg}$$

Solvent required for reducing the acid concentration from 30 % to 23.68 % in the single stage extraction = 1313 kg ... Ans. (ii)

Total solvent required for the three-stage crosscurrent extraction

$$= 3 \times 400 = 1200 \text{ kg}$$

So for the single stage extraction 1313 kg of the solvent is required instead of the total of 1200 kg for the three-stage extraction.

❑❑❑

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ADSORPTION AND ION EXCHANGE

When a solid surface is exposed to a fluid phase, the molecules from the bulk of the fluid phase have tendency to accumulate or concentrate at the surface of a solid. *The phenomenon of the enrichment (concentration) of chemical substances at the surface of a solid is called adsorption.*

- In adsorption, a fluid phase (a gas or a liquid phase) contacts a solid phase and mass is transferred from the gas/liquid to the solid.
- Adsorption involves the transfer of a solute from either a gas or a liquid/solution (i.e., a constituent of a fluid) to the surface of a solid. The solute is held to the surface of the solid as a result of/due to intermolecular attraction with the solid molecules. In order to complete the separation, the adsorbed solute (constituent) has to be removed from the solid. By simply changing the pressure or temperature of the operation, the solute may be removed from the solid (as the adsorption process is often reversible).
- It is a fluid-solid operation that involves the selective collection and concentration of a particular type of molecules contained in a fluid on the surface of a solid. Therefore, with this operation the components of a gaseous or liquid solution can be separated from each other.
- The substance on whose surface adsorption takes place (i.e., the enrichment of a chemical substance occurs) is called the *adsorbent* and the substance which is adsorbed is called as the *adsorbate (substrate)*. The adsorbate may be a gas or a solute in some solution while the adsorbent is a solid.
- The solid which takes up a gas or a solute from a solution is called the adsorbent whereas the gas or the solute which is held to the surface of the solid is called the adsorbate.

Applications of Adsorption :

Adsorption operation is used for :

(i) Separation of gases from gases (gaseous separations) : Removal of toxic gases and removal of odour from waste gases for air pollution control and deodourisation of air.

(ii) Separation of vapours from gases (gaseous separations) : Dehumidification of air and other gases, recovery of valuable solvent vapours from solvent-laden air and other gases (air leaving an evaporating process such as paint drying, news paper printing, etc.).

(iii) Separation of solutes from solutions (liquid separations) : Removal of moisture dissolved in gasoline, decolourisation of sugar solutions (crude sugar syrups), vegetable oils and petroleum products, removal of objectionable taste, odour and colour from municipal water supplies, removal of organic components from aqueous wastes for water pollution control.

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(iv) Removal of ions from solution : Demineralisation of water by ion exchange.
(v) Fractionation by selective adsorption of gases from gases, liquids from liquids, vapours from gases and vapours from vapours : To separate mixtures of light hydrocarbon gases (methane, ethylene, ethane, etc.), mixtures of aromatic and paraffinic hydrocarbons.
(vi) Odour removal from municipal illuminating-gas supplies.
(vii) Clarification of beverages.

Adsorbents :

Nature of Adsorbents (Characteristics of Adsorbent) :

1. Adsorbent solids are usually used in granular form with size ranging from 12 mm in diameter to 50 μm.
2. They must possess a large surface area per unit mass.
3. They should have preferential ability to take up the adsorbate (they should have sufficient specificity and adsorptive ability).
4. These are usually highly porous materials.
5. They should be free flowing for ease in handling.
6. They should offer a low pressure drop for flow of fluid when used in a fixed bed.
7. They should have an adequate strength and hardness.

The adsorbent should possess the above characteristics for adsorption to take place on its surface.

Adsorbents in general use :

(i) activated carbon, (ii) activated alumina, (iii) silica gel, (iv) molecular sieves (zeolites), (v) activated clays, (vi) bone char, (vii) Fuller's earths (natural clays).

1. Activated carbon - decolourising carbon is used for decolourising solutions of sugar, drugs, chemicals, water purification, etc.
2. Activated carbon - gas adsorbent carbon is used for recovery of solvent vapours from gas mixtures, gas masks, etc.
3. Silica gel is used for dehydration of air or other gases.
4. Molecular sieves are used for dehydration of gases and liquids, separation of gas and liquid hydrocarbon mixtures, etc.
5. Bone char is used for refining of sugar.
6. Activated clays are used for decolourising petroleum products.
7. Fuller's earths are useful for decolourising lubricating oils, kerosene, gasoline, and vegetable oils.

Adsorption takes place especially on the walls of pores or at specific sites inside the particle. When the capacity of an adsorbent is exhausted, the adsorbed material can be removed by methods such as desorption of adsorbate, displacement of adsorbate and combustion of adsorbate. When we remove the adsorbate by its desorption or displacement (from an adsorbent), then the operation is called regeneration. When the method used involves combustion of the adsorbate, the operation is called reactivation.

Types of Adsorption :

Physical Adsorption : It is a result of intermolecular forces of attraction between molecules of the adsorbent and the adsorbate (substance adsorbed). In this case, the adsorbate merely condenses in a thin on the surface of adsorbent solid. The intermolecular attractive forces that retain the adsorbent on the surface are purely physical and are called Van der Waals forces. As these forces are very weak, the adsorbate is loosely bound to the surface of the adsorbent.

Chemical Adsorption/Chemisorption : In this case, molecules of the adsorbate are retained on the surface of the solid by chemical bonding, i.e., it involves chemical

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interaction between the adsorbent solid and the adsorbed substance (fluid). Identifiable chemical compounds may not actually result but the adhesive force is much greater than that in physical adsorption. In many cases, it is irreversible. It is of particular importance in catalysis.

Physical adsorption	Chemisorption
1. Forces involved are weak.	1. Forces involved are stronger.
2. It is readily a reversible phenomenon.	2. It is frequently an irreversible phenomenon.
3. Heat of adsorption (i.e., heat evolved) is small, of the order of the heat of condensation. (4 to 40 kJ/mol)	3. Heat of adsorption is large, of the order of the heat of reaction. (40 to 400 kJ/mol)
4. Non-specific in nature, entire surface is available for adsorption.	4. Highly specific in nature and restricted to active sites on the surface.
5. Surface coverage is complete and can be extended to multilayers.	5. Surface coverage is incomplete and limited to a unimolecular layer.
6. Quantity of substance adsorbed per unit mass of adsorbent is high.	6. Quantity of substance adsorbed per unit mass of adsorbent is low.
7. Activation energy is low, nearly negligible.	7. Activation energy is high, corresponding to chemical reaction.
8. Rate of adsorption is very rapid at low temperature and decreases with increase in temperature.	8. Rate of adsorption is negligible at low temperature and increases with increase in temperature.
9. Rate of adsorption is controlled by resistance to mass transfer.	9. Rate of adsorption is controlled by resistance to surface reaction.
10. Adsorption at low partial pressure is small, and high at high partial pressures.	10. Adsorption at low partial pressure is large, slightly increases at high partial pressures.
11. Entire surface is available for physical adsorption.	11. Chemisorption is restricted to active sites on the surface.
12. Adsorption of O_2, H_2 gases on charcoal.	12. Adsorption of H_2 gas on nickel metal.

Characteristics of Adsorption by Solids :

(i) The amount of a substance adsorbed per unit mass of the adsorbent depends on the equilibrium concentration or pressure of the adsorbate and the temperature.

(ii) The larger the area of the surface, greater is the adsorption.

(iii) The amount of a substance adsorbed per unit mass of the adsorbent depends on the chemical nature of both the adsorbent and adsorbate and their physical state. In case of physical adsorption, more easily liquefiable is the gas, more readily it will be adsorbed.

(iv) Adsorption can be highly selective, e.g., activated carbon will adsorb butenes (butene-1 and 2) preferentially from a mixture containing ethylene and butenes.

(v) Adsorption is generally reversible.

(vi) Adsorption is invariably accompanied by decrease of enthalpy of the system, i.e., it is an exothermic process.

Adsorption Isotherms :

The equilibrium data for adsorption are usually presented in the form of adsorption isotherms. The adsorption isotherm is the equilibrium relationship between the concentration of the adsorbate held on the surface of the solid (adsorbent) and the

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concentration (or partial pressure) of the adsorbate in the fluid phase at a given temperature. The concentration of the adsorbate on the surface of the solid is expressed as the amount of a substance adsorbed per unit mass of the adsorbent. In case of gases, the concentration of the adsorbate in the fluid phase is expressed in terms of partial pressure or mole percent and in case of solutions, it is expressed as mass adsorbate per unit volume of solution or in mass units such as ppm [parts per million (parts)].

The Freundlich Equation : Data for physical adsorption can be expressed by an emperical equation. The Freundlich equation is an empirical relation that fits equilibrium adsorption data (equilibrium data for physical adsorption) in many cases.

The Freundlich adsorption isotherm is represented by the equation :

$$c^* = k\,(x/m)^n \quad \text{[for solutions]} \qquad \ldots (3.1)$$

$$p^* = k\,(x/m)^n \quad \text{[for gases]} \qquad \ldots (3.2)$$

where c^* is the equilibrium concentration of the adsorbate in the fluid phase and x/m is the mass of the substance adsorbed per unit mass of the adsorbent and p^* is the equilibrium partial pressure of the adsorbed gas (adsorbate) in the fluid phase (gas phase). x is the mass of the substance adsorbed and m is the mass of the original adsorbent. k and n are constants.

Equation (3.1) indicates that a plot of equilibrium concentration of the adsorbate against mass of the substance adsorbed per unit mass of the adsorbent, on logarithmic coordinates, is a straight line of slope n and intercept k.

Single Stage Adsorption :

This may be a batch or a continuous operation. The circle in Fig. 3.1 (a) represents all the arrangements constituting one stage.

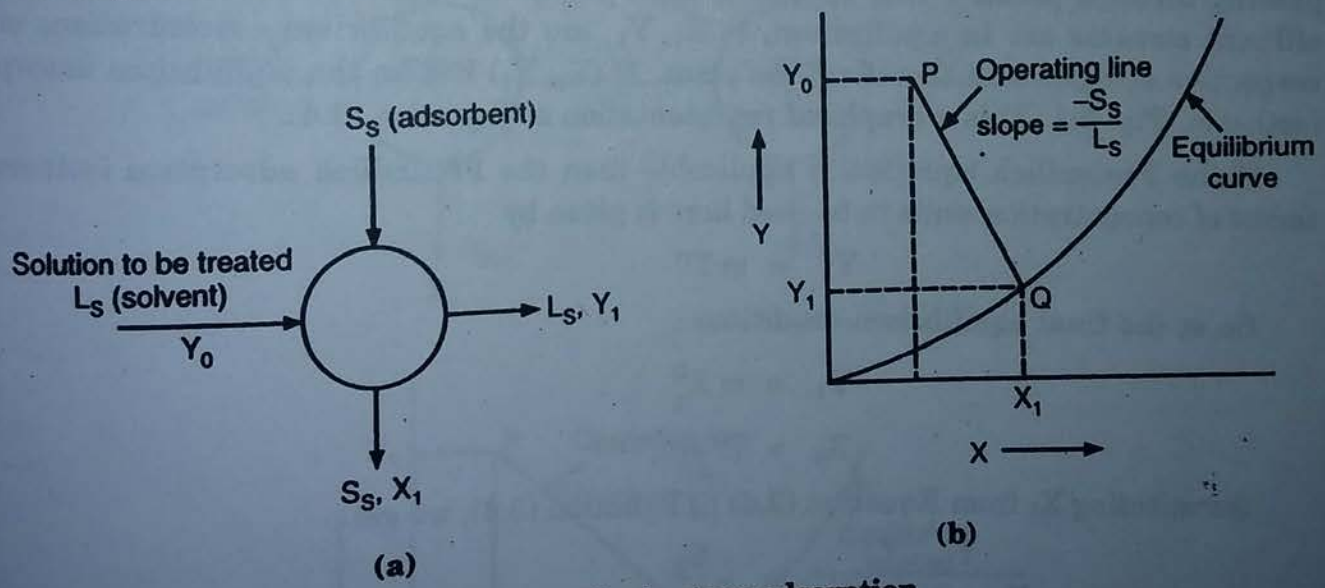

Fig. 3.1 : Single stage adsorption

Here the solution to be treated consists of an adsorbable solute (C) in a solvent (A). It is contacted with an adsorbent (B) either in a batch or continuous manner.

The amount of the adsorbent used is very small in comparison with the solution. Only the solute (C) is adsorbed and the adsorbent is insoluble in the solution.

Let L_S be the mass of the solvent (for batch operation) or mass of solvent/time (for continuous operation) in the solution to be treated. S_S be the mass or mass/time of the adsorbent added to the stage.

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Y_0 is the adsorbable solute concentration in the solution to be treated expressed as mass solute/mass solvent.

Y_1 is the adsorbable solute concentration in the solution leaving the stage expressed as mass solute/mass solvent.

X_0 is the solute concentration in the adsorbent added, in mass solute/mass adsorbent. For fresh adsorbent, $X_0 = 0$.

X_1 is the solute concentration in the adsorbent leaving the stage in mass solute/mass adsorbent.

Assume further that the amount of solution mechanically retained with the solid after phase separation (by filtration or settling) is negligible. This is quite satisfactory for most adsorption because the quantity of the adsorbent used is very small in comparison with the quantity of the solution treated.

So with this, the material balance of solute is :

Solute removed from the solution = Solute picked up by the solid

$$L_S (Y_0 - Y_1) = S_S (X_1 - X_0) \quad \ldots (3.3)$$

or

$$\frac{Y_0 - Y_1}{X_0 - X_1} = -\frac{S_S}{L_S} \quad \ldots (3.4)$$

This is the operating line equation for single stage adsorption.

On X-Y coordinates, Equation (3.4) is the equation of a straight line, of slope $-S_S/L_S$, passing through points P (X_0, Y_0) and Q (X_1, Y_1). For an equilibrium/theoretical stage, the effluent streams are in equilibrium, so X_1, Y_1 are the equilibrium concentrations of the respective streams and therefore, the point Q (X_1, Y_1) lies on the equilibrium adsorption isotherm. Fig. 3.1 (b) is a graphical representation of Equation (3.4).

If the Freundlich equation is applicable then the Freundlich adsorption isotherm in terms of concentration units to be used here is given by

$$Y^* = m X^n \quad \ldots (3.5)$$

So at the final equilibrium conditions :

$$Y_1 = m X_1^n$$

$$\therefore \quad X_1 = [Y_1/m]^{1/n} \quad \ldots (3.6)$$

Substituting X_1 from Equation (3.6) in Equation (3.4), we get

$$\frac{Y_0 - Y_1}{X_0 - (Y_1/m)^{1/n}} = -\frac{S_S}{L_S}$$

Usually, adsorbent used does not contain adsorbable solute initially, $X_0 = 0$ and with this, we get

$$\frac{Y_0 - Y_1}{[Y_1/m]^{1/n}} = \frac{S_S}{L_S} \quad \ldots (3.7)$$

With the help of Equation (3.7), we can calculate the adsorbent to solution ratio for a given change in solution concentration – Y_0 to Y_1.

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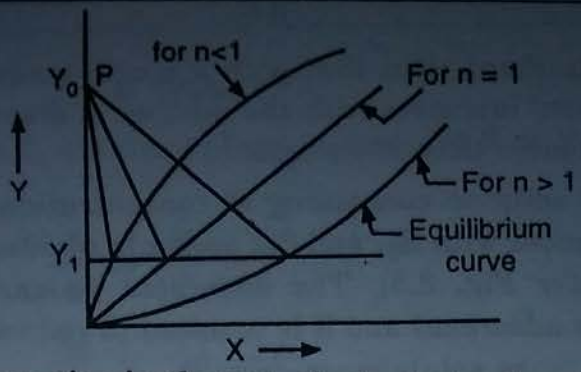
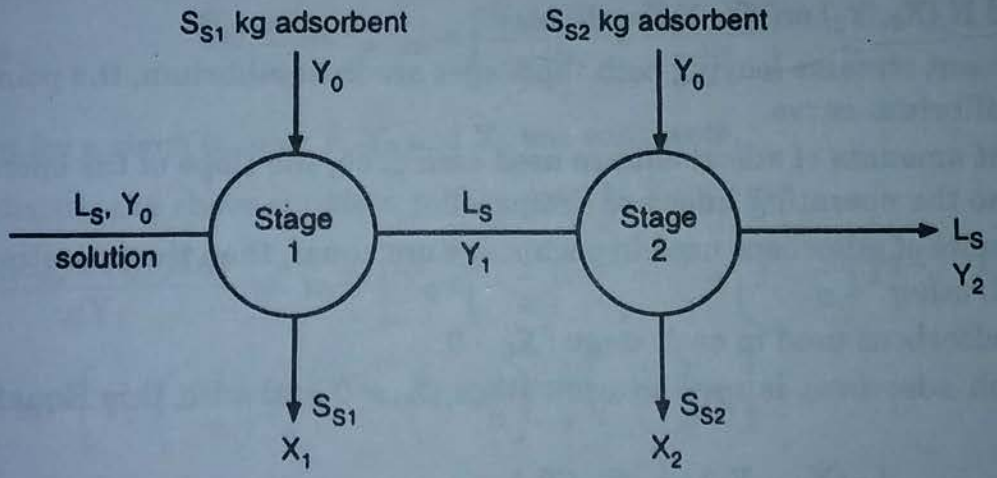


Fig. 3.2 : Freundlich adsorption isotherms - single stage adsorption, different 'n' values

The Freundlich adsorption isotherm is a straight line for n = 1, concave down for n < 1 and concave upward for n > 1. If in each case, the solution concentration is to be reduced from Y_0 to Y_1 in a single stage operation, then the three operating lines will radiate from point P with a slope proportional to the adsorbent to solution weight ratio. Generally, values of n in the range 2 to 10 represent good, 1 to 2 moderately difficult, and less than 1 poor adsorption characteristics.

Multistage Crosscurrent Adsorption Operation :

In this case, the solution is successively contacted with separate batches of adsorbent so that the effluent from any stage will be the feed to the next stage where it is contacted with a fresh batch of adsorbent. With this arrangement, the removal of solute is achieved with



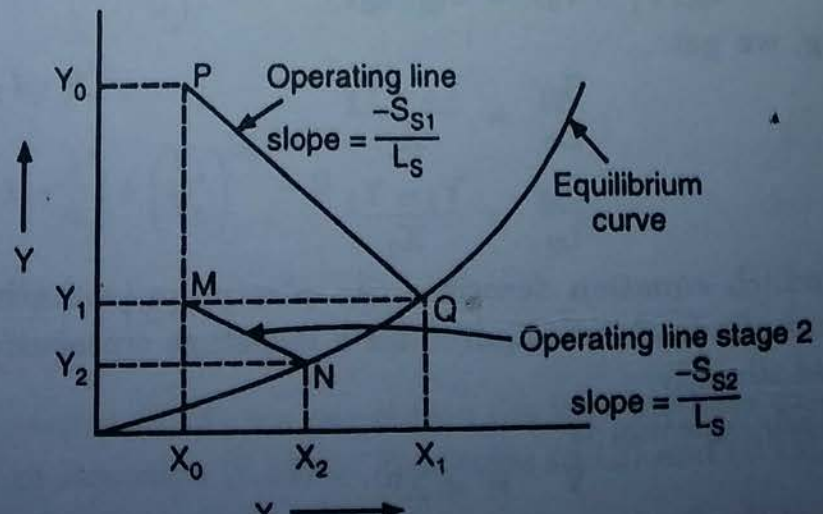


Fig. 3.3 : Two-stage crosscurrent adsorption (different amount of adsorbent in each stage)

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a greater economy of adsorbent than that with a single stage but at the same time the operating cost (on filtration) increases with the number of stages and considering this it is seldom economical to use more than two stages.

Consider that a feed solution containing at concentration Y_0 adsorbable solute, and L_S mass of a solvent is treated with S_{S1} and S_{S2} amounts of adsorbent in the two successive stages respectively. [Refer Fig. 3.3]. The adsorbent to each stage contains X_0 mass adsorbate solute per mass adsorbent and it is required to reduce the solute concentration of the solution from Y_0 to Y_2 mass solute/mass solvent.

The material balance of the solute [considering - only solute is adsorbed, negligible amount of solution with adsorbent and adsorbent is insoluble in the solution] is :

For stage - 1 : $$L_S (Y_0 - Y_1) = S_{S1} (X_1 - X_0) \quad \ldots (3.8)$$

For stage - 2 : $$L_S (Y_1 - Y_2) = S_{S2} (X_2 - X_0) \quad \ldots (3.9)$$

Equation (3.8) is an equation of the operating line for stage-1 and Equation (3.9) is an equation of the operating line for stage-2.

The operating line for stage-1, having a slope of $- S_{S1}/L_S$, passes through points P (X_0, Y_0) and Q (X_1, Y_1) on X, Y coordinates [Refer Fig. 3.3].

The operating line for stage-2, having a slope of $- S_{S2}/L_S$, passes through points M (X_0, Y_1) and N (X_2, Y_2) on X – Y coordinates.

As the effluent streams leaving both the stages are in equilibrium, the points Q and N lie on the equilibrium curve.

As different amounts of adsorbent are used each time, the slope of the operating line is different and so the operating lines are not parallel.

If the amounts of adsorbent used in each stage are equal, then the operating lines will be parallel to each other.

For fresh adsorbent used in each stage : $X_0 = 0$.

When fresh adsorbent is used in each stage, $X_0 = 0$ and with this Equations (3.8) and (3.9) reduce to

$$L_S (Y_0 - Y_1) = S_{S1} (X_1)$$

and $$L_S (Y_1 - Y_2) = S_{S2} (X_2)$$

Rearranging, we get

$$\frac{S_{S1}}{L_S} = \frac{Y_0 - Y_1}{X_1} \quad \ldots (3.10)$$

and $$\frac{S_{S2}}{L_S} = \frac{Y_1 - Y_2}{X_2} \quad \ldots (3.11)$$

If the Freundlich equation describes the adsorption isotherm satisfactorily then the minimum amount of adsorbent required for a two-stage crosscurrent adsorption operation can be calculated directly.

The Freundlich equation is

$$Y^* = m X^n$$

For stage-1, at the final equilibrium conditions, it becomes :

$$Y_1 = m X_1^n$$

$$X_1 = [Y_1/m]^{1/n}$$

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Similarly, for stage-2 :

$$X_2 = [Y_2/m]^{1/n}$$

Putting values of X_1 and X_2 obtained above in terms of Y in Equations (3.10) and (3.11) respectively, we get

For stage-1 :
$$\frac{S_{S1}}{L_S} = \frac{Y_0 - Y_1}{(Y_1/m)^{1/n}} \quad \ldots (3.12)$$

For stage-2 :
$$\frac{S_{S2}}{L_S} = \frac{Y_1 - Y_2}{(Y_2/m)^{1/n}} \quad \ldots (3.13)$$

The total amount of adsorbent used is

$$\frac{S_{S1} + S_{S2}}{L_S} = \frac{Y_0 - Y_1}{(Y_1/m)^{1/n}} + \frac{Y_1 - Y_2}{(Y_2/m)^{1/n}}$$

$$\frac{S_{S1} + S_{S2}}{L_S} = m^{1/n}\left[\frac{Y_0 - Y_1}{Y_1^{1/n}} + \frac{Y_1 - Y_2}{Y_2^{1/n}}\right] \quad \ldots (3.14)$$

For minimum total adsorbent required we have to set :

$$\frac{d\,[(S_{S1} + S_{S2})/L_S]}{dY_1} = 0$$

We have :
$$\frac{S_{S1} + S_{S2}}{L_S} = m^{1/n}\left[\frac{Y_0 - Y_1}{Y_1^{1/n}} + \frac{Y_1 - Y_2}{Y_2^{1/n}}\right]$$

Note that for a given case m, n, Y_0 and Y_2 are constants.

Differentiating the above equation with respect to Y_1, we get

$$\frac{d\,[(S_{S1} + S_{S2})/L_S]}{dY_1} = m^{1/n}\left[Y_0 \cdot \left(-\frac{1}{n}\right) \cdot Y_1^{-\frac{1}{n}-1} - \left(1 - \frac{1}{n}\right) Y_1^{1-\frac{1}{n}-1} + \frac{1}{(Y_2)^{1/n}}\right]$$

$$\therefore \quad 0 = m^{1/n}\left[Y_0\left(-\frac{1}{n}\right) Y_1^{-\frac{1}{n}-1} - \left(1 - \frac{1}{n}\right) Y_1^{-\frac{1}{n}} + \frac{1}{(Y_2)^{1/n}}\right]$$

$$-\frac{1}{n}\left(\frac{Y_0}{Y_1}\right) Y_1^{-1/n} - \left(1 - \frac{1}{n}\right) Y_1^{-1/n} + \frac{1}{(Y_2)^{1/n}} = 0$$

Multiply each term by $Y_1^{1/n}$.

$$-\frac{1}{n}\left(\frac{Y_0}{Y_1}\right) - \left(1 - \frac{1}{n}\right) + \left(\frac{Y_1}{Y_2}\right)^{1/n} = 0$$

$$\left(\frac{Y_1}{Y_2}\right)^{1/n} - \frac{1}{n}\left(\frac{Y_0}{Y_1}\right) = 1 - \frac{1}{n} \quad \ldots (3.15)$$

With the help of Equation (3.15), we can obtain the intermediate concentration Y_1 and using this value of Y_1 in Equation (3.14) or Equations (3.12) and (3.13), we can calculate the minimum total adsorbent required.

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Multistage Countercurrent Operation :

Here the solution to be treated and the adsorbent to be used, flow in opposite directions through a cascade of stages as shown in Fig. 3.4.

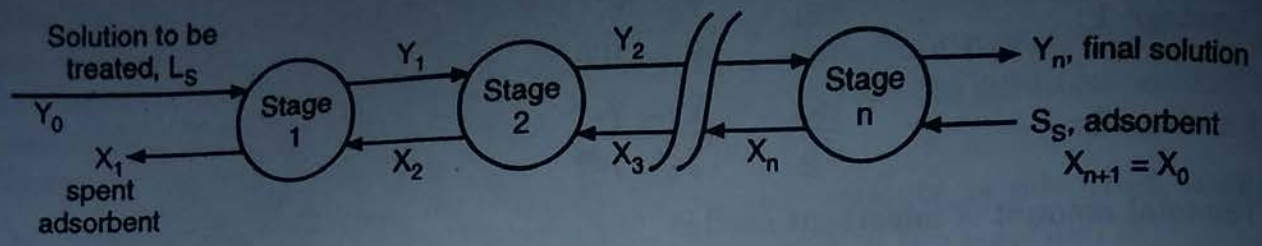

Fig. 3.4 : Multistage countercurrent adsorption

Overall material balance of the solute about the entire cascade :

$$L_S (Y_0 - Y_n) = S_S (X_1 - X_{n+1}) \quad \ldots (3.16)$$

or

$$\frac{Y_0 - Y_n}{X_1 - X_{n+1}} = \frac{S_S}{L_S} \quad \ldots (3.17)$$

This is the operating line equation for a countercurrent multistage adsorption. On X-Y coordinates, it is a straight line of slope S_S/L_S passing through points of coordinates of the terminal conditions (X_{n+1}, Y_n) and (X_1, Y_0).

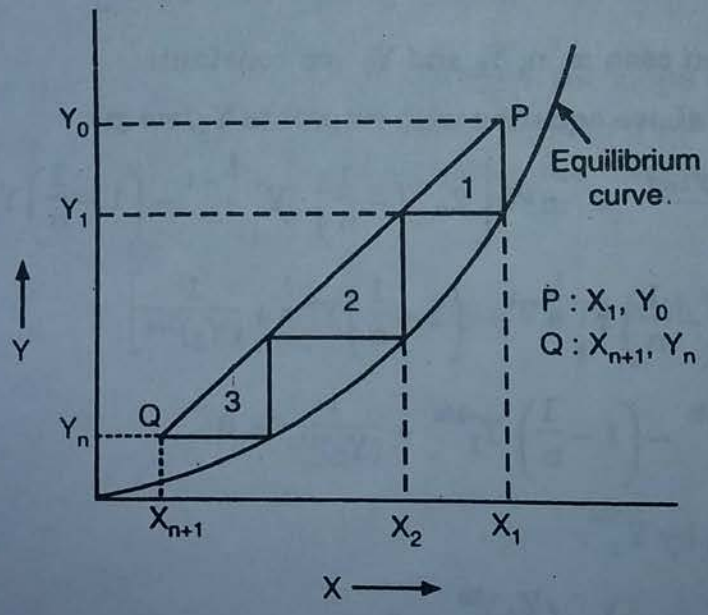

Fig. 3.5 : Graphical construction for number of stages (countercurrent adsorption)

On X-Y plot, locate P (X_1, Y_0) and Q (Y_n, X_{n+1}) and draw the operating line and construct the number of stages between the equilibrium curve and the operating line starting from the point P and ending exactly or crossing the point Q.

Determination of minimum adsorbent requirement :

The minimum adsorbent/solvent ratio requiring an infinite number of stages for the desired change in concentration of the solute in the solution corresponds to the operating line of the largest slope that touches the equilibrium curve.

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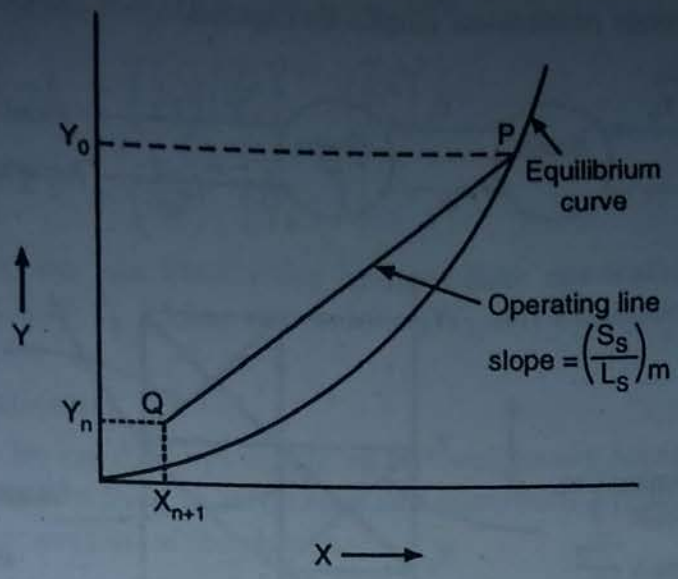

Fig. 3.6 : Minimum adsorbent requirement for infinite stages (X-Y plot concave upward)

(i) For the equilibrium curve/equilibrium isotherm - Straight or Concave upward : On X-Y plot, locate the point Q (X_{n+1}, Y_n) and through Y_0 draw a line parallel to x-axis which will cut the equilibrium curve at P (the coordinates of P are X_1, Y_0). Join PQ, which is the operating line for the minimum adsorbent to solvent ratio. The slope of this operating line gives the value of $(S_S/L_S)_m$, so measure the slope and then calculate the minimum amount of adsorbent required (as L_S is known).

(ii) For the equilibrium curve - Concave downward : In this case, locate the point Q (X_{n+1}, Y_n). Through the point Q, draw a line which will be a tangent to the curve and measure the slope of this line to get $(S_S/L_S)_m$ and then calculate the minimum adsorbent required from the value of slope and a known value of L_S.

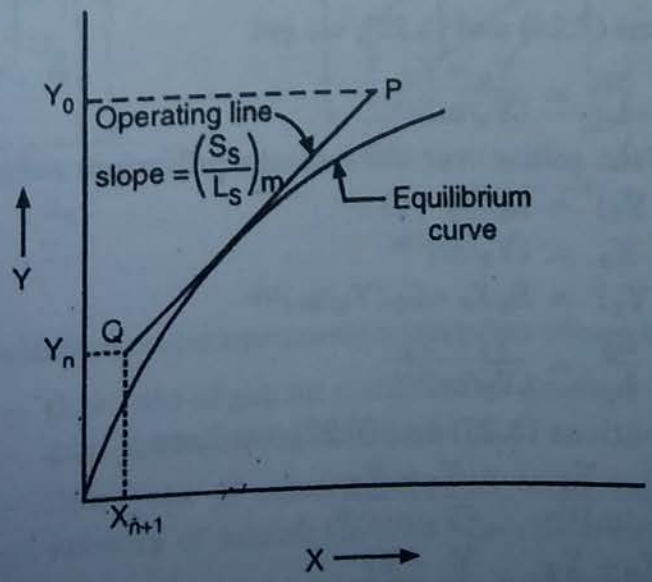

Fig. 3.7 : Minimum adsorbent to solvent ratio for infinite stages (Y v/s X plot, concave downward)

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Consider a two-stage cascade as shown in Fig. 3.8.

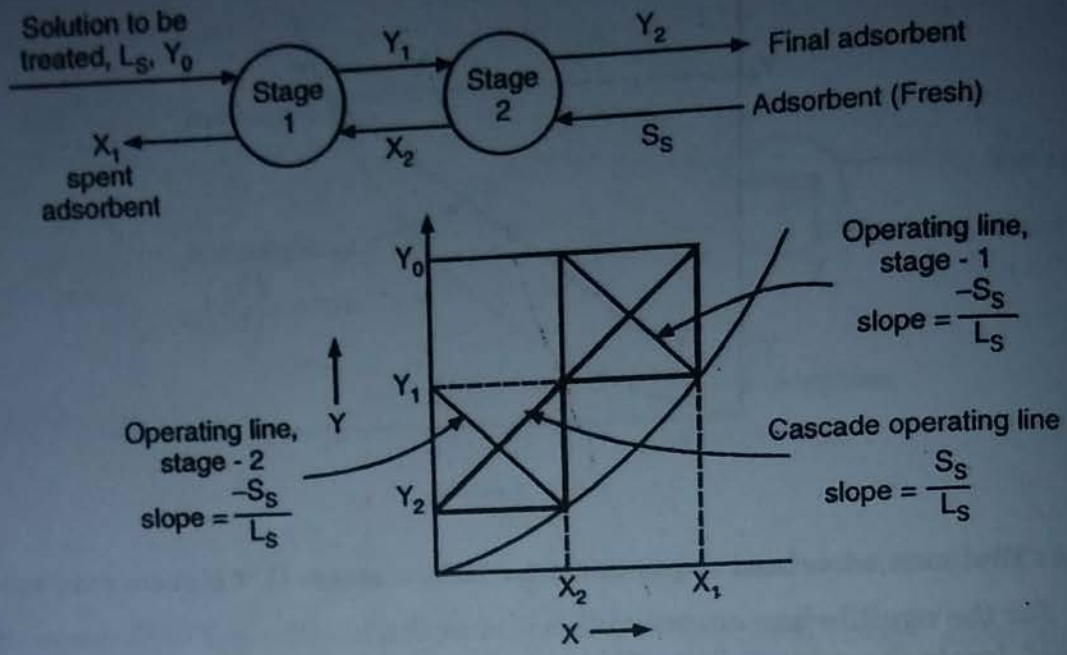

Fig. 3.8 : Two-stage countercurrent adsorption

For fresh adsorbent, $X_{n+1} = X_0 = 0$

Overall material balance of the solute :

$$S_S(X_1 - 0) = L_S(Y_0 - Y_2) \quad \ldots (3.18)$$

The Freundlich equation is

$$Y^* = m X^n$$

Applying the above Freundlich relation to the effluents from the first equilibrium stage, gives

$$Y_1 = m X_1^n$$

$$\therefore \quad X_1 = (Y_1/m)^{1/n} \quad \ldots (3.19)$$

Combining Equations (3.18) and (3.19), we get

$$\frac{S_S}{L_S} = \frac{Y_0 - Y_2}{(Y_1/m)^{1/n}} \quad \ldots (3.20)$$

Material balance of the solute over the second equilibrium stage :

$$L_S(Y_1 - Y_2) = S_S(X_2 - 0)$$

$$X_2 = (Y_2/m)^{1/n}$$

$$L_S(Y_1 - Y_2) = S_S X_2 = S_S(Y_2/m)^{1/n} \quad \ldots (3.21)$$

$$\frac{S_S}{L_S} = \frac{Y_1 - Y_2}{(Y_2/m)^{1/n}} \quad \ldots (3.22)$$

Therefore, from Equations (3.20) and (3.22), we have

$$\frac{Y_0 - Y_2}{(Y_1/m)^{1/n}} = \frac{Y_1 - Y_2}{(Y_2/m)^{1/n}}$$

$$\frac{Y_0 - Y_2}{Y_1^{1/n}} = \frac{Y_1 - Y_2}{Y_2^{1/n}}$$

$$Y_0 Y_2^{1/n} - Y_2 Y_2^{1/n} = Y_1 Y_1^{1/n} - Y_2 Y_1^{1/n}$$

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Dividing both the sides of the above equation by $Y_2\, Y_2^{1/n}$, we get

$$\left(\frac{Y_0}{Y_2}\right) - 1 = \left(\frac{Y_1}{Y_2}\right)\left(\frac{Y_1}{Y_2}\right)^{1/n} - \left(\frac{Y_1}{Y_2}\right)^{1/n}$$

$$\frac{Y_0}{Y_2} - 1 = \left(\frac{Y_1}{Y_2}\right)^{1/n}\left[\frac{Y_1}{Y_2} - 1\right] \quad \ldots (3.23)$$

From Equation (3.23), we can obtain the intermediate concentration Y_1 for given terminal compositions Y_0 and Y_2. Using this value of Y_1 and Equation (3.20), we can get S_S/L_S.

Continuous Contact Operations :

These operations can be carried out either in (i) continuous steady state fashion or (ii) semicontinuous fashion. The fluid to be treated and adsorbent are in contact throughout the unit without periodic separation of the phases.

In a continuous adsorber used for adsorption from gases or liquids, the solid and the fluid move through the bed countercurrently. The solid particles are fed from the top and flow down by gravity through the adsorption and regeneration sections and are returned to the top of the column by means of either an air lift or a mechanical conveyor.

Continuous countercurrent adsorption (one component adsorbed) :

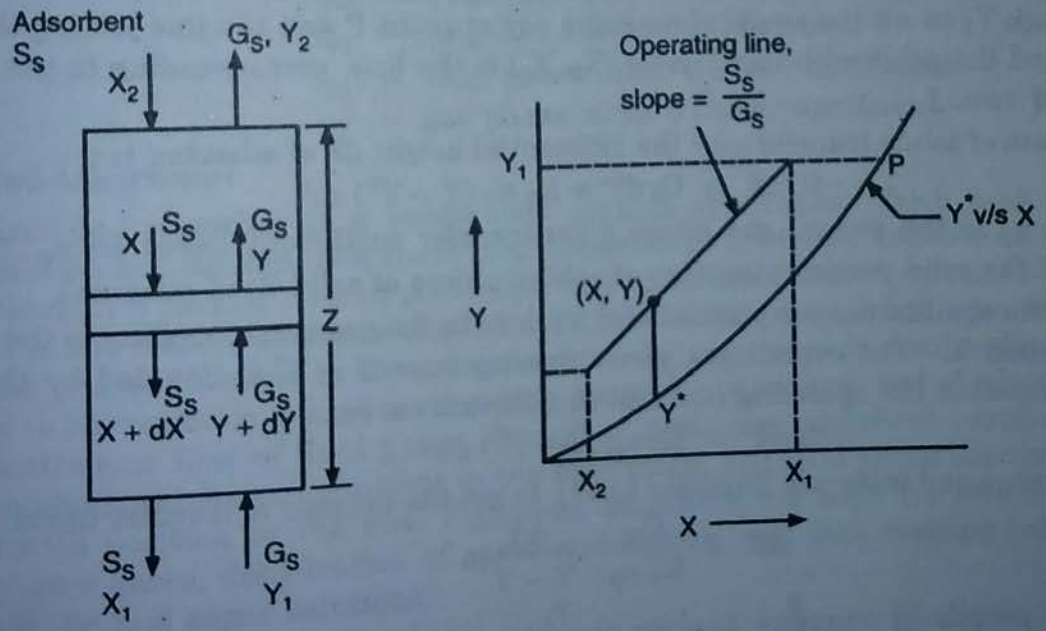

Fig. 3.9 : Continuous countercurrent adsorption of one component

G_S – flow rate of gas on a solute-free basis or mass velocity of solvent gas, mass/(area) · (time).

S_S – flow rate of adsorbent solid on a solute-free basis or mass velocity of adsorbate-free adsorbent solid, mass/(area) · (time).

Solute concentrations in the various streams are expressed in terms of mass ratio – mass solute/mass solute-free substance.

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Y_1 – mass solute/mass solute-free gas, i.e., mass solute/mass solvent gas at the entrance.

Y_2 – mass solute/mass solvent gas at the exit.

X_1 – mass solute/mass adsorbent at exit.

X_2 – mass solute/mass adsorbent at entrance.

We have considered here the case for treating of a gas, but the treatment can be equally applicable for treating a liquid by replacing G_S by L_S.

The overall material balance of solute over the entire column is

$$G_S (Y_1 - Y_2) = S_S (X_1 - X_2) \quad \text{... (3.24)}$$

Material balance of solute in the upper part of the column :

$$G_S (Y - Y_2) = S_S (X - X_2) \quad \text{... (3.25)}$$

On X, Y coordinates, Equation (3.24) is the equation of a straight line having a slope equal to S_S/G_S joining the terminal conditions (X_1, Y_1) and (X_2, Y_2). This operating line is plotted with the equilibrium curve in Fig. 3.9. The solute concentrations X and Y at any horizontal plane/height in the column lie on this line. The minimum solid/fluid ratio is given by the operating line of maximum slope which anywhere touches the equilibrium curve - which is tangent to the equilibrium curve. [The point (X_2, Y_2) in this case is fixed and a line is to be drawn through this point. In the present case of plot, draw a horizontal line through Y_1 to cut the equilibrium curve say at point P and the line joining this cutting point P and the point with coordinates (X_2, Y_2) is the line corresponding to the minimum solid/fluid ratio.]

The rate of solute transfer over the differential height dZ of adsorber is :

$$S_S \, dX = G_S \, dY = K_Y a_p (Y - Y^*) \, dZ \quad \text{... (3.26)}$$

where $K_Y a_p$ is the overall gas phase mass-transfer coefficient based on a_p, the outside surface of the solid particles (surface of solids/volume of solid fluid mixture).

Y^* is the equilibrium composition of the solute in the gas corresponding to the adsorbate concentration X. The overall gas phase driving force $Y - Y^*$ is denoted by the vertical distance between the operating line and equilibrium curve.

$$G_S \, dY = K_Y a_p (Y - Y^*) \, dZ \quad \text{... (3.27)}$$

Rearrange and integrate Equation (3.27) to get the number of transfer units.

$$dZ = \frac{G_S}{K_Y a_p} \cdot \frac{dY}{Y - Y^*}$$

$$\int_0^Z dZ = \frac{G_S}{K_Y a_p} \int_{Y_2}^{Y_1} \frac{dY}{Y - Y^*}$$

$$Z = \frac{G_S}{K_Y a_p} \int_{Y_2}^{Y_1} \frac{dY}{Y - Y^*} \quad \text{... (3.28)}$$

$$Z = H_{tOG} \cdot N_{tOG} \quad \text{... (3.29)}$$

The number of overall gas transfer units are :

$$N_{tOG} = \int_{Y_2}^{Y_1} \frac{dY}{Y^* - Y} \quad \text{... (3.30)}$$

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The height of an overall gas transfer unit is

$$H_{toG} = \frac{G_S}{K_Y a_p} \qquad ...(3.31)$$

$$N_{toG} = \int_{Y_2}^{Y_1} \frac{dY}{Y - Y^*} = \frac{Z}{H_{toG}} \qquad ...(3.32)$$

The value of the integral of Equation (3.30) is evaluated graphically and the active height Z is then determined using a known value the height of a transfer unit H_{toG}.

The relationship between the overall mass-transfer coefficient and the mass-transfer coefficients of the phases involved is

$$\frac{1}{K_Y a_p} = \frac{1}{k_Y a_p} + \frac{m}{k_s a_p}$$

$$\therefore \quad \frac{G_S}{K_Y a_p} = \frac{G_S}{k_Y a_p} + \frac{m G_S}{k_S a_p}$$

$$\frac{G_S}{K_Y a_p} = \frac{G_S}{k_Y a_p} + \frac{m G_S}{S_S} \cdot \frac{S_S}{k_S a_p}$$

$$\therefore \quad H_{toG} = H_{tG} + \frac{m G_S}{S_S} H_{tS} \qquad ...(3.33)$$

where $m = dY^*/dX$ = slope of the equilibrium curve.

H_{tG} – height of a gas transfer unit
H_{tS} – height of a solid transfer unit
k_S – solid phase mass transfer coefficient
k_Y – gas phase mass transfer coefficient.

Fixed-Bed Adsorbers :

A fixed-bed adsorber is a continuous contact apparatus, which may superficially resemble a packed column. But there is a fundamental difference between these two units. The packing in a packed column provides a large interfacial area between the fluid phases involved in the mass transfer, while the solid particles in a fixed bed adsorber are in fact one of the phases involved in the mass transfer. It is operated in semicontinuous fashion- the fluid to be treated is passed continuously through a stationary bed of adsorbent. In such units, continuous flow of fluid phase through a stationary adsorbent results in unsteady state conditions, where compositions in the fluid phase and solid phase change with time as well as with position in the bed. Fixed-bed adsorbers are used for recovery of solvent vapours from gases, dehydration of gases and liquids, and decolourising lubricating oils, vegetable oils and sugar solutions.

A typical system used for adsorption of solvent vapours is shown in Fig. 3.10. Generally two beds are used; when one is in operation, the other is regenerated. In order to operate the adsorber with a small pressure drop, adsorbents in granular form are used. The diameter of the adsorber is much greater than the bed depth. The bed length usually varies from 0.3 to 1.3 m and the bed of the adsorbent particles is supported by a screen or perforated plate. The feed gas is fed downward through the bed at a superficial gas velocity of order of 0.25 to 0.6 m/s. Upflow of feed is usually avoided because at high gas flow rates there is a tendency of fluidisation of adsorbent particles which results in attrition and loss of fines. When the concentration of the solute in the effluent gas reaches a certain value or at a

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schedule time, valves are automatically switched so that the feed is diverted to the second bed and regeneration of the first bed starts.

Regeneration can be carried out using either a hot inert gas or low pressure steam. The low pressure steam is usually used for regeneration purpose, if the solvent is not miscible with water. Steam condenses in the bed and provides the necessary heat of desorption (the hot inert gas or steam provides the necessary heat of desorption) and consequently the desorption process starts. The steam-vapour effluent from the bed is condensed in a condenser, the solvent is separated from water by decantation and dried before reuse. The bed may then be cooled and dried with the inert gas if the moisture interferes with vapour adsorption. If some water vapour is tolerated in the clean gas then the moisture is removed by the vapours during an adsorption cycle in which case the heat released in adsorption of vapour is used to desorb water.

For very large gas flow rates, a rectangular bed of certain depth may be installed in a horizontal cylinder.

The size of the bed depends upon the gas flow rate and the desired cycle time. The adsorption cycle is usually of 2 to 24 h. With a longer bed, the adsorption cycle could be extended to several days but it results in a large pressure drop and requires high capital cost. With small bed depth, the pressure drop is small but separation with a shallow bed is incomplete and it also requires more energy for regeneration.

In fixed-beds, an alternative procedure to provide the energy for regeneration is to heat the bed electrically (embedded electric heaters) or with embedded steam coils. The procedures which utilise heat as a regenerative means are called as thermal swing procedures.

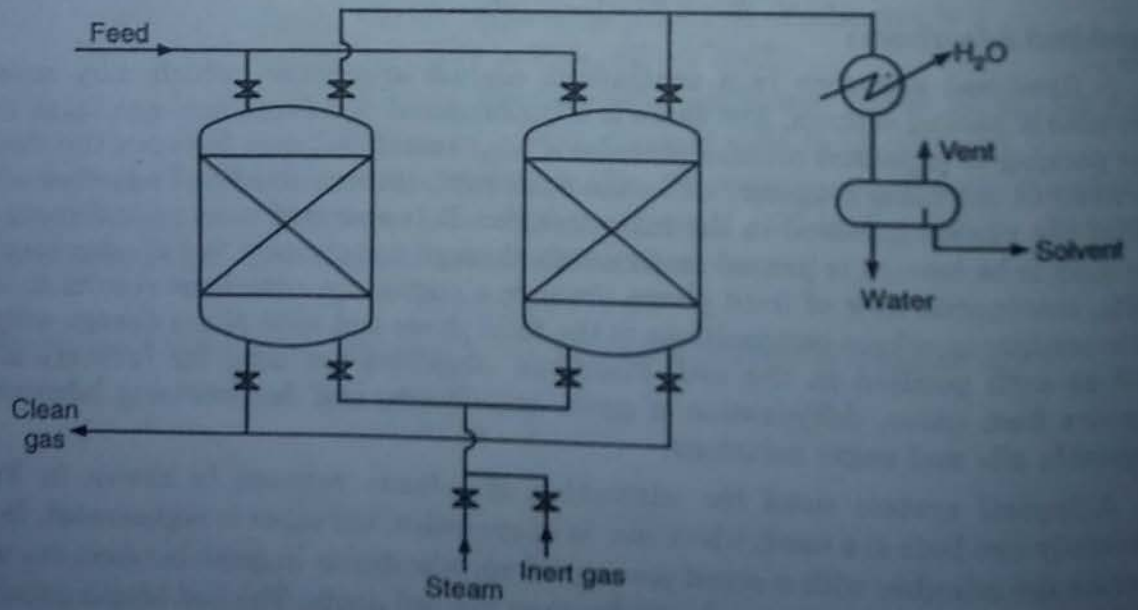

Fig. 3.10 : Vapour adsorption system

For adsorption from liquids, the bed of adsorbent may be tall upto 10 m and tall beds are needed because the rate of adsorption from liquids is much slower than from gases.

(e.g., the removal of trace organics from municipal water supplies using carbon - that improves taste and a chance of forming toxic compounds in the chlorination step.)

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Moist gases can be dried of their water content in a equipment similar to that shown in Fig. 3.10. In this case, a moist gas is passed through the bed of activated silica gel, alumina, or molecular sieves. The hot gas is used for regeneration. The moist gas leaving the bed being regenerated may be vented or much of the water may be removed in a condenser and the gas is recirculated through a heater to the bed.

Concentration profiles in fixed beds :

In two-component fixed-bed operation, the solute from the fluid phase is removed continuously and accumulated on the solid phase as the fluid phase moves through a stationary solid phase, and as a result the concentrations in the fluid phase and the solid phase change with time as well as with position in the bed. At first, most of the mass transfer takes place near the inlet of the bed where the fluid first comes in contact with the adsorbent. If the adsorbent solid is free of adsorbate initially, the concentration in the fluid drops exponentially with the bed length essentially to zero-before the end of the bed is reached. This concentration profile is shown by curve θ_1 in Fig. 3.11, where Y/Y_0 is the concentration in the fluid relative to that in the fluid entering the fixed-bed. After a few minutes of operation, the solid near the inlet becomes nearly saturated, and most of the mass transfer takes place further from the inlet. The concentration profile then becomes S shaped, as shown by curve θ_2. The region in which most of the change in concentration takes place is called the mass-transfer zone (adsorption zone) and the limits of which are often taken as Y/Y_0 values of 0.95 to 0.05. With passage of time, the adsorption zone moves down the bed, as shown by profiles θ_3 and θ_4. Similar profiles could also be drawn for the average concentration of adsorbate retained on the solid, indicating nearly saturated solid at the inlet, a large change in the adsorbate concentration in the mass transfer zone and zero at the end of the adsorbent bed.

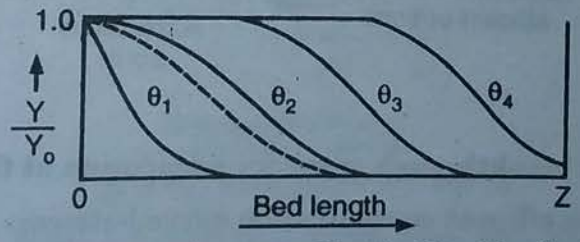

Fig. 3.11 : Concentration profiles in fixed-bed adsorber

Instead of plotting the adsorbate concentration, the concentration in the fluid phase in equilibrium with the adsorbent solid is plotted and indicated by a dashed line for time θ_2 in Fig. 3.11. The concentration in the fluid phase in equilibrium with the adsorbent solid has to be always less than the actual fluid concentration. The difference in these two concentrations (driving force) is large where the concentration profile is steep (mass transfer is rapid).

Breakthrough Curves :

When a binary solution (either gas or liquid) containing strongly adsorbed solute at concentration Y_0 mass solute/mass solvent flows continuously downward through a relatively deep bed of adsorbent initially free from adsorbate. The topmost layer of adsorbent which first comes in contact with the strong solution fed, at first adsorbs solute rapidly and effectively, and whatever a little amount of it left in the solution unadsorbed is then removed by the layers of adsorbent in the lower part of the bed. The effluent leaving the

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adsorbent bed from the bottom is practically a solute-free carrier fluid (i.e., concentration of solute in the effluent is zero). The topmost layer of the bed gets saturated with adsorbed solute (thus exhausted) and the bulk of the adsorption takes place over a relatively narrow adsorption zone in which the concentration changes very rapidly. If solution continues to flow, the adsorption zone moves gradually through the bed as a wave, with a rate ordinarily very much slower than the linear velocity of the solution in the bed. At the leading boundary of the adsorption zone, the advance solute is being adsorbed, while at the back of the zone the bed is just reaching saturation. As the zone moves downward, more and more portion of the bed gets saturated with the solute. The effluent leaving the bed is solute-free during the progress of the zone until the lower boundary of the adsorption zone coincides with the bottom of the adsorbent bed. At the moment of coincidence, the solute breaks through the bed, i.e., the solute appears in the effluent-for the first time the concentration of solute suddenly rises to an appreciable value Y_B (breakpoint/breakthrough point). Afterwards, the concentration of solute in the effluent rises rapidly as the adsorption zone passes through the bottom of the bed and reaches to Y_E (substantially close to Y_0) at which time the entire bed is practically saturated (the entire adsorption zone leaves the bed).

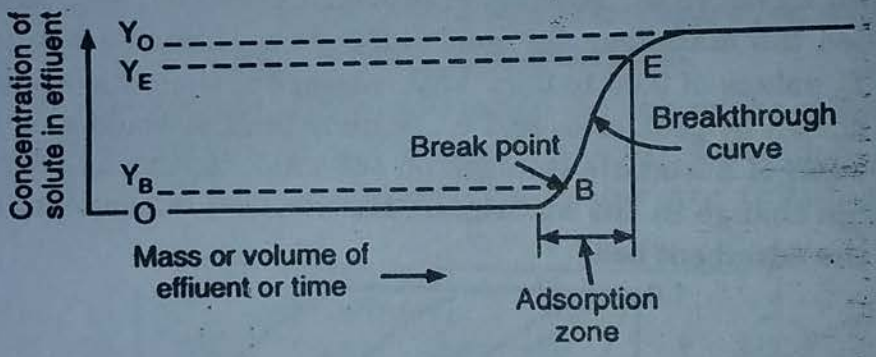

Fig. 3.12 : Adsorption breakthrough curve for adsorption at finite mass transfer rate

The portion/part of the effluent concentration curve between points B and E is termed as the breakthrough curve.

The portion of a plot of effluent concentration as a function of time between the breakpoint and time corresponding to practical saturation of the bed is termed as the breakthrough curve.

Generally, the breakthrough curve is S-shaped and it is often nearly symmetrical. The slope of the breakthrough curve is a function of the rate of adsorption. The higher the rate of adsorption, the steeper the breakthrough curve. If the adsorption process were instantaneous, i.e., infinitely rapid mass transfer rate (of adsorption) (ideal case), the adsorption zone would be of an infinitesimal width, and the breakthrough curve would be a straight vertical line from 0 to Y_0 when all the adsorbent solid was saturated.

The breakpoint - the point in time at which solute is first noticed in the effluent is often taken as a solute concentration of $Y_B = 0.05\ Y_0$. Similarly, the point at which the bed is practically saturated is often taken as a solute concentration of $Y_E = 0.95\ Y_0$. For the symmetrical curve, the ideal adsorption time (for the vertical breakthrough curve) is the time when $Y = 0.5\ Y_0$.

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Rate of Adsorption in fixed beds :

The design of a fixed-bed adsorber involves either determination of the bed length for a given cycle time or calculation of the breakpoint/breakthrough time for a given length of bed.

The simplified treatment given below is readily used though it is limited to isothermal adsorption from dilute feeds (with respect to adsorbable solute) and cases where the adsorption equilibrium curve is concave upward, where the adsorption zone is constant in height as it passes through the bed, and where the height of adsorption zone is short relative to the height of the bed. Many adsorption operations carried in industry fall within these restrictions.

Consider the breakthrough curve of Fig. 3.13 wherein the concentration of solute gas in terms of mass ratio, Y mass solute/mass solvent gas in effluent is plotted against the total mass of solute-free effluent accumulated at any time, from the start. This curve is resulted (say) from a continuous flow of a binary gas mixture through a relatively deep bed of adsorbent. The concentration of solute in a gas entering bed is Y_0 mass solute/mass solvent gas and the gas feed rate on the solute-free basis is G_S mass/(area) (time) [solvent gas mass velocity to bed].

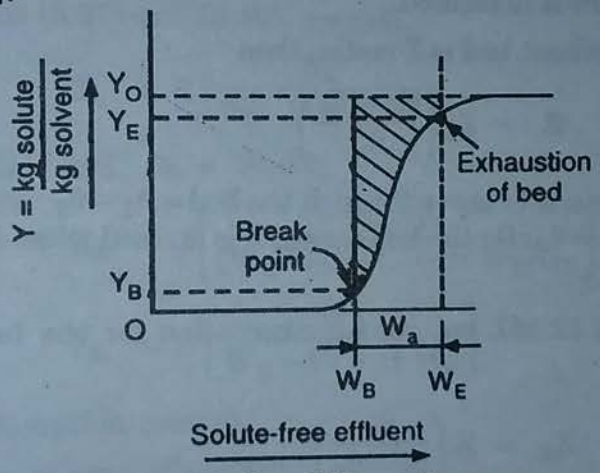

Fig. 3.13 : Breakthrough curve

W = total solute-free effluent after any time (cumulative solute-free effluent upto any time θ), mass/cross-sectional area of bed.

W_B = total solute-free effluent upto breakpoint, mass/area.

W_E = total solute-free effluent upto exhaustion of bed, mass/area.

$W_a = W_E - W_B$ = total solute-free effluent accumulated during the appearance of the breakthrough curve i.e. total solute-free effluent accumulated in time between the breakpoint and the exhaustion / saturation point, mass/area.

Z_a = adsorption zone height which is a part of the bed in which the concentration change Y_B to Y_E is occurring at any time.

Let θ_a be the time required for the adsorption zone to move a distance, equal to its own height (Z_a), down the bed, after establishment of zone. Then

$$\theta_a = \frac{\text{total solute-free effluent accumulated over breakthrough curve}}{\text{mass flow rate of solvent gas}}$$

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The total solute-free effluent accumulated in time between the breakpoint and the saturation/exhaustion point is

$$W_a = W_E - W_B$$

$$\therefore \quad \theta_a = \frac{W_a}{G_S} \qquad \ldots (3.34)$$

Let θ_E be the time required for the establishment of the adsorption zone and move it out of the bed. Then

$$\theta_E = \frac{\text{total solute-free effluent accumulated upto saturation}}{\text{mass flow rate of solvent gas}}$$

$$\theta_E = \frac{W_E}{G_S} \qquad \ldots (3.35)$$

Out of this, some time was required to form the adsorption zone at the beginning of the bed when the gas was introduced.

If θ_F is the time required to form the zone, then $\theta_E - \theta_F$ is the time available for the zone to move through the bed once it is formed.

If the height of the adsorbent bed is Z meter, then

$$Z = Z_a\left(\frac{\theta_E - \theta_F}{\theta_a}\right) \qquad \ldots (3.36)$$

[Time available for the zone to move through the bed = $\theta_E - \theta_F$ and time required for the zone to move a distance, $Z_a = \theta_a$. So the bed length Z is crossed when $(\theta_E - \theta_F)/\theta_a$ movements of length Z_a have occurred.]

Rearranging Equation (3.36), we get an expression for the height of the adsorption zone.

$$Z_a = Z\left(\frac{\theta_a}{\theta_E - \theta_F}\right) \qquad \ldots (3.37)$$

The quantity of solute removed from the gas in the adsorption zone from the breakpoint to exhaustion/saturation is U mass solute/area. It is represented by the shaded area of Fig. 3.13.

$$\therefore \quad U = \int_{W_B}^{W_E} (Y_0 - Y)\, dW \qquad \ldots (3.38)$$

If all the adsorbent in the adsorption zone were saturated, it would contain $Y_0 W_a$ mass of solute/area. As a result at the breakpoint, when the zone is still within the bed, the fractional ability of the adsorbent in the zone still to adsorb solute is given by

$$f = \frac{U}{Y_0 W_0} = \frac{\int_{W_B}^{W_E} (Y_0 - Y)\, dW}{Y_0 W_a} \qquad \ldots (3.39)$$

Limiting values of f :

f = 0 : all the adsorbent in the zone is essentially saturated with solute.

f = 1 : the adsorbent in the zone contains essentially no adsorbate.

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[For $f = 0$, $U/Y_0 W_a = 0$, $Y_0 W_a \neq 0$ $\therefore$ $U = 0$ i.e. the ability of the adsorbent in the zone still to adsorb solute is zero, it means that all the adsorbent in the zone is fully saturated.]

[For $f = 1$, $U = Y_0 W_a$, i.e., the ability of the adsorbent in the zone still to adsorb solute is equal to the amount of solute adsorbed corresponding to full saturation. The ability of the adsorbent still to adsorb solute means adsorption of solute further with some previous adsorption and therefore it is possible to adsorb solute corresponding to saturation only if the adsorbent does not contain any adsorbate. So for $f = 1$, the adsorbent in the zone contains essentially no adsorbate.]

Any value of f between 0 and 1 corresponds to a real (finite) adsorption rate.

The amount of solute retained in the adsorption zone is $(1 - f)$ times the amount of solute retained in the bed that reaches saturation while the zone is moving. As the flow of gas to the bed is assumed to be steady, the same fraction is applicable to time and effluent.

So θ_F is given by

$$\theta_F = (1 - f)\,\theta_a \qquad \ldots (3.40)$$

Combining Equations (3.37) and (3.40), we get,

$$Z_a = Z\left[\frac{\theta_a}{\theta_E - (1 - f)\,\theta_a}\right] \qquad \ldots (3.41)$$

We have : $\theta_a = W_a/G_S$ and $\theta_E = W_E/G_S$

$$\therefore \quad Z_a = Z\left[\frac{W_a/G_S}{(W_E/G_S) - (1 - f)\,W_a/G_S}\right]$$

$$Z_a = Z\left[\frac{W_a}{W_E - (1 - f)\,W_a}\right] \qquad \ldots (3.42)$$

Z is the height of adsorption column.

A is the cross-sectional area of column.

ρ_s is the apparent packed density of the adsorbent solid, mass adsorbent per unit volume of bed.

X_T is the adsorbate concentration in column, mass solute/mass adsorbent in equilibrium with the entering gas (of composition Y_0).

Mass of adsorbent in the column $= \rho_s A Z$.

If all the adsorbent in the column were in equilibrium with the entering gas and hence completely saturated at an adsorbate concentration X_T mass adsorbate/mass adsorbent then amount of solute retained is given by

Amount of adsorbate (adsorbed solute) when all the adsorbent in column were saturated $= \rho_s A Z X_T$

At the breakpoint :

The adsorption zone of height Z_a is still in the column at the bottom. Therefore, the remaining length of the column (i.e., bed length) $Z - Z_a$ is fully saturated with solute while the length Z_a is partially saturated.

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The amount of adsorbed solute/adsorbate in the length $(Z - Z_a)$ $= \rho_s A (Z - Z_a) X_T$

The adsorption zone Z_a is not fully saturated, so

The amount of adsorbed solute in length Z_a $= \rho_s A Z_a (1 - f) X_T$

$\therefore$ The total amount of adsorbed solute in the length Z (at breakpoint) $= \rho_s A (Z - Z_a) X_T + \rho_s A Z_a (1 - f) X_T$

So the fractional approach to saturation of the column (the fraction of saturation of the entire bed) at the breakpoint is

$$\text{Degree of saturation} = \frac{\text{Amount of solute adsorbed at the breakpoint}}{\text{Amount of solute adsorbed when the entire adsorbent bed were saturated}}$$

$$= \frac{\rho_s A (Z - Z_a) X_T + \rho_s A Z_a (1 - f) X_T}{\rho_s A Z X_T}$$

$$= \frac{Z - f Z_a}{Z} \quad \ldots (3.43)$$

Equation (3.43) can be used to calculate the total solute actually adsorbed when the breakpoint is reached.

The rate and extent of adsorption in the adsorption zone can be analysed using the concept of transfer units. To make the analysis, imagine that the solid adsorbent is moving upward through the column countercurrent to the flow of gas at a rate such that the adsorption zone remains stationary within the column – acting as a section of the continuous countercurrent column and offering continuous contact between the moving phases as shown in Fig. 3.14. It is shown in Fig. 3.14 that (i) the adsorbent leaving the column from the top is in equilibrium with the gas entering the column and (ii) the effluent gas leaving the column contains no solute (Y = 0) (i.e. all solute is removed from gas). The adsorbent entering the column contains no solute (X = 0), so that equilibrium exists between effluent gas leaving and adsorbent entering the column. This situation would require an infinitely tall column, but we are concerned primarily with the concentrations at the two extreme ends of the adsorption zone.

Material balance of solute over the entire column :

$$G_S (Y_0 - 0) = S_S (X_T - 0) \quad \ldots (3.44)$$

or

$$\frac{S_S}{G_S} = \frac{Y_0 - 0}{X_T - 0} \quad \ldots (3.45)$$

On X, Y coordinates, Equation (3.44) is the equation of a straight line through the origin, of slope S_S/G_S. This operating line is plotted in Fig. 3.14.

As the operating line passes through the origin, at any level in the column, the solute concentration in the gas (Y) and the adsorbate concentration on the adsorbent (X) are related by :

$$G_S Y = S_S X \quad \ldots (3.46)$$

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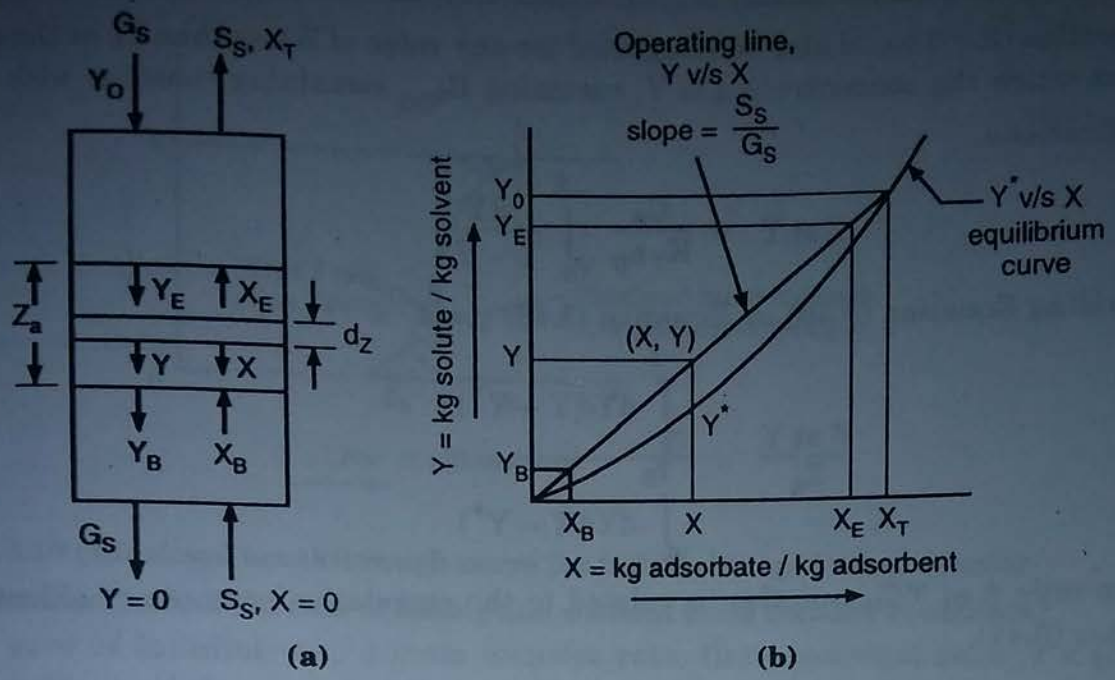

Fig. 3.14 : The Adsorption zone

The rate of adsorption can be related to a material balance over a differential volume element of height dZ.

$$\therefore \quad G_S\, dY = K_Y\, a_p\, (Y - Y^*)\, dZ \qquad \ldots (3.47)$$

Equation (3.47) can be integrated over the height of the adsorption zone.

$$dZ = \frac{G_S}{K_Y\, a_p} \frac{dY}{(Y - Y^*)}$$

$$\int_0^{Z_a} dZ = \frac{G_S}{K_Y\, a_p} \int_{Y_B}^{Y_E} \frac{dY}{(Y - Y^*)}$$

$$Z_a = \frac{G_S}{K_Y\, a_p} \int_{Y_B}^{Y_E} \frac{dY}{Y - Y^*} \qquad \ldots (3.48)$$

$$\therefore \quad Z_a = H_{tOG} \cdot N_{tOG}$$

$$N_{tOG} = \int_{Y_B}^{Y_E} \frac{dY}{Y - Y^*} = \frac{Z_a}{H_{tOG}} = \frac{Z_a}{G_S/K_Y\, a_p} \qquad \ldots (3.49)$$

where, N_{tOG} – overall number of gas transfer units in the adsorption zone

Z_a – height of the adsorption zone

H_{tOG} – height of an overall gas transfer unit

K_Y – overall gas phase mass transfer coefficient, mass/(area) · (time) (mass/mass)

G_S – mass velocity of solvent gas/unadsorbed gas based on empty column cross-section

a_p – surface of solids/volume solid-fluid mixture, area/volume.

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Equation (3.48) could also be integrated for any value of Z less than Z_a in the zone from Z = 0 at which the concentration is Y, assuming Ht_{OG} remaining constant with changing concentrations.

$$Z \text{ at } Y = \frac{G_S}{K_Y a_p} \int_{Y_B}^{Y} \frac{dY}{Y - Y^*} \quad \ldots (3.50)$$

Dividing Equation (3.50) by Equation (3.48) gives

$$\frac{Z \text{ at } Y}{Z_a} = \frac{\int_{Y_B}^{Y} dY/(Y - Y^*)}{\int_{Y_B}^{Y_E} dY/(Y - Y^*)} \quad \ldots (3.51)$$

The ratio Z at Y/Z_a can also be related to the cumulative amounts of effluents using Equation (3.42).

$$Z \text{ at } Y = Z\left(\frac{W - W_B}{W_E - (1 - f)\, W_a}\right) \quad \ldots (3.52)$$

W is the total solute-free effluent i.e. cumulative amount of solute-free effluent corresponding to Z at Y.

Dividing Equation (3.52) by Equation (3.42) gives

$$\frac{Z \text{ at } Y}{Z_a} = \frac{W - W_B}{W_a} \quad \ldots (3.53)$$

$$\therefore \quad \frac{Z \text{ at } Y}{Z_a} = \frac{W - W_B}{W_a} = \frac{\int_{Y_B}^{Y} dY/(Y - Y^*)}{\int_{Y_B}^{Y_E} dY/(Y - Y^*)} \quad \ldots (3.54)$$

We can plot the breakthrough curve. For plotting the breakthrough curve on dimensionless coordinates, Y/Y_0 v/s $(W - W_B)/W_a$, we have to evaluate the horizontal coordinate of the breakthrough curve, $(W - W_B)/W_a$, using the integrals of Equation (3.54). The breakthrough can be obtained without knowing the mass transfer rate, but the determination of Z_a requires a knowledge of $K_Y a$.

Above development is useful if both fluid-and solid-phase mass transfer coefficients are available. In many cases of industrial importance, the solid phase mass transfer coefficients are not usually available. In such cases, for systems with favourable adsorption isotherm, following extension can be adopted.

Consider Fig. 3.15. If adsorption were instantaneous (i.e. rate of mass transfer in adsorption is infinite) then the adsorption zone would be a plane of infinitesimal width, and the breakthrough curve would be a straight vertical line (from 0 to Y_0) at θ_s. (i.e. when all the adsorbent were saturated). θ_s can be located, so that the shaded areas are equal.

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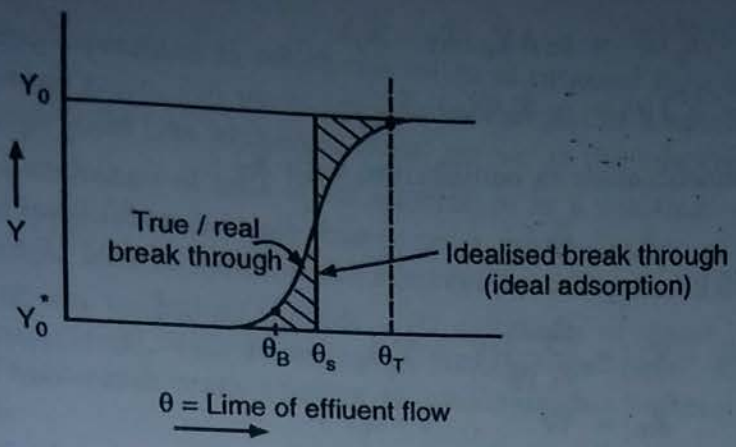

Fig. 3.15 : Idealised breakthrough curve for infinitely rapid mass transfer (breakthrough for ideal case of adsorption without mass transfer resistance)

For ideal case of infinitely rapid mass transfer rate, the adsorption zone is a plane - the bed on upstream of the plane is fully saturated and the bed on the downstream of the plane is unused.

(i) On the upstream of this plane, the length of bed Z_s is fully saturated with solute at adsorbate concentration X_T.

(ii) On the downstream of this plane, the remaining length of bed i.e. the length $Z - Z_s$ is equal to the length of unused bed (LUB).

At the breakpoint/breakthrough point :

$$\text{Length of bed} = \begin{matrix}\text{length saturated with solute} \\ \text{in equilibrium with} \\ \text{the feed stream}\end{matrix} + \begin{matrix}\text{length of unused} \\ \text{bed (LUB)}\end{matrix}$$

If V is the velocity of advancement of the adsorption plane, then at any time,

$Z_s = V\theta$, at θ_s, $Z = V\theta_s$, and at breakthrough, θ_B, $Z_s = V\theta_B$.

At breakthrough point :

$$Z = Z_s + LUB \quad \ldots (3.55)$$

$$LUB = Z - Z_s$$

$$LUB = V\theta_s - V\theta_B = V(\theta_s - \theta_B)$$

$$\text{but } V = Z/\theta_s$$

$$\therefore \quad LUB = \frac{Z}{\theta_s}[\theta_s - \theta_B] \quad \ldots (3.56)$$

$$(LUB)\,\theta_s = Z\theta_s - Z\theta_B$$

$$\frac{(LUB)\,\theta_s}{Z} = \theta_s - \theta_B$$

$$\theta_B = \theta_s - \frac{LUB\,\theta_s}{Z}$$

$$\therefore \quad \theta_B = \theta_s\left[1 - \frac{LUB}{Z}\right] \quad \ldots (3.57)$$

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Material balance of solute is :

$$G_S A (Y_0 - Y_0^*) \theta = \rho_S A Z_s (X_T - X_0)$$

$$G_S (Y_0 - Y_0^*) \theta = \rho_S Z_s (X_T - X_0) \quad \ldots (3.58)$$

where Y_0^* is the fluid concentration in equilibrium with X_0.

At θ_s : $Z = Z_s$

$\therefore$ Equation (3.58) becomes :

$$X_T - X_0 = \frac{G_S \theta_s}{Z_s \rho_S} (Y_0 - Y_0^*) \quad \ldots (3.59)$$

We have : $Z_s = V\theta \quad \therefore \quad \theta = Z_s / V$

Equation (3.58) becomes :

$$G_S (Y_0 - Y_0^*) Z_s/V = \rho_S Z_s (X_T - X_0)$$

$$V = \frac{G_S}{\rho_S} \frac{(Y_0 - Y_0^*)}{(X_T - X_0)} \quad \ldots (3.60)$$

Eliminating θ_S from Equations (3.59) and (3.57), we get

$$\theta_B = \frac{(X_T - X_0) Z_s \rho_S}{G_S (Y - Y_0^*)} \left[1 - \frac{LUB}{Z}\right]$$

$$\frac{G_S \theta_B (Y_0 - Y_0^*)}{(X_T - X_0) Z_s \rho_S} = 1 - \frac{LUB}{Z}$$

Multiplying each term of the above equation by Z, we get

$$\frac{G_S \theta_B (Y_0 - Y_0^*) Z}{(X_T - X_0) Z_s \rho_S} = Z - LUB$$

but at θ_s : $Z = Z_s$

$\therefore$ $$\frac{G_S \theta_B (Y_0 - Y_0^*)}{\rho_S (X_T - X_0)} = Z - LUB$$

$$LUB = Z - \frac{G_S \theta_B (Y_0 - Y_0^*)}{\rho_S (X_T - X_0)} \quad \ldots (3.61)$$

For a given breakpoint time, Z_s is calculated by using Equation (3.58) letting $\theta = \theta_B$. Equation (3.61) gives LUB.

[For real adsorption, i.e., adsorption with finite rate of mass transfer, the adsorbent between the inlet of the bed and the start of the adsorption zone is completely saturated. The adsorbent in the adsorption zone goes from nearly saturated to almost no adsorbate, and for a rough average, the adsorbent in the zone could be assumed to be half saturated and half unused. This is equivalent of having about half of the adsorbent in the zone fully saturated and half unused. So when the adsorption zone is reduced to a plane, the breakthrough curve is a vertical line when shaded areas on both sides of the real curve are equal.]

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Ion Exchange :

It is a solid-liquid operation in which the solute is retained by a chemical reaction with the solid ion-exchange resin. Ion exchange can be thought of a special case of adsorption, wherein an exchange solid (ion exchange resin) serves as the adsorbent.

The reversible exchange of ions of an electrolyte in a solution with ions of the same charge of a certain insoluble solid that thus permits the separation of the electrolytic solute is called ion exchange. Ions in solution are removed by this process.

Applications of ion exchange include – (i) softening of water, demineralisation of water (water treatment), (ii) recovery of antibiotics from fermentation broths, (iii) separation of rare-earth metals (using chromatographic techniques), (iv) desalting of brackish water, etc.

Principles of Ion Exchange :

An insoluble solid material used to remove ions in a solution contains immobile or fixed groups that carry an ionic charge (i.e., fixed ionic groups) either +ve or –ve in conjugation with free/diffusible ions of opposite charge that can be displaced. The charges of fixed ionic groups are balanced by the diffusible ions of opposite charge, so the exchange solid as a whole is electrically neutral. The insoluble solid material that is used to exchange like ions with the solution is called an ion exchanger. Zeolites (early exchanger) are naturally occurring clays that are capable of exchanging cations (sodium cation exchanger). Most ion exchangers currently in large scale use are synthetic resinous materials.

For example, cation exchange resin is a high molecular weight cross-linked polymer containing sulphonic, carboxylic or phenolic groups as an integral part of the resin (bound groups) and equivalent amounts of cations. In other words, cation exchanger consists of a polymeric anion and an active cation (displaceable cation). Anion exchange resin is a polymer containing quaternary ammonium or amino groups as an integral part of the resin and an equivalent amount of anions such as OH^-, Cl^-, etc. In other words, anion exchange resin consists of a polymeric cation and an active/diffusible anion.

Cation exchange takes place when the fixed charge groups (bound groups) of the exchanger are –ve and the solid is referred to as a cation or cationic exchanger. Their anion is responsible for insolubility in a solution and their cation (Na^+ or H^+) can take part under certain conditions in an exchange reaction with cations in the solution. So we have : sodium cation exchanger and hydrogen cation exchanger.

Consider the case of water softening to get the idea regarding an ion exchange process.

Sodium cation exchanger : Here the exchange cation is sodium. When hard water is passed through a bed of a sodium cationite (a cation exchanger), positively charged ions (cations) of water such as calcium and magnesium will be exchanged for the Na^+ ions of the cation exchanger. The exchange reaction is reversible and proceeds as follows :

$$Ca^{2+} + 2Na^+ R^- \rightleftharpoons Ca^{2+} R_2^- + 2Na^+$$

$$\underset{\text{(solution)}}{Mg^{2+}} + \underset{\text{(solid)}}{2Na^+ R^-} \rightleftharpoons \underset{\text{(solid)}}{Mg^{2+} R_2^-} + \underset{\text{(solution)}}{2Na^+}$$

where the symbol $Na^+ R^-$ represents the sodium cationite (cation exchanger) and R^- is a complex radical of the insoluble cationite that acts as a multivalent anion (may be of zeolite or synthetic ion exchange resin).

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So the hard water is softened by removing hardness causing ions, Ca^{2+} and Mg^{2+} from water (these ions are now immobilised in the cationite).

When the cation exchanger reaches a specific limit of exchange capability, it is regenerated with a solution of NaCl (wherein the cations captured by the cationite are removed and replaced by exchange sodium ions).

The regeneration of a cation exchanger with a solution of NaCl proceeds as follows :

$$\underset{\text{(solid)}}{CaR_2} + \underset{\text{(solution)}}{2\,NaCl} \rightleftharpoons \underset{\text{(solid)}}{2\,NaR} + \underset{\text{(solution)}}{CaCl_2}$$

Hydrogen cation exchanger : Here the exchange cation is hydrogen. In hydrogen cation exchange water softening, the following reactions take place :

$$\underset{\text{(solution)}}{Na^+} + \underset{\text{(solid)}}{H^+R^-} \rightleftharpoons \underset{\text{(solid)}}{Na^+R^-} + \underset{\text{(solution)}}{H^+}$$

$$Ca^{2+} + 2\,H^+R^- \rightleftharpoons Ca^{2+}R_2^- + 2\,H^+$$

$$Mg^{2+} + 2\,H^+R^- \rightleftharpoons Mg^{2+}R_2^- + 2\,H^+$$

$$HCO_3^- + H^+ \rightleftharpoons CO_2 + H_2O$$

It is evident from the above reactions that the hydrogen cation exchange process removes completely the alkalinity of water treated, with the result that total dissolved solids content of water diminish.

Here the salts of Ca, Mg, Na, etc. present in water pass into free acids since all cations are replaced by hydrogen cations. The total acidity of water softened by this exchanger will be equal to the sum of the anions of mineral acids Cl^-, SO_4^{2-}, NO_3^- present in the initial water, so water treated by this exchanger is not suitable for use as feed water.

The hydrogen cation exchanger is regenerated with a 1 to 1.5% solution of sulphuric acid.

$$\underset{\text{(solid)}}{CaR_2} + \underset{\text{(solution)}}{H_2SO_4} \rightleftharpoons \underset{\text{(solid)}}{2\,RH} + \underset{\text{(solution)}}{CaSO_4}$$

or $$Ca^{2+}R_2^- + n\,H^+ \rightleftharpoons 2\,R^-H^+ + Ca^{2+} + (n-2)\,H^+$$

$$MgR_2 + H_2SO_4 \rightleftharpoons 2\,RH + MgSO_4$$

$$2\,NaR + H_2SO_4 \rightleftharpoons 2\,NaR + Na_2SO_4$$

Water can also be deionised/demineralised by ion exchange, so that all dissolved solids are removed from it. For this we have to pass water first through a cation exchanger and then through an anion exchanger.

Anion exchange occurs when the fixed functional groups of the exchanger are positive and the solid is referred to as an anion or anionic exchanger. There is an exchange of anions between electrolytes in water and insoluble solid material. The anion, OH^-, is usually used as an exchanger ion/mobile ion in the anion exchanger.

When water is passed through the bed of an anion exchanger, the following anion exchange reactions occur :

$$R^+OH^- + Cl^- \rightleftharpoons R^+Cl^- + OH^-$$

$$2\,R^+OH^- + SO_4^{2-} \rightleftharpoons R_2^+SO_4^{2-} + 2\,OH^-$$

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Anion exchangers are usually regenerated with a 4% solution of sodium hydroxide.

$$\underset{\text{(solid)}}{RCl} + \underset{\text{(solution)}}{NaOH} \rightleftharpoons \underset{\text{(solid)}}{ROH} + \underset{\text{solution}}{NaCl}$$

$$R_2SO_4 + 2\,NaOH \rightleftharpoons 2\,ROH + Na_2SO_4$$

or $$R^+Cl^- + n\,OH^- \rightleftharpoons ROH^- + Cl^- + (n-1)\,OH^-$$

An important characteristic of ion exchange material is their exchange capacity - the ability of ion exchanger to acquire a certain number of ions under specified conditions. This exchange capacity determines how long the operating cycle of an ion exchange unit can last. When a specified limit is reached then the ion exchange material is subjected to regeneration - an ion exchange in the reverse direction.

The methods that are used to carry out adsorption operation can normally be adopted for conducting ion exchange operations. Thus, we have a single or multistage treatment of solution (batch or continuous), fluidised - or fixed bed operations. Fixed bed operations (involving percolation of electrolyte solution through ion exchange resins) are the most common.

Equilibria :

The phase equilibrium relationship (the equilibrium distribution of an ion between an ion exchange solid and a solution) in ion exchange process at a given temperature is given by the expression similar in type to that of adsorption isotherms. In addition, the mass-action relationships applied to the exchange reaction can also be used to describe the ion exchange equilibrium.

For example, for the cation exchange reaction,

$$\underset{\text{(solution)}}{Na^+} + \underset{\text{(solid)}}{R\text{-}H^+} \rightleftharpoons \underset{\text{(solid)}}{R\,Na^+} + \underset{\text{(solution)}}{H^+}$$

According to the law of mass-action, at equilibrium :

$$\alpha = \frac{(\text{conc. } R - Na^+)_{solid}\,(\text{conc. } H^+)_{solution}}{(\text{conc. } R - H^+)_{solid}\,(\text{conc. } Na^+)_{solution}}$$

$$= \left(\frac{\text{conc. } Na^+}{\text{conc. } H^+}\right)_{solid} \left(\frac{\text{conc. } H^+}{\text{conc. } Na^+}\right)_{solution} \quad \ldots (3.62)$$

where α is the equilibrium constant for the exchange reaction.

The quantity α is therefore seen to be an expression for the relative adsorptivity, in the present case it is the relative adsorptivity of Na^+ with respect to H^+ (The separation factor is the ratio of concentration ratio of A to B in one phase to that in the other phase).

For most engineering applications, α can be expressed as :

$$\alpha = \left(\frac{X}{X_0 - X}\right)\left(\frac{C_0 - C^*}{C^*}\right) = \left(\frac{X/X_0}{1 - X/X_0}\right)\left(\frac{1 - C^*/C_0}{C^*/C_0}\right) \quad \ldots (3.63)$$

where C^* is the equilibrium concentration of Na^+ after exchange in solution

C_0 is the initial concentration of $Na^+ + H^+$ in solution and so their total at any time.

$$\therefore \left(\frac{\text{conc. } H^+}{\text{conc. } Na^+}\right)_{solution} \text{ at equilibrium } = \frac{C_0 - C^*}{C^*}$$

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X is the equilibrium concentration of Na^+ in solid

X_0 is the concentration if all H^+ were replaced by Na^+.

$$\therefore \quad \left(\frac{\text{conc. } Na^+}{\text{conc. } H^+}\right)_{\text{solid}} = \frac{X}{X_0 - X}$$

Putting the values of the RHS terms of Equation (3.62) gives

$$\therefore \quad \alpha = \frac{X}{X_0 - X} \cdot \frac{C_0 - C^*}{C} = \left(\frac{X/X_0}{1 - X/X_0}\right)\left(\frac{1 - C^*/C_0}{C^*/C_0}\right) \quad \ldots (3.64)$$

All the quantities in the above equation are expressed as equivalents per unit volume or mass.

Rate of Ion Exchange :

Ion exchange is a complex phenomenon that involves the following steps in series and the rate of ion exchange depends on the rates of these steps.

(i) Diffusion of ions from the bulk liquid phase to the external surface of the resin.

(ii) Diffusion of ions from the surface of the resin through the resin to the site of exchange.

(iii) Exchange of ions at the active sites.

(iv) Diffusion of the released ions from inside to the surface of the resin.

(v) Diffusion of the released ions from the surface to the bulk of liquid.

Any one of the five steps is a rate controlling or rate determining. The rate of ion exchange is given by the rate of the slowest step in series. If the step (iii) is the slowest then the rate of ion exchange is given by the rate of the exchange reaction. If the exchange of ions [step (iii)] is very rapid compared to the diffusion steps then ion exchange equipment can be designed like conventional adsorbers. If the diffusional steps are rapid (exchange step is the rate controlling), then kinetic procedures are required.

SOLVED EXAMPLES

Ex. 3.1 : A volume of 1 m^3 contains a mixture of air and acetone vapour. The temperature is 303 K (30 °C) and the total pressure is 10^5 N/m^2.

If the relative saturation of the air by acetone vapour is 40%, determine the quantity of activated carbon that must be added to the space for reducing the relative saturation to 5% at 303 K (30 °C).

If, in fact, 1.6 kg carbon is added, what will be percent relative saturation of the equilibrium mixture assuming temperature to be the same ?

Data : The vapour pressure of acetone at 303 K = 37.9 kN/m^2.

Adsorption equilibrium data at 303 K for acetone on carbon are :

Partial pressure of acetone × 10^{-2}, N/m^2	0	5	10	30	50	90
kg acetone/kg carbon	0	0.14	0.19	0.27	0.31	0.35

Solution : Basis : 1 m^3 volume of acetone vapour-air mixture at 303 K.

Final relative saturation = 5%.

% Relative saturation = $p_A/p_S \times 100$

p_A – partial pressure of acetone

p_S – vapour pressure of acetone.

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$\therefore$ Final partial pressure of acetone $= \frac{5}{100} \times 37.9 \times 1000 = 1895\ N/m^2$.

Plot X v/s partial pressure of acetone.

From Fig. E 3.1 :

Acetone adsorbed on the carbon at equilibrium (corresponding to $p_A = 1895\ N/m^2$) $= 46.5\ mm \times \frac{0.10\ units}{20\ mm}$

$= 0.2325$ kg/kg carbon

Acetone initially present $= n_{A1} = p_{A1} V/RT$

Acetone finally present $= n_{A2} = p_{A2} V/RT$

Acetone removed from air (and adsorbed on carbon) $= n_{A1} - n_{A2}$

$p_{A1} = 0.4 \times 37.9 = 15.16\ kN/m^2 = 15.16\ kPa$

$p_{A2} = 0.05 \times 37.9 = 1.895\ kN/m^2 = 1.895\ kPa$

$R = 8.31451\ m^3 \cdot kPa/(kmol.K)$

$V = 1\ m^3,\ T = 303\ K$

$$\text{Acetone removed from air} = \frac{p_{A1} V}{RT} - \frac{p_{A2} V}{RT}$$

$$= \frac{V}{RT}[p_{A1} - p_{A2}]$$

$$= \frac{1}{8.31451 \times 303}[15.16 - 1.895]$$

$$= 5.265 \times 10^{-3}\ kmol$$

Molecular weight of acetone $= 58$

Acetone removed from air $= 5.265 \times 10^{-3} \times 58$

$= 0.3054$ kg

Acetone in carbon at equilibrium $= 0.2325$ kg/kg

i.e., for 0.2325 kg of acetone adsorption, 1 kg of carbon is required.

Activated carbon that must be added for 0.3054 kg of acetone adsorption $= \frac{1}{0.2325} \times 0.3054$

$= 1.314$ kg ... **Ans. (i)**

Case II : Determination of % relative saturation :

Amount of carbon added = 1.6 kg (given)

Let X be the kg acetone removed/adsorbed per kg of carbon.

Then acetone balance is

Acetone adsorbed = Acetone removed from gas mixture

$$1.6X = \frac{V}{RT}(p_{A1} - p_{A2}) \times 58$$

(RHS is multiplied by mol. wt. 58 for converting kmol in kg.)

$p_{A1} = 0.4 \times 37.9 = 15.16\ kN/m^2 = 15.16\ kPa$

p_{A2} = final partial pressure of acetone in kN/m^2

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$$1.6\,X = \frac{1}{8.31451 \times 303}\,[15.16 - p_{A2}] \times 58$$

$$\therefore \quad X = 0.22 - 1.44 \times 10^{-2}\, p_{A2}$$

We have to find out the point of intersection of the above line with the equilibrium curve to get the value of equilibrium partial pressure of acetone.

We have to plot this line.

Take $p_{A2} = 0$, then $X = 0.22$.

Take $p_{A2} = 6\ kN/m^2 = 6000\ N/m^2$, then $X = 0.1336$.

Refer Fig. E 3.1

Draw this line passing through the points (0, 0.22) and (6000, 0.1336). This line will cut the equilibrium curve at point P. Read the coordinates of the point P.

$$X = 0.2025 \text{ kg/kg carbon}$$

and $$p = p_{A2} = 1200\ N/m^2$$

Final relative saturation of air $= (p_{A2}/p_S) \times 100$

$$= \frac{1200}{37900} \times 100$$

$$= 3.17\% \qquad \text{... Ans. (ii)}$$

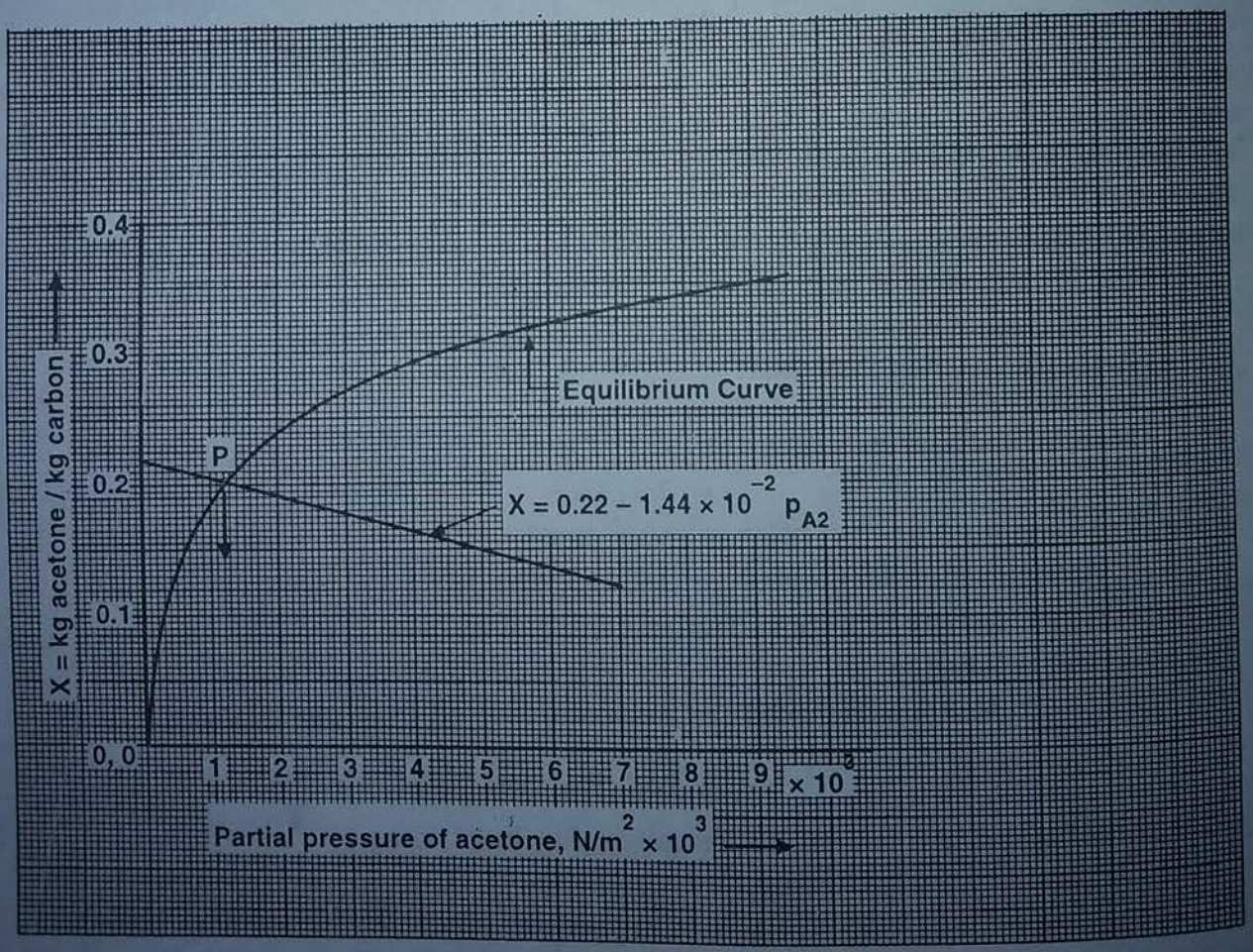

Fig. E 3.1 : Adsorption equilibrium curve

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Ex. 3.2 : The equilibrium adsorption of acetone vapour on an activated carbon at 303 K (30 °C) is given by the following data :

g adsorbed/g carbon	0	0.1	0.2	0.3	0.35
Partial pressure of acetone, mmHg	0	2	12	42	92

The vapour pressure of acetone at 303 K is 283 mmHg. A 1 litre flask contains air and acetone vapour at 1 atm and 303 K (30 °C), with a relative saturation of vapour of 35%. After 2 g of fresh activated carbon has been introduced into the flask, the flask is sealed. Calculate the final vapour concentration at 303 K and the final pressure neglecting the adsorption of air.

Solution : Basis : 1 litre flask containing air and acetone at 303 K and 1 atm pressure.

Molecular weight of acetone [CH_3COCH_3] = 58 kg/kmol.

Air contains 21% O_2 and 79% N_2 by volume.

Average molecular weight of air = $M_{avg} = 0.21 \times 32 + 0.79 \times 28 = 28.84$ kg/kmol.

Initial total pressure in the flask = 1 atm = 760 mmHg.

Relative saturation of acetone in air (initial) = 35%.

$$\text{Relative saturation} = \frac{p_A}{p_S} \times 100$$

where p_A is the partial pressure of acetone.

p_S is the vapour pressure of acetone.

At 303 K, p_S = 283 mmHg (given).

$$35 = \frac{p_A}{283} \times 100$$

$$p_A = 0.35 \times 283 = 99.05 \text{ mmHg}$$

$$\text{Partial pressure of air} = P - p_A$$

$$= 760 - 99.05 = 660.95 \text{ mmHg}$$

As the air is not adsorbed, its partial pressure will remain constant at 660.95 mmHg.

Now, we will convert partial pressure data into more useful form – g acetone/g air.

$$\text{Partial pressure of acetone} = \text{molefraction of acetone} \times P$$

$$\text{Partial pressure of air} = \text{molefraction of air} \times P$$

$$\frac{\text{Partial pressure of acetone}}{\text{Partial pressure of air}} = \frac{\text{molefraction of acetone}}{\text{molefraction of air}}$$

$$= \frac{\text{moles of acetone/total moles}}{\text{moles of air/total moles}}$$

$$\frac{\text{Partial pressure of acetone}}{\text{Partial pressure of air}} = \frac{\text{moles of acetone}}{\text{moles of air}} = \frac{\text{g acetone/58}}{\text{g air/28.84}}$$

$$\frac{\text{g acetone}}{\text{g air}} = \frac{\text{Partial pressure of acetone} \times 58}{\text{Partial pressure of air} \times 28.84}$$

Initially :

$$\text{partial pressure of acetone} = 99.05 \text{ mmHg}$$

$$\text{partial pressure of air} = 660.95 \text{ mmHg}$$

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$$\frac{\text{g acetone}}{\text{g air}} \text{ (initial)} = Y_0 = \frac{99.05 \times 58}{660.95 \times 28.84}$$
$$= 0.3014$$

As the adsorbent is pure, $X_0 = 0$.

Convert the partial pressure data in g acetone/g air and tabulate. For each time, partial pressure of air is 660.95 mmHg.

g adsorbed/g carbon = X	0	0.10	0.2	0.30	0.35
Partial pressure of acetone, mmHg	0	2	12	42	92
g acetone/g air = Y	0	6.085×10^{-3}	0.0365	0.1278	0.28

Sample calculation :

Partial pressure of acetone = 12 mmHg

$$\frac{\text{g acetone}}{\text{g air}} = \frac{\text{Partial pressure of acetone} \times 58}{\text{Partial pressure of air} \times 28.84}$$
$$= \frac{12 \times 58}{660.95 \times 28.84} = 0.0365$$

Plot the Y v/s X data : the equilibrium data.

Amount of air remained unchanged as it is not going to adsorb [i.e. it remains same before and after adsorption of acetone].

$$\text{Moles of air} = p_{air} \times V/RT$$
$$= \frac{(660.95/760) \times 1}{0.08206 \times 303} = 0.035 \text{ kmol}$$

$[R = 0.08206 \text{ m}^3 \cdot \text{atm/(kmol.K)}]$

$$\text{Mass of air} = L_S = 0.035 \times 28.84 = 1.0094 \text{ g}$$

$$\text{Mass of adsorbent} = S_S = 2 \text{ g}$$

The operating line for single stage adsorption is

$$L_S (Y_0 - Y_1) = S_S (X_1 - X_0)$$

On X-Y coordinates, it is a straight line passing through the points (X_0, Y_0) and (X_1, Y_1) with a slope equal to $-S_S/L_S$. The point (X_1, Y_1) for equilibrium/theoretical stage lies on the equilibrium adsorption isotherm.

$$\frac{-S_S}{L_S} = \frac{-2}{1.0094} = -1.9814$$
$$\tan\theta = -1.9814 \quad \therefore \quad \theta = -63.22^\circ$$

For pure adsorbent, $X_0 = 0$, and $Y_0 = 0.3014$.

So draw the operating line through the point (0, 0.3014) with an angle -63.22° which will cut the equilibrium curve at point P.

Read the coordinates of the point P :

y-coordinate : $Y_1 = 0.0135$ g acetone/g air

$X_1 = 0.144 \rightarrow$ not required for calculation

$$\frac{\text{g acetone}}{\text{g air}} = 0.0135$$

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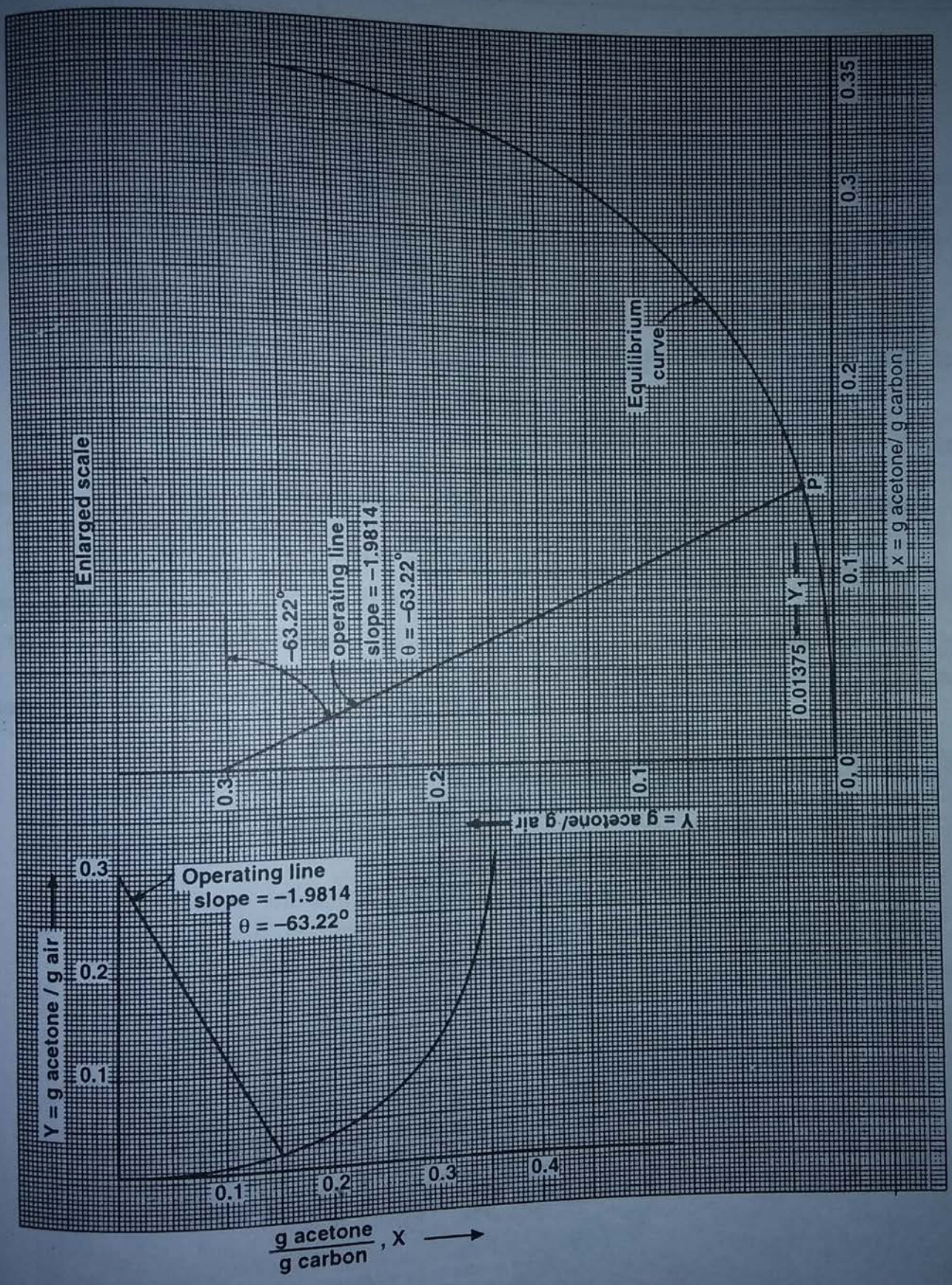

Fig. E 3.2 (a) : Single stage adsorption

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$$\frac{\text{gram mole acetone}}{\text{gram mole air}} = \frac{\text{partial pressure of acetone}}{\text{partial pressure of air}}$$

$$\frac{\text{g acetone/58}}{\text{g air/28.84}} = \frac{\text{partial pressure of acetone}}{\text{partial pressure of air}}$$

$$\frac{\text{partial pressure of acetone}}{\text{partial pressure of air}} = \frac{\text{g acetone}}{\text{g air}} \times \frac{28.84}{58} = Y_1 \times \frac{28.84}{58}$$

So final partial pressure of acetone :

$$\frac{\text{partial pressure of acetone}}{\text{partial pressure of air}} = 0.0135 \times \frac{28.84}{58}$$

$$\text{Final partial pressure of acetone} = \frac{0.0135 \times 28.84}{58} \times \text{partial pressure of air}$$

$$= \frac{0.0135 \times 28.84 \times 660.95}{58}$$

$$p_{Af} = 4.44 \text{ mmHg}$$

Final partial pressure of acetone after adsorption = 4.44 mmHg ... **Ans.**

$$\text{Final relative saturation of air after adsorption} = \frac{p_{Af}}{p_S} \times 100 = \frac{4.44}{283} \times 100$$

$$= 1.5689 \approx 1.57\%$$ ... **Ans.**

Final total pressure in the flask = partial pressure of acetone + partial pressure of air

$$= 4.44 + 660.95$$

$$= 665.39 \text{ mmHg}$$ ... **Ans.**

OR Alternatively :

Relative saturation (initial) = 35%

$$\text{Initial partial pressure of acetone in air} = 0.35 \times 283 = 99.05 \text{ mmHg}$$

Amount of carbon used = 2 g (pure)

Let X be the concentration of acetone in g acetone/g carbon after adsorption.

Material balance of acetone :

Acetone adsorbed = Acetone removed from gas mixture

$$2X = \frac{V}{RT}[p_{A1} - p_{A2}] \times 58$$

Molecular weight of acetone = 58 g/mol

p_{A1} = 99.05/760 atm

p_{A2} = final partial pressure of acetone in atm.

T = 303 K

R = 0.08206 l.atm/(mol.K)

V = 1 l

$$2X = \frac{1}{0.08206 \times 303}\left[\frac{99.05}{760} - p_{A2}\right] \times 58$$

$$X = 0.152 - 1.105\, p_{A2}, \quad (p_{A2} \text{ is in atm units})$$ ... (1)

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From Equation (1) :

For $p_{A2} = 0$ atm, $X = 0.152$

For $p_{A2} = 80/760$ atm, $X = 0.0294$.

i.e. for partial pressure of acetone = 80 mmHg, $X = 0.0294$.

Plot X v/s partial pressure of acetone – the equilibrium curve.

Draw a line ($X = 0.152 - 1.105\ p_{A2}$) through the points (0, 0.152) and (80, 0.0294) which will cut the equilibrium curve at point Q. (For theoretical stage, the effluent streams are in equilibrium, so point [X_1, Y_1 (i.e., Q)] lies on the equilibrium curve.) Read the coordinates of the point Q.

y-coordinate : $X_1 = 0.145$

x-coordinate : $p_1 = p_{A2} = 4.5$ mmHg

Partial pressure of acetone after adsorption = 4.5 mmHg ... Ans.

$$\text{Final relative saturation of air} = \frac{4.5}{283} \times 100$$

$$= 1.59\% \quad \text{... Ans.}$$

$$\text{Initial partial pressure of air} = \left(\begin{matrix}\text{Initial}\\ \text{total pressure}\end{matrix}\right) - \left(\begin{matrix}\text{Initial partial}\\ \text{pressure of acetone}\end{matrix}\right)$$

$$= 760 - 99.05$$

$$= 660.95 \text{ mmHg}$$

As air is not getting adsorbed, its partial pressure remains constant at 660.95 mmHg.

$\therefore$ Final pressure in the flask = 660.95 + 4.5

= 665.45 mmHg ... Ans.

There is a small variation in the results due to small error that may crept in plotting the graph.

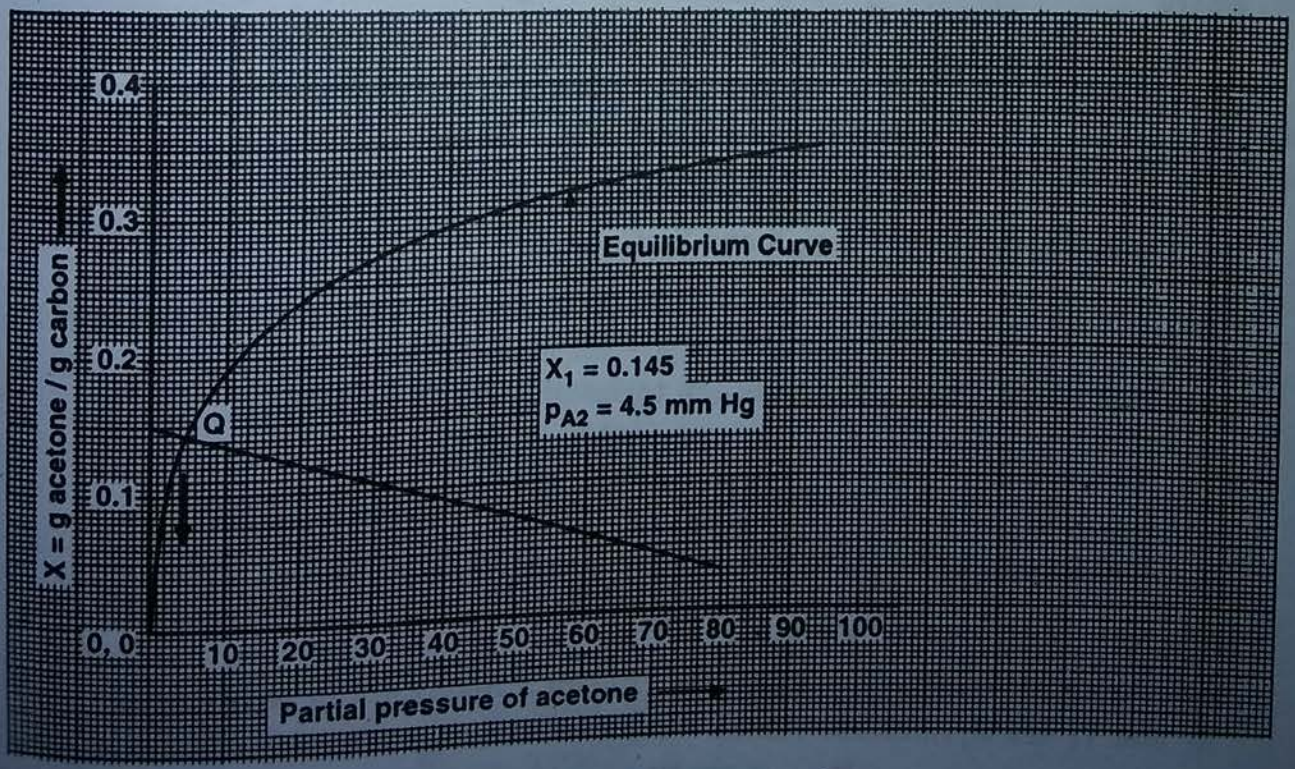

Fig. E 3.2 (b) : Adsorption isotherm

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Ex. 3.3 : Experiments on decolourisation of oil yielded the following equilibrium relationship :

$$y = 0.5\, x^{0.5}$$

where y = g colour removed/g adsorbent

and x = colour in oil, g colour/1000 g colour-free oil

100 kg oil containing 1 part of colour to 3 parts of oil is agitated with 25 kg of adsorbent. Calculate the % colour removed if

(i) all 25 kg of adsorbent is used in one step.

(ii) 12.5 kg of adsorbent is used initially, followed by another 12.5 kg of adsorbent.

Solution : Basis : 100 kg of feed oil.

Feed oil that is coloured oil contains 1 part of colour and 3 parts of oil by weight.

$$\therefore \quad \text{Colour in feed oil} = \frac{1}{(1+3)} \times 100 = 25 \text{ kg}$$

$$\text{Colour-free oil in feed oil} = \frac{3}{(1+3)} \times 100 = 75 \text{ kg}$$

L_S = colour-free oil (solvent) = 75 kg.

Case I : 25 kg of adsorbent is used in one step/single stage.

$$S_S = 25 \text{ kg}$$

Equilibrium relationship is given as

$$y = 0.5\, x^{0.5}$$

where y = g colour removed/g adsorbent

x = colour in oil, g colour/1000 g colour-free oil.

Transform this equilibrium relationship in terms of Y and X (usual notations).

where Y = g colour/g colour-free oil

X = g colour/g adsorbent

From above definitions of x, y, X and Y, we have,

$$y = X$$

and $$x = 1000\, Y$$

Therefore, equilibrium relationship in terms of X and Y becomes

$$X = 0.5\,(1000\, Y)^{0.5}$$

or $$Y = 0.004\, X^2$$

kg colour/kg colour-free oil in feed $= Y_0 = \frac{25}{75} = 0.3333.$

Adsorbent is pure, so $X_0 = 0.0$.

For theoretical/equilibrium stage, the effluent streams are in equilibrium.

X_1 and Y_1 are equilibrium values.

$$Y = 0.004\, X^2$$

$$\therefore \quad Y_1 = 0.004\, X_1^2$$

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Material balance of solute-colour :

$$L_S (Y_0 - Y_1) = S_S (X_1 - X_0)$$

$$75 (0.3333 - Y_1) = 25 (X_1), \text{ as } X_0 = 0.0$$

but $Y_1 = 0.004 X_1^2$

$$75 (0.3333 - 0.004 X_1^2) = 25 X_1$$

$$25 - 0.3 X_1^2 = 25 X_1$$

$$0.3 X_1^2 + 25 X_1 - 25 = 0$$

$$\therefore \quad X_1 = \frac{-25 \pm \sqrt{(25)^2 - 4 \times 0.3 (-25)}}{2 \times 0.3}$$

$$= 0.9883 \text{ (positive root)}$$

X_1 = kg colour adsorbed/kg adsorbent = 0.9883

i.e. For each kg of adsorbent used, 0.9883 kg of colour is removed.

Adsorbent used = 25 kg

$$\therefore \quad \text{Colour removed} = \frac{0.9883}{1} \times 25 = 24.7075 \text{ kg}$$

$$\% \text{ original colour removed} = \frac{24.7075}{25} \times 100$$

$$= 98.83 \quad \text{... Ans. (i)}$$

We have, $Y_1 = 0.004 X_1^2$

$$= 0.004 (0.9883)^2 = 3.90695 \times 10^{-3}$$

$$\% \text{ colour removed} = \frac{(Y_0 - Y_1)}{Y_0} \times 100$$

$$= \frac{(0.3333 - 3.90695 \times 10^{-3})}{0.3333} \times 100$$

$$= 98.8278 \approx 98.83 \quad \text{... Ans. (i)}$$

Case II : Two-stage crosscurrent operation with 12.5 kg of adsorbent used in each stage.

Stage-1 : Adsorbent used = 12.5 kg

S_S = 12.5 kg

Material balance of colour over stage-1 :

$$L_S (Y_0 - Y_1) = S_S (X_1 - X_0)$$

$$75 (0.3333 - Y_1) = 12.5 (X_1), \quad X_0 = 0.0$$

$$Y_1 = 0.004 X_1^2$$

$$75 (0.3333 - 0.004 X_1^2) = 12.5 X_1$$

$$0.3 X_1^2 + 12.5 X_1 - 25 = 0$$

$$\therefore \quad X_1 = \frac{-12.5 \pm \sqrt{(12.5)^2 - 4 \times 0.3 \times (-25)}}{2 \times 0.3}$$

$$= 1.912 \text{ kg colour/kg adsorbent}$$

We have, $Y_1 = 0.004 X_1^2$

$$Y_1 = 0.004 (1.912)^2 = 0.01462 \text{ kg colour/kg colour-free oil}$$

Stage-2 : $\therefore$

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Material balance for second stage using 12.5 kg of pure adsorbent :

$$L_S(Y_1 - Y_2) = S_S(X_2 - X_0)$$

X_2 and Y_2 are in equilibrium and $X_0 = 0.0$

$$Y_1 = 0.01462$$

$$Y_2 = 0.004\,X_2^2$$

$$75\,(0.01462 - 0.004\,X_2^2) = 12.5\,X_2$$

$$1.0965 - 0.3\,X_2^2 = 12.5\,X_2$$

$$0.3\,X_2^2 + 12.5\,X_2 - 1.0965 = 0$$

$$X_2 = \frac{-12.5 \pm \sqrt{(12.5)^2 - 4 \times 0.3 \times (-1.0965)}}{2 \times 0.3}$$

$$= 0.08754$$

We have, $Y_2 = 0.004\,X_2^2$

$\therefore$ $Y_2 = 0.004\,(0.08754)^2 = 3.0653 \times 10^{-5}$

$$\% \text{ colour removed} = \left(\frac{Y_0 - Y_2}{Y_0}\right) \times 100$$

$$= \left(\frac{0.3333 - 3.0653 \times 10^{-5}}{0.3333}\right) \times 100$$

$$= 99.99$$ ... **Ans. (ii)**

$$X_1 = 1.912$$

$$X_2 = 0.08754$$

$\therefore$ Colour removed per 1 kg of adsorbent = 1.912 kg in stage-1.

Colour removed per 1 kg of adsorbent = 0.08754 kg in stage-2.

For each stage, adsorbent used = 12.5 kg.

$$\text{Colour removed (totally)} = \frac{1.912}{1} \times 12.5 + \frac{0.08754}{1} \times 12.5$$

$$= 24.99425 \text{ kg}$$

$$\% \text{ of original colour removed from oil} = \frac{24.99425}{25} \times 100$$

$$= 99.98$$ ... **Ans. (ii)**

Ex. 3.4 : An aqueous solution containing valuable solute is coloured by small amounts of an impurity. It is to be decolourised by adsorption of an impurity on an adsorptive carbon. The equilibrium data obtained by stirring various amounts of adsorbent with original solution at constant temperature are as follows :

kg carbon/kg solution	0	0.001	0.004	0.008	0.02	0.04
Equilibrium colour	9.6	8.6	6.3	4.3	1.7	0.7

The original solution has a colour concentration of 9.6 measured on an arbitrary scale and it is desired to reduce the colour to 10% of its original value.

(i) Convert the data to a form suitable for plotting the equilibrium adsorption isotherm

Y = equilibrium colour, colour units/kg solution

X = colour units/kg carbon

Do they follow the Freundlich equation ?

If yes, what are the equation constants ?

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(ii) Determine the quantity of fresh carbon required per 1000 kg of solution for a single stage operation.

Solution : Basis : 1000 kg of solution.

Let us first convert the data to a suitable form for plotting the adsorption isotherm.

Y is defined as – colour units/kg of solution

X is defined as – colour units adsorbed/kg of carbon

It is not necessary to convert the data in Y, as Y values are directly provided in the data given as equilibrium colour.

$$\text{Conversion of the data to X} = \frac{\text{colour units}}{\text{kg carbon}}$$

Refer to the first two columns of the data given.

From the values given, it is clear that when 0.001 kg of carbon is added to 1 kg of initial solution, at equilibrium, the colour concentration of solution drops from 9.6 to 8.6 and will be adsorbed by carbon.

Therefore,

Colour adsorbed by 0.001 kg carbon = 1 (9.6 – 8.6)

So colour adsorbed by 1 kg carbon $= \frac{1\,(9.6 - 8.6) \times 1}{0.001} = 1000$ colour units.

The adsorbate concentration, X, expressed as, colour units/kg carbon = 1000.

OR
$$\frac{\text{Colour units}}{\text{kg of carbon}} = \frac{1\,(9.6 - 8.6)}{0.001} = 1000$$

Similarly, when
$$\frac{\text{kg carbon}}{\text{kg solution}} = 0.004$$

When 0.004 kg carbon is added to 1 kg of initial solution, the colour concentration drops from 9.6 to 6.3. The colour adsorbed by 0.004 kg carbon = 1 (9.6 – 6.3).

Therefore,
$$\frac{\text{colour units}}{\text{kg carbon}} = \frac{1\,(9.6 - 6.3)}{0.004} = 825$$

Adopt the same procedure for converting the remaining data and tabulate it.

$\frac{\text{kg carbon}}{\text{kg solution}}$	Y = Equilibrium colour, $\frac{\text{colour units}}{\text{kg solution}}$	X = Adsorbate concentration, $\frac{\text{colour units}}{\text{kg carbon}}$
0	9.6	–
0.001	8.6	(9.6 – 8.6)/0.001 = 1000
0.004	6.3	(9.6 – 6.3)/0.004 = 825
0.008	4.3	(9.6 – 4.3)/0.008 = 662.5
0.02	1.7	(9.6 – 1.7)/0.02 = 395
0.04	0.7	(9.6 – 0.7)/0.04 = 222.5

Plot this equilibrium data on logarithmic coordinates.

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As the plot results in a straight line, the Freundlich equation is applicable. ... **Ans. (i)**

Or convert the data : X as ln X and Y as ln Y and plot ln Y v/s ln X.

X	1000	825	662.5	395	222.5
ln X	6.91	6.715	6.496	5.98	5.405

Y	8.6	6.3	4.3	1.7	0.7
ln Y	2.152	1.84	1.46	0.531	–0.357

Refer Fig. E 3.4 (a)

Slope of line = $n = 1.67$

We have : at $X = 662.5$, $Y = 4.3$

$\therefore$ $m = Y/X^n = 4.3/(662.5)^{1.67}$

$= 8.36 \times 10^{-5}$

The Freundlich equation is therefore,

$Y = m X^n$

$Y = 8.36 \times 10^{-5} X^{1.67}$ ... **Ans. (i)**

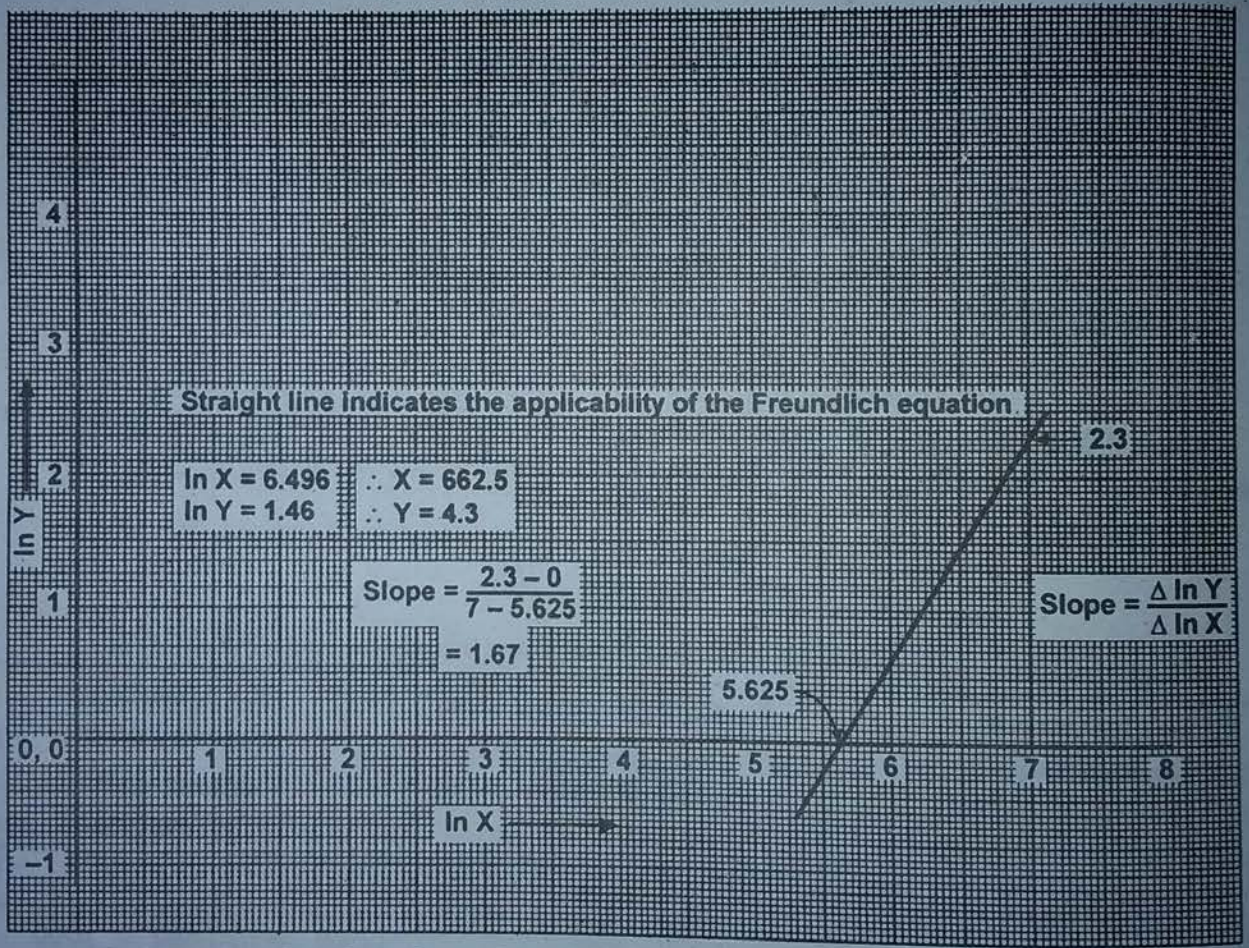

Fig. E 3.4 (a) : Log-log plot to determine constants of the Freundlich equation

For single stage operation :

L_S = 1000 kg of solution.

S_S = Amount of carbon (adsorbent) required.

Initial colour concentration of solution = 9.6

$\therefore$ Y_0 = 9.6 colour units/kg solution

Final colour concentration of solution = 10% of its original value

Solute concentration in single stage will be reduced from Y_0 to Y_1

$Y_1 = 0.1\ (9.6) = 0.96$ colour units/kg solution

As the fresh carbon is to be used, $X_0 = 0.0$.

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The operating line for single stage adsorption is

$$L_S (Y_0 - Y_1) = S_S (X_1 - X_0)$$

The operating line is a straight line with a slope $-S_S/L_S$ and passes through the points (X_0, Y_0) and (X_1, Y_1).

For an equilibrium stage, the effluent streams are in equilibrium, so that the point (X_1, Y_1) lies on the equilibrium adsorption isotherm.

Plot Y v/s X.

Locate point P (0, 9.6) representing the initial solution and fresh solvent.

Locate point Q on the equilibrium curve at the colour concentration of final solution, i.e., at $Y = Y_1 = 0.96$ and read the corresponding equilibrium value X_1, colour units adsorbed/kg carbon.

From the plot : $X_1 = 270$ colour units/kg carbon

$$L_S (Y_0 - Y_1) = S_S (X_1 - X_0)$$

$$\frac{S_S}{L_S} = \frac{(Y_0 - Y_1)}{(X_1 - X_0)}$$

$$S_S = \frac{(Y_0 - Y_1)}{(X_1 - X_0)} \times L_S$$

$$= \frac{(9.6 - 0.96)}{(270 - 0)} \times 1000$$

$$= 32 \text{ kg}$$

Fresh carbon required = 32 kg ... **Ans.**

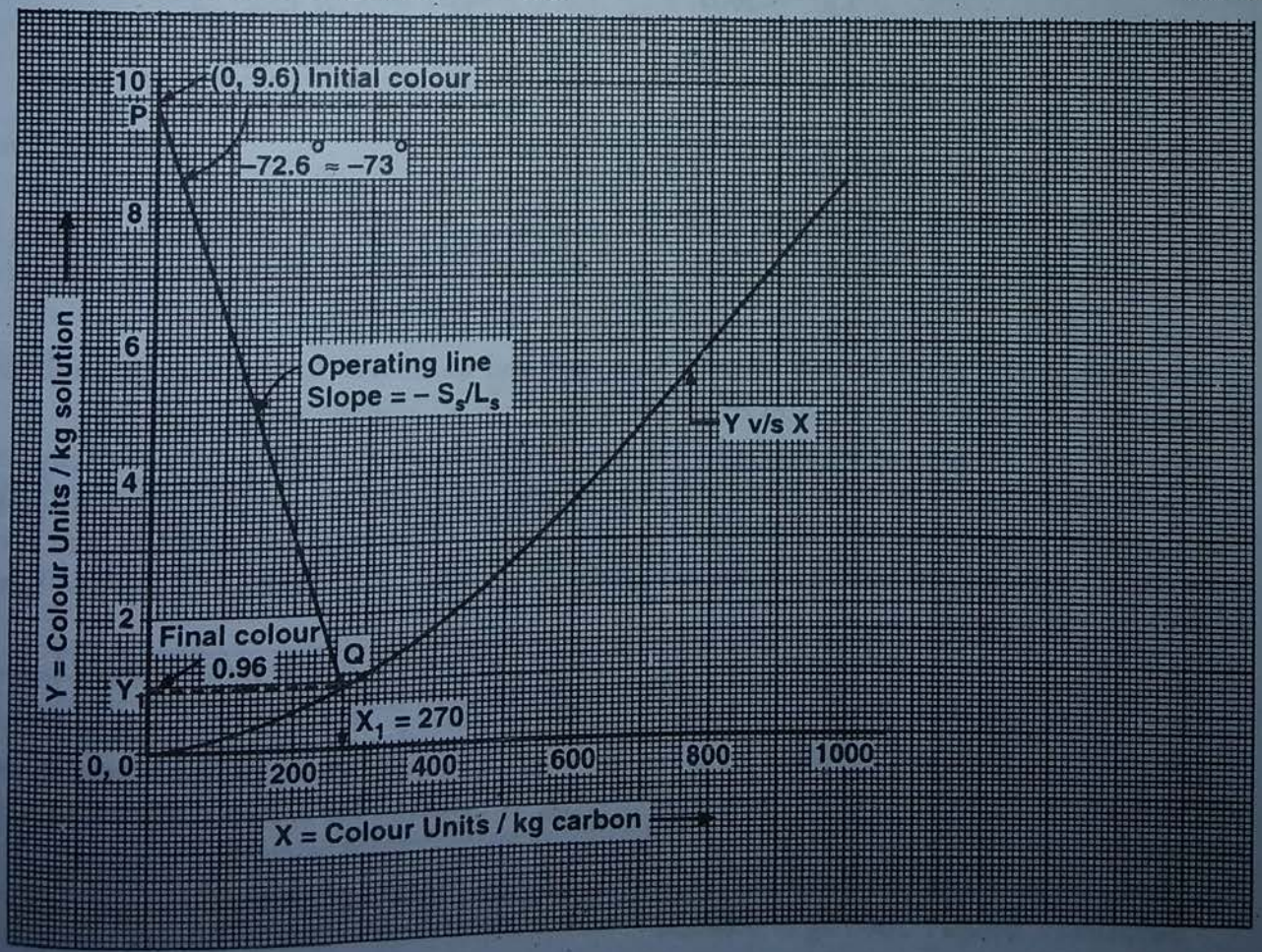

Fig. E 3.4 (b) : Single stage adsorption

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Extra :

Scales of y and x-axes are different.

scale of y-axis : 1 unit = 10 mm

scale of x-axis : 100 units = 10 mm

y-distance = 96 mm – 9.6 mm = 86.4 mm

x-distance = 27 mm – 0 mm = 27 mm

$$\text{Slope} = -\left(\frac{86.4 \text{ mm} \times 1/10}{27 \text{ mm} \times 100/10}\right)$$

$$= -0.032$$

$$\text{Slope} = -\frac{S_S}{L_S} = -0.032$$

$$\frac{S_S}{L_S} = 0.032$$

$$S_S = 0.032 \times 1000 = 32 \text{ kg}$$

If we know S_S as 32 kg and L_S as 1000 kg and we have to find Y_1 and X_1 then we have to draw a line through point P with an angle θ.

$$\frac{S_S}{L_S} = \frac{32}{1000} = 0.032$$

$$-\frac{S_S}{L_S} = -0.032$$

$$\text{slope} = \frac{-0.032 \times 10/1}{10/100} = -3.2$$

$$\text{slope} = -3.2$$

$$\tan\theta = -3.2$$

$$= -72.64^\circ$$

So through the point (0, 9.6) draw a line inclined at 72.64° to the line through (0, 9.6) parallel to x-axis and read Y_1 and X_1 at point Q.

Ex. 3.5 : A solution of washed raw cane sugar is coloured by the presence of impurities. It is to be decolourised by treatment with an adsorptive carbon in a contact filtration plant. The data for an equilibrium isotherm is given below. The original solution has a colour concentration of 9.6 measured on an arbitrary scale and it is desired to reduce colour to 10% of its original value.

kg carbon/kg solution	0	0.001	0.004	0.008	0.02	0.04
Equilibrium colour	9.6	8.6	6.3	4.3	1.7	0.7

Convert the data to a suitable form for plotting the equilibrium isotherm and determine the quantity of fresh carbon required per 1000 kg of solution for a two-stage countercurrent operation.

Hint : Equilibrium colour is expressed as colour units/kg solution.

Solution : Basis : 1000 kg of solution.

$\therefore$ $L_S = 1000$ kg

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First we will convert the data to a form in which it is suitable for plotting the equilibrium isotherm. For this :

Let Y be the colour units per kilogram of solution and X be the colour units adsorbed per kilogram of carbon.

$$Y = \text{equilibrium colour, } \frac{\text{colour units}}{\text{kg solution}}$$

So Y values are directly provided.

Conversion of the data to adsorbate concentration, $X = \frac{\text{colour units}}{\text{kg carbon}}$.

Refer to the first two columns of the data given.

kg carbon/kg solution	0	0.001
Equilibrium colour	9.6	8.6

From the values given, it is clear that when 0.001 kg of carbon is added to 1 kg of original solution, at equilibrium the colour concentration of the solution drops from 9.6 to 8.6 and will be adsorbed on the carbon.

Therefore,

Colour adsorbed by 0.001 kg carbon $= 1 (9.6 - 8.6)$

So colour adsorbed by 1 kg carbon $= \frac{1 (9.6 - 8.6)}{0.001} \times 1 = 1000$ colour units.

The adsorbate concentration, X, is expressed as :

$$\frac{\text{colour units}}{\text{kg carbon}} = 1000$$

or $$\frac{\text{colour units}}{\text{kg carbon}} = \frac{1 (9.6 - 8.6)}{0.001} = 1000$$

Similarly, when

kg carbon/kg solution = 0.004

i.e., when 0.004 kg of carbon is added to 1 kg of original solution, the colour concentration drops from 9.6 to 6.3. Therefore,

Colour adsorbed by 0.004 kg carbon $= 1 (9.6 - 6.3)$.

Hence,

$$\text{colour units/kg carbon} = \frac{1 (9.6 - 6.3)}{0.004} = 825$$

Find X for the remaining data and tabulate the X, Y data.

$\frac{\text{kg carbon}}{\text{kg solution}}$	Y = Equilibrium colour, $\frac{\text{colour units}}{\text{kg solution}}$	X = Adsorbate concentration, $\frac{\text{colour units}}{\text{kg carbon}}$
0	9.6	–
0.001	8.6	(9.6 – 8.6)/0.001 = 1000
0.004	6.3	(9.6 – 6.3)/0.004 = 825
0.008	4.3	(9.6 – 4.3)/0.008 = 662.5
0.02	1.7	(9.6 – 1.7)/0.02 = 395
0.04	0.7	(9.6 – 0.7)/0.04 = 222.5

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Initial colour concentration in solution $= 9.6 \frac{\text{colour units}}{\text{kg solution}}$.

$\therefore$ $Y_0 = 9.6$ colour units/kg solution

Final colour in solution = 10% of its original value

$Y_2 = 0.10\,(9.6) = 0.96$ colour units/kg solution

As the fresh carbon is used, $X_{n+1} = X_0 = 0$.

A solute balance for two-stage countercurrent adsorption is

$$L_S\,(Y_0 - Y_2) = S_S\,(X_1 - X_{n+1})$$

X_{n+1} is colour units/kg carbon in fresh carbon which is equal to zero.

This operating line passes through the points (X_{n+1}, Y_2) and (X_1, Y_0) and has a slope of S_S/L_S.

Plot Y v/s X.

The two ends of the operating line are :

(i) (X_{n+1}, Y_2) i.e. P (0, 0.96) and (ii) (X_1, Y_0) i.e. $(X_1, 9.6)$.

For finding S_S, we must know X_1 and for knowing X_1, we have to locate the operating line by a trial and error procedure so that exactly two stages can be constructed between equilibrium curve and operating line starting from $Y_0 = 9.6$ and reaching exactly $Y_2 = 0.96$.

We know one end of the operating line, the point P (0, 0.96).

The other end of the operating line should lie on a line parallel to X-axis at $Y = Y_0 = 9.6$. So draw the operating line through (0, 0.96), and see that whether it is possible to construct two stages starting from $Y_0 = 9.6$ and ending in $Y_2 = 0.96$. If not, then draw other line and so on till we can construct exactly two stages from Y_0 to Y_2.

Then read the value of X_1 for the correct trial.

Line PQ corresponds to the correct trial. Read x-coordinate of the point Q.

$$X_1 = 675 \text{ colour units/kg carbon}$$

$$S_S = \frac{L_S\,(Y_0 - Y_2)}{(X_1 - X_{n+1})}$$

$$= \frac{1000\,(9.6 - 0.96)}{(675 - 0)} = 12.8 \text{ kg}$$

Quantity of carbon required = 12.8 kg ... **Ans.**

Check : Alternatively,

Measure the angle made by the line PQ to x-axis.

$$\alpha = 52°$$

$$\text{Slope} = (\tan \alpha) \times \text{scale factors}$$

$$= (\tan \alpha) \times \left(\frac{1/10}{100/10}\right)$$

$$= 0.012799 \approx 0.0128$$

but $\text{slope} = S_S/L_S$

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$$\therefore \quad S_S/L_S = 0.0128$$

$$S_S = 0.0128 \times L_S = 0.0128 \times 1000$$

$$= 12.8 \text{ kg} \quad \text{... Ans.}$$

(So the construction is accurate)

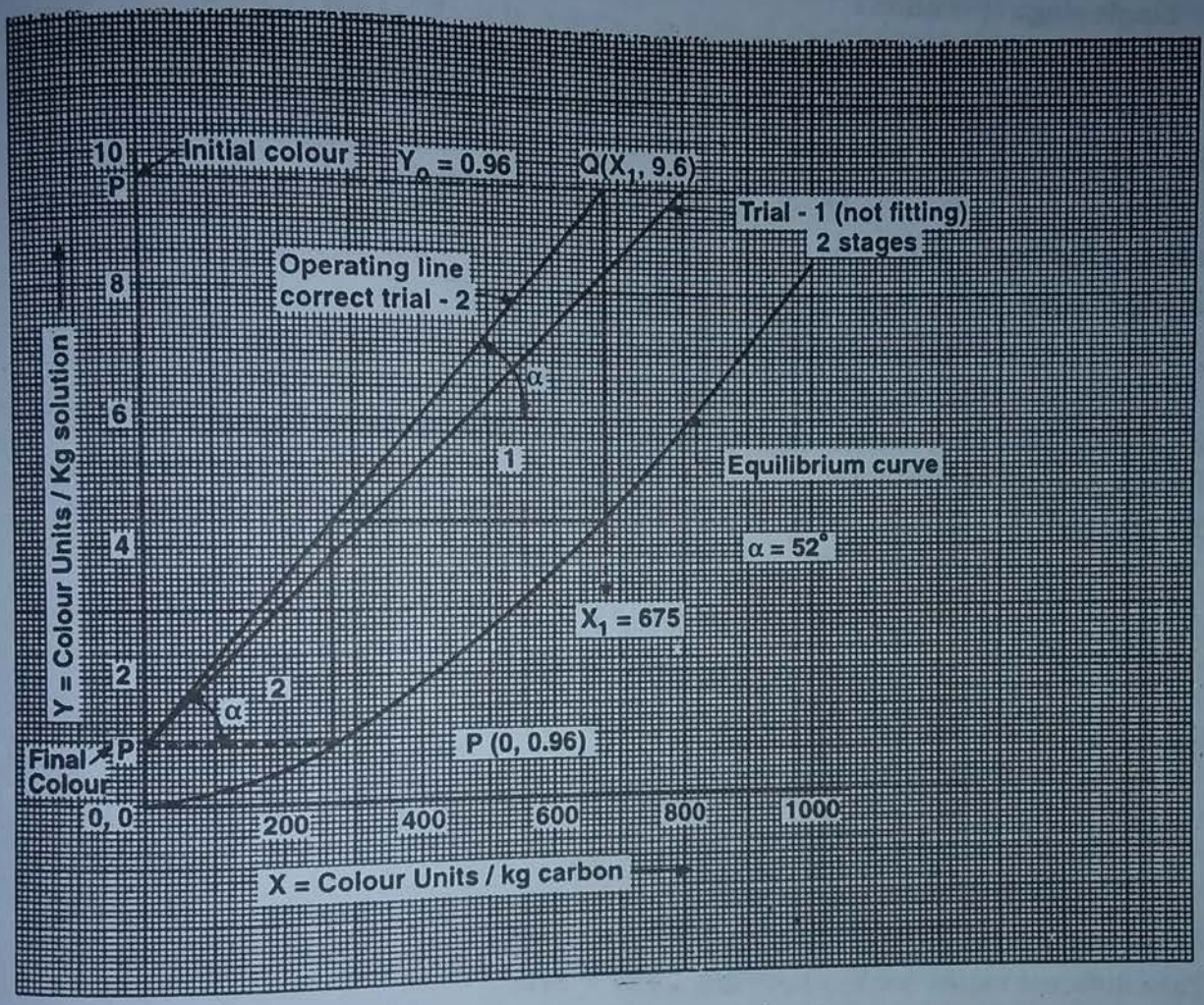

Fig. E 3.5 : Construction of stages to determine amount of adsorbent

Ex. 3.6 : An aqueous solution containing valuable solute is coloured by small amounts of an impurity. Decolourisation experiments of an aqueous solution yielded the following equilibrium relationship :

$$Y^* = 8.91 \times 10^{-5} \ X^{1.66}$$

where

Y = colour units/kg solution

X = colour units/kg carbon

1000 kg of initial solution with colour concentration of 9.6 colour units/kg solution is to be treated with an adsorbent.

(i) Calculate percent of original colour removed in single stage using 32 kg of fresh adsorbent.

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(ii) Calculate the quantity of fresh adsorbent required to reduce the colour to 10% of its original value in two-stage countercurrent operation assuming that colour concentration in the solution stream leaving first stage is 4.6 times the final colour of the solution.

Solution : Basis : 1000 kg of solution.

(i) Single stage operation :

$$L_S = 1000 \text{ kg}$$

$$\text{Adsorbent used} = S_S = 32 \text{ kg}$$

Initial concentration of colour is :

$$Y_0 = 9.6 \text{ colour units/kg solution}$$

As fresh adsorbent used, $X_0 = 0$.

For single stage, we have,

$$L_S (Y_0 - Y_1) = L_S (X_1 - X_0)$$

$$L_S (Y_0 - Y_1) = L_S X_1, \quad \text{as } X_0 = 0$$

For theoretical stage, the effluent streams are in equilibrium, so X_1 and Y_1 lie on the equilibrium adsorption isotherm.

The equilibrium relationship is

$$Y = 8.91 \times 10^{-5} X^{1.66}$$

$$\therefore \quad Y_1 = 8.91 \times 10^{-5} X_1^{1.66}$$

$$X_1 = [Y_1/8.91 \times 10^{-5}]^{1/1.66}$$

$$L_S (Y_0 - Y_1) = S_S X_1$$

$$\frac{S_S}{L_S} = \frac{Y_0 - Y_1}{X_1} = \frac{Y_0 - Y_1}{[Y_1/8.91 \times 10^{-5}]^{1/1.66}}$$

$$\frac{9.6 - Y_1}{[Y_1/8.91 \times 10^{-5}]^{1/1.66}} = \frac{32}{1000} = 0.032$$

Adopt a trial and error procedure to find Y_1.

Take different values for Y_1 till we get LHS = RHS

Take $Y_1 = 2$, LHS = 0.01818

Take $Y_1 = 3$, LHS = 0.0123

$\therefore$ Y_1 has to be less than 2.

$Y_1 = 1$, LHS = 0.03123

$Y_1 = 0.95$, LHS = 0.0324

$Y_1 = 0.96$, LHS = 0.03216 $\approx$ 0.032

$\therefore$ At $Y_1 = 0.96$, LHS = RHS

So final colour of solution = 0.96 colour units/kg solution

$$\% \text{ original colour removed} = \frac{Y_0 - Y_1}{Y_0} \times 100$$

$$= \frac{9.6 - 0.96}{9.6} \times 100$$

$$= 90$$

... Ans. (i)

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So the % original colour retained = 100 – 90 = 10.

So colour of the solution is reduced to 10% of its original value.

(ii) Two-stage countercurrent operation :

L_S = 1000 kg, S_S = ?

Y_0 = 9.6 colour units/kg solution

Given : Colour of the solution is reduced to 10% of its original value.

Final colour of the solution = $Y_2 = 0.1\,(9.6) = 0.96 \frac{\text{colour units}}{\text{kg solution}}$

As the fresh adsorbent is used, $X_{n+1} = 0$.

For two-stage countercurrent operation, we have,

$$L_S\,[Y_0 - Y_2] = S_S\,[X_1 - X_{n+1}]$$

$$L_S\,[Y_0 - Y_2] = S_S X_1, \qquad X_{n+1} = 0.$$

$$S_S = \frac{L_S\,(Y_0 - Y_2)}{X_1}$$

Given : Colour of solution from first stage = 4.6 final colour

$$\therefore \quad Y_1 = 4.6\,Y_2$$

The equilibrium relationship is

$$Y = 8.91 \times 10^{-5}\,X^{1.66}$$

$$Y_1 = 8.91 \times 10^{-5}\,X_1^{1.66}$$

$$X_1 = [Y_1/8.91 \times 10^{-5}]^{1/1.66}$$

But $Y_1 = 4.6\,Y_2$

$$\therefore \quad X_1 = [4.6\,Y_2 / 8.91 \times 10^{-5}]^{1/1.66}$$

$$\therefore \quad S_S = \frac{L_S\,(Y_0 - Y_2)}{[4.6\,Y_2/8.91 \times 10^{-5}]^{1/1.66}}$$

L_S = 1000 kg, Y_0 = 9.6, and Y_2 = 0.96

$$S_S = \frac{1000\,(9.6 - 0.96)}{[4.6 \times 0.96/8.91 \times 10^{-5}]^{1/1.66}}$$

$$= 12.83 \text{ kg}$$

Amount of adsorbent required = 12.83 kg ... Ans. (ii)

(for two-stage countercurrent operation)

Ex. 3.7 : A solution of washed raw cane sugar is coloured by the presence of small amounts of an impurity. The solution is to be decolourised by treatment with an adsorptive carbon. The original colour has a colour concentration of 9.6 measured on an arbitrary scale and it is desired to reduce colour to 10% of its original value.

The data for an equilibrium isotherm is as follows :

kg carbon/kg solution	0	0.001	0.004	0.008	0.02	0.04
Equilibrium colour, colour units/kg solution	9.6	8.6	6.3	4.3	1.7	0.7

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Determine the minimum total amount of fresh carbon required for two-stage cross-current operation per 1000 kg of solution.

Solution : Basis : 1000 kg of solution.

Initial colour of the solution = $Y_0 = 9.6 \frac{\text{colour units}}{\text{kg solution}}$.

Colour is to be reduced to 10% of the original colour.

$\therefore$ Final colour = $Y_2 = 0.10\,(9.6) = 0.96$ colour units/kg solution.

First we will convert the data in a form suitable for plotting the equilibrium adsorption isotherm.

We have :

kg carbon/kg solution	0	0.001
Equilibrium colour	9.6	8.6

i.e., when 0.001 kg carbon is added to 1 kg of initial solution, the concentration of colour drops from 9.6 to 8.6.

$\therefore$ Colour adsorbed by 0.001 kg carbon = $1\,(9.6 - 8.6)$.

Colour adsorbed by 1 kg carbon = $\frac{1\,(9.6 - 8.6)}{0.001} \times 1$

$\therefore$ The adsorbate concentration, $X = \frac{\text{colour units}}{\text{kg carbon}} = \frac{1\,(9.6 - 8.6)}{0.001} \times 1 = 1000.$

Similarly, for kg carbon/kg solution = 0.004.

When 0.004 kg of carbon is added to 1 kg of initial solution, colour drops from 9.6 to 6.3.

$\therefore$ Adsorbate concentration, $X = \frac{\text{colour units}}{\text{kg carbon}} = \frac{1\,(9.6 - 6.3)}{0.004} = 825$

Convert the data and tabulate.

$\frac{\text{kg carbon}}{\text{kg solution}}$	Equilibrium colour, $Y = \frac{\text{colour units}}{\text{kg solution}}$	Adsorbate concentration, $X = \frac{\text{colour units}}{\text{kg carbon}}$
0	9.6	–
0.001	8.6	(9.6 – 8.6)/0.001 = 1000
0.004	6.3	(9.6 – 6.3)/0.004 = 825
0.008	4.3	(9.6 – 4.3)/0.008 = 662.5
0.02	1.7	(9.6 – 1.7)/0.02 = 395
0.04	0.7	(9.6 – 0.7)/0.04 = 222.5

For two-stage crosscurrent operation, we have :

For stage-1 : $L_S (Y_0 - Y_1) = S_{S1} (X_1 - X_0)$

For stage-2 : $L_S (Y_1 - Y_2) = S_{S2} (X_2 - X_0)$

For both the stages, we are using fresh carbon.

$\therefore$ $X_0 = 0$

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The operating line of first stage has two end points : (X_0, Y_0) and (X_1, Y_1) and has a slope $-S_{S1}/L_S$ and that of second stage has two end points : (Y_2, X_2) and (X_0, Y_1) and has a slope $-S_{S2}/L_S$.

Refer Fig. E 3.7 :

Locate initial colour, the point P (0, 9.6) and final colour at $Y_2 = 0.96$ at point Q on y-axis. Draw horizontal through Q to cut the equilibrium curve at M and read X_2.

$X_2 = 270$ colour units/kg carbon.

We have to construct two crosscurrent stages i.e. two operating lines starting from $Y_0 = 0.96$ and end at final colour $Y_2 = 0.96$ and find the value of Y_1 and X_1.

For finding out the minimum total amount of carbon, we have to adopt a trial-and-error procedure. Thus, any point N on the equilibrium curve is assumed, the operating lines are drawn (not shown for point N). Read Y_1 and X_1 and compute the values of S_{S1} and S_{S2} using the above given equation. The position of the point N is changed until the sum of S_{S1} and S_{S2} is a minimum. In Fig. E 3.7, the operating lines are shown only for a correct trial, i.e., a trial giving the minimum total amount of carbon.

A trial-and-error procedure :

Point N on the equilibrium curve, read Y_1 and X_1, i.e., for $Y_1 = 4.5$ and $X_1 = 670$,

$$S_{S1} = \frac{L_S (Y_0 - Y_1)}{X_1 - X_0} = \frac{1000 (9.6 - 4.5)}{670 - 0} = 7.61$$

$$S_{S2} = \frac{L_S (Y_1 - Y_2)}{X_2 - X_0} = \frac{1000 (4.5 - 0.96)}{270 - 0} = 13.11$$

$$S_{S1} + S_{S2} = 7.61 + 13.11 = 20.72$$

For $Y_1 = 4$ ... $X_1 = 630$,

$$S_{S1} = \frac{1000 (9.6 - 4)}{630 - 0} = 8.9$$

$$S_{S2} = \frac{1000 (4 - 0.96)}{270 - 0} = 12.26$$

$$S_{S1} + S_{S2} = 8.9 + 12.26 = 21.16$$

For $Y_1 = 3$, $X_1 = 530$
$S_{S1} = 12.45$, $S_{S2} = 7.56$ Total = 20.01

For $Y_1 = 2.5$, $X_1 = 490$
$S_{S1} = 14.5$, $S_{S2} = 5.70$ Total = 20.2

For $Y_1 = 3.5$, $X_1 = 585$
$S_{S1} = 10.43$, $S_{S2} = 9.41$ Total = 19.84

For $Y_1 = 3.6$, $X_1 = 595$
$S_{S1} = 10.08$ $S_{S2} = 9.78$ Total = 19.86

For $Y_1 = 3.2$, $X_1 = 555$
$S_{S1} = 11.53$, $S_{S2} = 8.3$ Total = 19.83

For $Y_1 = 3.3$, $X_1 = 565$
$S_{S1} = 11.15$, $S_{S2} = 8.67$ Total = 19.82

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So for the point R, with coordinates $Y_1 = 3.3$ and $X_1 = 565$, minimum total amount of carbon is required.

So the operating lines corresponding to the minimum total amount of carbon are PR and LM.

$$\therefore \quad S_{S1} = 11.15 \text{ kg} \quad \text{and} \quad S_{S2} = 8.67 \text{ kg}$$

$$\text{Total} = S_{S1} + S_{S2} = 11.15 + 8.67 = 19.82 \text{ kg}$$

$$\text{Minimum total carbon required} = 19.82 \text{ kg} \quad \text{... Ans.}$$

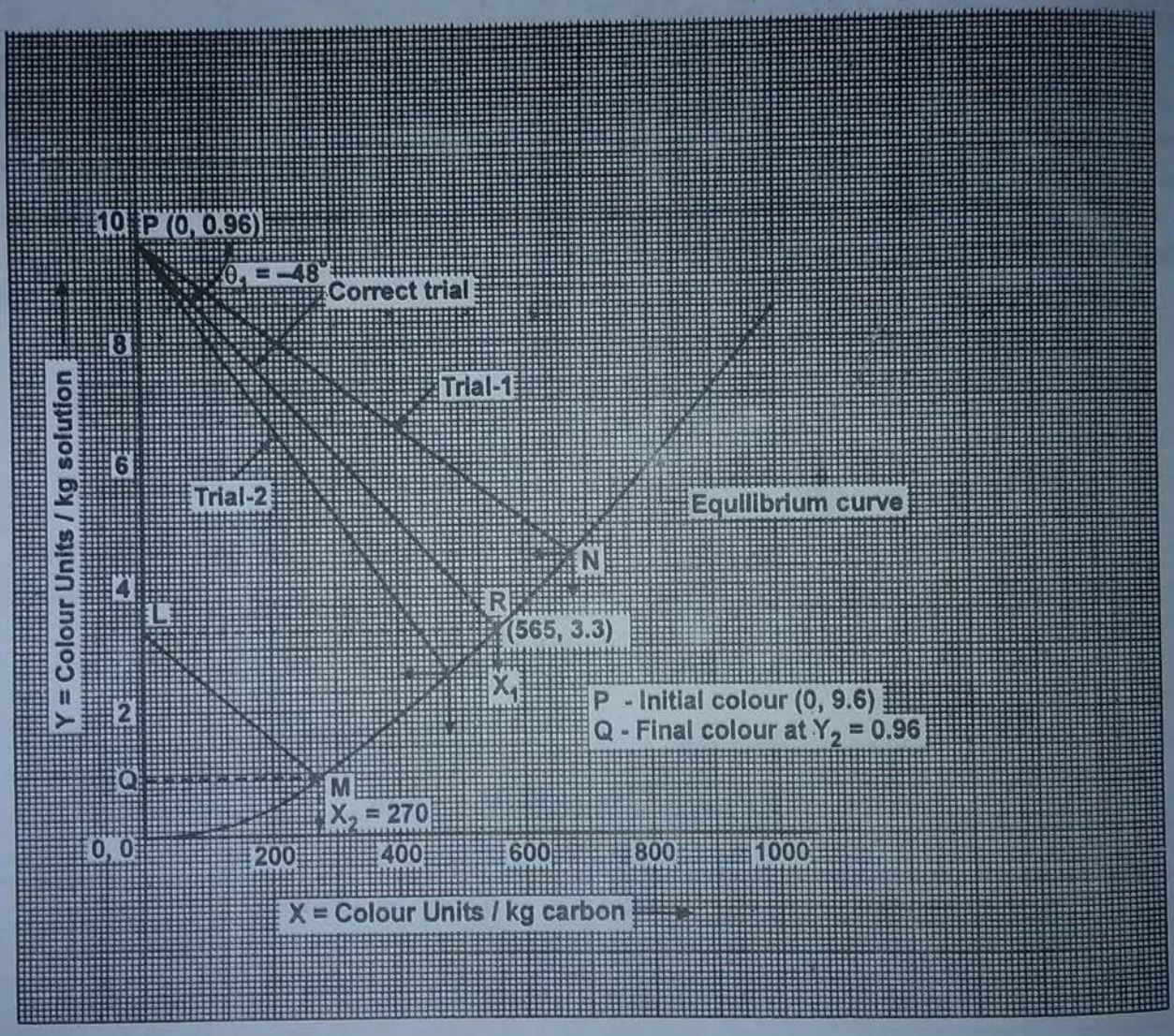

Fig. E 3.7 : Determination of minimum total adsorbent

Extra : PR and LM are the operating lines corresponding to the minimum total amount of carbon required.

$$\text{Slope of PR} = -S_{S1}/L_S$$

$$\text{Slope of LM} = -S_{S2}/L_S$$

Angle made by PR with line through P and parallel to x-axis = $\theta_1 = -48^\circ$.

Angle made by LM with line through L and parallel to x-axis = $\theta_2 = -41^\circ$.

$$\tan \theta_1 = \tan(-48) = -1.111$$

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Taking scale into consideration :

$$\therefore \quad -S_{S1}/L_S = \tan\theta_1 \times \left(\frac{\text{scale y-axis}}{\text{scale x-axis}}\right) = \frac{1.111 \times 1/10}{100/10} = -0.01111$$

$$S_{S1} = 0.01111 \times L_S$$

$$S_{S1} = 0.01111 \times 1000 = 11.11 \text{ kg}$$

We have, $\theta_2 = -41°$

$$\tan\theta_2 = \tan(-41) = -0.869$$

$$-S_{S2}/L_S = \tan\theta_2 \times \left(\frac{\text{scale y-axis}}{\text{scale x-axis}}\right) = -\frac{0.869 \times 1/10}{100/10} = -0.00869$$

$$\therefore \quad S_{S2}/L_S = 0.00869$$

$$S_{S2} = 0.00869 \times L_S$$

$$S_{S2} = 0.00869 \times 1000 = 8.69$$

Total minimum carbon required = 11.11 + 8.69 = 19.80 kg ... Ans.

We have from the trial and error procedure, $S_{S1} = 11.15$ kg and $S_{S2} = 8.67$

$$\text{Slope} = -\frac{S_{S1}}{L_S} = -\frac{11.15}{1000} = -0.01115$$

Taking scale into consideration :

$$\text{Slope} = -\frac{0.01115 \times 100/10}{1/10} = -1.115$$

$$\tan\theta = \text{slope} = -1.115$$

$$\theta = \tan^{-1}(-1.115) = -48.11° \quad \text{... (1)}$$

Measured angle of PR with line parallel to x-axis = – 48°. ... (2)

$$S_{S2} = 8.67 \text{ kg}$$

$$\text{Slope} = -\frac{S_{S2}}{L_S} = -\frac{8.67}{1000} = -0.00867$$

Taking scale into consideration,

$$\text{Slope} = -\frac{0.00867 \times 100/10}{1/10} = -0.867$$

$$\tan\theta = \text{slope} = -0.867$$

$$\theta = \tan^{-1}(0.867) = -40.92° \quad \text{... (1)}$$

... (2)

and measured angle of LM is – 41°.

Note that : All this is done to have check on the calculations and plotting accuracy.

Ex. 3.8 : A solution of washed, raw cane sugar, 48% sucrose by weight is coloured by the presence of small quantities of impurities. It is to be decolourised at 353 K (80 °C) by treatment with an adsorptive carbon in a contact filtration plant. The equilibrium adsorption isotherm data were obtained by adding various amounts of the carbon to the separate batches of the original solution and observing the equilibrium colour reached in each case. The data, with the quantity of carbon expressed on the basis of the sugar content of the solution, are as given below :

kg carbon/kg dry sugar	0	0.005	0.01	0.015	0.02	0.03
% colour removed	0	47	70	83	90	95

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The original solution has a colour concentration of 20, measured on an arbitrary scale. It is desired to reduce the colour to 2.5% of its original value.

(i) Convert the equilibrium data to Y^* = colour units/kg sugar, X = colour units/kg carbon. Do they follow the Freundlich equation ? If yes, what are the equation constants ?

(ii) Calculate the quantity of fresh carbon required per 1000 kg solution for a single-stage process.

Solution : Basis : 1000 kg of solution.

First we will convert the given data into a desired form.

Y = colour units/kg sugar

X = colour units/kg carbon

The original solution has a colour of 20.

Initial colour = 20 colour units/kg solution

Sugar solution contains 48% sucrose by weight i.e. 1 kg solution contains 0.48 kg sugar.

Expressing colour per kg sugar :

$$\text{Initial colour of solution} = Y_0 = \frac{20}{0.48} = 41.67 \text{ colour units/kg sugar}$$

For 0.48 kg sugar, colour concentration is 20 colour units.

$\therefore$ For 1 kg sugar, colour concentration is $20 \times \frac{1}{0.48} = 41.67$ colour units.

Take first two columns of the data provided :

kg carbon/kg sugar	0	0.005
% colour removed	0	47

When 0.005 kg carbon/kg sugar is used, colour removed is 47% of its original/initial value.

$\therefore$ Colour in solution remained at equilibrium (equilibrium colour) $= (1 - 0.47) \times 20$

$= 10.6$ colour units/kg solution.

Expressing colour per kg sugar,

$$\text{equilibrium colour} = Y = \frac{10.6}{0.48} = 22.1 \text{ colour units/kg sugar}$$

OR $\text{equilibrium colour} = Y = (1 - 0.47)\, 41.67 = 22.1$ colour units/kg sugar

When 0.005 kg carbon is added to the solution containing 1 kg of sugar, the colour concentration drops from 41.67 to 22.1.

Therefore, colour adsorbed by 0.005 kg carbon $= 1\,(41.67 - 22.1)$

$\therefore$ Adsorbate concentration $= X = \frac{\text{colour units adsorbed}}{\text{kg carbon}} = \frac{1\,(41.67 - 22.1)}{0.005} = 3914$

OR Adsorbate concentration $= X = \frac{\text{colour units}}{\text{kg carbon}} = \left(\frac{20 - 10.6}{0.48}\right) \times \frac{1}{0.005} = 3916.67$

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In this way, convert all the data in terms of Y and X and tabulate.

$\frac{\text{kg carbon}}{\text{kg sugar}}$	0	0.005	0.01	0.015	0.02	0.03
% colour removed	0	47	70	83	90	95
Equilibrium colour (remained), $\frac{\text{colour units}}{\text{kg solution}}$	20	(1 – 0.47) 20 10.6	(1 – 0.7) 20 6	(1 – 0.83) 20 3.4	(1 – 0.90) 20 2	(1 – 0.95) 20 1
Equilibrium colour, $Y = \frac{\text{colour units}}{\text{kg sugar}}$	20/0.48 = 41.67	10.6/0.48 = 22.1	6/0.48 = 12.5	3.4/0.48 = 7.08	2/0.48 = 4.17	1/0.48 = 2.083
Adsorbate concentration, $X = \frac{\text{colour units}}{\text{kg carbon}}$	–	$\frac{41.67 - 22.1}{0.005}$ = 3914	$\frac{41.67 - 12.5}{0.01}$ = 2917	$\frac{41.67 - 7.08}{0.015}$ = 2306	$\frac{41.67 - 4.17}{0.02}$ = 1875	$\frac{41.67 - 2.083}{0.03}$ = 1319.6
$X = \frac{\text{colour units}}{\text{kg carbon}}$	–	$\frac{20 - 10.6}{0.48 \times 0.005}$ = 3916.6	$\frac{20 - 6}{0.48 \times 0.01}$ = 2916.67	$\frac{20 - 3.4}{0.48 \times 0.015}$ 2305.55	← OR	

Plot the Y v/s X data on logarithmic coordinates [log-log paper].

OR : Convert Y into ln Y and X into ln X and plot ln Y against ln X.

X	3914	2917	2306	1875	1319.6
ln X	8.27	7.98	7.743	7.536	7.185
Y	22.1	12.5	7.08	4.17	2.083
ln Y	3.095	2.523	1.957	1.428	0.734

Refer Fig. E 3.8 (a) in which ln Y is plotted against ln X.

The plot provides a straight line, so that the Freundlich equation is applicable. ...Ans. (i)

$$Y = m X^n$$

The slope of the line = n = 2.133

From X, Y data, we have : At X = 2306, Y = 7.08

$$\therefore \quad m = Y/X^n = \frac{7.08}{(2306)^{2.133}}$$

$$= 4.754 \times 10^{-7}$$

The Freundlich equation is therefore :

$$Y = m X^n$$

$$Y = 4.754 \times 10^{-7} X^{2.133} \quad \text{...Ans. (i)}$$

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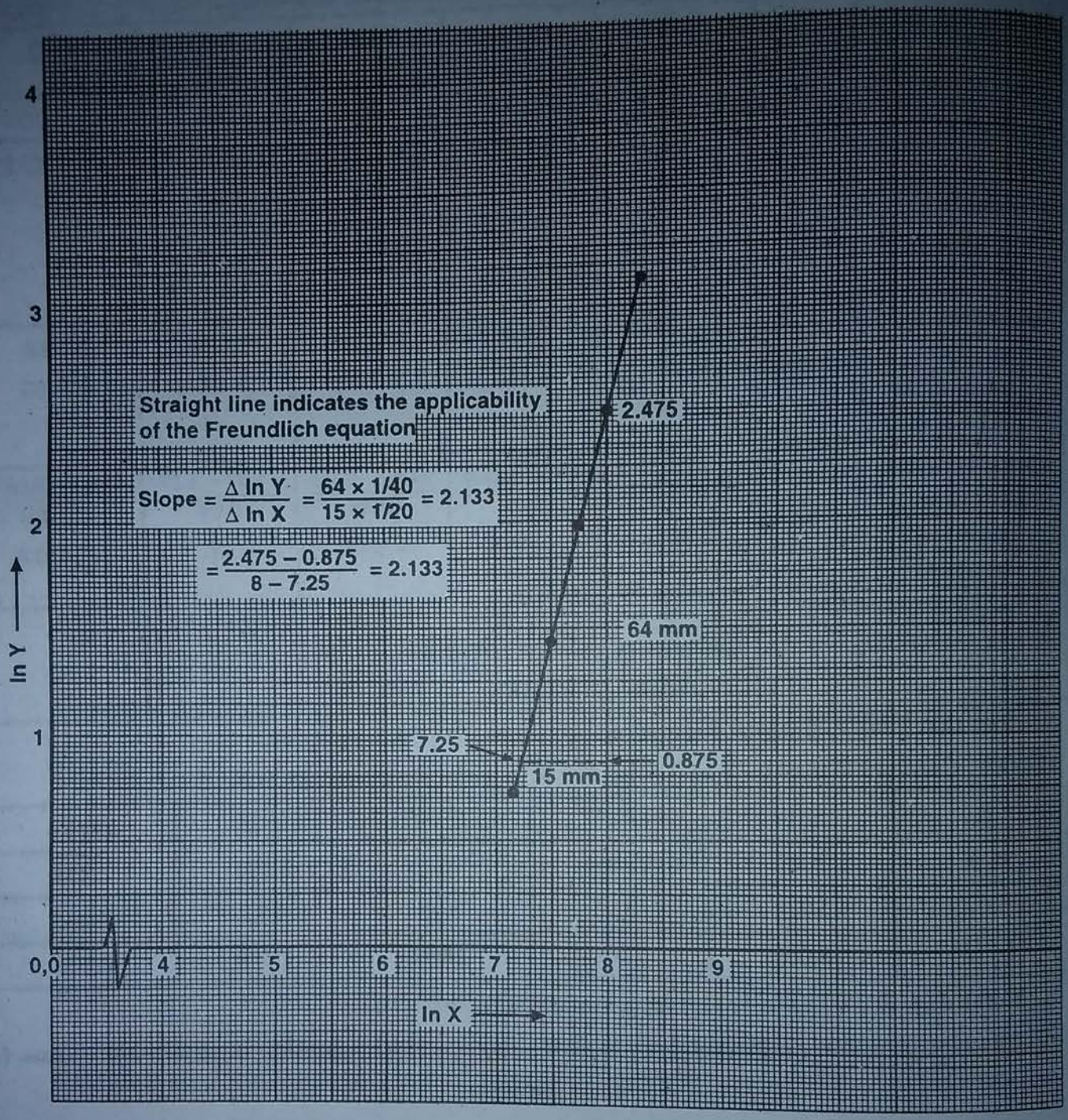

Fig. E 3.8 (a) : Determination of constants of the Freundlich equation

Single stage operation :

Initial colour = Y_0 = 41.67 colour units/kg sugar

Final colour = 2.5% of original colour

Y_1 = $0.025 \times 41.67 = 1.04175 \approx 1.042$ colour units/kg sugar

The operating line equation is :

$$L_S (Y_0 - Y_1) = S_S (X_1 - X_0)$$

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As the fresh carbon is to be used, $X_0 = 0$

$$\therefore \quad L_S (Y_0 - Y_1) = S_S X_1$$

$$S_S = \frac{L_S (Y_0 - Y_1)}{X_1}$$

Refer Fig. E 3.8 (b) :

Y_1 and X_1 are the equilibrium values, so from the point P at $Y_1 = 1.042$ on y-axis, draw a line parallel to x-axis which will cut the equilibrium curve at the point Q. Read X_1 at Q (i.e., the x coordinate of the point Q).

$$X_1 = 950 \text{ colour units/kg carbon}$$

1000 kg of solution is to be treated.

This solution contains 48% sugar.

$$\therefore \quad \text{Sugar in solution} = 0.48\,(1000) = 480 \text{ kg}$$

$$\therefore \quad L_S = 480 \text{ kg}$$

$$S_S = \frac{480\,(41.67 - 1.042)}{950} = 20.53 \text{ kg}$$

Quantity of fresh carbon required = 20.53 kg ... **Ans. (ii)**

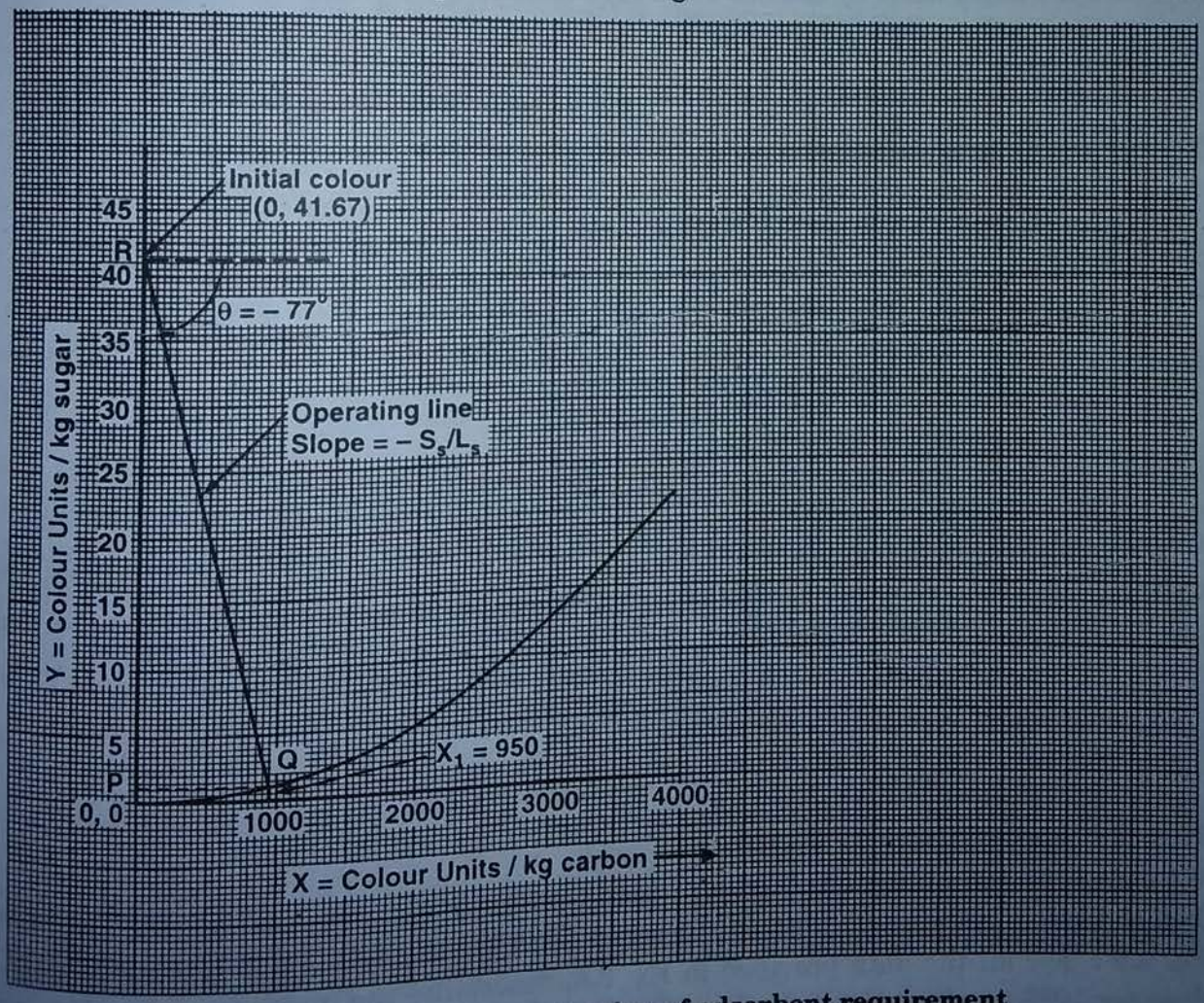

Fig. E 3.8 (b) : Determination of adsorbent requirement

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OR

Check :

The slope of operating line (i.e. line RQ) $= -S_S/L_S$.

Draw the operating line.

Two ends of the operating line :

Point R (0, 41.67)

Point Q (X_1, Y_1 = 1.042).

Locate point R at the initial colour (0, 41.67) i.e. (X_0,Y_0).

Locate point P on y-axis at the final colour, i.e., at Y_1 = 1.042.

For an equilibrium stage, X_1 and Y_1 lie on the equilibrium curve as effluent streams are in equilibrium for the stage. Draw a horizontal line through the point P which will cut the equilibrium curve at the point Q, the other end of the operating line. Draw the operating line RQ.

Measure the angle made by the line RQ with a horizontal line through the point R.

$$\theta = -77^\circ$$

$$\tan\theta = -4.33 \approx -4.3 \text{ (for measurement accuracy)}$$

Taking scales into consideration,

$$\text{slope} = \tan\theta\left(\frac{\text{scale y-axis}}{\text{scale x-axis}}\right) = \frac{-4.3 \times 5/10}{1000/20}$$

$$= -0.043$$

$$\text{slope} = -\frac{S_S}{L_S} = -0.043$$

$$S_S = 0.043 \times L_S$$

$$= 0.043 \times 480 = 20.64 \text{ kg}$$

Quantity of carbon required = 20.64 kg ... **Ans. (ii)**

Quantity of carbon required by this approach is almost same as that obtained previous one.

Present approach = 20.64 kg

by previous approach = 20.53 kg

$$\% \text{ error} = \frac{20.64 - 20.53}{20.64} \times 100 = 0.533$$

This indicates accuracy in construction work.

We cannot measure angle in fraction which creates error in results, so $\tan\theta = -4.33$ is taken as $\tan\theta = -4.3$.

Ex. 3.9 : For all the data provided in Ex. 3.8, determine the quantity of fresh carbon required per 1000 kg of solution for a two-stage countercurrent treatment.

Solution : Basis : 1000 kg of solution.

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Using the converted data (X, Y).

Initial colour = Y_0 = 41.67 colour units/kg sugar

Final colour = Y_2 = 1.042 colour units/kg sugar

L_S = 480 kg

For two-stage countercurrent operation, the solute material balance is

$$L_S (Y_0 - Y_2) = S_S (X_1 - X_0)$$

This operating line passes through the terminal points (X_0, Y_2) and (X_1, Y_0) and has a slope of S_S/L_S.

For fresh carbon, $X_0 = 0$

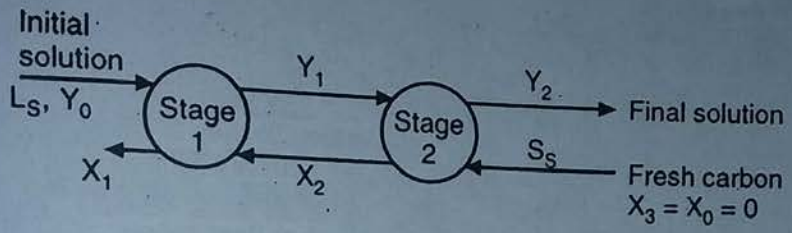

Fig. E 3.9 (a) : Two-stage countercurrent operation

$$\therefore \quad L_S (Y_0 - Y_2) = S_S X_1$$

Plot Y v/s X.

Mark initial colour on y-axis at Y_0 = 41.67 (point P). Draw a horizontal through the point P (drawn dotted).

Mark the point Q (X_0, Y_2) i.e. Q (0, 1.042) on y-axis.

The one end of operating line is at the point Q and we have to draw the operating line through the point Q in such a way that we can construct exactly two stages between equilibrium curve and operating line between the points R (X_1, Y_0) and Q (X_0, Y_2), i.e., (0, 1.042). The other end of the operating line should lie on a horizontal line through the point P $(0, Y_0 = 41.67)$. So we have to adopt a trial-and-error procedure till two stages can be drawn (can be constructed starting from R and ending exactly at Q). Then, read X_1.

Draw operating line randomly through the point Q which will cut dotted horizontal line (through P) say at point R and see whether we can construct exactly two stages between R and Q or not. Repeat the procedure till we construct exactly two stages starting from R and ending at Q.

Adopting a trial and error procedure - lines are not shown for incorrect trial - the line QR is shown for a correct/exact trial. So now read X_1.

X_1 = 3050 colour units/kg carbon

$$S_S = \frac{L_S (Y_0 - Y_2)}{X_1}$$

$$= \frac{480 (41.67 - 1.042)}{3050} = 6.394 \text{ kg} \approx 6.4 \text{ kg}$$

... Ans.

Quantity of carbon required = 6.4 kg

Please remember and always keep in mind : Small error in plotting points, curves and lines create a significant error in results.

Draw the operating line through the point Q randomly which will cut a horizontal dotted line through the point P say at point R.

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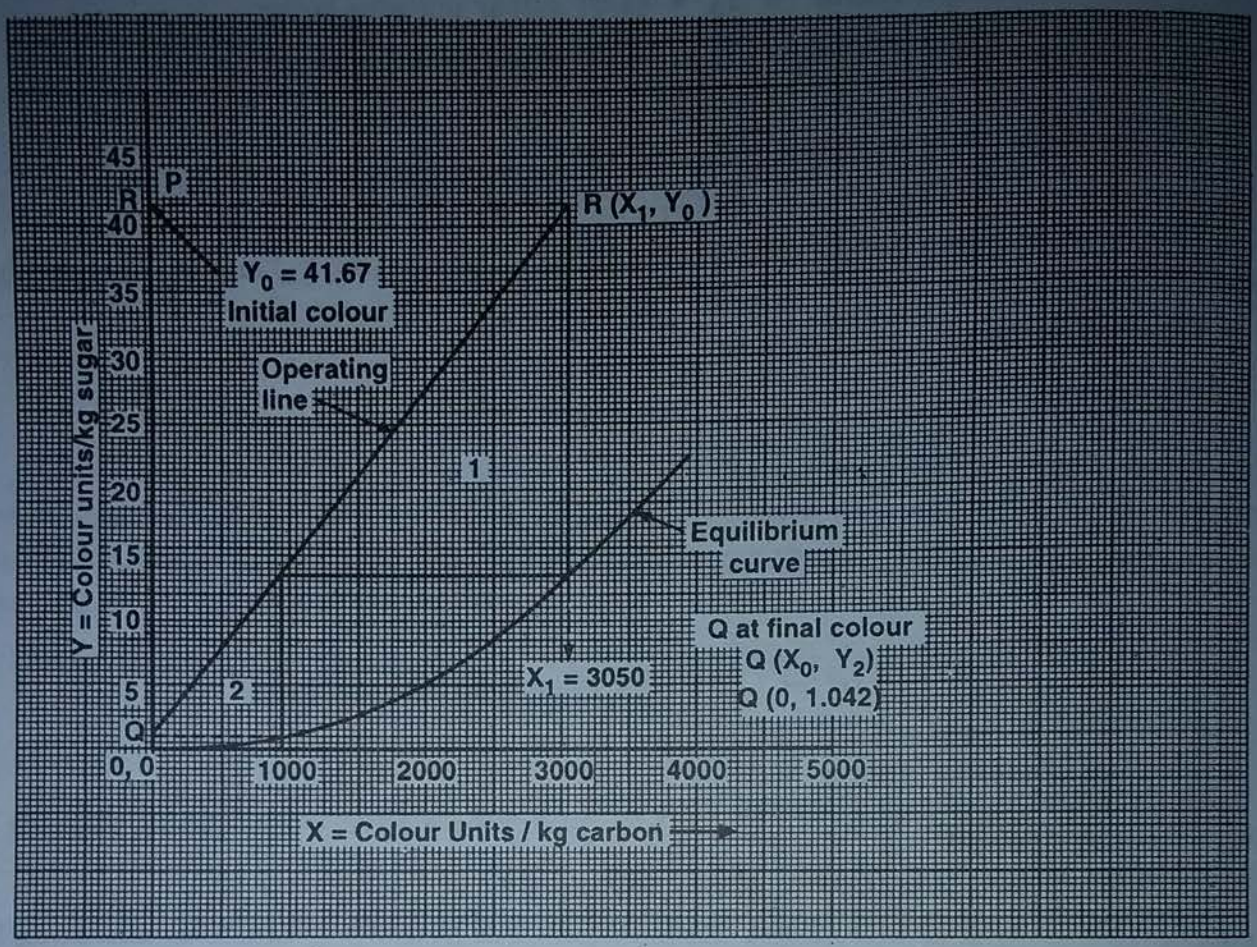

Fig. E 3.9 (b) : Construction of exact stages to determine the requirement of an adsorbent

Ex. 3.10 : The equilibrium water adsorbed by a silica-gel in contact with moist air varies linearly with the humidity of air.

$$Y = 3.4435 \times 10^{-2} X$$

where

X = kg water adsorbed/kg of dry gel

Y = humidity of air, kg moisture/kg dry air

0.5 kg of silica-gel containing 5% (dry basis) adsorbed water is placed in a collapsible vessel in which there are 10 m^3 of moist air, the partial pressure of water being 15 mmHg. The total pressure and temperature are kept at 1 atm and 298 K (25 °C) respectively.

What is the amount of water picked up from the moist air in the vessel by the silica-gel ? Also calculate the final partial pressure of the water vapour in the vessel and final total pressure in the vessel.

Solution : Basis : 10 m^3 of moist air at 298 K and 1 atm.

Quantity of silica-gel placed in vessel = 0.5 kg.

Given : Silica-gel contains 5% (dry basis) adsorbed water.

Let w_1 be the kg of adsorbed water and w_2 be the kg of dry silica-gel.

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Then,

$$w_1 + w_2 = 0.5$$

$$\text{Adsorbed water} = 5\% \text{ dry basis}$$

$$\therefore \quad \frac{w_1}{w_2} \times 100 = 5$$

we have,

$$w_1 = 0.05\, w_2$$

$$w_1 + w_2 = 0.50$$

$$\therefore \quad 0.05\, w_2 + w_2 = 0.5$$

$$w_2 = 0.4762 \text{ kg} \quad \text{and} \quad w_1 = 0.0238 \text{ kg}$$

Initial water content of silica-gel :

$$X_0 = \frac{\text{kg water}}{\text{kg dry silica-gel}} = \frac{w_1}{w_2} = \frac{0.0238}{0.4762} = 0.05$$

Partial pressure of water in moist air in vessel = 15 mmHg.

Total pressure in vessel = 1 atm = 760 mmHg.

Partial pressure of dry air = 760 – 15 = 745 mmHg.

As air is not going to adsorb on silica-gel, its partial pressure will remain constant at 745 mmHg.

We have,

$$\text{Partial pressure of A} = \text{molefraction of A} \times \text{total pressure}$$

$$= \frac{\text{moles A}}{\text{total moles}} \times P$$

$$\therefore \quad \text{Partial pressure of water} = \frac{\text{moles of water}}{\text{total moles}} \times P$$

$$\text{Partial pressure of dry air} = \frac{\text{moles of dry air}}{\text{total moles}} \times P$$

$$\therefore \quad \frac{\text{Partial pressure of water}}{\text{Partial pressure of dry air}} = \frac{\text{moles of water}}{\text{moles of dry air}}$$

Molecular weight of water = 18

Molecular weight of dry air = 28.84

[Air contains 21 mole % O_2 and 79 mole % N_2

$$\therefore \quad M_{avg.} \text{ of air} = 0.21 \times 32 + 0.79 \times 28 = 28.84]$$

Therefore, $\text{kmol of A} = \text{kmol of A/Mol.Wt. of A}$

$$\frac{\text{Partial pressure of water}}{\text{Partial pressure of dry air}} = \frac{\text{kg water}/18}{\text{kg dry air}/28.84}$$

$$\therefore \quad \frac{\text{kg water}}{\text{kg dry air}} = \frac{\text{Partial pressure of water} \times 18}{\text{Partial pressure of dry air} \times 28.84}$$

Initially : Partial pressure of water = 15 mmHg

Partial pressure of dry air = 745 mmHg

$$Y_0 = \frac{\text{kg water}}{\text{kg dry air}} = \frac{15 \times 18}{745 \times 28.84} = 0.012566$$

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Initial water vapour/moisture content of moist air i.e. concentration of moisture in moist air :

$$Y_0 = 0.012566 \text{ kg water/kg dry air}$$

We have, $p_A V = n_A RT$

$$n_A = p_A V/RT$$

$$\text{Moles of dry air} = \frac{p_{\text{dry air}} \times V}{RT}$$

where, $p_{\text{dry air}} = 745 \text{ mmHg} = (745/760) \text{ atm}$

V = volume of moist air (total) = 10 m^3

$R = 0.08206 \text{ m}^3 \cdot \text{atm/(kmol.K)}$

$T = 298 \text{ K}$

$$\text{Moles of dry air} = \frac{(745/760) \times 10}{0.08206 \times 298}$$

$$= 0.4 \text{ kmol}$$

$$\text{Mass of dry air} = 0.4 \times 28.84 = 11.536 \text{ kg}$$

So $L_S = 11.536$ kg

Mass of dry gel = 0.4762 kg

So $S_S = 0.4762$ kg

For single stage adsorption, we have, (i.e. material balance of moisture)

$$L_S [Y_0 - Y_1] = S_S (X_1 - X_0) \quad \ldots (1)$$

For an equilibrium stage, the effluent streams are in equilibrium, so X_1 and Y_1 are the equilibrium values.

The equilibrium relationship is

$$Y = 3.4435 \times 10^{-2} X, \text{ (Given)}$$

$$\therefore \quad Y_1 = 3.4435 \times 10^{-2} X_1$$

$$X_1 = Y_1/3.4435 \times 10^{-2} = 29.04 Y_1 \quad \ldots (2)$$

Putting the value of X_1 from Equation (2) into Equation (1),

$$\therefore \quad L_S (Y_0 - Y_1) = S_S (29.04 Y_1 - X_0)$$

$L_S = 11.536$ kg, $S_S = 0.4762$ kg

X_0 = initial adsorbate concentration = 0.05

and $Y_0 = 0.012566$

Putting the values of various terms in the above equation,

$$11.536 (0.012566 - Y_1) = 0.4762 (29.04 Y_1 - 0.05)$$

$$0.14496 - 11.536 Y_1 = 13.829 Y_1 - 0.02381$$

$$\therefore \quad Y_1 = 0.006654 \text{ kg water/kg dry air}$$

Final concentration of water in moist air :

$$Y_1 = 0.006654 \text{ kg water/kg dry air}$$

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Finally :

$$\frac{\text{Partial pressure of water}}{\text{Partial pressure of dry air}} = \frac{\text{moles of water}}{\text{moles of dry air}} = \frac{\text{kg of water/18}}{\text{kg of air/28.84}}$$

$$\frac{\text{Partial pressure of water}}{\text{Partial pressure of dry air}} = \left(\frac{\text{kg of water}}{\text{kg of air}}\right) \times \frac{28.84}{18}$$

kg water/kg dry air in moist air finally $= Y_1 = 0.006654$

$$\frac{\text{Partial pressure of water}}{\text{Partial pressure of dry air}} = (0.006654) \times \frac{28.84}{18}$$

$$= 0.01066$$

Partial pressure of water = 0.01066 (partial pressure of dry air)

Partial pressure of dry air finally is the same as its initial partial pressure of 745 mmHg (as it is not adsorbed).

$\therefore$ Final partial pressure of water $= 0.01066 \times 745$

$= 7.94$ mmHg ... **Ans.**

Final total pressure $= \Sigma p_i$

$= 745 + 7.94$

$= 752.94$ mmHg ... **Ans.**

We have, $Y_0 = 0.012566$ kg water/kg dry air

$Y_1 = 0.006654$ kg water/kg dry air

$L_S = 11.536$ kg dry air

Water balance is :

Water picked up by silica-gel = Water removed from moist air

$= L_S (Y_0 - Y_1)$

$= 11.536 (0.012566 - 0.006654)$

$= 0.0682$ kg ... **Ans.**

Ex. 3.11 : In a laboratory setup, toluene vapours in air are passed over a bed of activated carbon. In the first experiment, the bed depth is 50 cm, while in the second experiment, it is 100 cm. The breakthrough was determined at 5% of the inlet concentration. In the first experiment, the breakthrough occurred at 65 minutes while in the second test, it occurred at 140 minutes. Determine the length of the unused bed.

Solution :

Case – I : $\theta_{B1} = 65$ min, $Z_1 = 50$ cm

Case – II : $\theta_{B2} = 140$ min, $Z_2 = 100$ cm

θ_B = breakpoint/breakthrough time

Z = bed length

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In case - I, let x be the fractional utilisation of the bed at the breakpoint.

$\therefore$ Length of the fully used bed in case - I = 50 x cm

Length of the unused bed in case - I = 50 – 50 x cm

Length of unused bed does not change with the total bed length.

$\therefore$ Length of the unused bed in case - II = LUB in case – I

= 50 – 50 x cm

Length of the fully used bed in case - II = 100 – (50 – 50 x)

= 50 + 50 x cm

Fractional utilisation of the bed in case - II $= \dfrac{50 + 50x}{100}$

We have, $\theta_B = \theta_S \left[1 - \dfrac{LUB}{Z}\right]$

$\theta_B = \dfrac{\theta_S}{Z} [Z - LUB]$

Length of fully used bed = Z – LUB

$\theta_B = \dfrac{\theta_s}{Z}$ [length of the fully used bed]

$\therefore$ $\theta_B \propto$ length of the fully used bed

θ_B = k · length of the fully used bed

Case - I: $\theta_{B1} = k (\text{length of the fully used bed})_I$

$\theta_{B1} = k (50x)$

Case - II: $\theta_{B2} = k (\text{length of the fully used bed})_{II}$

$\theta_{B2} = k (50 + 50x)$

$\therefore$ $\dfrac{\theta_{B2}}{\theta_{B1}} = \dfrac{k (50 + 50x)}{k (50x)}$

$\dfrac{\theta_{B2}}{\theta_{B1}} = \dfrac{50 + 50x}{50x} = \dfrac{1 + x}{x}$

$\dfrac{140}{65} = \dfrac{1 + x}{x}$

Solving we get, $x = 0.867$

Fractional utilisation of the bed in case - I = x = 0.867

Length of the fully utilised bed in case - I = 0.867 × 50

= 43.35 cm

$\therefore$ Length of the unused bed in case - I (LUB) = 50 – 43.35

= 6.65 cm

Length of unused bed remains the same in both the cases.

$\therefore$ Length of unused bed = 6.65 cm ... Ans.

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Fractional utilisation of the bed in case - II $= \frac{50 + 50\,x}{100}$

$= \frac{50 + 50 \times 0.867}{100}$

$= 0.9335$

Length of the fully utilised bed in case - II $= 0.9335 \times 100$

$= 93.35$ cm

Length of the unused bed in case - II $= 100 - 93.35$

$= 6.65$ cm

$$\frac{\theta_{B2}}{\theta_{B1}} = \frac{(\text{length of fully utilised bed})_{II}}{(\text{length of fully utilised bed})_{I}}$$

$$\frac{\theta_{B2}}{65} = \frac{93.35}{43.35}$$

$$\theta_{B2} = \frac{65 \times 93.35}{43.35}$$

$\theta_{B2} = 139.97 \approx 140$ min (given value) ... **Ans.**

So the calculations performed are correct.

Ex. 3.12 : Nitrogen dioxide, NO_2, produced by a thermal process for fixation of nitrogen is to be removed from a dilute mixture with air by adsorbing NO_2 on silica gel using a continuous countercurrent adsorber. The gas entering the adsorber contains 1.50% NO_2 by volume and 90% of the NO_2 is removed. The mass flow rate of gas to the adsorber is 0.126 kg/s. Adsorption operation is to be isothermal at 298 K (25 °C), 760 mmHg. The gel entering the adsorber is free of NO_2. The equilibrium adsorption isotherm at 298 K (25 °C) is given by the following data.

Partial pressure of NO_2, mmHg	0	2	4	6	8	10	12
kg NO_2/100 kg gel	0	0.40	0.9	1.65	2.60	3.65	4.85

(i) Calculate the minimum gel flow rate required in kg/h.

(ii) For twice the minimum gel rate, calculate the number of transfer units required.

Solution : Basis : 0.126 kg/s of gas entering the adsorber.

Gas entering the adsorber contains NO_2 and air.

Molecular weight of NO_2 = 46

Molecular weight of air = 28.84 (assuming air contains 79% N_2 and 21% O_2 by volume).

Given : Gas entering the adsorber contains 1.5% NO_2 by volume.

For gas mixtures following ideal gas law :

Volume % of A = mole % of A = pressure % of A

Volume % of NO_2 = 1.5

Mole % of NO_2 = Volume % of NO_2 = 1.5

Molefraction of NO_2 = 1.5/100 = 0.015

Molefraction of air = 1 – 0.015 = 0.985

Total pressure = P = 760 mmHg

∴ Partial pressure of NO_2 in gas entering $= 0.015 \times 760 = 11.4$ mmHg

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Partial pressure of air in gas entering $= 0.985 \times 760 = 748.6$ mmHg $= 760 - 11.4 = 748.6$ mmHg

As the air is not going to adsorb, its partial pressure will remain constant throughout the adsorber.

Conversion of molefraction into weight fraction :

1 kmol of gas mixture contains 0.015 kmol of NO_2 and 0.985 kmol of air.

Amount of NO_2 in 1 kmol gas mixture $= 0.015 \times 46 = 0.69$ kg

Amount of air in 1 kmol gas mixture $= 0.985 \times 28.84$

$= 28.41$ kg

$\therefore$ Weight fraction of NO_2 in gas mixture $= \dfrac{0.69}{(0.69 + 28.41)}$

$= 0.0237$

or Weight fraction of NO_2 in the gas mixture $= y_1 = \dfrac{0.015 \times 46}{0.015 \times 46 + 0.985 \times 28.84}$

$= 0.0237$

Weight fraction of air in the gas mixture $= 1 - 0.0237 = 0.9763$

Gas entering the adsorber $= 0.126$ kg/s

NO_2 in gas entering $= 0.0237 \times 0.126 = 2.986 \times 10^{-3}$ kg/s

Air in gas entering $= 0.9763 \times 0.126 = 0.12304 \approx 0.123$ kg/s

We have, $y_1 = 0.0237$

kg NO_2/kg air in entering gas $= Y_1 = \dfrac{y_1}{1 - y_1} = \dfrac{0.0237}{1 - 0.0237} = 0.0243$

Mass flow rate of air (i.e., NO_2 free gas) entering the adsorber $= G_S = 0.123$ kg/s.

For continuous countercurrent adsorption, we have :

$$G_S (Y_1 - Y_2) = S_S (X_1 - X_2)$$

This is the operating line. On X, Y coordinates, it is a straight line of slope S_S/G_S that joins the terminal conditions (X_1, Y_1) and (X_2, Y_2).

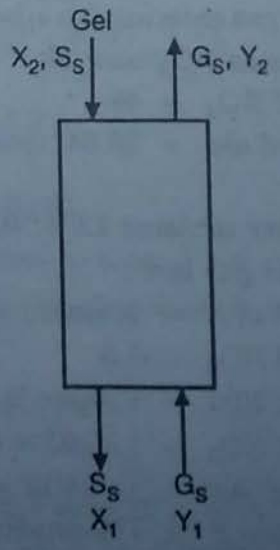

Fig. E 3.12 (a) : An adsorber

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Gel entering the adsorber is free of NO_2.

$\therefore$ $X_2 = 0$

NO_2 in entering gas $= 2.986 \times 10^{-3}$ kg/s

NO_2 removed from gas $= 90\%$

NO_2 removed from entering gas $= 0.90\,(2.986 \times 10^{-3})$

$= 2.6874 \times 10^{-3}$ kg/s

NO_2 in gas leaving adsorber $= 2.986 \times 10^{-3} - 2.6874 \times 10^{-3}$

$= 0.2986 \times 10^{-3}$ kg/s

As air is not adsorbed,

Air in gas leaving adsorber = air in gas entering adsorber

$= 0.123$ kg/s

$\therefore$ kg NO_2/kg air in gas leaving adsorber = $Y_2 = 0.2986 \times 10^{-3}/0.123$

$= 2.43 \times 10^{-3}$

$= 0.00243$

Now we will convert the data in terms of X, kg NO_2/kg gel and Y, kg NO_2/kg air (required to construct the plot of Y v/s X).

kg NO_2/100 kg gel	0	0.40	0.90	1.65	2.6	3.65	4.85
kg NO_2/kg gel	0	0.004	0.009	0.0165	0.026	0.0365	0.0485

Partial pressure of NO_2, mmHg	0	2	4	6	8	10	12

Now we will calculate kg NO_2/kg air, i.e., Y for each partial pressure.

For gas mixture following ideal gas law,

volume % = mole % = pressure %

$\therefore$ pressure fraction of A = molefraction of A

$$\frac{\text{pressure fraction of A}}{\text{pressure fraction of B}} = \frac{\text{molefraction of A}}{\text{molefraction of B}}$$

$$\frac{\text{partial pressure of A}}{\text{partial pressure of B}} = \frac{\text{moles of A}}{\text{moles of B}}$$

$$\frac{\text{partial pressure of A}}{\text{partial pressure of B}} = \frac{\text{kg of A/molecular weight of A}}{\text{kg of B/molecular weight of B}}$$

$$\frac{\text{kg of A}}{\text{kg of B}} = \frac{\text{partial pressure of A} \times \text{molecular weight of A}}{\text{partial pressure of B} \times \text{molecular weight of B}}$$

$$Y = \frac{\text{kg of } NO_2}{\text{kg of air}} = \frac{\text{partial pressure of } NO_2 \times 46}{\text{partial pressure of air} \times 28.84}$$

Partial pressure of air = 748.6 mmHg

(i) Partial pressure of NO_2 = 2 mmHg

$Y = 2 \times 46/748.6 \times 28.84 = 0.00426$

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(ii) Partial pressure of NO_2 = 4 mmHg

$$Y = 4 \times 46/748.6 \times 28.84 = 0.0085$$

Similarly, calculate Y at remaining partial pressure values (for NO_2) and tabulate.

Y = kg NO_2/kg air	0	0.00426	0.0085	0.0128	0.017	0.0213	0.0256
X = kg NO_2/kg gel	0	0.004	0.009	0.0165	0.0260	0.0365	0.0485

Plot Y v/s X.

Determination of minimum gel rate : Refer Fig. E 3.12 (b).

The point P (X_2, Y_2) is fixed and known.

For determining minimum gel rate :

Locate the point P (0, 0.00243) on the y-axis and through the point P, draw a line in such a way that it will be tangent to the equilibrium curve. The slope of this operating line $(S_S)_{min}/G_S$ then gives the value of minimum gel rate.

For minimum gel rate, a line PM is drawn as the line (i.e. operating line) of greatest slope which touches the equilibrium curve (tangent to the curve). The line PM is tangent to the curve at the point Q.

$$\text{Slope of line PM} = NL/PL = \frac{0.0288 - 0.00243}{0.04 - 0}$$

$$= 0.659 \approx 0.66$$

or Measure angle made by line PM with PL, $\alpha = 33°$

$$\text{Slope} = \tan\alpha = \tan(33) = 0.66$$

$$\therefore \quad \frac{(S_S)_{min}}{G_S} = 0.66$$

$$(S_S)_{min} = 0.66 \times G_S = 0.66 \times 0.123 = 0.08118 \text{ kg/s} \approx 0.0812 \text{ kg/s}$$

Minimum gel rate required = 0.0812 kg/s ... Ans. (i)

= 239.32 kg/h ... Ans. (i)

(iii) Determination of the number of transfer units :

$$\text{Operating gel rate} = S_S = 2 \times (S_S)_{min}$$

$$= 2 \times 0.0812$$

$$= 0.1624 \text{ kg/s}$$

$$\text{Slope of the operating line} = \frac{S_S}{G_S} = \frac{0.1624}{0.123} = 1.32$$

$$\text{Slope} = \tan\alpha = 1.32 \qquad \therefore \ \alpha = 52.85° \approx 53°$$

Draw the operating line through the point P (0, 0.00243) inclined at an angle of 53° to the horizontal line through the point P (line PR).

Now N_{tOG}, the number of overall gas transfer units are given by

$$N_{tOG} = \int_{Y_2}^{Y_1} \frac{dY}{Y - Y^*}$$

We have : $Y_1 = 0.0243$ and $Y_2 = 0.00243$.

Refer to Fig. E 3.12 (b) :

Now, the vertical distance between the operating line (line PR) and the equilibrium curve for any value of Y is $Y - Y^*$. So starting with $Y_1 = 0.0243$ and covering $Y_2 = 0.00243$ for different values of Y, get $Y - Y^*$ and tabulate.

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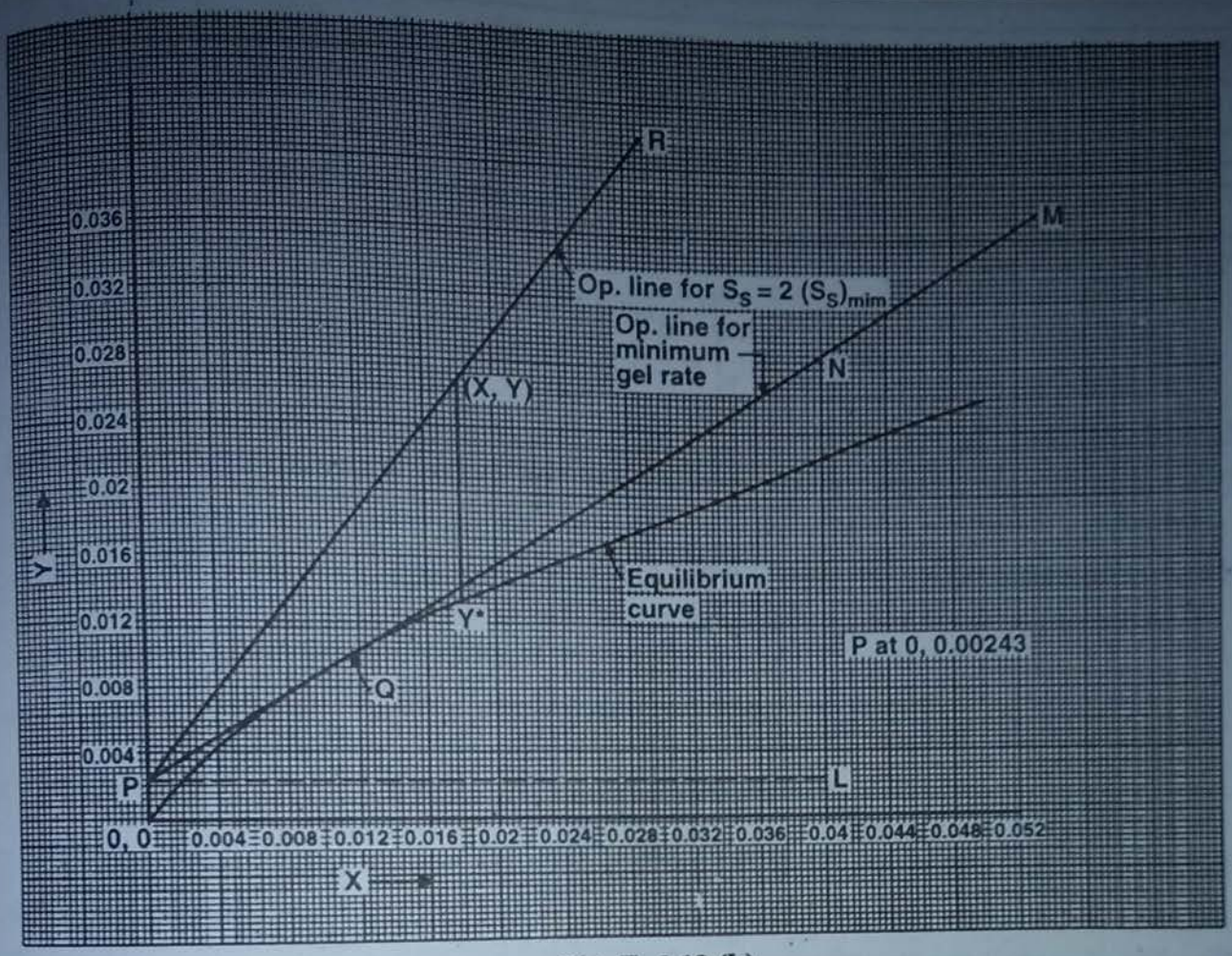

Fig. E 3.12 (b)

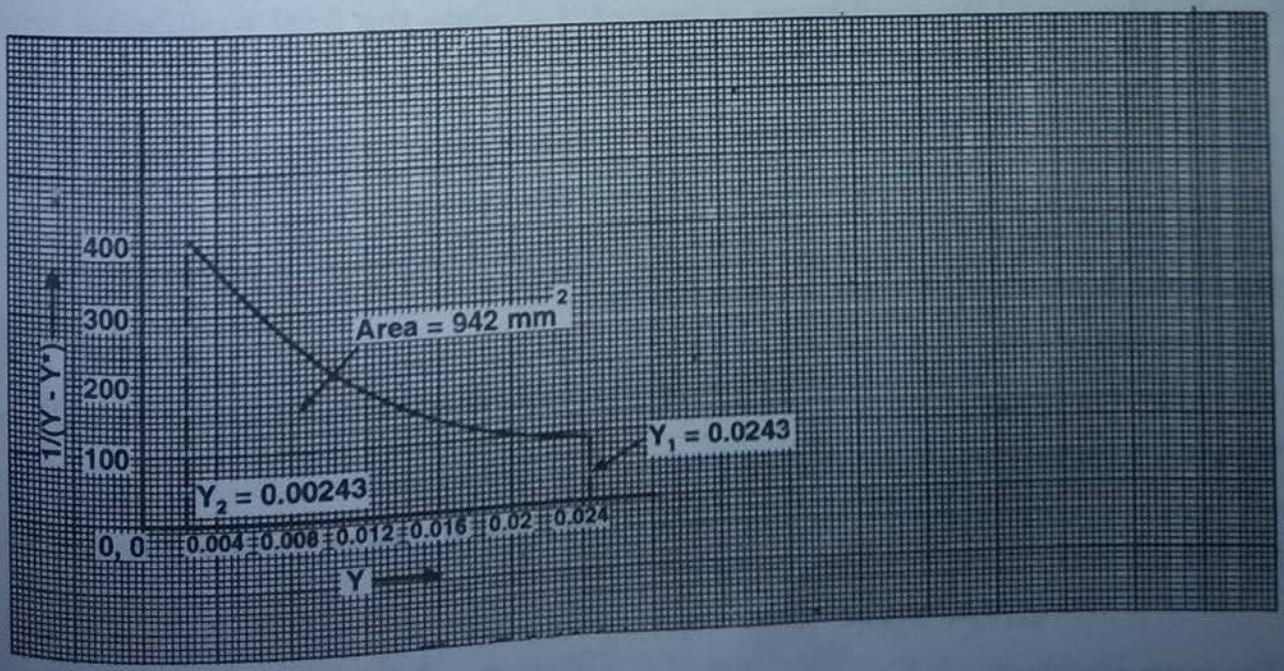

Fig. E 3.12 (c)

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Y	0.00243	0.008	0.012	0.016	0.02	0.0243
Y^*	0	0.00426	0.0066	0.009	0.0108	0.0124
$Y - Y^*$	0.00243	3.74×10^{-3}	5.4×10^{-3}	7×10^{-3}	9.2×10^{-3}	0.0119
$1/(Y - Y^*)$	411.5	267.4	185.2	143	108.7	84

Plot $1/(Y - Y^*)$ v/s Y.

$$N_{toG} = \int_{Y_2}^{Y_1} \frac{dY}{Y - Y^*}$$

$Y_1 = 0.0243$ and $Y_2 = 0.00243$

The value of the integral is the area under the curve between $Y_1 = 0.0243$ and $Y_2 = 0.00243$.

Measure the area under the curve : Refer Fig. E 3.12 (c).

Area under the curve $= 942 \text{ mm}^2 = 9.42 \text{ cm}^2$

$$N_{toG} = \int_{Y_1}^{Y_2} dY/(Y - Y^*)$$

= Area under the curve × (scale y-axis) × (scale x-axis)

$$= 9.42 \times \frac{100}{1} \times \frac{0.004}{1}$$

$$= 3.768 \approx 3.77$$

Number of overall gas transfer units = 3.77 **... Ans. (ii)**

Ex. 3.13 : A waste water solution having a volume of 1 m^3 contains 0.21 kg phenol/m^3 of solution. This solution is mixed thoroughly in a batch process with 1.4 kg of fresh granular activated carbon until the equilibrium is reached. Calculate the final equilibrium value and the percentage phenol removed from the solution.

The equilibrium data are :

kg phenol/m^3 solution	0.322	0.117	0.039	0.0061	0.0011
kg phenol/kg carbon	0.150	0.122	0.094	0.059	0.045

Solution : Basis : 1 m^3 of waste water solution.

Amount of adsorbent used = S_S = 1.4 kg

Amount of the solution to be treated = L = 1 m^3

Material balance statement for phenol :

The solute (phenol) removed from the solution is equal to phenol picked up by the solid (activated carbon).

$$L(Y_0 - Y_1) = S_S(X_1 - X_0)$$

For fresh adsorbent : $X_0 = 0$

$$\therefore \quad L(Y_0 - Y_1) = S_S(X_1)$$

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where, Y_0 = initial concentration of the adsorbable solute expressed as kg phenol/m^3 solution

Y_1 = final concentration of solution, kg phenol/m^3 solution

X_1 = solute (adsorbate) content of solids, kg phenol/kg activated carbon

$Y_0 = 0.21$ (given)

$$1\,(0.21 - Y_1) = 1.4\,X_1$$

$$0.21 - Y_1 = 1.4\,X_1$$

$$Y_1 = (-1.4)\,X_1 + 0.21$$

Plot the equilibrium data (Y v/s X).

Refer Fig. E 3.13.

We have, $Y_1 = (-1.4)\,X_1 + 0.21$ ...(1)

So as to find out the final equilibrium concentration values Y_1 and X_1, you can adopt the following procedure.

We have to plot the line given by Equation (1) on X-Y coordinates.

Let, $X_1 = 0$

$\therefore$ $Y_1 = (-1.4)\,0 + 0.21 = 0.21$

Let $X_1 = 0.15$

$\therefore$ $Y_1 = (-1.4) \times 0.15 + 0.21 = 0$

So draw the line [given by Equation (1)] passing through the points P (0, 0.21) and Q (0.15, 0).

The operating line PQ cuts the equilibrium curve at the point R.

As the final streams leaving the unit are in equilibrium with each other (given – equilibrium is reached), the coordinates of the point R are : Y_1 and X_1.

From Fig. E 3.13,

$Y_1 = 0.0625$ and $X_1 = 0.105$

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OR (Alternatively) :

We have, $L(Y_0 - Y_1) = S_S(X_1 - X_0)$

This is the operating line for single stage adsorption. This line passes through the points having coordinates (X_0, Y_0) and (X_1, Y_1) and has a slope of $-S_S/L$.

We have, $X_0 = 0,\ Y_0 = 0.21$

$$\text{Slope} = -\frac{S_S}{L} = -\frac{1.4}{1} = -1.4$$

$$\text{Slope} = \tan\alpha = -1.4 \qquad \therefore \alpha = \tan^{-1}(-1.4) = -54.46° \simeq -55°$$

So draw the operating line on the X-Y coordinates (plot) through the point P (0, 0.21) at an angle of –55°, i.e., at an angle of – 55° to the horizontal through the point P (measured in clockwise direction).

This operating line cuts the equilibrium curve at point R. As it is given that mixing is done till equilibrium is attained. The two streams will leave the stage/unit in equilibrium with each other and their concentrations will be given by the coordinates of the point R.

The coordinates of point R are :

$$Y_1 = 0.0625 \text{ and } X_1 = 0.105$$

Final (solute) concentration of solution $= Y_1 = 0.0625 \dfrac{\text{kg phenol}}{\text{m}^3 \text{ solution}}$

Final (solute) adsorbate content of activated carbon $= X_1 = 0.105 \dfrac{\text{kg phenol}}{\text{kg carbon}}$ **... Ans.**

Amount of phenol removed from the solution = Phenol adsorbed by the carbon $= X_1 \cdot S_S$

$= 0.105 \times 1.4$

$= 0.147$ kg

Amount of phenol in the solution (initial) $= \dfrac{0.21}{1} \times 1 = 0.21$ kg

[1] : % of phenol removed from the solution $= \dfrac{0.147}{0.21} \times 100 = 70$ **... Ans.**

OR

[2] : % of phenol removed from the solution $= \dfrac{Y_0 - Y_1}{Y_0} \times 100$

$= \dfrac{0.21 - 0.0625}{0.21} \times 100$

$= 70.23$ **... Ans.**

[Negligible difference in answer by (1) and (2) is incurred because of a very small error in plotting – indicates precision in plotting.]

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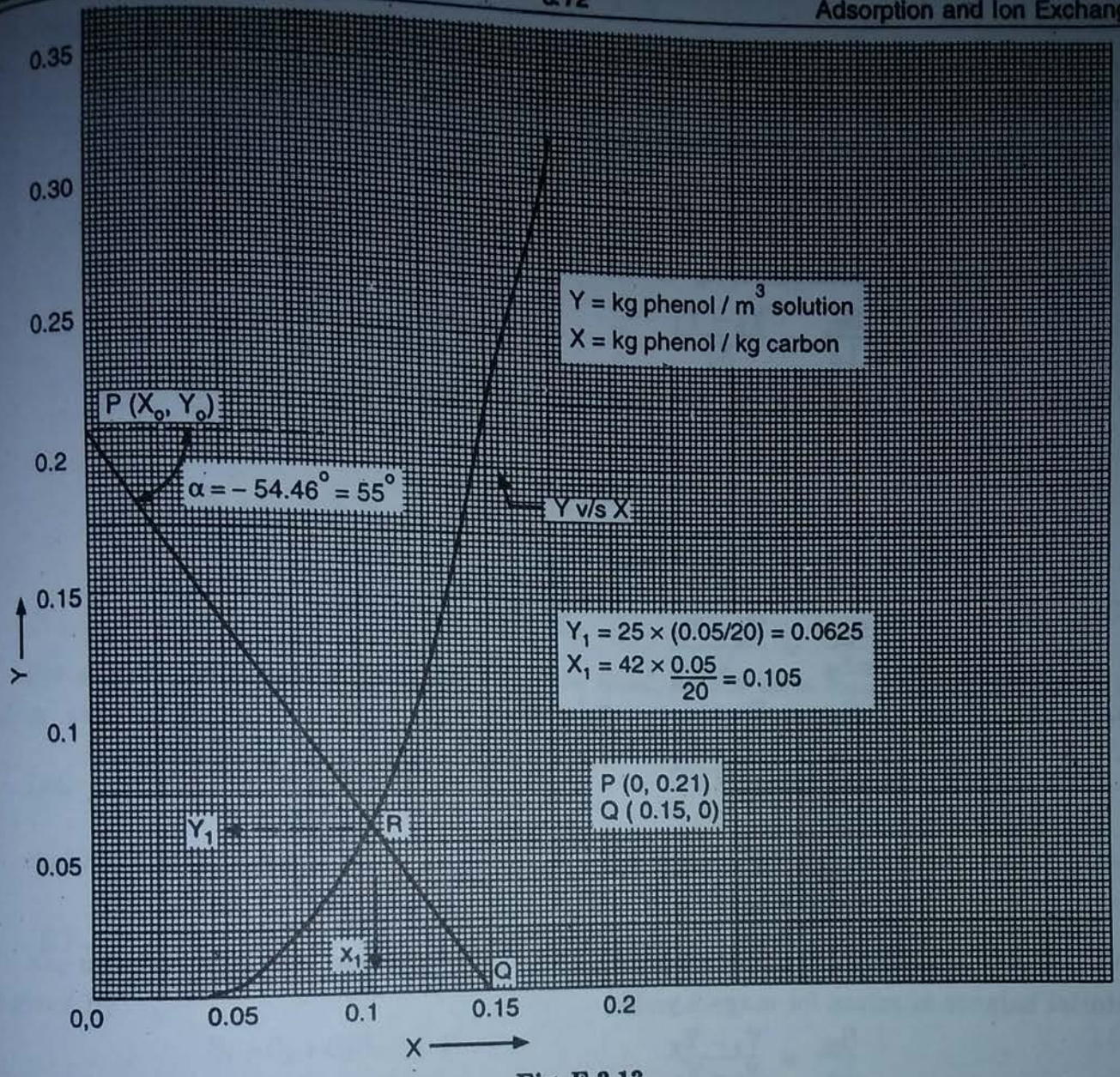

Fig. E 3.13

Ex. 3.14 : Prove that for two-stage crosscurrent treatment of liquid solutions by contact filtration, when the adsorption isotherm is linear, the least total adsorbent results if the amounts used in each stage are equal.

Solution :

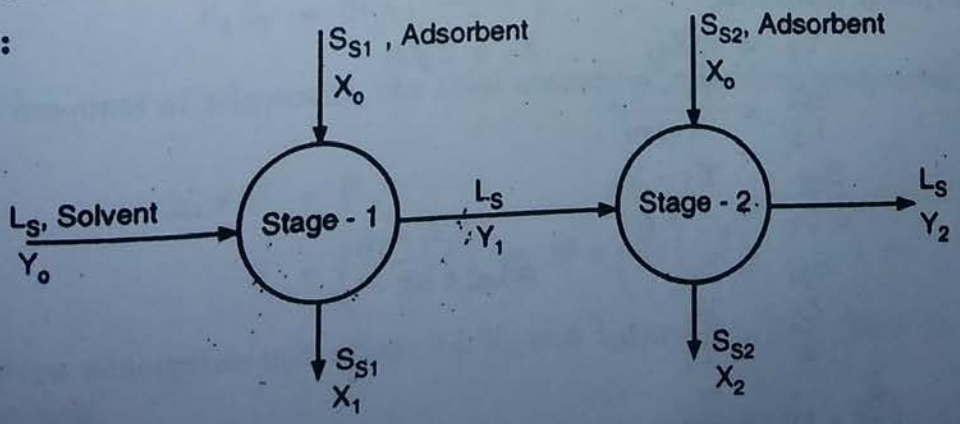

Fig. E 3.14 : Two-stage crosscurrent adsorption

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The solute concentration of the solution is reduced from Y_0 to Y_2 in two-stage crosscurrent adsorption operation.

Material balance of solute over stage-1 yields :

$$\frac{S_{S1}}{L_S} = \frac{Y_0 - Y_1}{X_1 - X_0}$$

For fresh adsorbent for each stage, $X_0 = 0$

$$\therefore \quad \frac{S_{S1}}{L_S} = \frac{Y_0 - Y_1}{X_1}$$

For linear adsorption isotherm, relationship between Y and X is :

$$Y = mX$$

$$\therefore \quad Y_1 = mX_1, \qquad X_1 = Y_1/m$$

$$\frac{S_{S1}}{L_S} = \frac{Y_0 - Y_1}{Y_1/m} = m\left(\frac{Y_0 - Y_1}{Y_1}\right)$$

$$\frac{S_{S1}}{mL_S} = \frac{Y_0 - Y_1}{Y_1}$$

Let, $S_{S1} = S_1, \qquad mL_S = k$

$$\therefore \quad \frac{S_1}{k} = \frac{Y_0 - Y_1}{Y_1}$$

$$\frac{S_1}{k} = \frac{Y_0}{Y_1} - 1$$

$$\left(\frac{S_1}{k} + 1\right) = \frac{Y_0}{Y_1} \qquad \therefore \; Y_1 = Y_0 / \left(\frac{S_1}{k} + 1\right) \qquad \ldots (1)$$

Material balance of solute for stage-2 yields :

$$\frac{S_{S2}}{L_S} = \frac{Y_1 - Y_2}{X_2 - X_0}$$

For fresh adsorbent, $X_0 = 0$

$$\frac{S_{S2}}{L_S} = \frac{Y_1 - Y_2}{X_2}$$

For linear adsorption isotherm,

$$Y_2 = mX_2, \qquad X_2 = Y_2/m$$

$$\frac{S_{S2}}{L_S} = \frac{Y_1 - Y_2}{Y_2/m}$$

$$\frac{S_{S2}}{mL_S} = \frac{Y_1 - Y_2}{Y_2}$$

Let, $S_{S2} = S_2, \qquad mL_S = k$

$$\frac{S_2}{k} = \frac{Y_1}{Y_2} - 1$$

$$\left(\frac{S_2}{k} + 1\right) = \frac{Y_1}{Y_2} \qquad \ldots (2)$$

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Put the value of Y_1 from Equation (1) into Equation (2).

$$\left(\frac{S_2}{k}+1\right) = \frac{Y_0}{\left(\frac{S_1}{k}+1\right)\cdot Y_2}$$

$$\left(\frac{S_2}{k}+1\right)\left(\frac{S_1}{k}+1\right) = \frac{Y_0}{Y_2}$$

$$\frac{S_2}{k}+\frac{S_1S_2}{k}+\frac{S_1}{k}+1 = \frac{Y_0}{Y_2}$$

$$\frac{S_2}{k}+\frac{S_1}{k}+\frac{S_1S_2}{k} = \frac{Y_0}{Y_2}-1 = \frac{Y_0-Y_2}{Y_2}$$

$$S_1+S_2+S_1S_2 = k\left[\frac{Y_0-Y_2}{Y_2}\right] = mL_S\left[\frac{Y_0-Y_2}{Y_2}\right]$$

$$S_1+S_2+S_1S_2 = mL_S\left[\frac{Y_0-Y_2}{Y_2}\right] \quad \ldots(3)$$

For a given adsorption, L_S, Y_0 and Y_2 are fixed, so the term $mL_S(Y_0 - Y_2)/Y_2$ can be taken as constant.

Let, $$mL_S\left(\frac{Y_0-Y_2}{Y_2}\right) = P$$

So Equation (3) becomes

$$S_1+S_2+S_1S_2 = P$$

For unequal amounts of adsorbent, the total amount of adsorbent required $(S_1 + S_2)$ is given by

$$S_1+S_2+S_1S_2 = P \quad \ldots(4)$$

For equal amounts of adsorbent :

$$S_1 = S_2 = S \quad \text{and} \quad S_1+S_2 = 2S$$

$$\therefore \quad S = (S_1+S_2)/2$$

$$S_1S_2 = S^2 = \left(\frac{S_1+S_2}{2}\right)^2$$

For equal amounts of adsorbent, the total amount of adsorbent required $(S_1 + S_2 = 2S)$ is given by, ...(5)

$$2S+S^2 = P$$

where $$S = \left(\frac{S_1+S_2}{2}\right), \text{ so } S^2 = \left(\frac{S_1+S_2}{2}\right)^2$$

For any given adsorption operation, Y_0, Y_2 and L_S are fixed, so the P is constant/fixed for given operation.

Refer to Equations (4) and (5).

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As the square of average of any two numbers [in Equation (5)] is always greater than the product of those two numbers [in Equation (4)], for any given value of P [i.e. for any given values of Y_0, Y_2, and L_S], the total amount of adsorbent calculated by using Equation (5) is always smaller than that calculated by using Equation (4).

Hence, for two-stage crosscurrent adsorption, when the adsorption isotherm is linear, the least total amount of adsorbent results if the amounts used in each stage are equal.

... Ans.

Let us have check for the above statement :

We have,

For unequal amounts

$$S_1 + S_2 + S_1S_2 = P$$

For given Y_0, Y_2 and L_S, i.e. for any given adsorption operation P is having fixed value.

Take P = 5 then :

$$S_1 + S_2 + S_1S_2 = P$$
$$S_1 + S_2 + S_1S_2 = 5$$
$$1 + 2 + 1 \times 2 = 5$$
$$\text{Total adsorbent} = S_1 + S_2 = 1 + 2 = 3 \text{ kg}$$

Take P = 53, then

$$S_1 + S_2 + S_1S_2 = P$$
$$S_1 + S_2 + S_1S_2 = 53$$
$$5 + 8 + 5 \times 8 = 53$$
$$\text{Total adsorbent} = S_1 + S_2$$
$$= 5 + 8 = 13 \text{ kg}$$

Take P = 175, then

$$S_1 + S_2 + S_1S_2 = P$$
$$S_1 + S_2 + S_1S_2 = 175$$
$$10 + 15 + 10 \times 15 = 175$$
$$\text{Total adsorbent} = S_1 + S_2 = 10 + 15$$
$$= 25 \text{ kg}$$

For equal amounts

$$2S + S^2 = P$$

$$2S + S^2 = P$$
$$2S + S^2 = 5, \; S^2 + 2S - 5 = 0$$
$$S = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-5)}}{2 \times 1} = 1.4495$$
$$\text{Total adsorbent} = 2S = 2 \times 1.4495$$
$$= 2.899 \text{ kg}$$

$$2S + S^2 = P$$
$$2S + S^2 = 53, \; S^2 + 2S - 53 = 0$$
$$\therefore S = 6.348 \approx 6.35$$

(positive root)

$$\text{Total adsorbent} = 2S = 2 \times 6.35$$
$$= 12.7 \text{ kg}$$

$$2S + S^2 = P$$
$$2S + S^2 = 175$$
$$S^2 + 2S - 175 = 0$$
$$\therefore S = 12.27 \text{ kg}$$
$$\text{Total adsorbent} = 2S = 2 \times 12.27$$
$$= 24.54 \text{ kg}$$

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Ex. 3.15 : A batch of water containing residual chlorine at a concentration of 12 ppm is to be treated with activated carbon at 25 °C to reduce the chlorine concentration to 0.5 ppm. Estimate the minimum mass of carbon per unit volume of water which can be used.

The equilibrium distribution coefficient $= \frac{c^*}{X} = 0.8 \frac{\text{kg } Cl_2/m^3 \text{ liquid}}{\text{kg } Cl_2/\text{kg C}}$

Solution : Basis : 1 m^3 of water.

1 ppm = 1 part in 10^6 parts (used to express the concentration of trace impurities).

For watery solution : 1 ppm = 1 mg/l = 1 kg/10^3 m^3

Let, v be the volume of water, v = 1 m^3

c_0 be the initial solute [Cl_2] concentration = 12 ppm = 12 kg/10^3 m^3 = $12 \times 10^{-3} \frac{kg}{m^3}$

c^* be the final equilibrium solute concentration

$$= 0.5 \text{ ppm}$$
$$= 0.5 \text{ kg}/10^3 \text{ m}^3$$
$$= 5 \times 10^{-4} \text{ kg}/\text{m}^3$$

Let S_S be the kg of carbon required.

For fresh carbon : $X_0 = 0$

Let X be the adsorbate concentration at equilibrium, kg Cl_2/kg C.

Material balance of Cl_2 :

$$V(c_0 - c^*) = S_S(X - X_0)$$
$$V(c_0 - c^*) = S_S X$$

We have, $$\frac{c^*}{X} = 0.80 \frac{\text{kg } Cl_2/m^3 \text{ liquid}}{\text{kg } Cl_2/\text{kg C}}$$

$$X = \frac{c^*}{0.80} = 1.25\, c^*$$

$\therefore$ $$X = 1.25 \times 5 \times 10^{-4} = 6.25 \times 10^{-4} \text{ kg}/\text{m}^3$$

$$1\,[12 \times 10^{-3} - 5 \times 10^{-4}] = S_S\,(6.25 \times 10^{-4})$$
$$S_S = 18.4 \text{ kg}$$

Minimum mass of carbon [per unit volume of water] to be used = 18.4 kg/m^3 **... Ans.**

❐❐❐

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LEACHING

- Leaching is an operation in which a solid mixture is contacted with a liquid solvent for the removal of one or more constituents of the solid mixture.
- The process of removing a solute (or solutes) from a solid by treating it with a liquid solvent is called leaching.
- This operation is also called as solid-liquid extraction or simply solid extraction.
- In leaching, a solid phase contacts a liquid phase, and mass is transferred from the solid phase to the liquid phase.
- Leaching involves the transfer of a soluble component of a solid phase (i.e., a soluble solid) into a relatively non-volatile liquid solvent. The soluble solid (the solute) from an insoluble solid/insoluble matter dissolves in the solvent used.
- The solid to be leached must be finely divided so that the liquid solvent contacts all parts of the solid. Usually, the desired component (the solute) is soluble while the undesired component is insoluble in the solvent used.

Leaching may be used either for the production of a concentrated solution of a valuable solid material, or in order to free an insoluble solid from a soluble material with which it is contaminated. This operation comprises of two steps : (i) contacting the solid with the liquid phase to effect transfer of solute to solvent and (ii) separation of the liquid phase from the solid. The streams resulting from the second step are : a solution of soluble constituent(s) in solvent, which is called as the overflow/strong leach solution and a mixture of insoluble solid/residue and solution - a wet solid comprising of insoluble solid material and adhering solution, which is called as the underflow or leached solids. The amount of the solution adhering to the solid generally depends upon the time provided for settling. The solution may contain some amount of an insoluble material if (i) solids do settle to give a clear solution and (ii) too much solution is withdrawn so that small amount of solid material is carried away. An every day example of leaching is making coffee in which case the soluble constituents of the ground coffee are separated from the insoluble grounds by solution in water.

Examples of leaching operation are :

(i) Leaching of gold from its ores by using sodium cyanide solutions.
(ii) Extraction of oil from oilseeds such as soyabeans by using hexane or petroleum ether as a solvent.
(iii) Extraction of sugar from sugar beats using hot water.
(iv) Extraction of tannin from tree barks using water.
(v) Extraction of medicinal compounds (fine pharmaceuticals) from plant roots, leaves and stems (in the pharmaceutical industry).
(vi) Extraction of perfumes from flowers.
(vii) Preparation of tea and coffee.

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Factors affecting the rate of leaching are :

(i) Particle size, (ii) solvent, (iii) temperature, and (iv) agitation.

Smaller size particles provide a greater interfacial area between the solid and liquid which in turn results in a higher rate of transfer of material. But the particle size should not be too small, otherwise the separation of the solid from the liquid and drainage of the solid residue will become more difficult.

The liquid used should be a good selective solvent and should have a low viscosity in order to circulate it freely.

Generally, the solubility of the soluble solute material to be leached increases with temperature, so high temperatures favour the leaching operation (results in a higher rate of extraction). For example, more tannin will be recovered out of barks by using hot water than cold water. Considering other factors there should be certain upper limit to the temperature (e.g., in the case of natural products, high temperatures results in chemical deterioration of the product).

Agitation increases the eddy diffusion and thus increases the rate of transfer of material from the solid surface to the bulk of the solution, i.e., agitation favours the leaching process. Agitation also prevents sedimentation/settling of solids, keeps the solid particles suspended in the liquid and thus helps to make effective use of the interfacial surface.

Prior to leaching so as to make the soluble portion of the solid more accessible to the solvent, a certain (prior) treatment should be given to the solids to be leached, e.g., crushing and grinding (ores), slicing (sugar beats) and flaking (vegetable seeds and beans).

The separation of a soluble constituent from a solid by means of a solvent, i.e., **leaching consists of two steps** :

(i) Contacting the solid with the selective solvent-the liquid in order to dissolve the soluble solute in the solvent.

(ii) Separation of the insoluble phases, i.e., separation of the liquid phase from the solid physically by settling, filtration, etc. (mechanical separation).

These two steps may be carried out in separate equipment or in the same piece of equipment. In actual practice, it is not possible to separate completely the liquid phase from the solid phase and so the solid will always retain some amount of solution. In normal operation, the streams leaving the second step are (i) a liquid phase – a clear solution of solute in solvent (called as the overflow/(strong) leach solution) and (ii) a slurry or a wet solid consisting of solid and adhering solution (called as the underflow/leached solids).

The above mentioned two steps – contacting solid with liquid solvent and separation of the overflow from the underflow constitute one stage and an ideal or equilibrium stage in leaching is a stage in which the solution retained by the solid in the underflow is of the same composition as the clear solution obtained as the overflow (so that $y^* = x$). (i.e., it is the one from which the resultant solution leaving is of the same composition as the solution adhering to the solids leaving from it.)

In case of leaching problems, we are generally concerned with calculations of the amount of soluble solute leached from a solid knowing the initial solute content of the solid, the amount of leaching solvent, etc. and determination of the number of stages required to reduce the solute content of the solid to some specified value knowing the amount of solvent, etc. We will be adopting a graphical procedure for determination of the number of stages, etc. and for the graphical solution we will make use of rectangular coordinates.

A three-component leaching system consists of : pure solvent (component A), insoluble solid/inert solid (B) and consists of soluble solute (C).

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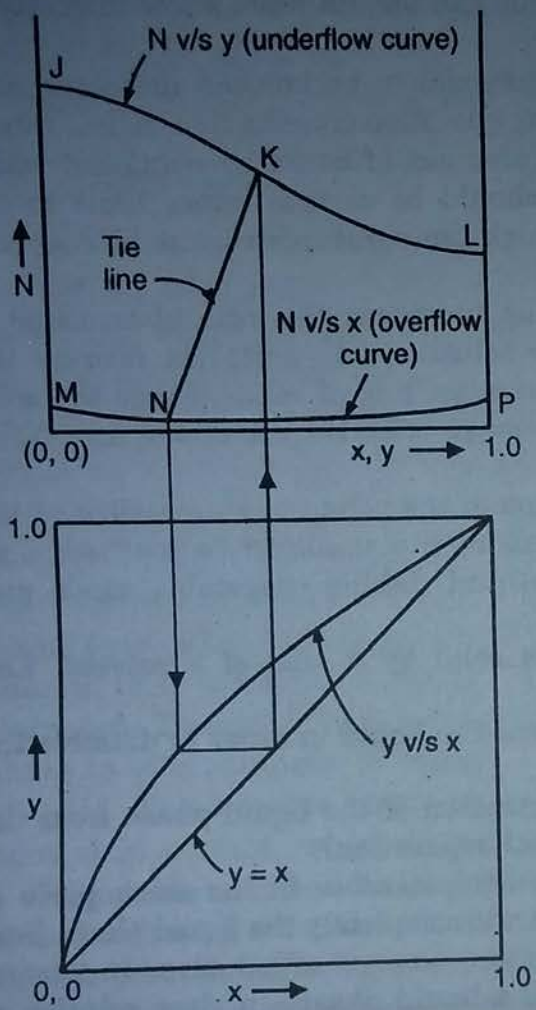
Graphical representation of equilibrium conditions in leaching :

In case of leaching operation, when the equilibrium data are plotted on rectangular coordinates with N (concentrations of insoluble solid B in any mixture – overflow and underflow) as ordinate and x, y (concentrations of solute C in the overflow and underflow) as abscissas, we may get one of the types of the equilibrium curves as shown in Fig. 4.1 (a) and (b) depending upon a given situation.



N : kg insoluble/kg solution

x : Wt. fraction C in overflow (B-free basis)

y^* : Wt. fraction C in underflow (B-free basis)

Fig. 4.1 (a) : Typical equilibrium diagram (solid in overflow, variable underflow)

Fig. 4.1 (a) represents a case where solute C is infinitely soluble in solvent A. In this case, therefore, x and y will have values over the entire range from 0 to 1.0. The curve JKL represents the separated solid (underflow/wet solid) as expected in actual practice. The curve JKL is a locus of underflow compositions. The curve MNP represents the clear solution withdrawn (overflow) and thus it is a locus of overflow compositions. Here the curve MNP lies above the N = 0 axis (x-axis here) which indicates that the overflow contains some amount of insoluble solid B. In this case, the presence of insoluble solid in the overflow indicates that either an insoluble solid is slightly soluble in the solvent or a partially settled solution is withdrawn so that some amount of insoluble solid is also carried away. Here, the tie lines (e.g., line KN) are not vertical and this will happen (i) if inadequate contact time with leaching solvent to dissolve all solute is provided, (ii) if adsorption of the solute occurs preferentially, (iii) if the solute C is soluble in the insoluble solid B.

Fig. 4.1 (b) represents a case where the insoluble solid is non-adsorbent.

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Fig. 4.1 (b) represents a case where adsorption of solute does not occur, so that clear solution is withdrawn (overflow) and solution associated with the solid in the underflow have the same composition as the overflow and the tie lines are vertical and thus an x - y curve coincides with the 45° diagonal and a distribution coefficient, $m = y^*/x$ equals unity. Here the overflow contains no insoluble solid B either dissolved or suspended so that the locus of overflow compositions is represented by the x-axis.

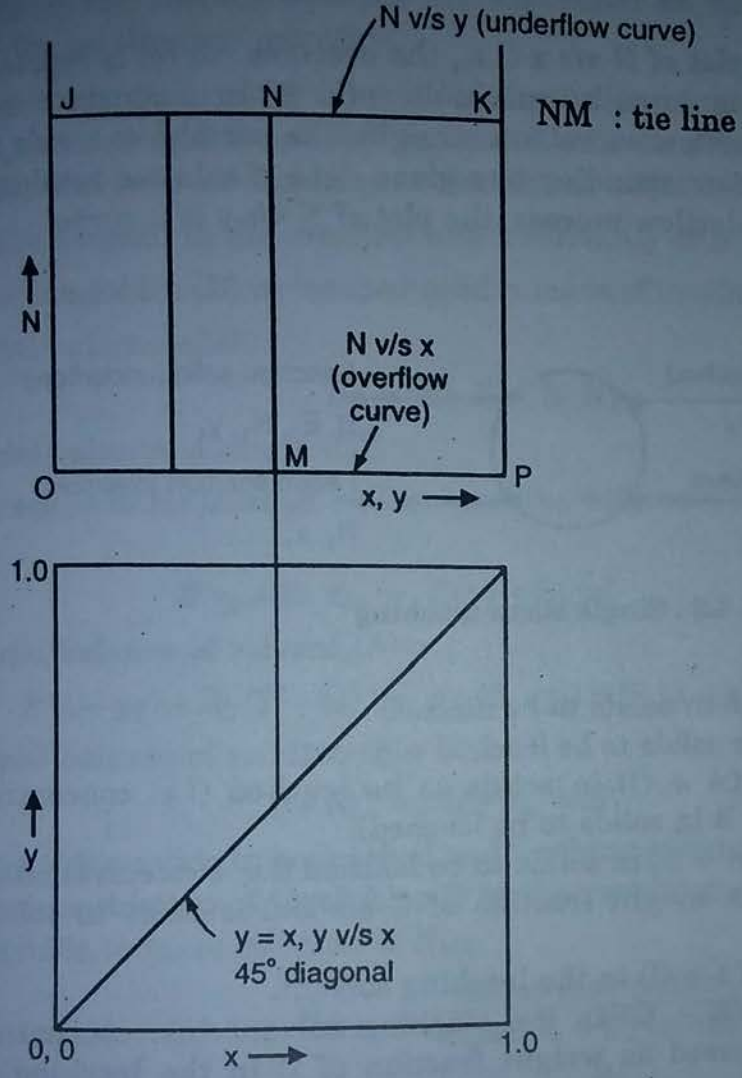

Fig. 4.1 (b) : Typical equilibrium diagram
(No solid in overflow, constant underflow)

Line JK representing the wet solids/underflow (locus of underflow compositions) is horizontal which indicates that the amount of solution retained by the insoluble solid B in the underflow is constant and independent of solute concentrations, i.e., solids are settled to the same extent at all solute concentrations. In practice, it is possible to operate thickener units in this manner.

Constant and Variable underflow :

The amount of solution retained by the insoluble solids in the underflow determines the locus of underflow compositions. We have to consider two cases – (i) constant underflow and (ii) variable underflow.

Variable underflow : When the amount of solution retained by the insoluble solid in the underflow varies and is a function of the solute concentration then we have variable underflow (i.e., the conditions are known as variable underflow). The ratio of insolubles

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to solution in the underflow is not constant and is a function of the solute concentration value. For the underflow from each stage of a cascade, N value is different and the plot of N v/s y is a curve.

Constant underflow : When the amount of solution retained by the insoluble solid in the underflow is constant and independent of the solute concentration then we have constant underflow (i.e., the conditions are known). The ratio of the insolubles to the solution in the underflow is constant from all the stages, i.e., we have a constant value of N for all the underflows so that a plot of N v/s y is a straight line parallel to x-axis at a given N (given B/(A + C) ratio).

In the graphical procedure, a plot of N v/s x (i.e., the overflow curve) is represented by x-axis as normally the overflow contains no insoluble solid B. In a constant underflow process, N v/s y plot, i.e., the underflow curve is a straight line parallel to x-axis and is to be drawn through an ordinate N corresponding to a given data of solution retained per kg of insoluble solid. In a variable underflow process, the plot of N v/s y is a curve.

Single Stage Leaching :

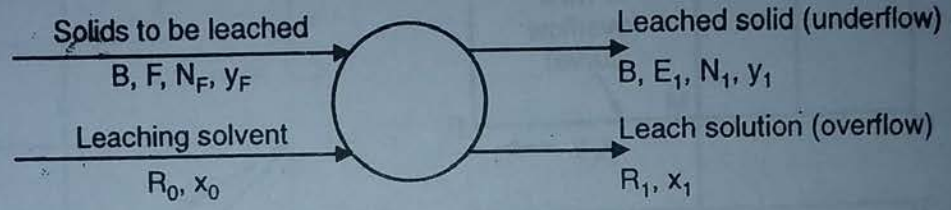

Fig. 4.2 : Single stage leaching

Notation Adopted :

- B – mass insolubles in solids to be leached
- F – mass (A + C) in solids to be leached
- N_F – mass B/mass (A + C) in solids to be leached (i.e. concentration of insoluble solid B in solids to be leached).
- y_F – mass C/mass (A + C) in solids to be leached (i.e. concentration of solute C expressed as weight fraction of C on B-free basis in solids to be leached).
- R_0 – mass solution (A + C) in the leaching solvent.
- x_0 – mass C/mass (A + C) in the leaching solvent (i.e. concentration of solute C expressed as weight fraction of C in the leaching solvent, B-free basis).
- E_1 – mass (A + C) in the leached solid i.e. mass solution associated with the leached solid (solution in the underflow).
- N_1 – mass B/mass (A + C) in the leached solid i.e. concentration of insoluble solid B in the leached solid i.e. underflow on B-free basis.
- B – mass insolubles in the leached solid.
- y_1 – mass C/mass (A + C) in the leached solid i.e. concentration of solute C in the leached solid (underflow) expressed as weight fraction of C on B-free basis.
- R_1 – mass solution (A + C) in the leach solution (overflow)
- x_1 – mass C/mass (A + C) in the leach solution i.e. concentration of solute C in the leach solution (overflow) expressed as weight fraction of C on B-free basis.

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Consider the single stage leaching operation of Fig. 4.2, in which the solids to be leached are contacted with leaching solvent and then the resulting insoluble phases are physically separated and the effluent streams are withdrawn as leached solid and leach solution.

The circle in Fig. 4.2 represents a stage in which the entire operation of leaching, i.e., mixing of solid with solvent and mechanical separation of the phases is carried out.

Weights of the various streams are expressed as mass for a batch operation or as mass/time for continuous operation.

Since in many cases a clear liquid leach solution is obtained, i.e., the overflow does not contain any solid B and the solid B is insoluble in the solvent used, mass of the insolubles (B) in the leached solids is the same as that in the solids to be leached.

∴ For no solid in the overflow and insolubility of B in A :

mass of insolubles (B) in leached solid = mass of insolubles in the solid to be leached.

Material balance of B :

$$B = N_F F = E_1 N_1 \qquad \ldots (4.1)$$

Material balance of solute (C) :

C in solids to be leached + C in leaching solvent = C in leached solids + C in leach solution

$$F y_F + R_0 x_0 = E_1 y_1 + R_1 x_1 \qquad \ldots (4.2)$$

Material balance of solvent (A) :

$$F(1 - y_F) + R_0 (1 - x_0) = E_1 (1 - y_1) + R_1 (1 - x_1) \qquad \ldots (4.3)$$

Material balance of solution (A + C) :

$$F + R_0 = E_1 + R_1 = M_1 \qquad \ldots (4.4)$$

Mixing of the solids to be leached and leaching solvent produces a B-free mixture M_1, so that the concentration of insolubles B in the resulting mixture expressed as the ratio of mass insoluble to mass solution is N_{M1}.

$$N_{M1} = \frac{\text{mass insolubles}}{\text{mass (A + C) totally}}$$

$$N_{M1} = \frac{B}{F + R_0} = \frac{B}{M_1} \qquad \ldots (4.5)$$

The concentration of the solute C in the resulting mixture expressed as the weight fraction of C on a B-free basis in the resulting mixture, y_{M1} is determined by the material balance of C :

$$F y_F + R_0 x_0 = E_1 y_1 + R_1 x_1 = y_{M1} M_1$$

$$F y_F + R_0 x_0 = y_{M1} M_1$$

$$y_{M1} = \frac{F y_F + R_0 x_0}{M_1} \qquad \ldots (4.6)$$

but $M_1 = F + R_0$

∴

$$y_{M1} = \frac{F y_F + R_0 x_0}{F + R_0} \qquad \ldots (4.7)$$

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Fig. 4.3 is a graphical representation of these relations.

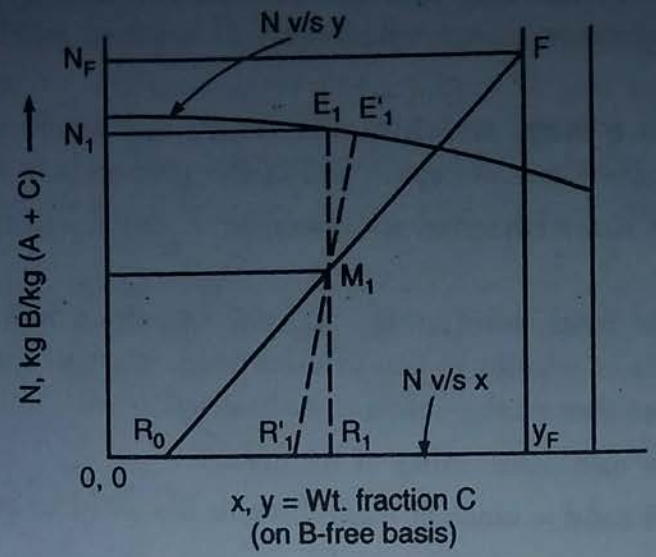

Fig. 4.3 : Single stage leaching

(variable underflow, no solid in overflow, with E_1 R_1 tie line vertical)

In Fig. 4.3, point F (y_F, N_F) represents the solids to be leached, i.e., feed to the leaching unit, point R_0 (x_0, 0) represents the leaching solvent and point M_1 (y_{M1}, N_{M1}) represents the overall mixture (of solid + solvent that can split into overflow and underflow).

Equation (4.4) indicates that F, R_0 and M_1 must lie on the same straight line in

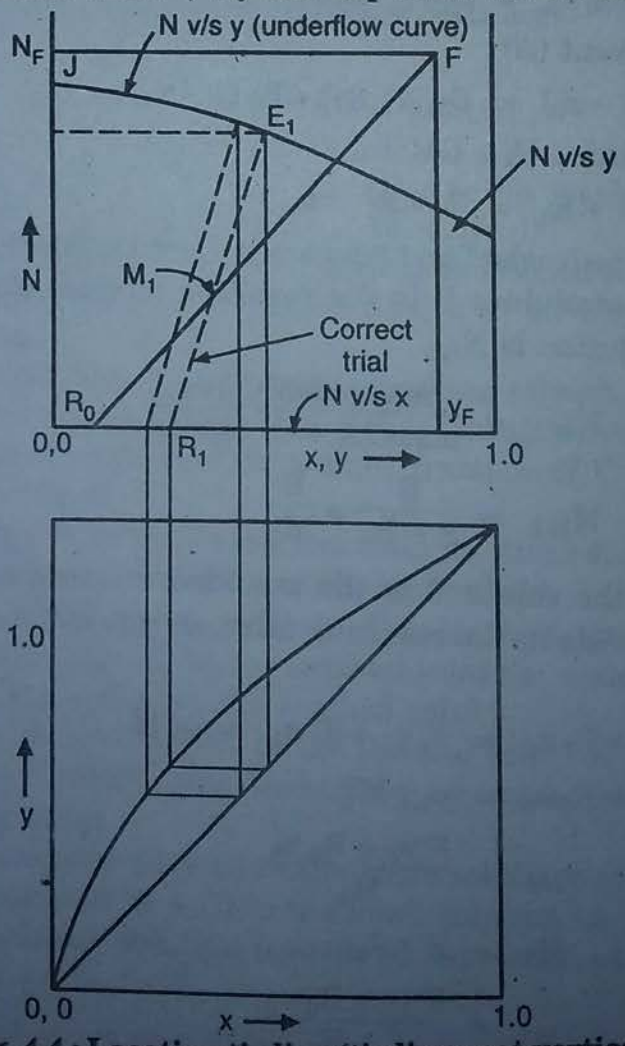

Fig. 4.4 : Locating tie line (tie lines not vertical)

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accordance with the characteristics of these diagrams – the ratio FM_1/M_1R_0 = mass R_0/mass F. Also the points E_1, R_1 and M_1 must lie on the same straight line. For an equilibrium stage, the effluent streams (E_1 – representing the leached solid/underflow and R_1 - representing the leach solution/overflow) leave the stage in equilibrium so that y_1 and x_1 are in equilibrium with each other (i.e., these are the equilibrium values). So M_1 falls on the line FR_0 and E_1 and R_1 representing the effluent streams are located at the opposite ends of the tie line through M_1. The tie line is located by a trial and error method (tie lines that are sloping), N v/s y is the equilibrium plot (locus of underflow compositions) for the leached solid/underflow/wet solid and N v/s x is the equilibrium plot for the leach solution/overflow. Coordinates of E_1 and R_1 give the compositions of the leached solid and leach solution. Read x_1 and y_1. Equations (4.2) and (4.4) are used to calculate E_1 and R_1. Equations (4.6) and (4.7) are also used to locate M_1 on FR_0.

Multistage Crosscurrent Leaching :

This is simply an extension of single-stage leaching in which the leached solids are successively contacted with fresh leaching solvent. This may be performed in a batch or a continuous fashion. Here the leached solids from any stage act as the feed solids to the next, so more and more amount of the solute is removed from the solids as we move along the cascade.

The material balance calculations that are done for a single-stage are now applicable here to the first stage. The calculations for the subsequent stages are to be performed in the same manner, except that the leach solids from any stage acting as the feed solids to the next.

Fig. 4.5 shows a three-stage crosscurrent leaching operation in which unequal amounts of the leaching solvent of the same composition are used in all the three stages.

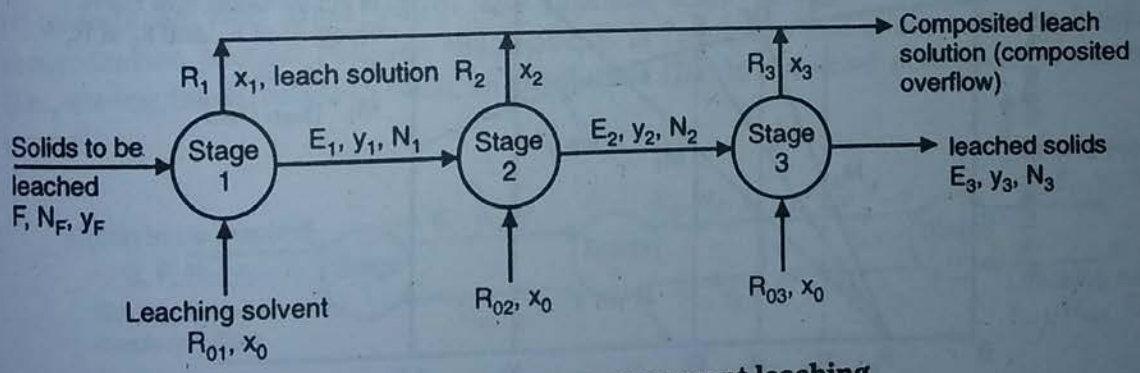

Fig. 4.5 : Three-stage crosscurrent leaching

Assume that there are no insolubles in the leach solution (overflow) so that B in solids to be leached is equal to B in leached solids from all the stages.

Material balance of insolubles (B) for any stage-n :

$$B = E_{n-1} N_{n-1} = E_n N_n \quad \ldots (4.8)$$

For stage-1, it becomes

$$B = E_1 N_1 = N_F F$$

Solution (A + C) balance for any stage-n :

$$E_{n-1} + R_{0n} = E_n + R_n = M_n \quad \ldots (4.9)$$

For stage-1 :

$$E_{n-1} = F \quad \ldots (4.10)$$

$$\therefore \quad F + R_{01} = E_1 + R_1 = M_1$$

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For stage-2, it becomes :

$$E_1 + R_{02} = E_2 + R_2 = M_2$$

Balance of solute (C) for any stage-n :

$$E_{n-1}\, y_{n-1} + R_{0n}\, x_0 = E_n\, y_n + R_n\, x_n = M_n\, y_{Mn} \qquad \ldots (4.11)$$

$$\therefore \quad y_{Mn} = \frac{E_{n-1}\, y_{n-1} + R_{0n}\, x_0}{M_n}$$

$$y_{Mn} = \frac{E_{n-1}\, y_{n-1} + R_{0n}\, x_0}{E_{n-1} + R_{0n}} \qquad \ldots (4.12)$$

But for stage-1 : $E_{n-1} = F$ and $y_{n-1} = y_F$

$$\therefore \quad y_{M1} = \frac{F\, y_F + R_{01}\, x_0}{F + R_{01}}$$

Concentration of insolubles :

The material balance of B for any stage-n is :

$$B = E_{n-1}\, N_{n-1} = E_n\, N_n = N_{Mn}\, M_n$$

$$\therefore \quad N_{Mn} = \frac{B}{M_n} = \frac{B}{E_{n-1} + R_{0n}} \qquad \ldots (4.13)$$

but for stage-1 : $E_{n-1} = F$

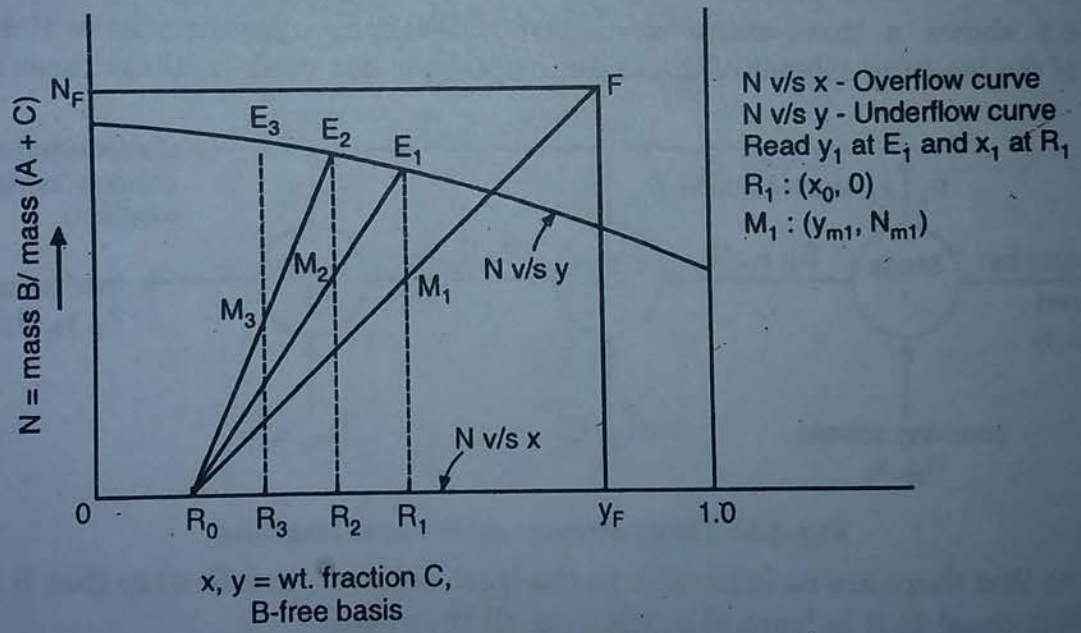

Fig. 4.6 : Three-stage crosscurrent leaching

(variable underflow, no insolubles in overflow and tie lines-vertical)

Fig. 4.6 is a graphical representation of the various streams for a three-stage crosscurrent leaching operation. The leaching solvents containing R_{01}, R_{02} and R_{03} mass of (A + C) are of the same composition x_0 (of solute C on B-free basis), so they are represented by the point R_0 (x_0, 0) in Fig. 4.6.

(i) Locate points F (y_F, F) representing the solids to be leached and R_0 (x_0, 0) representing the leaching solvent.

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(ii) As per Equation (4.10), the points F, $R_{01}(= R_0)$, and M_1 must lie on the same straight line, join FR_0 and locate M_1 on it either by making use of $FM_1/M_1R_0 = R_0/F$ or coordinates of $M_1 - y_{M1}$ and N_{M1}.

(iii) The effluent streams leaving the stage-1, i.e., E_1 and R_1 are in equilibrium, line E_1R_1 is the required tie line. E_1 lies on N v/s y curve (underflow curve) and R_1 lies on N v/s x curve (overflow curve). Equation (4.10) indicates that the points E_1, R_1 and M_1 must lie on the same straight line, so draw a tie line through M_1 by a trial and error procedure. Here the tie lines are drawn vertical assuming that no adsorption of solute occurs and all the solute is dissolved in solvent so that $y^* = x$. The coordinates of E_1 and R_1 give the compositions of the leached solids (y_1) and leach solution (x_1).

(iv) Now E_1 on the underflow curve represents the feed solids to the stage-2 (the leached solids from the stage-1 are now the feed solids to the stage-2). As per Equation (4.9), the points E_1, R_{02} (= R_0) and M_2 must lie on the same straight line, so join E_1 R_0 and locate M_2 on it. Through M_2 draw a vertical tie line of which the two ends are E_2 and R_2. The coordinates of E_2 and R_2 give the compositions of the leached solids (underflow) i.e. y_2 and leach solution (overflow) i.e. x_2.

(v) Adopt the same procedure for the stage-3 and get y_3 and x_3. R_1, R_2, and R_3 will be on the N v/s x curve and E_1, E_2 and E_3 will be on the N v/s y curve.

Multistage Countercurrent Leaching :

Consider a multistage countercurrent leaching system containing n stages as shown in Fig. 4.7. Stage-1 represents one end of the system at which the solids to be leached are fed and from which strong leach solution (overflow) of highest solute concentration leaves the system. Stage-n represents the another end of the system at which a fresh leaching solvent is introduced and from which the leached solid (underflow) with the lowest solute concentration leaves. Assume that solid B is insoluble and a clear strong leach solution [i.e., strong leach solution with no insolubles (B)] is obtained and steady-state conditions prevail.

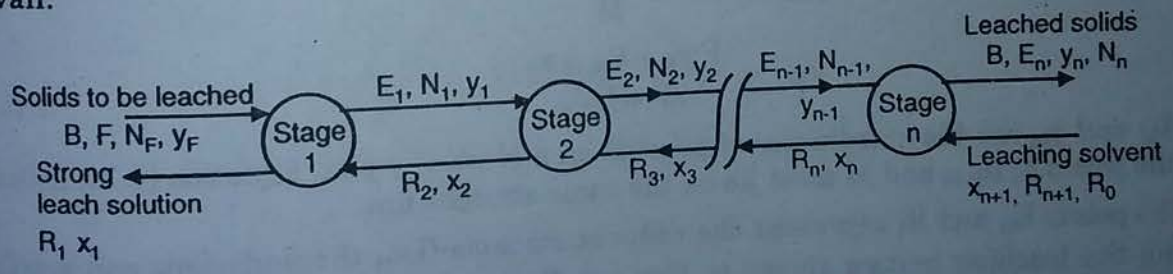

Fig. 4.7 : Multistage countercurrent leaching

F – mass (A + C)/time in solids to be leached

B – mass/time insolubles in solids to be leached

N_F – mass B/mass (A + C) in solids to be leached

y_F – mass C/mass (A + C) in solids to be leached
i.e. weight fraction of C in solids to be leached on B-free basis.

R_1 – mass (A + C)/time strong leach solution (overflow)

x_1 – mass C/mass (A + C) in the strong leach solution
Concentration of solute C in the strong leach solution expressed as weight fraction of C in the strong leach solution on B-free basis.

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R_{n+1} – mass (A + C)/time fresh leaching solvent

x_{n+1} – mass C/mass (A + C) in fresh leaching solvent i.e. weight fraction of C in leaching solvent on B-free basis.

B – mass insolubles/time in the leached solids

E_n – mass (A + C)/time in the leached solids (underflow)

N_n – mass B/mass (A + C) in the leached solids i.e. concentration of insolubles B as the ratio of mass B to mass (A + C) in the leached solids

y_n – mass C/mass (A + C) in the leached solids i.e. concentration of solute C in the leached solids (underflow) expressed as weight fraction of C on B-free basis.

The material balance of solution (A + C) over the entire unit is :

$$F + R_{n+1} = R_1 + E_n = M \qquad \ldots (4.14)$$

The material balance of solute (C) over the entire unit is :

$$F y_F + R_{n+1} x_{n+1} = R_1 x_1 + E_n y_n = M y_M \qquad \ldots (4.15)$$

As the overflow contains no insolubles, we have :

$$B = N_F F = E_n y_n \qquad \ldots (4.16)$$

Mixing of solids to be leached and leaching solvent produces a B-free mixture of mass M, so that in the resulting slurry mixture, the concentration of insolubles expressed as mass insoluble/mass solution is

$$N_M = \frac{B}{M} = \frac{B}{F + R_{n+1}} \qquad \ldots (4.17)$$

We have :

$$F y_F + R_{n+1} x_{n+1} = M y_M$$

$$\therefore \quad y_M = \frac{F y_F + R_{n+1} x_{n+1}}{M}$$

$$y_M = \frac{F y_F + R_{n+1} x_{n+1}}{F + R_{n+1}} \qquad \ldots (4.18)$$

N_M and y_M are the coordinates of point M on a N v/s x, y plot. Equation (4.14) indicates that the points F, R_{n+1} and M must lie on the same straight line.

The points E_n and R_1 represent the effluent streams (i.e., the underflow and overflow) leaving the leaching system shown in Fig. 4.6. Equation (4.14) also indicates that points E_n, R_1 and M must lie on the same straight line, i.e., line joining $E_n - R_1$ must pass through M.

Equation (4.14) can be rearranged as :

$$F - R_1 = E_n - R_{n+1} = \Delta_R \qquad \ldots (4.19)$$

Material balance of solution (A + C) over the first two stages :

$$F + R_3 = R_1 + E_2$$

$$\therefore \quad F - R_1 = E_2 - R_3 = \Delta_R \qquad \ldots (4.20)$$

Similarly, material balance of solution over the first three stages :

$$F + R_4 = R_1 + E_3$$

$$F - R_1 = E_3 - R_4 = \Delta_R \qquad \ldots (4.21)$$

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From Equations (4.19), (4.20) and (4.21), it is clear that the difference in flow E–R at a location between any two adjacent stages is constant and it is equal to Δ_R. Usually, Δ_R is a negative quantity (as E < R). Δ_R is the difference point and is the net flow outward at the last stage-n.

In the graphical representation, according to Equation (4.19), the extended lines FR_1 and $E_n R_{n+1}$ must intersect at the point Δ_R, in other words, Δ_R is the point of intersection of extended lines FR_1 and $E_n R_{n+1}$.

Equation (4.14) indicates that M lies on a line joining points F and R_{n+1}, so the point $M (y_M, N_M)$ is located on the line $F - R_{n+1}$. Also E_n, R_1 and M lie on the same straight line and E_n lies on the N v/s y equilibrium curve, so the line joining E_n and M is extended to meet the N v/s x curve at R_1.

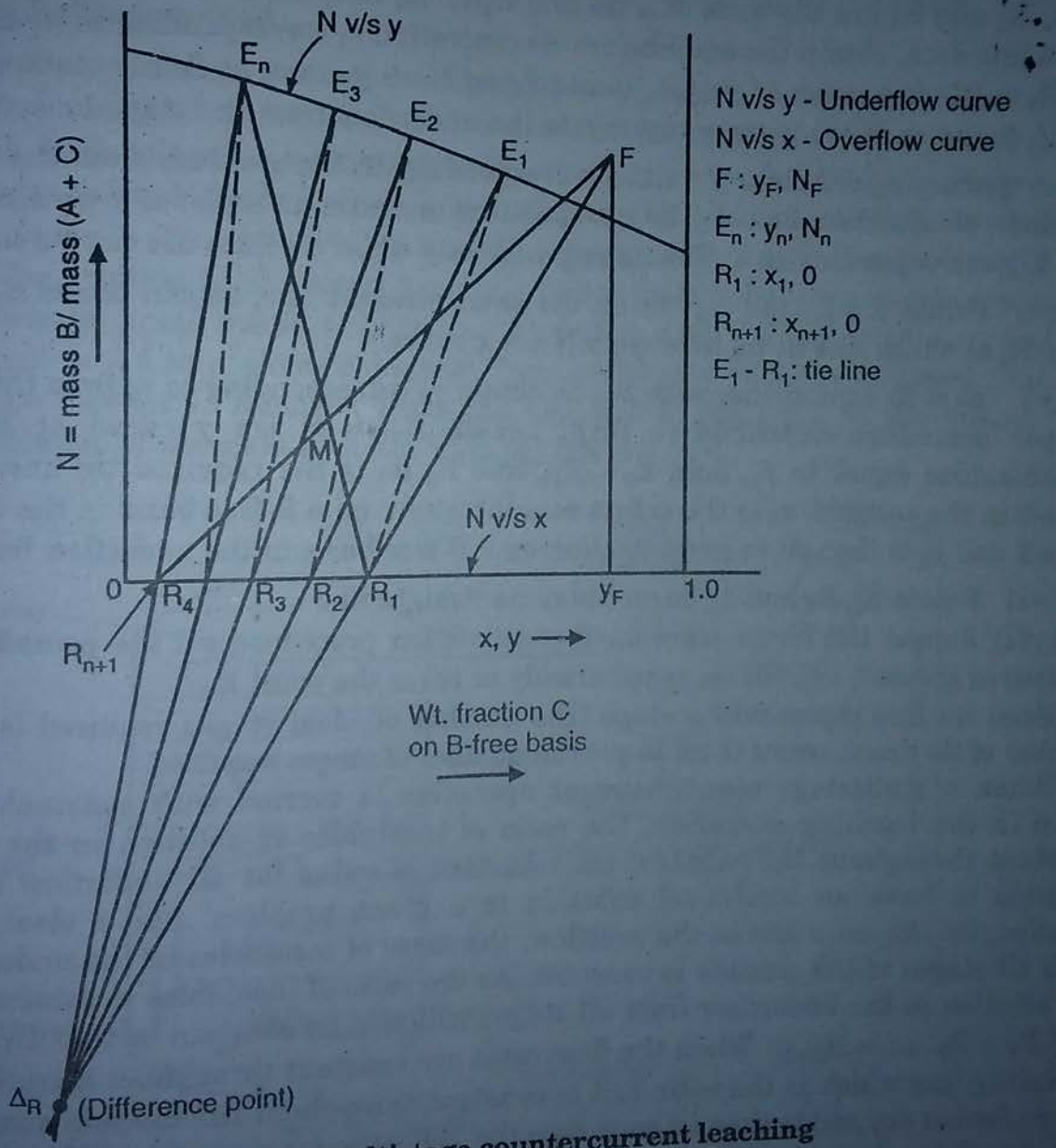

Fig. 4.8 : Multistage countercurrent leaching

[No solids in overflow, variable underflow and tie lines - not vertical (sloping)]

A graphical procedure for construction of the number of stages required is shown in Fig. 4.8. (N v/s x, y plot).

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(i) Locate points F (y_F, N_F), R_{n+1} (x_{n+1}, 0), R_1 (x, 0) and E_n (y_n, N_n). (We can locate R_1 by locating M on line F R_{n+1} and extending line E_n M to meet the N v/s x curve at R_1 if not known).

(ii) Draw a line joining the points F and R_1 and extend. Draw a line joining points E_n and R_{n+1} and extend to meet the line FR_1 extended at Δ_R (the difference point). The point of intersection of lines FR_1 and E_n R_{n+1} extended is the point Δ_R, so locate it.

(iii) Now E_1 and R_1 – the effluent streams from the stage-1 are in equilibrium with each other. E_1 lies on the N v/s y curve (the underflow curve) and R_1 on the N v/s x curve (the overflow curve).

As the effluent streams E_1 and R_1 are in equilibrium, x_1 and y_1 are the equilibrium values. E_1 and R_1 are the ends of a tie line E_1R_1. R_1 is already located, so locate E_1. From the tie line data, obtain the equilibrium concentration y_1 corresponding to x_1 and mark the point E_1 on N v/s y curve at y_1, i.e., locate E_1 on N v/s y curve at B-free concentration equal to y_1. E_1 lieing on N v/s y curve represents the underflow from the stage-1.

For getting equilibrium y values corresponding to the x values from the tie line data provided - obtain the value of y by interpolation or construct a plot of y v/s x and then read from it y corresponding to x. This is required only when tie lines are not vertical.

(iv) Points E_1, R_2 and Δ_R lie on the same straight line, so join points E_1 and Δ_R and mark R_2 at which line E_1 Δ_R intersects N v/s x curve.

(v) E_2 is in equilibrium with R_2. So obtain y_2 corresponding to x_2 from the tie line data [as per procedure described in (iii)]. Locate E_2 on N v/s y curve at B-free solute concentration equal to y_2. Join $E_2 - R_2$, line E_2 R_2 is the required tie line representing stage-2 in the cascade. x_2 is the solute concentration on a B-free basis in the overflow from stage-2 and y_2 is the solute concentration on a B-free basis in the underflow from stage-2.

(vi) Points E_3, R_3 and Δ_R lie on the same straight line .....

(vii) Repeat the above mentioned construction procedure till the point E_n is exactly reached or crossed, i.e., till we reach exactly or cover the point E_n.

Each tie line represents a stage (the number of ideal stages required is equal to the number of tie lines), count them to get the number of stages required.

When a multistage countercurrent operation is carried with constant underflow – when in the leaching operation, the ratio of insolubles to solution in the underflow is constant throughout the cascade, i.e., constant N value for all underflow streams, it is possible to have an analytical solution to a given problem. For a clear liquid leach solution, i.e., for no solids in the overflow, the mass of insolubles in the underflow streams from all stages of the cascade is constant. As the ratio of insolubles to solution is constant, the solution in the underflow from all stages will also be constant so that $E_1 = E_2 = \ldots = E_n$ and $R_1 = R_2 \ \ldots \ = R_{n+1}$. When the flow rates are constant throughout then the slope of the operating line which is the ratio E/R is constant throughout the cascade except for stage-1 (when feed is dry solids then in such case the E/R ratio for stage-1 is different from that for the rest of the stages in the cascade). If, in addition, the equilibrium distribution curve is a straight line passing through the origin - $y^* = mx$ then the following equation is used to calculate the number of stages required :

$$\frac{y_1 - y_n}{y_1 - m\,x_{n+1}} = \frac{(R/mE)^n - R/mE}{(R/mE)^n - 1} \qquad \ldots (4.22)$$

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When the feed has the same insolubles to solution ratio as that of the underflow from the rest of the stages, then the following equation is applicable :

$$\frac{y_F - y_n}{y_F - m\,x_{n+1}} = \frac{(R/mE)^{n+1} - R/mE}{(R/mE)^{n+1} - 1} \qquad ...(4.23)$$

Equipments for Leaching of Vegetable Seeds :

Oil-bearing seeds such as cotton seeds, soyabean, linseeds, etc. are leached with organic solvents for removing vegetable oils from them. Hexane is the most commonly used leaching solvent. In the oil industry, the solution of oil in solvent containing a small amount of finely divided suspended solids is called as miscella and leaching equipments are called as extractors. The dilute solution of oil in solvent is called as half miscella and the final strong/concentrated solution of oil in solvent is called as full miscella.

Batch plant for Extraction of oil :

It consists of a cylindrical vertical vessel which is divided into two sections by a slant partition as shown in the Fig. 4.9. The upper section of the vessel is filled with a charge of flaked seeds. A fresh solvent is sprayed over the seeds with the help of a distributor. The solvent percolates through the bed of solids and while doing so it extracts oil from the seeds and finally it drains in the lower section through a perforated plate (along with any water from seeds). The solution in the lower section is continuously boiled to produce vapours using steam heat (a steam coil is employed). The vapours are fed to a condenser and the condensed liquid is fed to a phase separator from which the water is removed from the bottom and the solvent is removed from the top and is fed back to the vessel. The concentrated solution of the oil is removed from the bottom.

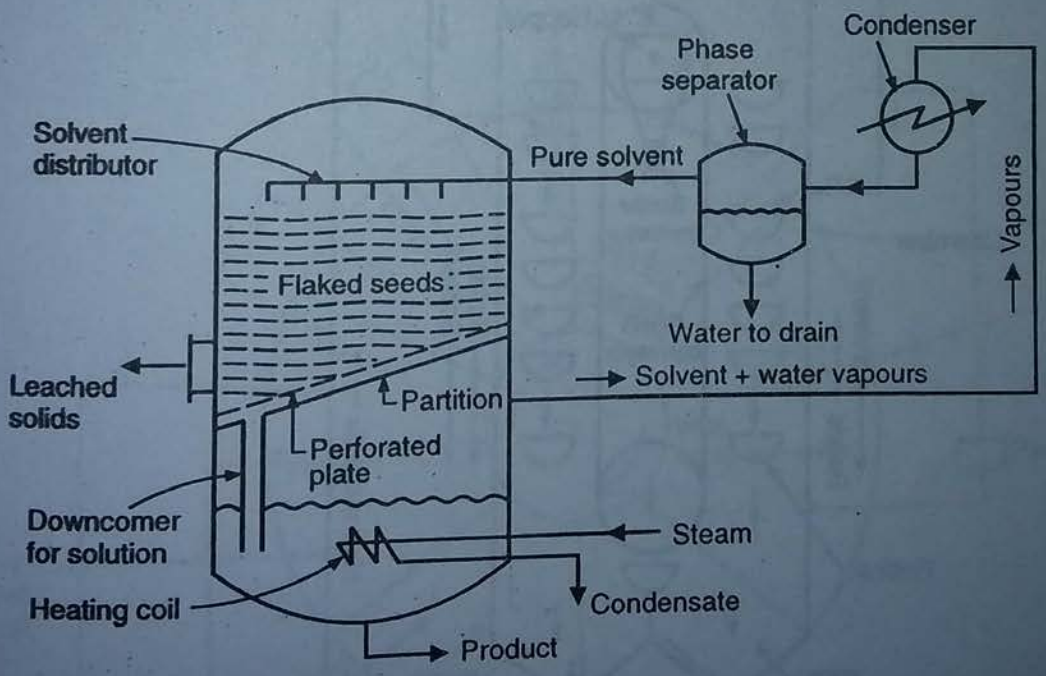

Fig. 4.9 : Batch plant for extraction of oil from seeds

Bollman Extractor/Basket Extractor :

It consists of a vapour-tight vertical chamber in which a series of perforated baskets are attached to a chain conveyor. The baskets are provided with perforations at the bottom.

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Solids are conveyed in the perforated basket down on the right and up on the left in the chamber as shown in the Fig. 4.10. The baskets are loaded with flaky solids (fresh meal) when each of the baskets comes at the top of the downward side (RHS). They are sprayed near the top of the chamber with the dilute solution of oil in solvent (half miscella) as they travel downward. As the baskets travel downward, the solids are leached with half miscella in parallel flow so that the solvent extracts more oil. The solution sprayed percolates through the solids from basket to basket and collects at the bottom of the chamber in the right hand sump as the final strong solution of the oil (full miscella) and is removed out of the chamber from the outlet provided at the bottom of the sump.

On the upward side of the extractor (left side of the chamber), the partially extracted solids (meal) travel upward and are leached further countercurrently with a spray of fresh solvent to obtain the dilute solution of the oil (half miscella) at the bottom of the chamber in the left hand sump. On this side of travel, the fresh solvent is sprayed on the meal near the top which then percolates through the meal from basket to basket. On this side, meal moves upward and solution flows downward and so the countercurrent flow is achieved. When the baskets are at top of their travel, they are inverted and fully exhausted solid (meal) is discharged into the hopper from which it is removed by screw conveyor. Usually conveyor speed is one revolution per hour and solvent to fresh meal (seeds) ratio is one.

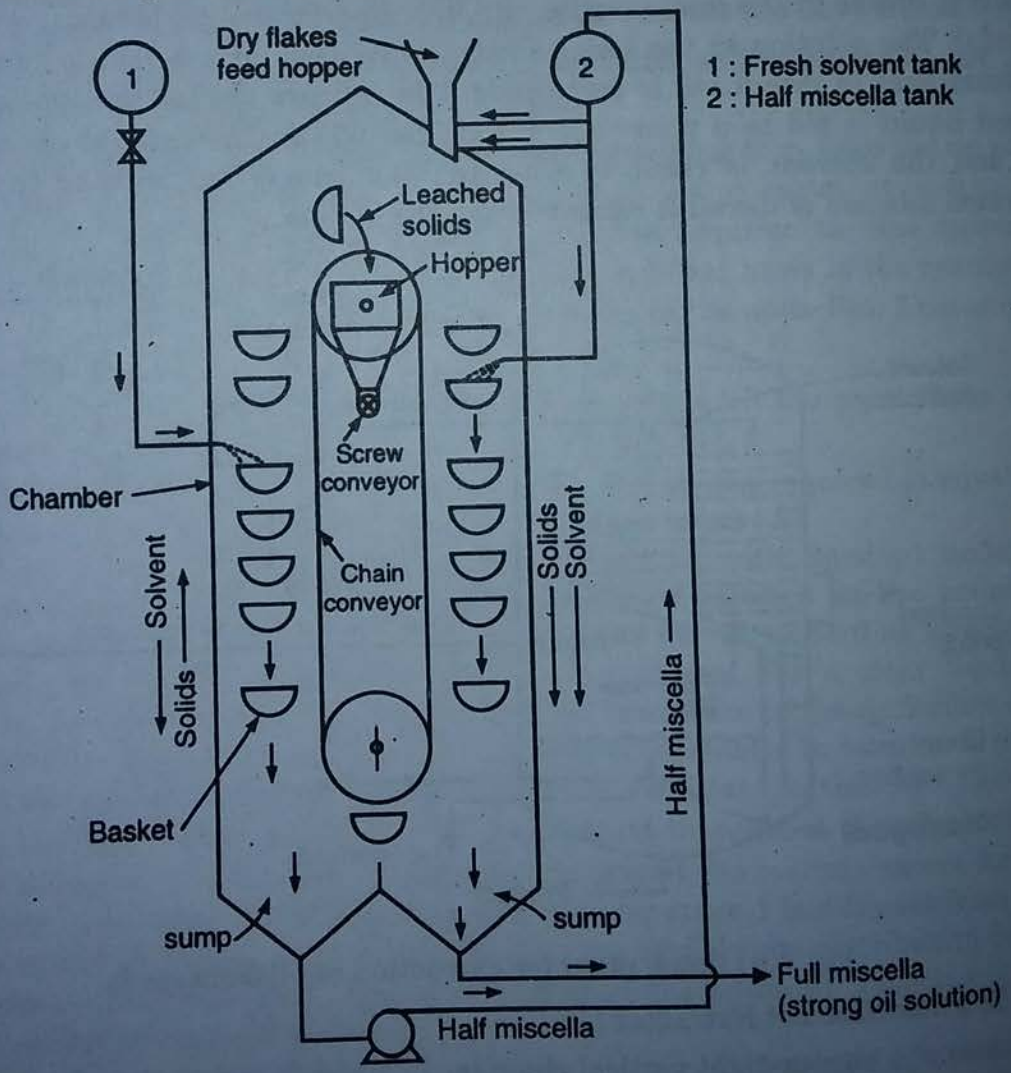

Fig. 4.10 : Bollman extractor/Basket extractor

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Rotocel Extractor :

It consists of a short, vertical cylindrical tank which is divided into 18 sector-shaped compartments. Each compartment is provided with a hinged, perforated bottom that supports the seeds and drains solution. This cylindrical tank slowly rotates above a stationary compartmented tank (acting as reservoir for solvent + oil) and both the tanks are enclosed in a vapour-tight housing in order to prevent escape (loss) of solvent vapours.

As the tank rotates, each compartment passes under a feeding hopper where it is filled with the prepared seeds and then under a series of sprays of solvent for leaching. Each successive spray of solvent is more dilute than the previous one and seeds are finally sprayed with pure solvent before they are discharged. The solvent from each spray percolates downward through seeds, extracts oil during percolation and then drained through the perforations into the appropriate compartment of the lower tank, from which it is continuously pumped to the next spray so that seeds and solvent flow countercurrently.

At the end of one rotation, the exhausted seeds from each compartment are dumped in one of the compartments of the lower tank from where they are continuously taken away. A full finished miscella is continuously removed from the second last compartment of lower tank at the feed end.

Miscella is not withdrawn from compartment-1 as it may contain suspended solids. These are filtered by passing through a bed of seeds in compartment-2, from where finished miscella is removed. The last compartment of the lower tank is of a large size to provide a sufficient time for the drainage of the solids before discharge.

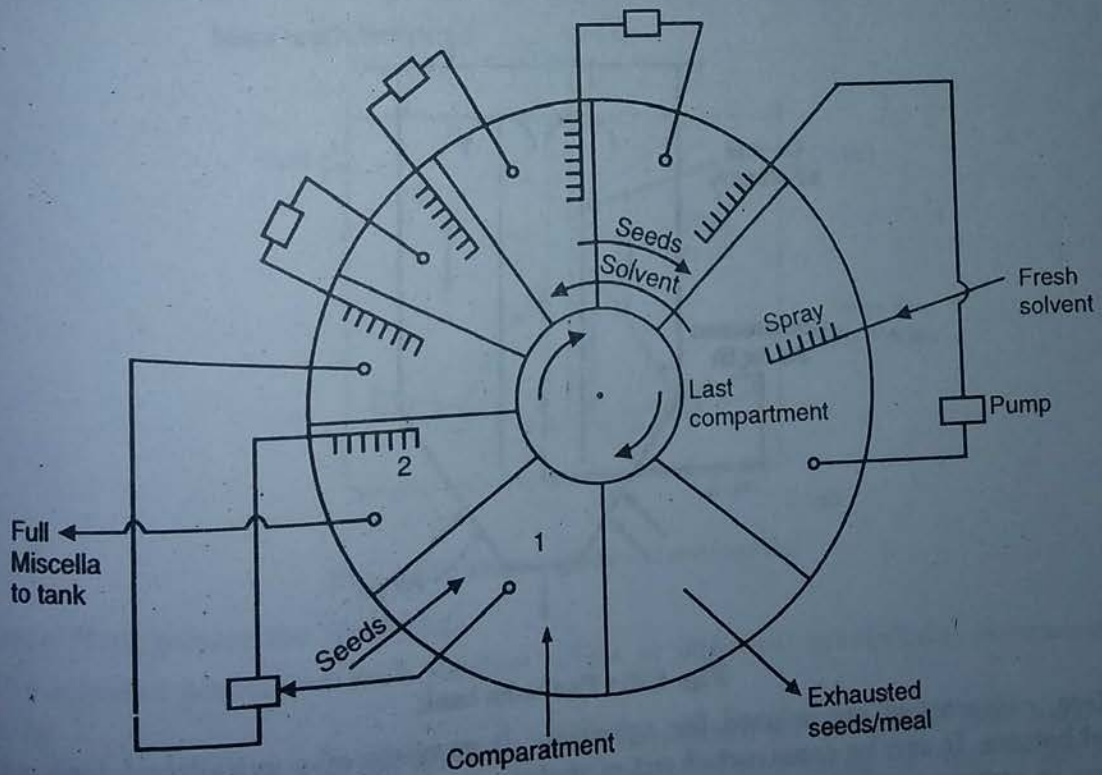

Fig. 4.11 : Operation of Rotocel extractor (Plan of lower compartmented tank)

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Leaching of Fine Solids :

Many solids must be ground to a fine size in order to make the soluble portions more accessible to a leaching solvent (e.g., extraction of gold from its ore using sodium cyanide solutions).

Leaching in such cases requires – (i) some sort of equipment in which finely divided solids can be kept in suspension in a liquid (wherein the solute is leached) and (ii) an arrangement for washing the solution off the (leached) solids.

The arrangements employed to wash chemical precipitates free of adhering solution (mother liquor) are quite similar to those used in the case of fine solids [e.g., washing of caustic soda solution from precipitated calcium carbonate following the reaction between slaked lime $Ca(OH)_2$ and soda ash (Na_2CO_3)].

Finely ground solids that can be kept in suspension in liquid solvents by agitation may be leached using mechanically agitated vessels (vertical cylindrical vessels incorporating an agitator).

Such units may be adopted for either a batch or a continuous leaching operation. In the case of continuous operations, the units must be provided with inlet and outlet connections at the opposite sides of the vessel.

Agitation can also be produced by making use of compressed air.

Pachuca Tank :

It is frequently used in the metallurgical industries. It is operated in a batchwise fashion and thus provides a single leaching stage (i.e., it is single stage in its action).

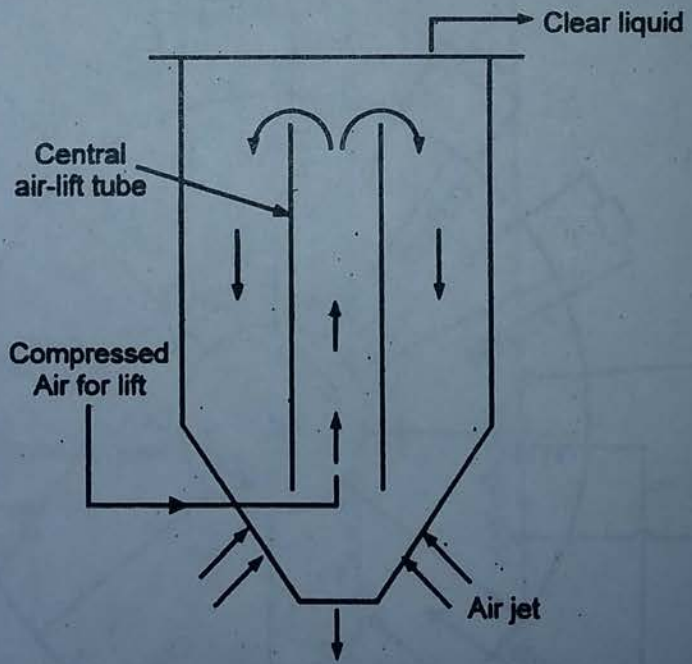

Fig. 4.12 : Pachuca tank

Here, compressed air is used for agitation. It consists of a cylindrical tank with a conical bottom. It can be constructed out of metal, wood or concrete. Lead lining may also be done in certain cases depending upon the nature of the leaching liquid. It is fitted with an arrangement for introducing compressed air as shown in Fig. 4.12. It is provided

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with a central tube that acts as an air lift, the bubbles of air rising through the central tube cause the upward flow of liquid and suspended solids in the tube that consequently result in continuous vertical circulation of the tank contents. Air jets are provided at the conical portion of the tank for dislodging any material that settles out. After leaching is complete, the agitation is stopped and the solids are allowed to settle. The clear supernatant liquid is decanted from the top by siphoning or by withdrawing through discharge connections located at an appropriate height in the side of the tank.

The Dorr Agitator :

It consists of a flat-bottomed tank provided with a central tube that acts as an air lift and also serves as a shaft and carries a set of arms both at the top and the bottom. The upper arms are in the form of launders. The launders are having a number of perforations for distribution of the suspension they receive. The lower arms [arms attached to the central tube (acting also as a shaft) at the bottom] are provided with scapper blades set at an angle so that they carry the settled solids toward the centre of the tank bottom from where they are lifted by the air lift through the central tube and discharged into the revolving launders. The launders then distribute the suspension evenly over the entire surface of the liquid in the tank. The lower arms are usually hinged and provided with arrangements for lifting them from the bottom of the tank whenever the power interruption is there, so that they may not be blocked in the settled solids that would otherwise prevent them from starting again.

For continuous operation, the Dorr agitator is provided with inlet and outlet connections at opposite sides of the tank.

The Dorr agitator is extensively used in both the metallurgical and the chemical industry for continuous leaching and washing of finely divided solids.

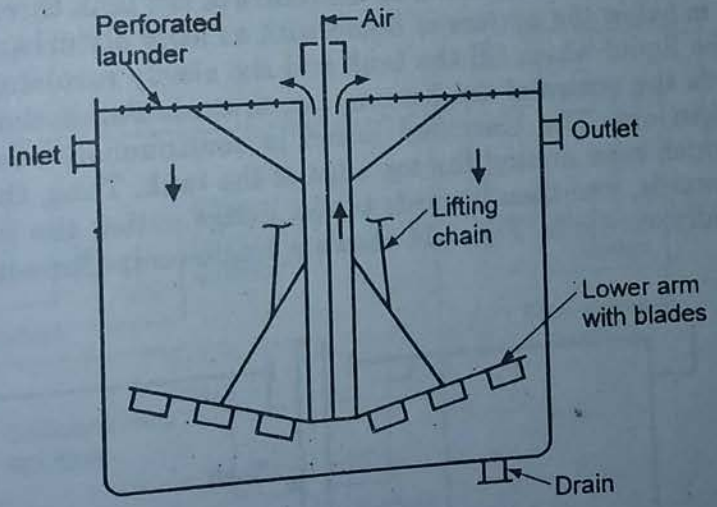

Fig. 4.13 : Dorr Agitator (schematic)

Washing (Thickeners for Washing) :

The solution associated with leached solids or chemical precipitates is removed by washing.

In case of leached solids, the solution must be washed off the solids. Here a solute is dissolved in a wash solvent in order to obtain a product containing as much of the solute material as possible. So, here washing involves a dissolving action.

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In case of chemical precipitates resulting from the chemical reactions, the precipitates must be washed free of the solution, i.e., solids in the form of precipitates suspended in the solution must be freed of the solution (precipitates are the desired products). Here, washing involves reducing the concentration of the solute in the film of solution adhering to the solid particles, and thus accomplishing a more or less complete separation of insoluble material from soluble material.

Thickeners are frequently used to wash leached solids (fine solids) and chemical precipitates free of adhering (associated) solution. In washing operation, thickeners are normally arranged to give continuous countercurrent flow of the solids and the wash solvent.

Thickener :

The thickener is the equipment used for carrying out sedimentation operation (settling operation) in a batch or a continuous fashion. It consists of a relatively shallow tank from the top of which the clear liquid is taken off and the thickened liquid is withdrawn from the bottom. Further, the thickener is frequently provided with a slow stirrer which helps in the consolidation of the sediment.

The Dorr thickener is simply a device for separating solids from liquids, but it is extremely useful in the washing of fine solids (e.g., leaching of gold from ores, causticizing of soda ash.)

Dorr Thickener :

It consists of a flat bottomed large diameter shallow tank. It is provided with slow moving radial rakes driven from a central shaft for removing the sludge. The thin slurry of liquid and suspended solids is fed at the centre of the tank through a feed well at a depth of 0.3 m to 1 m below the surface of liquid with as little disturbance as possible. The solids settle from the liquid which fill the tank and the slowly revolving rakes direct the settled sludge towards the centre of the bottom, i.e., towards the discharge cone from where the sludge is taken out. The clarrified liquid is continuously removed from an overflow (a launder) which runs around the top edge of the tank. Thus, the solids are continuously moving downwards, and then inwards to the sludge outlet; the liquid is moving upwards, and then rapidly outwards. Fig. 4.14 shows a single-compartment Dorr thickener.

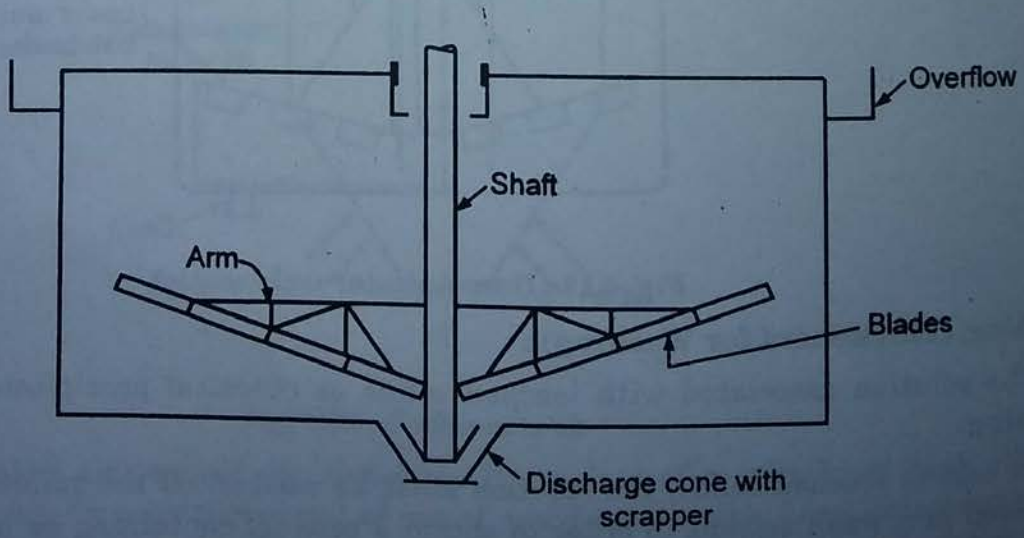

Fig. 4.14 : Dorr thickener

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Continuous Countercurrent Decantation :

The flow sheet of a simple countercurrent decantation system is shown in Fig. 4.15. This system comprises of one or more agitators arranged in series (for leaching of finely divided solids or carrying out some chemical reactions that produce precipitates, e.g., causticizing of soda ash) followed by a series of thickeners (for washing, i.e., for the removal of solution associated with leached solids or precipitates).

The solids to be leached (or the reactants in the case of a chemical reaction) together with solution from the second thickener are introduced into the leaching agitators arranged in series as shown in Fig. 4.15.

The mixture of solid and solution (in liquid form) from the last agitator is introduced into the first thickener through the feed well. The strong solution (i.e., the product of operation) is decanted from the solids by the first thickener (as an overflow from it). The agitators together with the first thickener thus form a single stage. The sludge (the underflow) from the first thickener then fed to a series of thickeners for washing it by the solvent in a countercurrent fashion and the washed solids are discharged as a sludge from the bottom of the last thickener.

The sludge from the first thickener is sent to a second thickener where it meets with the solution (the overflow) from the third thickener (acting as solvent). These two streams mix in the feed well of the second thickener. The solution decanted from the solids (the overflow) from the second thickener goes to the agitators and serves as solvent in them. The sludge (the underflow) from the second thickener goes to the third thickener where it meets the fresh wash and leaching solvent. The agitators may be replaced by a grinding mill.

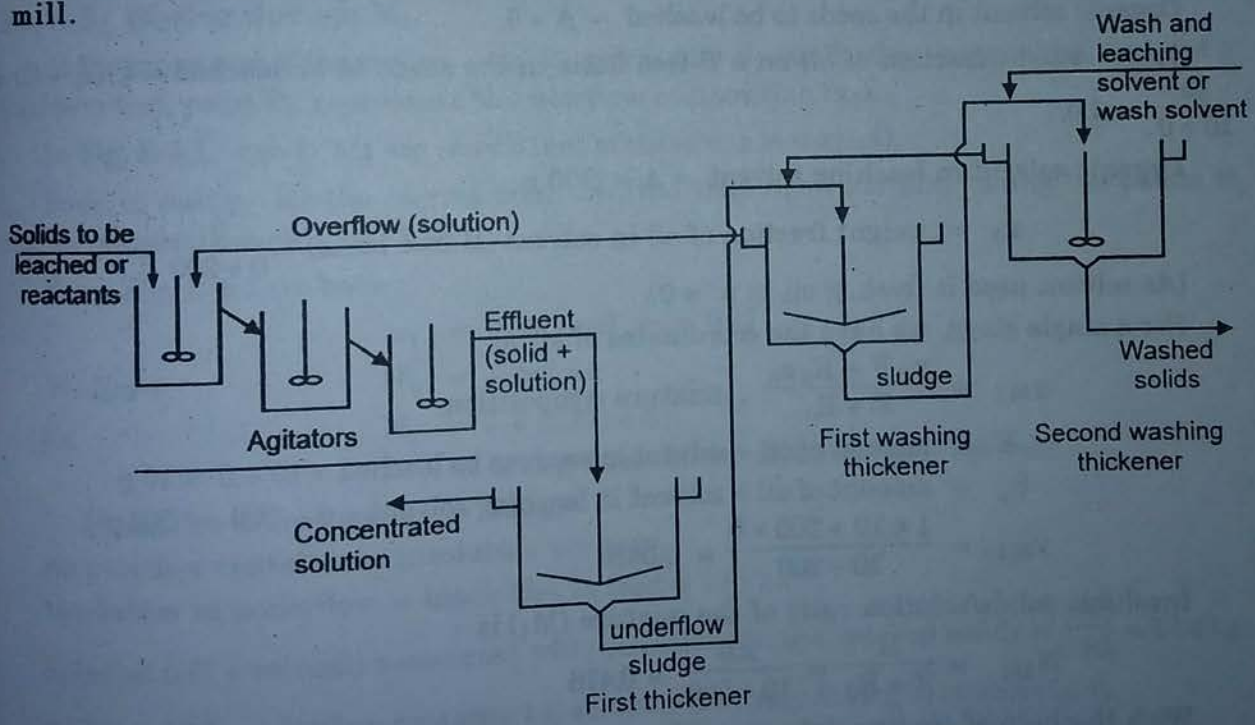

Fig. 4.15 : Continuous countercurrent decantation (continuous countercurrent leaching or reaction and washing using thickeners)

Wash and leaching solvent : For leaching and further washing of solution off the solids.

Washing solvent : For reaction-washing of precipitates to freed them of solution.

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SOLVED EXAMPLES

Ex. 4.1 : Vegetable oil seeds containing 100 g insoluble solids and 10 g oil are contacted with 200 g of organic solvent in a single stage leaching operation. The solvent used is fresh. Determine the amount of oil left in the oil seeds after leaching. The equilibrium data can be expressed as :

$$N = -4y + 8$$

where

$$N = \frac{\text{g insolubles}}{\text{g solvent + g oil}}$$

$$y = \frac{\text{g oil}}{\text{g solvent + g oil}}, \text{ in seed phase (underflow)}$$

$$x = \frac{\text{g oil}}{\text{g solvent + g oil}}, \text{ in solvent phase (overflow)}$$

The tie-line data are :

y	0.26	0.28	0.31	0.34
x	0.02	0.04	0.06	0.08

Solution : Basis : 100 g of insoluble solids and 10 g of oil in feed seeds.

B = insoluble solids, C = oil, A = organic solvent.

Oil in the seeds to be leached = C = 10 g

Organic solvent in the seeds to be leached = A = 0.

y_F = weight fraction of oil on a B-free basis in the seeds to be leached = C/(A + C) = $\frac{10}{10+0}$ = 1.0.

Organic solvent in leaching solvent = A = 200 g.

x_0 = weight fraction of oil in solvent (B-free basis) = $\frac{0}{0+200} = 0.$

(As solvent used is fresh, g oil in it = 0).

For a single stage, we have the coordinates of M_1 as :

$$y_{M1} = \frac{y_F F + R_0 x_0}{F + R_0}, \text{ mixture composition.}$$

F = amount of oil + solvent in seeds to be leached = 10 + 0 = 10 g

R_0 = amount of oil + solvent in leaching solvent = 0 + 200 = 200 g

$$y_{M1} = \frac{1 \times 10 + 200 \times 0}{10 + 200} = 0.0476$$

Insoluble solids/solution ratio of the mixture (M_1) is :

$$N_{M1} = \frac{B}{F + R_0} = \frac{100}{10 + 200} = 0.476$$

With the help of tie line data provided and equation relating N and y, find N values corresponding to x and y and tabulate.

e.g., For x = 0.02, y = 0.26

$$N = -4y + 8$$

$$= -4(0.26) + 8 = 6.96, \text{ do it for other data.}$$

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x	0.02	0.04	0.06	0.08
y	0.26	0.28	0.31	0.34
N	6.96	6.88	6.76	6.64

Plot N v/s x (overflow curve), N v/s y (underflow curve), i.e., N v/s x, y and also y v/s x.

For No insolubles in the overflow, plot of N v/s x is represented by x-axis.

Plot of N v/s x : overflow compositions.

Plot of N v/s y : underflow compositions.

Locate the point M_1 (0.0476, 0.476) on the N v/s x, y plot. We have to draw a tie line through M_1 cutting the overflow curve at R_1 and the overflow curve at E_1 (giving x_1 and y_1) as R_1 is in equilibrium with E_1. For this we have to adopt a trial and error procedure.

For the said procedure, we have to make use of y v/s x plot.

For any value of x represented by point R_1 on N v/s x, i.e., the x-axis, drop a vertical through the point R_1 on the y v/s x curve, cutting the curve at P. Through P, move horizontally to meet the diagonal (y = x) at point Q. Through the point Q, draw a vertical to cut N v/s y curve at E_1 (transferring y value on N v/s y). Points R_1, M_1 and E_1 should lie on the same line. Draw a line joining R_1 to E_1 and see whether M_1 lies on the same line or not. If not, then take another value of x (for point R_1) and repeat the procedure till we get a tie line R_1E_1 passing through M_1.

At the upper end of the tie line, point E_1 represents the underflow composition (y_1) and at the lower end, point R_1 represents the overflow composition (x_1).

In Fig. E 4.1, two trials are shown (out of these one is correct).

Read x_1 and y_1 for the correct trial, i.e., the trial in which line joining the points R_1 and E_1 passes through a point M_1.

From Fig. E 4.1 we have :

$$x_1 = 0.03 \quad \text{and} \quad y_1 = 0.27$$

We have : $N_1 = -4y_1 + 8$

So $N_1 = -4\,(0.27) + 8$

$= 6.92$ g insolubles/g solution (oil + solvent)

Insolubles in feed seeds = 100 g

As overflow contains no insolubles we have,

Insolubles in underflow = insolubles in feed = 100 g

Solution (oil + solvent) associated with insolubles in the leached solids = $\frac{100}{6.92} = 14.45$ g.

As $N_1 = 6.92$, underflow contains 1 g solution for every 6.92 g insolubles in it.

Solution associated with insolubles in the leached solids (underflow) = $\frac{1}{6.92} \times 100$

$E_1 = 14.45$ g

Amount of oil in leached solids = y_1E_1 = y_1 × amount of solution in leached solids.

$= 0.27 \times 14.45 = 3.9$ g

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Amount of oil left in the leached solids = 3.9 g ... Ans.

$$\% \text{ oil left in the solids (as unextracted)} = \frac{\text{oil left in leached solids}}{\text{oil in seeds}} = \frac{3.9}{10} \times 100 = 39$$

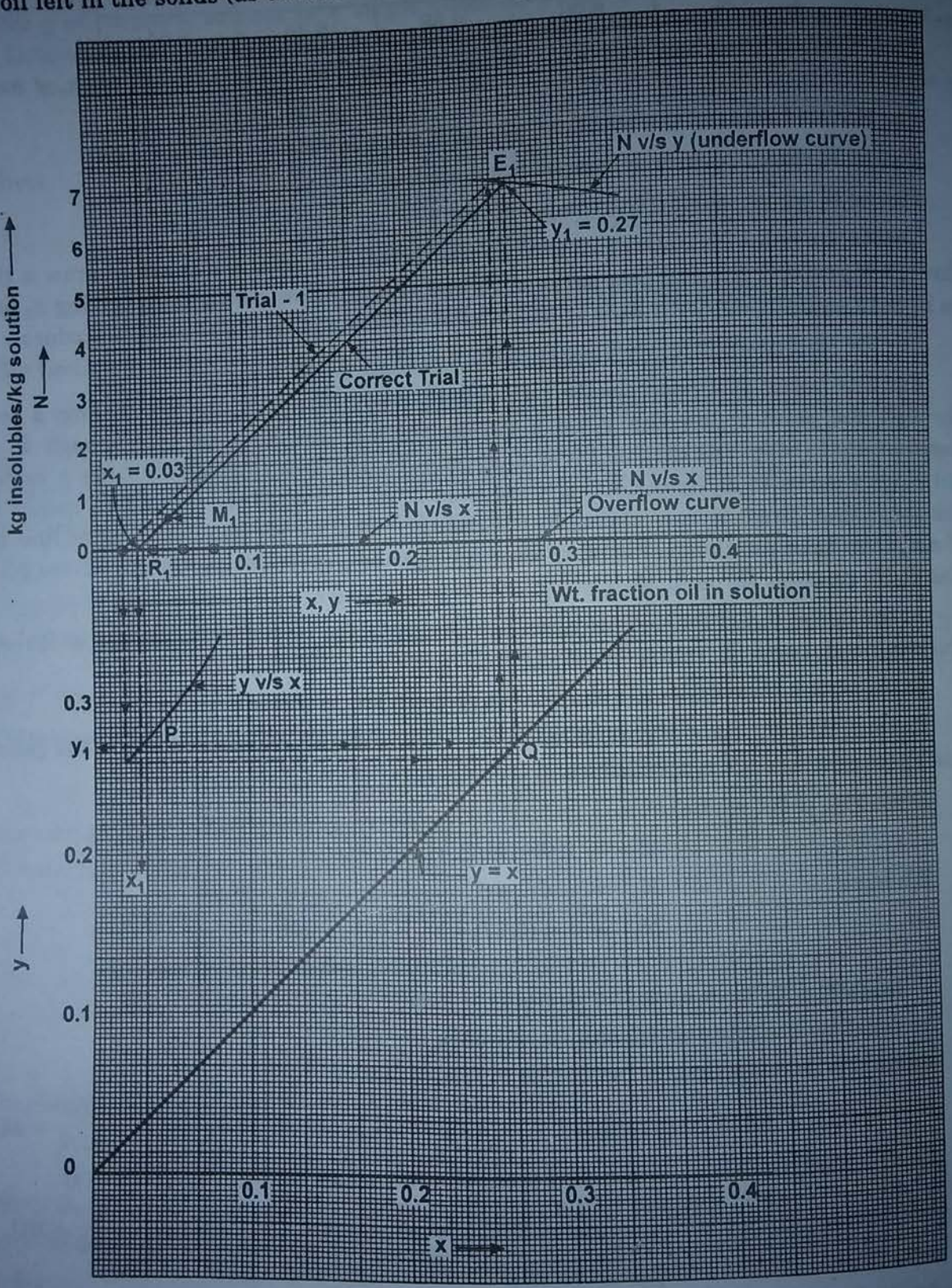

Fig. E 4.1 : Determination of underflow and overflow compositions

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Ex. 4.2 : Seeds containing 20% percent by weight oil are to be extracted in a counter-current plant and 90% of the oil is recovered in a solution containing 50% by weight oil. If the seeds are contacted with fresh solvent and 1 kg of solution is removed in the underflow in association with every 2 kg of insoluble matter, determine the theoretical stages required.

Solution : Basis : 100 kg seeds as a feed to the plant per unit time.

So per unit time :

Seeds to the plant = 100 kg

A - solvent (pure), B - insoluble matter, C - oil.

For determination of the number of stages required for achieving the desired degree of recovery, we have to find out the compositions of streams entering and leaving the plant by material balances.

Here, we will first make use of right-angled triangle for plotting (though not covered in theory).

So for the determination of the number of stages using triangular coordinates-one type-right-angled triangle we will express the concentrations of oil, solvent in terms of weight fraction based on the entire quantity of each stream, i.e., feed, underflow etc. (as required for such a diagram) and not on a B-free basis.

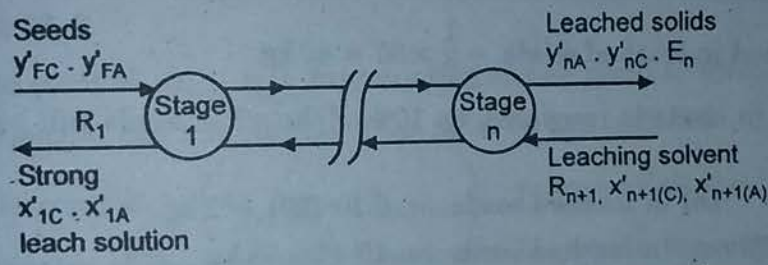

Fig. E 4.2 (a) : Countercurrent leaching operation

Feed : Seeds to be leached :

Seeds = 100 kg, containing no solvent.

Oil in seeds = C = $0.20 \times 100 = 20$ kg

Insoluble matter in seeds = B = $(1 - 0.2)\,100 = 80$ kg

Solvent in seed = A = 0

Weight fraction of oil in seeds to be leached = $y'_{FC} = 20/100 = 0.20$.

Weight fraction of solvent in seeds to be leached = $y'_{FA} = 0/100 = 0$.

Strong leach solution (overflow) :

The strong leach solution (overflow) leaving the unit contains 50% by weight oil and hence also 50% solvent by weight (as the overflow contains no insoluble matter).

Weight fraction of oil in strong leach solution = $x'_{1C} = 50/100 = \text{Wt. \% oil}/100 = 0.50$

Weight fraction of solvent in strong leach solution = $x'_{1A} = 50/100 = 0.50$.

Leaching solvent :

Oil present in leaching solvent = 0.

Solvent used is fresh and pure.

So, Weight fraction of solvent (A) in leaching solvent = $x'_{n+1\,(A)} = 1.0$.

Weight fraction of oil in leaching solvent = $x'_{n+1\,(C)} = 0$.

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Leached solids (underflow) :

It is given that 1 kg of solution (oil + solvent) is retained in association with every 2 kg of insoluble matter in the underflow from each stage.

As the quantity of solution removed in association with solids is constant, i.e., the ratio of insolubles to solution in the underflow is constant, it is a constant underflow process.

So the weight fraction of insoluble matter in the underflow from all stages is the same.

$$y'_{1B} = y'_{2B} = \ldots = y'_{nB}$$

Leached solids contain 1 kg of solution with every 2 kg of insolubles.

i.e., (2 + 1) 3 kg of leached solids contain 1 kg of solution and 2 kg of insoluble matter.

Weight fraction of insoluble matter in leached solids = $y'_{nB} = 2/(1+2) = 0.6667$.

As the overflow contains no insoluble matter :

Based on 100 kg seeds :

Insoluble matter in leached seeds = insoluble matter in feed seeds = 80 kg.

Every 2 kg of insoluble matter in underflow retain 1 kg of solution.

Solution retained in leached solids $= \frac{1}{2} \times 80 = 40$ kg.

90% of the oil in seeds is recovered, so 10% of the oil in seeds will be there in leached solids.

$\therefore$ Oil in leached seeds = 0.10 (20) = 2 kg

Solvent in leached seeds = 40 – 2 = 38 kg

Leached solids leaving the plant based on 100 kg seeds = (insolubles + solution) in it

= 80 + 40 = 120 kg

Weight fraction of oil in leached solids (underflow) = $y'_{nC} = 2/120 = 0.017$.

Weight fraction of solvent (A) in leached solids = $y'_{nA} = 38/120 = 0.3166$.

Use of right-angled triangle for determining the number of stages :

(1) **Overflow curve :** As there are no solids in the overflow, the overflow curve is represented by the hypotenuse of the triangle.

(2) **Underflow curve** (for constant underflow) : For constant underflow process, the locus of underflow compositions, i.e., the underflow curve is a straight line with an intercept on both the axes equal to $1 - y'_{nB}$.

Intercept on x and y-axes = $1 - y'_{nB} = 1 - 0.6667 = 0.3333$

OR

If p is the kg solution retained per kg of insoluble matter in the underflow then the intercept on the vertical axis (also on the x-axis) of the right-angled triangle is given by

Intercept = $p/(p+1)$

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In our case, 1 kg of solution is retained with every 2 kg of insoluble matter.

i.e., 0.5 kg solution is retained with every 1 kg insoluble matter. So $p = 0.5$.

$$\text{Intercept} = 0.5/(1 + 0.5) = 0.3333 = y'_A.$$

So the underflow curve is a straight line parallel to the hypotenuse having an intercept of (y'_A) 0.333 on the vertical axis.

Refer Fig. E 4.2 (b) for graphical construction details.

(i) Draw a right-angled triangle. Concentration of C (oil) in terms of weight fraction (x'_C, y'_C) in the overflow and underflow is plotted along x-axis. Concentration of A (solvent) in underflow and overflow (x'_A, y'_A) is plotted along y-axis.

(ii) Draw the underflow curve which is a straight line parallel to the hypotenuse with an intercept of 0.3333 on y-axis.

(iii) Locate point F (y'_{FC}, y'_{FA}) representing feed seeds on x-axis. Coordinates of F are : 0.20, 0.

(iv) Locate point R_1 (x'_{1C}, x'_{1A}) representing strong leach solution on the hypotenuse of the triangle. The coordinates of the point R_1 are : 0.5, 0.5.

(v) Locate point $R_{n+1}\left(x'_{n+1\,(C)}, x'_{n+1\,(A)}\right)$ representing the leaching solvent. The coordinates of R_{n+1} are : 0, 1.0, so locate R_{n+1} at the top vertex of the triangle.

(vi) Locate $E_n\left(y'_{nC}, y'_{nA}\right)$ representing the leached solids on the underflow curve. The coordinates of E_n are : 0.017, 0.3166.

(vii) Join R_1 to F and extend. Join R_{n+1} to E_n and extend to meet the extended line through R_1 F to get a difference point Δ_R. The point Δ_R is the point of intersection of the lines R_{n+1} E_n and R_1 F.

(viii) Draw a line BR_1, i.e., draw a line joining the origin of triangle with R_1. Line BR_1 represents constant C/A ratio.
The streams R_1 and E_1 are in equilibrium, E_1 should lie on the underflow curve.
The point of intersection of BR_1 with the underflow curve is E_1, so mark E_1.

(ix) Now points Δ_R, E_1 and R_2 should lie on the same line. R_2 to be on the hypotenuse.
So join Δ_R E_1 and extend to cut the hypotenuse at R_2.

(x) Join B to R_2. Mark E_2 at the point of intersection of line B R_2 with the underflow curve.

(xi) Now Δ_R, E_2 and R_3 should lie on the same line. So join Δ_R E_2 and extend to cut the hypotenuse at R_3 and so on.

(xii) Repeat the above mentioned procedure for further construction of stages till point E_n on the underflow is reached exactly or covered.
Each tie line represents a single stage. Count the stages.
Number of theoretical stages required for achieving required degree of separation = 5 ... Ans.

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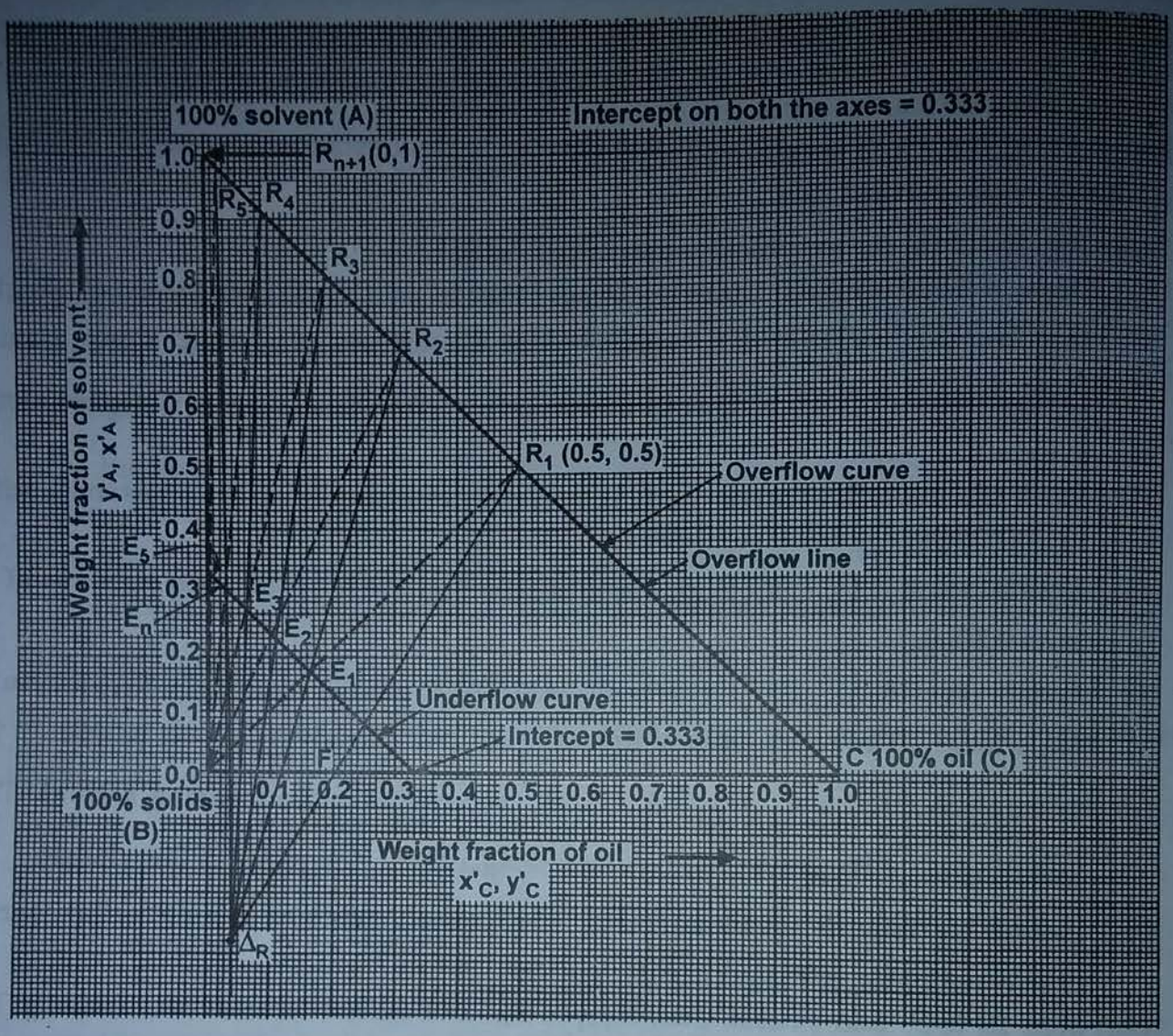

Fig. E 4.2 (b) : Determination of the number of stages
(countercurrent leaching-constant underflow)

OR

As per theory given (rectangular coordinates)

Solution with normal notation and meaning :

(This seems to be very easy approach.)

i.e., Making use of N v/s x and N v/s y plots for determining the number of theoretical stages.

Basis : 100 kg seeds as feed to the plant per unit time.

So per unit time :

Seeds to the plant = 100 kg

A – solvent (pure), B - insoluble matter, C – oil.

For determination of number of stages required for desired recovery of oil, we have to find out the compositions of the entering and leaving streams by performing simple material balances.

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Feed : Seeds to be leached.

oil in seeds = C = 0.20 (100) = 20 kg

insolubles in seeds = B = (1 – 0.8) × 100 = 80 kg

solvent in seeds = A = 0

oil + solvent in seeds = (A + C) = F = 20 + 0 = 20 kg

Weight fraction of oil in seeds on B-free basis $= y_F = \frac{\text{oil}}{\text{oil + solvent}} = \frac{20}{20} = 1.0$

kg insoluble matter/kg of (oil + solvent) in seeds to be leached $N_F = \frac{\text{insolubles}}{(\text{oil + solvent})} = \frac{80}{20} = 4$

Leaching solvent :

Fresh solvent is used

∴ oil in leaching solvent = C = 0

Pure solvent (A) in leaching solvent = p kg (say)

oil + A in leaching solvent = 0 + p = p kg

Weight fraction of oil in leaching solvent on B-free basis = $x_{n+1} = 0/p = 0$

kg insolubles/kg leaching solvent = N = 0.

Strong leach solution (overflow) :

The strong leach solution leaving the unit contains 50% by weight oil and as the overflow contains no insoluble matter, this solution contains (100 – 50) = 50% by weight solvent (A).

i.e., in 100 kg strong leach solution (oil + solvent), there are 50 kg of oil and 50 kg of pure solvent (A).

Weight fraction of oil in strong leach solution on B-free basis = $x_1 = 50/100 = 0.50$

kg insolubles/kg solution in strong leach solution = $N_1 = 0$.

Leached solids (underflow) :

It is given that 1 kg of solution (oil + solvent) is retained in the underflow in association with every 2 kg of insoluble matter from each stage.

As the quantity of solution removed in association with the solids is constant (from all stages), it is a Constant Underflow process and for the constant underflow conditions, the ratio of insolubles to solution in the underflow is constant.

Solids are settled or drained to the same extent at all solute concentrations, so the underflow curve, i.e., N v/s y is a horizontal line (parallel to x-axis) at $N = N_n$.

kg insolubles/kg solution in leached solids = $N_n = 2/1 = 2$.

Based on 100 kg seeds :

Insoluble matter in leached seeds = insoluble matter in feed seeds

= 80 kg

(assuming no insolubles are lost in overflow)

1 kg solution is retained with every 2 kg insoluble matter.

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Solution retained in leached solids $= \frac{1}{2} \times 80 = 40$ kg.

Oil recovered = 90% oil in seeds

Oil in leached solids = 10% oil in seeds

Oil in leached seeds = $0.1\,(20) = 2$ kg

Solvent in leached seeds = $40 - 2 = 38$ kg

Weight fraction of oil in leached seeds on B-free basis = $y_n = 2/(2 + 38) = 0.05$.

For no adsorption of solute, composition of the withdrawn solution and solution associated with the solid is the same ($y^* = x$) from each stage and so the tie lines are vertical on N v/s x, y plots.

Determination of the number of stages required.

Refer Fig. E 4.2 (c) for graphical construction details.

(i) Draw N v/s x (overflow curve) and N v/s y (underflow curve). As there are no insolubles in the overflow, N v/s x plot is represented by x-axis.

For constant underflow, underflow curve is a horizontal straight line through N_n on y-axis. So draw a horizontal line (parallel to x-axis) through $N_n = 2.0$ on y-axis.

(ii) Locate the point F (y_F, N_F) representing feed seeds. Coordinates of F are : 1, 4.

(iii) Locate the point R_{n+1} (x_{n+1}, 0) representing leaching solvent on N v/s x. The coordinates of R_{n+1} are : 0, 0, pure solvent.

(iv) Locate the point R_1 (x_1, 0) representing strong leach solution on N v/s x (overflow curve). The coordinates of R_1 are : 0.5, 0.

(v) Locate the point E_n (y_n, N_n) representing leached solids on N v/s y, i.e., underflow curve. The coordinates of E_n are : 0.05, 2.

(vi) Join F – R_1 and extend. Join $E_n - R_{n+1}$ and extend to meet the extended line F R_1 in point Δ_R (difference point).

(vii) Through R_1, draw a vertical tie line to cut N v/s y, the underflow curve, at E_1.

(viii) The points Δ_R, E_1 and R_2 lie on the same line. Join $E_1\,\Delta_R$ which cuts N v/s x plot, i.e., x-axis at R_2. Mark R_2 at the intersection of the line $E_1\Delta_R$ with x-axis.

(ix) Through R_2, draw a vertical tie line to cut the N v/s y, i.e., underflow curve, at E_2.

(x) Repeat the above mentioned procedure for further construction of stages till point E_n on the underflow curve (N v/s y) is reached exactly or covered. Every tie line represents a stage. Count the number of stages.

Number of stages required for achieving the desired recovery of oil = 5 ... Ans.

The method of construction using right angled triangle is given here only for knowing it if asked.

Answer by both the methods is the same but we have to adopt a method for which theory is given.

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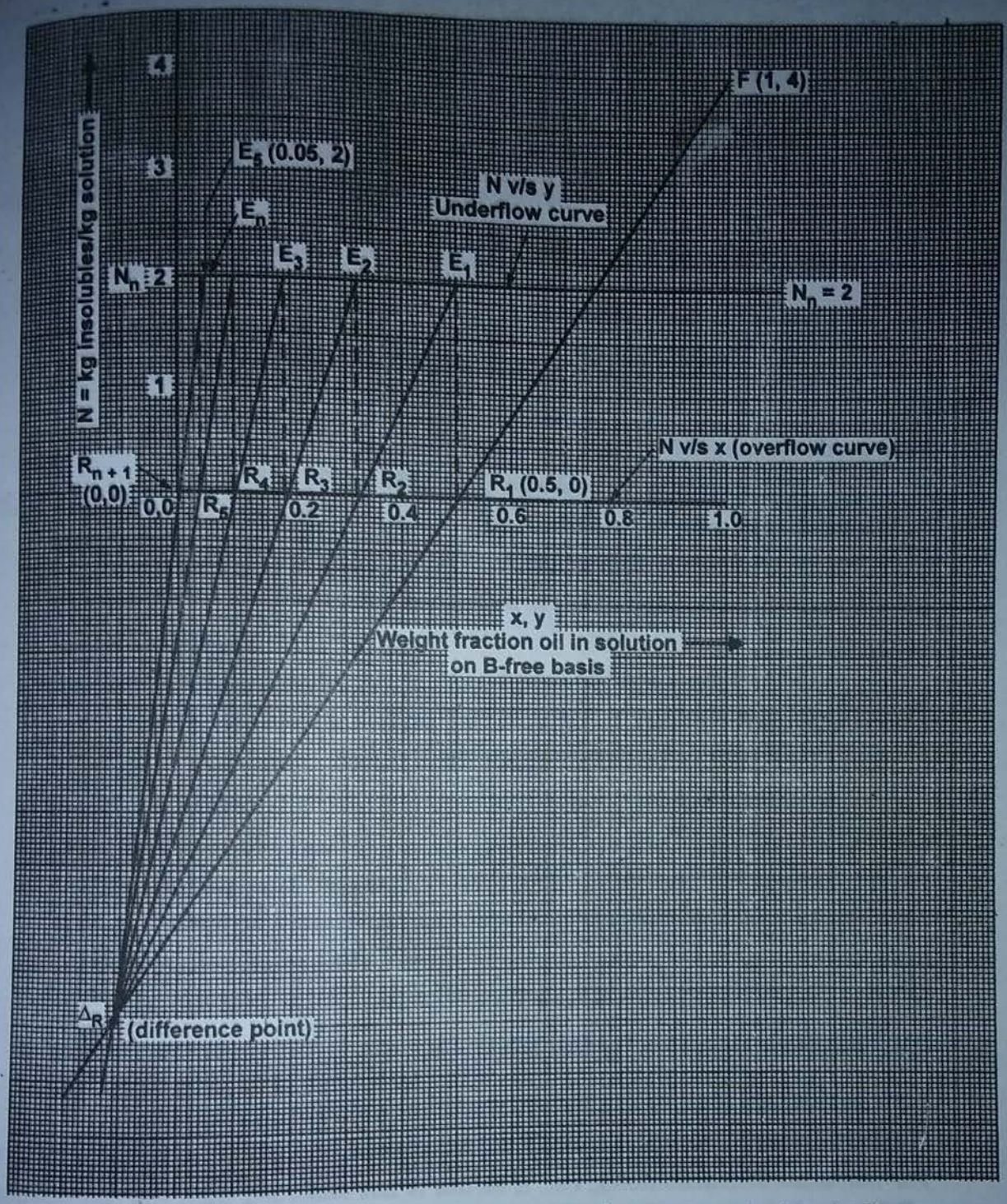

Fig. E 4.2 (c) : Determination of the number of stages-constant underflow (countercurrent leaching)

Ex. 4.3 : Crushed oil seeds containing 55% oil by weight are to be extracted at the rate of 4000 kg/h using 6000 kg/h of hexane containing 5% oil by weight as the solvent. A countercurrent two stage extraction system is used. The oil seeds retain 1 kg of solution per kg of oil-free cake. Calculate the percent recovery of oil (based on the original feed) obtained under the above conditions.

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Solution : Basis : 4000 kg/h of crushed oil seeds.

A – hexane, B – insolubles, C – oil

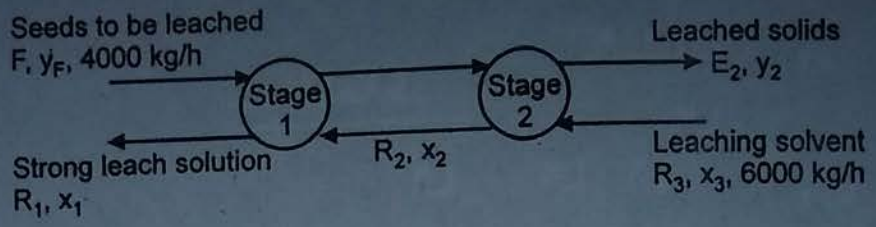

Fig. E 4.3 : Two-stage countercurrent operation

Feed :

Feed seeds contain 55% oil and 45% insolubles.

$\therefore$ Oil in seeds to be leached = C = 0.55×4000 = 2200 kg/h

Insolubles in seeds to be leached = B = 0.45×4000 = 1800 kg/h

Hexane in seeds to be leached = A = 0

(oil + hexane) i.e. solution in seeds to be leached = F = 2200 + 0 = 2200 kg/h.

Leaching solvent :

Leaching solvent to the unit = $R_{n+1} = R_3$ = 6000 kg/h

Leaching solvent contains 5% oil and 95% hexane (by difference)

Oil in leaching solvent = 0.5×6000 = 300 kg/h

Hexane in leaching solvent = 0.95×6000 = 5700 kg/h

Leached solids :

As the overflow contains no insolubles, we have :

Insolubles in leached solids (underflow) = insolubles in feed seeds

= 1800 kg/h

Given : Leached seeds retain 1 kg of solution per 1 kg of insolubles.

$\therefore$ Solution retained in leached solids = $E_n = E_2 = \frac{1}{1} \times 1800$ = 1800 kg/h.

Let z be the kg/h of oil in the leached seeds i.e. underflow from stage-2.

Weight fraction of oil in the underflow from stage-2 on B-free basis = $y_2 = z/E_2 = z/1800$

For non-adsorbent insolubles, i.e., if solute is not adsorbed then the composition of oil on a B-free basis in the overflow will be the same as that of the underflow, so that $y_1^* = x$. For equilibrium stage-2, R_2 is in equilibrium with E_2.

$\therefore$ $x_2 = y_2$, $y_2 = z/1800$

So, $x_2 = z/1800$

As the ratio of solution to insolubles in the underflow is constant, the quantity of solution removed in association with solid is constant, it is a constant underflow process. The mass of insolubles in the underflow is same from all stages (as solids are not lost in

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overflow) and as ratio of solution to insolubles is constant for all stages, the solution in the underflow for all stages will be the same, so

$$R_2 = R_3 = \ldots R_n$$

∴ We have, $R_2 = R_3 = 6000$ kg/h

$$R_2 = 6000 \text{ kg/h}$$

Oil in overflow/strong leach solution from stage-2 $= x_2 R_2$

$$= \frac{z}{1800} \times 6000$$

$$= 3.33\, z \text{ kg/h}$$

Hexane in the overflow from stage-2 $= R_2 - 3.33\, z$ kg/h (by difference)

$$= 6000 - 3.33\, z \text{ kg/h}$$

Total oil entering stage-1 = oil in seed + oil in overflow-2

$$= 2200 + 3.33\, z \text{ kg/h}$$

Total solution (oil + hexane) entering stage-1 $= F + R_2 = 2200 + 6000 = 8200$ kg/h

Weight fraction of oil in underflow or overflow leaving the stage-1 on B-free basis $= \dfrac{2200 + 3.33\, z}{8200}$

Therefore, $x_1 = y_1 = \dfrac{2200 + 3.33\, z}{8200}$

Overall material balance of solution (oil + solvent) over the entire unit :

$$R_3 + F = R_1 + E_2$$

$$R_1 = R_3 + F - E_2$$

$$= 6000 + 2200 - 1800 = 6400 \text{ kg/h}$$

R_1 is the strong leach solution leaving the unit i.e. overflow from stage-1.

Oil in strong leach solution (from stage-1) $= R_1 x_1$

$$= 6400 \left[\frac{2200 + 3.33\, z}{8200}\right]$$

$$= 0.78\,(2200 + 3.33\, z) \text{ kg/h}$$

Overall material balance of oil :

oil in seeds + oil in leaching solvent = oil in leached seeds + oil in strong leach solution

$$2200 + 300 = z + 0.78\,(2200 + 3.33\, z)$$

$$z = 217.93 \simeq 218 \text{ kg/h}$$

∴ Oil in leached solids (remained unextracted) = 218 kg/h

Oil recovered/extracted $= 2200 - 218 = 1982$ kg/h

% Recovery of oil (based on the oil in the original seeds)

$$= \frac{\text{oil recovered}}{\text{oil in seeds}} \times 100$$

$$= \frac{1982}{2200} \times 100 = 90.09$$

... Ans.

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Extra :

Seeds entering = 4000 kg/h

Leaching solvent entering = 6000 kg/h

Leached seeds leaving = 1800 kg/h

Solution with leached seeds leaving = 1800 kg/h

Strong leach solution leaving (underflow) = 6400 kg/h

Leached seeds + solution associated leaving i.e. underflow leaving = 1800 + 1800

= 3600 kg/h

Overall material balance over entire unit :

feed + solvent = overflow + underflow

4000 + 6000 = 6400 + 3600

10000 = 10000

Overall material balance over stage-1 :

Σ Input streams = Σ Output streams

4000 + 6000 (R_2) = [1800 + 1800] (underflow) + 6400 (R_1)

10000 = 10000

As RHS = LHS, the calculations done/performed are correct.

Ex. 4.4 : Oil is to be extracted from meal by means of benzene using continuous countercurrent extraction unit. The unit is expected to treat 1000 kg of meal (based on completely exhausted solid) per hour. The untreated meal contains 365 kg of oil and 30 kg of benzene. The solvent used contains 14 kg of oil and 590 kg of benzene. The exhausted solids are to contain 55 kg of unextracted oil. Experimental data on the extraction of oil from meal by means of benzene at a given operating temperature are as follows :

Solution composition, kg oil/kg solution	Solution retained (in underflow) kg solution/kg solids
0	0.500
0.10	0.505
0.20	0.515
0.30	0.530
0.4	0.550
0.5	0.571
0.6	0.595
0.7	0.620

Find the number of ideal stages required.

Solution : Basis : 1000 kg/h of meal (based on completely exhausted solids)

Meal based on completely exhausted solid = insolubles in meal.

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Untreated meal contains insolubles, oil and benzene.

Untreated meal contains 1000 kg of insolubles, 365 kg of oil and 30 kg of benzene per hour.

A : benzene, B : insolubles, C : oil

Feed : Oil in meal to be leached = C = 365 kg/h

Benzene in meal to be leached = A = 30 kg/h.

Insolubles in meal to be leached = B = 1000 kg/h.

Weight fraction of oil in meal to be leached on B-free basis $= y_F = 365/(365 + 30)$

$$y_F = 0.924$$

kg insolubles/kg solution in meal to be leached $= N_F = \frac{B}{A + C} = \frac{1000}{365 + 30} = 2.53.$

Oil + benzene in meal to be leached = A + C = F = 365 + 30 = 395 kg/h

Leaching solvent :

Leaching solvent contains 14 kg of oil and 590 kg of benzene.

Weight fraction of oil in leaching solvent on B-free basis $= x_{n+1} = \frac{14}{14 + 590} = 0.023$

Oil + benzene in leaching solvent $= R_{n+1} = 590 + 14 = 604$ kg.

Leaching solvent does not contain insolubles

$\therefore$ kg insolubles/kg solution in leaching solvent $= N_{n+1} = 0.$

We have provided with solute (oil) concentration of the solution. Assuming that solids (insolubles in meal) does not contain any adsorbed or undissolved solute, the concentration of the solute on B-free basis in the underflow and the overflow will be same i.e. $y^* = x$.

Convert the data in the form suitable for plotting :

Solution composition, kg oil/kg solution, y^*	Solution retained, kg solution/kg solids (1/N)	N $\frac{\text{kg solid}}{\text{kg solution}}$	$\frac{\text{kg oil}}{\text{kg insoluble solids}}$ y^*/N
0.0	0.500	2.0	0
0.1	0.505	1.98	0.051
0.2	0.515	1.942	0.103
0.3	0.530	1.887	0.159
0.4	0.550	1.820	0.220
0.5	0.571	1.751	0.286
0.6	0.595	1.681	0.367
0.7	0.620	1.613	0.434

Leached solids (underflow) :

As the overflow contains no insolubles, we have,

Insolubles in leached solids = insolubles in meal = 1000 kg/h

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Oil in leached solids = 55 kg/h

kg oil/kg insolubles in leached solids = $y^*/N = \frac{55}{1000} = 0.055$

Find out N_n for $y^*/N = 0.055$.

We have from the data :

y^*/N	0.051	0.055	0.103
N	1.98	p (say)	1.942

$$\frac{1.98 - p}{0.055 - 0.051} = \frac{p - 1.942}{0.103 - 0.055}$$

$\therefore$ p = 1.977 by interpolation.

So for $y^*/N = 0.055$, $N_n = 1.977$.

kg insolubles/kg solution in leached solids = N_n = 1.977

Insolubles in leached solids = 1000 kg/h

(oil + benzene) solution retained in leached solids = $E_n = B/N_n$

E_n = 1000/1.977

= 505.8 ≈ 506 kg/h

N_n = 1.977

i.e., 1.977 kg insolubles retains with it 1 kg solution in the underflow.

$\therefore$ Solution retained in association with insolubles in leached solids $= \frac{1}{1.977} \times 1000$

E_n = 506 kg/h

Oil in leached solids = 55 kg/h

Benzene in leached solids = (solution – oil) in leached solids

= 506 – 55 = 451 kg/h

Weight fraction of oil in leached solids on B-free basis = y_n = 55/506 = 0.1087

Strong leach solution :

Overall material balance of oil :

oil in meal + oil in leaching solvent = oil in leached solids + oil in strong leach solution

Oil in strong leach solution leaving the unit = 365 + 14 – 55 = 324 kg/h

Overall material balance of benzene (A) :

Benzene in meal + Benzene in leaching solvent = Benzene in leached solids + Benzene in strong leach solution.

Benzene (A) in strong leach solution leaving the unit = 590 + 30 – 451 = 169 kg/h

$\therefore$ (oil + benzene) in strong leach solution = R_1 = 324 + 169 = 493 kg/h

Weight fraction of oil in strong leach solution leaving the unit on a B-free basis = $x_1 = \frac{324}{493} = 0.657$.

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Determination of the number of stages : The graphical solution for the number of stages is shown in Fig. E 4.4.

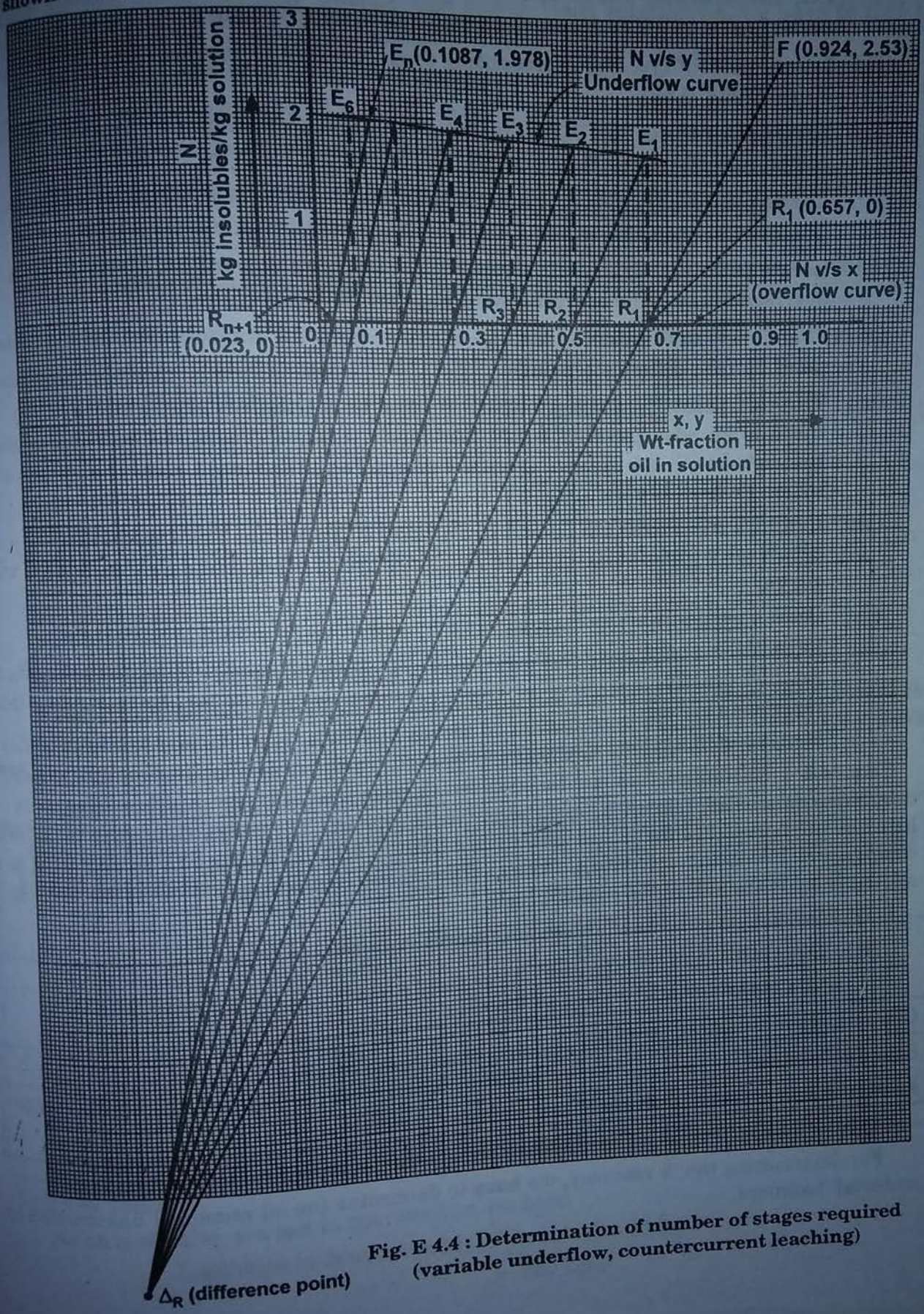

Fig. E 4.4 : Determination of number of stages required (variable underflow, countercurrent leaching)

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(i) From the data given, plot : N v/s x, y. With no solids in overflow, N v/s x i.e., the overflow curve is a straight line represented by x-axis. N v/s y is a underflow curve (the locus of all underflow compositions).

(ii) Locate point F (y_F, N_F) representing feed - meal to be leached. The coordinates of F are (0.924, 2.53).

(iii) Locate point R_1 (x_1, 0) representing strong leach solution, i.e., overflow on N v/s x. The coordinates of R_1 are (0.657, 0).

(iv) Locate point E_n (y_n, N_n) representing leached solids on N v/s y, underflow curve. The coordinates of E_n are (0.1087, 1.978).

(v) Locate point R_{n+1} (x_{n+1}, 0) representing leaching solvent on N v/s x curve i.e. overflow curve. The coordinates of R_{n+1} are (0.023, 0).

(vi) Join $R_1 - F$ and extend. Join $E_n - R_{n+1}$ and extend to meet extended R_1 F line at point Δ_R (the difference point).

(vii) Through point R_1, draw a vertical tie line to cut the underflow curve at E_1. As there is no adsorption of solute, the leach solution (overflow) and the solution associated with solids have the same composition from each stage, i.e., $y^* = x$ and so the tie lines are vertical.

(viii) Join $E_1 - \Delta_R$. Points E_1, Δ_R and R_2 lie on the same line. The line joining $E_1 - \Delta_R$ cuts N v/s x curve at R_2. So mark R_2 at the cutting point of line $E_1 - \Delta_R$ with N v/s x curve.

(ix) Draw a vertical tie line through R_2 to cut the N v/s y, underflow curve at E_2.

(x) Repeat the above mentioned procedure for further construction till point E_n is reached exactly or covered and count the number of stages. Each tie line thus represents a stage.

Number of theoretical stages required = 6 ... Ans.

Ex. 4.5 : Halibut oil is to be extracted from granulated halibut livers in a countercurrent extraction unit using ether as a solvent. The mass flow rate of halibut livers to the unit is 350 kg/h. The fresh livers contain 20% oil and are to be extracted to a composition 1% oil (on solvent-free basis). The ether containing 2% oil to be used as a solvent is fed to the unit at the rate 250 kg/h.

(i) Determine number of theoretical stages required.

(ii) Determine % recovery of the oil.

The entrainment data are as follows :

Concentration, kg oil/kg solution	0	0.1	0.2	0.3	0.4	0.5	0.6
Solution retained, kg solution/kg exhausted livers	0.28	0.34	0.40	0.47	0.55	0.66	0.80

Solution : Basis : 350 kg/h of halibut livers to the countercurrent extraction unit.

For determining the % recovery, we have to determine the oil remained unextracted by material balances.

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For determining the number of stages graphically, we have to find out the compositions of feed, leached solids, leaching solvent, and strong leach solution i.e. we have to find the terminal compositions by material balances.

A : Ether, B : Exhausted livers (insolubles), C : Oil.

Feed : Oil in the feed livers (C) = 0.20 (350) = 70 kg/h

Exhausted livers in the feed livers (B) = 0.80 (350) = 280 kg/h

Ether/solvent in the feed livers (A) = 0.

(Oil + ether) i.e. solution in feed livers = 70 + 0 = 70 kg/h

Weight fraction of oil in feed livers on B-free basis $= y_F = \frac{\text{oil}}{\text{oil + solvent}} = \frac{70}{70+0} = 1.0.$

kg insolubles/kg solution in feed livers = N_F = Insolubles/solution = 280/70 = 4.

Leaching solvent :

Leaching solvent used = R_{n+1} = 250 kg/h

Solvent contains 2% oil by weight.

Oil in leaching solvent = 0.02×250 = 5 kg/h

Ether in leaching solvent = $(1 - 0.02) \times 250$ = 245 kg/h

Insolubles in leaching solvent = 0

(oil + ether) in leaching solvent = 5 + 245 = 250 kg/h.

Weight fraction of oil in leaching solvent on B-free basis = x_{n+1}.

x_{n+1} = 5/250 = 0.02

kg insolubles/kg solution in leach solvent = N_{n+1} = 0/250 = 0.

We are provided with the solute concentration of the solution. If we assume that the solids does not contain any adsorbed solute then the concentration of solute on a B-free basis in the underflow and the overflow will be same, i.e., $y^* = x$.

Convert the data in the required form :

Concentration, kg oil/kg solution (y^*)	Solution retained, $\frac{\text{kg solution}}{\text{kg exhausted livers}}$ (1/N)	$\frac{\text{kg exhausted livers}}{\text{kg solution}}$ (N)	$\frac{\text{kg oil}}{\text{kg exhausted livers}}$ (y^*/N)
0	0.28	3.57	0
0.1	0.34	2.94	0.034
0.2	0.40	2.50	0.080
0.3	0.47	2.13	0.141
0.4	0.55	1.82	0.22
0.5	0.66	1.51	0.33
0.6	0.80	1.25	0.48

Leached solids :

As no insolubles are lost in the overflow, we have

Insolubles in leached solids = insolubles in feed livers

= 280 kg/h

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Given : Leached solids (underflow) contain 1% oil by weight on a A-free basis.

Therefore, on the A-free basis :

oil in leached solids = 1%

insolubles in leached solids = 100 – 1 = 99%

i.e., 1 kg of oil is associated with 99 kg of insolubles in every 100 kg leached solids.

Oil associated with insolubles in leached solids $= \frac{1}{99} \times 280 = 2.83$ kg/h

oil in feed livers = 70 kg/h

oil recovered = oil in feed livers – oil in leached solids

= 70 – 2.83 = 67.17 kg/h

% recovery of oil $= \frac{67.17}{70} \times 100$

= 95.96 ≈ 96 **... Ans. (i)**

kg oil/kg exhaust livers in leached solids = 2.83/280 = 0.0101.

i.e., y^*/N in leached solids = 0.0101

For y^*/N = 0.0101, from the table prepared by interpolation,

$N = N_n = 3.383$

y^*/N	0	0.0101	0.034
N	3.57	p	2.94

$$\frac{3.57 - p}{0.0101 - 0} = \frac{p - 2.94}{0.034 - 0.0101}, \therefore p = 3.383$$

kg exhausted livers (insolubles)/kg solution in leached solids = N_n = 3.383.

i.e., 3.383 kg insolubles retain 1 kg of solution (in underflow)

Insolubles in leached solids = 280 kg/h

Solution retained in leached solids $= \frac{1}{3.383} \times 280 = 82.77$ kg/h

Ether in solution retained in leached solids = 82.77 – 2.83 = 79.94 kg/h

Weight fraction of oil in leached solids on B-free basis $= y_N = \frac{\text{oil}}{\text{solution}} = \frac{2.83}{82.77} = 0.0342$

(oil + solvent) i.e. solution in leached solids = E_n = 82.77 kg/h.

Strong leach solution :

Overall material balance of oil :

oil in feed livers + oil in leaching solvent = oil in leached solids + oil in strong leach solution.

Oil in strong leach solution (overflow) = 70 + 5 – 2.83 = 72.17 kg/h.

Overall material balance of ether :

Ether in leaching solvent + ether in feed livers = ether in strong leach solution + ether in leached solids.

Ether in strong leach solution = 245 + 0 – 79.94 = 165.06 kg/h

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Oil + ether in strong leach solution = R_1 = 237.23 kg/h

Weight fraction of oil in strong leach solution on B-free basis

$= x_1 = 72.17/(72.17 + 165.06)$

$= 0.304$

Determination of number of stages :

Refer Fig. E 4.5 (b) for construction details.

(i) Plot N v/s x, y (overflow, underflow compositions).

N v/s x – overflow curve and N v/s y - underflow curve.

N v/s x plot is represented by x-axis as the overflow contains no insolubles.

(ii) Locate point F (y_F, N_F) representing feed livers. The coordinates of F are (1, 4).

(iii) Locate point R_1 (x_1, 0) representing strong leach solution on N v/s x. The coordinates of R_1 are (0.304, 0). Actually R_1 (x_1, N_1) with $N_1 = 0$.

(iv) Locate point E_n (y_n, N_n) representing final leached solids on N v/s y. The coordinates of E_n are (0.0342, 3.383).

(v) Locate point R_{n+1} (x_{n+1}, 0) representing leaching solvent on N v/s x. The coordinates of R_{n+1} are (0.02, 0).

(vi) Join point $R_1 - F$ and extend. Join $E_n - R_{n+1}$ and extend to meet extended R_1 line at point Δ_R (the difference point).

(vii) Through point R_1, draw a vertical tie line to cut the underflow curve at E_1.

The tie lines are vertical as $y^* = x$.

(viii) Join $E_1 - \Delta_R$. Points E_1, Δ_R and R_2 lie on the same line. The line joining $E_1 - \Delta_R$ cuts N v/s x (overflow) curve at R_2. Mark R_2 at the cutting point of line $E_1 - \Delta_R$ with N v/s x curve, i.e., x-axis.

(ix) Draw a vertical tie line through R_2 to cut the underflow curve at E_2.

(x) Repeat the above mentioned procedure for further construction till we reach exactly or cover point E_n and count the number of stages. Each tie line represents a stage.

Number of theoretical stages required = 4 **... Ans.**

Extra :

Total net flow at any point = Material flow to RHS – Material flow to LHS

Total net flow at the entrance of first stage = Flow of A + B + C to RHS – Flow of A + B + C to LHS

$= 350 - 237.33 = 112.77$ kg/h.

$\therefore$ Total net flow is towards RHS of the extraction unit.

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Material flow to RHS at the entrance of first stage :

Flow of A : 0 kg/h ... ether

Flow of B : 280 kg/h ... exhausted livers

Flow of C : 70 kg/h ... oil

350 kg/h

Material flow to LHS at the entrance of first stage.

Flow of A : 165.06 kg/h

Flow of B : 0 kg/h

Flow of C : 72.17 kg/h

237.23 kg/h

Net flow of components :

Net flow of B = B to RHS – B to LHS

= 280 – 0 = 280 kg/h

Net flow of A = 0 – 165.06 = – 165.06 kg/h

Net flow of C = 70 – 72.17 = – 2.17 kg/h

So net flow of B is to the left whereas the net flows of A and that of C are to the right.

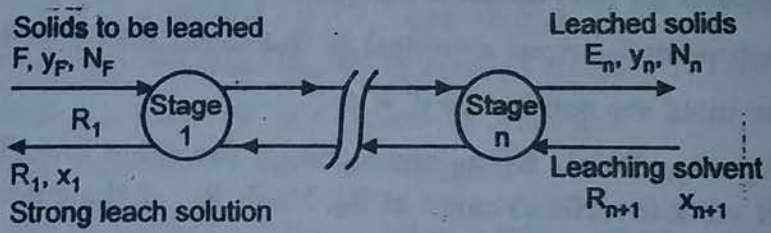

Fig. E 4.5 (a) : Countercurrent extraction unit

Overall material balance :

Σ Input streams = Σ Output streams

kg/h feed livers + kg/h leaching solvent = kg/h leached solids + kg/h strong leach solution.

350 + 250 = kg/h insolubles in leached solids + E_n + R_1

600 = 280 + 82.77 + 237.23

600 = 600

As RHS = LHS, therefore the calculations performed are correct.

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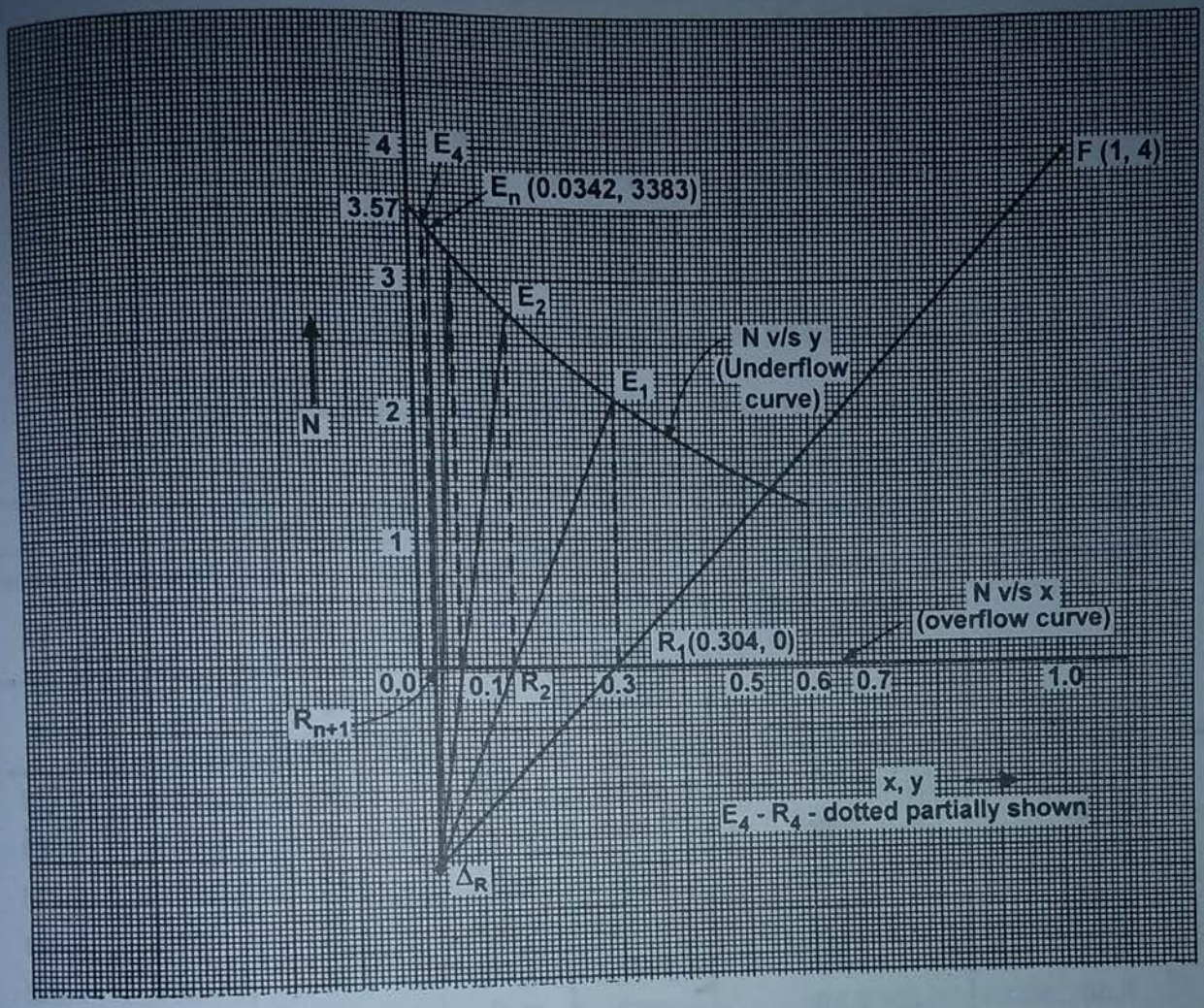

Fig. E 4.5 (b) : Determination of the number of stages-variable underflow (countercurrent leaching)

Extra : Given if asked to determine the number of stages using a right-angled triangle (a triangular diagram).

Use of right-angled triangular diagram for determining the number of stages :

Data :

Concentration, $y^* = x$ kg oil/kg solution	0	0.10	0.2	0.30	0.4	0.5	0.60
Solution retained, kg solution/kg exhausted livers	0.28	0.34	0.40	0.47	0.55	0.66	0.80

Convert the data into weight fractions of C and A in underflow.

We have : kg oil/kg solution = 0.10

kg solution/kg livers = 0.34

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So per 1 kg of livers, we have :

amount of solution = 0.34 kg

amount of oil = $0.1 \times 0.34 = 0.034$ kg

amount of solution = kg oil + kg ether

amount of ether = $0.34 - 0.034 = 0.306$ kg

amount of underflow = kg livers + kg solution

$= 1 + 0.34 = 1.34$ kg

$\therefore$ Weight fraction of oil in the underflow = $y'_C = 0.034/1.34 = 0.0254$.

Weight fraction of ether in the underflow = $y'_A = 0.306/1.34 = 0.228$.

In this way, convert the data and tabulate.

Concentration, kg oil/kg solution, y^*	Solution retained, $\frac{\text{kg solution}}{\text{kg livers}}$	Underflow compositions				
		Per 1 kg of livers			Weight fractions	
		kg oil	kg ether	kg underflow	y'_C	y'_A
0	0.28	0	0.28	1.28	0	0.219
0.10	0.34	0.034	0.306	1.34	0.0254	0.228
0.20	0.40	0.08	0.32	1.40	0.0571	0.228
0.30	0.47	0.141	0.329	1.47	0.096	0.224
0.40	0.55	0.22	0.33	1.55	0.142	0.213
0.50	0.66	0.33	0.33	1.66	0.199	0.199
0.60	0.80	0.48	0.32	1.80	0.267	0.177

Find the terminal compositions required for plotting.

Feed :

Weight fraction of oil in feed livers = $y'_{FC} = 20/100 = 0.20$.

Weight fraction of livers (insolubles) in feed livers = $y'_{FB} = 80/100 = 0.80$.

Leaching solvent :

Weight fraction of oil in leaching solvent = $x'_{n+1, C} = 2/100 = 0.02$.

Weight fraction of ether in leaching solvent = $x'_{n+1, A} = 98/100 = 0.98$.

Leached solids :

As no insolubles are lost in overflow,

insolubles in underflow/(leached solids) = insolubles in feed livers

$= 0.8 \times 350 = 280$ kg/h

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Leached solids contain 1% oil by weight on a A-free basis.

On the A-free basis : oil → 1%

insolubles → 99%

i.e., 1 kg of oil is associated with 99 kg of insolubles in every 100 kg of leached solids.

$\therefore$ Oil in leached solids $= \frac{1}{99} \times 280 = 2.83$ kg/h

kg oil/kg livers in leached solids $= 2.83/280 = 0.0101$

Determination of the kg solution in leached solids by interpolation :

From the data provided :

kg solution/kg livers	kg oil/kg livers
0.28	0
0.34	0.034
p	x = 0.0101

$$\frac{0.28 - p}{0.0101 - 0} = \frac{p - 0.34}{0.034 - 0.0101}$$

$\therefore$ $p = 0.2978$

$\therefore$ kg solution/kg livers in leached solids $= 0.2978$

Solution retained in leached solids $= 0.2978 \times 280 = 83.38$ kg/h

Ether in leached solids $= 83.38 - 2.83 = 80.55$ kg/h

Underflow/Leached solids leaving the unit $= 280 + 83.38 = 363.38$ kg/h.

Weight fraction of oil in leached solids $= y'_{nC} = 2.83/363.38 = 0.00778 \approx 0.008$.

Weight fraction of ether in leached solids $= y'_{nA} = 80.55/363.38 = 0.222$.

Strong leach solution/overflow :

Material balance of oil over the entire unit :

Oil in strong leach solution = oil in livers + oil in leaching solvent – oil in leached solids.

$= 70 + 5 - 2.83 = 72.17$ kg/h

By material balance of ether over entire unit :

Ether in strong leach solution = ether in feed livers + ether in leaching solvent – ether in leached solids.

$= 0 + 245 - 80.55 = 164.45$ kg/h

Strong leach solution $= 72.17 + 164.45 = 236.62$ kg/h

Weight fraction of oil in strong leach solution $= x'_{1C} = 72.17/236.62 = 0.305$.

Weight fraction of ether in strong leach solution $= x'_{1A} = 1 - 0.305 = 0.695$.

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Determination of the number of stages :

Refer Fig. E 4.5 (c) for graphical construction.

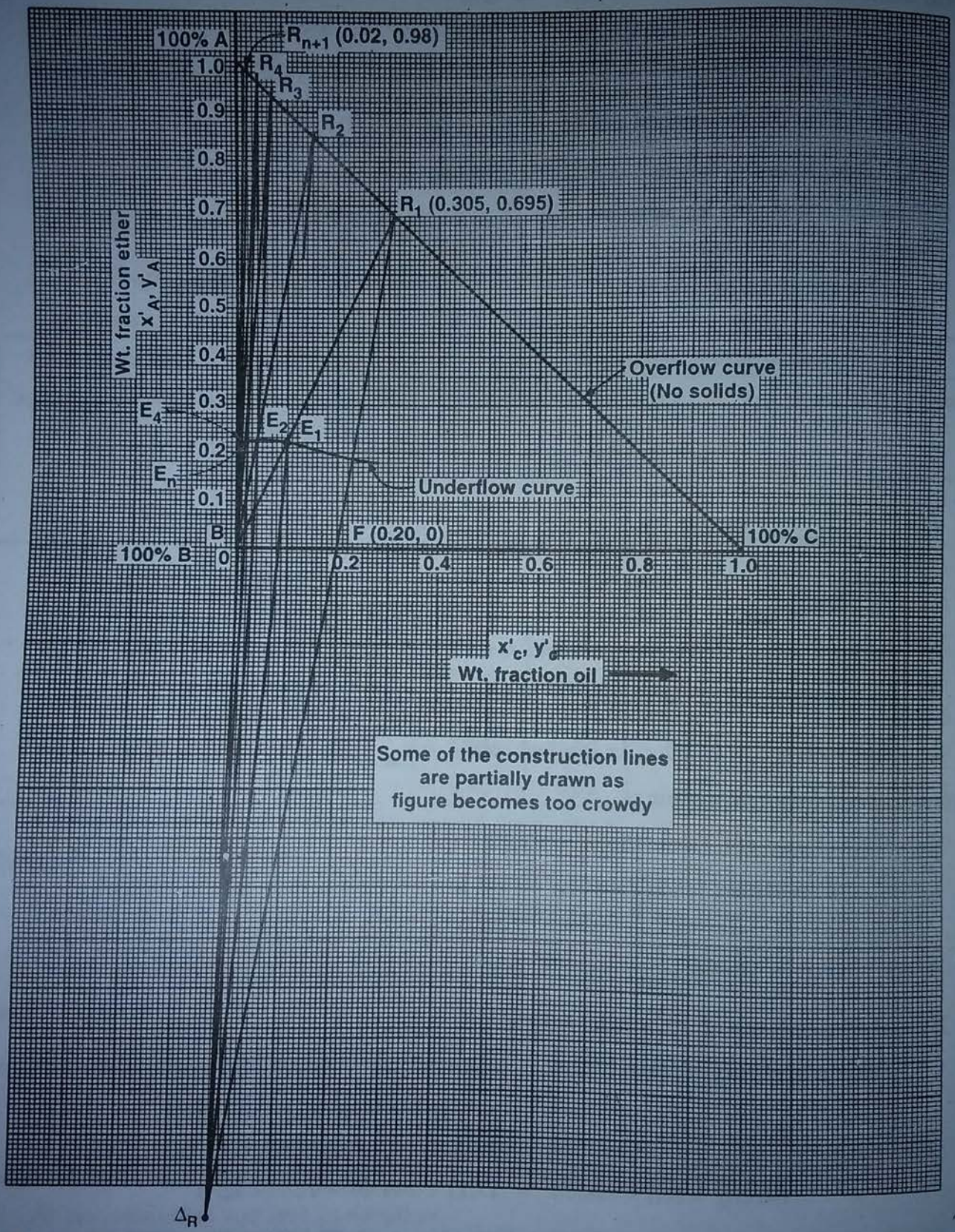

Fig. E 4.5 (c) : Determination of the number of stages using right-angled triangle (triangular diagram) – variable underflow (countercurrent leaching)

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(i) Draw a right-angled triangle.

(ii) The hypotenuse of the triangle represents the overflow curve.

(iii) With the help of the data provided y'_C, y'_A draw the underflow curve.

(iv) Locate point F (y'_{FC}, 0) representing feed on x-axis. Coordinates of F are : 0.20, 0.

(v) Locate point R_1 on the hypotenuse/overflow curve.

The coordinates of R_1 are : x'_{1C}, x'_{1A} : 0.305, 0.695.

(vi) Locate point R_{n+1} ($x'_{n+1, C}$, $x'_{n+1, A}$) on the hypotenuse.

The coordinates of point R_{n+1} are : 0.02, 0.98.

(vii) Locate point E_n (y'_{nC}, y'_{nA}) on the underflow curve.

The coordinates of point E_n are : 0.008, 0.222.

(viii) Join $R_1 - F$ and extend. Join $R_{n+1} - E_n$ and extend to meet the extended R_1 F line at point Δ_R (the difference point).

(ix) Join $B - R_1$. Mark E_1 at which a line B R_1 cuts the underflow curve.

(x) Points Δ_R, E_1 and R_2 lie on the same line. So join $\Delta_R - E_1$ and extend to cut the overflow curve at R_2. Join $B - R_2$ and mark E_2 at which line B R_2 cuts the underflow curve.

(xi) Repeat the above mentioned procedure for further construction work till we reach exactly or cover point E_n on the underflow curve. $R_1 - E_1$, $R_2 - E_2$, etc. are tie lines. Each tie line represents a stage. Count the stages.

Number of stages required = 4 ... **Ans.**

So answer by both the methods is same.

Ex. 4.6 : Halibut oil is to be extracted from granulated halibut livers using ether as a solvent in countercurrent batch extraction unit. The solid charge contains 0.35 kg oil per kg exhausted livers and it is desired to have 90% oil recovery. For 50 kg of pure ether used per 100 kg untreated solids, determine the number of theoretical stages required.

The entrainment data are as follows :

Concentration, kg oil/kg solution	0	0.1	0.2	0.3	0.4	0.5	0.6	0.67
Entrainment, kg solution/kg livers	0.28	0.34	0.40	0.47	0.55	0.66	0.80	0.96

Solution : Basis : 100 kg of untreated solids.

It contains 0.35 kg oil per 1 kg exhausted livers i.e. in 1.35 kg untreated solids there are 0.35 kg of oil and 1 kg of exhausted livers.

A - ether, B - livers (insolubles), C - oil.

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Feed :

$$\text{Livers in feed i.e. insolubles in feed} = (B) = \frac{1}{1.35} \times 100$$

$$= 74.07 \approx 74.1 \text{ kg}$$

$$\text{oil in feed} = C = 100 - 74.1 = 25.9 \text{ kg}$$

$$\text{oil + ether in feed} = A + C = 25.9 + 0 = 25.9 \text{ kg}$$

$$\text{Weight fraction of oil in feed on B-free basis} = y_F = \frac{25.9}{25.9} = 1.0$$

$$\text{kg insolubles/kg (oil + ether) in feed} = N_F = 74.1/25.9 = 2.86.$$

Leaching solvent :

$$\text{oil in leaching solvent} = 0$$

$$\text{ether in leaching solvent} = 50 \text{ kg}$$

$$\therefore \quad \text{leaching solvent used} = R_{n+1} = 0 + 50 = 50 \text{ kg}$$

$$\text{Weight fraction of oil on B-free basis in leaching solvent} = x_n = 0/50 = 0.$$

$$\text{insolubles in leaching solvent} = 0$$

$$\text{kg insolubles/kg oil + ether in leaching solvent} = N = \frac{0}{50} = 0.$$

If we assume that the solids does not contain any adsorbed or undissolved solute then the solute composition on B-free basis of the underflow and the overflow will be the same. So $y^* = x$.

Convert the data in required form and tabulate.

Concentration, kg oil/kg solution (y^*)	Entrainment, $\frac{\text{kg solution}}{\text{kg exhausted livers}}$ (1/N)	$\frac{\text{kg livers}}{\text{kg solution}}$ (N)	$\frac{\text{kg oil}}{\text{kg livers}}$ (y^*/N)
0	0.28	3.57	0
0.1	0.34	2.94	0.034
0.2	0.40	2.50	0.080
0.3	0.47	2.13	0.141
0.4	0.55	1.82	0.22
0.5	0.66	1.51	0.33
0.6	0.80	1.25	0.48
0.67	0.96	1.04	0.64

Leached solids :

As no insolubles are lost in overflow, we have,

$$\text{Insolubles in leached solids/underflow} = \text{insolubles in feed}$$

$$= 74.1 \text{ kg}$$

Given :

$$\text{oil recovery} = 90\%$$

$$\text{oil recovered in solvent} = 0.9 \text{ (oil in untreated solids)}$$

$$= 0.9 \times 25.9 = 23.31 \text{ kg}$$

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oil remained in leached solids = 25.9 – 23.31

= 2.59 kg

kg oil/kg insolubles in leached solids = $y^*/N = \frac{2.59}{74.1} = 0.03495 \approx 0.035$.

For $y^*/N = 0.035$ by interpolation :

$N = N_n = 2.93$

From the tabulated data :

y^*/N	N
0.034	2.94
0.035	p
0.080	2.50

$$\frac{2.94 - p}{0.035 - 0.034} = \frac{p - 2.50}{0.08 - 0.035}$$

$p = 2.93$

kg exhausted livers/kg solution in leached solids = $N_n = 2.93$.

i.e., 2.93 kg exhausted livers (insolubles) retain 1 kg solution.

Insolubles in leached solids = 74.1 kg

∴ Solution retained (entrainment) in leached solids = $\frac{1}{2.93} \times 74.1 = 25.29$ kg

oil in leached solids = 2.59 kg

ether in leached solids = 25.29 – 2.59 = 22.7 kg

Weight fraction of oil in leached solids on B-free basis $= y_n$ = oil/solution

= 2.59/25.29 = 0.102

(oil + ether) i.e. solution in leached solids = E_n = 25.29 kg

Strong leach solution :

Overall material balance of oil :

oil in untreated solids = oil in leached solids + oil in strong leach solution

+ oil in leaching solvent

Oil in strong leach solution = 25.9 + 0 – 2.59 = 23.31 kg

Overall material balance of ether :

ether in leaching solvent + ether in feed livers = ether in leached solids + ether in strong leach solution

ether in strong leach solution = 50 + 0 – 22.7 = 27.3 kg

Oil + ether, i.e., solution in strong leach solution = R_1 = 23.31 + 27.3 = 50.61 kg.

Weight fraction of oil in strong leach solution (overflow) on B-free basis $= x_1$ = oil/(oil + ether)

= 23.31/50.61

$x_1 = 0.461$

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Determination of the number of stages :

Refer Fig. E 4.6 for the graphical construction of stages.

(i) Plot N v/s x, y (overflow and underflow compositions – underflow and overflow curves). N v/s x is represented by x-axis as the overflow contains no insolubles.

Make use of converted and tabulated data to plot N v/s y curve.

(ii) Locate point F (y_F, N_F) representing feed livers/untreated solids. The coordinates of F are : 1, 2.86.

(iii) Locate point R_1 (x_1, 0) representing strong leach solution. The coordinates of R_1 are : 0.461, 0 (on N v/s x i.e. overflow curve).

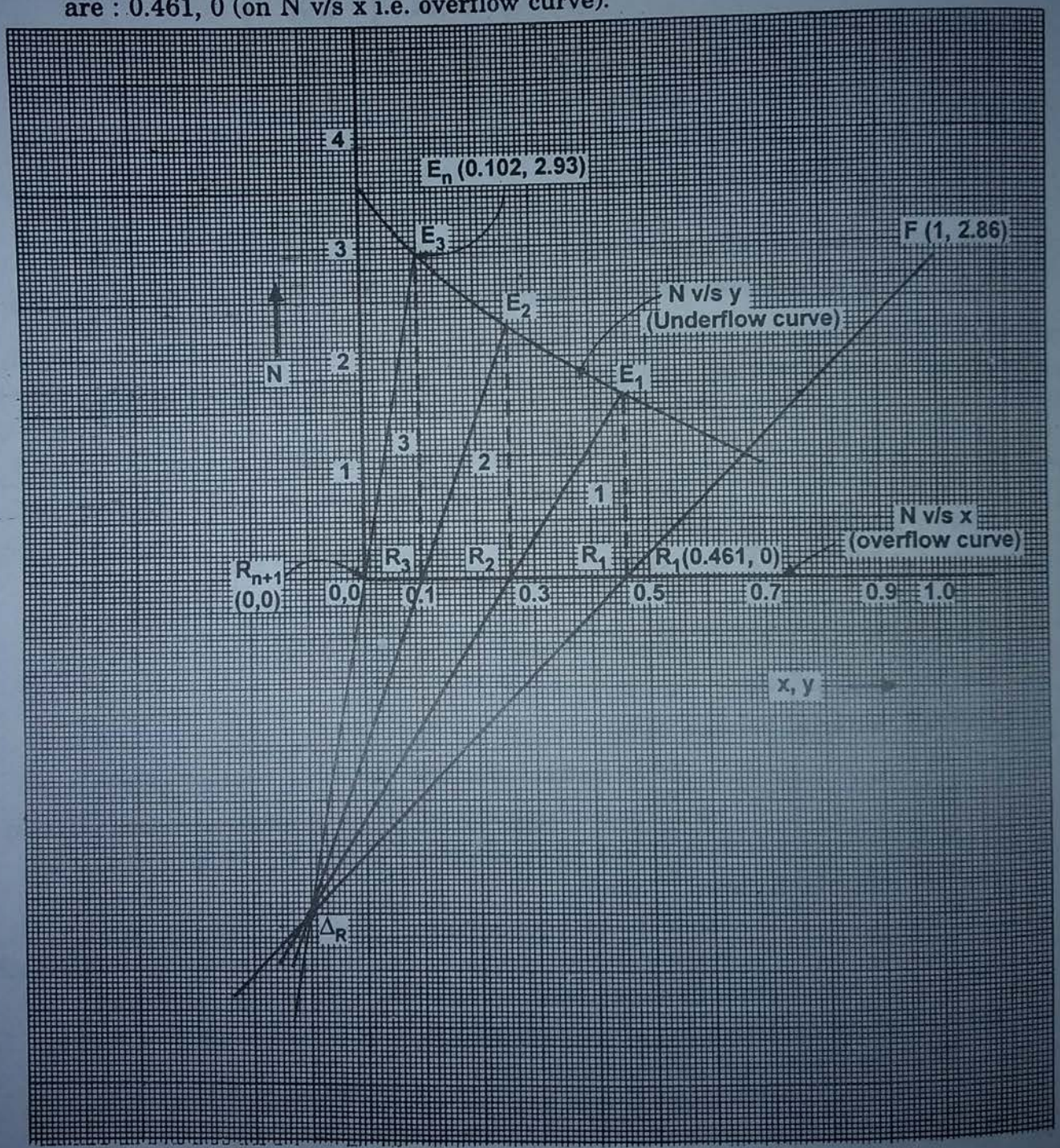

Fig. E 4.6 : Determination of the number of stages – variable underflow (countercurrent operation)

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(iv) Locate point E_n (y_n, E_n) representing leached solids obtained on N v/s y (underflow curve). The coordinates of E_n are : 0.102, 2.93.

(v) Locate point R_{n+1} (x_{n+1}, 0) representing leaching solvent on N v/s x curve i.e., x-axis. Coordinates of R_{n+1} are : 0, 0.

(vi) Join $R_1 - F$ and extend. Join $E_n - R_{n+1}$ and extend to meet extended line F R_1 at point Δ_R (difference point).

(vii) The tie lines are vertical. Through R_1 (as $y^* = x$), draw a vertical tie line (shown dotted) to cut the underflow curve at E_1.

(viii) Join $E_1 - \Delta_R$. Points E_1, Δ_R and R_2 lie on the same line. Mark R_2 at the intersection of $E_1 - \Delta_R$ with N v/s x curve i.e. x-axis.

(ix) Draw the a vertical tie line through R_2 to cut the underflow curve at E_2.

(x) Repeat above mentioned procedure for further construction till we reach exactly or cover point E_n. Count the number of stages. Each tie line represents a stage.

Number of theoretical stages required = 3 ... **Ans.**

Ex. 4.7 : Flaked soyabeans are to be leached with hexane to remove the soyabean oil. The flakes in layer form are fed onto a slowly moving perforated endless belt that passes under a series of continuously operating sprays. As the solid passes under each spray, it is showered with liquid. The showered liquid percolates through a solid, collects in a trough below and is recycled to the spray by means of pump. The sprays are spaced in such a way that the solid leaving a spray is permitted to drain for a period of 6 min before it reaches the next spray. The solvent also passes from trough to trough in countercurrent fashion with respect to a moving belt and hence maintaining truely continuous countercurrent stagewise operation with each spraying and draining constituting a stage.

It is found that the flakes retain solution after 6 min drain time to an extent depending upon the oil content of the solution. The solution retained as function of oil concentration (entrainment) data are as follows :

Solution composition, weight % oil in solution	0	20	30
Solution retained, kg solution/kg insoluble solid	0.58	0.66	0.70

The soyabean flakes entering the unit contain 20% oil by weight and are to be leached to 1% oil (on solvent-free basis). The net forward flow of solvent to be 1 kg hexane introduced as a fresh solvent per kg of flakes. The fresh solvent is free of oil. The solvent draining from the flakes is generally free of solid except in the first stage : the rich miscella contains 10% of the insoluble solid in the feed as a suspended solid, which falls through the perforations of the belt during loading. Determine graphically theoretical stages required.

Solution : Basis : 1 kg of flakes to be leached.

A – hexane, B – insoluble solid, C – oil.

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For determining number of stages required, we have to rearrange/convert the data in required form and find out the terminal compositions (of stream entering and leaving) by material balances and making use of data.

Solids to be leached : Flakes : Feed :

Flakes contain 20% oil and 80% insolubles.

$$\text{oil in flakes} = 0.20 \times 1 = 0.2 \text{ kg}$$

$$\text{insoluble solid in flakes} = B = 0.80 \times 1 = 0.8 \text{ kg}$$

$$\text{hexane solvent in flakes} = A = 0$$

$$\text{oil + solvent in feed} = A + C = 0 + 0.2 = 0.2 + 0 = 0.2 \text{ kg}$$

Weight fraction of oil in feed on B-free basis = $y_F = 0.2/0.2 = 1.0$

kg insolubles/kg (oil + solvent) in feed to be leached = $N_F = B/(A + C) = 0.8/0.2 = 4$

Leaching solvent : 1 kg fresh hexane solvent per kg flakes (given)

$$\text{hexane in leaching solvent (A)} = 1 \text{ kg}$$

$$\text{oil in leaching solvent (C)} = 0$$

$$\text{oil + hexane in leaching solvent} = R_{n+1} = 1 + 0 = 1 \text{ kg}$$

Weight fraction of oil in leaching solvent on B-free basis = $x_{n+1} = 0/1 = 0$.

We are provided with solute concentration.

The solute concentration on B-free basis in the overflow and underflow will be the same, so $y^* = x$, assuming that solids does not contain any adsorbed or undissolved solute. The tie lines are therefore vertical on the N v/s x, y plot.

Rearranging the data in the following required form :

Solution composition, Weight % oil in solution (100 y^*)	Solution retained, $\frac{\text{kg solution}}{\text{kg insoluble solid}}$ (1/N)	Solution composition, $\frac{\text{kg oil}}{\text{kg solution}}$ (y^*)	$\frac{\text{kg solid}}{\text{kg solution}}$ (N)	$\frac{\text{kg oil}}{\text{kg insoluble solid}}$ (y^*/N)
0	0.58	0	1.724	0
20	0.66	0.20	1.515	0.132
30	0.70	0.30	1.428	0.21

Leached solids :

It is given that rich miscella from first stage contains 10% insoluble solid in the feed.

$$\therefore \quad \text{Insoluble solid lost in miscella} = 0.1\,(0.8) = 0.08 \text{ kg}$$

$$\text{Insoluble solid in feed} = 0.8 \text{ kg}$$

Out of 0.8 kg solids, 0.08 kg is lost in miscella and remainder is there in underflow stream.

$$\text{Insoluble solid in leached solid} = 0.8 - 0.08 = 0.72 \text{ kg}$$

Leached solids contain 1% oil and hence 99% insoluble solid on A-free basis (given).

i.e., 1 kg of oil is associated with every 99 kg of insoluble solid in every 100 kg of leached solids.

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$\therefore$ oil in leached solids $= \frac{1}{99} \times 0.72 = 7.273 \times 10^{-3}$ kg

kg oil/kg insoluble solid in leached solids $= y^*/N = \frac{7.273 \times 10^{-3}}{0.72} = 0.0101.$

or kg oil/kg insoluble solid in leached solids $= y^*/N = \frac{1}{99} = 0.0101$

We have to find out N corresponding to $y^*/N = 0.0101$ from the rearranged data by interpolation.

y^*/N	N
0	1.724
0.0101	p
0.132	1.515

$$\frac{1.724 - p}{0.0101 - 0} = \frac{p - 1.515}{0.132 - 0.0101}$$

$$p = 1.708$$

$\therefore$ For $y^*/N = 0.0101$, $N_n = N = 1.708$

kg insoluble solid/kg solution in leached solid $= N_n = 1.708$

i.e., 1.708 kg insoluble solid retains 1 kg of solution in leached solids.

(oil + hexane), i.e., solution retained in leached solids $= E_n = \frac{1}{1.708} \times 0.72 = 0.421$ kg

Hexane retained in leached solids = (solution – oil) retained
$= 0.42 - 7.273 \times 10^{-3} = 0.4127$ kg

Weight fraction of oil retained in leached solids on B-free basis $= y_n$ = oil / solution $= 7.273 \times 10^{-3} / 0.421$
$= 0.0173$

Miscella from stage-1 :

Hexane balance :

Hexane in miscella + hexane in leached solids = hexane fed as a solvent

Hexane in miscella $= 1 - 0.4127 = 0.5873$ kg

Oil balance :

oil in flakes = oil in miscella + oil in leached solids

oil in miscella $= 0.2 - 7.273 \times 10^{-3} = 0.1927$ kg

(oil + hexane) i.e. clear solution in miscella $= R_1 = 0.5873 + 0.1927 = 0.78$ kg.

Weight fraction of oil in miscella on B-free basis $= x_1 = 0.1927/0.78 = 0.247.$

Insolubles solid in miscella = 0.08 kg

kg insoluble solid/kg solution in miscella $= N_1 = 0.08/0.78 = 0.1026.$

Determination of the number of stages :

Refer Fig. E 4.7 for graphical construction work.

(i) Plot N v/s x, y – overflow and underflow curve. The overflow curve N v/s x is represented by x-axis.

(ii) Locate point F (y_F, N_F) representing feed flakes. The coordinates of F are : 1, 4.

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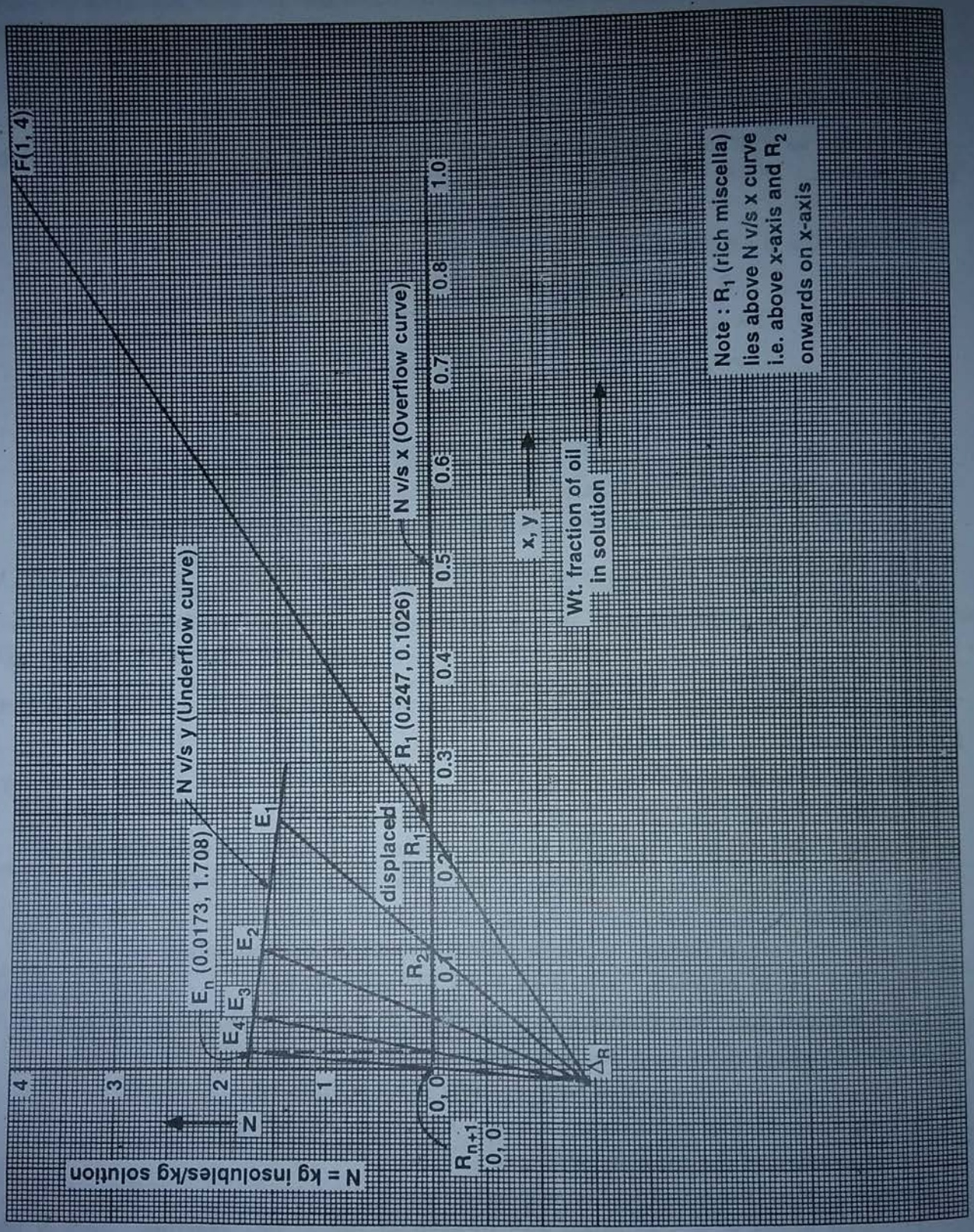

Fig. E 4.7 : Determination of the number of stages (variable overflow, solids in rich miscella - countercurrent operation)

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(iii) Locate point R_1 (x_1, N_1) representing the miscella (overflow) from stage-1. The coordinates of R_1 are : 0.247, 0.1026. Point R_1 is displaced from the x-axis as it represents cloudy miscella.

(iv) Locate point E_n (y_n, N_n) representing leached solids. The coordinates of E_n are : 0.0173, 1.708.

(v) Locate point R_{n+1} (x_{n+1}, 0) representing leaching solvent. The coordinates of R_{n+1} are : 0, 0, $N_{n+1} = 0$ as no solids in solvent used.

(vi) Join F – R_1 and extend. Join $E_n - R_{n+1}$ and extend to meet extended line FR_1 at Δ_R (difference point).

(vii) The tie lines are vertical (as compositions of underflow and overflow are same, $y^* = x$). Through point R_1, draw a vertical tie line (shown dotted) to cut the underflow curve at E_1.

(viii) Join $E_1 - \Delta_R$. Points E_1, Δ_R and R_2 lie on same line. So mark R_2 at the intersection of $E_1 - \Delta_R$ line with x-axis (N v/s x, overflow curve).

(ix) Draw a vertical tie line through R_2 to cut N v/s y, i.e., underflow curve at E_2.

(x) Repeat above mentioned procedure for further construction till we reach exactly or cover point E_n. Count the stages. Each tie line represents a stage.

Number of theoretical stages required = 4 ... **Ans.**

Extra :

oil in flakes = 0.80 kg

oil in leached solids = 7.273×10^{-3} kg

oil recovered = $0.80 - 7.273 \times 10^{-3} = 0.79273$ kg

$$\% \text{ recovery of oil} = \frac{0.79273}{0.8} \times 100 = 99.09\%$$

Check :

leaching solvent = R_{n+1} = 1 kg

Solids to be leached = 1 kg

Miscella = R_1 + insoluble solid = 0.78 + 0.08 = 0.86

Leached solids = insoluble solid + solution = 0.72 + 0.421 = 1.141 kg

Material balance over entire unit :

Leaching solvent + solids to be leached = Miscella + leached solids

1 + 1 = 0.86 + 1.141

2 = 2.001

LHS = RHS

So the calculations done/performed are correct.

Ex. 4.8 : Seeds containing 22 % by weight oil is extracted countercurrently with oil-free hexane as a solvent. Calculate the number of theoretical stages required if 90 % of the oil is recovered in extract with 40 % oil by weight and the amount of liquid (solvent + oil) in the underflow from each stage is 0.60 kg per kg of insoluble matter.

Use triangular coordinates or rectangular coordinates.

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Solution : Basis : 100 kg seeds as a feed to the unit per unit time.

So per unit time :

Seeds to the unit = 100 kg

A - hexane, B - insolubles/insoluble matter, C - oil

For determining the number of theoretical stages required, we have to find out the terminal compositions, i.e., the compositions of streams entering and leaving, by making material balances.

We will first make use of right-angled triangular diagram for plotting.

Feed : Seeds to be leached :

Seeds = 100 kg, containing no solvent

Oil in seeds = C = $0.22 \times 100 = 22$ kg

Insoluble matter in seeds = B = $(1 - 0.22) \times 100 = 78$ kg

Hexane solvent in seeds = A = 0

Keep in mind :

Here we will express the concentrations of oil, hexane in terms of weight fraction based on the entire quantity of feed, overflow, underflow, etc. and not on B-free basis.

Weight fraction of oil in seeds to be leached = $y'_{FC} = \frac{22}{100} = 0.22$

Weight fraction of insoluble matter in seeds to be leached = $y'_{FB} = \frac{78}{100} = 0.78$

Weight fraction of hexane in seeds to be leached = $y'_{FA} = \frac{0}{100} = 0$

Strong leach solution :

The strong leach solution (overflow) leaving the unit contains 40 % oil by weight.

As the overflow does not contain solids, strong leach solution contains only oil and hexane.

Weight % oil = 40

So weight % hexane = 100 – 40 = 60

Weight fraction of oil in strong leach solution = $x'_{1C} = \frac{40}{100} = 0.40$

[as weight % of C = weight fraction of C × 100]

Weight fraction of hexane in strong leach solution = $x'_{1A} = \frac{60}{100} = 0.60$

Leaching solvent :

Oil in leaching solvent = 0 as hexane is oil-free

Hexane in leaching solvent = p kg (say)

Leaching solvent = 0 + p = p kg

Weight fraction of hexane in leaching solvent = $x'_{n+1(A)} = \frac{p}{p} = 1.0$

Weight fraction of oil in leaching solvent = $x'_{n+1(C)} = \frac{0}{p} = 0$

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Leached solids :

It is given that 0.6 kg of solution (oil + hexane) is retained in association with every 1 kg of insoluble matter in underflow/leached solids from each stage.

The ratio of kg solution/kg insoluble from each stage is constant.

As quantity of solution removed in association with solids is constant, it is a constant underflow process.

So the weight fraction of insoluble matter in the underflow (leached solids) from all stages is the same/constant.

$$\therefore \quad y'_{1B} = y'_{2B} = \ldots\ldots = y'_{nB}$$

Leached solids contain 0.6 kg of solution with every 1 kg of insoluble matter.

i.e. (0.6 + 1) = 1.6 kg of leached solids contain 1 kg of insoluble matter and 0.6 kg of solution.

Weight fraction of insoluble matter in leached solids $= y'_{nB} = \frac{1}{1.6} = 0.625$

As the overflow contains no insoluble matter, we have :

Insoluble matter in leached solids = insoluble matter in feed seeds
= 78 kg (based on 100 kg seeds)

Given : 1 kg of insoluble matter retains with it 0.6 kg of solution.

$\therefore$ Solution retained in leached solids $= \frac{0.6}{1} \times 78 = 46.8$ kg (based on 100 kg seeds)

Given : 90 % of oil in feed seeds is recovered. Oil recovered $= 0.9 \times 22 = 19.8$ kg

Oil unextracted and appearing in leached solids $= 22 - 19.8 = 2.2$ kg

or oil in leached solids = 100 – 90 = 10 % of the oil in seeds

oil in leached solids $= 0.1 \times 22 = 2.2$ kg

(oil + hexane) solution in leached solids = 46.8 kg

$\therefore$ Hexane in leached solids = solution – oil
= 46.8 – 2.2 = 44.6 kg

Leached solids leaving the unit per 100 kg seeds
= insoluble matter + solution retained
= 78 + 46.8 = 124.8 kg

Weight fraction of oil in leached solids/underflow $= y'_{nC} = \frac{2.2}{124.8} = 0.0176$

Weight fraction of hexane in leached solids/underflow $= y'_{nA} = \frac{44.6}{124.8} = 0.3574$

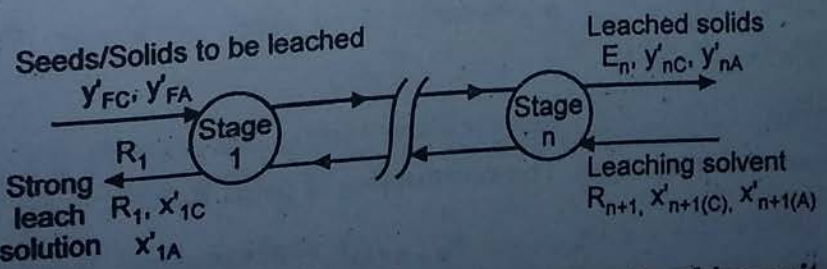

Fig. E 4.8 (a) : Multistage countercurrent leaching unit

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1. Underflow curve :

For constant underflow process, the locus of underflow compositions, i.e., the underflow curve is a straight line with an intercept on both the axes of a right-angled triangle equal to $1 - y'_{nB}$.

$$\therefore \text{ Intercept on x and y axes } = 1 - y'_{nB}$$
$$= 1 - 0.625 = 0.375$$

OR : If p is the kg solution retained per 1 kg of insoluble matter in the underflow then intercept on two axes of right-angled triangular diagram is given by :

$$\text{Intercept} = \frac{p}{p+1}$$

In our case, 0.6 kg solution is retained per kg of insoluble matter.

$$\therefore \quad p = 0.6 \text{ kg}$$

$$\text{Intercept} = \frac{0.6}{0.6+1} = 0.375$$

$$\text{Intercept} = y'_A = 0.375$$

So the underflow curve is a straight line parallel to the hypotenuse of the triangle and intersecting a vertical axis at (intercept value) $y'_A = 0.375$.

2. Overflow curve :

As there are no solids in overflow, the overflow curve is represented by the hypotenuse of a right-angled triangular diagram.

Procedure for determining number of stages using a right-angled triangle :

Refer Fig. E 4.8 (b) :

1. First step is to draw the right-angled triangle. Concentrations of C (oil) in terms of weight fractions, x'_C, y'_C (in overflow and underflow) are plotted along x-axis. Concentrations of A (hexane) in terms of weight fractions, x'_A, y'_A are plotted along y-axis. The hypotenuse represents overflow and is a locus of all possible mixtures of A and C (No B).
2. Draw the underflow curve which is a straight line parallel to the hypotenuse of the triangle having an intercept of 0.375 on vertical axis.
3. Locate point F (y'_{FC}, y'_{FA}) representing feed seeds on x-axis. The coordinates of F are : 0.22, 0.
4. Locate point R_1 (x'_{1C}, x'_{1A}) representing strong leach solution i.e. overflow on the hypotenuse of the triangle. The coordinates of point R_1 are : 0.40, 0.60.
5. Locate point R_{n+1} ($x'_{n+1\,(C)}$, $x'_{n+1\,(A)}$) representing leaching solvent. The coordinates of R_{n+1} are · 0, 1.0. So locate R_{n+1} at the top vertex of the triangle.

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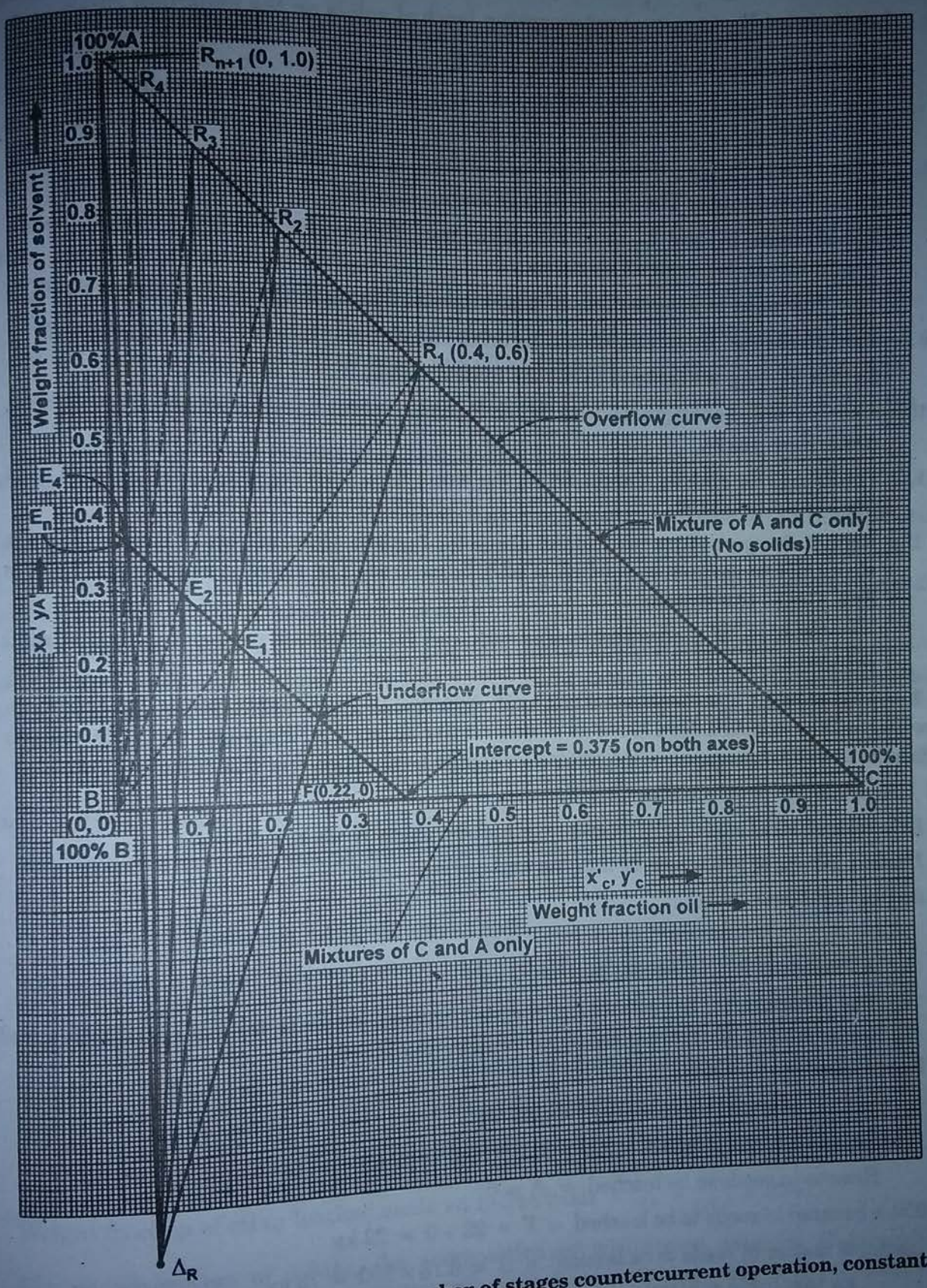

Fig. E 4.8 (b) : Determination of the number of stages countercurrent operation, constant underflow (right angled-triangular diagram)

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6. Locate point E_n (y'_{nC}, y'_{nA}) representing leached solids on the underflow curve.

   The coordinates of E_n are : 0.0176, 0.3573.

7. Join $R_1 - F$ and extend. Join $R_{n+1} - E_n$ and extend to meet the extended line R_1F to get the difference point Δ_R. Δ_R is the point of intersection of lines $R_1 F$ and $R_{n+1} E_n$.

8. Draw line B R_1, i.e., draw a line through the origin of the triangle joining point R_1 on the hypotenuse. The line B R_1 represents constant C/A ratio.

   The streams R_1 and E_1 are in equilibrium. The underflow from stage-1 i.e. E_1 should lie on the underflow curve.

   The point of intersection of line B R_1 with the underflow curve represents E_1.

9. Now points Δ_R, E_1 and R_2 should lie on the same line. R_2 to be on the hypotenuse.

   So join $E_1 - \Delta_R$ and extend to cut the hypotenuse at R_2.

10. Join $B - R_2$ i.e. origin and R_2. Mark E_2 at the point of intersection of $B - R_2$ with underflow curve.

11. Points Δ_R, E_2 and R_3 should lie on the same line. Join $\Delta_R - E_2$ and extend to meet the hypotenuse at R_3.

12. Repeat the procedure mentioned above for further construction of stages till point E_n on the underflow is reached exactly or covered. The stage is represented by a tie line. Count the stages.

    Number of theoretical stages required for achieving required degree of solid extraction = 4 ... **Ans.**

Keep in mind : Construction work with triangular coordinates generally results in crowding of the construction lines into one corner of such a diagram. So it is usually preferable to use rectangular coordinates as adopted in the theory of this topic.

OR Use of normal Rectangular Coordinates.

Determination of number of stages required by using N v/s x, y plots :

Basis : 100 kg of seeds as a feed to the unit per unit time.

So per unit time :

Seeds to the unit = 100 kg

A - hexane, B - insoluble matter, C - oil

For determining the number of stages required, we have to find out the terminal compositions of streams entering and leaving the unit.

Feed : Seeds to be leached.

Seeds to be leached = 100 kg

Oil in seeds to the leached = C = 0.22 (100) = 22 kg

Hexane in seeds to be leached = A = 0

(Oil + hexane) in seeds to be leached = F = 22 + 0 = 22 kg

Insoluble matter in seeds to be leached = B = 0.78×100 = 78 kg

$$\frac{\text{kg insoluble matter}}{\text{kg solution in seeds to be leached}} = N_F = \frac{\text{insolubles}}{\text{oil + hexane}} = \frac{78}{(22+0)} = 3.545$$

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Leaching solvent :

Oil in leaching solvent = 0, as oil-free hexane is used.

Hexane in leaching solvent = p kg (say).

Oil + hexane in leaching solvent = 0 + p = p kg

Weight fraction of oil in leaching solvent on B-free basis $= x_{n+1} = \frac{0}{p} = 0$

Leaching solvent contains no solids. So $N_{n+1} = 0$.

Strong leach solution (Overflow) :

The strong leach solution contains no insolubles.

The strong leach solution contains 40 % oil and hence 100 – 40 = 60 % hexane

In 100 kg solution, there are 40 kg oil and 60 kg hexane.

$\therefore$ Weight fraction of oil in strong leach solution on B-free basis $= x_1 = \frac{40}{100} = 0.40$

kg insolubles/kg solution in strong leach solution $= N_1 = 0$

Leached solids (Underflow) :

It is given that 0.6 kg solution (oil + hexane) is retained in the underflow in association with every 1 kg of insoluble matter from each stage.

As the quantity of solution removed in association with the solids is constant throughout the unit, it is a **constant underflow process.**

For constant underflow conditions, the ratio of insolubles to solution in underflow is constant and solids are settled or drained to the same extent at all solute concentrations. So the underflow curve (i.e., N v/s y) is a horizontal line (parallel to x-axis) at $N = N_n$.

kg insoluble matter/kg solution in leached solids $= N_n = \frac{1}{0.6} = 1.667$

Based on 100 kg seeds :

Insoluble matter in leached solids = Insoluble matter in feed seeds

= 78 kg

(assuming no insolubles are lost in overflow)

0.6 kg solution is retained with every 1 kg insolubles in the leached solids.

$\therefore$ Solution retained in leached solids $= \frac{0.6}{1} \times 78 = 46.8$ kg

oil recovered = 90 % of the oil in feed seeds

oil in leached solids = 10 % of the oil in feed seeds

= 0.1 (22) = 2.2 kg

Weight fraction of oil in leached seeds on B-free basis $= y_n = \frac{2.2}{46.8} = 0.047$

For no adsorption of solute, the composition of withdrawn solution and solution associated with the solids is the same (i.e., $y^* = x$) from each stage and so the tie lines are vertical on the N v/s x, y plot.

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Refer to Fig. E 4.8 (c).

1. Draw N v/s x (overflow curve) and N v/s y (underflow curve). As there are no insolubles in the overflow, N v/s x curve is represented by x-axis.

For constant underflow, the underflow curve is a straight line parallel to x-axis (horizontal line) through N_n on y-axis.

So draw a straight line parallel to x-axis through N_n = 1.667 on y-axis.

2. Locate point F (y_F, N_F) representing the feed seeds. The coordinates of F are : 1, 3.545.
3. Locate point R_1 (x_1, 0) representing the strong leach solution on N v/s x. Coordinates of R_1 are : 0.4, 0.
4. Locate point R_{n+1} (x_{n+1}, 0) representing leaching solvent. The coordinates of R_{n+1} are : 0, 0.

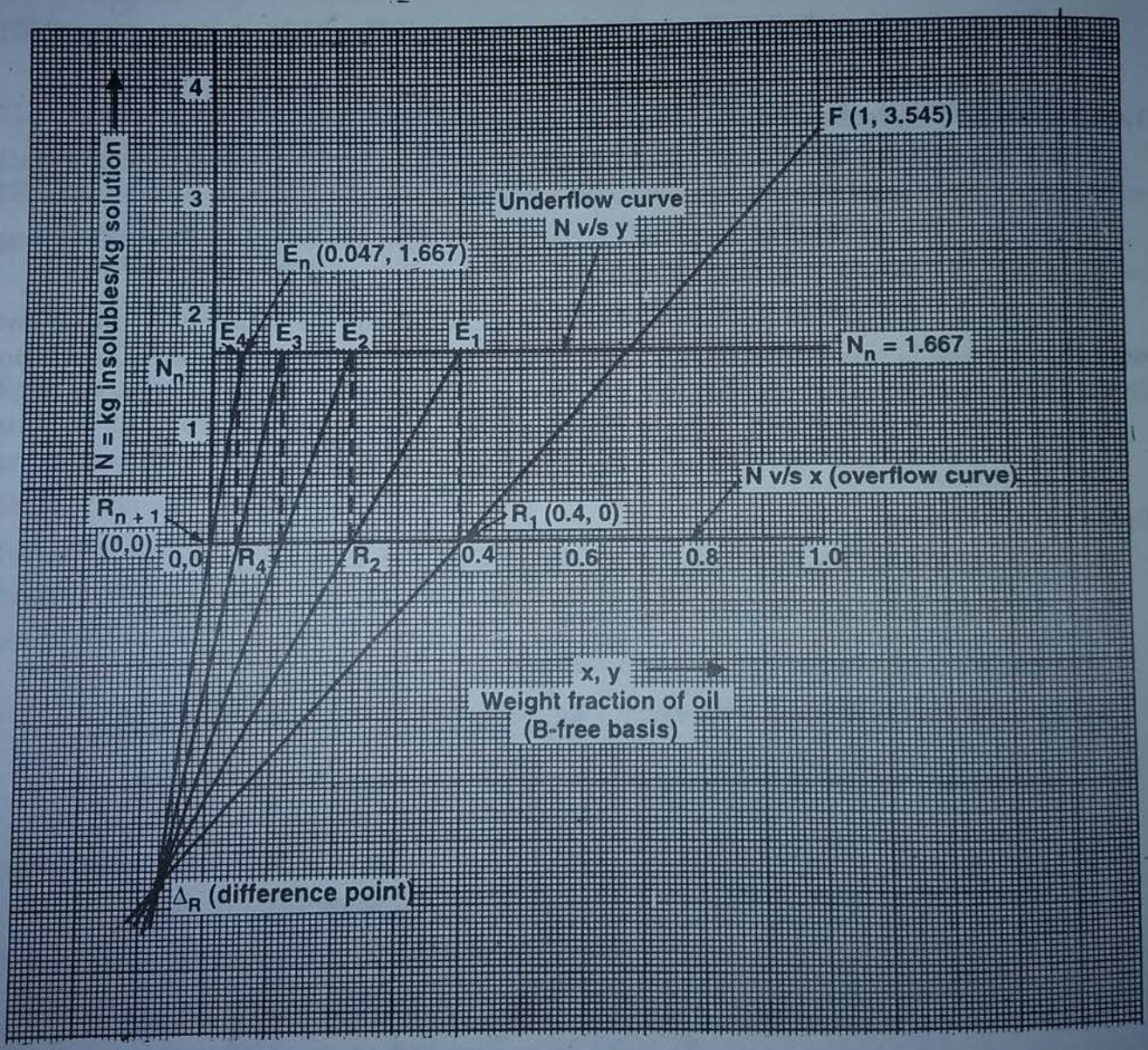

Fig. E 4.8 (c) : Determination of the number of stages (constant underflow, countercurrent operation and rectangular coordinates)

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5. Locate E_n (y_n, N_n) representing the leached solids on N v/s y (underflow curve). The coordinates of E_n are : 0.047, 1.667.
6. Join F to R_1 and extend. Join E_n to R_{n+1} and extend to meet extended line F R_1 at Δ_R (the difference point). Δ_R is the point of intersection of F R_1 and E_n R_{n+1} lines.
7. Through point R_1, draw a vertical tie line (as $y^* = x$) to cut N v/s y curve at E_1.
8. Points Δ_R, E_1 and R_2 lie on the same line. Join $\Delta_R - E_1$ and mark R_2 at which line Δ_R E_1 cuts the x-axis (N v/s x curve).
9. Through R_2, draw a vertical tie line to cut the N v/s y i.e. underflow curve at E_2.
10. Points Δ_R, E_2 and R_3 lie on the same line. Join $E_2 - \Delta_R$ and mark R_3 at which line E_2 Δ_R cuts the x-axis. Draw a vertical tie line through R_3 to cut N v/s y curve at E_3.
11. Repeat the above mentioned procedure for further construction work till the point E_n (on N v/s y) is reached exactly or covered. Each tie line represents a stage. Count the number of stages.

Number of theoretical stages required for achieving desired recovery of oil = 4 ... **Ans.**

Answer by both the methods is same. The second approach is recommended.

Ex. 4.9 : One of the steps in the production of caustic soda by the lime-soda process gives a slurry containing 9 kg water and 0.8 kg sodium hydroxide (NaOH) per kg calcium carbonate particles. The slurry is washed countercurrently with water in four stages. The solid discharged from each stage contains 3 kg water per kg calcium carbonate. Calculate the amount of wash water needed when the discharged calcium carbonate after drying contains a maximum of 0.0075 kg sodium hydroxide per kg calcium carbonate.

Solution : Basis : 1 kg calcium carbonate ($CaCO_3$) in feed to the unit.

A - water, C - NaOH and B - calcium carbonate

Feed to the unit is a slurry containing water, NaOH and $CaCO_3$.

Feed : Slurry :

Water in feed = A = 9 kg

NaOH in feed = C = 0.8 kg

$CaCO_3$ in feed = B = 1 kg

$CaCO_3$ is insoluble matter in feed.

Solution (Water + NaOH) in feed = A + C = F = 9 + 0.8 = 9.8 kg

Leached solids (Underflow) :

As there are no insolubles in overflow, we have :

$CaCO_3$ in leached solids = $CaCO_3$ in feed

= 1 kg

Given : Maximum NaOH content of leached solids is 0.0075 kg per kg $CaCO_3$.

$\therefore$ NaOH in leached solids = $\frac{0.0075}{1} \times 1 = 0.0075$ kg

Given : Water associated with leached solids is 3 kg per 1 kg $CaCO_3$ in it.

Water in leached solids = $\frac{3}{1} \times 1 = 3$ kg

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(NaOH + Water) i.e. solution in leached solids = E_n

$$E_n = 0.0075 + 3 = 3.0075 \text{ kg}$$

Weight fraction of NaOH in leached solids on B-free basis = $y_n = \frac{0.0075}{3.0075} = 0.0025$

Leaching solvent : Pure water is used.

∴ Wash water (pure) in leaching solvent (required) = leaching solvent

$$= R_{n+1} = p \text{ kg (say)}$$

(as water contains no NaOH)

Strong leach solution :

Material balance of NaOH :

NaOH in feed = NaOH in strong leach solution (overflow) + NaOH in leached solids (underflow)

NaOH in strong leach solution = 0.8 – 0.0075 = 0.7925 kg

(A + C) solution material balance :

$$F + R_{n+1} = R_1 + E_n$$

$$\therefore \quad R_1 = F + R_{n+1} - E_n$$

$$= 9.8 + p - 3.0075 = 6.7925 + p \text{ kg}$$

R_1 is the solution.

Weight fraction of NaOH in strong leach solution on B-free basis

$$x_1 = \frac{0.7925}{6.7925 + p}$$

Each stage is an equilibrium stage, so R_1 is in equilibrium with E_1. The composition of the solution in the sludge (underflow) is the same as that of the clear liquid withdrawn (the overflow) for the case of no adsorption of solute, i.e., when insoluble solid is non-adsorbent.

$$\therefore \quad y^* = x$$

$$\therefore \quad y_1 = x_1 = \frac{0.7925}{6.7925 + p}$$

The mass of insolubles from all the stages is equal as the solid is not lost in the overflow. Also the insolubles to solution ratio is constant and therefore the solution in the underflow from all stages is the same.

$$E_1 = \ldots = E_n = E \text{ and } R_1 = \ldots\ldots = R_{n+1} = R$$

$\therefore$ $R/E = p/3.0075 = 0.3325\,p$, $y^* = x$ $\therefore$ $m = 1$ $\therefore$ $R/mE = 0.3325\,p$.

For constant underflow process, we have,

$$\left(\frac{y_1 - y_n}{y_1 - m\,x_{n+1}}\right) = \frac{(R/mE)^n - R/mE}{(R/mE)^n - 1}$$

Given : n = number of stages = 4, $x_{n+1} = 0$, $y_n = 0.0025$, $m = 1$

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Putting values of various terms, we get,

$$\frac{\left(\frac{0.7925}{6.7925 + p} - 0.0025\right)}{\left(\frac{0.7925}{6.7925 + p} - 0\right)} = \frac{(0.3325\ p)^4 - 0.3325\ p}{(0.3325\ p)^4 - 1}$$

$$\frac{0.7925 - 0.0025\ (6.7925 + p)}{0.7925} = \frac{0.0122\ p^4 - 0.3325\ p}{0.0122\ p^4 - 1}$$

$$\frac{0.7725 - 0.0025\ p}{0.7925} = \frac{0.0122\ p^4 - 0.3325\ p}{0.0122\ p^4 - 1}$$

$$0.335\ p - 0.24 \times 10^{-3}\ p - 3.05 \times 10^{-5}\ p^5 = 0.7755 \ldots (A)$$

To find out p, we have to adopt a trial and error procedure. Assume a value of p and evaluate the LHS and see whether LHS is equal to RHS or not. Proceed till we get LHS ≈ RHS. For p = 2.3, the first term of Equation (A) itself is less than 0.7755, so values less than 2.3 are not feasible.

For p = 3, LHS = 0.978 ≠ RHS (0.7755)

For p = 2.4, LHS = 0.79 ≠ RHS

For p = 2.34, LHS = 0.77456 ≈ RHS

For p = 2.342, LHS = 0.7752 = 0.7755

So p = 2.342 kg

p is the wash water required.

Amount of wash water needed (per 1 kg $CaCO_3$) = 2.342 kg ... Ans.

Ex. 4.10 : Roasted copper ore containing copper as $CuSO_4$ is to be extracted in multistage countercurrent unit. The feed to be handled per hour consists of 10000 kg of gangue, 1200 kg of $CuSO_4$ and 500 kg water. The strong solution produced contains 90 % H_2O and 10 % $CuSO_4$ by weight. 98 % of $CuSO_4$ in ore is recovered using pure water as a solvent. After each stage, 1000 kg of inert gangue retains 2000 kg water, plus the copper sulphate dissolved in that water. Equilibrium is attained in each stage. Determine number of stages required.

Solution : Basis : 10000 kg/h of gangue in the feed ore to the unit.

A – water, B – inert gangue, C - $CuSO_4$

Feed : Water in feed = A = 500 kg/h

$CuSO_4$ in feed = C = 1200 kg/h

Inerts in feed = B = 10000 kg/h

Solution (A + C) in feed = F = 1200 + 500 = 1700 kg/h

Weight fraction of $CuSO_4$ in feed on B-free basis = $y_F = \frac{1200}{1700} = 0.706$

kg gangue/kg solution in feed = $N_F = \frac{10000}{1700} = 5.88 = 5.9$

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Leaching solvent : Pure water is used.

Weight fraction of $CuSO_4$ in leaching solvent on B-free basis $= x_{n+1} = 0$

(as leaching solvent contains no $CuSO_4$)

As the solvent does not contain inert gangue, $N_{n+1} = 0$

Leached solids (Underflow) :

$CuSO_4$ recovered = 98 % of that in ore (feed)

Unrecovered $CuSO_4$ appearing in leached solids = 100 – 98 = 2 % of that in ore (feed)

$CuSO_4$ in leached solids = 0.02 (1200) = 24 kg/h

Given : 1000 kg of gangue retains 2000 kg of water in association with it in the underflow.

As overflow contains no gangue :

Gangue in leached solids = Gangue in feed

= 10000 kg/h

$$\text{Water in leached solids} = \frac{2000}{1000} \times 10000 = 20000 \text{ kg/h}$$

Solution (Water + $CuSO_4$) in leached solids

= 20000 + 24 = 20024 kg/h

Weight fraction of $CuSO_4$ in leached solids on B-free basis,

$$y_n = \frac{24}{20024} = 1.198 \times 10^{-3} = 1.2 \times 10^{-3} = 0.0012$$

$$\text{kg gangue/kg solution in leached solids} = N_n = \frac{10000}{20024} = 0.4994 \approx 0.5$$

Strong leach solution :

Weight % $CuSO_4$ in strong leach solution = 10

Weight % water in strong leach solution = 90

i.e. 100 kg solution contains 10 kg $CuSO_4$ (C) and 90 kg water (A).

$$\text{Weight fraction of } CuSO_4 \text{ in strong leach solution on a B-free basis} = x_1 = \frac{10}{(10+90)}$$

= 0.10

It is given that 1000 kg of gangue retains 2000 kg of water and some quantity of $CuSO_4$ in the underflow. So we will find out the values of N at various solute concentrations. It is a case of variable underflow.

Equilibrium is attained : $y^* = x$

Water retained in the underflow = 2000 kg per 1000 kg gangue.

(1) For x = 0.01, y = 0.01

Weight fraction of $CuSO_4$ on a B-free basis in the underflow = 0.01

Weight fraction of water on a B-free basis in the underflow = 1 – 0.01 = 0.99

i.e., 0.01 kg $CuSO_4$ is dissolved in 0.99 kg water (in 1 kg solution) in the underflow.

Water in underflow per 1000 kg gangue = 2000 kg

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$\therefore$ $CuSO_4$ in underflow $= \frac{0.01}{0.99} \times 2000 = 20.20$ kg

Solution in the underflow = Water + $CuSO_4$

$= 2000 + 20.20 = 2020.20$ kg

kg gangue/kg solution in the underflow (for $y^* = 0.01$) $= N = \frac{1000}{2020.20} = 0.495$

(2) For $x = 0$, $y^* = 0$

Weight fraction of $CuSO_4$ in the underflow on B-free basis = 0

$\therefore$ $CuSO_4$ in the underflow = 0

Water in the underflow = 2000 kg

Gangue in the underflow = 1000 kg

Solution in the underflow = 2000 + 0 = 2000 kg

kg gangue/kg solution in the underflow (for $y^* = 0$) $= N = \frac{1000}{2000} = 0.5$

In this way, generate N - x, y data.

OR :

If y^* is a weight fraction of $CuSO_4$ on B-free basis

$1 - y^*$ is a weight fraction of water on B-free basis

then solution in the underflow $= \frac{y}{1-y}(2000) + 2000 = 2000\left[\frac{y}{1-y} + 1\right]$

$= \frac{2000}{(1-y)}$

kg gangue/kg solution in the underflow $= N = \frac{1000}{2000/(1-y)}$

$N = \frac{1000\,(1-y)}{2000}$

x	0	0.01	0.02	0.03	0.04	0.05	0.10	0.20
y^*	0	0.01	0.02	0.03	0.04	0.05	0.10	0.20
$N = \frac{1000\,(1-y^*)}{2000}$	0.5	0.495	0.49	0.485	0.48	0.475	0.45	0.40

Determination of the number of stages :

Refer Fig. E 4.10 (b) for graphical construction of stages.

1. Plot N v/s x and N v/s y, i.e., overflow curve and underflow curve. The overflow curve is represented by x-axis (for no solids in overflow).
2. Locate point F (y_F, N_F) representing feed. The coordinates of F are : 0.706, 5.9.
3. Locate point R_1 (x_1, 0) representing strong leach solution. The coordinates of R_1 are : 0.10, 0. Locate on N v/s x curve.
4. Locate point E_n (y_n, N_n) representing leached seeds.

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The coordinates of point E_n are : 0.0012, 0.4994 (0.5) on N v/s y, i.e., the underflow curve.

5. Locate R_{n+1} (x_{n+1}, 0) representing leaching solvent. The coordinates of R_{n+1} are : 0, 0.
6. As $y^* = x$, solute composition on B-free basis in the underflow and the overflow is the same, so the tie lines are vertical.
7. Join F – R_1 and extend. Join E_n – R_{n+1} and extend. Mark Δ_R at the point of intersection of extended lines F R_1 and E_n R_{n+1}. Δ_R is the difference point.
8. Through R_1, draw a vertical tie line to cut N v/s y curve at E_1.
9. Points Δ_R, E_1 and R_2 lie on the same line. R_2 represents overflow from stage - 2. Join Δ_R – E_1 and mark R_2 at which line Δ_R E_1 cuts the N v/s x curve, i.e., x - axis.
10. Draw a vertical tie line through R_2 to cut the underflow curve (N v/s y) at E_2.
11. Points Δ_R, E_2 and R_3 lie on the same line. Join Δ_R – E_2 and mark R_3 at which line Δ_R E_2 cuts the x-axis.
12. Through R_3, draw a vertical tie line to cut N v/s y curve at E_3.
13. Repeat the above mentioned procedure for further construction work till we reach exactly or cover point E_n (on the underflow curve). Each tie line represents a stage. Count the number of stages required.

Number of stages required for achieving the desired degree of recovery = 10 ... Ans.

Required for plotting : line joining Δ_R – R_1 – F (the point F going out of graph paper due to scale chosen) :

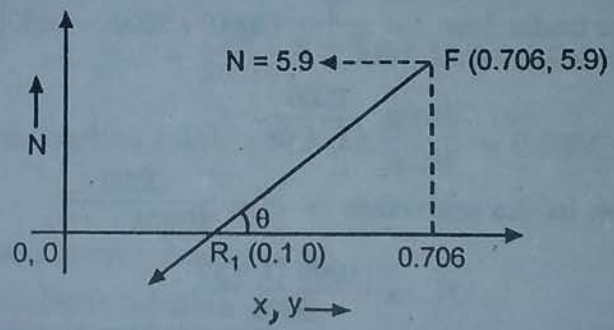

Fig. E 4.10 (a) : Not to scale

Scales for Fig. E 4.10 (b) : y-axis : 10 mm = 0.1 units
x-axis : 10 mm = 0.01 units

$$\text{Slope of line } R_1F = \frac{5.9 - 0}{0.706 - 0.10} = \frac{5.9}{0.606}$$

Taking scales into consideration,

$$\tan\theta = \frac{5.9 \times \text{scale y-axis}}{0.606 \times \text{scale x-axis}} = \frac{5.9 \times (10/0.1)}{0.606\,(10/0.01)} = \frac{590}{606}$$

$$\tan\theta = 0.9736$$

$$\theta = 44.23°$$

So the line R_1 to F is drawn through R_1 (0.1, 0) at an angle of 44.23° to x-axis in Fig. E 4.10 (b). As it is not possible to cover the point F with the scale taken on a given graph paper. Draw a line through R_1 at an angle 44.23° to x-axis and extend the same below x-axis to cut the extended line E_n R_{n+1} at point Δ_R. In this way draw line R_1F, extend below x-axis and get the difference point Δ_R required for the construction work.

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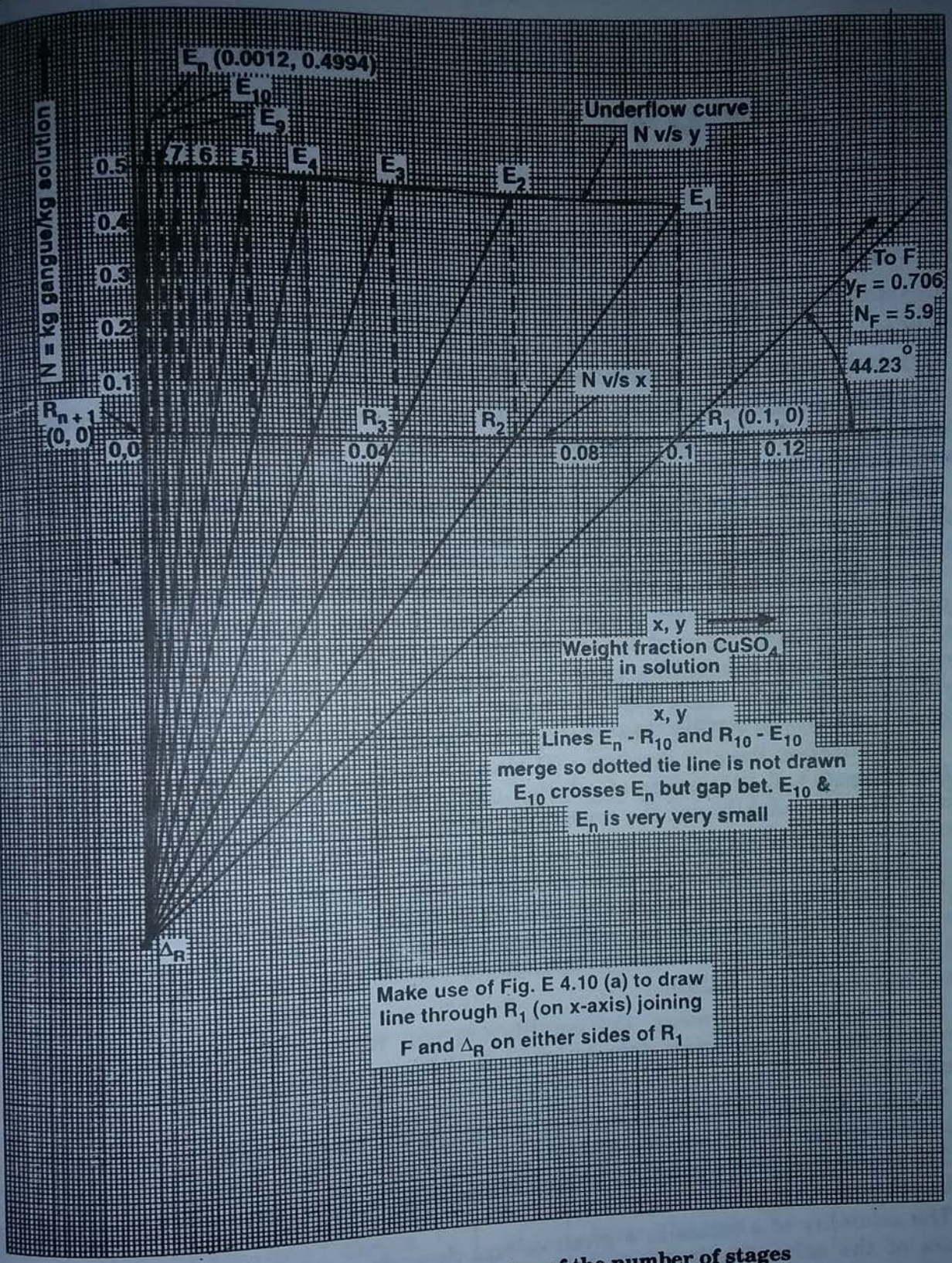

Fig. E 4.10 (b) : Determination of the number of stages (variable underflow, countercurrent operation)

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Chapter ... 5

CRYSTALLISATION

In this chapter, our discussion on crystallisation is restricted to crystallisation from liquid solution.

Crystallisation is a unit operation in which solid particles are formed from a liquid solution.

Crystallisation involves simultaneous heat and mass transfer.

Crystallisation is a solid-liquid operation used to separate solutes from solution in the form of crystals. In this operation, mass is transferred from the liquid phase (solution) to the solid phase/the crystal surface.

Crystallisation is normally carried out in process industry, either from solution or melt (polymer melts, etc.), the former being very common (crystallisation from solution is important as an industrial process since a number of salable products from plants have to be in the form of crystals). This operation gives almost pure product from relatively impure solutions in a single processing step in the form of crystals of the desired size range and is generally carried out at relatively low temperatures. From the energy point of view, crystallisation requires much low energy for separation as compared to other purification methods (e.g., distillation, etc.). It may be carried out at relatively low temperatures and on a scale ranging from a few grams to thousands of tons per day.

Crystallisation usually involves - (i) concentration of solution (by evaporating a part of the solvent) and (ii) cooling of solution until the concentration of the solute becomes greater than its solubility at the prevailing temperature. Then the solute comes out of the solution i.e., precipitates in the form of pure crystals.

It is more convenient to use any material in the crystalline form because it facilitates packing and storing. The performance of a crystallisation process is evaluated in terms of size, shape, structure, yield and purity of crystals. So, in commercial crystallisation, the size and shape of the crystals are as important as the yield and purity of the crystals.

Equilibrium data (solubilities) :

In crystallisation process, the equilibrium will be reached when the solution is saturated and the equilibrium data are given in the form solubility curves.

Solubility :

The solubility of a solute in a given solvent is the *concentration of the solute in a saturated solution at a given temperature.*

The concentration of a solute in a saturated solution is called the solubility of the solute in the solvent.

The solubility of a solute in a given solvent depends on the nature of the solute, the nature of the solvent and the prevailing temperature (solubility mainly depends on temperature). Solubility data are generally given as parts by weight of anhydrous solute per 100 parts by weight of solvent, e.g., the solubility of $MgSO_4$ in water at 293 K (20° C) is 35.5 kg $MgSO_4$ per 100 kg water.

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Solubility data are plotted as solubility curves - curves wherein solubilities are plotted against temperature.

The solubility of the solute in a given solvent is different at different temperatures and this forms the basis of crystallisation by cooling.

Saturation :

Consider a process of dissolving ammonium sulphate in water (solvent) at a given temperature. Initially, when some amount of ammonium sulphate is added, all of it get dissolved in water forming a solution of ammonium sulphate in water. Additional amount can be dissolved further till a stage comes when no more ammonium sulphate can be dissolved in the given amount of the solvent. At this stage, the solution is called as a saturated solution (The solution is said to be saturated with respect to the solute.)

A saturated solution is defined as the *one which is in equilibrium with an excess of solid solute at a given temperature.*

If the temperature of the solution is increased, more solute (Na_2SO_4) can be dissolved. Therefore, a saturated solution will contain different amounts of solute dissolved in it at different temperatures. When a saturated solution at higher temperature (say at T_1) is cooled (to say T_2, $T_1 > T_2$) then theoretically the amount of the solute corresponding to the difference in solubilities at these two temperatures will come out of the solution in the form of crystals.

Solubility Curves :

When any solid is in contact with its solution for sufficient time then : (i) the solid will dissolve if the solution is not saturated or (ii) the solid will grow at the expense of the solution if the solution is supersaturated. In any case, the solubility curve represents equilibrium conditions.

A graphical relationship between the solubility and temperature is termed as the solubility curve.

The concentration necessary for crystal formation and chemical species that separate can be determined from solubility curves. Such curves are obtained by plotting-the solubility of a solute (in some convenient unit) as a function of temperature. It shows the effect of temperature on the solubility of the solute substance. The solubility of solutes in a given solvent may increase, decrease, or more or less remains constant with temperature. Solubility curves have no general shape or slope.

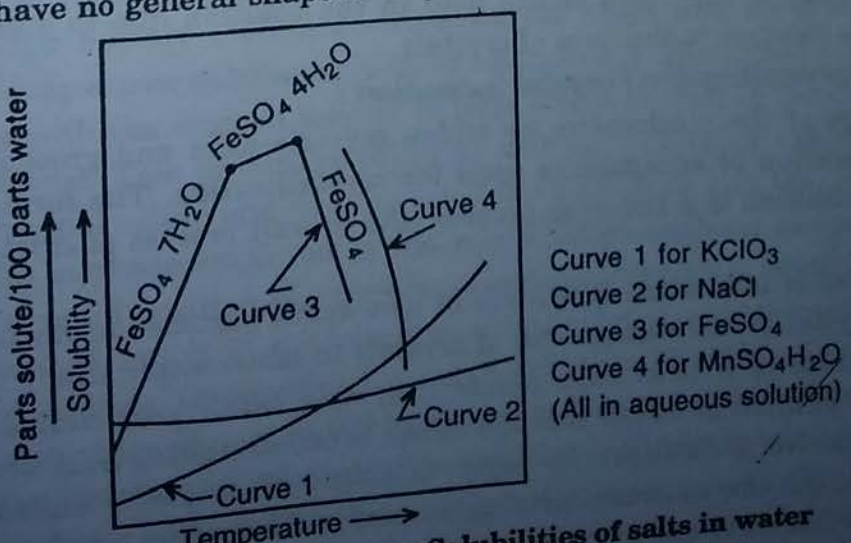

Fig. 5.1 : Solubility curves : Solubilities of salts in water

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Solubility curves of potassium chlorate, sodium chloride are continuous solubility curves as they show no sharp breaks anywhere. Sometimes, the solubility curve exhibits sudden changes of direction and these curves are therefore referred to as discontinuous solubility curves, e.g. that of $FeSO_4$, Na_2SO_4, etc.

For some substances, their solubility decreases with increase in temperature and in such cases their solubility curves are called as inverted solubility curves (e.g., that of $MnSO_4 \cdot H_2O$ in H_2O).

Usually increase in the temperature of the solution increases the solubility of the solute when no true compounds are formed between the solute and solvent., e.g., $KClO_3$, KNO_3 in H_2O. In case of hydrated salt, the solubility increases with increase in temperature over a certain temperature range and then decreases. In Fig. 5.1, the solubility curves are given for a number of salts in water. In case of potassium chlorate, the solubility increases with temperature, so it can be readily crystallised by cooling its saturated solution. The solubility of NaCl in water is almost independent of temperature (a slight increase in the solubility results by a large increase in temperature), so for crystallisation to occur, some portion of the solvent should be evaporated.

Supersaturation :

Consider an equilibrium solution (solid solute + liquid solvent). If we disturb this equilibrium either by cooling the solution or evaporating a part of the solvent, the quantity of dissolved solid solute will exceed the equilibrium concentration and the system will try to attain a new state of equilibrium by expelling (precipitating) excess solid solute present in it in the form of crystals. The resulting process is called crystallisation from solution and the concentration gradient driving force is called supersaturation.

Supersaturation : It is the *quantity of solute present in solution in which crystals are growing compared with the quantity of solute that is in equilibrium with the solution.* Supersaturation is expressed as a coefficient 's' and is given by

$$s = \frac{\text{Parts solute / 100 parts solvent at prevailing condition}}{\text{(Parts solute / 100 parts solvent) at equilibrium}} \geq 1.0$$

Crystallisation cannot take place/occur (i.e., crystals can neither form nor grow) unless the solution is supersaturated (i.e.,without supersaturation). The amount of crystals formed however depends upon the difference in saturation concentrations since once the crystallisation begins, the extra solute held in the solution due to supersaturation also comes out of the solution in the form of crystals.

Mechanism of crystallisation / crystal formation :

A knowledge of the mechanism by which crystals form and grow is required in the design and operation of equipments used for crystallisation. The formation of a crystal matter from a solution is a two-step process (the overall process of crystallisation consists of two steps).

The first step is nucleation (the birth of new small particles or nuclei) and the second one is called crystal growth-the growth of crystals to macroscopic size.

The generation of a new solid phase (i.e., new small particles) either on an inert material in the solution or in the solution itself, is called **nucleation** while *the increase in size of these nuclei with layer by layer addition of solute is called crystal growth.* Supersaturation is the common driving force for nucleation and crystal growth. The crystals can neither form nor grow unless a solution is supersaturated.

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The number of nuclei and rate of growth of crystals does depend upon the temperature of operation. The number of nuclei and growth rate of crystals increases with increase in temperature upto a certain point and then decreases. The temperature corresponding to a maximum number of nuclei formation is different than the temperature at which a growth rate is maximum.

If initially large number of nuclei are formed, then the yield of the process contains large number of tiny crystals and if few nuclei are formed initially, then the yield of the process contains large size crystals. Slow cooling results in the formation of a less number of nuclei and hence large size crystals are formed (as the material deposits on a relatively few nuclei), whereas rapid cooling results in the/leads to the formation of a large number of nuclei, giving yield containing a large number of tiny crystals.

Methods of Supersaturation :

Unless a solution is supersaturated, neither nucleation nor crystal growth occurs (i.e., crystals can neither form nor grow). Thus, for crystallisation to occur/take place, supersaturation can be achieved by any one of the following methods :

(a) Supersaturation by cooling a concentrated, hot solution through indirect heat exchange.

(b) Supersaturation by evaporation of a part of the solvent/By evaporating a solution.

(c) Supersaturation by adiabatic evaporation and cooling (i.e., by vacuum cooling). By flashing a feed solution adiabatically to a lower temperature and inducing/causing crystallisation by simultaneous cooling and evaporation of the solvent.

(d) Supersaturation by adding a new substance (i.e., a third substance) which reduces the solubility of the original solute, i.e., by salting.

(e) Supersaturation by chemical reaction with a third substance.

When the solubility of the solute increases with increase in temperature, a saturated solution becomes supersaturated, i.e., supersaturation is achieved by cooling or temperature reduction. This is the case with many inorganic salts and organic substances (e.g. potassium nitrate, potassium chlorate, oxalic acid, etc.). When the solubility of the solute is relatively independent of temperature, as is the case with common salt (NaCl), supersaturation is generated by evaporating a part of the solvent. When the solubility of solute is very high neither cooling nor evaporation helps and supersaturation may be generated by adding a new substance (a third component). The added component may get mixed physically with the original solvent to form mixed solvent in which the solubility of the solute is reduced. This process is called as salting. The third component added may react with the original solute and form an insoluble substance and the process is called as precipitation. The methods employed in wet quantitative analysis are prime examples of precipitation. The rapid creation of a very large supersaturation is possible by the addition of a third component but it is not common in industry. Supersaturation by adiabatic evaporation and cooling is adopted for large scale production. The hot solution is introduced into a vacuum where the total pressure is less than the vapour pressure of the solvent (at the temperature of solution). This results in flashing of the solvent which in turn produces adiabatic cooling. The combination of evaporation and cooling produces the desired supersaturation.

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The Mier's Supersaturation theory :

According to Mier's theory there is a definite relationship between the concentration and temperature at which crystals will spontaneously form in a pure solution. This relationship is represented by supersolubility curve which is approximately parallel to the solubility curve. Both the curves are shown in Fig. 5.2. The curve AB is the solubility curve and the curve PQ is the supersolubility curve. The curve AB represents the maximum concentration of a solution that can be achieved by bringing a solid solute into equilibrium with solvent. If a solution having composition and temperature indicated by the point 'C' is cooled in the direction shown by the arrow, it first crosses the solubility curve AB and we would expect here crystallisation to start. Actually if we start with initially unseeded solution, crystal formation will not begin until the solution is super cooled considerably past the curve AB. According to the Mier's theory, crystallisation will start in the neighbourhood of the point D and the concentration of the solution then follows roughly along the curve DE. For an initially unseeded solution, the curve PQ represents the limit at which spontaneous nuclei formation begin and consequently, crystal formation starts. According to the Mier's theory, under normal conditions, nuclei cannot form and crystallisation cannot then occur in the area between solubility curve and the supersolubility curve, i.e., at any position short of point D along the line CD in Fig. 5.2.

Mier's theory is useful in the discussion of qualitative aspects of nucleation from seeded and unseeded solution.

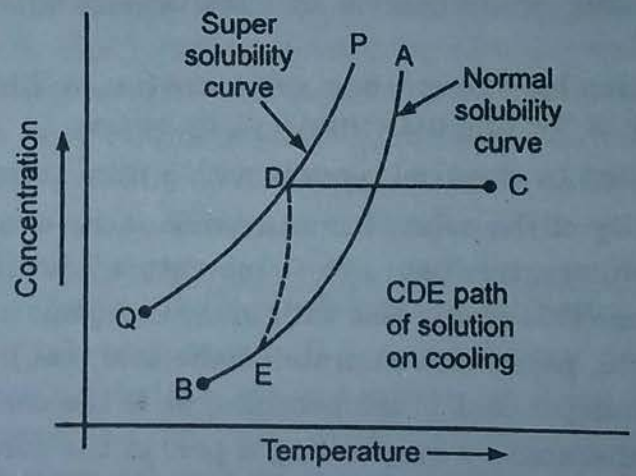

Fig. 5.2 : The Mier's Supersaturation Theory

Yield of crystallisation process :

Usually, crystallisation processes are carried out slowly and the mother liquor is in contact with sufficiently large crystals so that at the end of the process, the mother liquor is saturated at the final temperature. In such cases, the yield of crystallisation process is calculated from the initial solution composition and the solubility of the solute material at the final temperature. If an appreciable evaporation of the solvent occurs during the crystallisation process, then the solvent evaporated must be taken into account in determining the yield. Whenever crystals are anhydrous, the yield is obtained by taking the difference between the initial composition of the solution and the solubility of the solute corresponding the final temperature of the process. In cases where the material precipitates as a hydrated salt, we have to take into account the water of crystallisation (in crystals), as this water is not available for retaining solute in solution. Under these circumstances, the

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key to calculations of yield is to express all compositions in terms of hydrated salt and excess water, as excess water remains constant during crystallisation operation and composition or amounts thus expressed on the basis of excess water can be deducted to obtain correct results.

Crystal form :

The constituent particles of a crystal are arranged in an orderly and repetitive manner. The constituent particles may be atoms, molecules or ions. They are arranged in orderly three dimensional arrays called space lattices. Crystals are classified according to the angle between the faces. Different forms of crystals based upon the angle between faces and lengths of axes are cubic, tetragonal, orthorhombic, hexagonal, monoclinic, triclinic and trigonal.

Caking of Crystals :

Caking of crystalline materials is caused by a small amount of dissolution occurring at the surface of crystals and subsequent re-evaporation of the solvent. Due to caking, the crystals can get very tightly bonded together.

Since the vapour pressure of a saturated solution of a crystalline solid is less than that of pure water at a given temperature, condensation can take place on the surface of the crystals even though the atmospheric relative humidity is less than 100 per cent. The solution thus formed enters/penetrates into the pack of crystals as a result of capillary action of the small gaps between the crystals and caking can result due to subsequent evaporation of moisture when the atmospheric humidity falls. Crystalline materials can also cake at a constant relative humidity, because of vapour pressure of solution is less in small capillary as compared to in large capillary as a result of temperature effect. As condensation occurs, the small particles get first dissolved and therefore average size of capillaries increases and the vapour pressure of the solution may increase sufficiently for the evaporation to take place. When the particle size is non-uniform, a crystalline material will cake more rapidly since the porosity of a bed of particles of mixed sizes is less and fine particles are more readily soluble. Thus the tendency of crystalline materials to cake can be reduced by forming crystals of relatively large and uniform sizes or by adding a water repellent agent, such as stearic acid.

Magma : It is a two-phase mixture of mother liquor and crystals that occupy the crystalliser and in turn withdrawn as product.

In order to reduce the load on a crystalliser, evaporation may be carried out for the removal of excess of solvent. Crystallisation is generally followed by filtration for the separation of crystals from mother liquor.

Effects of impurities on crystal formation :

(i) Soluble impurities may get adsorbed on the surface of the nuclei or crystals-nucleation sites and retard the rate of nucleation and crystal growth.

(ii) The shape of crystal may get modified as adsorption of impurities may occur preferentially on a particular face.

The impurities may decrease the rate of crystal growth. In some cases, it is desirable e.g., addition of a small quantity of glue or tannin to boiler feed water prevents nucleation and growth of calcium carbonate crystals and thus reduces scaling.

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Classification of Crystallisers :

Crystallisers may be classified on the basis of mode of operation. These may be operated batch-wise or continuously. Generally, continuous operation is preferred, however, batch operation is used for certain special applications.

1. Batch crystalliser : stir-tank crystalliser
2. Continuous crystalliser : Swenson-Walker crystalliser.

Crystallisers may also be classified according to the method by which supersaturation is achieved in actual practice. Thus, agitated tank crystalliser, Swenson-Walker crystalliser are examples of crystallisers wherein supersaturation is achieved by cooling which is the usual practice for materials of which the solubility decreases with decrease in temperature. Krystal crystalliser is the example of evaporative crystalliser wherein supersaturation is achieved by evaporating a part of the solvent and is the usual practice for materials whose solubility remains almost constant with variation in temperature. Vacuum crystalliser is the example of crystalliser wherein supersaturation is achieved by an adiabatic evaporation and cooling which is the most suitable for heat sensitive materials. This is used for large scale production in which supersaturation is achieved by introducing a hot solution into a vacuum in which pressure is less than the vapour pressure of solvent at the temperature at which it is fed, the solvent thus flashes or evaporates and the solution is cooled adiabatically. Salting out with the help of a third substance is not often in use. An indirect application of salting method is found out in the evaporation of glycerine soap lyes. In this case, the presence of glycerine in high concentration reduces the solubility of the solute, so that as the concentration of very soluble component increases, the solubility of less soluble component (in this case sodium chloride) decreases to a point where it crystallises out. At present, deliberate introduction of a foreign substance to decrease the solubility is rarely found.

Classification based on the method of achieving supersaturation :

1. Supersaturation by cooling alone :
   (a) Batch – Agitated tank crystallisers
   (b) Continuous – Swenson-Walker crystalliser
2. Supersaturation by adiabatic evaporation and cooling
   (a) Vacuum crystallisers with and without external classifying seed bed.
3. Supersaturation by evaporation.
   (a) Krystal crystallisers
   (b) Draft-tube crystallisers.

Crystallisers may also be classified according to the method of suspending the growing product crystals, as (a) where suspension is agitated in a tank, (b) it is circulated through a heat exchanger, (c) it is circulated through a scraped surface exchanger.

Agitated Tank Crystalliser :

It is also known as stir-tank crystalliser or agitated batch crystalliser. This is the simplest and perhaps the most economical unit. In this crystalliser, supersaturation is generated by cooling, so it is a cooling crystalliser.

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This type of crystalliser is commonly employed in small scale production or batch processing due to several advantages such as low initial cost, simple in construction and flexibility of operation. These are having capacities more than tank crystallisers.

Construction : Agitated tank crystalliser consists of a cylindrical tank provided with a low speed agitator and a cooling coil (through which a coolant is circulated). The tank is having a conical bottom through which the product of crystallisation is withdrawn. The agitator improves the heat transfer rate, keeps the temperature of the solution uniform and keeps the fine crystals in a suspension which is essential for uniform growth of the crystals, otherwise we get large crystals or aggregates.

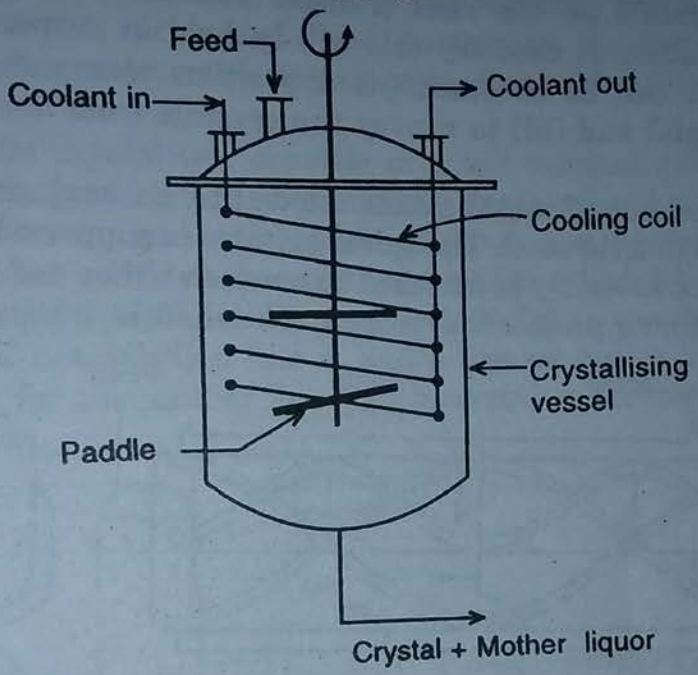

Fig. 5.3 : Agitated tank crystalliser

Working : A known quantity of a hot solution is charged to the crystalliser, cooling is applied by circulating a coolant through the coil and agitator is started. The mass in the crystalliser cools due to heat transfer to the circulated coolant and as the temperature decreases, crystals are formed due to decrease in the solubility of the solute in the solvent. The mass is cooled to a predecided temperature and finally product stream containing crystals plus mother liquor is withdrawn from the bottom of the crystalliser.

One of the drawbacks of this crystalliser is that the deposited solids on the surface of the coil add resistance to heat transfer so that it ceases to function efficiently. High supersaturation at a cooling surface is unavoidable and because of which the cooling surface gets rapidly fouled with adhering crystals so surface might therefore require frequent washing and scrapping. Other disadvantages include difficulty in controlling nucleation and size of crystals, and high labour costs.

This type of crystalliser is used to produce fine chemicals, pharmaceutical products and dye intermediates.

Scrapped surface crystalliser :

(i) Swenson-Walker crystalliser and (ii) Double pipe crystalliser.

Swenson-Walker Crystalliser :

The Swenson-Walker crystalliser is a cooling type, continuous, jacketed trough crystalliser. It is an example of the scrapped surface crystalliser and is probably the most widely used crystalliser.

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Construction :

It consists of a long open rectangular trough with a semi-cylindrical bottom, i.e., a U-shaped trough, of width 0.6 m and length 3 to 6 m. The trough is jacketed externally for circulating cooling water during operation. A spiral agitator rotating at about 7 rpm is incorporated in the trough in such a way that it is as close to the bottom of the trough as possible. The capacity of this crystalliser can be increased by use of a number of units in series (four troughs of 3 metre length each may be arranged). At one end of the crystalliser, an inlet for the hot solution is provided and at other end of the crystalliser an overflow gate for the crystals and mother liquor discharge are provided. The functions of the spiral agitator include : (i) to scrap the crystals for the cooling surface (i.e., to prevent accumulation of the crystals on the heat transfer surface), (ii) to lift and shower the crystals through the solution so that crystals will be held in suspension and thereby the crystals of uniform size can be obtained (as deposition occurs mainly by build-up on previously formed crystals) and (iii) to convey the crystals from one end of equipment to the other end.

The trough is formed out of metal sheets welded to an angle-iron frame at the top. A jacket is usually made of mild steel. The spiral agitator is supported at either end outside the trough. A stuffing box assembly is attached to cover on either end that prevents leakage of the liquor from the opening provided for a shaft. The shaft is driven by belt.

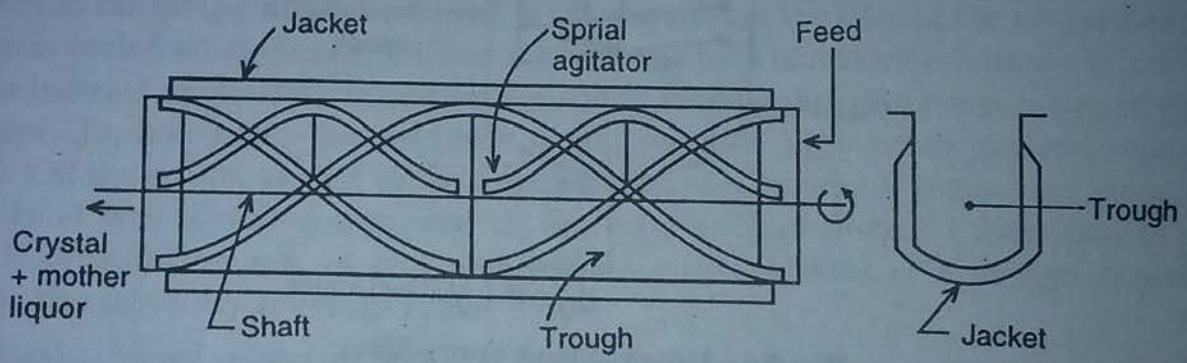

Fig. 5.4 : Swenson – Walker Crystalliser

Working :

The hot, concentrated solution is fed at one end of the open trough and flows slowly towards the other end of the trough. Water is fed to the jacket in such a way that it flows in a countercurrent fashion with respect to the solution flow. The solution while flowing through the trough of the crystalliser cools by heat transfer to water. Once the solution becomes supersaturated, crystals start forming and building. A spiral agitator keeps the crystals in suspension so that previously formed crystals grow instead of formation of new crystals and ultimately the two phase mixture of crystals and liquor leaves the crystalliser through an overflow gate.

Advantages of this crystalliser are : saving in floor space, in material in process and saving in labour.

This crystalliser is suitable only when supersaturation can be achieved by cooling alone.

Double pipe scrapped surface crystalliser :

Less common type of a continuous scrapped surface crystalliser that is somewhat similar to the jacketed trough crystalliser, i.e., Swenson-Walker crystalliser is a double pipe crystalliser. It consists of a concentric double pipe, the outer pipe acting as a jacket. Cooling water flows through the annular space between the two pipes and a long pitch spiral agitator rotates in the inner pipe at 5 to 30 rpm. These are also arranged in series (3 pipes

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each of 3 m long). Other constructional features and method of operation are same as in the Swenson-Walker crystalliser.

It is used in crystallising ice cream and plasticizing margarine.

Vacuum Crystalliser :

In this crystalliser, supersaturation is achieved by adiabatic evaporative cooling. A hot solution is introduced into a vessel wherein a vacuum is maintained that corresponds to the boiling point of the solution lower than the feed liquor temperature. Evaporation will result due to flashing of the hot solution. The energy needed for vaporisation is taken from the feed (sensible heat), so that the temperature of a liquor-vapour mixture after flashing becomes much lower than the temperature of the liquor before flashing.

Vacuum crystallisers often operated continuously but they can also be operated batch-wise. These crystallisers are very simple and contain no moving parts so they can be constructed out of corrosion resistant materials or lead or rubber-lined mild steel.

Construction :

A continuous vacuum crystalliser consists of a tall vertical cylindrical vessel with a conical bottom, a circulating pump (screw pump) of a low head and a vertical tubular heater on the shell side of which, steam is condensing. A low pressure (i.e., vacuum) is maintained in the vertical cylindrical vessel (crystallising body) by a condenser usually with the help of steam jet ejector, etc. A tangential inlet is provided on the cylindrical vessel for introducing a hot solution into it and a vapour outlet is provided on the top. A discharge connection for the mother liquor and crystal is provided on a down-pipe just above the feed connection.

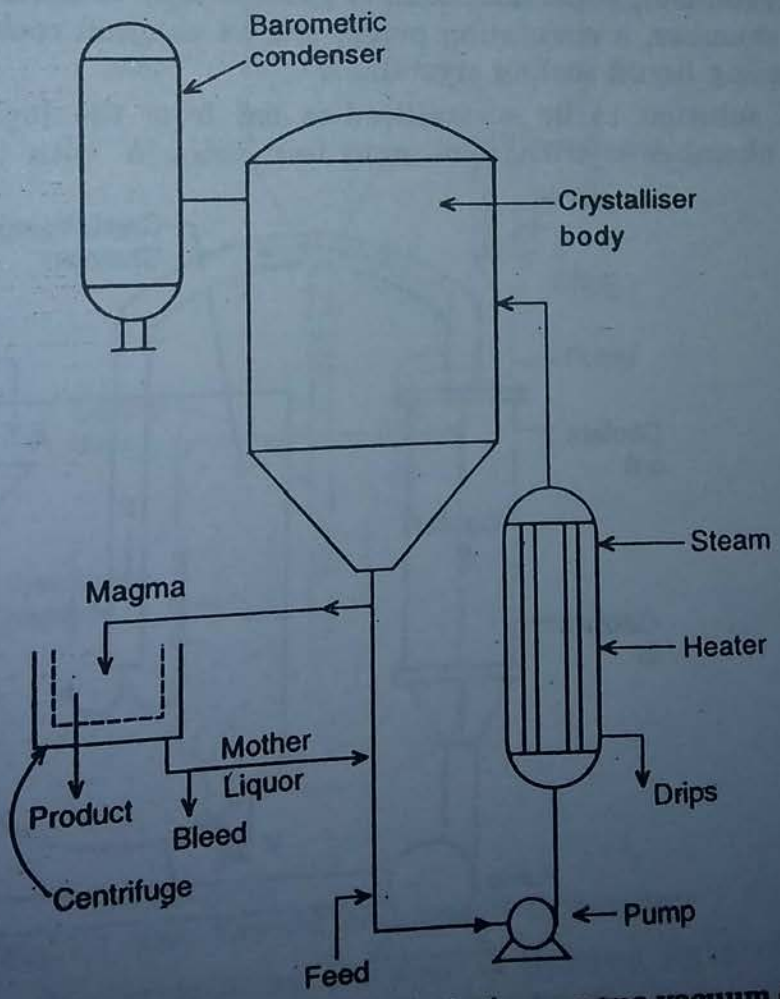

Fig. 5.5 : Continuous Vacuum Crystalliser / Circulating magma vacuum crystalliser

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Working :

The magma from the bottom of the cylindrical vessel goes to a pump via a down-pipe and is pumped through a vertical tubular heater where it is heated by means of condensing steam and finally the hot stream enters the cylindrical vessel tangentially just below the level of the magma surface. Flash evaporation of the solution takes place and produces rapid cooling, resulting into supersaturation, which is the driving force for nucleation and growth. Fresh solution enters the down-pipe just before suction of the circulating pump and a suspension of crystals is continuously taken out from a discharge pipe located above the feed inlet in the down-pipe. The suspension of crystals is fed to a centrifuge machine, crystals are taken out as product, and the mother liquor is recycled to the down-pipe with small part of it, continuously bled.

It is used for the production of large crystals.

Krystal or Oslo Crystalliser :

It is widely used whenever large quantities of crystals of controlled size are to be produced.

In this crystalliser, a supersaturated solution is passed upward through a bed of crystals which are maintained in a fluidised state whereby uniform temperature is maintained and the crystals segregate in the bed with large ones at the bottom and small ones at the top.

Oslo Cooling Crystalliser :

In this crystalliser, supersaturation is generated by indirect cooling. It consists of a crystallising chamber, a circulating pump, and an external cooler for cooling a solution. It is a circulating liquid cooling crystalliser.

The feed solution to be crystallised is fed from the top. Mother liquor from a crystallising chamber is withdrawn near feed point 'A' with the help of a circulating

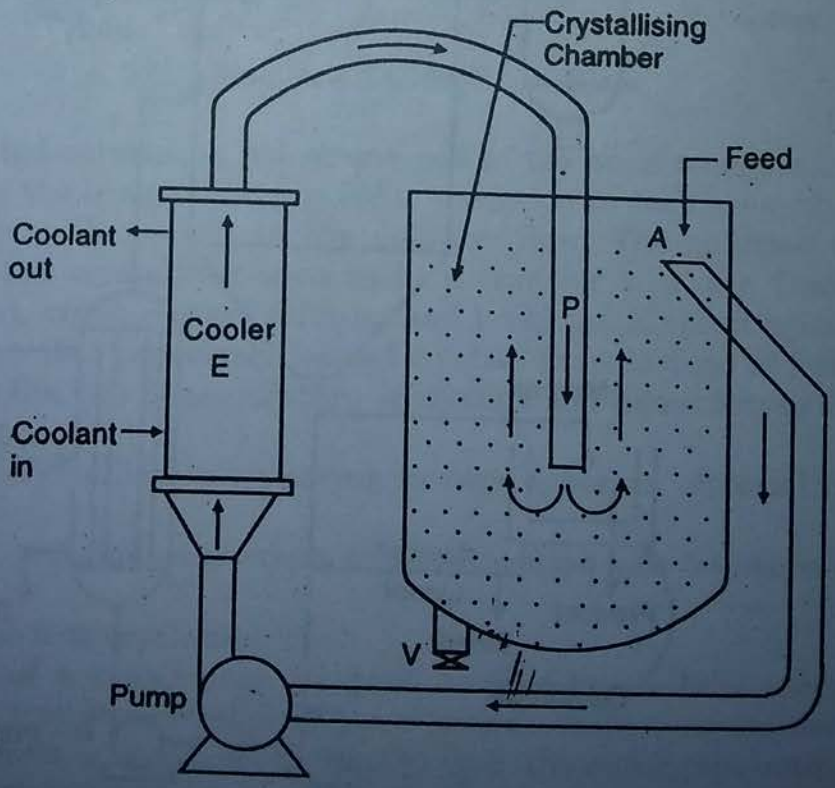

Fig. 5.6 : Oslo / Krystal Cooling Crystalliser

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pump, and it is then admitted to a cooler (E) wherein supersaturation is achieved by cooling. The supersaturated solution from the cooler is finally fed back to the bottom of the crystallising chamber through a central pipe (P). Usually, nucleation takes place in the bed of crystals in crystallising chamber. The nuclei formed circulate with mother liquor and once they grow sufficiently large, they will be retained in the fluidised bed. Once the crystals grow to the required size, they are removed as product from the bottom of the crystallising chamber through a valve 'V' as these crystals cannot be retained in the fluidised bed by the circulation velocity.

Oslo Evaporative Crystalliser :

In this crystalliser, supersaturation is generated/achieved by evaporation. It is a circulating liquid evaporative crystalliser.

It consists of a crystallising chamber containing bed of forming and growing crystals, a circulating pump, an external heater for heating the solution with the help of condensing steam and a vapour - head wherein reduced pressure is maintained by a vacuum generating equipment. The heater is maintained under sufficient hydrostatic head to avoid boiling on the heating surface.

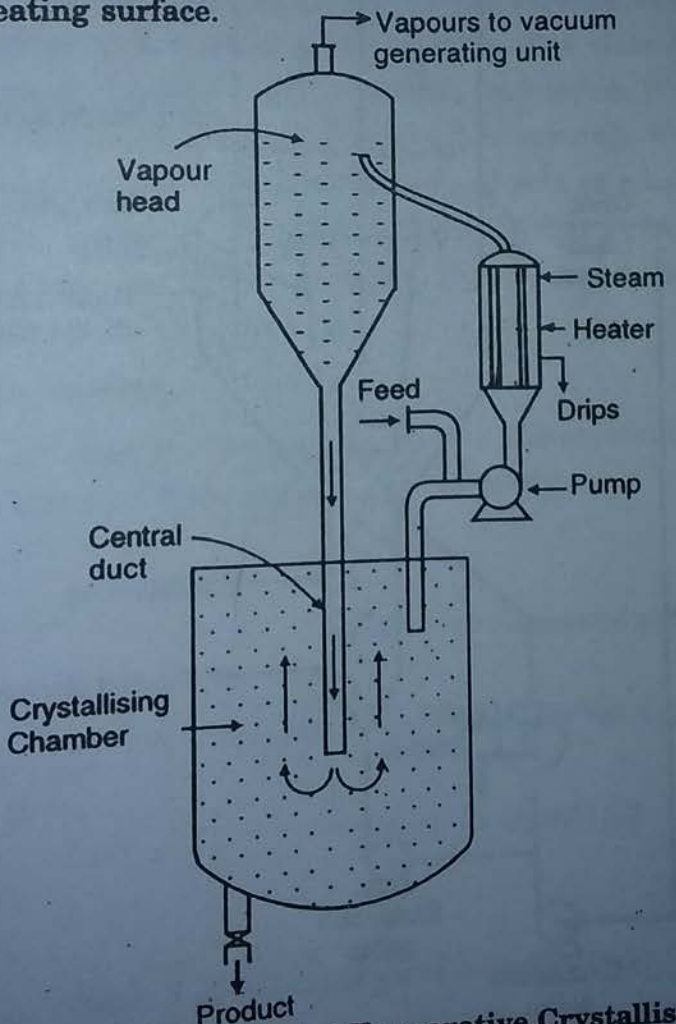

Fig. 5.7 : Oslo/Krystal Evaporative Crystalliser

The solution from the crystallising chamber is pumped by a circulating pump, on the suction side of which the feed solution forming a small part of total circulating liquid is introduced into a heater, where it is heated by means of condensing steam (on the

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shell side of the heater), and then fed to a vapour head wherein some of the solution flashes into vapour resulting into some degree of supersaturation. The supersaturated solution is then returned to the bottom of the crystallising chamber through a central duct prolonged from the vapour head into the crystallising chamber. Nucleation takes place in the crystal bed which is maintained in a fluidised state by means of a upward flowing stream of liquid through the duct. The nuclei formed circulate with the mother liquor and once they grow sufficiently large they will be retained in the fluidised bed. When the crystals grow to a required size, they will be withdrawn as product from the bottom of the crystallising chamber as they will not be maintained in the fluidised bed by the circulation velocity.

Draft tube-baffle crystalliser :

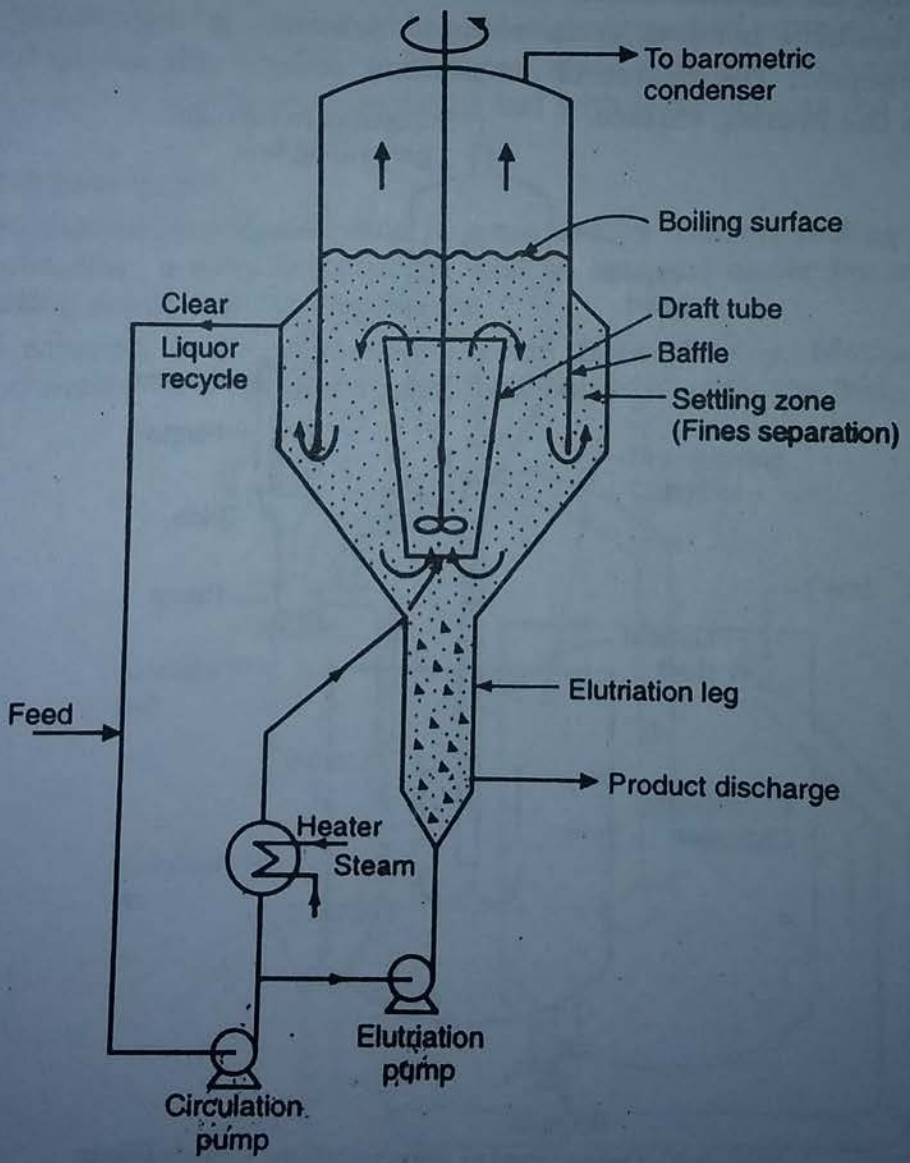

Fig. 5.8 : Draft tube-baffle crystalliser with elutriation leg and fines separation

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Draft tube-baffle (DTB) crystallisers are commonly used in the production of granular materials (e.g. ammonium sulphate, potassium chloride).

It consists of a crystalliser body, draft tube, propeller, circulating pump, heating element (heater), elutriation leg, circulation and elutriation pumps and a settling zone (formed). The annular space acting as a settling zone surrounding the crystalliser body is formed by enlarging the cone bottom and using the lower wall of the crystalliser body as a baffle. The heater supplies the heat to meet the evaporation requirements and to raise the temperature of solution removed from the settling zone so as to dissolve the tiny crystals withdrawn. The essentially clear liquid from the heater is rapidly mixed with the slurry circulating in the main body of the crystalliser. The suspension of product crystals is maintained by a slow moving propeller surrounded by a draft tube within the body. The propeller directs the slurry to the liquid surface in order to prevent the solids from short-circuiting the zone of most intense supersatuation. The cooled slurry is returned to the bottom of the vessel and recirculated through the propeller. At the propeller, the heated solution is mixed with the recirculating slurry. In the settling zone, coarse crystals are separated from the fines by gravitational sedimentation. The clear liquor is removed from the top of the settling zone and mixed with the fresh feed before it enters the heater. The elutriation leg below the crystalliser body classifies crystals by size. A part of the circulating liquor is pumped to the bottom of the leg and acts as a hydraulic sorting fluid to carry small crystals back to the crystallising zone for further growth. The product slurry is withdrawn from the lower part of the elutriation leg and sent to a centrifuge from where the mother liquor is returned to the crystalliser.

Material balances of crystalliser :

Material balances are used to calculate the yield of the crystallisation operation which is the mass of crystals obtained from a given mass of solution.

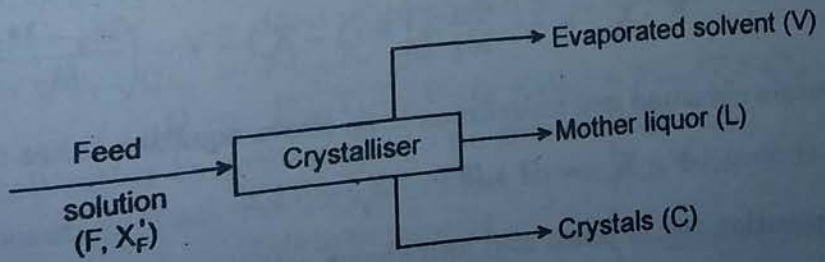

Fig. 5.9 : Block diagram of a crystalliser

Consider the crystallisation process carried out under steady state shown in Fig. 5.9.

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Let

F be the mass flow rate of feed solution in kg/h

X_F' be the weight fraction of anhydrous solute in the feed solution

C be the kg/h of crystals obtained

V be the mass flow rate of evaporated solvent in kg/h

X' be the solubility of the solute at the final temperature of the operation expressed as a weight ratio of anhydrous salt to solvent

M_1 be the molecular weight of anhydrous solute salt

M_2 be the molecular weight of hydrated salt

L' be the kg/h of solvent in the mother liquor

L be the kg/h of mother liquor leaving the crystalliser

Overall material balance :

$$F = C + L + V \quad \ldots (5.1)$$

Material balance of solvent :

Solvent in feed = Solvent evaporated + Solvent in mother liquor + Solvent in hydrated crystals

$$\therefore \quad F(1 - X_F') = V + L' + \left(\frac{M_2 - M_1}{M_2}\right)C \quad \ldots (5.2)$$

Rearranging the above equation, we get :

$$L' = F(1 - X_F') - V - \left(\frac{M_2 - M_1}{M_2}\right)C \quad \ldots (5.3)$$

Material balance of solute :

Solute in feed = Solute in product crystals + Solute in mother liquor

$$F \cdot X_F' = C \cdot \frac{M_1}{M_2} + \left[F(1 - X_F') - V - \left(\frac{M_2 - M_1}{M_2}\right)C\right]X' \quad \ldots (5.4)$$

If the crystals obtained are anhydrous then above equation takes the following form :

$$F \cdot X_F' = C + [F(1 - X_F') - V]\,X' \quad \ldots (5.5)$$

If no evaporation takes place and anhydrous salts are crystallised then :

$V = 0$ and equation (5.5) becomes :

$$F \cdot X_F' = C + [F(1 - X_F')]\,X' \quad \ldots (5.6)$$

C – kg/h of crystal obtained (anhydrous/hydrated) and is the yield of the process.

$$\% \text{ yield of crystals} = \frac{\text{Amount of solute crystallised}}{\text{Amount of solute in feed solution}} \times 100$$

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Energy balances :

Energy balance calculations are necessary in crystallisation process in order to determine the cooling requirements or to determine the final conditions. In these calculations, heat of crystallisation is important and it is the latent heat evolved when solid forms from a solution. Ordinarily, crystallisation process is exothermic and the heat of crystallisation varies with temperature and concentration. Heats of crystallisation are not available but heat of solution data are available in the literature. The process of crystallisation is the reverse of dissolution. Therefore, the heat of solution (i.e., the heat of dissolution) with a reverse sign is taken as heat of crystallisation, i.e., the heat of the crystallisation is equal to the negative heat of solution for a particular compound.

Heat of crystallisation = – (Heat of solution)

In case of cooling crystalliser :

(i) With no evaporation of solvent, the energy balance is

$$\begin{pmatrix}\text{Total heat to be removed}\\ \text{from the system}\end{pmatrix} = \begin{pmatrix}\text{Heat to be removed}\\ \text{from the feed solution}\\ \text{over the given cooling range}\end{pmatrix} + (\text{Heat of crystallisation})$$

$$Q = F\,C_{p_F}\,(T_1 - T_2) + C \cdot \lambda_c \qquad \ldots (5.7)$$

where F is the feed rate or feed in kg/h or kg.

C_{p_F} is the specific heat/heat capacity of feed solution, kJ/(kg.K)

$T_1 - T_2$ is the cooling range for solution, K $[T_1 > T_2]$

C is the kg or kg/h of crystals formed

λ_c is the heat of crystallisation, kJ/kg.

Heat of crystallisation = – (Heat of solution)

When specific heat data are available for the initial feed solution over a range of temperatures then the heat to be removed is equal to the heat to be removed to cool the feed from the initial temperature (T_1) to the final temperature (T_2) such that $T_1 > T_2$) without any crystal precipitating plus the heat liberated/evolved due to the formation of crystals from the supersaturated solution at the final temperature.

Heat with solution at T_1 + Heat liberated due to crystallisation =

= Heat with solution at T_2 + Heat to be removed

$$F\,C_p\,T_1 + C\lambda_c = F\,C_p\,T_2 + Q$$

$$\therefore \quad Q = FC_pT_1 - FC_pT_2 + C\lambda_C = FC_p\Delta T + C\lambda_c$$

(valid for crystallisation by cooling alone - no evaporation)

(ii) With evaporation of solvent during cooling, the energy balance is

$$\begin{pmatrix}\text{Net heat to be removed}\\ \text{from the system}\end{pmatrix} = \begin{pmatrix}\text{Heat to be removed}\\ \text{from the feed solution}\\ \text{over the given cooling range}\end{pmatrix} + (\text{Heat of crystallisation}) - (\text{Heat associated with the evaporated solvent})$$

$$Q = F\,C_{p_F}\,(T_1 - T_2) + C\,\lambda_c - \dot{m}\lambda \qquad \ldots (5.8)$$

where m is kg or kg/h of solvent evaporated.

λ is the latent heat of vaporisation of the solvent, kJ/kg.

In most of the cases, water is the solvent.

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Heat to be removed from a vessel for cooling the vessel initially at T_1 to T_2 may also be considered in the equation given above but it is small in comparison with the other terms $[M\ C_p\ (T_1 - T_2)]$, where M is the mass of vessel, C_p is the heat capacity of material of the vessel in kJ/(kg.K). This term is to be added (+) in the above equations.

We have,

Heat removed from the system = Heat gained by the coolant

$$Q = UA\ \Delta Tlm = \dot{m}_c\ C_{p_c}\ (t_2 - t_1) \qquad \ldots (5.9)$$

where U is the overall heat transfer coefficient, kJ/(m^2.K).

A is the heat transfer surface, m^2

ΔTlm is the log mean temperature difference, K

$\dot{m}_c$ is the coolant flow rate, kg/h

t_2 is the outlet temperature of coolant, K

t_1 is the inlet temperature of coolant, K

C_{p_c} is the heat capacity of coolant, kJ/(kg.K).

ΔTlm assuming countercurrent flow,

$$\Delta Tlm = \frac{(T_1 - t_2) - (T_2 - t_1)}{\ln\ [(T_1 - t_2)/(T_2 - t_1)]} \qquad \ldots (5.10)$$

So, using the above equations, one can calculate the coolant flow rate required if the coolant temperatures are provided or the heat transfer surface required when other parameters are provided.

SOLVED EXAMPLES

Ex. 5.1 : Calculate the yield of $MgSO_4 \cdot 7\ H_2O$ crystals when 1000 kg saturated solution of $MgSO_4$ at 353 K (80° C) is cooled to 303 K (30° C) assuming 10 % of the water is lost by evaporation during cooling.

Data : Solubility of $MgSO_4$ at 353 K (80° C) = 64.2 kg/100 kg water

Solubility of $MgSO_4$ at 303 K (30° C) = 40.8 kg/100 kg water

At. Wt. : Mg = 24, S = 32, H = 1 and O = 16

Solution : Basis : 1000 kg of solution at 353 K (80° C)

F = 1000 kg

X'_F = weight fraction of $MgSO_4$ in feed

$$= \frac{64.2}{164.2} = 0.391$$

Water in feed solution = $F\ (1 - X'_F)$

$= 1000\ (1 - 0.391) = 609$ kg

V = kg of water evaporated during cooling

$= 609 \times 0.1 = 60.9$ kg

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X' = solubility of $MgSO_4$ at 303 K = 40.8 kg/100 kg water

M_1 = Molecular weight of $MgSO_4$ = 120

M_2 = Molecular weight of $MgSO_4 \cdot 7\,H_2O$ = 246

C = Yield of $MgSO_4 \cdot 7\,H_2O$ crystals from feed solution

$M_2 - M_1$ = water of crystallisation

= 246 – 120 = 126 kg

Material balance of $MgSO_4$:

$$X_F' F = C \cdot \frac{M_1}{M_2} + \left[F(1 - X_F') - V - C\left(\frac{M_2 - M_1}{M_2}\right) \right] X'$$

$MgSO_2$ in feed = $MgSO_4$ in hydrated crystal + $MgSO_4$ in mother liquor at 303 K

$$\therefore \quad 0.391\,(1000) = C\left(\frac{120}{246}\right) + \left[1000\,(1 - 0.391) - 60.9 - \frac{C\,(126)}{246}\right] \frac{40.8}{100}$$

$$391 = 0.488\,C + [609 - 60.9 - 0.512\,C]\,(0.408)$$

$$C = 599.7 \text{ kg} \approx 600 \text{ kg}$$

Yield of $MgSO_4 \cdot 7\,H_2O$ crystals = 600 kg ... **Ans.**

Ex. 5.2 : A hot solution containing 5000 kg of Na_2CO_3 and water with a concentration of 25 % by weight Na_2CO_3 is cooled to 293 K (20° C) and crystals of $Na_2CO_3 \cdot 10\,H_2O$ are precipitated. At 293 K (20° C), the solubility is 21.5 kg anhydrous Na_2CO_3 per 100 kg of water. Calculate the yield of Na_2CO_3 crystals obtained if 5 % of the original water in the system evaporates on cooling.

At. Wt. : Na = 23, C = 12, O = 16, H = 1.

Solution : Basis : 5000 kg of hot solution of Na_2CO_3.

It contains :

Na_2CO_3 = $0.25 \times 5000 = 1250$ kg

Water = $0.75 \times 5000 = 3750$ kg

M_1 = Mol. Wt. of $Na_2CO_3 = 106$

M_2 = Mol. Wt. of $Na_2CO_3 \cdot 10\,H_2O = 286$

$M_2 - M_1$ = water of crystallisation = $10\,H_2O = 180$

F = 5000 kg

X_F' = weight fraction of Na_2CO_3 in feed

$= \frac{25}{100} = 0.25$

V = kg of water evaporated during cooling

$= 0.05 \times 3750 = 1875.5$ kg

Material balance of water :

Water in feed = Water of crystallisation + Water in mother liquor + Water evaporated

$$0.75\,(5000) = C\left(\frac{M_2 - M_1}{M_2}\right) + L' + 187.5$$

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$$3750 = C\left(\frac{286-106}{286}\right) + L' + 187.5$$

$$3562.5 = C\left(\frac{180}{286}\right) + L'$$

$$L' = 3562.5 - C\left(\frac{180}{286}\right)$$

Material balance of solute :

Solute in feed = Solute in crystals produced + Solute in mother liquor

$$X_F' F = C\frac{M_1}{M_2} + L' X'$$

C = yield of hydrated crystals

L' = kg of solvent water in mother liquor

X' = solubility of anhydrous solute as kg solute/kg solvent.

$$X_F' F = C\frac{M_1}{M_2} + \left[3562.5 - C\left(\frac{180}{286}\right)\right] \times X'$$

$$0.25 \times 5000 = C\frac{106}{286} + \left[3562.5 - C\left(\frac{180}{286}\right)\right]\frac{21.5}{100}$$

$$1250 = 0.37\,C + [3562.5 - 0.629\,C]\,0.215$$

$$1250 = 0.37\,C + 765.94 - 0.135\,C$$

$$0.235\,C = 484.06$$

$$C = 2059.83 \text{ kg}$$

∴ Yield of $Na_2CO_3 \cdot 10\,H_2O$ crystals = 2059.83 kg ... **Ans.**

Amount of mother liquor = kg of feed – kg of crystals – kg water evaporated

= 5000 – [2059.83 + 187.5]

= 2752.67 kg

Ex. 5.3 : A hot solution containing 2000 kg of $MgSO_4$ and water at 330 K (57° C) and with a concentration of 30 weight % $MgSO_4$ is cooled to 293 K (30° C) and $MgSO_4 \cdot 7H_2O$ crystals are removed. The solubility at 293 K (30° C) is 35.5 kg $MgSO_4$ per 100 kg water. Calculate the yield of crystals. Assume that no water is evaporated.

At. Wt. : Mg = 24, S = 32, O = 16, H = 1.

Solution : Basis : 2000 kg of hot solution (or simply 2000 kg hot solution)

$$F = 2000 \text{ kg}$$

$$X_F' = \frac{30}{100} = 0.30$$

M_1 = Mol. Wt. of $MgSO_4 = 120$

M_2 = Mol. Wt. of $MgSO_4 \cdot 7\,H_2O = 246$

Material balance of water :

Water in feed = Water of crystallisation + Water in mother liquor

$$(1 - X_F')\,F = C\left(\frac{M_2 - M_1}{M_2}\right) + L'$$

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$$(1-0.30)\,2000 = C\left(\frac{246-120}{246}\right) + L'$$

$$L' = 1400 - \frac{126}{246}\,C = 1400 - 0.5122\,C$$

where C is the yield of hydrated crystals.

Material balance of solute :

$MgSO_4$ in feed = $MgSO_4$ in crystals produced + $MgSO_4$ in mother liquor

$$X_F'\,F = C\frac{M_1}{M_2} + L'\,X'$$

where X' = solubility of $MgSO_4$ as kg $MgSO_4$/kg solvent

$$= \frac{35.5}{100} = 0.355$$

$M_1 = 120$

$M_2 = 246$

$X_F' = 0.355, \quad F = 2000$ kg

$$0.3\,(2000) = C\left(\frac{120}{246}\right) + [1400 - 0.5122\,C]\;0.355$$

$$600 = 0.488\,C + 497 - 0.182\,C$$

$$C = 336.6 \text{ kg}$$

Yield of $MgSO_4 \cdot 7\,H_2O$ crystals = 336.6 kg ... Ans.

Ex. 5.4 : A crystalliser is charged with 7500 kg of an aqueous solution at 377 K (104° C), 29.6 % by weight of which is anhydrous sodium sulphate. The solution is cooled. During the cooling operation, 5 % of the initial water is lost by evaporation. As a result, crystals of $Na_2SO_4 \cdot 10\,H_2O$ crystallise out. Calculate the yield of crystals and the quantity of mother liquor if the mother liquor is found to contain 18.3% by weight anhydrous Na_2SO_4.

At. Wt. : Na = 23, S = 32, O = 16, H = 1.

Solution : Basis : 7500 kg of aqueous solution

M_1 = Mol. Wt. of $Na_2SO_4 = 142$

M_2 = Mol. Wt. of $Na_2SO_4 \cdot 10\,H_2O = 322$

$F = 7500$ kg

Amount of $MgSO_4$ in feed $= X_F'\,F$

$$= \frac{29.6}{100} \times 7500 = 2220 \text{ kg}$$

Amount of water in feed $= (1 - X_F')\,F$

$$= (1 - 0.296)\,7500 = 5280 \text{ kg}$$

V = Amount of water lost by evaporation $= \frac{5}{100} \times 5280 = 264$ kg

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Material balance of water :

$$(1 - X_F') F = C\left(\frac{M_2 - M_1}{M_2}\right) + L' + V$$

$$5280 = C\left(\frac{322 - 142}{322}\right) + L' + 264$$

$$5016 = 0.56\,C + L'$$

$$L' = 5016 - 0.56\,C$$

where C is the yield of hydrated crystals.

Mother liquor contains 18.3 % by weight Na_2SO_4.

$$X' = \text{kg } Na_2SO_4/\text{kg water (solvent)} = \frac{0.183}{1 - 0.183} = 0.224$$

Material balance of Na_2SO_4 :

Na_2SO_4 in feed = Na_2SO_4 in hydrated crystals + Na_2SO_4 in mother liquor

$$X_F' F = C\frac{M_1}{M_2} + L' X'$$

$$2220 = \frac{142}{322}\,C + 0.224\,L'$$

$$2220 = 0.441\,C + 0.224\,L'$$

We have $L' = 5016 - 0.56\,C$

$$\therefore \quad 2220 = 0.441\,C + 0.224\,(5016 - 0.56\,C)$$

Solving, we get $C = 3474.5$ kg

Overall material balance :

Feed = Hydrated crystals + Mother liquor + Water evaporated

7500 = 3474.5 + Mother liquor + 264

$\therefore$ Amount of mother liquor = 3761.5 kg ... **Ans.**

Yield of $Na_2SO_4 \cdot 10\,H_2O$ crystals = 3474.5 kg ... **Ans.**

Ex. 5.5 : A saturated solution of $MgSO_4$ at 353 K (80° C) is cooled to 303 K (30° C) in a crystalliser. During cooling, 4 % solution is lost by evaporation of water. Estimate the quantity of the original saturated solution to be fed to the crystalliser per 1000 kg of $MgSO_4 \cdot 7\,H_2O$ crystals.

Data : Solubility of $MgSO_4$ at 303 K (30° C) = 40.8 kg/100 kg water

Solubility of $MgSO_4$ at 353 K (80° C) = 64.2 kg/100 kg water

At. Wt. : Mg = 24, S = 32, O = 16, and H = 1.0

Solution : Basis : 100 kg of feed solution

M_1 = Mol. Wt. of $MgSO_4$ = 120

M_2 = Mol. Wt. of $MgSO_4 \cdot 7\,H_2O$ = 246

$M_2 - M_1$ = Mol. Wt. of $7\,H_2O$ = 126

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Solubility of $MgSO_4$ at 353 K = 64.2 kg/100 kg water

Solubility of $MgSO_4$ at 303 K = 40.8 kg/100 kg water

$$MgSO_4 \text{ in feed solution} = \frac{64.2}{164.2} \times 100 = 39.1 \text{ kg}$$

Loss of solution by evaporation = $0.04 \times 100 = 4$ kg

The solute balance equation for hydrated salt is :

$$F \cdot X_F' = C \cdot \frac{M_1}{M_2} + \left[F(1 - X_F') - V - \left(\frac{M_2 - M_1}{M_2}\right)C\right] X'$$

where $F = 100$ kg

$X_F' = 39.1/100 = 0.391$

$V = 4$ kg

$X' = \frac{40.8}{100} = 0.408$

$$100 \times 0.391 = \frac{120}{246} C + \left[100(1 - 0.391) - 4 - \left(\frac{246 - 120}{246}\right)C\right] \times 0.408$$

$$39.1 = 0.4878\,C + [60.9 - 4 - 0.512\,C] \times 0.408$$

$$0.2788\,C = 15.89$$

$$C = 56.99 \text{ kg yield of crystals}$$

C is the kg of hydrated crystals obtained.

In order to obtain 56.99 kg of crystals, 100 kg of the feed solution is required.

∴ For a yield of 1000 kg crystals, the charge to the crystalliser (i.e., the quantity of the feed solution) required $= \frac{100}{56.99} \times 1000$

$= 1754.56$ kg ... Ans.

Ex. 5.6 : A batch of 500 kg of KCl is dissolved in water to make a saturated solution at 350 K (77 °C) where the solubility is 30% by weight KCl in water. The solution is cooled to 293 K (20 °C) at which the solubility is 25.4% by weight.

Determine the quantity of crystals obtained if 3.5% of the original water evaporates on cooling.

Also, determine the capacity of the vessel in which this solution is cooled if it is 1.20 times the volume of the solution. Data : the density of solution is 1200 kg/m³.

Solution : Basis : 500 kg of KCl in required amount of water at 350 K.

At 350 K, the solubility of KCl is 30%.

The weight fraction of KCl in feed solution = $X_F' = 30/100 = 0.30$

Amount of feed solution = $F = 500/0.30$

$= 1666.67$ kg ≈ 1667 kg

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$$\therefore \quad \text{Weight \% KCl in solution} = \frac{\text{Amount of KCl}}{\text{Amount of solution}} \times 100$$

$$\therefore \quad \text{Amount of feed solution} = F = \frac{\text{Amount of KCl} \times 100}{\text{Weight \% KCl}}$$

$$= \frac{500}{30} \times 100 = 1667 \text{ kg}$$

$$\text{Water in feed solution} = F(1 - X_F') = 1667(1 - 0.3)$$

$$= 1166.9 \text{ kg}$$

$$\approx 1167 \text{ kg}$$

3.5% of the original water is evaporated.

$$\therefore \quad \text{Amount of water evaporated during cooling} = V = \frac{3.5}{100} \times 1167$$

$$= 40.845 \text{ kg}$$

$$= 40.85 \text{ kg}$$

Solubility of KCl at 293 K = 25.4% by weight

i.e., 25.4 kg KCl in 100 kg solution.

i.e., 25.4 kg KCl per (100 – 25.4) kg solvent (water).

X' = solubility of KCl as kg KCl/kg solvent

$$= \frac{25.4}{100 - 25.4} = 0.3405$$

C is the yield of crystals obtained in kg

Material balance of KCl (solute) :

KCl in feed = KCl as crystals + KCl in mother liquor

$$F X_F' = C + [F(1 - X_F') - V] X'$$

$$C = F X_F' - [F(1 - X_F') - V] X'$$

$$= 1667(0.30) - [1167 - 40.85] \times 0.3405$$

$$= 116.55 \text{ kg}$$

Quantity of crystals obtained = 116.55 kg ... Ans.

$$\text{\% yield of crystals} = \frac{\text{KCl crystallised}}{\text{KCl in feed solution}} \times 100$$

$$= \frac{116.55}{500} \times 100$$

$$= 23.31 \quad \text{... Ans.}$$

Amount of feed solution = 1667 kg

Density of feed solution = 1200 kg/m^3

Volume of feed solution = 1667/1200 = m/ρ

$$= 1.389 \text{ m}^3 \approx 1.4 \text{ m}^3$$

Capacity of the vessel = 1.2 (Volume of feed solution)

$$= 1.2 \times 1.4$$

$$= 1.68 \text{ m}^3 = 1680 \, l \quad \text{... Ans.}$$

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OR Amount of KCl in feed solution = 500 kg

Solubility of KCl in water at 350 K = 30% by weight

(i.e., 100 kg feed contains 30 kg KCl)

∴ Amount of feed solution = 500/0.30

= 1667 kg

Water in feed solution = Feed solution – Amount of KCl in it

= 1667 – 500 = 1167 kg

Water evaporated on cooling = 3.5% of original water

Amount of water evaporated = 0.035 (1167)

= 40.85 kg

Material balance of water :

Water in mother liquor = Water in feed solution – Water evaporated

= 1167 – 40.85

= 1126.15 kg

Final solution (mother liquor) contains 1126.15 kg of water.

Mother liquor is at 293 K at which the solubility of KCl in water is 25.4% by weight.

Mother liquor contains 25.4% KCl and 100 – 25.4 = 74.6% water.

$$\text{Weight \% water in the mother liquor} = \frac{\text{Amount of water}}{\text{Amount of mother liquor}} \times 100$$

$$\therefore \quad \text{Amount of mother liquor} = \frac{\text{Amount of water}}{\text{Weight \% water in it}} \times 100 = \frac{1126.15}{74.6} \times 100$$

= 1509.6 kg

Mother liquor contains KCl and water. Therefore,

KCl in mother liquor = mother liquor – water in mother liquor

= 1509.6 – 1126.15

= 383.45 kg

Material balance of KCl :

KCl in feed solution = KCl crystallised + KCl in mother liquor

∴ KCl crystallised = 500 – 383.45

= 116.55 kg ... **Ans.**

(The rest of the calculations are the same as the previously done.)

Ex. 5.7 : A solution containing 500 kg of Na_2SO_4 and 2500 kg of water is cooled from 333 K (60 °C) to 283 K (10 °C) in an agitated mild steel vessel weighing 750 kg. At 283 K (10 °C), the solubility of anhydrous salt is 8.9 kg per 100 kg water. Na_2SO_4 crystallises as $Na_2SO_4 \cdot 10\ H_2O$. If 2% of the original water is lost by evaporation during cooling, calculate the heat that must be removed and yield of hydrated crystals in kg.

Data : heat capacity of the solution = 3.6 kJ/(kg.K)

heat capacity of the mild steel = 0.5 kJ/(kg.K)

The heat of solution = – 78.5 MJ/kmol

latent heat of vaporisation of water = 2395 kJ/kg

Hint : Heat of crystallisation = – (heat of solution)

Atomic weight – Na = 23, S = 32, H = 1, O = 16.

Solution : Basis : A solution containing 500 kg of Na_2SO_4 and 2500 kg of water at 333 K.

We have to calculate the net heat to be removed to reduce the temperature from 333 K to 283 K, so everything - solution and vessel will be at 283 K. The word net is used as some of the heat is lost by evaporation of water.

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Net heat to be removed = heat to be removed from solution + heat of crystallisation + heat to be removed from vessel – heat lost by evaporation

First perform the material balance calculations to find the quantities of various parameters.

Amount of Na_2SO_4 in feed solution = 500 kg.

Amount of water in feed solution = 2500 kg.

Total amount of feed solution = F = 500 + 2500 = 3000 kg.

Weight fraction of Na_2SO_4 in feed solution = X_F' = 500/3000 = 0.1667.

2% of original water is lost by evaporation.

∴ Amount of water lost by evaporation = V = 0.02 × 2500 = 50 kg.

Molecular weight of Na_2SO_4 = M_1 = 142.

Molecular weight of $Na_2SO_4 \cdot 10\ H_2O$ = M_2 = 322.

At 283 K, the solubility of anhydrous salt (Na_2SO_4) is 8.9 kg per 100 kg water.

X' is the solubility of solute expressed as kg solute per kg solvent.

$$X' = \frac{\text{kg } Na_2SO_4}{\text{kg water}} = 8.9/100 = 0.089$$

Material balance of Na_2SO_4 :

Na_2SO_4 in feed = Na_2SO_4 in hydrated crystal + Na_2SO_4 in mother liquor

$$X_F' \cdot F = C \cdot \frac{M_1}{M_2} + \left[F(1 - X_F') - V - \left(\frac{M_2 - M_1}{M_2}\right) C X'\right]$$

$$0.1667\,(3000) = \frac{142\,C}{322} + \left[3000\,(1 - 0.1667) - 50 - \frac{(322 - 142)\,C}{322}\right] \times 0.089$$

$$500 = 0.441\,C + [2500 - 50 - 0.56\,C] \times 0.089$$

$$500 = 0.441\,C + 218.05 - 0.04984\,C$$

$$C = 721 \text{ kg}$$

Yield of hydrated Na_2SO_4 crystals i.e., of $Na_2SO_4 \cdot 10\ H_2O$ crystals = 721 kg ... Ans.

Amount of water evaporated = 50 kg

Latent heat of vaporisation of water = λ = 2395 kJ/kg

Heat lost by evaporation = Vλ

= 50 × 2395 = 119750 kJ

Heat of crystallisation = – (Heat of solution)

= – (– 78.5)

= + 78.5 MJ/kmol

= 78500 kJ/kmol

Molecular weight of $Na_2SO_4 \cdot 10\ H_2O$ = 322 kg/kmol

$$\text{Heat of crystallisation} = \frac{78500 \text{ kJ/kmol}}{322 \text{ kg/kmol}} = \frac{78500}{322} = 243.8 \text{ kJ/kg}$$

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Heat of crystallisation for 721 kg yield of crystals (i.e., total heat of crystallisation)

$= 243.8 \times 721$

$= 175779.8 \text{ kJ} = 175780 \text{ kJ}$

Heat to be removed from the solution $= F C_p (T_2 - T_1)$

where $F = 3000$ kg, $C_p = 3.6$ kJ/(kg.K), $T_2 = 333$ K and $T_1 = 283$ K.

Heat to be removed from the solution $= 3000 \times 3.6 (333 - 283)$

$= 540000$ kJ

Mass of mild steel vessel $= 750$ kg

C_p of mild steel $= 0.5$ kJ/(kg.K)

Heat to be removed from the mild steel $= 750 \times 0.5 (333 - 283)$

$= 18750$ kJ

Net heat to be removed for cooling the solution from 333 K to 283 K (by energy balance) :

Net heat to be removed = Heat removed from the solution + Heat removed from the mild steel vessel + Heat of crystallisation − Heat lost by evaporation of water

$= 540000 + 18750 + 175780 - 119750$

$= 614780$ kJ ... Ans.

Simple procedure for calculating the various quantities involved :

OR : You can do the material balance calculations as follows :

Amount of Na_2SO_4 in feed solution = 500 kg.

Amount of water in feed solution = 2500 kg.

Water lost by evaporation during cooling = 2%

Water lost during cooling = 0.02 (2500) = 50 kg.

Molecular weight of $Na_2SO_4 = 142$

Molecular weight of $Na_2SO_4 \cdot 10 H_2O = 322$.

Molecular weight of $10 H_2O = 180$.

At 283 K, the solubility of Na_2SO_4 in H_2O (in mother liquor) is 8.9 kg per 100 kg H_2O.

Let x be the kg of hydrated crystals formed and y be the kg of mother liquor.

Overall material balance :

Feed = Mother liquor + hydrated crystals + water lost by evaporation

$500 + 2500 = y + x + 50$

$x + y = 2950$ ... (1)

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Material balance of Na_2SO_4 :

Na_2SO_4 in feed = Na_2SO_4 in mother liquor + Na_2SO_4 in hydrated crystals

$$500 = \left(\frac{8.9}{100+8.9}\right)y + \frac{142}{322}x$$

$$500 = 0.0817\,y + 0.441\,x \quad \ldots(2)$$

$$x + y = 2950 \qquad \therefore\ y = 2950 - x$$

Putting the value of y in terms of x in Equation (2),

$$500 = 0.0817\,(2950 - x) + 0.441\,x$$

$$\therefore \quad x = 720.8 \approx 721 \text{ kg}$$

Yield of hydrated crystals ($Na_2SO_4 \cdot 10\,H_2O$) = 721 kg ... Ans.

Ex. 5.8 : A 35% by weight aqueous $MgSO_4$ solution is originally present at 360 K (87 °C). If the solution is cooled (without evaporation) to 294 K (21 °C), what solid phase hydrate will form ? If the crystalliser is operated at 4540 kg/h of feed, how many kilograms of crystals will be produced per hour ? What will be the solid phase yield ? At 294 K (21 °C), the saturated aqueous solution containing 27 weight % $MgSO_4$ is in equilibrium with $MgSO_4 \cdot 7\,H_2O$ hydrated solid. Atomic weight : Mg = 24, S = 32, H = 1, O = 16.

Solution : Basis: 4540 kg/h of feed solution at 360 K,

$$F = 4540 \text{ kg/h.}$$

Weight % $MgSO_4$ in feed solution = 35.

Weight fraction of $MgSO_4$ in feed solution = X'_F = 35/100 = 0.35.

Let C be the kg/h of yield of hydrated crystals [$MgSO_4 \cdot 7\,H_2O$].

Molecular weight of $MgSO_4 = M_1 = 120$.

Molecular weight of $MgSO_4 \cdot 7\,H_2O = M_2 = 346$.

Let L' be the kg/h water in mother liquor.

Material balance of water :

Water in feed = Water of crystallisation – Water in mother liquor

$$(1 - X'_F)\,F = C\left(\frac{M_2 - M_1}{M_2}\right) + L'$$

$$(1 - 0.35)\,4540 = C \cdot \frac{(246 - 120)}{246} + L'$$

$$L' = 2951 - 0.5122\,C$$

At 294 K, the hydrated crystals are in equilibrium with the mother liquor and the solubility of $MgSO_4$ in water at 294 K is expressed as weight % $MgSO_4$ in the saturated solution.

Weight % $MgSO_4$ in mother liquor = 27, i.e., 100 kg mother liquor (saturated solution of $MgSO_4$) will contain 27 kg of $MgSO_4$ and 73 kg of water.

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Let X' be the solubility of solute expressed as kg solute per kg solvent, then

$$X' = \frac{\text{kg } MgSO_4}{\text{kg water}} = \frac{27}{73} = 0.3698$$

Material balance of solute :

$MgSO_4$ in feed = $MgSO_4$ in hydrated crystals + $MgSO_4$ in mother liquor

$$X'_F \cdot F = C \cdot \frac{M_1}{M_2} + L' X'$$

but $$L' = 2951 - 0.512\,C$$

$$\therefore \quad X'_F F = C \cdot \frac{M_1}{M_2} + [2951 - 0.5122\,C]\,X'$$

$$0.35 \times 4540 = \frac{120}{246} C + [2951 - 0.5122\,C] \times 0.3698$$

$$C = 1667.89 \text{ kg/h} \approx 1668 \text{ kg/h}$$

$MgSO_4 \cdot 7\,H_2O$ crystals produced = 1668 kg/h. ... Ans.

Check :

Overall material balance :

Feed = Mother liquor + Hydrated crystals

Mother liquor = Feed – Hydrated crystals

= 4540 – 1668 = 2872 kg/h

$MgSO_4$ in hydrated crystals = $\frac{120}{246} \times 1668 = 813.66$ kg/h

$MgSO_4$ in feed = 0.35×4540 = 1589 kg/h.

$MgSO_4$ in mother liquor = 1589 – 813.66 = 775.34 kg/h.

Given : Weight % $MgSO_4$ in saturated solution at 294 K = 27.

Weight % $MgSO_4$ in mother liquor = $\frac{775.34}{2872} \times 100 = 26.996 \approx 27$.

So the calculations performed are correct.

OR Without Notations :

Feed solution = 4540 kg/h

$MgSO_4$ in feed solution = 0.35 (4540) = 1589 kg/h

Water in feed solution = 4540 – 1589 = 2951 kg/h

Molecular weight of $MgSO_4$ = 120

Molecular weight of $MgSO_4 \cdot 7\,H_2O$ = 246

The final solution, i.e., the mother liquor at 294 K contains 27 % $MgSO_4$ by weight is in equilibrium with hydrated crystals.

Let x be the kg/h of $MgSO_4 \cdot 7H_2O$ crystals formed (rate of crystal formation) and y be the kg/h of mother liquor.

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Overall material balance :

$$\text{Feed} = \text{mother liquor} + \text{hydrated crystals}$$

$$4540 = y + x \quad \ldots (1)$$

$$y = 4540 - x$$

Material balance of $MgSO_4$:

$$MgSO_4 \text{ in feed} = MgSO_4 \text{ in mother liquor} + MgSO_4 \text{ in hydrated crystals}$$

$$1589 = 0.27\,y + \frac{120}{246}\,x$$

$$1589 = 0.27\,y + 0.4878\,x \quad \ldots (2)$$

Solving (1) and (2) simultaneously, we get

$$1589 = 0.27\,(4540 - x) + 0.4878\,x$$

$$1589 = 1225.8 - 0.27\,x + 0.4878\,x$$

$$x = 1667.6 \text{ kg/h}$$

Rate of $MgSO_4 \cdot 7H_2O$ crystals production = 1667.6 kg/h ... Ans.

Ex. 5.9 : A Swenson-Walker crystalliser is used to produce hydrated crystals of Na_2SO_4 (as $Na_2SO_4 \cdot 10\,H_2O$) by cooling a solution from 300 K (27 °C) to 290 K (17 °C) with the help of cooling water which enters at 280 K (7 °C) and leaves at 290 K (17 °C). Assuming evaporation to be negligible, determine the number of sections of the crystalliser, each 3 m long, required to process 0.25 kg/s of product.

Data : Solubility of Na_2SO_4 at 300 K = 40 kg/100 kg water

Solubility of Na_2SO_4 at 290 K = 14 kg/100 kg water.

Mean heat capacity of liquor = 3.8 kJ/(kg.K)

Heat of crystallisation = 230 kJ/kg.

Available area for heat transfer in the crystalliser = $3m^2$ per 1 m length of crystalliser.

Overall heat transfer coefficient = 0.15 kW/(m^2.K)

Atomic weight : Na = 23, S = 32, H = 1 and O = 16.

Solution : Let us solve this problem without using notation F, V, etc.

Basis : 0.25 kg/s of production rate of hydrated ($Na_2SO_4 \cdot 10\,H_2O$) crystals.

Molecular weight of $Na_2SO_4 = 2\,(23) + 1\,(32) + 4\,(16) = 142$.

Molecular weight of $Na_2SO_4 \cdot 10\,H_2O = 322$

1 kmol $Na_2SO_4 \cdot 10\,H_2O \equiv$ 1 kmol Na_2SO_4

322 kg $Na_2SO_4 \cdot 10\,H_2O \equiv$ 142 kg Na_2SO_4

$\therefore$ In 322 kg of $Na_2SO_4 \cdot 10\,H_2O$ there are 142 kg of Na_2SO_4 and 180 kg of H_2O.

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At 300 K the solubility of Na_2SO_4 is 40 kg/100 kg water, i.e., 140 kg of the solution contains 40 kg of Na_2SO_4 and 100 kg water. Feed is at 300 K, so in 140 kg of the feed (saturated solution), there are 40 kg of Na_2SO_4.

At 290 K, the solubility of Na_2SO_4 is 14 kg/100 kg water, i.e., 114 kg saturated solution (mother liquor) contains 14 kg of Na_2SO_4 and 100 kg water.

Let x be the kg/s of feed to the crystalliser and y be the kg/s of mother liquor leaving it.

Overall material balance :

$$\text{Feed} = \text{Mother liquor} + \text{Hydrated crystals}$$

(Assuming evaporation to be negligible).

$$x = y + 0.25$$

$$\therefore \quad y = x - 0.25$$

Material balance of Na_2SO_4 :

$$Na_2SO_4 \text{ in feed} = Na_2SO_4 \text{ in mother liquor} + Na_2SO_4 \text{ in hydrated crystals}$$

$$\frac{40}{140} x = \frac{14}{114} y + \frac{142}{322} (0.25)$$

$$0.2857\, x = 0.1228\, y + 0.11025$$

We have, $y = x - 0.25$

$$0.2857\, x = 0.1228\,(x - 0.25) + 0.11025$$

$$x = 0.488 \text{ kg/s}$$

Feed rate of Na_2SO_4 solution to the crystalliser (mass flow rate of the feed solution) $= \dot{m} = 0.488$ kg/s.

C_p of solution = 3.8 kJ/(kg.K)

T_2 = 300 K and T_1 = 290 K

Heat of crystallisation = 230 kJ/kg (given)

Rate of crystal production = 0.25 kg/s

Q_1 = Heat of crystallisation for required production = 0.25×230 = 57.5 kJ/s

Q_2 = Heat to be removed from the solution = $\dot{m}\, C_p\, (T_2 - T_1)$

$= 0.488 \times 3.8\,(300 - 290)$

$= 18.544$ kJ/s

Total heat to be removed = $Q = Q_1 + Q_2$

$= 57.5 + 18.544$

$= 76.044$ kJ/s

$= 76.044$ kW (1 J/s = 1 W)

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Assuming countercurrent flow :

$$300\,K \xrightarrow{\text{Solution}} 290\,K$$

$$290\,K \xleftarrow{\text{Water}} 280\,K$$

$\Delta T_1 = 300 - 290 = 10\,K$, $\Delta T_2 = 290 - 280 = 10\,K$

$\therefore \quad \Delta T_{lm} = \Delta T_1 = 10\,K$

$Q = UA\,\Delta T_{lm}$

where $Q = 76.044\,kW$, $\Delta T_{lm} = 10\,K$

$U = 0.15\ kW/(m^2.K)$

$\therefore \quad A = Q/U \cdot \Delta T_{lm}$

$= 76.044/(0.15 \times 10)$

$= 50.696 \approx 50.7\,m^2$

Heat transfer area required $= 50.7\,m^2$

Given : Heat transfer area = $3m^2$ per 1 m length of crystalliser.

Length of the crystalliser required to have heat transfer surface of 50.7 $m^2 = 50.7 \times \frac{1}{3}$ = 16.9 m

Length of the crystalliser = 16.9 m.

Length of each section of the crystalliser = 3 m

Number of sections of the crystalliser $= 16.9 \times \frac{1}{3} = 5.633.$

Number of sections should not be in fraction.

$\therefore$ Number of sections required = 6 ... **Ans.**

6 sections - each of 3 m length are required for 0.25 kg/s of production rate of hydrated crystals ($Na_2SO_4 \cdot 10\ H_2O$).

Ex. 5.10 : A hot solution containing 2000 kg of $MgSO_4$ and water at 330 K and with a concentration of 30% by weight $MgSO_4$ is cooled to 293 K and $MgSO_4 \cdot 7\ H_2O$ crystals are formed. The solubility at 293 K is 35.5 kg $MgSO_4$/100 kg water. The average heat capacity of the feed solution is 2.93 kJ/(kg.K). The heat of solution at 291 K is -13.31×10^3 kJ/kmol $MgSO_4 \cdot 7\ H_2O$. Calculate the yield of crystals and make the heat balance. Assume no water is vaporised. At. Wt. Data : Mg = 24, S = 32, O = 16, H = 1.0.

Solution : Let us solve this problem without using notation such as F, V, C, etc.

Basis : 2000 kg of hot solution at 330 K.

Amount of $MgSO_4$ in hot solution $= 0.30 \times 2000 = 600$ kg.

Amount of water in feed solution = 2000 – 600 = 1400 kg.

M_1 = Molecular weight of $MgSO_4 = 120$

M_2 = Molecular weight of $MgSO_4 \cdot 7\ H_2O = 246$

Let x be the kg of hydrated crystals formed and y be the kg of mother liquor obtained.

Overall material balance :

Feed = mother liquor + hydrated crystals

$2000 = y + x$

$x + y = 2000$ ... (1)

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Material balance of $MgSO_4$:

$MgSO_4$ in feed = $MgSO_4$ in mother liquor + $MgSO_4$ in hydrated crystals

At 293 K the solubility of $MgSO_4$ is 35.5 kg per 100 kg water, i.e., 135.5 kg mother liquor contains 35.5 kg $MgSO_4$ and 100 kg water.

1 kmol $MgSO_4$ ≡ 1 kmol $MgSO_4 \cdot 7\,H_2O$

120 kg $MgSO_4$ ≡ 246 kg $MgSO_4 \cdot 7\,H_2O$

246 kg $MgSO_4 \cdot 7\,H_2O$ contains 120 kg $MgSO_4$ and 126 kg water.

∴ Material balance of $MgSO_4$:

$MgSO_4$ in hot solution = $MgSO_4$ in mother liquor + $MgSO_4$ in hydrated crystals

$$600 = \frac{35.5}{135.5}\,y + \frac{120}{246}\,x$$

$$600 = 0.262\,y + 0.49\,x \quad \ldots (2)$$

We have, $y = 2000 - x$

$$600 = 0.262\,(2000 - x) + 0.488\,x$$

$$x = 336.3 \text{ kg}$$

and $y = 2000 - 336.3 = 1663.7$ kg

Yield of hydrated $MgSO_4$ crystals (i.e. $MgSO_4 \cdot 7\,H_2O$ crystals) = C = 336.3 kg ... Ans.

Total heat to be removed from the system = Q

$$Q = F\,C_{PF}\,(T_1 - T_2) + C\,\lambda_C$$

... for cooling crystalliser with no evaporation

$Q = Q_1 + Q_2$

Q_1 = Heat to be removed from the solution

$Q_1 = F\,C_{PF}\,(T_1 - T_2)$

where F = 2000 kg, C_{PF} = 2.93 kJ/(kg.K)

T_1 = 330 K, T_2 = 293 K

$Q_1 = 2000 \times 2.93\,(330 - 293)$

= 216820 kJ

Q_2 = Total heat of crystallisation (for 336.3 kg crystals)

λ_C = Heat of crystallisation = – (Heat of solution)

$\lambda_C = -(-13.31 \times 10^3)$ kJ/kmol = 13.31×10^3 kJ/kmol

Molecular weight of $MgSO_4 \cdot 7\,H_2O$ = 246 kg/kmol

∴ $\lambda_C = 13.31 \times 10^3/246 = 54.10$ kJ/kg

$Q_2 = C\,\lambda_C$

$= 336.3 \times 54.1 = 18194$ kJ

$Q = Q_1 + Q_2 = 216820 + 18194 = 235014$ kJ ... Ans.

Total heat to be removed from the system (crystalliser) = 235014 kJ

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Ex. 5.11 : An adiabatic vacuum crystalliser is to be used to produce 10 tonnes per hour of $FeSO_4 \cdot 7H_2O$ crystals. The feed is a solution containing 38.9 parts $FeSO_4$ per 100 parts water and it enters the crystalliser at 343 K (70 °C). The crystalliser vacuum is such that it produces a crystallisation temperature of 299.7 K (26.7 °C). Find the feed rate to the crystalliser. The saturated solution at 299.7 K (26.7 °C) contains 30.2 parts $FeSO_4$ per 100 parts water. The following data are available.

Enthalpy of mother liquor = – 5.57 kJ/kg

Enthalpy of crystals leaving the crystalliser = – 211 kJ/kg

Enthalpy of feed solution = 108.9 kJ/kg

Enthalpy of vapour leaving the crystalliser = 2546 kJ/kg

Mol. wt. $FeSO_4$ = 152 and $FeSO_4 \cdot 7H_2O$ = 278.

Solution : Basis : 1000 kg/h of $FeSO_4 \cdot 7H_2O$ crystals produced.

Let, F be the feed solution per hour.

V be the water evaporated per hour.

C be the crystals obtained per hour.

L be the mother liquor obtained per hour.

Let X_F' be the weight fraction of $FeSO_4$ in feed solution, X_C' be the weight fraction of $FeSO_4$ in hydrated crystals and X be the weight fraction of $FeSO_4$ in mother liquor.

Let H_F be the enthalpy of feed solution, H_V be the enthalpy of vapour leaving, H_C be the enthalpy of crystals leaving the crystalliser and H_L be the enthalpy of mother liquor leaving.

Overall material balance :

Feed = mother liquor + water evaporated + hydrated crystals

$F = L + V + C$, but $C = 10000$ kg/h

$$F = L + V + 10000 \quad \ldots (1)$$

Material balance of $FeSO_4$:

$FeSO_4$ in feed = $FeSO_4$ in mother liquor + $FeSO_4$ in water evaporated + $FeSO_4$ in hydrated crystals.

$$X_F' F = XL + 0 + X_C' C$$

Feed solution contains 38.9 parts $FeSO_4$ per 100 parts water.

$$\therefore \quad X_F' = \frac{38.9}{38.9 + 100} = 0.28$$

1 kmol $FeSO_4 \cdot 7H_2O$ ≡ 1 kmol $FeSO_4$

278 kg $FeSO_4 \cdot 7H_2O$ ≡ 152 kg $FeSO_4$

i.e., 278 kg $FeSO_4 \cdot 7H_2O$ contains 152 kg $FeSO_4$ and 126 kg water.

$$\therefore \quad X_C' = \frac{152}{278} = 0.547$$

The saturated solution, i.e., mother liquor contains 30.2 parts $FeSO_4$ in 100 parts water.

$$\therefore \quad X = \frac{30.2}{30.2 + 100} = 0.232$$

The material balance of $FeSO_4$ becomes :

$$\therefore \quad 0.28 F = 0.232 L + 0.547 C \quad \ldots (2)$$

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For an adiabatic operation, no heat is transferred to or from the system, therefore, the energy balance is :

Heat with feed = Heat with vapour + Heat with mother liquor + Heat with crystals

$$F H_F = V H_V + L H_L + C H_C$$

$H_F = 108.9$ kJ/kg, $H_L = -5.57$ kJ/kg

$H_V = 2546$ kJ/kg, $H_C = -211$ kJ/kg

$$108.9\,F = 2546\,V - 5.57\,L - 211\,C$$

$$C = 10000 \text{ kg/h}$$

$$108.9\,F = 2546\,V - 5.57\,L - 211 \times 10000$$

$$108.9\,F = 2546\,V - 5.57\,L - 2110000 \quad \ldots (3)$$

Equation (2) is,

$$0.28\,F = 0.232\,L + 0.547\,C$$

$$C = 10000 \text{ kg/h}$$

$$\therefore \quad 0.28\,F = 0.232\,L + 0.547 \times 10000$$

$$0.28\,F = 0.232\,L + 5470$$

$$\therefore \quad L = \frac{0.28\,F - 5470}{0.232}$$

$$L = 1.207\,F - 23577.6 \quad \ldots (4)$$

From Equation (1), we have :

$$V = F - L - 10000$$

Putting the value of L from Equation (4) in the above equation gives

$$V = F - (1.207\,F - 23577.6) - 10000$$

$$\therefore \quad V = -0.207\,F + 13577.6 = 13577.6 - 0.207\,F \quad \ldots (5)$$

Put the values of L from Equation (4) and of V from Equation (5) in Equation (3).

$$108.9\,F = 2546\,(13577.6 - 0.207\,F) - 5.57\,(1.207\,F - 23577.6) - 2110000$$

$$108.9\,F = 34568569.6 - 527\,F - 6.723\,F + 131327.2 - 2110000$$

$$642.623\,F = 32589896.8$$

$$\therefore \quad F = 50714 \text{ kg/h}$$

Feed rate to the crystalliser = 50714 kg/h ... **Ans.**

Check :

Vapour rate from the crystalliser = $V = 13577.6 - 0.207\,F$

$= 13577.6 - 0.207\,(50714)$

$= 3079.8 \approx 3080$ kg/h

Mother liquor rate = $L = 1.207\,F - 23577.6$

$= 1.207\,(50714) - 23577.6 = 37634$ kg/h.

Overall material balance :

Feed = mother liquor + vapour + crystals

$50714 = 37634 + 3080 + 10000$

$50714 = 50714$

LHS = RHS – the calulations done are correct.

As

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DRYING

- Drying refers to the removal of moisture of a substance by thermal means (i.e., with the help of thermal energy).
- Drying refers to an operation in which the moisture of a substance (usually a solid) is removed by thermal means.
- Drying usually refers to the removal of relatively small amounts of water from a solid or nearly solid material. It involves the transfer of liquid from a wet solid into an unsaturated gas phase (drying medium).
- In drying, a gas phase contents a solid phase and mass is transferred from the solid phase to the gas phase.

It is an operation in which a small amount of water is removed from a wet solid with the help of heat. In majority of drying operations, heat is provided by a hot air or any other gas.

During drying operation, mass and heat transfer occur simultaneously. Heat is transferred from the bulk of the gas phase (drying medium) to the solid phase and mass is transferred from the solid phase to the gas phase in the form of liquid and vapour through various resistances. The material (liquid) that is transferred is called solute and transfer takes place as the gas phase is always unsaturated with respect to the solute material.

In case of drying, relatively small amounts of water or other liquid is removed from a solid or semi-solid material, whereas in case of evaporation, relatively large amounts of water are removed from solutions. Drying involves the removal of water at a temperature below the boiling point, while evaporation involves the removal of water as a vapour at its boiling point. Drying involves the circulation of hot air or other gas over a material for the removal of water while in evaporation use of steam heat is done for removal of water (i.e., for evaporator).

To obtain products almost in a dried form is the purpose of drying operation whereas concentration of a solution is the main purpose of evaporation. In drying operation, the emphasis is usually on the solid product and in evaporation, the emphasis is on concentrated liquor.

As the removal of moisture by thermal means is more costly than mechanical means (e.g., filtration), the liquid content of solids must be reduced to the minimum possible level by the latter means before the material is fed to drying equipments.

In most of the drying operations, the heat, required to evaporate where, is provided by hot air or any other gas-drying medium.

Drying is frequently the last operation in a manufacturing process and is usually carried after evaporation, filtration, or crystallisation.

Drying operations are mostly encountered in food, chemical, agricultural, pharmaceutical and textile industries.

Drying operation is carried out for a number of reasons given below :

(i) For reducing the transport cost.

(ii) For purifying a crystalline product so that the solvent adhering to the crystals is removed.

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(iii) For making a material more suitable for handling and storage. Handling and storage of dry solids is easy.

(iv) To meet the market specifications of solid products set by the customers.

(v) For providing definite properties to materials.

(vi) In some cases for preventing corrosion arising due to the presence of moisture. Dry chlorine gas is not corrosive but traces of moisture make it very corrosive.

(vii) Sometimes it is an essential part of the process e.g. drying of paper in paper making.

General Definitions :

Moisture content of a solid is usually expressed as moisture amount per unit weight of the dry or wet solid.

Moisture content, wet basis :

It is expressed as the *ratio of the weight of the moisture to the weight of the wet feed material.* If X is the kg moisture associated with one kg of dry solids, then

Moisture content on the wet basis : $\frac{X}{1+X}$

The percent moisture on the wet basis is the moisture associated with a feed material expressed as a percentage of the weight of the feed material (i.e., the wet solids).

$$\text{Weight \% moisture content (on the wet basis)} = \frac{\text{kg moisture}}{\text{kg wet solid}} \times 100 = 100\left[\frac{X}{1+X}\right]$$

Moisture content, dry basis :

It is expressed as the *ratio of the weight of the moisture to the weight of the dry solids present in a wet feed material.* If the feed material contains X kg moisture and 1 kg of dry solids, then

$$\text{Moisture content (on the dry basis)} = \frac{\text{kg moisture}}{\text{kg dry solid}} = \frac{X}{1} = X$$

The percentage moisture content on the dry basis = 100 X

Dry-weight basis : It expresses the moisture content of wet solid as kilograms of water per kilogram of bone-dry solid.

Percent dry-weight basis expresses the moisture in a material as a percentage of the weight of the dry solid.

Wet-weight basis : It expresses the moisture content of wet solid as kilograms of water per kilogram of wet solid.

Percent wet-weight basis expresses the moisture as a percentage of the weight of the wet solid.

Use of a dry-weight basis or simply a dry basis is recommended because the percentage of moisture is constant for all moisture levels.

Equilibrium moisture content (X^*) :

The moisture content of a substance that is in thermodynamic equilibrium with its vapour (given partial pressure of vapour) in the gas phase under specified humidity and temperature of the hot gas or air is termed as equilibrium moisture content. It represents the limiting moisture content to which a given material can be dried under constant drying conditions.

Bound moisture content :

It is that *moisture in a substance which exerts a vapour pressure less than that of the pure liquid at the same temperature.*

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Unbound moisture content :

It is the *moisture held by a substance in excess of the equilibrium moisture content corresponding to saturation in the surrounding atmosphere.* It is primarily held in the voids of solid.

It is that *moisture in a substance that exerts an equilibrium vapour pressure equal to that of the pure liquid at the given temperature.*

Free moisture content :

It is the *moisture in a substance which is in excess of the equilibrium moisture content* $(X - X^*)$. At a given temperature and humidity, it is the moisture of a material that can be removed by drying. It may include bound and unbound moisture.

Critical moisture content :

It is the moisture content of a material at which the constant rate period ends and the falling rate period starts. It is a function of constant drying rate, material properties and particle size.

Constant rate period :

It is that part of a *drying process during which the rate of drying, expressed as the moisture evaporated per unit time per unit area of drying surface, remains constant.* (It is a drying period during which the rate of drying is constant.)

Falling rate period :

It is that part of a *drying process during which the rate of drying varies with time and the instantaneous drying rate expressed as the amount of moisture evaporated per unit time per unit area of drying surface continuously decreases.* (It is a drying period during which the instantaneous rate of drying continually decreases.)

Properties of air-water system :

The moisture removed (from a wet solid) during drying operation gets added in the hot gas or air which in turn depends upon the temperature and humidity of the gas or air. Usually in drying operation, the hot air is used as a drying medium, so it is essential to know some of the properties of the air-water system.

Relative humidity (R.H.) :

It is defined as the *ratio of the partial pressure of water vapour in the air-water vapour mixture to the vapour pressure of pure water at the temperature of the mixture (at DB).*

$$\% \text{ R.H.} = (p_A / p_A^0) \times 100$$

where p_A = partial pressure of water-vapour in the mixture

p_A^0 = vapour pressure of pure water

when $p_A = p_A^0$ then air is said to be saturated with water vapour.

Humidity (H) :

It is the *ratio of the mass of water vapour to the mass of dry air present in the air-water vapour mixture under any given set of conditions.*

$$H = \frac{\text{kg of water vapour}}{\text{kg of dry air}}$$

Dry bulb temperature (DB) :

The temperature of the vapour-gas mixture recorded by a thermometer (immersed in the mixture) whose bulb is kept dry, is called as dry bulb temperature.

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Wet bulb temperature (WB) :

The temperature recorded by a thermometer whose bulb is kept wet by wrapping a wet cloth in the open air or *air-water vapour mixture* is called as the wet bulb temperature.

The temperature of the bulb goes down as the latent heat of vaporisation required for the natural evaporation of water from the cloth will be supplied from the bulb. The evaporation is continued until the air surrounding the bulb becomes saturated. Some of the heat will flow from the surrounding air to the bulb by the temperature difference, even then the temperature of the bulb will not rise as that heat is consumed in the evaporation of water. At one particular point, the temperature becomes constant and is recorded as the wet bulb temperature.

When the air is more unsaturated, then the difference between dry bulb temperature and wet bulb temperature is more and is less for a more humid air. The relative humidity of the air is found out from a psychrometric chart knowing wet bulb and dry bulb temperature.

Saturation humidity :

It is the *humidity of air when saturated with water vapour* and is denoted by the symbol H_s.

$$H_s = \frac{18}{29}(p_A^\circ/P - p_A^\circ)$$

Percentage humidity (H_0) :

It is the *ratio of the actual absolute humidity of air (H) to the humidity of saturated air (H_s).*

$$\text{Percentage humidity, } H_0 = p_A(P - p_A^\circ)/p_A^\circ(P - p_A)$$

Dew point (DP) :

When the air-water vapour mixture is cooled at constant pressure, it becomes saturated and further cooling results in condensation of water vapour. The *temperature at which the condensation will first occur* is known as dew point. For saturated air (i.e., at 100% saturation), dew point, wet bulb and dry bulb temperatures are identical. Dew point is always lower or equal to the dry bulb temperature.

Humid heat : It is the amount of heat required to raise a unit mass of dry air and the associated moisture through 1 degree K at constant pressure (i.e. it is the heat capacity of 1 kg of dry air and the moisture contained in it.)

$$C_s = 1.005 + 1.88\ H \text{ kJ/(kg dry air.K)}$$

Humid volume : It is the volume of a mixture of dry air and water vapour per kg of dry air.

$$V_H = RT/p_A M_A \quad \text{and} \quad V_{Hs} = RT/p_A^\circ M_A$$

where R = 8.31451 kJ/(kmol.K).

Molal humidity is expressed as kmol water vapour/kmol dry air. It is the ratio of partial pressure of water to the partial pressure of air.

Equilibrium :

The moisture of wet solids exerts a definite vapour pressure depending upon the temperature and the nature of solid and the moisture. Consider that the wet solids containing water which exerts a vapour pressure of p_A° are exposed to a continuous supply of a fresh gas (usually air) with a fixed partial pressure of the vapour (p_A). If p_A° is greater than p_A, then the

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solids will lose moisture (reverse is true for $p_A^\circ < p_A$) by evaporation till the vapour pressure of moisture of the solids equals the partial pressure of the vapour in the gas. The solid and gas are then said to be in equilibrium with each other and the corresponding moisture content is referred to as equilibrium moisture content. The equilibrium data in case of drying operations are given as the *relationship between the moisture content of a solid (expressed on a dry basis) and the relative humidity of a gas (usually air) in contact with the solid.*

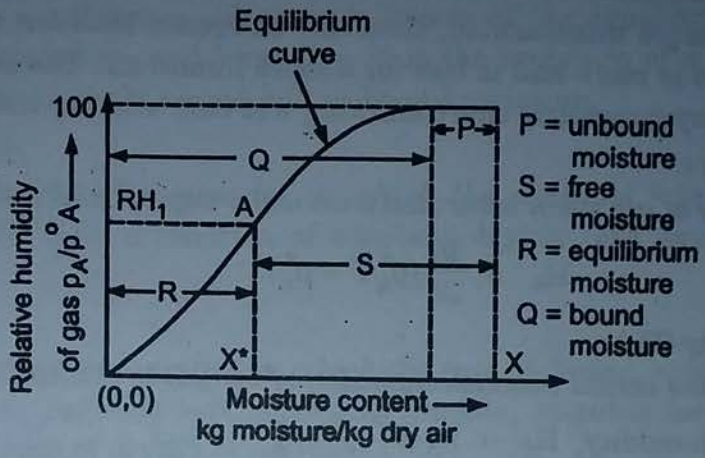

Fig. 6.1 : Equilibrium Moisture Curve

When the humidity of air is less as compared with the moisture content of solids, then solid will lose moisture by evaporation and dry to equilibrium and if the air is more humid than the solids, then solids will gain moisture until the equilibrium is attained. A typical equilibrium curve for drying a certain wet solid is shown in Fig. 6.1 where the ordinate is the relative humidity of the gas and the abscissa is the moisture content on dry basis. When solids having very high initial moisture content (X) is exposed to a continuous supply of the gas with relative humidity of (RH_1), the solids will lose moisture by evaporation and thus go on drying until the moisture content corresponding to the point A is reached (which is the equilibrium moisture content X^*).

Beyond this, no drying takes place even if solids are exposed to this air for an infinitely long period. The moisture content of solids can be reduced below X^* (below that corresponding to point A) only by exposing them to air of lower humidity and to obtain solids bone-dry, we have to expose it to perfectly dry air which corresponds to the origin of curve.

Constant drying conditions :

These conditions mean the *conditions under which the temperature, humidity, velocity and direction of flow of the hot air or gas across the drying surface are constant during drying operation.*

Rate of batch drying and drying test :

A knowledge of the time required for drying a substance from one particular level of moisture content (initial) to another level (desired) under specified drying conditions is needed for determining the size of dryer and also for setting up of drying schedules. In respect

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of time required to dry a substance using one particular type of dryer, one is also interested to estimate the effect of different drying conditions upon the time needed for drying. For the said purposes in many cases we have to run a drying test of the material in question in a model dryer of the type that is desired. Conducting the test and measuring the rates of batch drying is relatively simple and thus provides information useful for batch as well as continuous drying operation.

In the drying test, for determining the rate of drying (needed for further calculations) we have to keep a sample of a given wet material uniformly distributed over a tray (of uniform thickness) suspended from a weighing balance in a cabinet. The bottom and edges of the tray should be well insulated. Then we have to blow air over the surface of the wet material. The air should be blown over the material at constant velocity and should be of constant temperature and humidity. Then, all that we have to do is to measure the weight of the sample as a function of time and also the dry weight of the sample. The test can be performed on samples of different thickness.

Rate-of-drying curve :

The drying characteristics of wet solids are generally described by the drying rate curves obtained under constant drying conditions. Figs. 6.2 and 6.3 are such curves with moisture content v/s time and drying rate v/s moisture content. Generally, the experimental evaluation of these curves is done before performing design calculations.

Consider that the wet solids with an initial moisture content (X_1) are exposed to air of constant temperature and humidity. If we then measure the moisture content as a function of time (i.e., moisture content of the material is measured at various values of time), then a curve such as shown in Fig. 6.2 (a) is obtained from the collected data. The curve relates the moisture content on a dry basis with time. It is clear from the curve that the moisture content of solids decreases with time and after some time it remains constant at X^*, which is the equilibrium moisture content.

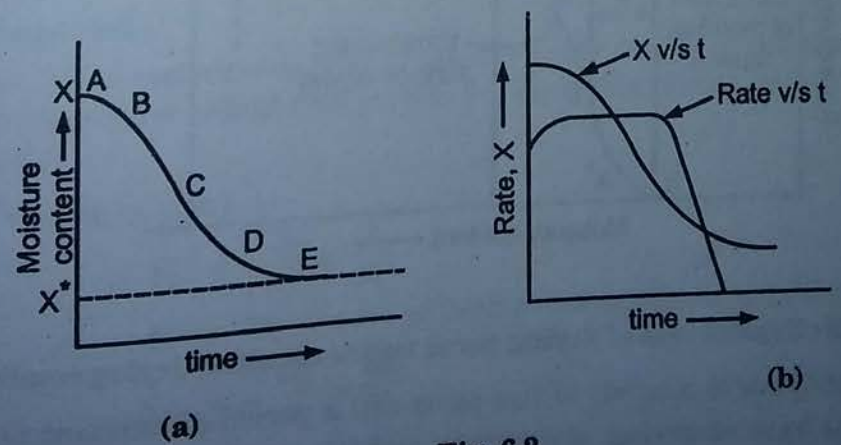

Fig. 6.2

From this curve, we can draw another type of curve which is known as rate of drying curve. This curve is much more descriptive of drying process. The rate of drying curve gives a relationship between rate of drying, expressed as the moisture evaporated per unit time per unit

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area of the drying surface and moisture content on a dry basis. This curve can be constructed by measuring the slopes of tangents drawn to the curve of Fig. 6.2 (a) at various values of the moisture content and then calculating rate as $N = -S_S \frac{dX}{d\theta} \times \frac{1}{A}$, where S_S is the weight of dry solids and A is the area of drying surface.

Fig. 6.3 shows the **rate of drying curve**. The section AB of the curve represents a warming up period during which the temperature of the solid is becoming equal to the temperature of drying air. From B to C, the curve is a straight line parallel to x-axis representing the constant rate of drying, thus the straight section BC is called the **constant rate period** during which a layer of water on the surface of solid is being evaporated. The rate of drying is constant from B to C as the drying takes place from a saturated surface. The section (CE) of the curve represents the **falling rate period** composed of first falling rate period (CD) and second falling rate period (DE). From point 'C' onwards some dry patches have started forming on the surface of the solid. The rate of drying decreases for the unsaturated portion and hence rate for total surface decreases. The section CD of the curve represents the period corresponding to the zone of unsaturated surface drying. The moisture content at which the constant rate period ends (i.e. at which unsaturated surface drying starts) and at which the drying rate starts to fall is known as the **critical moisture content**. After point D, the surface of the solid is completely dry and now internal moment of moisture starts coming to the surface and this is continued upto point E, where the equilibrium is attained. The rate of drying over section DE is governed by the rate of internal moisture movement. The second falling rate period (DE) represents the zone where internal moisture movement controls.

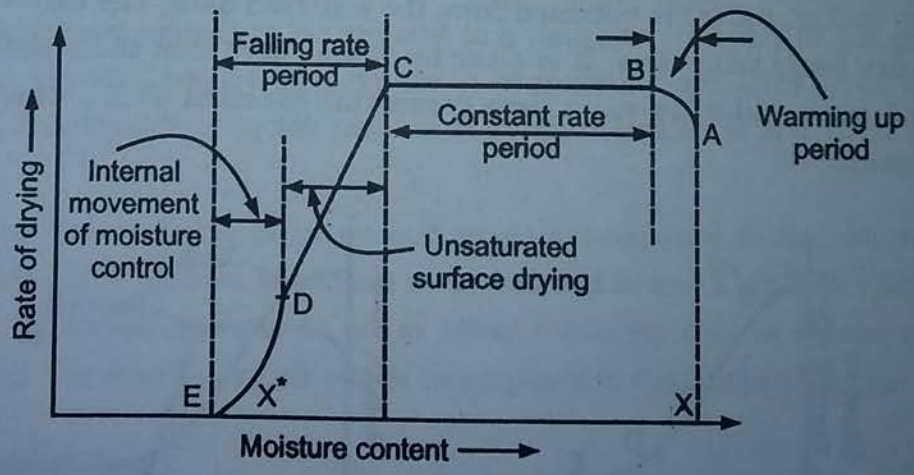

Fig. 6.3 : Typical rate of drying curve under constant drying conditions

The rate of drying curve consists of two parts : (i) a period of constant rate/a constant rate period, wherein the rate of drying is constant and (ii) a period of falling rate/a falling rate period, wherein the rate of drying is falling.

The rate of drying as a function of time is given in Fig. 6.4, which indicates how long each drying period lasts.

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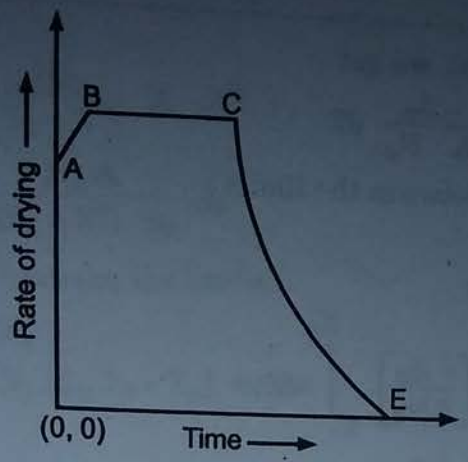

Fig. 6.4 : Drying rate v/s time

Factors on which the rate of drying depends :

(a) Gas velocity : When the velocity of gas or air is high, the rate of drying will also be high.

(b) Humidity of gas : Lesser the relative humidity, the more will be the rate of drying.

(c) Area of drying surface : If the area of wet surface exposed to gas or air is more, the rate of drying will also be more.

(d) Temperature : If the temperature of the gas is increased, its relative humidity decreases (i.e., gas becomes more unsaturated) and it thus increases driving force (The driving force is a concentration difference of the moisture between the solid and gas) and so the rate of drying increases.

Time of drying under constant drying conditions :

Consider that the wet solids are to be dried by passing hot air over them under constant drying conditions. The time of drying a solid material from the initial moisture content to the final moisture content is the sum of the time of drying required during the constant rate period and the time required during the falling rate period (in cases where the final moisture is less than the critical).

(a) Constant rate period :

Let X_1 be the initial moisture content of the wet solids and X_2 be the final moisture content of the wet solids during the constant rate period. Let X_c be the critical moisture content of the wet solids. $[X_1, X_2 > X_c]$.

The rate of drying is given as :

$$N = -\frac{S_S}{A} \times \frac{dX}{d\theta} \quad \ldots (6.1\ a)$$

$N = N_c$ = rate of drying during constant rate period

$$N_c = -\frac{S_S}{A} \times \frac{dX}{d\theta} \quad \ldots (6.1\ b)$$

where S_S = mass of dry solid in kg

A = area of drying surface in m^2

N_c = rate in $kg/(m^2 \cdot h)$

θ = time in hours (h)

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Rearranging Equation (6.1 b), we get

$$d\theta = \frac{-S_S}{A \cdot N_c} dX \quad \ldots (6.2)$$

Integrating Equation (6.2) between the limits :

$\theta = 0, \quad X = X_1$

and $\theta = \theta, \quad X = X_2$, we get

$$\int_0^{\theta} d\theta = \left[\frac{-S_S}{AN_c}\right] \int_{X_1}^{X_2} dX \quad \ldots (6.3)$$

$$\theta = \frac{-S_S}{A \cdot N_c} [X_2 - X_1] \quad \ldots (6.4)$$

$$\theta = \frac{S_S}{A \cdot N_c} [X_1 - X_2] \quad \ldots (6.5)$$

Equation (6.5) gives the time of drying for drying material from X_1 to X_2 in the constant rate period.

If the material is to be dried to the moisture content of X_c, then the time required during the entire constant rate period is given by :

$$\theta_c = \frac{S_S}{A N_c} [X_1 - X_c] \quad \ldots (6.6)$$

(b) Falling rate period :

During this period, the rate of drying is proportional to the free moisture content.

$$\frac{-S_S}{A} \times \frac{dX}{d\theta} = m [X - X^*] \quad \ldots (6.7)$$

where X^* is equilibrium moisture content and X is the moisture content of wet solids less than the critical moisture content.

Let X_1 be the initial moisture content and X_2 be the final moisture content such that $X_1, X_2 < X_c$.

Assume that the entire falling rate period is represented by a straight line CE, then :

$$m = \frac{N_c}{[X_c - X^*]} \quad \ldots (6.8)$$

m = slope of line CE

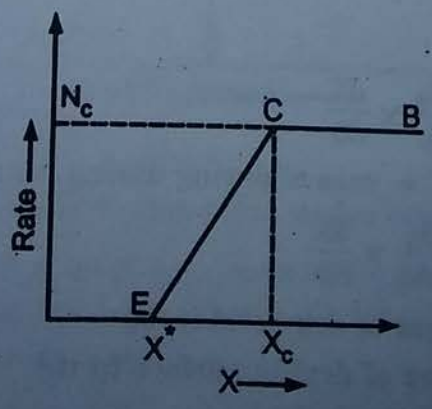

Fig. 6.5 (a) : Rate v/s moisture content

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Equation (6.7) then becomes :

$$\frac{-S_S}{A}\frac{dX}{d\theta} = \frac{N_c}{[X_c - X^*]}[X - X^*] \quad \ldots (6.9)$$

$$\frac{-dX}{[X - X^*]} = \frac{N_c A}{[X_c - X^*]\, S_S}\, d\theta \quad \ldots (6.10)$$

Integrating Equation (6.10) between the limits :

$X = X_1$

$X = X_2$ $[X_1, X_2 < X_c]$, we get

$$-\int_{X_1}^{X_2} \frac{dX}{[X - X^*]} = \frac{N_c A}{[X_c - X^*]\, S_S} \int_0^{\theta} d\theta \quad \ldots (6.11)$$

$$\theta = \frac{S_S (X_c - X^*)}{N_c A} \ln\left[\frac{X_1 - X^*}{X_2 - X^*}\right] \quad \ldots (6.12)$$

Equation (6.12) gives the time of drying required during the falling rate period to dry any material from X_1 to X_2.

If a material is to be dried from the critical moisture content X_c to the final moisture content X_2 $(X_2 < X_c)$ then the time required for drying during the entire falling rate period is given by θ_f as :

$$\theta_f = \frac{S_S (X_c - X^*)}{N_c \cdot A} \ln\left[\frac{X_c - X^*}{X_2 - X^*}\right] \quad \ldots (6.13)$$

[As X_1 becomes X_c]

θ_f – drying time during the entire falling rate period.

The total drying time $= \theta = \theta_c + \theta_f$

Appropriate equations and values of X_1, X_2 and X_c are to be used to calculate the time required during a particular period.

The time required for drying a solid from the initial moisture content X_1 where $(X_1 > X_c)$ to the final moisture content X_2 $(X_2 < X_c)$ is given by

$$\theta = \theta_c + \theta_f = \frac{S_S}{A\, N_c}\left[(X_1 - X_c) + (X_c - X^*) \ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right] \quad \ldots (6.14)$$

Cross-circulation drying (batch drying) :

Constant rate period :

In the constant rate period, a solid material in contact with air remains completely wetted and the evaporation takes place from the surface. The process of drying is independent of the mechanism by which the moisture transfer occurs at the surface and thus the process of drying reduces to the case of mass transfer from the wetted surface to the air stream (diffusion of vapour from the film of liquid) and heat transfer from the air to the surface by convection provided that radiation and conduction of heat to the surface are negligible.

For cross-circulation drying, the rate of evaporation per unit area and the rate of heat transfer per unit area are related by

$$N_c \lambda = Q$$

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If conduction through the solid and radiation effects are absent then the rate of drying is given as

$$\therefore \quad N_c = \frac{h\,(T_G - T_S)}{\lambda} = K_G\,(p_S - p_G) = K_Y\,(Y_S - Y) \qquad \ldots (6.15)$$

where T_G - dry bulb temperature of air, T_S – temperature of wetted (drying) surface,

h – convective heat transfer coefficient from air to the wet surface,

λ – latent heat of vaporisation of water at T_S, K_G – mass transfer coefficient from wetted surface through the gas film,

p_S – the vapour pressure of water at the surface temperature

p_G – the partial pressure of water vapour in the air stream.

When no radiation to the surface or conduction of heat through the solid material occurs (i.e., for no radiation and conduction through the solid) then the drying surface assumes wet-bulb temperature of air, i.e., T_S = WB of air.

Y_S is the saturated humidity at T_S, Y is the humidity of air at the prevailing conditions of air.

Factors affecting the constant drying rate :

(i) Air temperature : N_c is directly proportional to $(T_G - T_S)$ so increase in a air temperature, increases $T_G - T_S$ and thus, increases N_c (as for constant air velocity h will be unaffected).

(ii) Air humidity : N_c is directly proportional to $Y_S - Y$. So increasing the humidity, decreases the quantity $Y_S - Y$ or $p_S - p_G$ and hence lowers the rate of drying.

(iii) Air velocity : The rate of drying is proportional to the mass velocity of air stream, raised to a power of about 0.80 (for negligible radiation and conduction through the solid).

$$N_c \propto (G)^{0.8}, \text{ so } N_c \propto (u)^{0.8}$$

where u is the linear velocity of air. When the radiation and conduction through the solid are negligible then only the effect of air velocity is important. When radiation and conduction effects are present then the effect of gas rate will be less important.

(iv) Thickness of drying solid : When conduction of heat through the solid occurs, then N_c decreases with an increase in the thickness of the drying solids.

Material and Enthalpy Balance Equations in Continuous Drying Operation :

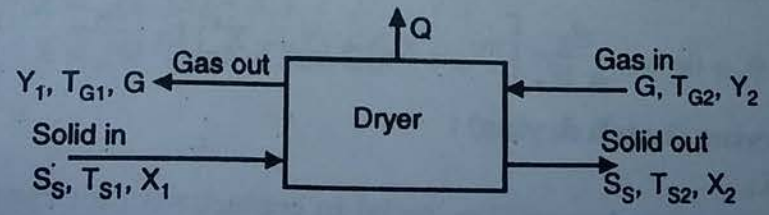

Fig. 6.5 (b) : Material and enthalpy balances, continuous dryer

In Fig. 6.5 (b) location-1 is at the entrance of the solid and location-2 is at the entrance of the drying gas to the dryer.

Fig. 6.5 (b) shows the block diagram of a continuous dryer where the drying gas and solid flow countercurrent to each other. The solid enters at a rate of S_S kg dry solid per hour and are to be dried from X_1 to X_2 kg moisture/kg dry solid. In the dryer, the solid undergoes a

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temperature change T_{S1} to T_{S2}. The gas enters at a rate of G kg dry gas (air) per hour. In the dryer, the humidity of the gas changes from Y_2 to Y_1 kg moisture/kg dry gas (air), and its temperature changes from T_{G2} to T_{G1}.

Material balance of the moisture over the dryer :

Moisture in solid in + moisture in gas in = moisture in gas out + moisture in solid out

$$S_S + X_1 + G\,Y_2 = G\,Y_1 + S\,X_2$$

$$\therefore \quad S_S(X_1 - X_2) = G(Y_1 - Y_2) \qquad \ldots (6.16)$$

For an enthalpy balance let T_0 °C be the datum temperature and in most cases, it is selected as 0 °C.

The equation for the enthalpy of the solid :

The enthalpy of the wet solid H_S in kJ/kg dry solid is the sum of the enthalpy of the dry solid and that of the free moisture associated with it and is given by the following equation :

$$H_S = C_{pS}(T_S - T_0) + X\,C_{pA}(T_S - T_0) \qquad \ldots (6.17)$$

where C_{pS} = heat capacity of the dry solid, kJ/(kg dry solid.K)

C_{pA} = heat capacity of the moisture (H_2O), kJ/(kg moisture.K)

The equation for the enthalpy of the gas is

The enthalpy of the gas in kJ/kg dry gas (air) is given by

$$H_G = C_S(T_G - T_0) + Y\lambda_0 \qquad \ldots (6.18)$$

where λ_0 is the latent heat of water at 0 °C and C_S is the humid heat in kJ/kg dry gas (air).

The humid heat for the air-water system is given by

$$C_S = 1.005 + 1.88\,Y$$

λ_0 at 0 °C (i.e., the latent heat of water at 0 °C) is 2501 kJ/kg.

If Q is the loss of heat from the dryer (in kJ/h) then the steady state enthalpy balance over the dryer is given is

$$\begin{pmatrix}\text{Enthalpy in}\\ \text{[with streams}\\ \text{entering the dryer]}\end{pmatrix} = \begin{pmatrix}\text{Enthalpy out}\\ \text{[with streams}\\ \text{leaving the dryer]}\end{pmatrix} + \begin{pmatrix}\text{loss of heat}\\ \text{from the dryer}\end{pmatrix}$$

$$G\,H_{G2} + S_S\,H_{S1} = G\,H_{G1} + S_S\,H_{S2} + Q \qquad \ldots (6.19)$$

For an adiabatic operation, $Q = 0$.

DRYING EQUIPMENTS :

Dryers used in industry may be classified on the basis of (a) mode of operation, (b) physical properties and handling characteristics of the material and (c) the method of supplying heat to the material to be dried, i.e., the method of heat transfer.

(i) Mode of operation :

On the basis of mode of operation that is based on the nature of the production schedule, there are two types of drying equipments : (i) batch dryers and (ii) continuous dryers.

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In case of batch dryers, a definite size of a batch of the wet feed is charged to the dryer and drying is carried out over a given time period. These dryers operate under unsteady state conditions.

Drying in batches is relatively expensive operation and consequently batch dryers are preferred for small-scale production, pilot plant and for drying valuable materials.

In case of continuous dryers, the material flows in and out continuously and drying is carried out under steady state conditions continuously. These are generally used for large-scale production.

Advantages of continuous drying include :

Equipment required is small relative to the quantity of product.

Product has more uniform moisture content.

Cost of drying per unit of product is relatively small.

(ii) Nature of the material to be dried :

The wet feed material to be dried may be a liquid solution, a thin slurry, a thick paste, a sludge, a free flowing powder, a granular, a crystalline or fibrous solid. The physical form of the material and the handling methods have the greatest influence on the type of dryer to be used.

(iii) Method of heat transfer :

On the basis of method of heat transfer, dryers are classified as : (a) direct dryers, and (b) indirect dryers. Dryers in which the wet material is directly exposed to a hot gas (usually air) are called direct or adiabatic dryers. Dryers in which heat is transferred to the solid from an external heating medium (such as condensing steam) through a metal surface that keeps the solid and drying medium separated are called indirect or non-adiabatic dryers.

In case of direct contact dryers when the hot gas is blown across the surface of a bed of solids or across face(s) of a continuous sheet then the process of drying is called as cross-circulation drying and when the hot gas is blown through the bed of solids that are supported on the screen then the process of drying is called as through-circulation drying.

Direct dryers :

In such dryers, the heat necessary for drying (i.e., the heat necessary for evaporation of the moisture) is supplied by direct contact of the wet feed material with the hot gas. The moisture evaporated from the wet feed is carried by the hot gas.

Dryers in which the wet solids (or the substance to be dried) are directly exposed to a hot gas (generally air) are called adiabatic or direct dryers. In the adiabatic dryers, heat for evaporation of the moisture is supplied by sensible heat of a hot gas.

Indirect dryers :

In such dryers, the heat necessary for drying is supplied to the wet feed by conduction through a metal surface/wall in contact with the feed material to be dried. The moisture evaporated from the wet feed is carried by the air or other gas independently of a heating medium.

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Dryers in which the heat necessary for evaporation of the moisture is transferred to the solids/substance from an external medium (condensing steam) through a metal wall in contact with the substance by conduction are called non-adiabatic or indirect dryers.

Production rates of 5000 kg per day are best handled by batch dryers and rates over 50000 kg in a continuous dryer.

Tray Dryer :

Construction : It is the simplest of the batch dryer and also known as cabinet or compartment dryer. The tray dryer shown in Fig. 6.6 is a batch operated direct dryer. It consists of an enclosed insulated cabinet or a large compartment into which the material to be dried is placed on a number of removable trays. The trays may either be fabricated from sheets or from screens. The trays may be stacked on racks or loaded on trucks. It is provided with the inlet and outlet connections for air. A heating coil either electrical or steam-heating is incorporated in it. In these dryers, steam, or electrically heated air is used as the drying medium. The air is circulated in the dryer over the trays by means of a fan fitted at the top or at the side (on one of the sides, opposite to the coil).

Working : The material to be dried is spread over the trays and put into the cabinet and then it is closed. Steam is continuously passed through the coil and the fan is started. The air taken in is heated by heating coils, its relative humidity decreases (i.e., its capacity to evaporate the moisture is increased) and the hot air then passes over the trays.

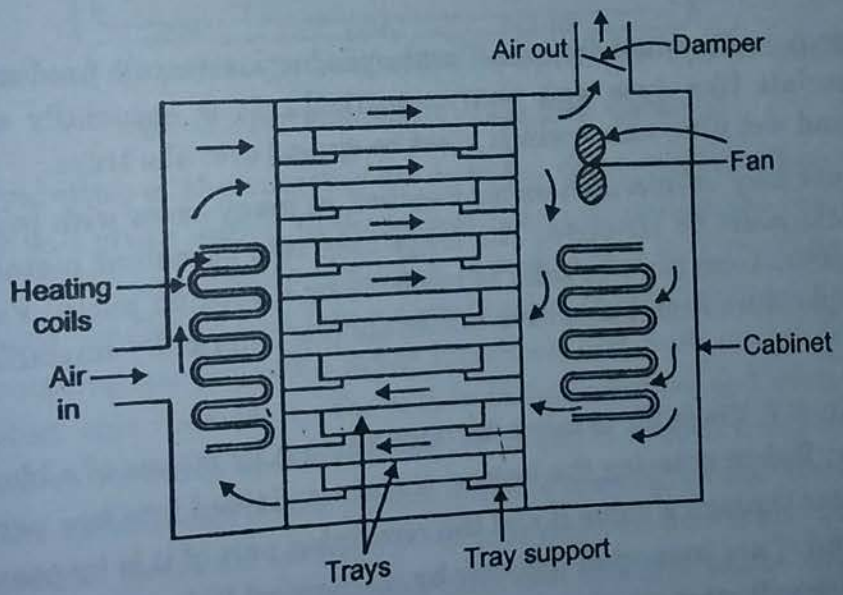

Fig. 6.6 : Tray Dryer

The moisture is evaporated from the wet feed, gets added in the air and finally the air leaves the dryer through the outlet. The process is continued until the solids are dried. The cabinet is then opened, dried material is removed from the trays and a fresh batch is charged.

For getting good drying, the air after drying should be thrown out completely but by this way, a large portion of heat associated with the hot air will be lost and the operation will become costly. To avoid this heat loss (i.e., to conserve heat) about 80 – 95 % of the air is recirculated by adjusting a damper provided at the outlet, and the remaining portion is exhausted out, and the same amount of fresh air is taken in through the inlet.

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The overall rate of drying of such dryer is 0.2 to 2.0 kg water/(m^2·h) material surface and the thermal efficiency is of the order of 20 – 25 %.

The trays are generally 600 mm wide, 900 to 1500 mm long and 30 to 40 mm deep. They are made of mild steel stainless steel, enamelled iron etc. and are fabricated from sheets of 3 mm to 6 mm thick.

Advantages :

(i) Relatively cheap and easy to construct/build.

(ii) Low space requirement.

(iii) Ease in cleaning.

(iv) Low maintenance (low maintenance cost).

(v) No loss of product during drying.

Disadvantages :

(i) Expensive to operate due to high labour requirements for loading and unloading (high labour costs and low heat economy).

(ii) Long drying cycles (4 to 48 h per batch).

(iii) Small quantities are handled.

Applications :

Tray dryer is well suited for small scale production (small production rate) and drying valuable materials like dyes and pharmaceuticals. It is especially useful for drying wet lumpy solids and wet filter cakes which must be spread over the trays.

These dryers may be operated under vacuum, in many cases with indirect heating. In such dryers, all joints must be air-tight. The trays may rest on hollow metal plates supplied with steam and vapours from the wet solid is removed by a vacuum pump. Vacuum-tray dryers are suited for temperature / heat sensitive materials (i.e., thermally unstable materials).

Dryer control :

Refer to Fig. 6.7. Fresh air is taken through valve A by means of a blower or fan and then it enters a heater. Before entering the heater, the air is divided into two parts, one part of the air goes to the heater through a valve B and the remaining part of it is by-passed through a valve C. The valves B and C are connected together by mechanical links in such a way that if one starts closing, the other will start opening. Both the valves are operated by a diaphragm control valve which is getting signals from a dry and wet bulb recorder and controller. Hot air from the heater and cold by-passed air are mixed together and sent to the dryer for drying. For heat economy, a major part of the air leaving the dryer (moist air) can be recirculated in the system through a valve D and the remaining air is vented to the atmosphere through a valve E. The valves A, D and E are connected together by mechanical links in such a way that if the valve E starts opening, the valve D will start closing, and the valve A will start opening. All these three valves are operated by a diaphragm control valve which is getting signals from a dry and wet bulb recorder and controller. The dry and wet bulb thermometer assembly is kept in an outlet line from the dryer.

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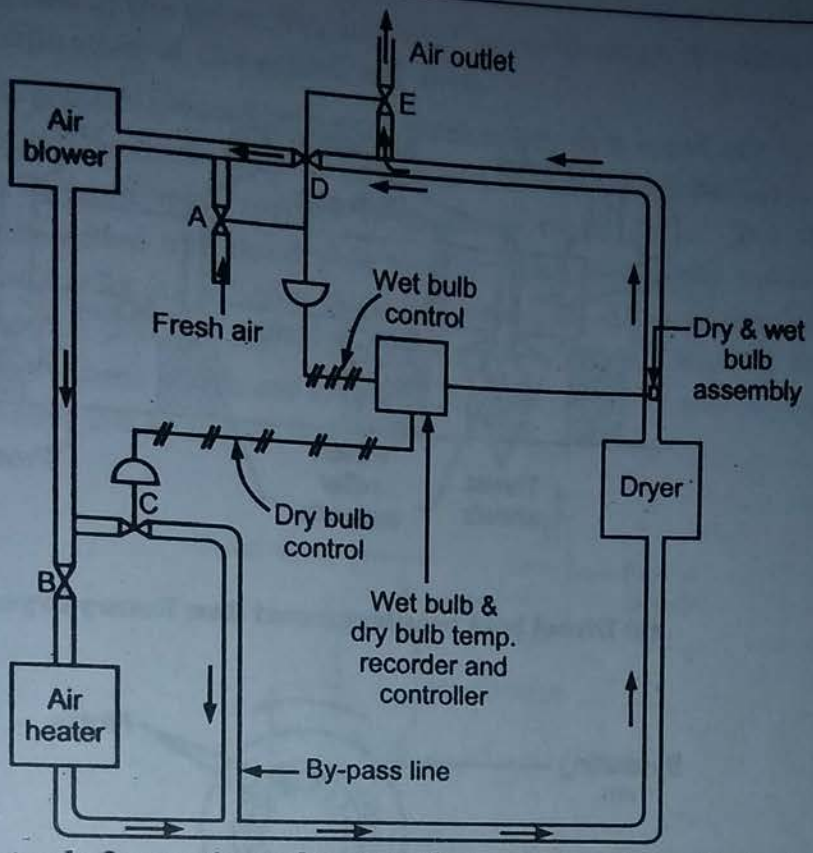

(control valves instead of operating valve stem they operate a lever that actuates dampers)

Fig. 6.7 : Instrumentation diagram for dryer

When the temperature of the outgoing air from the dryer increases beyond a certain limit then the controller will give a signal to the valves B and C through the control valve which is actuated by the dry bulb temperature controller in such a way that the valve B starts opening and the valve C starts closing so that less hot air is going to the dryer and vice versa. When the humidity of the outgoing air from the dryer increases, wet bulb temperature will increase. Controller will detect this increase in the wet bulb temperature and accordingly will give a signal to the control valve which operates valves A, D and E in such a way that when the valve E starts opening, the valve D starts closing and A starts opening, so that more fresh air is taken into the system and the moist air is thrown out and recirculation is also reduced. Because of this action, the humidity is lowered. The dryer control of this type can be used for batch as well as for continuous dryers.

Rotary Dryer :

This type of dryer, that may be directly or indirectly heated, is adopted for the continuous drying of free-flowing granular materials on a large scale. Fig. 6.8 shows one form of rotary dryer. It consists of a relatively long cylindrical shell (having a diameter of 1 m to 3 m and length 3 m to 30 m) set with its axis at a slight angle to the horizontal (slightly inclined towards the outlet), so that the material fed is consequently advanced through the dryer from one end to the another end (under gravity) from where it is discharged.

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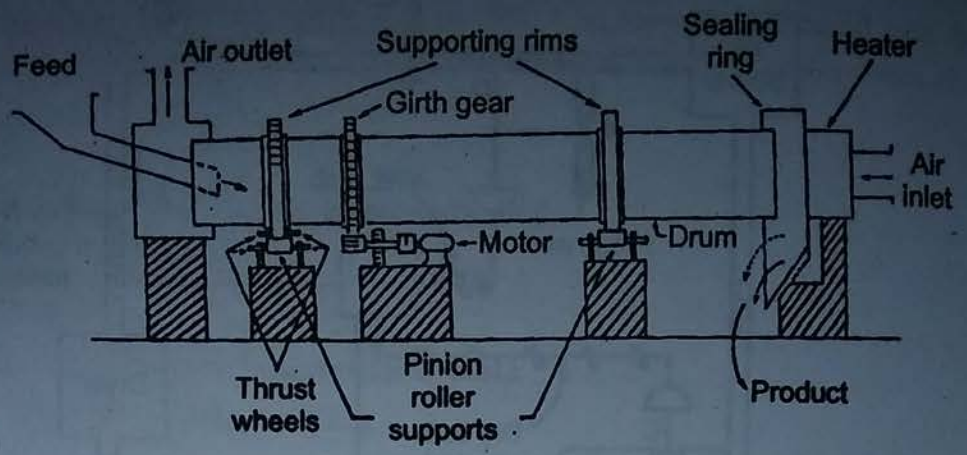
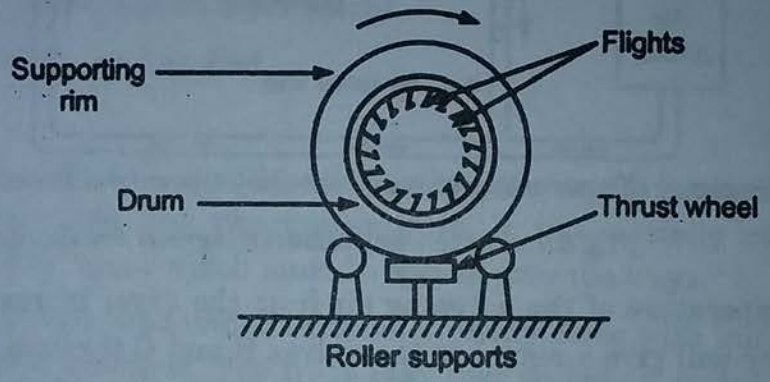
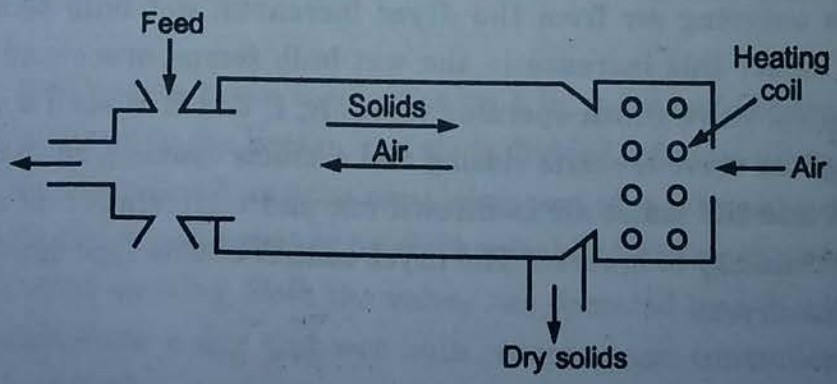


(a) Direct heat countercurrent flow Rotary dryer



(b) Rotary dryer, side view



(c) Schematic view of rotary dryer

Fig. 6.8

The shell is mounted on rollers so that it can be rotated. To avoid its slipping over the rollers, it is fitted with thrust wheels. It is fitted inside with flights which lift the material upward and shower it down from the top. A few spiral flights are fitted near feed end which help in giving the initial forward motion to the material before principle flights are reached. The material to be dried (feed) is fed at the high end of the dryer by a hopper and the product is taken

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out from the lower end of the dryer. The material moves through the dryer by virtue of rotation, heat effect and inclination of the cylindrical shell.

The cylindrical shell is rotated by a gear mechanism at a speed of 2 to 25 rpm. Air is taken into the dryer from the product end. It is heated in a heater, and then moves through the dryer in a countercurrent fashion with respect to material to be dried. The moisture of the feed evaporates and gets added into the drying medium, and finally the moist air leaves the dryer at the feed end. Generally, an exhaust moist air leaves the dryer at the feed end. Generally, an exhaust fan is used to pull the air through the dryer. The air leaving the dryer will contain some dust particles (fines) which are removed by interposing a cyclone separator between the dryer and the exhaust fan as shown in Fig. 6.9.

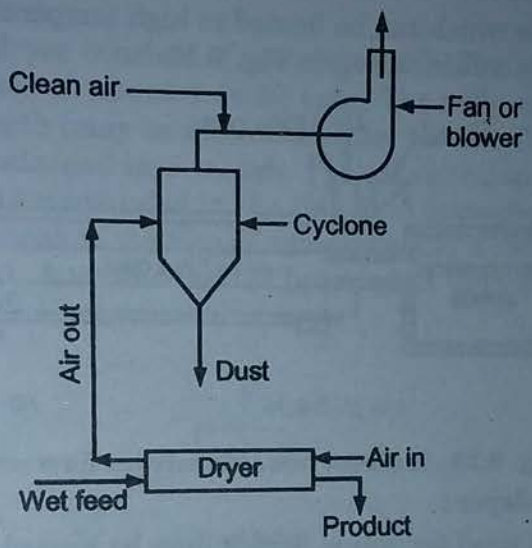

Fig. 6.9 : Rotary dryer with dust collector (cyclone)

The mode of operation of this dryer is usually continuous. In the case of direct contact dryers, the hot gas is passed over the material in a countercurrent fashion, while in the case of indirect contact dryers, heat is transferred through a wall of the cylindrical shell.

The thermal efficiency of rotary dryers is about 50 – 80 % and drying rate ranges between 10 – 50 kg/(h·m^3 of shell volume).

Advantages :

- good gas contacting
- moderate drying time
- low capital cost
- drying and calcining in the same unit
- high thermal efficiency
- greater flexibility of control of the gas velocity.

Disadvantages :

- difficulty of sealing
- product build-up on interior walls
- high structural load
- non-uniform residence time.

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Rotary dryers are grouped into four categories :

1. Direct heat countercurrent flow :

This type of dryer is shown in Fig. 6.8 and employed for the materials which may be heated to high temperature like minerals, sand clays, etc. using the hot flue gas as a drying medium. It is also used for products like ammonium sulphate and cane sugar wherein the hot air is used as a drying medium.

2. Direct heat parallel flow :

In this type, both the material and hot gas/air move in the same direction through the dryer. It is used for drying materials like gypsum, iron pyrites, etc., which should not be heated to destructive temperatures.

3. Indirect heat countercurrent flow :

It is used for materials which can be heated to high temperatures out of contact with flue gas such as white pigments, etc. It is shown in Fig. 6.10.

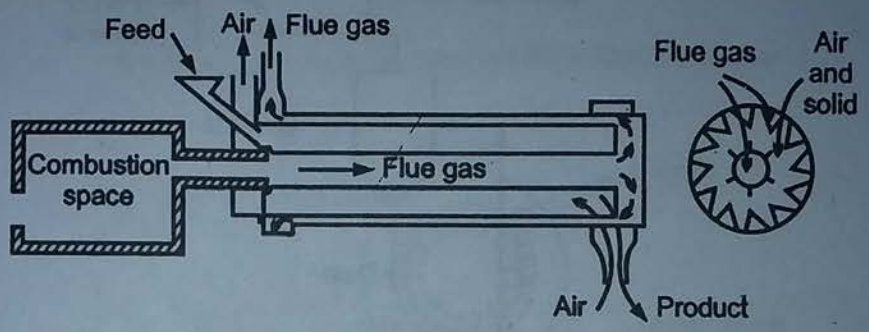

Fig. 6.10 : Indirect countercurrent flow rotary dryer

4. Direct-indirect type dryer :

This type of dryer is used for solids, which may be heated to very high temperatures by a flue gas, such as lignite, coal, and coke. It is shown in Fig. 6.11.

It contains an inner hollow tube through which the flue gas passes from a combustion chamber to the lower end of the dryer. At this end, the direction of the flue gas is reversed and made to pass through hollow flights, through an annular space and leaves the dryer from the feed end. The central tube is provided with longitudinal fins to increase the outer area of heat

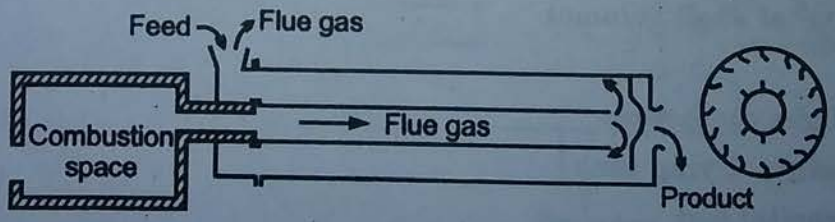

Fig. 6.11 : Direct - Indirect type dryer

transfer. The wet material is admitted at the high end of the dryer and travels down to the lower end and taken out as dried product. Air enters from product end and travels over the solid in the reverse direction. The moisture gets added in the air and finally leaves the dryer from the feed end. The material is heated indirectly by the flue gas by conduction and radiation and the moisture is taken out by air, flowing in a countercurrent fashion with respect to the feed. Another type of indirect dryer is the steam tube dryer employing a number of tubes and finds application where the material to be dried must not be heated to a high temperature, e.g., drying of cattle feed, etc.

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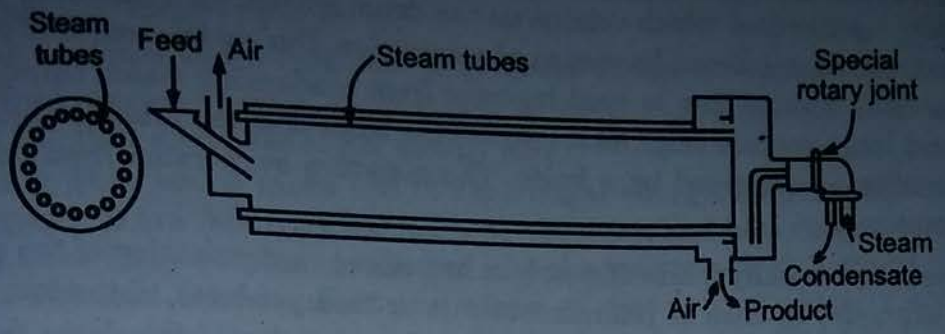

Fig. 6.12 : Indirect steam-tube dryer

Drum Dryer :

Construction : A drum dryer consists of one or more hollow metal rolls that are heated internally by steam and rotates at about 1 to 10 rpm. The rolls are usually 0.6 to 3.5 m in diameter and 0.6 to 5 m in length (may be fabricated from plates). The drum is submerged into a pool of solution or slurry contained in a trough. For an agitation purpose, i.e., to avoid the setting of solids, an agitator is incorporated in the trough. A spreader is provided on one side of the drum to regulate the thickness of the film of substance and a knife is provided on the other end to scrap the dried material. A vapour-hood is provided at the top of the drum for collection and removal of moisture. Fig. 6.13 shows such a dryer.

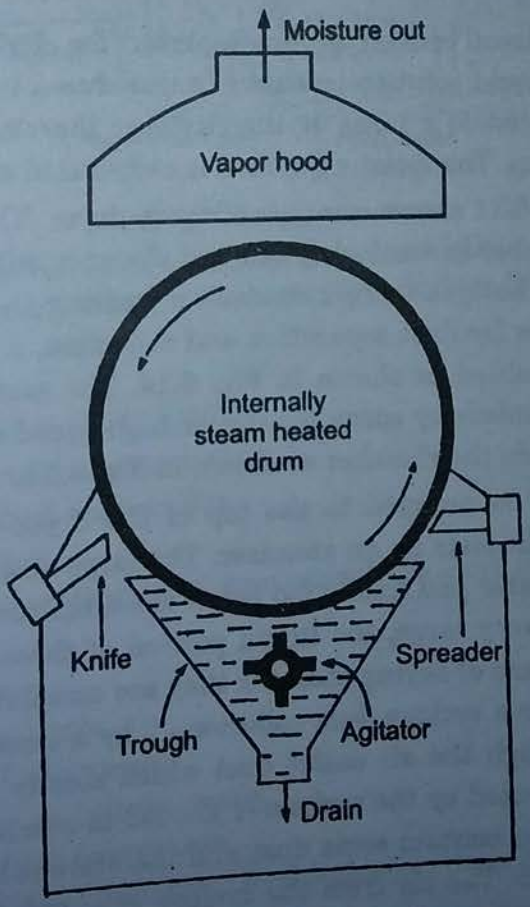

Fig. 6.13 : Single drum dryer

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Working : A slowly revolving internally steam heated drum continuously dips into a trough and picks up the feed which retains on the drum surface as a thin film. The thickness of this film of material is regulated by means of a spreader. During the course of revolution of the drum, the material is dried due to heat transfer from condensing steam through a metal wall of the drum and large surface area. As it reaches the other end, the dried product (in the form of flakes) of operation is scrapped by a knife. The moisture evaporated from the feed material goes to a vapour-hood.

Drum dryers are usually made of cast iron but where contamination of the product must be avoided, for example, in case of pharmaceuticals or food products, chromium plated steel or alloy steel is used as material of construction. The capacity of the drum dryers is less as compared to the spray dryer but their operating cost is low.

Drum dryers are suitable for handling fluid and semifluid materials such as solutions, slurries, pastes of solids in fine suspension and dilute or concentrated solutions of highly soluble materials. These are not suited for solution of salts with limited solubility and for slurries of abrasive solids that have tendency to settle-out.

Cylindrical dryers are drum dryers which are commonly employed for drying of a material in the continuous sheet form, such as paper and cloth. The wet material is fed continuously over the revolving drum or a series of drums each heated internally by steam (or by other heating fluid).

Spray Dryer :

It is a continuous direct contact dryer employed for drying of solutions, slurries, and pastes. In this dryer, a liquid solution or slurry is introduced in the form of very fine droplets into a stream of hot gas inside a large drying chamber thereby a large contact area becomes available for perfect drying. The moisture of feed is evaporated and gets added into the hot gas.

Construction : Fig. 6.14 shows one type of spray dryer. The essential components of this dryer are : a drying chamber (a vertical cylindrical chamber with a conical bottom), where the feed is contacted with a hot gas (air), a heater for heating the fresh air sucked by a fan or blower, cyclone separators for dust separation and collection, a pneumatic conveying duct and blowers, which are assembled as shown in Fig. 6.14. The material is usually spread in the form of a mist of fine droplets by spray nozzles or high speed rotating spray discs (atomiser) into a hot gas stream inside the chamber as shown in Fig. 6.14.

Working : The feed is pumped to the top of the dryer (drying chamber) where it is disintegrated into small droplets by an atomiser. The large quantity of fresh air is taken in by a fan, it is heated in a heater and finally fed below the atomiser in the drying chamber. As the surface area of drops is very large, the liquid portion of these drops rapidly evaporates and before they touch the bottom of drying chamber they are completely dried. This dried product is taken out and conveyed to a cyclone dust collector - 2 by a stream of air. The major portion of the air is taken out through the air outlet duct which mostly contains dust and is sent to a cyclone -1. The solids collected by the cyclone -1 are fed to a pneumatic conveying duct. The air leaving the cyclone - 2 may contain some dust and therefore it is sent to the cyclone - 1 by the fan, for further separation. The air from the cyclone -1 is thrown out to the atmosphere by a blower. The dried product from the cyclone - 2 is collected in a dry product collector.

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An atomiser is a device which causes the liquid to disintegrate into the fine drops. The atomisers commonly employed are :

1. Pressure nozzles which make use of pressure energy for atomisation.
2. Two fluid nozzles wherein air or steam at certain pressure are used to tear the liquid into droplets, i.e., they make use of gas energy and
3. Rotating discs which make use of centrifugal energy for atomisation. Spray nozzles (type -1 and type - 2) are relatively inflexible in operation and also subject to erosion and tear. The rotating discs (may be plane, vaned or cup-shaped) rotate at a speed of about 3000 to 12000 r.p.m. The feed introduced at the centre of disc is centrifugally accelerated to the periphery and ultimately thrown in an umbrella-shaped spray. The rotating discs are very flexible in their operating characteristics and can handle thick slurries without danger of clogging.

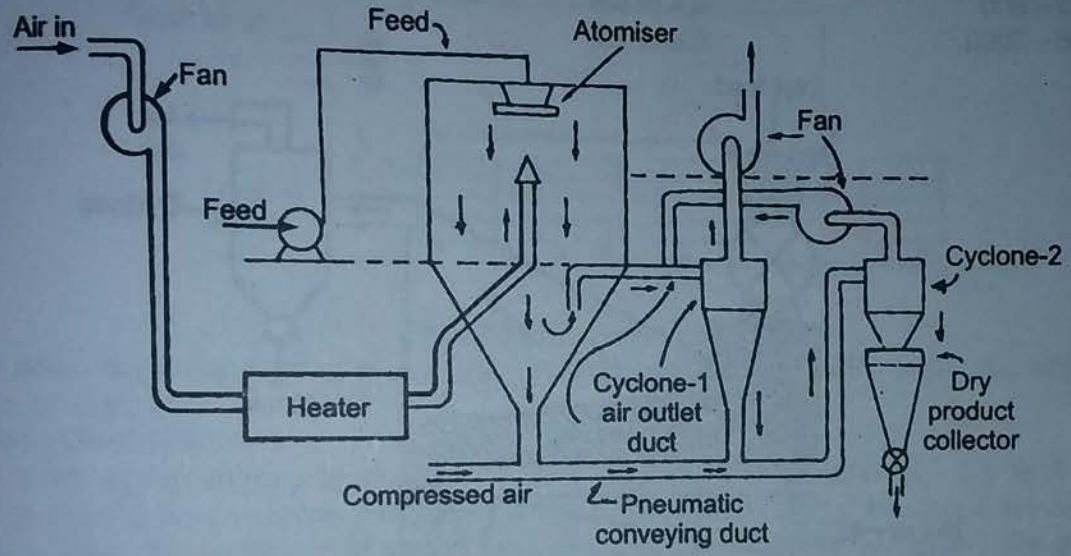

Fig. 6.14 : Spray dryer

In this dryer, care must be taken to ensure that the droplets or wet particles of solids do not strike and stick to the dryer surface before complete drying. So as to avoid this, large drying chambers are used.

Advantages :

(i) Very short drying times (2 – 20 s).
(ii) Handle heat-sensitive materials.
(iii) Control of a product particle size.
(iv) Rapid dehydration.
(v) Relatively low operating costs, particularly in large capacity units.

Disadvantages :

(i) Low solids content.
(ii) Relatively large units.
(iii) Maintenance of atomisers.
(iv) Product built-up on interior walls.
(v) Insufficient in (its) energy use.

These are popular in dairy industry, food industry, detergent industry, chemical and dyes industry.

Spray dryers extensively used for products such as milk powder, detergents, dyes, coffee, pharmaceuticals, etc.

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Fluidised Bed Dryer :

It is also known as a fluid bed dryer. Fluid bed drying systems are becoming popular because of the following reasons :

(i) absence of moving parts results in ease of maintenance

(ii) high heat transfer rates

(iii) rapid mixing indicating more or less isothermal operation leading to uniform drying.

In this dryer, hot gas / air is passed through a wet material at a velocity sufficiently high to fluidise the wet material but not too high enough to cause pneumatic conveying.

Typical recommended gas/air velocities :

Particle size (μm)	Velocity (m/s)
300 – 800	0.4 to 0.8
800 – 2000	0.8 to 1.2

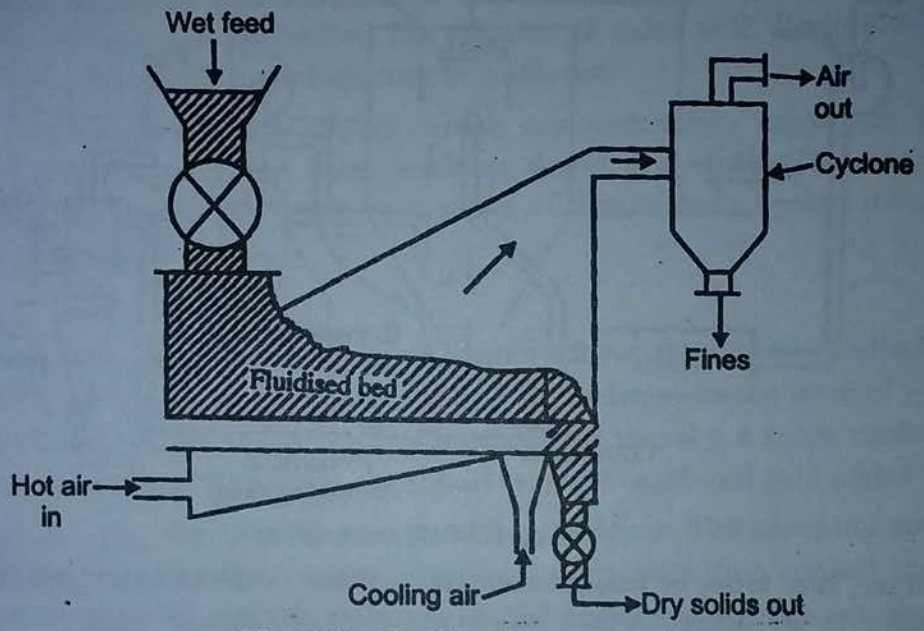

Fig. 6.15 : Fluidised bed dryer

A fluidised bed system in addition to a fluidising chamber also needs an air blower, a hot air generator, a feed conveyor, a cyclone separator and a product conveyor.

In this dryer, hot air is used to keep a wet feed in the fluidised state. In the dryer shown in Fig. 6.15, a material is dried and cooled in the same bed. Wet feed is admitted to the top of the fluidised bed through a hopper via a rotary valve and hot air is distributed at the bottom of the bed through a diffuser plate and dry product is taken out from the side or near the bottom. Heat and mass transfer coefficients are high because of turbulence created in the bed. The material to be dried and hot air are in cross flow with respect to each other. The residence time can be controlled from seconds to hours. The moist air from the dryer containing fines is admitted to a cyclone separator for the recovery of fines.

It is used for drying very fine size free flowing materials. It is well suited for the heat sensitive materials.

These dryers may also be operated batchwise. A charge of wet feed material in a perforated container attached to the bottom of a fluidising chamber is fluidised, heated until dry and then discharged. Units of this type have replaced tray dryers in many processes.

Tunnel Dryer :

The continuously operated direct type tunnel dryer is shown in Fig. 6.16. The dryer is built in the form of a long tunnel. It is provided with inlet and outlet for air. A fan and heating coils

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are incorporated in the tunnel as shown in Fig. 6.16. The material to be dried is filled in trays and trucks loaded with these trays move progressively through the tunnel in contact with a current of hot gas to evaporate the moisture. The air flow in a tunnel dryer can be totally concurrent, countercurrent or a combination of both with respect to the material flow. The wet material enters at one end and dried product leaves at the other end. These dryers are generally employed for drying of all forms of particulate solids and large solid objects, on a large scale. For relatively low temperature operation, steam heated air is used as the drying medium/drying gas, while for high temperatures, flue gas is used as the drying medium/drying gas.

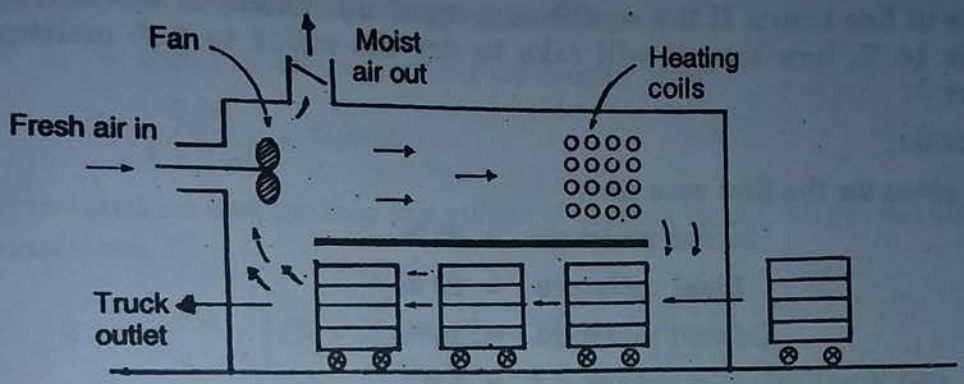

Fig. 6.16 : Tunnel dryer

It is often used for drying of pottery, ceramic products, paraffin wax, etc. which require slow rate of drying.

Pneumatic (flash) Dryer :

In a flash dryer, drying is carried out in a very short span of time - 0.5 to 3 seconds. A pneumatic (flash) dryer consists of a long duct, a suitable feeder, a heater, a cyclone collector and a bag filter for recovery of solids from the gas and blowers to propel the gas/air. The pneumatic conveying duct is the heart of the system wherein drying operation is carried out. The hot gas is a conveying medium-flowing rapidly with velocity of the order ~ 25 m/s in which granular free flowing solid materials are dispersed.

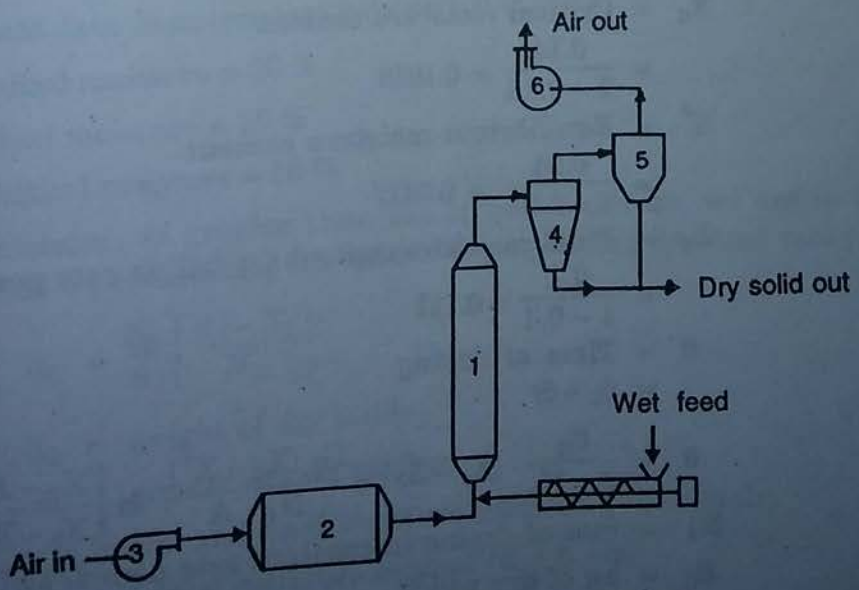

Fig. 6.17 : Pneumatic (Flash) dryer

1 : drying duct, 2 : heater, 3 and 6 : blower, 4 : cyclone and 5 : bag filter

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Finely powdered wet material is introduced into the hot gas (air stream) with the help of a screw conveyor. The material is pneumatically conveyed through the duct and during its conveyance, moisture is removed. The dried material is separated from the air stream in a cyclone as well as in a bag filter. The product from the bottom of collecting equipments is transported to a silo with the help of a screw conveyor (not shown).

Due to very short residence time, it is used for drying of heat-sensitive materials.

SOLVED EXAMPLES

Ex. 6.1 : A wet solid is to be dried from 35 % to 10 % moisture under constant drying conditions in five hours. If the equilibrium moisture content is 4 % and the critical moisture content is 14 %, how long it will take to dry the solids to 6 % moisture under the same conditions ?

Solution :

Data given for the first case are :

Initial moisture = 35 %

Final moisture = 10 %

Critical moisture = 14 %

and Equilibrium moisture = x = 4 %

Time required for drying solids from 35 % to 10 % moisture = 5 h

It is clear from the values of the moisture content given above (final moisture and critical moisture) that the period of five hours will be the time required during the constant rate period plus the time required during the falling rate period.

Initial moisture of solid = 35% by wt.

Weight fraction of moisture = 35/100 = 0.35

X_1 = Initial moisture content (on the dry basis)

$$X_1 = \frac{x}{1-x}$$

$$= \frac{0.35}{1-0.35} = 0.5385$$

X_c = Critical moisture content

$$= \frac{0.14}{1-0.14} = 0.1628$$

X^* = Equilibrium moisture content

$$= \frac{0.04}{1-0.04} = 0.0417$$

X_2 = Final moisture content (in falling rate period)

$$= \frac{0.1}{1-0.1} = 0.111$$

θ = Time of drying

$= \theta_c + \theta_f$

$$\theta = \frac{S_S}{A \cdot N_c} [X_1 - X_c] + \frac{S_S (X_c - X^*)}{N_c \cdot A} \ln \left[\frac{X_c - X^*}{X_2 - X^*}\right]$$

where,

N_c = rate of drying in constant rate period, kg/(m^2.h)

S_S = kg of dry solids

A = area of drying surface, m^2

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The above equation can be written as

$$\theta = \frac{S_S}{A \cdot N_c}\left[(X_1 - X_c) + (X_c - X^*) \ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right]$$

Putting the values of θ, X_1, X_2, X_c and X^*,

$$5 = \frac{S_S}{N_c \cdot A}\left[(0.5385 - 0.1628) + (0.1628 - 0.0417) \ln\left(\frac{0.1628 - 0.0417}{0.111 - 0.0417}\right)\right]$$

$$\therefore \quad \frac{S_S}{N_c \cdot A} = 11.28$$

For the second case, the final moisture is given as 6 %

X_2 = Final moisture content for second case

$$= \frac{0.06}{1 - 0.06} = 0.0638$$

Using the above equation and putting the values of various terms, we get the time of drying from 35 % to 6 % moisture,

$$\theta = \frac{S_S}{N_c \cdot A}\left[(X_1 - X_c) + (X_c - X^*) \ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right]$$

$$\theta = (11.28)\left[(0.5385 - 0.1628) + (0.1628 - 0.0417) \ln\left(\frac{0.1628 - 0.0417}{0.0638 - 0.0417}\right)\right]$$

$$\theta = 6.56 \text{ h} \quad \text{... Ans.}$$

Ex. 6.2 : A 100 kg bath of granular solids containing 30 % moisture is to be dried in a tray dryer to 16 % moisture by passing a current of air at 350 K across its surface at a velocity of 1.8 m/s. If the constant rate of drying under these conditions is 0.7×10^{-3} kg/(m^2·s) and the critical moisture content is 15 %, calculate the drying time.

Drying surface = 0.03 m^2/kg dry weight.

Solution :

Moisture content data from constant rate period is :

Initial moisture = 30 %

Final moisture = 16 %

Critical moisture = 15 %

As the final moisture is greater than the critical moisture, we are in the constant rate period. The equation for calculating the time of drying during constant rate period is :

$$\theta_c = \frac{S_S}{A}\left[\frac{X_1 - X_2}{N_c}\right]$$

where, S_S = weight of dry solids

A = area of drying surface

and X_1 and X_2 are initial and final moisture contents respectively.

X_1 = Initial moisture content (dry basis)

$$= \frac{0.3}{1 - 0.3} = 0.428$$

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X_2 = Final moisture content

$$= \frac{0.16}{1-0.16}$$

$$= 0.19 \frac{\text{kg moisture}}{\text{kg dry solid}}$$

Surface of drying is given as :

= 0.03 m^2 per kg dry solid weight, therefore,

$$\frac{A}{S_S} = 0.03$$

$$\frac{S_S}{A} = \frac{1}{0.03} = 33.33$$

N_c = Rate of drying during constant rate period

$= 0.7 \times 10^{-3}$ kg/(m^2·s)

$= 0.7 \times 10^{-3} \times 3600$ kg/(m^2·h)

$= 2.52$ kg/(m^2·h)

$$\theta_c = \frac{S_S}{A} \times \left[\frac{X_1 - X_2}{N_c}\right]$$

Putting the values of X_1, X_2, N_c and $\frac{S_S}{A}$ in the above equation,

$$\theta_c = 33 \times \left[\frac{0.428 - 0.19}{2.52}\right] = 3.15 \text{ h}$$

Time of drying = 3.15 h ... Ans.

Ex. 6.3 : Slabs of paper pulp 100 cm × 100 cm × 1.5 cm are to be dried under constant drying conditions from 67 % to 30 % moisture. The value of equilibrium moisture for the material is 0.5 %. If the critical moisture content is 60 % and the rate of drying at the critical point is 1.5 kg/(m^2·h), calculate the drying time. The dry weight of each slab is 2.5 kg. All moisture contents are on weight basis. The falling rate may be assumed to be linear.

Solution : Considering drying from the two big faces,

$A = 1 \times 1 \times 2 = 2 \text{ m}^2$

S_S = dry weight of slab = 2.5 kg

Initial moisture = 67%, i.e., x = 0.67 (wet basis)

$$\text{Initial moisture content} = X_1 = \frac{x}{1-x} = \frac{0.67}{1-0.67}$$

$$= 2.03 \frac{\text{kg moisture}}{\text{kg dry solid}}$$

$$\text{Final moisture content} = X_2 = \frac{0.30}{1-0.30} = 0.428$$

$$\text{Equilibrium moisture content} = X^* = \frac{0.005}{1-0.005} = 0.005025$$

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$$\text{Critical moisture content} = X_c = \frac{0.60}{1-0.60} = 1.5$$

N_c = Constant rate of drying

$= 1.5 \text{ kg/(m}^2\text{·h)}$

$$\text{Total time of drying} = \theta = \theta_c + \theta_f$$

$$\theta = \frac{S_S}{A \cdot N_c}\left[(X_1 - X_c) + (X_c - X^*) \ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right]$$

$$= \frac{2.5}{2 \times 1.5}\left[(2.03 - 1.5) + (1.5 - 0.005025) \ln \frac{(1.5 - 0.005025)}{(0.428 - 0.005025)}\right] = 2.01 \text{ h}$$

Time of drying = 2.01 h ... **Ans.**

Ex. 6.4 : 1000 kg (dry mass) of a non-porous solid is dried under constant drying conditions with an air velocity of 0.75 m/s. The area of drying surface is 55 m^2. If the initial rate of drying is 0.3 g/(m^2.s), how long it will take to dry a material from 0.15 to 0.025 kg water/kg dry solid ? The critical moisture content is 0.125 kg water/kg dry solid. Assume that the falling rate is linear. The equilibrium moisture content may be assumed to be zero.

If the air velocity is increased to 4 m/s, what will be the anticipated saving in drying time ?

Assume that the rate of evaporation in constant rate period is proportional to the air velocity raised to the power of 0.80.

Solution : Basis : 1000 kg of dry mass of non-porous solid.

$S_S = 1000 \text{ kg}$

Drying surface = $A = 55 \text{ m}^2$

Initial moisture content of the solid = $X_1 = 0.15$.

Final moisture content of the solid = $X_2 = 0.025$.

Critical moisture content of the solid = $X_c = 0.125$.

From the values of moisture content cited above, it is clear that the solid is to be dried in both the constant rate period and falling rate period.

In the constant rate period, drying is done from $X = X_1 = 0.15$ to $X = X_c = 0.125$ and in the falling rate period, drying is done from $X_c = 0.125$ to $X_2 = 0.025$.

As the falling rate is linear, we can calculate the time required for drying by the following equation :

$$\theta = \theta_c + \theta_f = \frac{S_S}{A N_c}\left[(X_1 - X_c) + (X_c - X^*) \ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right] \quad \ldots (1)$$

Initial rate of drying = 0.3 g/(m^2.s)

$= 3 \times 10^{-4}$ kg/(m^2.s)

The initial rate of drying is the rate of drying solid valid from $X = X_1 = 0.15$ to $X = X_c = 0.125$.

$\therefore$ $N_c = 3 \times 10^{-4}$ kg/(m^2.s)

$S_S = 1000$ kg, $X_1 = 0.150$, $X_c = 0.125$

$X^* = 0$, $X_2 = 0.025$, $A = 55 \text{ m}^2$

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$$\theta = \frac{1000}{55 \times 3 \times 10^{-4}} \left[(0.15 - 0.125) + (0.125 - 0.0) \ln \left(\frac{0.125 - 0.0}{0.025 - 0.0} \right) \right]$$

$$= 13708 \text{ s} = 3.808 \text{ h}$$

Time required for drying the solid when velocity of air is 0.75 m/s = 3.808 h ≈ 3.81 h ... Ans.

Given : In the constant rate period :

Rate of evaporation $\propto$ (velocity)$^{0.8}$

Rate of drying = Rate of evaporation per unit area

$$N_{c1} \propto u_1^{0.80} \qquad N_{c1} = k u_1^{0.80}$$

$$N_{c2} \propto u_2^{0.80} \qquad N_{c2} = k u_2^{0.8}$$

$$\frac{N_{c2}}{N_{c1}} = \frac{u_2^{0.8}}{u_1^{0.8}} = \left(\frac{u_2}{u_1} \right)^{0.8}$$

$$N_{c2} = N_{c1} \cdot \left(\frac{u_2}{u_1} \right)^{0.8}$$

$$= 3 \times 10^{-4} \times \left(\frac{4}{0.75} \right)^{0.8} = 1.145 \times 10^{-3} \text{ kg/(m}^2\text{.s)}$$

Rate of drying in constant rate period : When the air velocity is 4 m/s, the rate of drying in the constant rate period is 1.143×10^{-3} kg/(m^2.s).

Calculate θ in this case, using Equation (1) with the same moisture content data.

$$\therefore \quad \theta = \frac{1000}{55 \times 1.143 \times 10^{-3}} \left[(0.15 - 0.125) + (0.125 + 0.0) \ln \left(\frac{0.125 - 0.0}{0.025 - 0.0} \right) \right]$$

$$= 3598 \text{ s} \approx 1 \text{ h}$$

Saving in time by using air at velocity of 4 m/s instead of 0.75 m/s = 3.81 – 1 = 2.81 h ... Ans.

Ex. 6.5 : A batch of solid is to be dried from 28 % to 6 % moisture, on wet basis. The initial weight of the solid is 380 kg and the drying surface is 0.15 m^2/40 kg dry weight. The critical moisture content is 18% on a dry basis and the constant drying rate is 0.32 kg/(m^2.h). For the falling rate period, the following data are available.

Moisture content, % dry basis	25	21.9	19	16	13.6	11	8.2	7.5	6.4
Rate of drying, kg/(m^2.h)	0.30	0.27	0.24	0.21	0.18	0.15	0.07	0.044	0.025

Solution : Basis : 380 kg of wet solid.

Initial moisture in wet solid = $0.28 \times 380 = 106.4$ kg

Dry solid in wet solid = $S_S = 380 - 106.4 = 273.6$ kg

Initial moisture content of the wet solid = $X_1 = 106.4/273.6 = 0.39$

Final moisture in dried solid = 6 % on wet basis

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Let x be the moisture in dried solid.

Amount of dried solid = $273.6 + x$ kg

The weight % moisture on a wet basis is given by

$$6 = \left(\frac{x}{273.6 + x}\right) \times 100$$

$$\therefore \quad x = 17.46 \text{ kg}$$

Final moisture content of dried solid = $X_2 = 17.46/273.6 = 0.0638 \approx 0.064$.

Given : Drying surface = 0.15 m^2/40 kg dry weight.

$$\therefore \quad \text{Drying surface} = A = \frac{0.15}{40} \times 273.6 = 1.026 \text{ m}^2$$

Critical moisture on a dry basis = 28 %

$$\frac{\text{kg moisture}}{\text{kg dry solid}} \times 100 = 28$$

kg moisture/kg dry solid of wet solid = 28/100 = 0.28

Critical moisture content of solid = $X_C = 28/100 = 0.28$

Rate of drying in the constant rate period = $N_C = 0.32$ kg/(m^2.h)

Time required for drying = $\theta_C + \theta_f$

where, θ_C = time of drying during the constant rate period

θ_f = time of drying during the falling rate period

θ_C = time required to dry the solid from X_1 to X_c

$$\theta_C = \frac{S_S}{AN_C} [X_1 - X_C]$$

$$= \frac{273.6}{1.026 \times 0.32} [0.39 - 0.18]$$

$$= 175 \text{ h}$$

To determine the time required from X_C to X_2, we will adopt a procedure of graphical integration.

We have, $$d\theta = \frac{-S_S}{A} \frac{dX}{N}$$

$$\therefore \quad \theta_f = \frac{-S_S}{A} \int_{X_C}^{X_2} dX/N = \frac{S_S}{A} \int_{X_2}^{X_C} dX/N = \frac{S_S}{A} \int_{0.064}^{0.18} dX/N$$

Convert the data and tabulate.

Moisture content, % dry basis	25	21.9	19	16	13.6	11	8.2	7.5	6.4
X, % dry basis/100	0.25	0.219	0.19	0.16	0.136	0.11	0.082	0.075	0.064
N	0.30	0.27	0.24	0.21	0.18	0.15	0.07	0.044	0.025
1/N	3.33	3.7	4.17	4.76	5.56	6.67	14.3	22.73	40

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From Fig. E 6.5 :

Area under the curve between $X_C = 0.18$ and $X_2 = 0.064$ is 840 mm² = 8.40 cm².

The time required in the falling rate period to dry the solid from X_C to X_2 is

$$\theta_f = \frac{S_S}{A} \text{ (Area} \times \text{scale y-axis} \times \text{scale x-axis)}$$

$$= \frac{273.6}{1.026}\left(8.40 \times \frac{10}{2} \times \frac{0.05}{2}\right) = 280\,\text{h}$$

$$\text{Total time} = \theta_c + \theta_f = 175 + 280 = 455\,\text{h}$$

Time required for drying the solids from $X_1 = 0.39$ to $X_2 = 0.064$ (for the batch of 380 kg) = 455 h ... Ans.

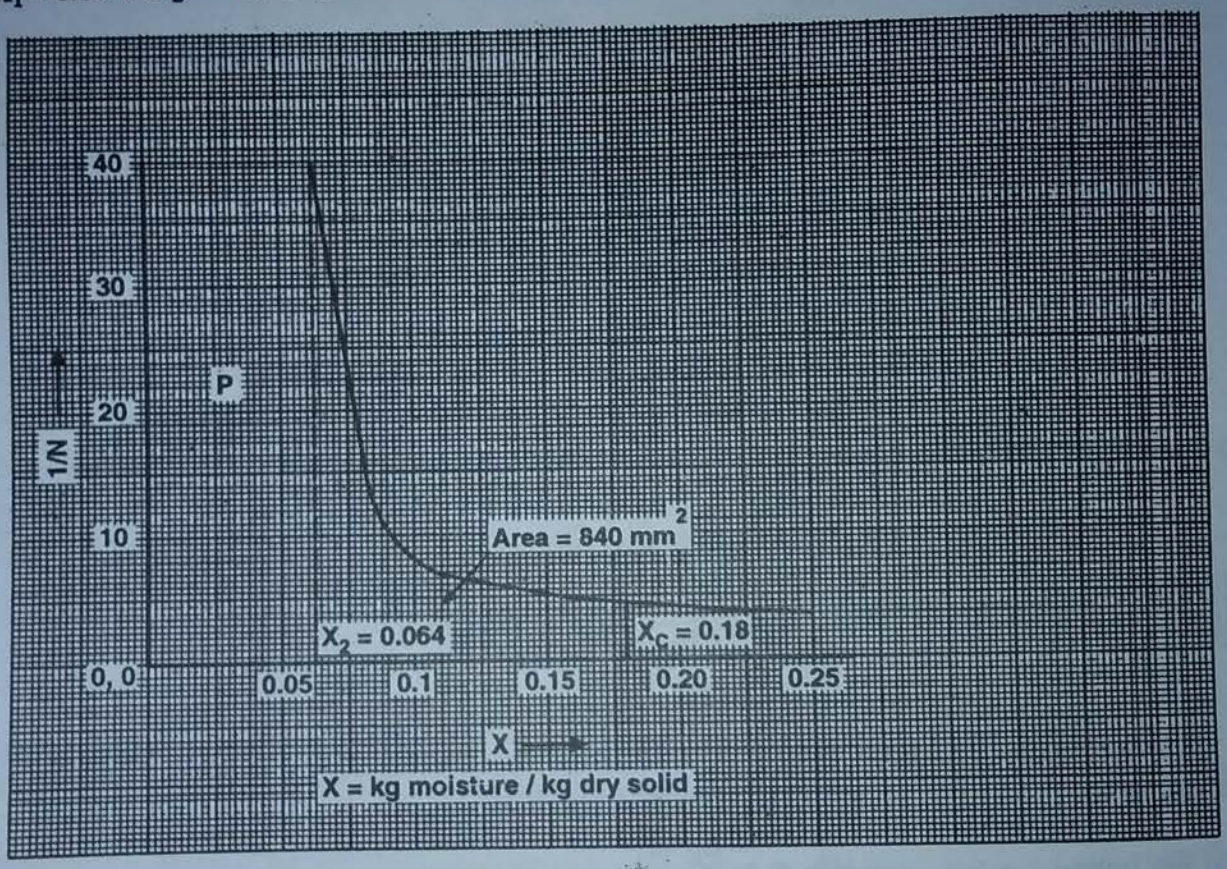

Fig. E 6.5

Ex. 6.6 : It is desired to dry a sheet material from 55 % to 4 % moisture content. The sheets are 125 cm by 140 cm by 5 cm. The drying rate during the constant rate period is 1.5 g/(cm².h). The critical moisture content is 26 % and the equilibrium moisture content is negligible. If the material is dried from both sides and has a bone dry density of 4000 kg/m³, estimate the time required for drying assuming the falling rate to be linear.

Solution : Considering drying from two sides of sheets,

$$A = 125 \times 140 \times 2 = 35000\,\text{cm}^2 = 3.5\,\text{m}^2$$

Sheet size : 125 cm × 140 cm × 5 cm.

$$\therefore \quad \text{Volume of sheet} = 1.25 \times 1.40 \times 0.5 = 0.0875\,\text{m}^3$$

Bone dry density of material of sheet = 4000 kg/m³

S_S = dry weight of the sheet material = 4000 × 0.0875 = 350 kg

Initial moisture of sheet = 55 %

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$$\text{Initial moisture content} = X_1 = \frac{0.55}{1-0.55} = 1.222$$

$$\text{Final moisture of sheet} = 4\%$$

$$\text{Final moisture content} = X_2 = \frac{0.04}{1-0.04} = 0.0417$$

$$\text{Equilibrium moisture of sheet} = \text{negligible}$$

$$\text{Equilibrium moisture content} = X^* = 0/1 = 0$$

$$\text{Critical moisture of sheet} = 26\,\%$$

$$\text{Critical moisture content} = \frac{0.26}{1-0.26} = 0.351$$

Drying rate in the constant rate period $= N_c = 1.5\ g/cm^2.h = 15\ kg/(m^2.h)$

Considering the falling rate to be linear, the total time required for drying is given by

$$\theta = \theta_c + \theta_f$$

$$\theta = \frac{S_S}{A\,N_c}\left[(X_1 - X_c) + (X_c - X^*)\ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right]$$

$$= \frac{350}{3.5 \times 15}\left[(1.222 - 0.351) + (0.351 - 0)\ln\left(\frac{0.351 - 0}{0.041 - 0}\right)\right]$$

$$= 10.8\ h \quad \text{... Ans.}$$

Ex. 6.7 : A slab with wet weight of 5 kg originally contains 50 % moisture (on wet basis). The slab is 600 by 900 by 75 mm thick. The equilibrium moisture content is 5% of the total weight when in contact with air at 20 °C and 20 % humidity. The drying rate is given below for contact with air of the above quality at a definite velocity. Drying is from one face only. How long it will take to dry the slab to 15 % moisture content (wet basis) ?

Wet slab, kg	9.1	7.2	5.3	4.2	3.3	2.8	2.5
Drying rate, kg/(m².h)	4.9	4.9	4.4	3.9	3.4	2.0	1.0

Solution : Basis : 5 kg of wet slab.

To determine the time required for drying, we have to make use of the data given.

Moisture originally present in the slab = 50 % (wet basis)

∴ Moisture in the slab = $0.50 \times 5 = 2.5$ kg

Dry solid in the slab, by difference = $100 - 50 = 50\%$

Dry solid in the slab = $0.50 \times 5 = 2.5$ kg (or $5 - 2.5$).

We are provided with the rate of drying as a function of weight of the wet slab. The dry solid of the wet slab is constant throughout and is equal to 2.5 kg throughout the run. Over the entire run conducted only water content varies with time of drying. So convert the data in a required form, i.e., rate of drying as a function of moisture content of the slab on a dry basis (X). In our case, only weight of the wet slab is to be converted to X.

Data no. 1 : Weight of wet slab = 9.1 kg and rate of drying = 4.9 kg/(m².h)

Weight of wet slab = 9.1 kg

Dry solid in slab = 2.5 kg

∴ Weight of moisture in slab = $9.1 - 2.5 = 6.6$ kg

∴ X = kg moisture/kg dry solid, in slab = $6.6/2.5 = 2.64$

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Data no. 2 : Weight of wet slab = 7.2 kg and rate of drying = 4.9 kg/(m².h)

Weight of wet slab = 7.2 kg

Dry solid in slab = 2.5 kg

Weight of moisture in slab = 7.2 – 2.5 = 4.7 kg

∴ X = kg moisture/kg dry solid in slab = 4.7/2.5 = 1.88

In this way, convert the data and tabulate.

Weight of wet slab, kg	9.1	7.2	5.3	4.2	3.3	2.8	2.5
Weight of dry solid in slab, kg	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Weight of moisture in slab, kg	6.6	4.7	2.8	1.7	0.8	0.3	0
X = kg moisture/kg dry solid	2.64	1.88	1.12	0.68	0.32	0.12	0
Rate of drying, kg/(m².h)	4.9	4.9	4.4	3.9	3.4	2.0	1.0

We have to find out the time required to dry the slab to 15 % moisture content on wet basis.

$$\text{Final moisture content of the slab on dry basis} = X_2 = \frac{0.15}{1-0.15} = 0.176$$

Initial moisture content of the slab $= X_1 = 1.0$.

From the values of X_1 and X_2 and data tabulated, it seems that drying takes place in the falling rate period only.

For calculating the time required during this period, we can adopt an analytical procedure if the falling rate is linear. If the falling rate period is not linear then we have to adopt a procedure of graphical integration.

So for doing this, first plot a graph of rate of drying v/s moisture content (X).

It is clear from the Fig. E 6.7 (a) that

(i) X_1 to X_2 are in falling rate period – so drying is in the falling rate period.

(ii) Falling rate is not linear– since the graph is curved in this period and not a straight line.

So for evaluating the time for drying, we have to adopt a procedure of graphical integration.

$$\text{Rate of drying} = \frac{-S_S}{A}\frac{dX}{d\theta}$$

$$d\theta = \frac{-S_S}{A}\frac{dX}{N}$$

$$\theta = \frac{-S_S}{A}\int_{X_1}^{X_2}\frac{dX}{N} = \frac{S_S}{A}\int_{X_2}^{X_1}\frac{dX}{N} = \frac{S_S}{A}\int_{0.176}^{1.0}\frac{dX}{N}$$

So plot 1/N against X and measure the area under the curve between X = 1.0 and X = 0.176 and evaluate θ- the time required for drying the wet slab.

N	4.9	4.9	4.4	3.9	3.4	2.0	1.0
1/N	0.204	0.204	0.23	0.256	0.294	0.5	1.0
X	2.64	1.88	1.12	0.68	0.32	0.12	0

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Refer Fig. E 6.7 (b) :

Area under the curve between $X_1 = 1.0$ and $X_2 = 0.176$ is 893 mm^2 = 8.93 cm^2

$$\text{Area} = 8.93 \text{ cm}^2$$

We have, S_S = weight of dry solid in slab = 2.5 kg

Size of slab : 600 mm × 900 mm × 75 mm

Given : Drying is from one face only.

A = area of drying surface = $600 \times 900 \text{ mm}^2$

$$A = 0.54 \text{ m}^2$$

$$\theta = \frac{-S_S}{A} \int_{1.0}^{0.176} dX/N = \frac{S_S}{A} \int_{0.176}^{1.0} dX/N$$

$$= \frac{2.5}{0.54} \text{ [Area} \times \text{scale y-axis} \times \text{scale x-axis]}$$

$$= \frac{2.5}{0.54} [8.93 \times (0.1/1) \times (0.5/2)]$$

$$= 1.033 \text{ h}$$

Time required for drying = 1.033 h ... **Ans.**

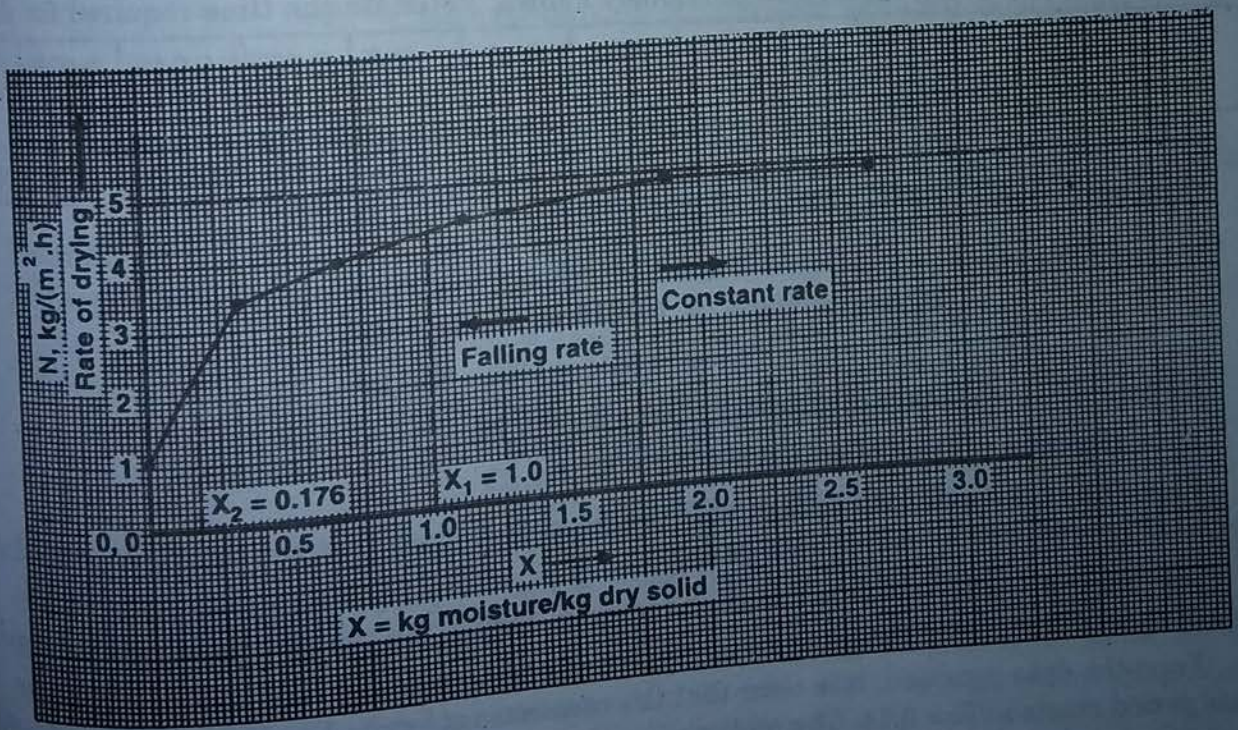

Fig. E 6.7 (a)

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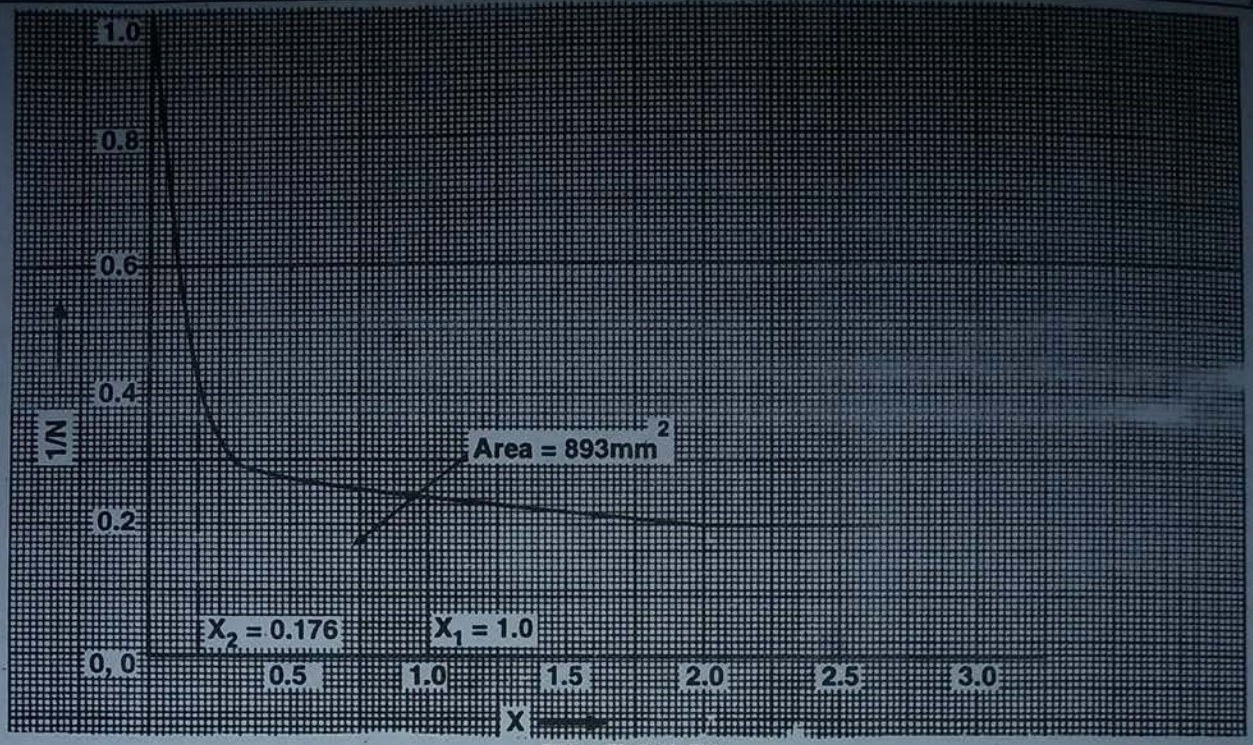

Fig. E 6.7 (b)

Ex. 6.8 : 1400 kg (bone dry) of a granular solid is to be dried under constant drying conditions from a moisture content of 0.2 kg/kg dry solid to a final moisture content of 0.02 kg/kg dry solid. The drying surface is given as 0.0615 m^2/kg dry solid. Under the same conditions, the following rates were previously known. Estimate the time required for drying.

Moisture content, X = kg/kg dry solid	0.30	0.20	0.14	0.096	0.056	0.046	0.026	0.016
Rate of drying, kg/(m^2.h)	1.71	1.71	1.71	1.46	1.29	0.88	0.54	0.376

Solution : Basis : 1400 kg bone dry solid.

$$S_S = 1400 \text{ kg}$$

$$\text{Drying surface} = 0.0615 \text{ m}^2\text{/kg dry solid}$$

$$\therefore \quad A = 0.0615 \times 1400 = 86.1 \text{ m}^2$$

Initial moisture content of granular solid = $X_1 = 0.20$

Final moisture content of granular solid = $X_2 = 0.02$

From the values of X_1 and X_2 and data provided, it is clear that the drying is carried out in constant rate as well as falling rate period.

(1) We can easily calculate the time required during the constant rate period by making use of equation for the same.

From the data provided, it is clear that the constant rate period ends at and thus the falling rate period starts at X = 0.14. The moisture content at which this occurs is called as the critical moisture content.

So the critical moisture content of granular solid = $X_c = 0.14$

Rate of drying during constant rate period (from the data) = $N_c = 1.71$ kg/(m^2.h).

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Time required during the constant rate period :

$$\theta_c = \frac{S_S}{AN_c}[X_1 - X_c]$$

$$= \frac{1400}{86.1 \times 1.71}[0.20 - 0.14]$$

$$= 0.57 \text{ h}$$

Now we have to determine the time required for drying the granular solid to reducing the moisture content from 0.14 to 0.02, i.e., the time required during the falling rate period.

We can determine the same by making use of known equation only if the falling rate is linear. If the falling rate is not linear then we have to adopt a procedure of graphical integration for the evaluation of time.

Check by plotting N v/s X whether the falling rate is linear or not.

Refer Fig. E 6.8 (a).

The plot of N v/s X indicates that the falling rate is not linear, i.e., rate of drying in the falling rate period is not linear.

So we have to evaluate the time required for drying in the falling rate period by a graphical integration.

We have, $$d\theta = \frac{-S_S}{A}\frac{dX}{N}$$

$$\therefore \quad \theta_f = \frac{-S_S}{A}\int_{X_c}^{X_2}\frac{dX}{N} = \frac{S_S}{A}\int_{X_2}^{X_C}\frac{dX}{N} = \frac{S_S}{A}\int_{0.02}^{0.14} dX/N$$

θ_f – time required to dry the solid to $X_2 = 0.02$ in falling rate period.

Convert N into 1/N, tabulate 1/N and X, and plot 1/N against X.

X	0.30	0.20	0.14	0.096	0.056	0.046	0.026	0.016
N	1.71	1.71	1.71	1.46	1.29	0.88	0.54	0.376
1/N	0.585	0.585	0.585	0.685	0.775	1.136	1.852	2.66

From Fig. E 6.8 (b).

The area under the curve between $X = X_c = 0.14$ and $X = X_2 = 0.02$ is 815 mm^2 (8.15 cm^2).

$$\therefore \quad \theta_f = \frac{S_S}{A}(\text{Area} \times \text{scale y-axis} \times \text{scale x-axis})$$

$$= \frac{1400}{86.1}\left(8.15 \times \frac{0.5}{1} \times \frac{0.1}{4}\right)$$

$$= \frac{1400}{86.1}(0.101875) = 1.6565 \approx 1.66 \text{ h}$$

Total time required for drying $= \theta_c + \theta_f = 0.57 + 1.66 = 2.23$ h ... **Ans.**

Extra : OR : Refer Fig. E 6.8 (b) :

We can evaluate the time required for drying the granular solid from the initial moisture content $X = X_1 = 0.20$ to the final moisture content $X = X_2 = 0.02$ by a graphical integration by measuring the area under the curve between $X = X_1 = 0.2$ and $X = X_2 = 0.02$.

Area under the curve between $X = X_1 = 0.20$ and $X = X_2 = 0.02$

$$\text{Area} = 815 + 280 = 1095 \text{ mm}^2 = 10.95 \text{ cm}^2$$

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Total time required for drying $= \frac{S_S}{A}$ (Area × scale y-axis × scale x-axis)

$$= \frac{1400}{86.1}\left(10.95 \times \frac{0.5}{1} \times \frac{0.10}{4}\right)$$

$$= 2.226 \text{ h}$$ ...Ans.

So answer by both methods is almost same.

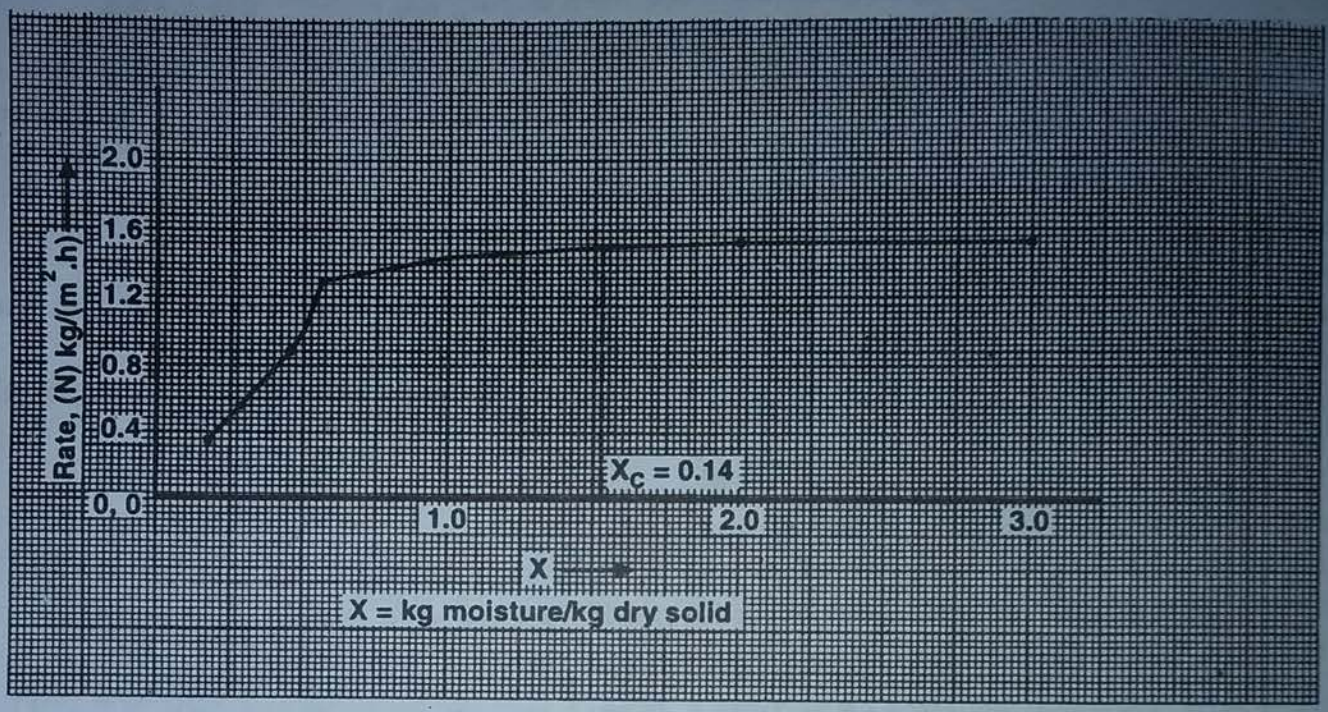

Fig. E 6.8 (a)

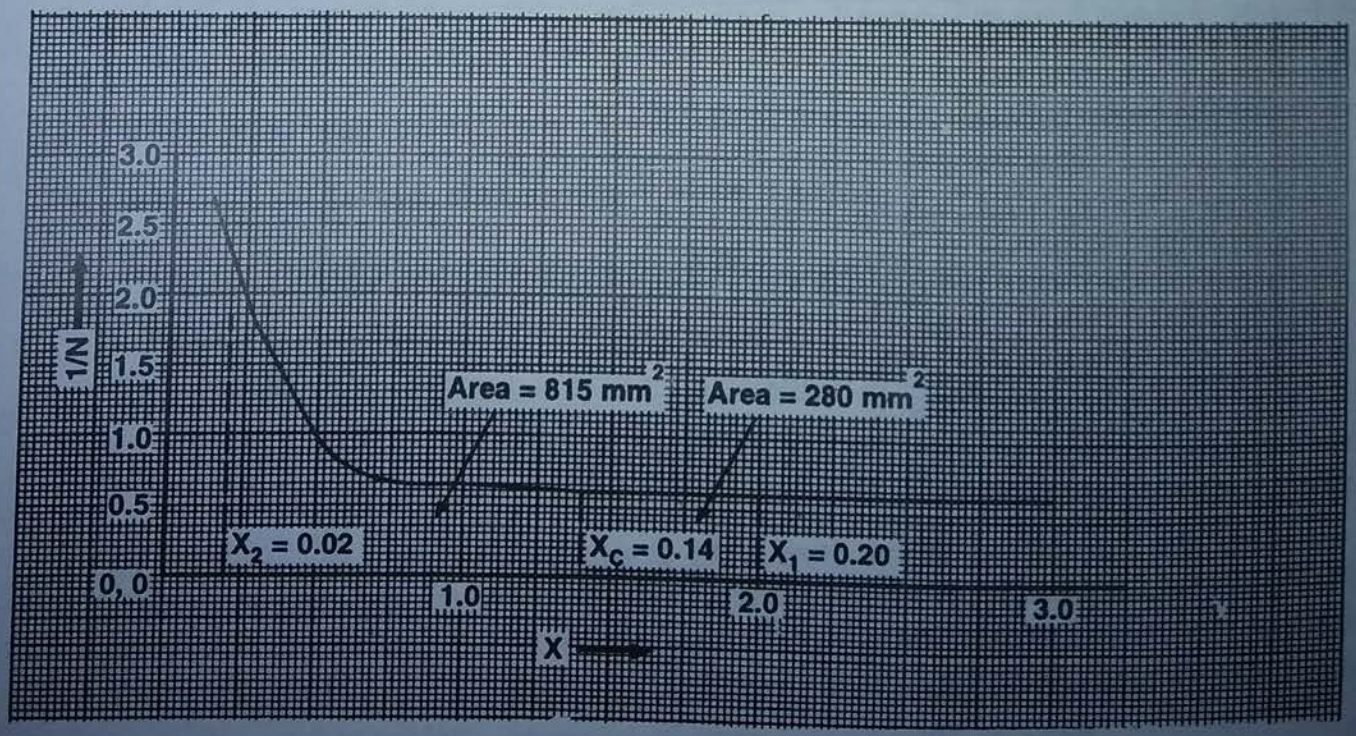

Fig. E 6.8 (b)

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Ex. 6.9 : A sample of a porous, manufactured sheet material of mineral origin was dried, in a laboratory dryer, from both sides by cross-circulation of air. The sample was 0.3 m square and 6 mm thick. The edges of the sample were sealed. The air velocity over the surface was 3 m/s. The dry bulb temperature of air was 52 °C and wet bulb temperature was 21 °C. There were no radiation effects. The solid lost moisture at a constant rate of 7.5×10^{-5} kg/s until the critical moisture content of 15 % (on wet basis) was reached. The rate of drying in the falling rate period was found to be linear with moisture content until the sample was dry. The equilibrium moisture was negligible and the dry weight of the sheet was 1.8 kg.

Calculate the time of drying sheets of this material 0.6 by 1.2 m by 12 mm thick from both sides from 25 to 2 % moisture on a wet basis using air having a dry bulb temperature of 66 °C but of the same absolute humidity at a linear velocity of 5 m/s. Assume that there is no change in the critical moisture content of sheet with changed drying conditions.

Hint : In constant rate period, rate of drying $\propto$ (velocity of air)$^{0.80} \cdot \Delta T$.

where ΔT = temperature difference between the air and surface.

For no radiation to the wet surface, the drying surface assumes the wet bulb temperature. The wet bulb temperature of air at 66 °C (of the same humidity as that of the air used for the lab test) is 24 °C.

Solution : (i) The data obtained with laboratory dryer :

Air velocity = u_1 = 3 m/s

Temperature of air = dry bulb temperature = 52 °C

Wet bulb temperature of air = 21 °C

For no radiation effect :

Temperature of drying surface = Wet bulb temperature of air = 21 °C

$\therefore$ $\Delta T_1 = T_{G1} - T_{S1}$ = $DB_1 - WB_1 = 52 - 21 = 31$ °C

Rate of evaporation = rate of moisture loss = 7.5×10^{-5} kg/s

Sample sheet size : 0.3 m square × 6 mm thick.

Area of one surface of sheet = $0.3 \times 0.3 = 0.09$ m^2

Given : Drying is form both sides

$\therefore$ Area of drying surface = $A_1 = 2 \times 0.09 = 0.18$ m^2

In the constant rate period :

Rate of drying = rate of evaporation/area

$N_{c1} = 7.5 \times 10^{-5}/0.18 = 4.167 \times 10^{-4}$ kg/(m^2.s)

Critical moisture content on wet basis = 15 %

Critical moisture content = $X_c = 0.15/0.85 = 0.1765$

Weight of dry sheet = dry weight of sheet = 1.8 kg

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Calculation of X_c (alternatively) :

Let x be the kg moisture in sample sheet then

$$\left(\frac{x}{1.8+x}\right)\times 100 = 15$$

$$x = 0.31765 \text{ kg}$$

$\therefore$ Critical moisture content $= X_c = 0.31765/1.8 = 0.1765$

You can adopt any method to get X_c.

We have to make use of X_c, N_{c1} etc. to calculate the time required for drying in the actual case (practical case).

(ii) Drying of sheet in practice : Here the critical moisture content is the same as that in the laboratory test.

Critical moisture of sheet $= X_c = 0.1765$

The sheet is to be dried from 25 to 2 % moisture content on wet basis.

Initial moisture content of sheet $= X_1 = 25/(100-25) = 0.3333$

Final moisture content of sheet $= X_2 = 2/(100-2) = 0.02041$

Given : sheet size : 0.6 m × 1.2 m × 12 mm thick.

Sample sheet size : 0.3 m × 0.3 m × 6 mm thick.

$$\text{Dry weight of sheet to be dried} = \frac{\text{dry weigh of sample sheet}}{\text{volume of sample sheet}} \times \text{volume of sheet to be dried}$$

$\therefore$ Dry weight of sheet to be dried $= 1.8\left(\frac{0.6\times 1.2\times 0.012}{0.3\times 0.3\times 0.006}\right)$

$$S_{S2} = 28.8 \text{ kg}$$

Given : Drying is from both sides.

$\therefore$ Area of drying surface $= 2\,(0.6\times 1.2) = 1.44 \text{ m}^2$

Air velocity $= u_2 = 5$ m/s

Given : Air temperature $= T_{G2} = 66\,^{\circ}\text{C}$ and

Wet-bulb temperature of air $= T_{S2} = 24\,^{\circ}\text{C}$

$$\Delta T_2 = T_{G2} - T_{S2} = DB_2 - WB_2 = 66 - 24 = 42\,^{\circ}\text{C}$$

(for no radiation effect, i.e., for no radiation to the wet surface, the temperature of drying surface is equal to the wet bulb temperature of the air.)

Let N_{c2} – rate of drying in the constant rate period (in the actual case)

We have provided with :

In the constant rate period,

Rate of drying $\propto$ (velocity of air)$^{0.80} \cdot \Delta T$

i.e. $N_c \propto (u)^{0.80}\,\Delta T$

$\therefore$ $N_{c2} = k\,(u_2)^{0.80}\,\Delta T_2$

and $N_{c1} = k\,(u_1)^{0.80}\,\Delta T_1$

$\therefore$ $$N_{c2} = N_{c1}\left(\frac{u_2}{u_1}\right)^{0.80}\frac{\Delta T_2}{\Delta T_1}$$

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$$N_{c1} = 4.167 \times 10^{-4} \text{ kg/(m}^2\text{.s)}, \; u_1 = 3 \text{ m/s}, \; u_2 = 5 \text{ m/s}$$

$$\Delta T_1 = 31\,^{\circ}\text{C}, \; \Delta T_2 = 42\,^{\circ}\text{C}$$

$$N_{c2} = 4.167 \times 10^{-4} \, (5/3)^{0.80} \times (42/31)$$

$$= 8.4955 \times 10^{-4} \text{ kg/(m}^2\text{.s)}$$

Given : Equilibrium moisture content is negligible $\therefore X^* = 0.0$.

We have $X_1 = 0.3333$, $X_2 = 0.02041$, and $X_c = 0.1765$.

As $X_2 < X_c$, the drying of sheet takes place in both the constant-and falling-rate periods.

So, the time required for drying the sheet from 25 to 2% moisture on wet basis is given by

$$\theta = \frac{S_S}{AN_c}\left[(X_1 - X_c) + (X_c - X^*)\ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right]$$

$A = 1.44 \text{ m}^2$, $N_c = N_{c2} = 8.4955 \times 10^{-4} \text{ kg/(m}^2\text{.s)}$

$S_S = S_{S2} = 28.8 \text{ kg}$, $X^* = 0$, $X_c = 0.1766$, $X_1 = 0.3333$, $X_2 = 0.02041$

$$\theta = \frac{28.8}{1.44 \times 8.4955 \times 10^{-4}}\left[(0.3333 - 0.1765) + (0.1765 - 0)\ln\left(\frac{0.1765 - 0}{0.02041 - 0}\right)\right]$$

$$= 12655 \text{ s} = 3.515 \text{ h}$$

Time of drying (from 25 to 2 % wet basis) = 3.515 h ... Ans.

Ex. 6.10 : A laboratory drying test was made on a 0.1 m^2 sample of a fibrous board like material. The sample was suspended from a balance, its edges were sealed, and drying occurred from the two large surfaces. The air had a dry-bulb temperature of 65 °C and a wet-bulb temperature of 29 °C. The air velocity past the sample was 1.5 m/s. During the test, the following weights are recorded at various times.

Time, h	Weight, kg	Time, h	Weight, kg	Time, h	Weight, kg
0	4.820	3	4.269	7	3.885
0.1	4.807	3.4	4.206	7.5	3.871
0.2	4.785	3.8	4.150	8.0	3.859
0.4	4.749	4.2	4.130	9.0	3.842
0.8	4.674	4.6	4.057	10	3.832
1.0	4.638	5.0	4.015	11	3.825
1.4	4.565	5.4	3.979	12	3.821
1.8	4.491	5.8	3.946	14	3.819
2.2	4.416	6.0	3.933	16	3.819
2.6	4.341	6.5	3.905		

The sample was then dried in an oven at 110 °C and dry mass was 3.765 kg.

(i) Plot the rate-of-drying curve.

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(ii) Estimate the time required to dry the same sheets form 20 to 2% moisture on wet basis using air of the same temperature and humidity but with 50% greater velocity, assuming that the critical moisture content remains unchanged. In the constant rate period, rate of drying is proportional to air velocity raised to the power of 0.80.

Solution : Basis : Fibrous sheet with 0.1 m^2 face surface with dry weight of 3.765 kg.

$$S_S = 3.765 \text{ kg}$$

Convert the data in the required form :

(i) The moisture content of the sheet on the dry basis is obtained from weights of sample at various values of time by dividing by the weight of dry material.

For $\theta = 0.0$ h : weight of sample = 4.820 kg

weight of moisture = $4.820 - 3.765 = 1.055$ kg

weight of dry sample = 3.765 kg

So, the moisture content of the sheet is $X = 1.055/3.765 = 0.2802$

For $\theta = 0.1$ h : weight of the sample = 4.807 kg

$\therefore$ $X = (4.807 - 3.765) = 0.27676$

So, the average moisture content during this $(0.10 - 0) = 0.1$ h of time is

$$\text{Average } X = \frac{0.2802 + 0.27676}{2} = 0.27848 = 0.2785$$

The rate of drying may be defined as :

$$\text{Rate} = N = \frac{W_\theta - W_{\theta + \Delta\theta}}{A\,\Delta_\theta}$$

where $\theta = 0.0$ h, $\theta + \Delta\theta = 0.10$ h $\therefore$ $\Delta_\theta = 0.10 - 0.0 = 0.10$ h

W_θ = weight of sample at $\theta = 4.820$ kg

$W_{\theta + \Delta\theta}$ = weight of sample at $\theta + \Delta\theta = 4.807$ kg

A = drying surface $= 2 \times 0.1 = 0.2$ m^2 (drying from two big faces)

$$N = \frac{4.8020 - 4.807}{0.2 \times 0.10} = 0.65 \text{ kg/(m}^2\text{.h)}$$

Or you can calculate the rate by

$$N = -\frac{S_S}{A}\frac{dX}{d\theta} = \frac{-3.765\,(0.27676 - 0.2802)}{0.2 \times 0.10} = 0.64758 \approx 0.6476$$

We will adopt this formula to calculate the rate.

In this way convert the data into (i) average moisture content and (ii) rate and tabulate.

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Time, h	Weight, kg	X		Average X	Rate (N)	1/N
0	4.820	0.2802				
			⇒	0.2875	0.6476	1.544
0.10	4.807	0.27676				
			⇒	0.2738	1.103	0.907
0.20	4.785	0.2709				
			⇒	0.2661	0.9036	1.107
0.40	4.749	0.2613				
			⇒	0.2513	0.9365	1.07
0.80	4.674	0.2414				
			⇒	0.2366	0.8942	1.12
1.0	4.638	0.2319				
			⇒	0.2222	0.9130	1.095
1.4	4.565	0.2125				
			⇒	0.2026	0.9271	1.0786
1.8	4.491	0.1928				
			⇒	0.1829	0.9365	1.068
2.2	4.416	0.1729				
			⇒	0.1630	0.9365	1.068
2.6	4.341	0.1530				
			⇒	0.1434	0.8990	1.112
3.0	4.269	0.1339				
			⇒	0.1255	0.7906	1.265
3.4	4.206	0.1171				
			⇒	0.1147	0.7012	1.426
3.8	4.150	0.1022				
			⇒	0.09957	0.2475	–
4.2	4.130	0.09694				
			⇒	0.0873	0.9121	1.096
4.6	4.057	0.07756				
			⇒	0.072	0.5271	1.897
5.0	4.015	0.06640				
			⇒	0.0616	0.450	2.222
5.4	3.979	0.05684				
			⇒	0.05245	0.4127	2.423
5.8	3.946	0.04807				
			⇒	0.0463	0.3247	3.0797
6.0	3.933	0.04462				
			⇒	0.0409	0.2801	3.57
6.5	3.905	0.03718				
			⇒	0.03454	0.1988	5.03
7.0	3.885	0.03190				
			⇒	0.0300	0.1412	7.082
7.5	3.871	0.02815				
			⇒	0.02656	0.1197	8.354
8.0	3.859	0.02497				
			⇒	0.0227	0.0425	23.53
10.0	3.842	0.02045				
			⇒	0.01912	0.04988	20.05
11	3.832	0.01780				
			⇒	0.01633	0.0551	18.149
12	3.821	0.01487				
			⇒	0.0146	0.04988	20.48
14	3.819	0.01434				
			⇒	0.01434	0.0	∞
16	3.819	0.01434				

From Fig. E 6.10 (a).

(i) The constant rate period ends at $X = X_c = 0.106$.

(ii) Rate of drying in constant rate period $= N_c = 0.92$ kg/(m^2.h).

(iii) The falling rate is linear from $X_c = 0.106$ to $X_f = 0$.

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(iv) Equilibrium moisture content = $X^* = 0.01434$ from the X values (see the tabulated X values). At $\theta = 14$ h as well as $\theta = 16$ h we have a constant value of X (0.01434 w.r.t. time so it represents the equilibrium value). OR Read X^* from Fig. E 6.10 (a), $X^* = 0.0143$. It is almost the same as X^* from the tabulated data.

(v) Estimation of time required for drying the same sheets from 20 to 2 % moisture (wet basis) in the actual practice :

Initial moisture content of sheet $= X_1 = 20/(100 - 20) = 0.20/0.80 = 0.25$

Final moisture content of sheet $= X_2 = 0.02/0.98 = 2/(100 - 2) = 0.02041$.

Here the critical moisture content is the same as that determined in the laboratory test.

Critical moisture content of sheet $= X_c = 0.106$.

For the laboratory test : air velocity $= u_1 = 1.5$ m/s.

For drying of sheet in the practice, air velocity is 50 % greater than that was used in the laboratory test.

$$\therefore \quad u_2 = 1.5\, u_1 = 1.5 \times 1.5 = 2.25 \text{ m/s}$$

(all other conditions of air are the same in both the cases)

In the constant rate period :

$$\text{The rate of drying} \propto (\text{velocity of air})^{0.80}$$

$$N_c \propto (u)^{0.80}$$

$$\therefore \quad N_{c1} = k\, u_1^{0.8} \text{ and } N_{c2} = k\, u_2^{0.8}$$

$$N_{c2} = N_{c1}\, (u_2/u_1)^{0.80}$$

$$= 0.92\, (2.25/1.5)^{0.80} = 1.2725 \text{ kg/(m}^2\text{.h)}$$

The rate of drying in the constant rate period in the actual practice with a air velocity of 50 % greater than that used in the laboratory test (with all other air conditions unchanged) is 1.2725 kg/(m^2.h).

As $X_2 < X_c$ and $X_1 > X_c$, the drying of sheet was done in the constant-as well as in the falling drying rate period.

In the actual practice suppose that drying is done from one side.

$\therefore$ Area of drying of surface = $A = 0.10$ m^2.

Time required for drying is given by,

$$\theta = \theta_c + \theta_f = \frac{S_S}{A N_c}\left[(X_1 - X_c) + (X_c - X^*) \ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right]$$

S_S = dry weight of sheet = 3.765 kg, $A = 0.10$ m^2

$N_c = N_{c2} = 1.2725$ kg/(m^2.h), $X_1 = 0.25$, $X_c = 0.106$

$X_2 = 0.02041$, $X^* = 0.01434$

$$\theta = \frac{3.765}{0.1 \times 1.2725}\left[(0.25 - 0.106) + (0.106 - 0.01434) \ln\left(\frac{0.106 - 0.01434}{0.02041 - 0.01434}\right)\right]$$

$$= 11.62 \text{ h} \quad \text{... Ans.}$$

If the sheet was dried from both sides, then

A = drying surface $= 2 \times 0.1 = 0.2$ m^2

$$\therefore \quad \theta = 11.62/2 = 5.81 \text{ h} \quad \text{... Ans.}$$

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In such type of problems a very high accuracy in the drawing of lines for the constant and falling-rate periods is required. The error encountered is generally large.

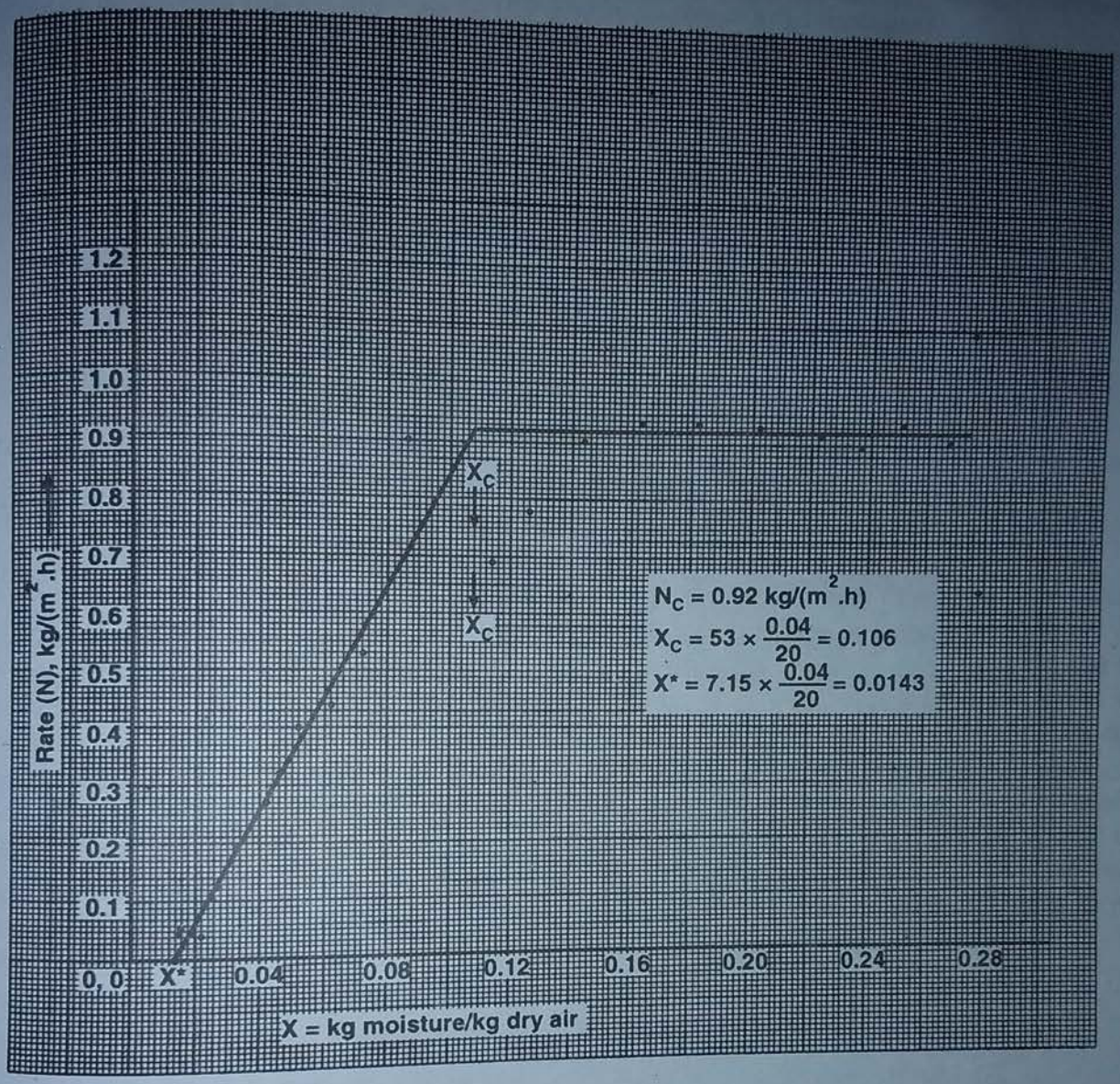

Fig. E 6.10 (a)

Extra : An alternate approach to find the time required for drying :

We have, $$d\theta = \frac{-S_S}{A}\frac{dX}{N}$$

$$\theta = \frac{-S_S}{A}\int_{X_1}^{X_2}\frac{dX}{N} = \frac{S_S}{A}\int_{X_2}^{X_1}\frac{dX}{N}$$

For the graphical determination of θ we have to plot 1/N v/s X and determine the area under the curve to evaluate the integral on the RHS of the above equation.

Find the area under the curve between $X_1 = 0.25$ and $X_2 = 0.02041$.

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From Fig. E 6.10 (b) :

Area under the curve = 21.27 cm^2

For drying from both sides = A = $2 \times 0.1 = 0.2\ m^2$

$$\therefore \quad \theta = \frac{3.765}{0.20} \text{ (Area} \times \text{scale y-axis} \times \text{scale x-axis)} = \frac{3.765}{0.20} \times 21.27 \times \frac{1}{1} \times \frac{0.04}{2}$$

= 8.008 h (for lab test) for A = 0.20 m^2

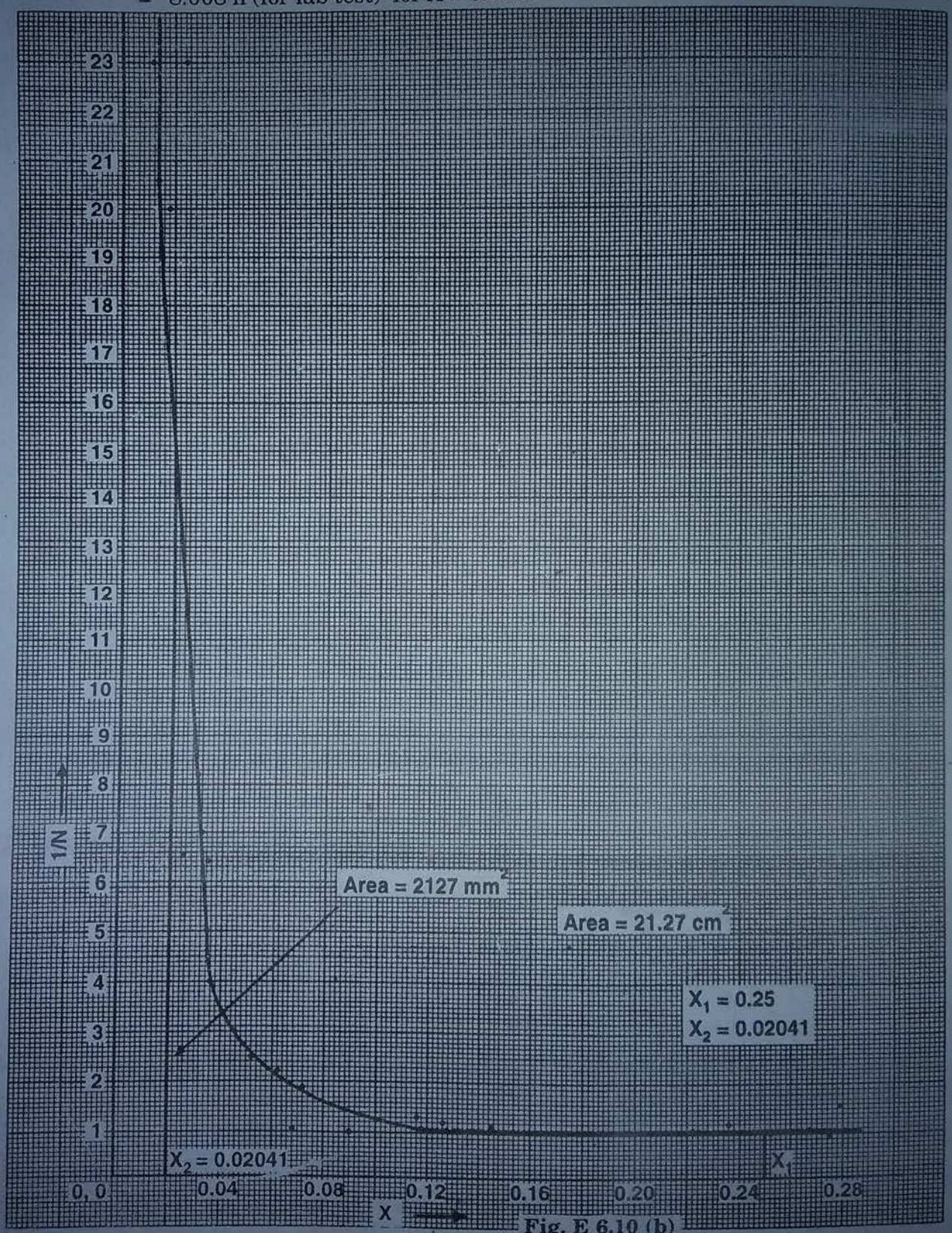

Fig. E 6.10 (b)

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This is the time required for drying the sheet with air velocity of 1.5 m/s. In the actual practice, drying of sheet air velocity used was 50 % grater than 1.5 m/s. So, we have to convert this time to determine the drying time in the actual practical case of drying :

We have $\theta \propto 1/N_c$ and $N_c \propto u^{0.8}$

$\therefore$ $\theta = k/(u)^{0.80}$

$\theta_1 = k/(u_1)^{0.8}$ and $\theta_2 = k/(u_2)^{0.80}$

θ_1 = time for lab test = 8.008 h

θ_2 = time for practical drying of sheet

$\theta_2 = \theta_1(u_1/u_2)^{0.80}$

$= 8.008\,(1.5/2.25)^{0.80} = 5.7896 \approx 5.790$ hr

With air velocity of 50% greater of 1.5 m/s :

$\therefore$ Time required for drying sheet (drying from both sides) = 5.790 h

$\therefore$ Time required for drying sheet (drying from one side) = $2 \times 5.790 = 11.58$ h

(pervious obtained time of drying by calculations from rate of drying curve = 11.62 h)

So, these values of time obtained by rate v/s X and 1/rate v/s X curves are almost same.

Ex. 6.11 : It is desired to dry a certain type of fiber board in sheets. The sheet size is 1.2 m by 2 m by 12 mm. In order to determine the drying characteristics of these sheets, a 0.3 - by 0.3 - m sample of the board, with the edges sealed, so that the drying took place from the two large faces, was suspended from a balance in a laboratory cabinet dryer. In the dryer it was exposed to a current of hot, dry air. The initial moisture content was 75%. The sheet lost weight at the constant rate of 1×10^{-4} kg/s until the moisture content fell to 60%, after which the drying rate fell. Measurements of the rate of drying were discontinued, but after a long period of exposure of the sheet to this hot air it was established that the equilibrium moisture content was 10 %. The dry mass of the sample was 0.90 kg. All moisture contents are on the wet basis. Determine the time for drying the large sheets from 75 to 20 % under the same drying conditions. Assume that the falling rate to be linear with free moisture content.

Solution : (i) Laboratory test : Sample sheet drying

Sample sheet size = 0.3 m × 0.3 m × 12 mm

Area of one face = $0.3 \times 0.3 = 0.09$ m^2

As drying took place from two faces of the sample sheet :

Area of drying surface = $A = 2 \times 0.09 = 0.18$ m^2

In the constant rate period :

Mass rate of evaporation = rate of moisture lost = 1×10^{-4} kg/s

Rate of drying = N_c = mass rate of evaporation/area

$= 1 \times 10^{-4}/0.18 = 5.556 \times 10^{-4}$ kg/(m^2.s)

Dry weight of sample sheet = $S_{S1} = 0.90$ kg

Critical moisture : 60 %, Initial moisture = 75 %

Equilibrium moisture : 10 % (all on the wet basis)

On the dry basis, moisture is expressed as kg/kg dry sheet :

Critical moisture content of sample sheet = $X_c = 0.60/(1 - 0.60) = 1.5$

Equilibrium moisture content of sample sheet = $X^* = 0.10/(1 - 0.10) = 0.111$.

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(ii) Drying of large sheet in practice :

Sheet size : 1.2 m × 2 m × 12 mm

Sample sheet size : 0.3 m × 0.3 m × 12 mm

Dry weight of sample sheet = S_{S1} = 0.90 kg

$$\therefore \text{ Dry weight of large sheet} = S_{S2} = S_{S1} \times \left(\frac{1.2 \times 2 \times 0.012}{0.3 \times 0.3 \times 0.012}\right)$$

$$= 0.90\left(\frac{1.2 \times 2 \times 0.012}{0.3 \times 0.3 \times 0.012}\right)$$

$$= 24 \text{ kg}$$

In the constant rate period, i.e., from moisture of 75 to 60%.

Rate of drying large sheet = rate of drying sample sheet

$$N_c = N_{c2} = 5.556 \times 10^{-4} \text{ kg/(m}^2\text{.s)}$$

(As the drying conditions here are the same as that were with drying of the sample sheet.)

Area of one face of large sheet = $A = 1.2 \times 2 = 2.4 \text{ m}^2$

For drying from one face = $A_1 = 2.4 \text{ m}^2$

For drying from both faces = $A_2 = 2 \times 2.4 = 4.8 \text{ m}^2$

Initial moisture = 75% on wet basis

Final moisture = 20% on wet basis.

Initial moisture content of large sheet $= X_1 = 0.75/(1 - 0.75) = 3$

Final moisture content of large sheet $= X_2 = 0.20/(1 - 0.20) = 0.25$

Critical moisture content of large sheet $= X_c = 1.5$

Equilibrium moisture content of large sheet $= X^* = 0.111$

(Here equilibrium and critical moisture contents are same as those with the sample sheet).

As $X_1 > X_c$ and $X_2 < X_c$, the drying takes place in constant-as well as falling-rate period.

Considering drying from one face and linear rate of drying in falling-rate period :

The time required for drying the large sheet from 75 to 20 % moisture on the wet basis from one face is :

$$\theta = \frac{S_S}{AN_c}\left[(X_1 - X_c) + (X_c - X^*) \ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right]$$

$S_S = S_{S2} = 24$ kg, $A_1 = A = 2.4 \text{ m}^2$ (one face)

$N_c = 5.556 \times 10^{-4} \text{ kg/(m}^2\text{.s)}$, $X_1 = 3$, $X_c = 1.5$, $X_2 = 0.25$, $X^* = 0.111$.

$$\theta = \frac{24}{2.4 \times 5.556 \times 10^{-4}}\left[(3 - 1.5) + (1.5 - 0.111) \ln\left(\frac{1.5 - 0.111}{0.25 - 0.111}\right)\right]$$

$= 85544 \text{ s} = 23.48 \text{ h}$ ... (from one face) ... Ans.

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For drying from both faces of the large sheet :

$$A = A_2 = 2 \times (1.2 \times 2) = 4.8 \text{ m}^2$$

$$\theta = \frac{24}{4.8 \times 5.556 \times 10^{-4}} \left[(3 - 1.5) + (1.5 - 0.111) \ln \left(\frac{1.5 - 0.111}{0.25 - 0.111} \right) \right]$$

$$= 42272 \text{ s} = 11.74 \text{ h}$$

... Ans.

Time required for drying the large sheet from both sides to reduce moisture from 75 to 20 % = 11.74 h ... Ans.

Ex. 6.12 : The experimental data given below were obtained during a test on the tray drying of sand.

A drying test was conducted at constant drying conditions (air velocity = 6 m/s, air dry-bulb temperature = 71 °C; air wet-bulb temperature = 32 °C) gave a drying rate for constant-rate period of 3.08 kg/(m².h).

(i) Determine the corresponding rate at the same air velocity and wet-bulb temperature if the air dry-bulb temperature is 52 °C. The radiation and conduction to the wetted surface were negligible.

(ii) Determine the corresponding rate if air velocity is 4 m/s, air properties DB and WB remained unchanged.

(iii) Determine the drying rate during the constant rate period under constant drying conditions ;

air velocity = 3 m/s

air dry-bulb temperature = 66 °C and air wet-bulb temperature = 38 °C.

Solution : (i) For no radiation and conduction effects, the temperature of the drying surface (wetted surface) will be the same as the wet-bulb temperature of the air (32 °C).

During constant rate period :

$$\text{Rate of drying} \propto (T_G - T_S)$$

$$N_C \propto \Delta T, \quad \Delta T = T_G - T_S$$

$$\therefore \quad N_{C1} = k \Delta T_1$$

$$N_{C2} = k \Delta T_2$$

$$N_{C2} = N_{C1} \frac{\Delta T_2}{\Delta T_1}$$

where, $N_{C1} = 3.08 \text{ kg/(m}^2\text{.h)}$

$\Delta T_1 = 71 - 32 = 39 \text{ °C}$, $\Delta T_2 = 52 - 32 = 20 \text{ °C}$

$$N_{C2} = 3.08 \times \frac{20}{39} = 1.5795 \approx 1.58 \text{ kg/(m}^2\text{.h)}$$

... Ans. (i)

In the constant rate period :

∴ Rate of drying with air of DB = 52 °C (all else remaining same) = 1.58 kg/(m².h)

(ii) In constant rate period :

$$\text{Rate of drying} \propto (\text{velocity of air})^{0.80}$$

$$N_C \propto (u)^{0.80}$$

$$N_{C1} = k (u_1)^{0.80}$$

$$N_{C2} = k (u_2)^{0.80}$$

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$$N_{c2} = N_{c1}\,(u_2/u_1)^{0.80}$$

where $N_{c1} = 3.08$ kg/(m².h), $u_1 = 6$ m/s

$N_{c2} = ?$, $u_2 = 4$ m/s

$$N_{c2} = 3.08\,(4/6)^{0.80}$$

$$= 2.2267 \simeq 2.23 \text{ kg/(m}^2\text{.h)}$$

Rate of drying in the constant rate period with air velocity of 4 m/s = 2.23 kg/(m².h)

...Ans. (ii)

(i) N_c : 3.08 kg/(m².h) DB = 71 °C, WB = 32 °C, u_1 = 6 m/s

N_c : 1.58 kg/(m².h) DB = 52 °C, WB = 32 °C, u_2 = 6 m/s

So, the rate increases with temperature of air

(ii) N_c : 3.08 kg/(m².h), DB = 71 °C, WB = 32 °C, u_1 = 6 m/s

N_c : 2.23 kg/(m².h), DB = 71 °C, WB = 32 °C, u_2 = 4 m/s

So, the rate increases with air velocity.

(iii) Determination of the rate of drying in the constant rate period using air velocity of 6 m/s, DB of air = 66 °C, WB of air = 38 °C.

$N_{c1} = 3.08$ kg/(m².h), $DB_1 = 71$ °C, $WB_1 = 32$ °C, $u_1 = 6$ m/s

$N_{c2} = ?$, $DB_2 = 66$ °C, $WB_2 = 38$ °C, $u_2 = 6$ m/s

For no radiation effects, temperature of the drying surface is equal to the wet-bulb temperature of the air used.

$$N_{c2} = N_{c1}\frac{\Delta T_2}{\Delta T_1}$$

$$\Delta T_2 = T_{G2} - T_{S2} = DB_2 - WB_2 = 66 - 38 = 28\ ^\circ\text{C}$$

$$\Delta T_1 = DB_1 - WB_1 = 72 - 32 = 39\ ^\circ\text{C}$$

$$N_{c2} = 3.08 \times 28/39 = 2.21 \text{ kg/(m}^2\text{.h)}$$

Rate of drying in constant rate period (with air velocity 6 m/s, DB = 66 °C, WB = 38 °C) = 2.21 kg/(m².h) ...Ans. (iii)

Ex. 6.13 : A porous solid is dried in a batch dryer under constant drying conditions. Seven hours are needed for reducing the moisture content from 35 to 10 %. The critical moisture content was found to be 20 % and the equilibrium moisture content was 4 %. All moisture contents are on the dry basis. Assuming that the rate of drying during the falling rate period is proportional to the free moisture content, find the time required to dry a sample of the same solid from 35 to 5 % under the same drying conditions ?

Solution : Case - I :

Here moisture content data are given on dry basis (Note it)

Initial moisture content of solid = 35 % dry basis

Initial moisture content of solid = X_1 = 35/100 = 0.35

Final moisture content of solid = X_2 = 10/100 = 0.10

Critical moisture content of solid = X_c = 20/100 = 0.20

Equilibrium moisture content of solid = X^* = 4/100 = 0.04

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We have provided the data of moisture content of the solid on the dry basis. So for converting it into moisture content expressed as X kg moisture per kg dry solid divide the % values on the dry basis by 100.

OR e.g., initial moisture content of solid is given as 35 % on the dry basis.

With x kg moisture and 1 kg dry solid :

$$\left(\frac{x}{1}\right) \times 100 = 35 \qquad \therefore \; x = 0.35 \text{ kg}$$

$\therefore$ Initial moisture content of the solid $= X_1 = 0.35/1 = 0.35$

As the final moisture content ($X_2 = 0.10$) is less than the critical moisture content ($X_c = 0.20$) the drying is done in both the constant-and falling-rate periods.

Given : The rate of drying in the falling rate period is proportional to the free moisture content, i.e., the falling rate is linear. So we can use the equation derived for the same assumption for calculating the total time for drying.

Given : θ = time required for drying = 7 h

$$\theta = \theta_c + \theta_f = \frac{S_S}{AN_c}\left[(X_1 - X_c) + (X_c - X^*) \ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right]$$

$$\frac{S_S}{AN_c} = \frac{\theta}{(X_1 - X_c) + (X_c - X^*) \ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)}$$

$$= \frac{7}{(0.35 - 0.20) + (0.20 - 0.04) \ln\left(\frac{0.20 - 0.04}{0.10 - 0.04}\right)}$$

$$= 22.81 \text{ h/(kg/kg dry solid)}$$

Case II : Here drying is carried out under the same conditions as those were in the case-I, so we make use of the S_S/AN_c calculated above for the estimation of the time for drying in this case.

Here, final moisture content $= X_2 = 5/100 = 0.05$

Initial moisture content $= X_1 = 35/100 = 0.35$

$X_c = 0.20, \quad X^* = 0.04$

$$\theta = \frac{S_S}{AN_c}\left[(X_1 - X_c) + (X_c - X^*) \ln\left(\frac{X_c - X^*}{X_2 - X^*}\right)\right]$$

$$= 22.81\left[(0.35 - 0.20) + (0.20 - 0.40) \ln\left(\frac{0.20 - 0.04}{0.05 - 0.04}\right)\right]$$

$$= 13.54 \text{ h}$$

Time required to dry the solid from 35 to 5% (dry basis) = 13.54 h ... Ans.

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Ex. 6.14 : A continuous countercurrent dryer is used to dry 425.6 kg dry solid/h from a moisture content of 0.035 to 0.0017 kg moisture/kg dry solid. The granular solids enters at 25°C and leaves at 60°C. Air is used as a heating medium and it enters at 84.2°C and leaves the dryer at 32.8°C. The humidity of the air entering the dryer is 0.0175 kg water/kg dry solid. Calculate the air flow rate and the outlet humidity, assuming the heat losses from the dryer to be 9300 kJ/h.

Data : Heat capacity of the dry solid is 1.465 kJ/(kg·K)

Heat capacity of the moisture is 4.187 kJ/(kg·K)

Latent heat of water at 0°C is 2501 kJ/kg

Solution : Basis : 425.6 kg dry solid per hour to a dryer.

$$S_S = 425.6 \text{ kg/h}$$

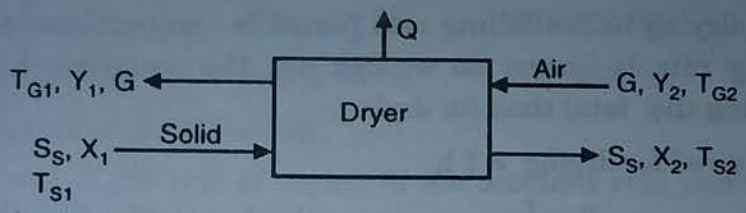

Fig. E6.14 : Continuous dryer

Initial moisture content of solid = X_1 = 0.035 kg/kg dry solid

Moisture content of solid leaving dryer = X_2 = 0.0017

Solid inlet temperature = T_{S1} = 25°C

Solid outlet temperature = T_{S2} = 60°C

Air (Gas) inlet temperature = T_{G2} = 84.2°C

Air outlet temperature = T_{G1} = 32.8°C

Humidity of inlet air = Y_2 = 0.0175 kg/kg dry air

Humidity of outlet air = Y_1 = ?

Dry air flow rate to dryer = G kg/h = ?

Latent heat of water at 0°C = λ_o = 2501 kJ/kg

Heat capacity of dry solid = C_{PS} = 1.465 kJ/(kg·K)

Heat capacity of moisture (H_2O) = C_{PA} = 4.187 kJ/(kg·K)

Here $(T_S - T_o)$ °C = $(T_S - T_o)$ K, and T_o = 0°C (reference temperature)

Material balance of moisture :

$$GY_2 + S_S X_1 = S_S X_2 + GY_1$$

$$G(0.0175) + 425.6(0.035) = 425.6(0.0017) + GY_1$$

$$0.0175\,G + 14.1725 = GY_1 \quad \ldots(1)$$

Enthalpy balance over the dryer :

$$GH_{G2} + S_S H_{S1} = GH_{G1} + S_S H_{S2} + Q \quad \ldots(2)$$

Q = heat loss from the dryer = 9300 kJ/h

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Now we will determine the enthalpies of streams entering and leaving the dryer.

(i) Enthalpy of the air (gas) entering the dryer = H_{G2}

H_{G2} is given by

$$H_{G2} = C_{S2}(T_{G2} - T_o) + Y_2 \lambda_o$$

and humid heat is given by

$$C_{S2} = (1.005 + 1.88\, Y_2)$$

$$\therefore \quad H_{G2} = (1.005 + 1.88\, Y_2)(T_{G2} - T_o) + Y_2 \lambda_o$$

$$= [1.005 + 1.88\,(0.0175)]\,(84.2 - 0) + 0.0175 \times 2501$$

$$= 131.16 \text{ kJ/kg dry air}$$

(ii) Enthalpy of the air leaving the dryer = H_{G1}

$$H_{G1} = C_{S1}(T_{G1} - T_o) + Y_1 \lambda_o$$

and

$$C_{S1} = (1.005 + 1.88\, Y_1)$$

$$H_{G1} = (1.005 + 1.88\, Y_1)(T_{G1} - T_o) + Y_1 \lambda_o$$

$$= (1.005 + 1.88\, Y_1)(32.8 - 0) + 2501\, Y_1$$

$$= 32.964 + 2562.664\, Y_1$$

$$\approx 32.964 + 2562.7\, Y_1 \text{ kJ/kg dry air}$$

(iii) Enthalpy of the solid entering the dryer = H_{S1}

H_{S1} is given by

$$H_{S1} = C_{PS}(T_{S1} - T_o) + X_1 C_{PA}(T_{S1} - T_o)$$

C_{PS} and C_{PA} are taken as constant over the temperature range.

$$H_{S1} = 1.465\,(25 - 0) + 0.035 \times 4.187\,(25 - 0)$$

$$= 40.2886 \approx 40.29 \text{ kJ/kg dry solid}$$

(iv) Enthalpy of the solid leaving the dryer = H_{S2}

$$H_{S2} = C_{PS}(T_{S2} - T_o) + X_2 C_{PA}(T_{S2} - T_o)$$

$$= 1.465\,(60 - 0) + 0.0017 \times 4.187\,(60 - 0)$$

$$= 88.33 \text{ kJ/kg dry solid}$$

Putting the values of H_{G2}, H_{G1}, H_{S2}, H_{S1} etc. in Equation (2), we get

$$G\,(131.16) + 425.6\,(40.29) = G\,(32.964 + 2562.7\, Y_1) + 425.6\,(88.33) + 9300$$

$$131.16\, G + 17147.4 = 32.964\, G + 2562.7\, GY_1 + 46893.25 \quad \ldots (2)$$

$$98.196\, G = 2562.7\, GY_1 + 29745.85 \quad \ldots (1)$$

We have : $0.0175\, G + 14.1725 = GY_1$

$$\therefore \quad 98.196\, G = 2562.7\,(0.0175\, G + 14.1725) + 29745.85$$

$$\therefore \quad G = 1238.4 \text{ kg dry air/h} \quad \ldots \text{Ans.}$$

Flow rate of dry air to the dryer = G = 1238.4 kg/h

Put the value of G in Equation (2) to get Y_1

$$98.196\,(1238.4) = 2562.7\,(1238.4)\, Y_1 + 29745.85$$

$$\therefore \quad Y_1 = 0.02894 \text{ kg } H_2O\text{/kg dry air}$$

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Amount of water (moisture) in air entering the dryer

$$= \frac{0.0175}{1} \times 1238.4$$

$$= 21.67 \text{ kg/h}$$

$\therefore$ Air flow rate to the dryer (as is basis) = 1238.4 + 21.67

= 1260.07 kg/h ... Ans.

Humidity of the air leaving the dryer = 0.02894 kg water/kg dry air ... Ans.

Ex. 6.15 : An uninsulated hot air countercurrent rotary dryer (1200 mm diameter × 6.7 m long) is used to dry ammonium sulphate from 3.5 to 0.2 % moisture. The atmospheric air at 25°C, 50 % relative humidity will be heated by passing it over steam coils to 90°C before it enters the dryer. The air leaves the dryer at 32 °C. The solid enters at 25°C and leaves the dryer at 60°C. The dried ammonium sulphate, i.e., product rate from the dryer is 900 kg/h. Determine the air flow rate to the dryer (on dry basis) and steam consumption per kg of water evaporated (into the air).

Data : (i) latent heat of condensation of steam = 2215 kJ/kg

(ii) Latent heat of water at 0°C = 2501 kJ/kg

(iii) Heat capacity of dry ammonium sulphate = 1.507 kJ/(kg.K)

(iv) Heat capacity of water = 4.187 kJ/(kg.K)

(v) Absolute humidity of air at 25°C, 50 % RH = 0.0101 kg H_2O/kg dry air

(vi) Combined convection and radiation heat transfer coefficient from the dryer to the surroundings = 12 W/(m^2·K)

(vii) The average ΔT between dryer and surroundings = 36°C.

Solution : Basis : 900 kg/h of product rate

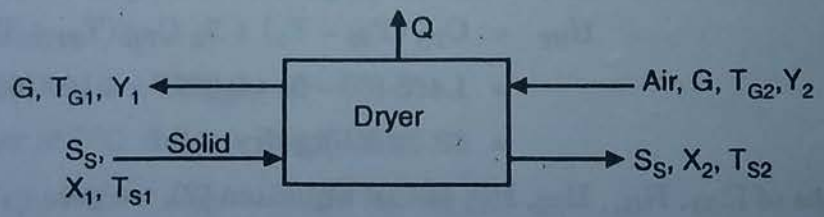

Fig. E 6.15 (a) : Rotary dryer

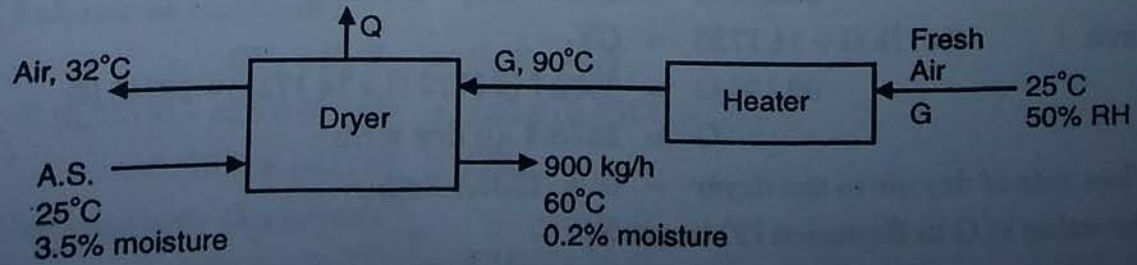

Fig. E 6.15 (b) : Rotary dryer and air heater

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S_S = kg/h dry solid (dry ammonium sulphate)

Product rate = 900 kg/h

The product contains 0.2 % moisture on the wet basis, i.e., 1 kg product contains 0.002 kg moisture and (1 – 0.002) kg dry solid.

or the product contains 0.2 % moisture and 99.8 % dry solid (by difference)

$\therefore$ Dry solid in product = $0.998 \times 900 = 898.2$ kg/h

S_S = 898.2 kg/h

Initial moisture of solid = 3.5 (wet basis)

Final moisture of solid = 0.2 % (wet basis)

Moisture content of solid at the inlet = $X_1 = \dfrac{3.5}{(100 - 3.5)} = 0.03627$

Moisture content of solid at the exit of dryer = $X_2 = \dfrac{0.20}{(100 - 0.20)}$

$= 0.002004$

≈ 0.002

Mass rate of evaporation of water (from solid)

$= S_S (X_1 - X_2)$

$= 898.2 (0.03627 - 0.002)$

$= 30.78$ kg/h

Solid inlet temperature = $T_{S1} = 25°C$

Solid outlet temperature = $T_{S2} = 60°C$

Moisture content of solid entering the dryer = $X_1 = 0.03627$

Moisture content of solid leaving the dryer = $X_2 = 0.002$

Air inlet temperature = $T_{G2} = 90°C$

Air outlet temperature = $T_{G1} = 32°C$

Humidity of inlet air = $Y_2 = 0.0101$ kg H_2O/kg dry air

Humidity of outlet air = $Y_1 = ?$

Dry air flow rate to the dryer = G kg/h

Heat capacity of dry solid = $C_{PS} = 1.507$ kJ/(kg dry solid·K)

Heat capacity of moisture = $C_{PA} = 4.187$ kJ/(kg·K)

Latent heat of water at 0·C = $\lambda_o = 2501$ kJ/kg

Here : $(T_S - T_o)$ °C = $(T_S - T_o)$ K

and reference temperature is $T_o = 0°C$

Material balance of moisture :

(Moisture in inlet air + moisture in inlet solid)

= (moisture in exit air + moisture in exit solid)

$$GY_2 + S_S X_1 = S_S X_2 + GY_1$$

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$$G(0.0101) + 898.2(0.03627) = 898.2(0.002) + GY_1$$

$$0.0101\,G + 30.78 = GY_1 \quad \ldots(1)$$

As the dryer is not insulated there will be loss of heat from the dryer to the surrounding air by convection and radiation. Let it be Q, kJ/kg.

So, the enthalpy balance over the entire dryer is :

$$GH_{G2} + S_S H_{S1} = GH_{G1} + S_S H_{S2} + Q \quad \ldots(2)$$

Now evaluate the enthalpies of the streams entering and leaving the dryer.

(i) Enthalpy of air entering the dryer = H_{G2}

H_{G2} is given by

$$H_{G2} = C_{S2}(T_{G2} - T_o) + Y_2 \lambda_o$$

and humid heat is given by

$$C_{S2} = (1.005 + 1.88\,Y_2)$$

$$\therefore \quad H_{G2} = (1.005 + 1.88\,Y_2)(T_{G2} - T_o) + Y_2 \lambda_o$$

$$= (1.005 + 1.88 \times 0.0101)(90 - 0) + 0.0101 \times 2501$$

$$= 117.42 \text{ kJ/kg dry air}$$

(ii) Enthalpy of air leaving the dryer = H_{G1}

$$H_{G1} = C_{S1}(T_{G1} - T_o) + Y_1 \lambda_o, \qquad C_{S1} = 1.005 + 1.88\,Y_1$$

$$= (1.005 + 1.88\,Y_1)(T_{G1} - T_o) + Y_1 \lambda_o$$

$$= (1.005 + 1.88\,Y_1)(32 - 0) + Y_1 \times 2501$$

$$H_{G1} = 32.16 + 1561.16\,Y_1 \text{ kJ/kg dry air}$$

(iii) Enthalpy of solid entering the dryer = H_{S1}

H_{S1} is given by

$$H_{S1} = C_{PS}(T_{S1} - T_o) + X_1 C_{PA}(T_{S1} - T_o)$$

$$= 1.507(25 - 0) + 0.03627 \times 4.187(25 - 0)$$

$$= 41.47 \text{ kJ/kg dry solid}$$

(iv) Enthalpy of solid leaving the dryer = H_{S2}

$$H_{S2} = C_{PS}(T_{S2} - T_o) + X_2 C_{PA}(T_{S2} - T_o)$$

$$= 1.507(60 - 0) + 0.002 \times 4.187(60 - 0)$$

$$= 90.92 \text{ kJ/kg dry solid}$$

(v) Heat loss from the dryer = Q

Heat transfer coefficient = 12 W/(m²·K) = 12 J/(s·m²·K)

$$= 43.20 \text{ kJ/(m}^2\text{·h·K)}$$

Avg. $\Delta T = 36°C$

Air temperature : in = 90°C, out = 32°C

Surroundings : 25°C, 25°C

$\Delta T_1 = 90 - 25 = 65°C$, $\Delta T_2 = 7°C$

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Temperature difference at inlet $= \Delta T_1 = 65°C$

Temperature difference at exit $= \Delta T_2 = 7°C$

Average temperature difference $= \Delta T = (\Delta T_1 + \Delta T_2)/2$

$= (65 + 7)/2 = 36°C$

Length of dryer $= L = 6.7$ m

Diameter of dryer $= D = 1200$ mm $= 1.2$ m

Area of heat transfer $= A = \pi DL$

$= \pi (1.2)(6.7) = 25.26 \text{ m}^2$

$\therefore \quad Q = h A \Delta T$

$= 43.20 \times 25.20 \times 36 = 39284.35$ kJ/h

Putting the values of H_{G2}, H_{G1}, H_{S1}, H_{S2}, S_S and Q in equation (2), we get

$$G(117.42) + 898.2(41.47) = G(32.16 + 2561.16\, Y_1) + 898.2(90.92) + 39284.35$$

$$85.26\, G = 83700.34 + 2561.16\, GY_1 \quad \ldots (3)$$

We have $0.0101\, G + 30.78 = GY_1 \quad \ldots (1)$

$\therefore \quad 85.26\, G = 83700.34 + 2561.16(0.0101\, G + 30.78)$

$G = 2736.6$ kg/h

Putting value of G in Equation (1) gives

$$0.0101(2736.6) + 30.78 = 2736.6\, Y_1$$

$\therefore \quad Y_1 = 0.02135$ kg water/kg dry air

$\therefore$ Flow rate of dry air to the dryer $= 2736.6$ kg/h ... Ans.

Humidity of air leaving the dryer $= 0.02135$ kg water/kg dry air ... Ans.

Flow rate of atmospheric air (as is basis) $= 2736.6 + \dfrac{0.0101}{1} \times 2736.6 = 2764.24$ kg/h

Determination of steam requirements :

Latent heat of condensing steam $= \lambda_s = 2215$ kJ/kg

Mass flow rate of steam $= \dot{m}_s$ kg/h

Rate of water evaporation $= 30.78$ kg/h

Heat content of the air entering dryer $= G H_{G2}$

$= 2736.6 \times 117.42$

$= 321331.6$ kJ

H_2 = Heat content of the air entering dryer at 90°C $= 321331.6$ kJ

H_1 = Heat content of the air entering heater at 25°C

Y = humidity of air at 25°C $= 0.0101$, $T_1 = 25°C$

$H_1 = G[(1.005 + 1.88\, Y)(T_1 - T_o)]$

$= 2736.6[(1.005 + 1.88 \times 0.0101)(25 - 0) + 0.0101 \times 2501]$

$= 139183$ kJ

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Heat load of heater = Heat required to raise the temperature of air from 90°C to 25°C

$= H_2 - H_1$

$= 321331.6 - 139183$

$= 182148.6$ kJ

This heat load of the heater is provided by steam condensing in the coils.

$\therefore \quad \dot{m}_s \lambda_s$ = Heat load

$m_s (2215) = 182148.6$

$\dot{m}_s = 82.234 = 82.23$ kg/h

Steam consumption $= \dot{m}_s = 82.23$ kg/h

Rate of water evaporation = 30.78 kg/h

$\therefore$ Steam consumption per kg of water evaporated

$$= \frac{82.23}{30.78} = 2.67 \text{ kg}$$ ... Ans.

Ex. 6.16 : A granular material containing 40 % moisture enters a countercurrent rotary dryer at a temperature of 295 K (22°C) and leaves the dryer at 305 K (32°C) containing 5 % moisture. The air enters the dryer at 385 K (112°C) with 0.006 kg water/kg dry air and leaves at 310 K (37°C). The dryer handles wet stock at the rate of 450 kg/h.

Determine the mass flow rate of dry air to the dryer and the humidity of the exit air assuming that radiation losses amount to 20 kJ/kg dry air used.

Latent heat of water at 0°C = 2501 kJ/kg

Heat capacity of dried material = 0.88 kJ/(kg.K)

Heat capacity of water = 4.187 kJ/(kg.K)

Solution : Basis : 450 kg/h of wet stock to the dryer.

Moisture in wet stock at inlet = 40 %

$\therefore$ Moisture in wet stock $= 0.40 \times 450 = 180$ kg/h

Dry solid in wet stock $= 450 - 180 = 270$ kg/h

$\therefore \quad S_S = 270$ kg/h

Moisture content of solid at inlet $= X_1 = \frac{40}{(100 - 40)} = 0.6667$

Moisture content of solid at exit $= X_2 = \frac{5}{(100 - 5)} = 0.05263$

Temperature of solid at inlet $= T_{S1} = 295$ K

Temperature of solid at exit $= T_{S2} = 305$ K

Temperature of air at inlet $= T_{G2} = 385$ K

Temperature of air at exit $= T_{G1} = 310$ K

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Humidity of air at inlet = $Y_2 = 0.006$

Humidity of air at exit = $Y_1 = ?$

Dry air flow rate at inlet to dryer = G kg/h

Latent heat of water at 0°C = $\lambda_o = 2501$ kJ/kg

Heat capacity of water = $C_{PA} = 4.187$ kJ/(kg-K)

Heat capacity of dry solid = $C_{PS} = 0.88$ kJ/(kg-K)

Heat losses from the dryer = 20 kJ/kg dry air

Material balance of moisture :

$$S_S X_1 + GY_2 = GY_1 + SX_2$$

$$270\,(0.6667) + G\,(0.006) = GY_1 + 270\,(0.05263)$$

$$0.0064 + 165.80 = GY_1 \qquad \ldots (1)$$

Enthalpy balance over the dryer :

$$GH_{G2} + S_S H_{S1} = GH_{G1} + S_S H_{S2} + Q \qquad \ldots (2)$$

The datum temperature is 0°C (i.e., $T_0 = 273$ K)

(i) Enthalpy of air entering the dryer = H_{G2}

$$H_{G2} = C_{S2}\,(T_{G2} - T_o) + Y_2\,\lambda_o, \quad C_S = 1.005 + 1.88\,Y_2$$

$$= (1.005 + 1.88\,Y_2)\,(T_{G2} - T_o) + Y_2\,\lambda_o$$

$$= (1.005 + 1.88 \times 0.006)\,(385 - 273) + 0.006 \times 2501$$

$$= 128.83 \text{ kJ/kg dry air}$$

(ii) Enthalpy of air leaving the dryer :

$$H_{G1} = C_{S1}\,(T_{G1} - T_o) + Y_1\,\lambda_o$$

$$= (1.005 + 1.88\,Y_1)\,(T_{G1} - T_o) + Y_1\,\lambda_o$$

$$= (1.005 + 1.88\,Y_1)\,(310 - 273) + Y_1\,(2501)$$

$$= 37.185 + 2570.56\,Y_1$$

(iii) Enthalpy of solid entering the dryer = H_{S1}

$$H_{S1} = C_{PS}\,(T_{S1} - T_o) + X_1\,C_{PA}\,(T_{S1} - T_o)$$

$$= 0.88\,(295 - 273) + 0.6667 \times 4.187\,(295 - 273)$$

$$= 80.77 \text{ kJ/kg dry solid}$$

(iv) Enthalpy of solid leaving the dryer = H_{S2}

$$H_{S2} = C_{PS}\,(T_{S2} - T_o) + X_2\,C_{PA}\,(T_{S2} - T_o)$$

$$= 0.88\,(305 - 273) + 0.05263 \times 4.187\,(305 - 273)$$

$$= 35.21 \text{ kJ/kg dry solid}$$

(v) Heat loss from dryer = Q

Heat loss is given as 20 kJ/kg dry air

$\therefore$ $Q = 20 \times G$ kJ/h

Putting the values of H_{G2}, H_{G1} etc. in Equation (2), we get

$$G\,(128.83) + 270\,(80.77) = G\,[37.185 + 2570.56\,Y_1] + 270\,(35.21) + 20G$$

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$$128.83\,G + 21807.9 = 37.185\,G + 2570.56GY_1 + 20G + 9506.7$$

$$71.645\,G + 12301.2 = 2570.56\,GY_1$$

We have, $$0.006\,G + 165.80 = GY_1$$

$$\therefore \quad 71.645\,G + 12301.2 = 2570.56\,(0.006\,G + 165.80)$$

$$G = 7361.9 \text{ kg/h}$$

Putting value of G in Equation (1) we get

$$0.006\,(7361.9) + 165.80 = (7361.9)\,Y_1$$

$$Y_1 = 0.02852$$

$\therefore$ Dry air flow rate to the dryer = 7361.9 kg/h ... Ans.

= 2.045 kg/s

Humidity of the exit air = 0.02852 kg water/kg dry air ... Ans.

Heat loss to the surroundings = $20\,G = 20 \times 7361.9$

= 147238 kJ ... Ans.

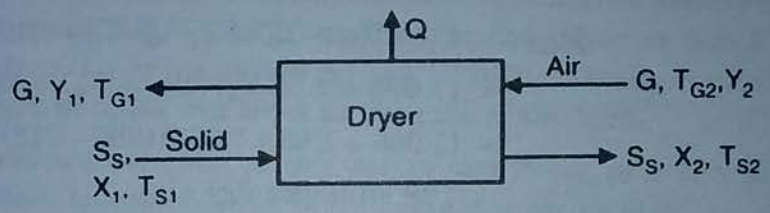

Fig. E 6.16 : Rotary dryer

Extra : In case of the above example if we are provided with the following data :

(i) Latent heat of water vapour at 295 K (22°C) = 2449 kJ/kg

(ii) Heat capacity of dry material/solid = 0.88 kJ/(kg.K)

(iii) Heat capacity of dry air = 1.0 kJ/(kg.K)

(iv) Heat capacity of water vapour = 2.01 kJ/(kg.K)

Hat capacity of water = 4.187 kJ/(kg.K)

then the problem can be solved as :

Mass flow rate of wet stock to dryer = 450 kg/h

Water in wet stock to dryer = 0.40×450 = 180 kg/h

Solid in wet stock to dryer = S_S = 450 – 180 = 270 kg/h

(a) Heat content of the solid stream (wet stock) entering the dryer

$= \Sigma \dot{m}_i\, C_{pi}\, \Delta T$

= Heat content of dry solid + Heat content of associated moisture (liquid water)

$= 270 \times 0.88\,(295 - 273) + 180 \times 4.187\,(295 - 273)$

= 21807.7 kJ/h

The wet stock entering the dryer is at temperature of 295 K and the reference temperature is

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The product leaving the dryer is at 305 K. The product solid leaving the dryer contains 5 % moisture.

Dry solid in the product leaving the dryer = 270 kg/h

Let the moisture in it be x kg/h, then

$$\left(\frac{x}{270+x}\right)\times 100 = 5, \quad x = 14.21 \text{ kg/h}$$

Moisture in the product leaving the dryer = 14.21 kg/h

(b) Heat content of the product leaving the dryer = $\Sigma \dot{m}_i C_{pi} \Delta T$

= Heat content of dry solid + moisture (liquid water)

= $270 \times 0.88 (305 - 273) + 14.21 \times 4.187 (305 - 273)$

= 9507.1 kJ/h

Note that the moisture in a solid is in liquid form

Water evaporated/removed from wet stock

= Initial moisture – Final moisture

= 180 – 14.21 = 165.79 kg/h

Water added to the air = Water removed from stock = 165.79 kg/h

Let G be the kg/h of dry air entering the dryer.

Air enters the dryer at 385 K with 0.006 kg water vapour/kg dry air

Dry air in air entering the dryer = G kg/h

Water vapour (moisture) in the air entering = $\frac{0.006}{1} \times G = 0.006\ G$ kg/h

In air, moisture is in the form of water vapour (Note this)

(c) Heat content of the air entering dryer = $\Sigma \dot{m}_i C_{pi} \Delta T$

= heat in dry air + heat in moisture (water vapour)

= $G \times 1 (385 - 273) + 0.006\ G \times 2.01 (385 - 273)$

= 113.35 G kJ/h

Water as water vapour added in the exit air = 165.79 kg/h

Air leaving the dryer = dry air + initial water vapour + added water vapour

= G + 0.006 G + 169.79 kg/h

Air leaves the dryer at 310 K.

Note : Water vapour added to the exit air is evaporated at 295 K and then its temperature is increased to 310 K.

(d) Heat content of the exit air = Heat in dry air + heat in initial moisture (water vapour) + heat in added water vapour

(i) Heat in dry air at 310 K and above 273 K = $G \times 1 (310 - 273)$ = 37 G kJ/h

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(ii) Heat content of initial water vapour at 310 K and above 273 K

$$= 0.006\,G \times 2.01\,(310 - 273)$$

$$= 0.446\,G \text{ kJ/h}$$

(iii) Heat content of added water vapour at 310 K and above 273 K = Latent heat at 295 K + Sensible heat at 310 K

$$= 165.79\,(2449) + 165.79 \times 2.01\,(310 - 273)$$

$$= 418349.5 \text{ kJ/h}$$

(e) Heat content of exit air $= 37\,G + 0.446\,G + 418349.5 = 37.446\,G + 418349.5$ kJ/h

Heat losses from dryer = 20 kJ/kg dry air

(f) Heat loss from dryer = $Q = \frac{20}{1} \times G = 20\,G$ kJ/h

Energy balance over dryer is :

$$\Sigma \text{ Heat in} = \Sigma \text{ Heat out}$$

Heat with air + Heat with solid = Heat with exit air + Heat with exit solid product + Heat loss from dryer

$$113.35\,G + 21807.7 = 37.446\,G + 418349.5 + 9507.1 + 20\,G$$

$$\therefore \quad G = 7263.3 \text{ kg/h}$$

Previously calculated : $G = 7361.9$ kg/h

$$\% \text{ error} = \frac{7361.9 - 7263.3}{7361.9} \times 100$$

$$= 1.34 \text{ (tolerable)}$$

Dry air leaving the dryer = 7263.3 kg/h ... Ans.

Water vapour in air leaving the dryer

= Initial water vapour of air (inlet air) + Water evaporated and added to air

$$= 0.006\,G + 165.79$$

$$= 0.006\,(7263.3) + 165.79 = 209.37 \text{ kg/h}$$

Humidity of air leaving the dryer $= \frac{209.37}{7263.3} = 0.0288$ kg water vapour/kg dry air.

Previously calculated humidity of air leaving the dryer = 0.02852 kg/kg dry air .

(Results are almost same)

Ex. 6.17 : Sheets are to be dried by blowing hot air at a velocity of 10 m/s. The air is at 60°C and its relative humidity is 10%. The critical moisture content of solid sheet was 0.35 kg water/kg dry solid. The rate of drying in the falling rate period can be considered as linear. The sheets must be dried from 55 to 12% moisture (on wet basis). The equilibrium moisture content is 5% on wet basis. The sheets are kept in layers 6 cm thick, in insulated trays. The dry density of solid is 1.38 g/cc. Calculate the time required for drying.

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Data : (i) Sheet size : 1 m by 1 m.

(ii) The rate of drying in g/(cm^2·h) in constant rate period is given by

$$N_C = 0.004\,(u)^{0.80}\,(p_i - p_g)$$

where u is the air velocity in m/s, p_i is the vapour pressure of water in mmHg at WBT of air and p_g is the vapour pressure of water in mmHg at the dew point (DPT) of air.

(iii) For DBT = 60 °C and RH = 10 % from Psychrometric chart :

(a) DPT = 17.5 °C, H = Humidity = 0.0125 kg water per kg dry air.

(b) WBT = 29 °C and saturation humidity = 0.026 kg water per kg dry air.

(It is expected that students should read the above cited properties of air-water system from the Psychrometric chart corresponding to DBT of 60 °C and RH of 10 %.)

Solution : Dry density of solid = 1.38 g/cc

Sheet size : 1 m by 1 m i.e. 100 cm by 100 cm

Thickness of the sheet = 6 cm

Volume of the sheet = $100 \times 100 \times 6 = 60000$ cm^3

Dry weight of the sheet = S_S = volume of the sheet × dry density

$= 60000 \times 1.38 = 8.28 \times 10^4$ g

Drying surface = A = $100 \times 100 = 10^4$ cm^2

(Drying from one surface only as the sample is in the insulated tray.)

At DBT = 60 °C and 10 % RH

DP temperature = 17.5 °C and

the corresponding absolute humidity is

H = 0.0125 kg water/kg dry air

H_m = molal humidity

We have, $H = 0.622\, H_m$

or $$H_m = \frac{H \times 1/18}{1/29} = \frac{29\,H}{18} = \frac{29 \times 0.0125}{18} = 0.02014 \ \frac{\text{kmol water}}{\text{kmol dry air}}$$

(Mol. wt. of water = 18 and Mol. wt. of dry air = 29)

We have, $$H_m = \frac{p_A}{P - p_A}$$

where, P = 101.325 kPa (atmospheric pressure)

p_A = partial pressure of water vapour

$P - p_A$ = partial pressure of dry air

$0.02014 = p_A/(101.325 - p_A)$

$\therefore$ $p_A = 1.9979$ kPa $= 14.986$ mmHg

At DPT (dew point) :

Vapour pressure of water = partial pressure of water vapour

$\therefore$ $p_g = 14.986$ mmHg

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Check : At DPT of 17.5 °C, from the vapour pressure data of water as a function of temperature (from any standard reference book),

Vapour pressure of water = 1.9994 kPa = 14.9967 mmHg

So, the calculated value of vapour pressure (14.986 mmHg) from the absolute humidity seems to be correct as it is very close to the value cited in a standard reference book.

At WBT of 29 °C, saturation humidity = 0.026 kg water vapour/kg dry air.

$$\text{Molal saturation humidity} = \frac{0.026 \times 1/18}{1 \times 1/29}$$

$$= 0.04189 \frac{\text{kmol water vapour}}{\text{kmol dry air}}$$

or $H_S = 0.622\, H_{S\,(m)}$

$H_{S\,(m)} = H_S/0.622 = 0.026/0.622 = 0.0418$

Molal saturation humidity $= p_A^o/(P - p_A^o)$

where, p_A^o = vapour pressure of water

P = total pressure (atmospheric) = 101.325 kPa

$\therefore \quad 0.04189 = p_A^o/(101.325 - p_A^o)$

$\therefore \quad p_A^o = 4.0738 \text{ kPa} = 4.0738 \times \frac{760}{101.325} = 30.556 \text{ mmHg}$

Vapour pressure of water at WBT of 29 °C $= p_A^o = p_i = 30.556$ mmHg.

Check : At WBT of 29 °C, from the vapour pressure data from standard reference,

Vapour pressure of water at 29 °C = 4.004 kPa = 30.032 mmHg

So, the value of vapour pressure of water calculated from the humidity data is very close to the value cited in a standard reference.

We will use the values of vapour pressures obtained from absolute humidity values.

$\therefore$ For further calculations :

$p_g = 14.986$ mmHg

$p_i = 30.556$ mmHg

The rate of drying in g/(cm^2.h) in constant rate period is given by

$$N_C = 0.004\,(u)^{0.8}\,(p_i - p_g)$$

where $u = 10$ m/s, $p_i = 30.556$ mmHg, $p_g = 14.986$ mmHg

$$N_C = 0.004\,(10)^{0.80}\,(30.556 - 14.986)$$

$$= 0.39296 \approx 0.393 \text{ g/(cm}^2\text{.h)}$$

Critical moisture content of sheet $= X_C = 0.35$ kg water/kg dry solid

Initial moisture of sheet = 55 % (wet basis)

Initial moisture content of sheet $= X_1 = 55/(100 - 55) = 1.222$

Final moisture of sheet = 12 % (wet basis)

Final moisture content of sheet $= X_2 = 12/(100 - 12) = 0.1364$

Equilibrium moisture of sheet = 5 % (wet basis)

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We have $N_C = 0.393$ g/(cm^2.h)

$S_S = 8.28 \times 10^4$ g, $A = 10^4$ cm^2

$X_C = 0.35$, $X_1 = 1.222$, $X_2 = 0.1364$ and $X^* = 0.0526$

As the final moisture content (X_2) is less than critical moisture content (X_C), drying of the sheet takes place in both the constant-and falling-rate period.

$\therefore$ Time of drying $= \theta = \theta_c + \theta_f$

$$\theta = \frac{S_S}{AN_C}\left[(X_1 - X_C) + (X_C - X^*)\ln\left(\frac{X_C - X^*}{X_2 - X^*}\right)\right]$$

$$= \frac{8.28 \times 10^4}{1 \times 10^4 \times 0.393}\left[(1.222 - 0.35) + (0.35 - 0.0526)\ln\left(\frac{0.35 - 0.0526}{0.1364 - 0.0526}\right)\right]$$

$$= 26.3756 \approx 26.4 \text{ h}$$

Time required for drying of sheet = 26.4 h ... Ans.

Ex. 6.18 : Filter cake is dried in a pan 30 cm × 30 cm × 2.5 cm. Air velocity is 75 cm/s. The initial moisture content is 0.2 kg water/kg dry solid and critical moisture content is 0.09 kg water/kg dry solid. The time of drying is 0.5 h. What will be the drying time required if the air velocity is increased to 150 cm/s ?

Solution : Case I :

Air velocity $= u_1 = 75$ cm/s $= 0.75$ m/s

$X_1 = 0.2$ kg water/kg dry solid (initial moisture content)

$X_C = 0.09$ kg water/kg dry solid (critical moisture content)

θ_c = time of drying = 0.5 h = 1800 s.

Drying of sample (filter cake) is done from X_1 to X_C i.e. in entire constant drying rate period.

The rate of drying is thus given by

$$N_{C1} = \frac{S_S}{A}\left(\frac{X_1 - X_C}{\theta_c}\right)$$

where, S_S – mass of dry solid not known.

So, assume $S_S/A = k$ (some constant).

$$N_{C1} = \frac{k\,(X_1 - X_C)}{\theta_c}$$

$$= \frac{k\,(0.20 - 0.09)}{1800} = 6.11 \times 10^{-5}\,K \text{ kg/(area·s)}$$

Case II : Air velocity $= u_2 = 150$ cm/s $= 1.5$ m/s

Here also the filter cake sample is to be dried from $X_1 = 0.20$ to $X_C = 0.09$ and we have to determine the time required for drying when the air velocity is doubled (i.e., 75 cm/s to 150 cm/s).

$\theta_{C2} = ?$

$N_{C2} = ?$

In the constant rate period, the rate of drying depends upon air velocity. If all other drying conditions are the same, dependency of the rate of drying on the air velocity is given by,

$$N_C \propto (u)^{0.80}$$

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When this is applied to cases (I and II) and combined, we get

$$\frac{N_{C2}}{N_{C1}} = \left(\frac{u_2}{u_1}\right)^{0.80}$$

$$\therefore \quad N_{C2} = N_{C1}\,(u_2/u_1)^{0.80}$$

$$= 6.11 \times 10^{-5}\,k\,(1.5/0.75)^{0.80}$$

$$= 1.064 \times 10^{-4}\,k \quad \text{kg/(area·s)}$$

We have : $$N_C = \frac{S_S}{A}\left(\frac{X_1 - X_C}{\theta_C}\right)$$

$$\therefore \quad N_C \propto 1/\theta_C$$

$$\therefore \quad N_{C1} \propto 1/\theta_{C1}$$

and $$N_{C2} \propto 1/\theta_{C2}$$

Combining these two equations, we get

$$\frac{N_{C2}}{N_{C1}} = \frac{\theta_{C1}}{\theta_{C2}}$$

$$\therefore \quad \theta_{C2} = \frac{N_{C1}}{N_{C2}} \cdot \theta_{C1}$$

$$= \frac{6.11 \times 10^{-5}\,k}{1.064 \times 10^{-4}\,k}(1800) = 1034\ s$$

$\therefore$ Time required for drying when the air velocity is 150 cm/s $= \theta_{C2} = 1034\ s = 0.287\ h$ **... Ans.**

Ex. 6.19 : Drying test of a food product was carried out in a insulated tray. The temperature of the dryer was 60 °C and a dew point was 20 °C. The dry weight of the material was 3.765 kg and the surface area 0.186 m^2.

Following data were obtained :

Time, h	0	0.4	0.8	1.40	2.2	3.0	4.2	5.0	7.0	9.0
Weight, kg	4.94	4.88	4.81	4.70	4.55	4.40	4.24	4.15	4.02	3.98

(a) Determine the drying rate in kg water/(m^2.h) as a function of free moisture content in the constant rate and in the falling rate period.

(b) Calculate the time required under the same conditions to dry the sample from the total moisture content of 0.25 to 0.10 kg/kg dry matter.

Solution :

Dry weight of the material = S_S = 3.765 kg

Drying surface = A = 0.186 m^2

For plotting the rate-of-drying curve, we have to calculate the moisture content, average moisture content and rate of drying from the data given.

(i) At θ = 0.0 h

Weight of the sample = 4.94 kg

Dry weight of the material = 3.765 kg

Weight of the moisture in sample = 4.94 – 3.765 = 1.175 kg

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(ii) At $\theta = 0.40$ h

Weight of the sample = 4.88 kg

Weight of the moisture in the sample = 4.88 – 3.765 = 1.115 kg

Moisture content of the sample = X = 1.115/3.765 = 0.296 kg/kg dry solid.

Average moisture content of the sample (during 0.0 and 0.40 h) $= X_{avg} = \dfrac{0.312 + 0.296}{2} = 0.304$ kg/kg dry solid.

$$\text{Rate of drying} = N = -\frac{S_S}{A}\frac{dX}{d\theta}$$

$$dX = 0.296 - 0.312 = -0.016 \text{ kg/kg dry solid}$$

$$d\theta = 0.40 - 0 = 0.4 \text{ h}$$

Keep in mind :

Difference in any parameter = final value – initial value

$$\therefore \quad N = -\frac{3.765}{0.186}\frac{(-0.016)}{(0.40)} = 0.8096 \approx 0.81 \text{ kg/(m}^2\text{·h)}$$

In this manner, calculate X, X_{avg} and N for remaining data and tabulate.

Time, h	Weight, kg	X	Avg.	X	Rate, N, kg/(m².h)
0	4.94	0.312			
			→	0.304	
0.4	4.88	0.296			0.81
			→	0.2867	
0.8	4.81	0.2775			0.936
			→	0.2629	
1.40	4.70	0.2483			0.985
			→	0.2284	
2.2	4.55	0.2085			1.01
			→	0.1885	
3.0	4.40	0.1686			1.01
			→	0.1474	
4.2	4.24	0.1262			0.715
			→	0.1142	
5.0	4.15	0.1022			0.607
			→	0.08495	
7.0	4.02	0.0677			0.349
			→	0.0624	
9.0	3.98	0.0571			0.107

Plot N v/s X_{avg}.

From Fig. E 6.19 :

Critical moisture content = $X_C = 0.1575$

Rate of drying in the constant rate period = $N_C = 1.0$ kg/(m²·h)

The rate of drying in the falling rate period as a function of free moisture is given as,

$$\text{Rate of drying in the falling rate period} = N = m(X - X^*)$$

$$= \frac{N_C (X - X^*)}{(X_C - X^*)}$$

$$= \frac{1.0 (X - X^*)}{(0.1575 - 0.0525)}$$

$$= 9.524 (X - X^*)$$

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Initial moisture content = X_1 = 0.25 kg water/kg dry solid

Final moisture content = X_2 = 0.10 kg water/kg dry solid

From Fig. E 6.19 : Read a value of X^* at which the falling rate curve meets the x-axis.

Equilibrium moisture content = X^* = 0.0525

We have : S_S = 3.765 kg, A = 0.186 m^2

As X_2 (0.10) is less than X_C (0.1575), the drying takes place in both constant-and falling-rate period. Therefore, the time required for drying is given by

$$\theta = \theta_c + \theta_f$$

$$\theta = \frac{S_S}{A\,N_C}\left[(X_1 - X_C) + (X_C - X^*)\ln\left(\frac{X_C - X^*}{X_2 - X^*}\right)\right]$$

$$= \frac{3.765}{0.186 \times 1.0}\left[(0.25 - 0.1575) + (0.1575 - 0.0525)\ln\left(\frac{0.1575 - 0.0525}{0.10 - 0.0525}\right)\right]$$

$$= 3.56\ \text{h}$$

Time required for drying = 3.56 h ... **Ans.**

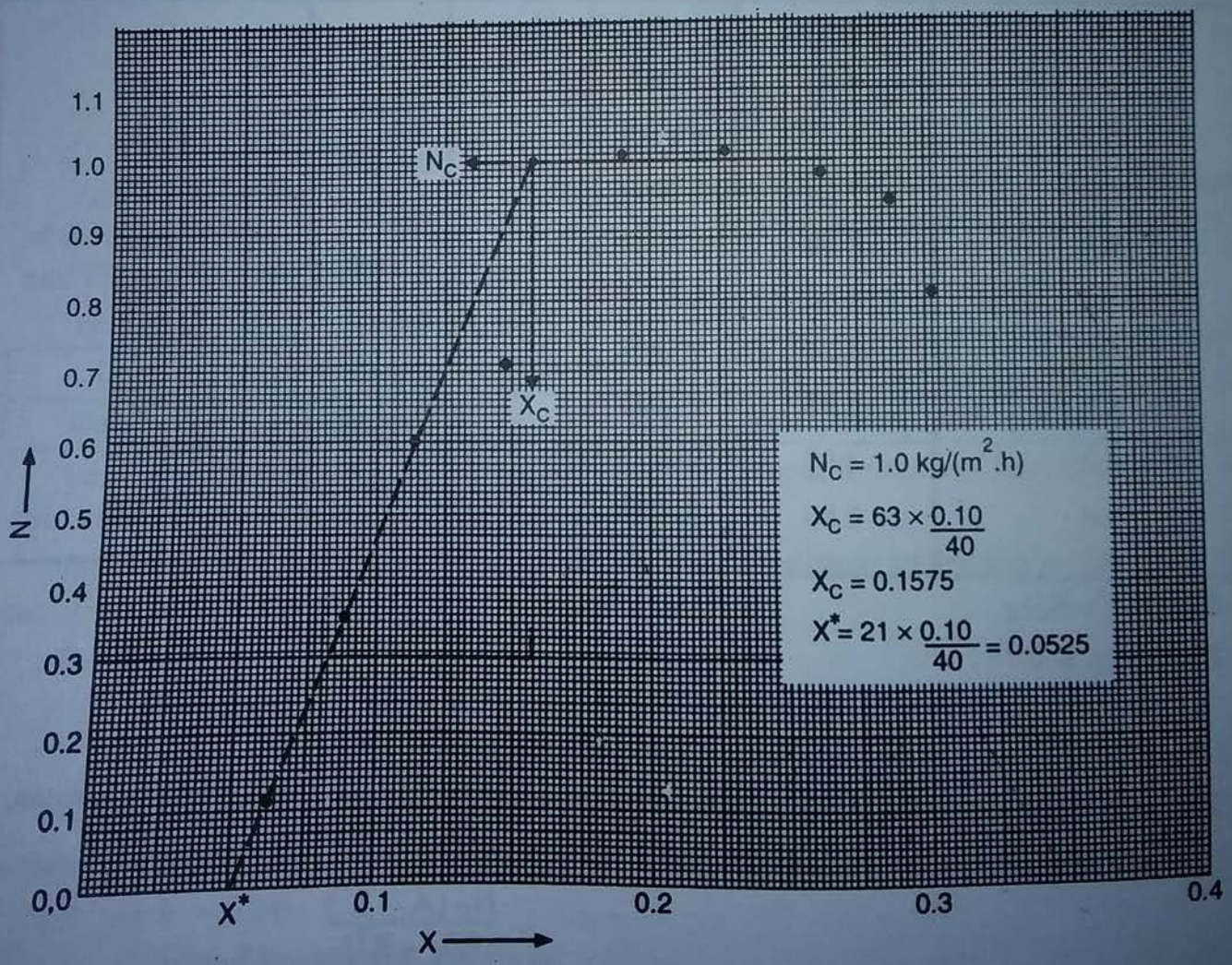

Fig. E 6.19

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APPENDIX - I

Extractive Distillation :

Extractive distillation is a distillation technique wherein a high boiling solvent is added to alter/change the relative volatilities of the components of a feed mixture. The alteration of volatilities is needed due to similarities in the vapour pressures of the components or the presence of an azeotrope. Since the boiling point of the solvent is much higher than those of components to be separated, the formation of new azeotropes is impossible. Furthermore, if any azeotropes present in the feed, it disappears in the presence of the solvent.

As there is no possibility of formation of azeotropes and the solvent can be recovered by simple distillation, extractive distillation is much simpler as well as more useful technique than azeotropic distillation.

The typical flow diagram for an extractive distillation for separation of toluene from methylcyclohexane [close boiling components] by using phenol as a solvent is shown in Fig. A. The volatility of the methylcyclohexane relative to the toluene is enhanced by the addition of phenol. This leads to separation of these components using fewer plates compared to an ordinary distillation. Methylcyclohexane is taken as an overhead product from the extractive distillation column.

For maintaining high concentration of solvent throughout most of the column (to maximise the difference in volatilities between the components to be separated), the solvent should be fed to the column above the feed plate, usually at a location few plates below the top plate. The solvent phenol does not form an azeotrope with toluene in the bottom product from the extractive distillation column. The mixture of phenol and toluene is then fed to the solvent recovery column where toluene is taken as a top product and phenol is withdrawn as a bottom product and recycled.

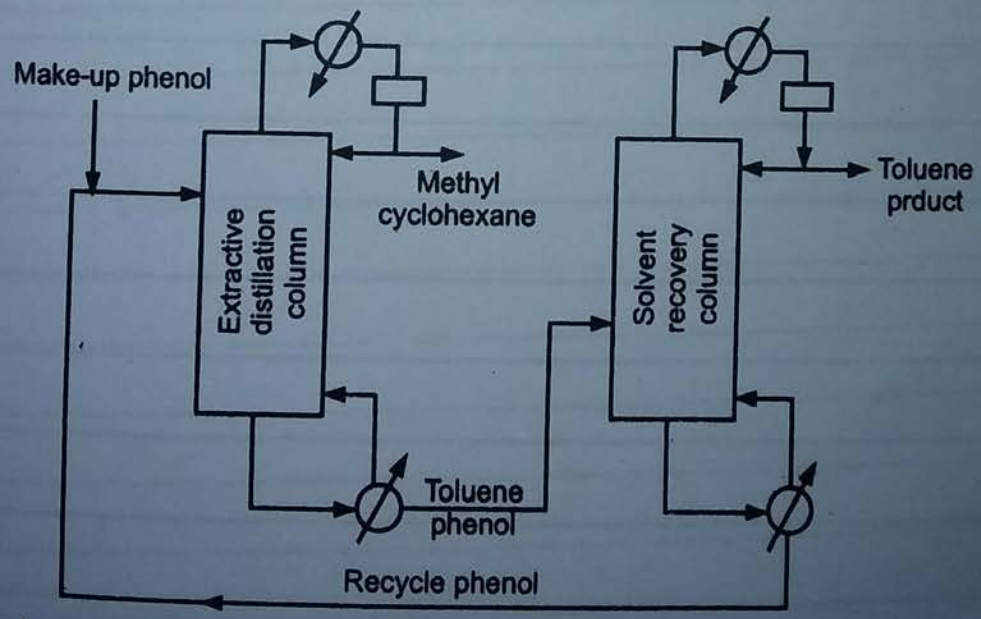

Fig. A : Flow diagram for an extractive distillation to separate toluene and methylcyclohexane

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